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1 Quant Course — Arbitrage Pricing & Derivatives

A fifteen-chapter study guide covering arbitrage pricing, stochastic calculus, equity derivatives, short-rate models, and rate derivatives. The guide is written for a self-study reader who is comfortable with undergraduate probability and real analysis and who wants a single coherent path from the one-period binomial model through Heston and swaptions. Formulas are typeset in LaTeX; baseline figures are regenerated from a single Python script and live inline next to the concepts they illustrate.

1.1 Table of Contents

1.1.1 Part I — Discrete-Time Models

#	Chapter	Description
1	One-Period Binomial	Single-period FTAP, replication, indifference pricing.
2	Multi-Period Binomial and the FTAP	Backward induction, CRR parameterisations, FTAP, American options.

1.1.2 Part II — Continuous-Time Models

#	Chapter	Description
3	Stochastic-Calculus Primer (Brownian Motion, Itô, SDE Solutions)	BM construction, quadratic variation, Itô integrals, Itô's lemma, OU/GBM SDEs.
4	Feynman-Kac and the SDE-PDE Bridge	FK zero-drift, discounting, state-dependent coefficients; martingale derivation.
5	Measure Changes, Radon-Nikodym, and Girsanov	RN derivative, density process, Girsanov, two-numeraire switching, T-forward measure.

1.1.3 Part III — Equity Derivatives

#	Chapter	Description
6	Dynamic Hedging I — Self-Financing + Black-Scholes PDE	Self-financing strategies, market price of risk, BS PDE and formula.
7	Dynamic Hedging II — Greeks, Delta-Gamma, Vega, Dividends	Delta-gamma hedging, vega, dividend-paying assets, put-call parity.
8	Forwards, Futures, and the Black PDE	Forward/futures pricing, Black PDE, Margrabe, convexity adjustment.
9	Risk Measures — VaR, CTE, Coherent Risk	VaR (historical/parametric/MC), CTE, coherent risk, Basel, backtesting.
10	Monte Carlo and Path-Dependent Options	SLLN, GBM generator, forward-starting and barrier options, variance reduction.

1.1.4 Part IV — Interest-Rate Models

#	Chapter	Description
11	Calibration — Lattices, Short Rates, Yield Curve	FTAP on lattices, RN on multinomial trees, rate-tree bootstrap, governance.
12	Short-Rate Models — Vasicek, Ho-Lee, Hull-White, Affine	Canonical Vasicek derivation, Ho-Lee, Hull-White, affine term structure.
13	Rate-Derivative Applications — Swaps, CDS, Bond Options, Callables	IRS, par swap rate, CDS, callable bonds, bond options via T-forward measure.

1.1.5 Part V — Stochastic Vol & Rate Derivatives

#	Chapter	Description
14	Heston Model (Stochastic Volatility)	Heston SDE, characteristic-function pricing, Feller condition, Fourier inversion.
15	Caps, Floors, and Swaptions	Caplets under T-forward measure, Black-76, swaptions under annuity measure.

1.2 Course Arc

```
Part I   Discrete Time
  CH01 One-Period Binomial


  CH02 Multi-Period Binomial + FTAP

Part II   Continuous Time

  CH03 Stochastic-Calculus Primer     foundation for all that follows



  CH04 FK      CH05 Measure Change / Girsanov


Part III   Equity Derivatives

        CH06 BS PDE (Dynamic Hedging I)



  CH07   CH08    CH09         CH10
  Greeks Fut/Fwd Risk/VaR    Monte Carlo

Part IV   Interest-Rate Models

  CH11 Calibration   CH12 Short-Rate Models   CH13 Rate Derivatives

Part V   Stoch Vol & Rate Derivatives

                  CH14 Heston   CH15 Caps/Floors/Swaptions
```

CH03 is the single prerequisite spine: every chapter from CH04 onward rests on its Itô calculus. CH05's Girsanov machinery is cited (not re-derived) by CH06, CH08, CH12, and CH13. Vasicek is derived once, in CH12, and cross-referenced elsewhere.

1.3 Regenerating the Figures

```
pip install matplotlib numpy scipy
python docs/guide/build_figures.py
```

Writes baseline PNGs under `docs/guide/figures/`.

2 Chapter 1 — One-Period Binomial: Utility + No-Arbitrage

This chapter merges two overlapping lectures on the one-period binomial model. The first thread runs through utility theory and indifference pricing — showing that naive expected-value pricing fails for risk-averse agents, and that the familiar risk-neutral pricing formula emerges as the vanishing-risk-aversion limit of indifference pricing. The second thread is the no-arbitrage / replication argument that makes no reference to preferences, and pushes through to multi-period trees, calibration to log-normal drift/variance, and the CLT limit that gives geometric Brownian motion. The two derivations are deliberately placed side by side: utility first (because it motivates why we want a "risk-neutral" measure at all), then no-arbitrage (because it delivers the same measure cleanly and extends to the rest of the book).

> Narrative spine. A derivatives price cannot be the discounted real-world expectation of its payoff; that would require a utility function, a risk premium, and disagreement across investors. The binomial model shows that a *different* probability measure — the one making the hedging portfolio break even on average — is the only pricing measure that rules out free lunches. Everything in continuous time (Black–Scholes, Heston, HJM, Margrabe) is a limiting case of what you see on the next few pages.

A word on the mindset this chapter demands. Derivatives pricing is a subject that rewards slow reading, because every term in every equation carries economic meaning that is easy to skip past and hard to recover later. We will revisit the same identity from three or four different angles — as a utility limit, as a no-arbitrage consequence, as a replication cost, and as a martingale property — not because the identity changes but because each viewpoint teaches a different lesson about *why* the identity holds. A reader who grasps all four viewpoints simultaneously has internalised the entire conceptual engine of modern derivatives pricing. Everything after this chapter is, in a precise sense, implementation detail.

It is also worth saying explicitly what we are *not* doing in this chapter. We are not using stochastic calculus, measure-theoretic probability, or Itô's lemma. All the heavy-lifting theorems will appear later. The binomial model is deliberately constructed to sidestep these tools while still delivering every conceptual ingredient of the continuous-time theory. If you have seen Black–Scholes derived via a PDE and found the argument opaque, reading the corresponding binomial derivation first tends to clarify enormously: the PDE version is the same argument, rewritten in the language of infinitesimals. The binomial version lets you inspect each step with arithmetic rather than analysis.

2.0.1 Why start with a binomial tree at all

Before we plunge into formulas it is worth stating, in plain language, what this chapter is buying us. Real-world derivatives markets are full of continuous prices, continuous time, calendar curves, volatility surfaces, interest rate term structures, and traders who speak entirely in Greeks. It is tempting to leap directly to Black–Scholes, Heston, or a full affine-jump-diffusion model. We resist that temptation for three reasons.

First, a two-state, one-step model is the smallest non-trivial example where every conceptual ingredient of modern derivatives pricing already appears: a source of randomness, a bond, a risky asset, a contingent claim, a replicating portfolio, a risk-neutral measure, and a no-arbitrage bound. Reducing the machinery to this minimum lets us see the *skeleton* before the muscles are attached. In continuous-time treatments the skeleton is buried under the technicalities of stochastic integration, filtrations, and measurable sets; the binomial model strips all that away and lets us stare directly at the structural bones.

Second, the argument that price equals replication cost is *discrete* in spirit, even in continuous time. When we later prove Black–Scholes by stochastic calculus, the step "the payoff is replicable by a self-financing strategy, so its price equals the initial value of that strategy" is the exact same step we establish here. Get it right in one period and the multi-period and continuous-time extensions become bookkeeping, not conceptual leaps. This point cannot be overstated: the majority of confusion students feel when they first encounter Black–Scholes comes from missing this equivalence. Once you know that the PDE is just a continuous-time restatement of "price equals replication cost," the formula loses its mystery and becomes, at worst, a calculation.

Third, the binomial model is how almost everyone in the industry still thinks about exotic options, especially American-style and path-dependent structures. Production pricing engines still use binomial and trinomial lattices for Bermudan swaptions, callable bonds, and employee stock options. Understanding the one-period tree is not a pedagogical toy — it is the atom out of which those industrial tools are built. A quant who has internalised the one-period logic can walk into a trading desk and immediately understand why the lattice code is structured the way it is: the backward induction, the early-exercise comparison, the numeraire choice, the discounting at each node —

all of these are direct consequences of the material covered in this single chapter.

There is a fourth reason that is harder to articulate but is worth naming. The binomial tree is *beautiful* in the mathematical sense: every assumption has a clear role, every formula has a geometric interpretation, and the transition from discrete to continuous is smooth and illuminating. Subjects that are beautiful tend to be the ones you come back to when you are confused about a harder problem. In the decades following your first encounter with the binomial tree, you will repeatedly find yourself sketching a two-node tree on a napkin to sanity-check a much more complicated calculation. The tree is therefore not just a pedagogical device but a tool you keep in your pocket for the rest of your career.

2.0.2 How this chapter is organized

We proceed in four movements. The first movement (§1–§4) takes the naïve approach to pricing a single risky payoff: assume you have preferences, write down a utility function, and demand that the investor be indifferent between paying the price now and keeping their wealth certain. This produces the *indifference price*. We will see that it depends on an agent's risk aversion parameter, that it produces a bid–ask spread, and that it collapses to discounted expected value only in the limit of zero risk aversion. A market cannot run on this price — every agent has a different one. But studying it first gives us a crucial calibration: it tells us *what* the no-arbitrage price is solving, so that when we meet the no-arbitrage price in §5 we understand why it is the right answer.

The second movement (§5–§9) abandons preferences and replaces them with no-arbitrage. We introduce a second asset (a numeraire, typically a bond), define arbitrage precisely, and show that the mere absence of arbitrage implies the existence of a synthetic probability — the risk-neutral measure — under which relative prices have zero drift. Any contingent claim can then be priced by computing its expected payoff under this measure and discounting. This *replication* argument is the engine of all derivatives pricing. The surprise, which we will emphasise repeatedly, is that by demanding something weaker than full preference specification (only that free lunches are impossible), we obtain something stronger (a price that is unique and agent-independent).

The third movement (§10–§12) scales the tree up: we extend to many periods, calibrate the step size to match a log-normal drift and variance, and show that the terminal distribution converges to a Gaussian log-return — equivalently, that the asset becomes geometric Brownian motion in the continuous-time limit. This is how continuous models emerge as limits of discrete ones. The reader unfamiliar with the central limit theorem's role in finance will see it here in its most important application: carefully scaled sums of coin flips produce, in the limit, the log-normal distribution that underlies every standard option pricing model.

The fourth movement (§13–§14) closes the circle. We give a clean explicit proof that the existence of a risk-neutral measure implies no-arbitrage, and we close the calibration argument with a clean central-limit theorem argument using characteristic functions. By the end of the chapter you will have derived, from scratch and without invoking stochastic calculus, every ingredient needed to read an option chain: the meaning of the implied volatility that underlies every quote, the hedge ratios every trading desk uses, the martingale structure every risk system assumes, and the log-normal benchmark against which all real-world pricing anomalies are measured.

A final note on reading order. The four movements are logically sequential, but they are not equally difficult. Movements one and two are elementary in their mathematical demands but deep in their conceptual content — the hard work there is absorbing what the formulas *mean*. Movement three is more demanding algebraically (Taylor expansions, careful limits) but conceptually simpler: it is a bridge from the discrete to the continuous. Movement four is the most technical, using characteristic functions to nail down the limit rigorously. Readers short on time can read §1–§9 with great care and skim §10–§14, and will have absorbed ninety percent of the conceptual content of the chapter.

2.1 1. The One-Period Binomial Tree

Before writing anything down formally, let us set the scene. We want the simplest possible model in which time passes, prices can change, and a rational agent might want to take a position. "Simplest" here means: one decision point, one uncertain outcome, a minimum of moving parts. A *single* risky asset with *two* possible future values, observed over *one* period of time, is the minimum. We will refer to this setup as the one-period binomial tree, or sometimes just "the tree."

One-period binomial tree

Figure 1: One-period binomial tree

Consider a single risky asset A observed at two dates $t = 0$ and $t = 1$. At $t = 1$ the price takes one of two values, A_u (up) or A_d (down), with real-world up-probability

$$p \in (0, 1). \tag{1.1}$$

Why this matters. The condition $p \in (0, 1)$ — not $p \in [0, 1]$ — means both states are genuinely possible under the real-world measure $\mathbb{P}$. If $p = 0$ or $p = 1$ the "tree" collapses to a deterministic path and arbitrage becomes trivial. Strict positivity of both branch probabilities is what makes the market risky, and what later forces the equivalent martingale measure $\mathbb{Q}$ to assign strictly positive probability to both branches as well.

It is important to dwell on what this condition *does not* say. It does not say that the two probabilities are equal. It does not say that the up-move and down-move are of equal magnitude. It does not say anything about how rational or irrational any agent's beliefs about p might be. All it says is that, under whatever measure we decide to call $\mathbb{P}$, both futures are reachable. The usefulness of this minimal condition is that it already rules out deterministic models disguised as random ones. A "risky" asset with $p \in \{0, 1\}$ is just a bond in disguise, and attempting to price options on a bond leads nowhere. An option written on a deterministic process has a deterministic payoff, and "pricing" it reduces to discounting a known number — not a subject worth a chapter.

There is a deeper philosophical point buried here. Classical probability theory is comfortable with events of probability zero; measure theory is built around negligible sets. But for asset pricing we will insist on *equivalent* measures — measures that agree on which events are negligible. If $\mathbb{P}$ gives probability zero to the down-move, then no equivalent measure can give it positive probability. This would prevent the down payoff from contributing to any price under $\mathbb{Q}$, and the pricing theory would collapse. The condition $p \in (0, 1)$ is our insurance policy against that collapse.

An observation about subjectivity. The probability p is a modeller's choice. Two different traders, looking at the same asset, can have genuinely different p. One might be a permabull who estimates $p = 0.7$, another a permabear at $p = 0.3$. Remarkably, as we will see, the *fair price* of any derivative on A does not depend on p at all — only on the payoff values A_u and A_d and the structure of the market. This is the most counterintuitive and most useful feature of no-arbitrage pricing. It says that agreement on fundamentals (the tree structure) trumps disagreement about likelihoods (the probabilities). We will spend several sections motivating and exploiting this feature.

A remark on the one-period horizon. Saying the asset is "observed" only at $t = 0$ and $t = 1$ is a modelling choice. Nothing forbids the price from fluctuating wildly *between* those two dates — it is only at the two endpoints that we care about its value. This is adequate for a European-style claim, whose payoff depends only on the terminal price. Path-dependent claims, and American-style claims with early exercise rights, will require multi-period trees (§10) to capture the intermediate behaviour.

ASCII diagram of the A-tree:

```
                A_u   (prob. p)


    A_0


                A_d   (prob. 1-p)

  t = 0          t = 1
```

One-period binomial tree.

The (random) time-1 price can be written as a function of a Bernoulli indicator $x_1 \in \{1, 0\}$:

$$A_1 = A_u\, x_1 + A_d\,(1 - x_1), \qquad x_1 = \begin{cases} 1 & \text{w.p. } p \\ 0 & \text{w.p. } 1 - p. \end{cases} \tag{1.2}$$

Reading the indicator form. Writing A_1 as a linear combination $A_u x_1 + A_d(1-x_1)$ makes A_1 a deterministic function of a single random coordinate x_1. That single coordinate generates the whole filtration $\mathcal{F}_1$, which means the market is one-dimensional in its uncertainty: one source of randomness, one degree of freedom to hedge it with. That is exactly the condition that later yields a *unique* risk-neutral measure (market completeness).

The notion of "one source of randomness, one hedging instrument" is one of the most important conceptual observations in this whole chapter. In continuous time it becomes the statement "one Brownian motion, one tradable risky asset" and reappears as the reason Black–Scholes delivers a unique price. When you enter markets with more sources of risk than hedging instruments — stochastic volatility, credit migration, interest-rate smiles — uniqueness fails and an ambient model choice must step in to select a particular $\mathbb{Q}$. But here, in the simplest setting, one coin and one risky asset are perfectly matched, and completeness falls out for free.

A good way to see why completeness hangs on dimension-matching: imagine replacing the Bernoulli x_1 with a three-valued random variable, so that $A_1 \in \{A_u, A_m, A_d\}$ at $t = 1$. Now you have three unknown payoffs to match, but still only two asset positions (α, β) to move around. The system is under-determined; generically you cannot replicate a claim with arbitrary payoffs on three states. Multiple risk-neutral measures exist, pricing is no longer unique, and you have just written down the simplest incomplete market. Completeness is therefore a *counting argument* at its core, and the binomial tree is set up precisely so the count works out.

Remark on the algebraic form. The expression $A_u x_1 + A_d(1-x_1)$ is a convex combination of the two possible values whenever $x_1 \in [0,1]$, but here x_1 only takes the extreme values 0 or 1. This means A_1 is always equal to one endpoint or the other — never anything in between. A continuous-time model, by contrast, smooths A_1 across the interval through the diffusion process. Both models converge to the same large-scale behaviour (§14), but they have very different microscopic structure. The binomial model discretises the randomness into a single coin flip; the Brownian model spreads it across infinitely many infinitesimal flips. The former is easier to understand; the latter is easier to compute with.

A small observation for later. If we write $A_1 - A_d = (A_u - A_d)x_1$, the random part of A_1 is a scalar multiple of x_1. The scalar $A_u - A_d$ is the *payoff range* of the asset, and it will appear in the denominator of nearly every hedging formula in this chapter. Whenever you see a hedge ratio in the form "something over $(A_u - A_d)$", you are seeing a discrete derivative: "how much does the target move per unit of underlying move." This is the finite-difference cousin of the continuous-time delta, and it is the reason the binomial tree is an intuitive stepping-stone to the Greeks of Black–Scholes.

The instructor annotates an intuition formula on the tree (in purple):

$$A_0 \;=\; \frac{p\,A_u + (1-p)\,A_d}{1+r}, \qquad A_0 \in (A_d, A_u), \tag{1.3}$$

i.e. the naïve "discounted expectation under $\mathbb{P}$" guess for what A_0 should be. Here r is the one-period risk-free rate and $1/(1+r)$ discounts back to today. We will see that (1.3) is *not* how prices are actually determined — $\mathbb{P}$ does not price; only a risk-neutral measure $\mathbb{Q}$ does. A real-world expected discount works only if investors are risk-neutral. Any risk-averse investor would pay *less* than (1.3) for a volatile asset, and any risk-seeking one would pay more.

Why the formula is "seductive but wrong". Equation (1.3) has every cosmetic feature one expects of a pricing rule: it is linear in the payoffs, it discounts the future by the risk-free rate, it collapses to trivial in the deterministic limit, and it looks like a textbook expectation. Yet it cannot be a pricing rule, because applying it consistently would imply that every investor — regardless of their risk tolerance, wealth, or horizon — agrees on exactly the same price. Empirically they do not. Even absent differences in preferences, equation (1.3) is silent on how the asset's own required return is determined. If A is equity and the expected return is 10% with $r = 3\%$, then (1.3) *under-prices* A relative to where it trades, because $\mathbb{E}[A_1]/(1+r)$ implicitly assumes A earns the risk-free rate. It does not — it earns the risk-free rate plus a premium. That premium's size is what distinguishes the physical measure from the risk-neutral measure.

Limiting-case sanity. Set $A_u = A_d = A^\star$. Then (1.3) returns $A_0 = A^\star/(1+r)$ regardless of p. In other words, a deterministic payoff must be the discounted bond price — the one case in which naïve expected-discounted-value is correct. What (1.3) cannot do is tell you how to interpolate between the deterministic case and the genuinely risky case; the correct interpolation requires a second asset and the concept of replication.

Another illuminating limit. Let $p \to 1$ with A_u and A_d held fixed. Equation (1.3) would give $A_0 \to A_u/(1+r)$, implying that an asset that is *almost surely* going to end up at A_u trades at its discounted value — the forward price.

Similarly $p \to 0$ gives $A_0 \to A_d/(1+r)$. Both of these limiting prices are sensible (they match the deterministic limit from earlier), but observe that the formula *interpolates linearly* between them as p varies. Linearity in p is exactly what makes the formula suspicious: the amount an agent is willing to pay for a risky payoff is emphatically *not* linear in the probability of a good outcome, once risk aversion enters. We need a tool that breaks this spurious linearity, and that tool is utility theory (§2) — or, equivalently, the no-arbitrage argument (§5).

Who this formula is for. There is a specific agent for whom (1.3) is exactly right: the risk-neutral agent, for whom expected payoff is all that matters. Such an agent exists only in textbooks (and in toy models), but the fiction is enormously useful. It is, as we will see, the agent whose preferences match the "synthetic" probability measure $\mathbb{Q}$ that no-arbitrage pricing extracts from prices. The real-world probability $\mathbb{P}$ and the real-world agent are messy and disagree with each other; the risk-neutral agent under $\mathbb{Q}$ is clean and unique. Pricing theory is in large part the art of substituting the latter for the former.

2.1.1 1.1 Two numerical examples — why expected value is not enough

Two symmetric lotteries with identical $\mathbb{P}$-expectation but very different "felt" risk:

Example 1 (symmetric small bet):

$$p = \tfrac{1}{2}, \qquad A_u = 100, \qquad A_d = -100. \tag{1.4}$$

Example 2 (symmetric huge bet):

$$p = \tfrac{1}{2}, \qquad A_u = 10^6, \qquad A_d = -10^6. \tag{1.5}$$

Both lotteries have identical expected payoff (zero, ignoring r for the illustration), yet almost no one would price them the same — the second is existentially ruinous. This motivates utility theory: *expected value is not a satisfactory pricing rule for a risk-averse agent.*

Take a moment to sit with the absurdity of treating these two gambles as equivalent. Example 1 is a quiet evening's entertainment at a casino — the kind of bet someone with a reasonable salary would think nothing of. Example 2 is a bet that, if lost, wipes out most or all of an individual's lifetime wealth; if won, it lifts them into an entirely different social and economic stratum. Yet *both* have zero expected payoff under $p = 1/2$. If you accepted (1.3) as a pricing rule, both gambles would trade at zero. Nobody behaves that way. The reason is that *consequences scale nonlinearly with wealth*: the marginal utility of a dollar lost when you are already near zero is vastly larger than the marginal utility of a dollar gained when you are already very rich.

This is the empirical content that utility theory will capture. Whether you are pricing an equity, an index option, or an exotic credit derivative, the price you quote depends not just on the distribution of payoffs but on the *curvature* of your utility function over wealth. The miracle of no-arbitrage pricing, which we reach in §5 and beyond, is that this curvature *cancels out* as soon as the payoff can be replicated by trading — because replication forces the price to depend only on hedging cost, not on feelings about wealth. Derivatives pricing is therefore a partial escape from the mess of preferences, valid precisely to the extent that hedging is possible.

A thought experiment to sharpen the intuition. Imagine that you are offered Example 2 and told: "to decide whether to accept, you have thirty seconds and no access to information about whether you would survive the downside." Most people's stomachs would refuse the gamble outright, regardless of its zero-expected-value property. Now imagine the same gamble offered ten thousand times in sequence, independently, with the constraint that after each draw the agent keeps their current cash. Now the law of large numbers applies, cancelling out on average, and the gambles aggregate to something close to zero at almost no risk. *The same lottery has dramatically different pricing consequences depending on whether it is one-shot or repeated*, and this is not captured by its expected value alone.

Why this matters for derivatives pricing. An option on a stock is typically a *one-shot* gamble for the holder (if they are an end-user, not a market maker hedging a book). This is why real-world option prices reflect risk aversion, even after all the theory in this chapter says they should be preference-free. The "real price" deviates from the model price in ways that encode market-level risk aversion; quantifying that deviation is the subject of the *risk premium* literature, and it is why the implied volatility surface is higher than the realised volatility in most markets. Market makers, who hold the mirror image of the end-user's book, tend to hedge away much of their exposure — bringing their *effective* preference closer to risk-neutral. The theoretical risk-neutral price is the market-maker's fair value; the deviation is the spread the end-user pays for offloading risk.

2.2 2. Utility Functions and the Monotonicity Axiom

Having convinced ourselves that expected value alone cannot price gambles, we need a way to formalize how an agent evaluates random wealth. The natural mathematical object is a *utility function*: a scalar map that takes a level of wealth and returns a number expressing how much the agent "likes" that wealth. Preferences between random outcomes are then encoded through expected utility. The axioms below are the minimal conditions that make this machinery coherent.

Let X_1, X_2 be two random outcomes. The fundamental preference axiom is:

$$X_1 \leq X_2 \quad \Rightarrow \quad \mathbb{E}[\,u(X_1)\,] \;\leq\; \mathbb{E}[\,u(X_2)\,], \tag{2.1}$$

where $u(\cdot)$ is the utility function.

Unpacking the axiom. Equation (2.1) is almost tautological: if one random outcome pointwise dominates another, then no rational agent should prefer the dominated one. The subtlety is the word "pointwise" — $X_1 \leq X_2$ means that in *every* state of the world, the realized value of X_1 is at most the realized value of X_2. This is strong. It does *not* say that the mean or median or any other summary statistic of X_1 is less than that of X_2; it says state-by-state dominance. For any such pair, the axiom demands that expected utility respect the dominance. If it did not, we could construct an argument for preferring strictly less money in every state — a transparently absurd preference.

This axiom, sometimes called *first-order stochastic dominance preservation*, is extremely weak and yet does most of the work in utility theory. Any u that violates it is wrong on economic grounds, regardless of mathematical elegance. Any u that satisfies it is a candidate utility function, subject to a few additional shape constraints discussed next.

A subtle point about what the axiom does not do. Monotonicity at the outcome level (more money in every state is better) does not immediately translate into monotonicity at the distribution level (a stochastically dominating lottery is better). The latter requires the agent to compute expected utility correctly; it does not emerge from mere coherence. For example, if an agent uses a non-expected-utility framework with cumulative prospect weighting, they can violate stochastic dominance in certain carefully constructed situations. Within the expected-utility framework, however, (2.1) does translate smoothly: if $X_1 \leq X_2$ almost surely, then $\mathbb{E}[u(X_1)] \leq \mathbb{E}[u(X_2)]$ follows from monotonicity of u combined with monotonicity of expectation. The elegance is that one very mild shape constraint on u delivers consistent behaviour across *all* payoff distributions.

Why the axiom is a starting point, not an ending point. Monotonicity rules out absurd preferences but does not pin down a utility function. Infinitely many functions u satisfy it — indeed any strictly increasing function does. The additional structure we need for a useful theory is specified next: concavity, to capture risk aversion; a particular functional form (exponential, power, log), to make the algebra tractable. Each layer narrows the class of admissible utilities while preserving the monotonicity baseline.

2.2.1 2.1 Properties of u

With the axiom in hand, we now specialize to utility functions that capture the empirical content of risk aversion. Two shape constraints turn out to be both natural and sufficient for most modeling:

$$u(x) : \mathbb{R} \longmapsto \mathbb{R}, \qquad u \text{ increasing, } u \text{ concave.} \tag{2.2}$$

- *Increasing* $\Leftrightarrow$ more wealth preferred to less.
- *Concave* $\Leftrightarrow$ diminishing marginal utility $\Leftrightarrow$ risk aversion.

The two conditions, decoded. Monotonicity (increasing) is the weakest possible rationality condition: we would never model a rational agent as preferring poverty to wealth, all else equal. It is really the calculus version of the axiom (2.1) applied to deterministic outcomes. Concavity is more subtle and does more work. It says that each additional dollar of wealth is worth less to the agent than the previous one. Equivalently, by Jensen's inequality, concavity implies $\mathbb{E}[u(X)] \leq u(\mathbb{E}[X])$: the utility of the expectation exceeds the expected utility. The certain version of the random wealth is always weakly preferred to the random version. That, precisely, is risk aversion.

Why diminishing marginal utility is equivalent to risk aversion. If the agent is willing to accept a small premium to avoid a fair bet, then they must value each additional dollar slightly less than the previous one; otherwise a perfectly

fair coin toss would be a free utility win. Running this argument the other direction: if marginal utility is strictly decreasing, then small coin tosses around the current wealth level always have negative expected utility, so the agent will pay to avoid them. The two statements — "risk averse" and "concave utility" — are economically equivalent.

A remark on unboundedness. Most of the classical utility functions — logarithmic, power, exponential — are unbounded on at least one side of their domain. This has real consequences: unbounded utilities allow for pathological constructions like the St. Petersburg paradox, in which a gamble with finite price has infinite expected utility. Practitioners usually avoid these pathologies by restricting the support of wealth to a reasonable interval, or by using utilities bounded above. For our purposes the binomial model keeps wealth bounded and the pathologies never surface.

Curvature measures and risk-aversion coefficients. Once you have a concave u, the natural question is "how concave?" The standard curvature measures are the *absolute* risk aversion $A(x) = -u''(x)/u'(x)$ and the *relative* risk aversion $R(x) = -xu''(x)/u'(x)$. Absolute risk aversion governs how the agent reacts to *dollar* gambles (a coin toss for $\pm\$100$); relative risk aversion governs how they react to *percentage* gambles (a coin toss for $\pm 10\%$ of wealth). Empirical evidence suggests that relative risk aversion is roughly constant across wealth levels for typical agents (values of R between 1 and 10 are commonly reported), while absolute risk aversion decreases with wealth (richer people take bigger dollar bets). The exponential utility of (2.3) has constant *absolute* risk aversion, which is the opposite of what wealthy people exhibit, but it is used because it makes the math clean. Power utility $u(x) = x^{1-\rho}/(1-\rho)$ has constant *relative* risk aversion ρ, more empirically realistic, but the indifference price depends on wealth and has no closed form. This modelling tradeoff — realism versus tractability — is pervasive.

Concavity, geometrically. Draw the graph of u and pick any two points $(x_1, u(x_1))$ and $(x_2, u(x_2))$ on it. Concavity says the chord connecting these points lies *below* the curve, whereas a convex function's chord lies above. The height of the curve above the chord at a point is exactly the quantity Jensen's inequality measures: the "utility benefit of certainty." If the chord is nearly touching the curve (nearly linear u), the benefit is small — the agent is nearly risk-neutral. If the chord is far below the curve (very curved u), the benefit is large — the agent is very risk-averse. This geometric picture is the cleanest way to internalise what concavity is doing in (2.2).

Diagram (concave, increasing u) with two deterministic wealth levels $x_1 < x_2$ evaluated on the curve:

```
u(x)

                                    u(x_2)
                          .''
                      .''
                   .'
                .'
             .'          u(x_1)
          .'
        .'
      .'
    .'
                        → x
                x_1          x_2
```

```
Concave u: gains diminish with wealth (u'' < 0).
The chord from (x_1, u(x_1)) to (x_2, u(x_2)) lies below the curve;
Jensen's inequality measures that gap.
```

2.2.2 2.2 Canonical exponential utility

Among all concave increasing utilities, one family is so analytically convenient that it dominates theoretical treatments of indifference pricing: the (negative) exponential utility. The running example used throughout the

$$u(x) \;=\; -e^{-\gamma x}, \qquad \gamma > 0. \tag{2.3}$$

Here γ is the (constant absolute) risk-aversion coefficient. Larger $\gamma \Rightarrow$ more risk averse.

Why exponential utility? It is the unique utility (up to affine transformations) with *constant absolute risk aversion*: $-u''(x)/u'(x) = \gamma$ independent of x. Equivalently, the agent's behaviour over small bets does not change as wealth changes. This is not quite realistic — real agents typically exhibit *decreasing* absolute risk aversion, becoming less timid as they get richer — but it has a crucial analytical virtue: for normal (or, here, binomial) wealth increments, expected exponential utility factorizes into a mean-plus-variance formula that makes explicit pricing tractable. Almost every indifference-pricing result in mathematical finance that admits a closed form uses exponential utility for this reason.

Reading the sign. The leading minus sign in $-e^{-\gamma x}$ is not a convention-choice, it is forced. Without it, u would be decreasing in x (higher wealth $\to$ smaller value of $e^{-\gamma x}$ $\to$ smaller utility), violating monotonicity. The negative sign flips monotonicity back, and the exponential form guarantees strict concavity: $u''(x) = -\gamma^2 e^{-\gamma x} < 0$.

Limiting cases. As $\gamma \to 0$, the utility becomes (after rescaling) approximately linear — the risk-neutral limit in which the agent only cares about expected wealth. As $\gamma \to \infty$, the agent becomes infinitely risk averse, willing to pay arbitrarily much to eliminate even tiny uncertainty. These two limits will reappear in §4 as the endpoints of the bid–ask spread induced by indifference pricing.

A practical use. The exponential utility also admits a clean decomposition of the indifference price into a drift term and a variance penalty, which is the binomial-tree analogue of the mean-variance frontier. Once you internalize the exponential-utility formula, the leap from static gambling to dynamic portfolio optimization becomes natural: you are always trading off expected growth against variance exposure, weighted by γ.

The numerical meaning of γ. What does it mean, concretely, to have $\gamma = 0.001$ vs. $\gamma = 0.1$? A useful rule of thumb: the agent is approximately indifferent between a sure loss of $\gamma\sigma^2/2$ and a mean-zero gamble with variance σ^2. So at $\gamma = 0.001$, a coin toss with standard deviation \$1000 is roughly equivalent to a sure loss of \$500. At $\gamma = 0.1$, the same coin toss is equivalent to a sure loss of \$50,000 — the agent would pay dearly to avoid it. These numbers are useful for building intuition: one can dial in a plausible γ for a given agent by asking what certain-loss-equivalent they would assign to a typical uncertain outcome.

Calibrating γ to real humans. Empirical studies of risk-aversion typically estimate γ implicitly via lottery choices, financial decisions, or insurance premia. For an agent with total wealth on the order of hundreds of thousands of dollars, estimates of the absolute risk aversion γ typically fall in the range 10^{-4} to 10^{-6} per dollar. Lower than this, and the agent is essentially indifferent to small gambles; higher than this, and they refuse even tiny bets. The small numerical magnitude is why exponential-utility results often look uninformative for typical wealth levels: the gamble has to be enormous to produce a visible effect on the indifference price. This is partly why we discuss exponential utility mostly for pedagogical reasons, and rely on no-arbitrage pricing in practice.

Why we bother with utility at all. Given that the next few sections will largely render utility obsolete by replacing it with no-arbitrage arguments, a fair question is: why spend any time on utility? The answer is threefold. First, for non-replicable claims (insurance, private equity, illiquid positions), no-arbitrage pricing does not apply, and utility-based indifference pricing is the correct tool. Second, utility provides the *motivation* for choosing $\mathbb{Q}$ over $\mathbb{P}$ — it shows that $\mathbb{P}$ pricing cannot be universal, creating the appetite for something better. Third, the utility perspective reveals that the risk-neutral measure is the $\gamma \to 0$ limit of a much richer family of prices; this connection is essential for understanding risk premia in real markets.

2.3 3. Indifference Pricing (Buy Price and Sell Price)

The utility axioms give a *preference*-based price that does not rely on any other traded asset. We will recover the no-arbitrage price from this in §4.

Why start with indifference pricing? The approach answers a question that no-arbitrage pricing cannot: "what is the most a risk-averse agent should pay for this gamble, absent any hedging opportunity?" It is the correct tool for illiquid or non-replicable claims (insurance contracts, private equity stakes, employee stock options). Even when replication *is* possible, studying indifference pricing first makes vivid *why* the no-arbitrage price happens to be independent of preferences — it is the miracle enabled by trading. Without the comparison, risk-neutral pricing looks like a trick; with the comparison, it looks like a consequence.

Setup. Agent currently holds deterministic wealth $X_1 = x$ (no play, no pay). Alternative: pay the price A_0 now and receive the random payoff A_1 at $t = 1$, so wealth becomes $X_2 = x - A_0 + A_1$. We verbalise these as:

- $X_1 \sim$ no play
- $X_2 \sim$ play but pay

$$X_1 = x, \qquad X_2 = x - A_0 + A_1. \tag{3.1}$$

The setup is fully general. We are not assuming anything about how A_1 is generated, nor that the market contains a bond or any other asset. The agent is being asked a fundamentally binary question: do you prefer the status quo of x certain, or the lottery of $x - A_0 + A_1$? The indifference price A_0 is the cash amount that makes them exactly undecided. This is the *reservation price* in the micro-economic sense: above it the agent will not buy, below it they will.

Comment on the single-agent assumption. Nothing in (3.1) says anything about what the rest of the market is doing. Two agents with different γ will produce different A_0, and a third agent with a different utility family (say logarithmic) will produce yet another. This is why indifference pricing by itself cannot generate a market price — it only generates a personal reservation price. The market price, if one is to exist at all, must reconcile these different reservation prices. In illiquid markets that reconciliation happens through bargaining; in liquid markets it happens through replication, and the personal price becomes irrelevant.

A remark on the timing of cash flows. In (3.1) the agent pays A_0 at $t = 0$ and receives A_1 at $t = 1$. If we were being fully general we would discount A_1 by some interest rate r. For the indifference-pricing derivation we suppress r to keep the algebra clean — either set $r = 0$ or imagine that A_0 is already expressed in $t = 1$ dollars. The concept is unchanged; only the arithmetic changes. When we introduce the bond in §5 the time-value-of-money will be handled explicitly.

What "random wealth" means here. The variable X_2 is a random variable taking one of two values: $x - A_0 + A_u$ in the up-state (with probability p) and $x - A_0 + A_d$ in the down-state (with probability $1-p$). The agent's preference over X_2 is defined by the expected utility $\mathbb{E}[u(X_2)]$, which is a weighted average of the utilities at these two specific wealth levels. Wealth, in the utility sense, is *final* wealth — what you have after all transactions are complete. The agent cares about how much money they end up with, not how they got there.

2.3.1 3.1 Expected utilities on each side

Having framed the agent's choice, we now compute the expected utilities on both sides of the indifference.

(i) No play:

$$\mathbb{E}[\, u(X_1)\,] \;=\; -e^{-\gamma x}. \tag{3.2}$$

Note: $-e^{-\gamma x}$ is a constant (non-random) since X_1 is deterministic.

Why this is trivial. When wealth is non-random, expectation does nothing: $\mathbb{E}[u(x)] = u(x)$. The equation is merely restating this. The value of (3.2) is that it gives us a benchmark level of utility — the level the alternative must meet for the agent to accept.

(ii) Play but pay:

$$\mathbb{E}[\, u(X_2)\,] \;=\; \mathbb{E}\big[\, -e^{-\gamma(x-A_0+A_1)}\,\big] \;=\; -\,e^{-\gamma(x-A_0)}\,\mathbb{E}\big[\, e^{-\gamma A_1}\,\big]. \tag{3.3}$$

The factor $e^{-\gamma(x-A_0)}$ factors out because it is deterministic.

What the factorization reveals. The expected utility splits into a purely deterministic part, $e^{-\gamma(x-A_0)}$, which is the utility level of having paid A_0 out of deterministic wealth x, and a stochastic part, $\mathbb{E}[e^{-\gamma A_1}]$, which is the *moment generating function* (MGF) of $-A_1$ evaluated at γ. The whole reason exponential utility is analytically tractable is that this MGF is a single scalar summary of the payoff distribution, which means the indifference equation reduces to a one-variable problem. No integrals to invert, no fixed points to chase — just logarithms.

The MGF as a "risk summary". For any bounded random variable Y, the MGF $M_Y(\theta) = \mathbb{E}[e^{\theta Y}]$ encodes every moment of Y and in particular captures both its mean and its variance. Small-θ expansions give $M_Y(\theta) \approx 1 + \theta\mathbb{E}[Y] + \frac{1}{2}\theta^2\mathbb{E}[Y^2] + \cdots$. This is exactly what §4 exploits: the small-γ expansion of the indifference price is a Taylor expansion of the MGF, producing a clean mean-plus-variance-correction form.

2.3.2 3.2 The indifference equation

Define the indifference price A_0 such that

$$\mathbb{E}[\,u(X_1)\,] \;=\; \mathbb{E}[\,u(X_2)\,]. \tag{3.4}$$

What "indifference" really means. Economically, equation (3.4) states that the agent is exactly on the fence: if you nudged the price up by one penny, they would refuse to buy; if you nudged it down by one penny, they would accept. Mathematically, it is the defining equation of the *certainty equivalent* of the gamble $x - A_0 + A_1$, shifted and rearranged to read off a cash price. This is how indifference pricing always operates: solve for the cash amount that makes the utility of the gamble equal to the utility of cash.

Substituting (3.2) and (3.3):

$$-e^{-\gamma x} \;=\; -e^{-\gamma(x-A_0)}\,\mathbb{E}\big[\,e^{-\gamma A_1}\,\big].$$

Cancelling the $-e^{-\gamma x}$ factor gives $1 = e^{\gamma A_0}\,\mathbb{E}[e^{-\gamma A_1}]$, hence

$$\boxed{A_0^{\text{buy}} \;=\; -\frac{1}{\gamma}\,\ln \mathbb{E}\big[\,e^{-\gamma A_1}\,\big]} \tag{3.5}$$

with the domain constraint

$$A_0^{\text{buy}} \in (A_d, A_u). \tag{3.6}$$

Reading (3.5) term by term. The right-hand side is $-\ln \mathbb{E}[e^{-\gamma A_1}]/\gamma$, the so-called *entropic risk measure* of the payoff $-A_1$. It is a smooth, convex, increasing-in-γ object that collapses to $\mathbb{E}[A_1]$ at $\gamma = 0$ and approaches the worst-case payoff A_d as $\gamma \to \infty$. The $-\gamma A_1$ inside the logarithm reflects that the buyer's risk is *losing* on A_1; the MGF of the negated payoff is the appropriate penalty functional.

The domain constraint (3.6) deserves attention. Why must the indifference price lie strictly between A_d and A_u? Because any price at or above A_u would guarantee a loss in every state, while any price at or below A_d would guarantee a gain in every state. In the first case the agent strictly prefers no play, contradicting indifference. In the second case the agent strictly prefers to play, again contradicting indifference. The indifference price is therefore sandwiched strictly between the worst and best outcomes, which is a mini no-arbitrage condition built into the preference-based formula.

Why the initial wealth x cancels. One of the remarkable properties of exponential utility is that the indifference price is *independent* of the agent's current wealth x. Look at the algebra: the $e^{-\gamma x}$ factor appeared on both sides of (3.4) and cancelled cleanly. This is the constant-absolute-risk-aversion property in action. For other utilities (logarithmic, power) this cancellation fails, and the indifference price depends on wealth. That wealth-independence is what makes exponential utility the workhorse of indifference pricing theory.

The entropic risk measure in disguise. The right-hand side of (3.5) can be read as "the entropic risk measure of $-A_1$ with parameter γ." Risk measures are a more modern reformulation of indifference pricing: rather than asking "what price makes the agent indifferent?", they ask "what capital must the agent hold to render the position acceptable?" For exponential utility these questions have the same answer, which is why the formula shows up in both literatures. Entropic risk measures have the advantage of being *convex* (sum of two positions has risk at most the sum of individual risks — diversification benefits), which many other risk measures lack. This technical property makes them useful in regulatory capital modelling.

Limits of the buy price formula. Before we differentiate it in §4, let us check what (3.5) does at the edges. If $A_1 = A_u$ with certainty, then $\mathbb{E}[e^{-\gamma A_1}] = e^{-\gamma A_u}$ and the log gives $A_0^{\text{buy}} = A_u$. Trivially correct: a risk-free payoff of A_u is worth exactly A_u to any agent. Conversely, if the payoff is extremely risky and catastrophic (say, $A_d \to -\infty$), then $\mathbb{E}[e^{-\gamma A_1}]$ blows up and $A_0^{\text{buy}} \to -\infty$: the agent would need to be paid an arbitrarily large amount to take the gamble. Both limits are economically sensible.

A micro-example of (3.5) plugged in. Take $A_u = 10$, $A_d = 0$, $p = 1/2$, $\gamma = 0.1$. Then $\mathbb{E}[e^{-\gamma A_1}] = \frac{1}{2}(e^{-1} + 1) = \frac{1}{2}(0.368 + 1) = 0.684$. The buy price is $A_0^{\text{buy}} = -\frac{1}{0.1}\ln(0.684) = 10 \cdot 0.380 \approx 3.80$. The physical expected payoff is 5. The buyer is willing to pay 3.80 — a 24% discount to $\mathbb{E}[A_1]$ — to compensate for the risk. Increasing γ to 1 changes the calculation: $\mathbb{E}[e^{-\gamma A_1}] = \frac{1}{2}(e^{-10} + 1) \approx 0.5$, so $A_0^{\text{buy}} = -\ln(0.5) \approx 0.693$. A tenfold increase in risk aversion dropped the price from 3.80 to 0.69. This is the steep sensitivity of exponential utility to γ: it captures, roughly, the fact that at high risk aversion, the agent treats any exposure to the down-state as nearly catastrophic.

2.3.3 3.3 Sell price (symmetric derivation)

From the counter-party perspective (sell rather than buy, so the sign on A_1 flips in the wealth update),

$$\boxed{A_0^{\text{sell}} \;=\; \frac{1}{\gamma}\,\ln \mathbb{E}\big[\, e^{\gamma A_1}\,\big]} \tag{3.7}$$

The asymmetry between buy and sell. Formulas (3.5) and (3.7) look symmetric — swap the sign of A_1 inside the exponent and negate the leading coefficient — but they are not equal. Consider an agent approached by two different counterparties: one offering to sell them A_1, one offering to buy it from them. The buy price (3.5) is the *most* they would pay; the sell price (3.7) is the *least* they would accept. These two numbers differ because the agent is risk averse: to accept the gamble as a buyer they demand a discount relative to expected value, and to part with the gamble as a seller they demand a premium. The gap between A_0^{buy} and A_0^{sell} is the *bid–ask spread induced by risk aversion*, sometimes called the "indifference spread."

A numerical illustration. Take the binomial $A_u = 2$, $A_d = 0$, $p = 1/2$, $\gamma = 1$. Then $\mathbb{E}[e^{-\gamma A_1}] = \frac{1}{2}(e^{-2} + 1) = \frac{1}{2}(0.135 + 1) \approx 0.568$, so $A_0^{\text{buy}} = -\ln(0.568) \approx 0.566$. Similarly $\mathbb{E}[e^{\gamma A_1}] = \frac{1}{2}(e^{2} + 1) = \frac{1}{2}(7.389 + 1) \approx 4.194$, giving $A_0^{\text{sell}} = \ln(4.194) \approx 1.434$. The mid-point is $\mathbb{E}[A_1] = 1$ — exactly as expected — but the bid–ask runs from roughly 0.57 to 1.43. A single agent, facing the same gamble, quotes a spread nearly as wide as the range of outcomes. This is the cost of risk aversion.

Why no rational agent quotes a two-sided market without a spread. Take any agent with any concave utility. The concavity directly implies that their buy price is strictly below their sell price — they are better off *not trading* than trading at a single "fair" price that ignores their risk aversion. An agent forced to quote a single number for both buy and sell would lose expected utility on every trade. This is a deep economic observation: the existence of bid–ask spreads in real markets is not a friction or an inefficiency but a *necessary consequence* of market makers being risk averse. You could try to eliminate the spread by making the market maker risk neutral, but no institution is genuinely risk neutral, so the spread survives.

The role of market makers. In practice, a market maker does not use (3.5)–(3.7) directly because their γ is low and, crucially, they plan to *hedge* the position after taking it on. Hedging reduces their effective exposure, which lowers the risk-aversion penalty and narrows the spread they quote. In the limit of perfect hedging — replication of the kind we study in §8 — the market maker can quote a zero spread around the no-arbitrage price. In practice, hedging is imperfect (transaction costs, discrete rebalancing, model risk), so the spread is positive but small. The tight bid–ask spreads we see in liquid markets are a direct consequence of market makers' ability to hedge well.

A puzzle to chew on. If every agent has their own γ and hence their own indifference spread, how does the market ever clear? The answer is: the market clears at a price where some agent's buy price meets another agent's sell price. The *transaction price* is somewhere in the intersection of different agents' intervals, and it is pinned down by whoever has the narrowest spread (typically the market maker) plus some competitive tension. This is why the no-arbitrage price (§5) is such a useful benchmark: it is the unique price that is *consistent with everyone's* indifference intervals simultaneously, under the additional assumption that the payoff can be replicated.

2.4 4. Small-γ Expansion: Recovering Expected Value

We now show that indifference pricing collapses to $\mathbb{P}$-expected value in the risk-neutral (no-risk-aversion) limit, and that a bid–ask spread opens up purely from risk aversion.

The question this section answers. We have two competing candidate pricing rules: the naïve discounted expectation of (1.3), and the indifference price of (3.5). Is there any limit in which they agree? The answer is yes — the limit

$\gamma \to 0$, where the agent becomes risk-neutral. Demonstrating this agreement does two things. First, it shows that (1.3) is not wrong per se, but is a *special case* (the risk-neutral limit) of a more general pricing theory. Second, it foreshadows how, via no-arbitrage, we will later recover the *same* formula without ever introducing a utility function — a kind of logical shortcut that sidesteps the preference apparatus altogether.

Taylor-expand $e^{-\gamma A_1}$ around $\gamma = 0$:

$$e^{-\gamma A_1} \;\sim\; 1 - \gamma A_1 + o(\gamma). \tag{4.1}$$

The mechanics of the expansion. We are expanding the exponential in its power series and keeping only the first-order term in γ. The $o(\gamma)$ term hides $\frac{1}{2}\gamma^2 A_1^2$ and higher; at this order they don't matter. The approximation is good for small γ and for bounded A_1.

Take expectations:

$$\mathbb{E}\big[e^{-\gamma A_1}\big] \;\sim\; 1 - \gamma\,\mathbb{E}[A_1] + o(\gamma). \tag{4.2}$$

Linearity of expectation. The step from (4.1) to (4.2) is just $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$. The randomness has been reduced to a single scalar summary, $\mathbb{E}[A_1]$, because we only kept the first-order term. At higher orders, variances and skewness would appear.

Apply $\ln(1 + z) \sim z + o(z)$:

$$\ln\big(\mathbb{E}[e^{-\gamma A_1}]\big) \;\sim\; -\gamma\,\mathbb{E}[A_1] + o(\gamma). \tag{4.3}$$

(Auxiliary identity used: $\ln(1 + z) \sim z + o(z)$*.)*

The inverse trick. We want to solve for A_0, which requires inverting the MGF via a logarithm. The small-z expansion of $\ln(1 + z)$ is the cleanest way to do this in a Taylor sense: near the base case $\mathbb{E}[e^{-\gamma A_1}] \approx 1$, the logarithm is linear in its deviation from 1. This is a generic trick that reappears throughout finance whenever you need to invert small perturbations of multiplicative quantities.

Therefore

$$A_0^{\text{buy}} \;\sim\; \mathbb{E}[A_1] + \gamma\,(\,\cdots\,) \tag{4.4}$$

(the $(\,\cdots\,)$ being a variance-like correction at next order), and in the limit $\gamma \downarrow 0$:

$$A_0^{\text{buy}} \;\underset{\gamma\downarrow 0}{\longrightarrow}\; \mathbb{E}[A_1], \qquad A_0^{\text{sell}} \;\underset{\gamma\downarrow 0}{\longrightarrow}\; \mathbb{E}[A_1]. \tag{4.5}$$

Risk-neutral pricing is the vanishing-risk-aversion limit of indifference pricing.

The variance correction, informally. Pushing the expansion one more order yields $A_0^{\text{buy}} \approx \mathbb{E}[A_1] - \frac{1}{2}\gamma\,\mathrm{Var}[A_1] + \cdots$, the celebrated mean-variance approximation of the certainty equivalent. The sign is intuitive: the buyer demands a discount proportional to the variance of the payoff, scaled by their risk aversion. For the seller, the sign flips: $A_0^{\text{sell}} \approx \mathbb{E}[A_1] + \frac{1}{2}\gamma\,\mathrm{Var}[A_1]$. The bid–ask spread, to first non-trivial order in γ, is $\gamma \cdot \mathrm{Var}[A_1]$ — proportional to risk aversion times the variance of the payoff. Double the risk aversion, double the spread; double the variance, double the spread.

Why this matters pedagogically. The mean-variance correction is the bridge between utility theory and Markowitz portfolio optimization, between behavioural finance and pricing theory. Even though we will not use mean-variance explicitly in the rest of this chapter (because no-arbitrage delivers a cleaner result), the appearance of $\mathrm{Var}[A_1]$ at second order is a hint that volatility is the dominant risk measure in finance for exactly the reason that utility curves are nearly quadratic near the current wealth level.

An observation about convergence rates. The convergence $A_0^{\text{buy}} \to \mathbb{E}[A_1]$ as $\gamma \to 0$ is *linear* in γ at leading order (the correction term is $O(\gamma \cdot \mathrm{Var}[A_1])$). This means that even for modest values of γ, the indifference price is close to the expected value if the variance is small, but deviates sharply when the variance is large. Applied to

real financial markets: for highly diversified portfolios with low residual variance, the risk-neutral price and the indifference price of individual agents are close. For concentrated, high-variance bets, the prices diverge significantly. This is why idiosyncratic risk matters: holding a single stock subject to substantial variance is qualitatively different from holding an index, and the price at which an agent accepts the single stock reflects their risk aversion far more than the price of the index does.

Higher-order terms. If you push the Taylor expansion beyond second order, you pick up skewness and kurtosis corrections. For a generic bounded A_1, the cumulant expansion of the MGF gives $\ln \mathbb{E}[e^{-\gamma A_1}] = -\gamma \mathbb{E}[A_1] + \frac{1}{2}\gamma^2 \mathrm{Var}[A_1] - \frac{1}{6}\gamma^3 \kappa_3 + \cdots$ where κ_3 is the third cumulant (essentially skewness times σ^3). So the indifference price becomes $A_0^{\mathrm{buy}} = \mathbb{E}[A_1] - \frac{1}{2}\gamma \mathrm{Var}[A_1] + \frac{1}{6}\gamma^2 \kappa_3 - \cdots$. Negative skew (a heavy left tail) makes the buyer even more conservative: they discount the mean not only by variance but also by a skewness penalty. This is the formal content of the preference for positively skewed payoffs that empirical finance consistently documents. Higher-order tail risks propagate into prices through these higher cumulants.

2.4.1 4.1 Buy/sell price as functions of γ

```
  price

  A_u
           ._____
                        ._____            ← A_0^{sell}( )


 E[A_1]                                  (risk-neutral level)
                                 --
                     --                  ← A_0^{buy}( )
      ---
  A_d
                          →
    0

   = 0:    both curves meet at E[A_1].
   → ∞:    buy → A_d  (worst-case),  sell → A_u  (best-case).
  Bid-ask spread A_0^{sell} - A_0^{buy} widens monotonically in  .
```

- At $\gamma = 0$: both curves meet at $\mathbb{E}[A_1]$.
- As $\gamma \uparrow \infty$: $A_0^{\mathrm{buy}} \to A_d$ (buyer only pays worst-case); $A_0^{\mathrm{sell}} \to A_u$ (seller demands best-case).
- $A_0^{\mathrm{buy}} \leq \mathbb{E}[A_1] \leq A_0^{\mathrm{sell}}$ always — a bid–ask spread induced purely by risk aversion.

Reading the picture. The graph gives a complete visual summary of how risk aversion morphs a price. On the $\gamma = 0$ axis, both curves pass through the physical expectation $\mathbb{E}[A_1]$: a risk-neutral agent is indifferent to risk and prices at the mean. As γ grows, the buy and sell curves splay outwards, with the buy price asymptoting to A_d (the agent will pay at most the guaranteed worst-case value of the payoff) and the sell price asymptoting to A_u (the agent demands at least the best-case value to part with it). The gap between the curves is the bid–ask, and it fills almost the entire range of outcomes for very risk-averse agents.

Why infinite risk aversion produces worst- and best-case pricing. When γ is enormous, the agent treats any non-negligible chance of losing as catastrophic. They will therefore only buy if they are guaranteed not to lose — which requires paying no more than A_d. Similarly they will only sell if they are guaranteed not to miss out — which requires charging no less than A_u. This is *worst-case* or *super-replication* pricing, and it will reappear in the literature as the *upper and lower hedging prices* for incomplete markets.

The monotonicity of the spread in γ. A basic sanity check: the bid–ask spread $A_0^{\mathrm{sell}} - A_0^{\mathrm{buy}}$ should increase with γ — more risk-averse agents quote wider spreads. A direct calculation using (3.5) and (3.7) confirms this: the MGF-based expressions are each monotone in γ (in the appropriate direction), and their difference is therefore monotone too. This matches our intuition about how a nervous market maker widens spreads during a crisis: effectively, their γ has spiked, so their two-sided quote gets pushed apart.

A geometric picture. The buy curve and sell curve can be drawn as the lower and upper envelopes of a family of chords on the graph of $-e^{-\gamma x}$. The buy price is where the chord connecting $(A_u, -e^{-\gamma A_u})$ and $(A_d, -e^{-\gamma A_d})$, weighted

by $(p, 1-p)$, crosses the utility curve on the way up. The sell price is the symmetric construction from the seller's viewpoint. The splay of the curves as γ grows is a visual encoding of the curve's increasing concavity.

When indifference pricing is the right tool. There are many situations where no-arbitrage pricing is simply not available, and the indifference framework we have just built is the appropriate one. Examples: (1) employee stock options (the employee cannot trade them, so they cannot replicate them away); (2) insurance contracts (the insured's individual exposure is not replicable in general); (3) real options on development projects (no liquid market for the underlying); (4) private equity investments (no mark-to-market, no hedge available). In each case, the agent's risk preferences are genuinely relevant to the price, and the indifference formulas (3.5)–(3.7) give the right answer. The rest of this chapter, however, will focus on situations where hedging *is* available, because that is where the no-arbitrage shortcut becomes transformative.

Bridge to the next section. Indifference pricing produced a *single* preference-dependent price per agent. In a market with many agents, only prices that no single agent can exploit for a free lunch survive. That no-arbitrage requirement turns out to pin down pricing uniquely, *without any reference to γ or u* — and the formula it delivers is precisely the $\gamma \downarrow 0$ limit above, but with the physical measure $\mathbb{P}$ replaced by a synthetic measure $\mathbb{Q}$. We derive this next.

The transition to §5 is not just a technical move — it is a philosophical one. Up to now, pricing has been a *personal* calculation: each agent's γ produces their own price. Starting with §5, pricing will become a *market* calculation: a single number that no agent can exploit, derived from the tree structure alone. The shift from personal to impersonal pricing is what makes derivatives markets possible as institutions. A market cannot function on a thousand different prices for the same claim; it needs a single transaction price, and no-arbitrage provides exactly that.

2.5 5. Two-Asset Model and Arbitrage

The pivot we are about to make is the most important conceptual step in the chapter. We drop preferences entirely, introduce a second asset, and argue purely from the absence of arbitrage. The result is a pricing theory that applies to *any* rational agent, not just to some particular utility function. The surprise — and the deep theorem of mathematical finance — is that this preference-free approach *still* produces a formula that looks like an expected value with respect to a carefully chosen probability measure. The measure is synthetic; nobody believes in it; yet it prices every replicable claim correctly.

Two assets observed at $t = 0$ and $t = 1$. Asset A is risky; asset B is a numeraire (e.g. a bond / bank account). A starts at A_0 and moves to A_u or A_d; B starts at B_0 and moves to B_u or B_d, both driven by the *same* Bernoulli coin.

In the bond case $B_u = B_d = B_0(1 + r)$; in the more general setting B has its own leaves:

```
        A_u                 B_u

A_0           B_0

        A_d                 B_d

     risky A             numeraire B
  (driven by same Bernoulli coin x_1)
```

Why the "same coin" is critical. We are not saying that A and B are deterministically related — we are saying that the *state of the world* at $t = 1$ is specified by a single coin flip, and both A and B are measurable functions of that coin. If each asset had its own independent coin, we would have four possible $t = 1$ states and only two hedging instruments — an incomplete market, where not every claim can be replicated. The "same coin" assumption is the discrete-time avatar of "one Brownian motion" in continuous time, and it is what forces the market to be complete.

Practical interpretation. In practice, a bond B is nearly deterministic over short horizons, which is why the bond case $B_u = B_d$ is the default working assumption in introductory treatments. But keeping $B_u \neq B_d$ in the setup is useful: it lets us pass seamlessly to foreign exchange pricing, where "the numeraire" is a risky foreign bond, or to stochastic-interest-rate models, where even the domestic bank account has state-contingent value.

A note on the word "numeraire". The term comes from French and means "accounting unit" — the good in which other goods are priced. In everyday language, the dollar is a numeraire for the U.S. economy; everything is priced in dollars. In mathematical finance, the numeraire is any strictly positive asset that we choose to denominate other

prices in. Changing numeraires changes the *units* in which prices are quoted but does not change the underlying economic content. Different numeraire choices lead to different risk-neutral measures (the "bond measure," the "stock measure," the "forward measure," etc.), but each measure prices the same set of claims consistently. This freedom to change numeraires is one of the most flexible tools in derivatives pricing and will be exploited heavily in later chapters.

A remark on the bond's determinism. When we say "the bond is nearly deterministic," we mean that over short horizons its payoff is known up to credit risk. An overnight Treasury bill, for example, is about as close to a deterministic $t = 1$ payoff as the financial world offers. The approximation becomes worse for longer-maturity or lower-quality bonds, where interest-rate risk and default risk introduce genuine randomness. For the purposes of this chapter, we take the bond as deterministic in the bond case and leave the general numeraire case for readers who want the broader picture.

2.5.1 5.1 Portfolio value

The point of introducing a second asset is to create *combinations.* A long-short portfolio of A and B can, in principle, produce any linear combination of payoffs in the up- and down-states — and that is exactly what we will exploit when pricing a third asset in §7 and §8. For now we just write down the portfolio value.

Introduce a portfolio (α, β) holding α units of A and β units of B:

$$V_0 \;=\; \alpha A_0 + \beta B_0, \tag{5.1}$$

$$V_1 \;=\; \alpha A_1 + \beta B_1. \tag{5.2}$$

Self-financing is automatic here. Between $t = 0$ and $t = 1$ there is no rebalancing event, so the portfolio inherits whatever the two assets do. In the multi-period extension of §10 we'll need to impose self-financing explicitly; for now it is baked into the two-date clock.

Short positions are allowed. The weights α and β are real numbers, not just non-negative ones. A negative α means the agent has *sold short* one or more units of A — borrowed the asset, sold it in the market, and is on the hook to return it at $t = 1$ (along with any intervening dividends, which we suppress in this chapter). A negative β means the agent has *borrowed* at the bond rate. These negative positions are essential for arbitrage construction and for replicating payoffs that lie outside the positive cone of the two assets.

A quick sanity check. If A and B are both positive and the portfolio has only positive weights, then $V_t \geq 0$ in all states — but this is too weak a position to be interesting, because no bounded positive linear combination can match a general $t = 1$ payoff. Interesting portfolios almost always mix signs.

A word about transaction costs and frictions. In reality, short selling is not free — there are borrow fees, margin requirements, and risk of recall. Trading is not free either — there are bid–ask spreads, exchange fees, and price impact. The no-arbitrage model we are building ignores all of these, which is sometimes justified by the claim that "large arbitrageurs face much smaller frictions than retail traders." There is truth to this, but in dislocated markets even large arbitrageurs may be unable to trade at their theoretical prices. The frictionless model is therefore a benchmark, not a description. The deviations between benchmark prices and observed prices tell us something about market frictions; the benchmark prices themselves come from the frictionless model.

The dimension of the portfolio space. A portfolio is specified by two real numbers (α, β), so the portfolio space is $\mathbb{R}^2$. At $t = 1$ the payoff is a vector (V_1^u, V_1^d) in $\mathbb{R}^2$. The map $(\alpha, \beta) \mapsto (V_1^u, V_1^d)$ is linear, and its matrix has rows (A_u, B_u) and (A_d, B_d). If this matrix is invertible — i.e. if $A_u B_d \neq A_d B_u$ — then every $t = 1$ payoff is reachable by some portfolio, which is exactly the completeness property. We will revisit this linear-algebra reading in §8 when we solve for the replicating portfolio explicitly.

2.5.2 5.2 Definition of arbitrage

The central concept of this section is the formal definition of arbitrage — a strategy that creates money from nothing. The definition below is the standard one; every pricing theorem in the book hangs on it.

An arbitrage strategy (portfolio) (α, β) is one such that

$$\text{(i)} \quad V_0 = 0, \tag{5.3}$$

$$\text{(ii)} \quad \exists\, t \text{ s.t.} \quad \text{(a)} \;\; \mathbb{P}(V_t \geq 0) = 1 \quad \text{(never lose)}, \tag{5.4}$$

$$\text{(b)} \;\; \mathbb{P}(V_t > 0) > 0 \quad \text{(sometimes win)}. \tag{5.5}$$

In plain English. (i) You enter the trade with zero capital. (ii a) You never lose money in any scenario. (ii b) In at least one scenario you make strictly positive money. The combination is "free lunch": strictly better than owning cash, with zero downside. If such a portfolio existed, rational investors would scale it unboundedly; the whole of mathematical finance is the study of prices at which such strategies *cannot* exist.

The three clauses are logically distinct. Clause (i) is the "self-financing at zero cost" condition — the arbitrageur has no skin in the game. Clause (ii a) is a *no-downside* condition: in no state of the world does the portfolio produce a loss. Clause (ii b) is the *strict-upside-sometimes* condition: the portfolio is not identically zero. Omit any one of these and the definition becomes trivial or vacuous. Requiring all three simultaneously is what makes arbitrage rare: it is the intersection of three stringent constraints on the portfolio weights.

A stronger alternative. Some treatments require $\mathbb{P}(V_t > 0) = 1$ — a strict gain in every state. This is sometimes called a *strong arbitrage*. It is a more restrictive condition, and a model might be strong-arbitrage-free while still admitting the weaker form (5.5). The weaker definition is standard and is the one that produces the martingale characterization we are after. Both in theory and in practice, the weaker notion is what matters for pricing.

Why arbitrages, if they exist, are exploited to infinity. If the portfolio produces a guaranteed non-negative payoff for zero up-front cost, and a strictly positive payoff in some state, then doubling the position size doubles the payoff without doubling the cost (the cost is zero either way). Scaling upward is free, so any rational agent would scale without bound. Real markets have such scaling frictions — balance-sheet constraints, funding costs, collateral — but in the frictionless limit of the model, unbounded scaling is the natural response to an arbitrage. The only self-consistent market is one that precludes arbitrage.

A philosophical remark. The definition of arbitrage is a *negative* definition — it specifies what we do *not* want a market to have. Much of mathematical finance can be read as: "assume markets do not admit arbitrage; what consequences follow?" The answer, as the next few sections show, is a surprisingly rich set of constraints on prices, hedging ratios, and the existence of a synthetic pricing measure. The conclusion is that arbitrage is a structural assumption with far-reaching consequences, not just a technical hygiene condition. This style of argument — assume a minimal property, derive maximal consequences — is characteristic of mathematical finance.

The weak vs strong arbitrage distinction, revisited. We noted above that (5.3)–(5.5) defines a *weak* arbitrage: non-negative terminal value with positive probability of strict gain. The alternative strong definition — strictly positive terminal value in every state — is economically more stringent but leads to a strictly weaker theory. The First Fundamental Theorem (stated in §8.4 and §10.2) uses the weak definition, which is why the theorem delivers an *equivalent* martingale measure rather than just an absolutely continuous one. The distinction matters for advanced readers; for our purposes, we work with the weak definition throughout.

2.5.3 5.3 Sign-sanity toy trees

The definition is abstract; making it concrete with small examples helps. Three toy trees illustrate exactly when (and why) the definition fires.

- Tree (a): $0 \to \{1, 0\}$ — is an arbitrage (cost 0 today, outcome ≥ 0 always, positive with probability > 0).
- Tree (b): $0 \to \{1, -1\}$ — not an arbitrage (outcome can be negative).
- Tree (c): $0 \to \{1, 1\}$ — is an arbitrage (pays 1 for sure at cost 0).

Parsing the three trees. Tree (a) is the canonical arbitrage: zero now, non-negative future, positive in some state. Tree (b) fails clause (ii a): the down-state pays -1, a loss. Tree (c) is actually a stronger arbitrage than (a): it pays $+1$ in both states, a risk-free profit. Intuition check: (c) violates the absence-of-free-lunches by generating money unconditionally, so any market that admits (c) is absurd. Tree (a) is more subtle — the arbitrageur might win, might break even, but never loses. Still absurd, because the expected win is strictly positive for zero cost.

And is the ratio A_0/B_0 inside $(A_d/B_d, A_u/B_u)$?):

Tree label	A_0	A_d	A_u	Check $A_d/B_d < A_0/B_0 < A_u/B_u$? (with $B \equiv 1$)	Verdict
$(-)$	10	5	20	$5 < 10 < 20$	No arb
$(+)$	2	4	15	$4 \not< 2$	Arb
$(14\ldots)$	≈ 14	0	0	violates	Arb

Specific solve for tree $(-)$: $q = (A_0 - A_d)/(A_u - A_d) = (10-5)/(20-5) = 5/15 = 1/3$; cross-check $20/4 = 5$.

Teaching point. If you can draw a horizontal line at A_0/B_0 and it passes *between* the two leaves of the relative-price tree, the model is viable; if the line lies outside that interval, one leaf strictly dominates it and a free lunch appears.

An intuition pump for the second tree. In tree $(+)$, the current price $A_0 = 2$ is below the down-leaf $A_d = 4$. That means every scenario at $t = 1$ has $A_1 \geq 4 > 2$. Buying one share at $t = 0$ costs 2 and pays at least 4 at $t = 1$ — a guaranteed doubling (at least). Financed by shorting two units of the bond, the portfolio has cost zero today and strictly positive value tomorrow. Classic arbitrage.

2.5.4 5.4 Scaled-tree arbitrage illustration

Consider a bond tree $1 \to \{1+r, 1+r\}$ vs. a second asset tree $1 \to \{1, 1\}$. With α in the second asset and $-\alpha$ in the bond, the zero-cost portfolio pays $\{(1-r)\alpha,\ -\alpha r\}$. Arbitrage would require

$$\begin{pmatrix} 1-r > 0 \\ -r < 0 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1-r < 0 \\ -r > 0 \end{pmatrix}, \tag{5.6}$$

which simplify to the impossible sign-combinations

$$\boxed{r > 0 \text{ and } r < 1} \quad \| \quad \boxed{r < 0 \text{ and } r > 1} \tag{5.7}$$

(i.e. a genuine arbitrage would require an internally-inconsistent sign-combination — confirming no such arb exists under reasonable r). transcription uncertain — these sign conditions are the instructor's running computation; the practical upshot is simply the standard ordering in (5.8) below.

What the exercise actually shows. The second asset is a *non-growing* claim — it pays one dollar at $t = 1$ regardless of state — while the bond grows to $1 + r$. If $r > 0$, the non-growing asset is dominated by the bond in every state, and we can arbitrage by shorting the asset and buying the bond. The sign analysis is a mechanical way to see this: the only arbitrage-free regime is when the asset's guaranteed return matches the bond's, which in this degenerate case means $r = 0$. More realistically, non-bond assets differ by scenario, and the absence-of-arbitrage condition becomes a non-trivial bound on the asset's possible trajectories.

2.5.5 5.5 The standard ordering

Highlighted in yellow:

$$A_0 \in (A_d, A_u), \qquad A_u > A_d, \qquad B_0 > 0. \tag{5.8}$$

Why this orientation is WLOG. If $A_u < A_d$ we relabel "up" and "down" — nothing in the model hangs on which leaf is which. The positivity assumption on B is what lets us form ratios A/B without dividing by zero.

A subtlety: strict vs. weak inequalities. The condition $A_0 \in (A_d, A_u)$ — open interval, strict inequalities — is stronger than $A_0 \in [A_d, A_u]$. The strict version rules out *exact* equality at either endpoint, which would correspond to a degenerate tree where one of the two states has zero effective probability. Keeping strict inequalities throughout is the discrete analogue of requiring the volatility in Black–Scholes to be strictly positive: without it the model collapses, and every claim becomes trivially replicable by the bond alone.

Connection to the no-arbitrage theorem. We will see in §6 that $A_d < A_0(1+r) < A_u$ is precisely the no-arbitrage condition in the bond case. It is striking that this condition is already implicit in the *ordering* assumption (5.8) combined with reasonable growth on the bond — the model setup is half-way to ruling out arbitrage, with the pricing identity (6.5) completing the argument.

2.6 6. No-Arbitrage Bound on A_0

With the two-asset setup and the definition of arbitrage in place, we now derive the first nontrivial theorem: in the absence of arbitrage, the current price of A is constrained to lie strictly between its discounted up and down values. This is the *no-arbitrage bound.* It is a necessary condition — violating it produces a trading strategy that is provably an arbitrage — but it will turn out to also be sufficient for the existence of a risk-neutral measure, linking the preference-free price to the martingale picture.

We now derive the same identity from two sides: first in absolute prices with a bond , then in B-relative prices in full generality .

2.6.1 6.1 Absolute-price derivation (bond case)

The bond case is the conceptually simplest and historically the original. We impose $V_0 = 0$, compute the terminal payoffs, and ask when their signs must be compatible with non-arbitrage. The result is a two-sided inequality on A_0.

With the bond convention $B_u = B_d = B_0(1+r)$, impose $V_0 = 0$:

$$\beta \;=\; -\alpha\,\frac{A_0}{B_0}. \tag{6.1}$$

What (6.1) is doing. Starting from $V_0 = \alpha A_0 + \beta B_0 = 0$ and solving for β shows that a zero-cost portfolio in (A, B) is always "buy α units of A, fund by shorting $\alpha A_0/B_0$ units of B." This is exactly how one would describe the trade to a trader: "for every share of A I buy, I borrow cash worth A_0." The key observation is that once α is chosen, β is determined by the zero-cost constraint. The arbitrage question reduces to a one-dimensional question in α.

Substitute into the terminal values:

$$\text{up:}\quad \alpha A_u + \beta B_0(1+r) \;=\; (A_u - (1+r)A_0)\alpha, \tag{6.2}$$

$$\text{down:}\quad \alpha A_d + \beta B_0(1+r) \;=\; (A_d - (1+r)A_0)\alpha. \tag{6.3}$$

Reading the two brackets. The bracketed expressions $A_u - (1+r)A_0$ and $A_d - (1+r)A_0$ are the payoffs *per unit of* α, i.e. per unit of long exposure to A combined with the appropriate short bond hedge. If both brackets are strictly positive, setting $\alpha > 0$ produces strictly positive terminal wealth in both states — an arbitrage. If both are strictly negative, setting $\alpha < 0$ does the same. To rule out arbitrage, the two brackets must have *opposite signs.*

To avoid arbitrage, the two bracketed coefficients cannot both be ≥ 0 (nor both ≤ 0) — choosing α of the right sign would produce a free lunch. Concretely the only viable case is

$$A_u - (1+r)A_0 > 0 \quad \text{and} \quad A_d - (1+r)A_0 < 0. \tag{6.4}$$

Why only this ordering is possible. Since $A_u > A_d$ by (5.8), the bracket involving A_u is always at least as large as the bracket involving A_d. If the two brackets have opposite signs, the larger (up) bracket must be positive and the smaller (down) must be negative. The reverse ordering (up negative, down positive) is forbidden by the monotonicity of the brackets. Hence (6.4) is the unique viable sign configuration.

These combine to

$$\boxed{A_d \;<\; A_0(1+r) \;<\; A_u} \qquad \text{(no-arb).} \tag{6.5}$$

No-arbitrage requires d < 1+r < u

Figure 2: No-arbitrage requires d < 1+r < u

I.e. the forward price $A_0(1+r)$ must lie strictly between the two future spot values:

```
                    → future value
A_d      A_0(1+r)       A_u

No-arbitrage forces the forward A_0(1+r) to lie strictly inside [A_d, A_u].
```

No-arbitrage requires $d < 1 + r < u$.

Economic reading of (6.5). The quantity $A_0(1+r)$ is the *forward price* of A: the value of A_0 rolled forward at the bond rate. The no-arbitrage condition says this forward price must be inside the interval of possible future spots. If the forward were at or above A_u, then cash invested in the bond would dominate the risky asset in every state; arbitrage by long-bond, short-A. If the forward were at or below A_d, then A would dominate cash in every state; arbitrage by long-A, short-bond. The "sandwich" condition is the only way to have neither dominance.

Limiting cases. If $A_u = A_d$, the inequality collapses to a single point and A is deterministic: the forward price must equal that single future value, pinning $A_0 = A_u/(1+r) = A_d/(1+r)$. If $r = 0$, the bound simplifies to $A_d < A_0 < A_u$: the current spot must lie between the two future spots. If r becomes very large, the bound $A_0 < A_u/(1+r) \to 0$ forces the current spot toward zero — the risk-free alternative crowds out the risky one.

A useful rearrangement. Writing (6.5) as $d \equiv A_d/A_0 < 1 + r < A_u/A_0 \equiv u$, we get the familiar "$d < 1 + r < u$" form that every introductory text quotes. The letters u and d are *gross returns*: $u = A_u/A_0$, $d = A_d/A_0$. No-arbitrage requires the bond's gross return $1 + r$ to lie strictly between the risky asset's gross returns in the two states.

An equivalent framing via dominance. The condition $d < 1 + r$ says "the down-state return on A is strictly less than the bond's return." If this failed, even the worst outcome for A would beat the bond, and A would dominate the bond. Conversely, $1 + r < u$ says "the up-state return on A is strictly greater than the bond's return." If this failed, even the best outcome for A would be no better than the bond, and the bond would dominate A. The no-arbitrage condition is precisely the non-dominance condition: neither asset dominates the other in every state. This is the simplest possible statement of what it means for two assets to *genuinely* differ in their risk profile.

A geometric picture of (6.5). Imagine a horizontal line labelled with the possible $t = 1$ prices from left to right. Plot the three numbers A_d, $A_0(1 + r)$, A_u on the line. The no-arbitrage condition is the assertion that $A_0(1 + r)$ sits strictly between A_d and A_u — the forward lies inside the interval of future possibilities. If it lies outside, there is a free lunch. The picture makes clear why no-arbitrage is so often the default assumption: it is geometrically obvious that the forward *should* be inside the interval of future outcomes. Any other configuration produces an arbitrage that would quickly be exploited away.

The no-arbitrage bound as a consistency condition. Another way to read (6.5) is as a statement of internal consistency among the parameters (A_0, A_u, A_d, r). If you try to specify a model with $A_0(1 + r) > A_u$, the model is inconsistent — it contradicts its own no-arbitrage assumption. Model-builders in practice respect this by ensuring that their calibrated up and down factors bracket the forward. The CRR calibration in §11 does this automatically by setting $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$, which for reasonable σ safely bracket any reasonable r.

2.6.2 6.2 Relative-price derivation (general numeraire)

Having done the bond case, we now generalize. The key observation is that the bond's role is purely as a *clock* — a way to compare cash at $t = 0$ with cash at $t = 1$. Any strictly positive asset can play this role. Replacing the bond with an arbitrary numeraire B with its own leaves B_u, B_d yields a more general but structurally identical derivation.

Impose $V_0 = \alpha A_0 + \beta B_0 = 0$, so $\beta = -\alpha A_0/B_0$, and substitute into each leaf of V_1:

$$\alpha A_u + \beta B_u \;=\; \alpha\left(A_u - \frac{A_0}{B_0}\,B_u\right), \tag{6.6}$$

$$\alpha A_d + \beta B_d \;=\; \alpha\left(A_d - \frac{A_0}{B_0}\,B_d\right). \tag{6.7}$$

The pattern. Each bracket has the same structure: the actual future payoff of A in that state, minus what the initial price of A would grow to if invested at the numeraire's rate of return in that state. If the first term exceeds the second, holding A beats holding B in that state; if less, B wins. Arbitrage is ruled out precisely when neither asset dominates the other in both states simultaneously.

For no arbitrage, the two bracketed factors must have opposite signs:

$$\left.\begin{aligned} A_u - \frac{A_0}{B_0} B_u &> 0 \\ A_d - \frac{A_0}{B_0} B_d &< 0 \end{aligned}\right\} \quad \text{all other cases lead to arb.} \tag{6.8}$$

(Implicit non-degeneracy assumption: the 2\$×\$2 payoff matrix $\left(\begin{smallmatrix} A_u & A_d \\ B_u & B_d \end{smallmatrix}\right)$ is non-singular, equivalently $A_u/B_u \neq A_d/B_d$. This is the standard market-completeness hypothesis: if the two assets' relative payoffs coincide in the up and down states, the two bracketed coefficients are linearly dependent and the "both non-zero, opposite signs" dichotomy collapses. For a bond numeraire $B_u = B_d$ automatically the brackets are monotone in the same direction of α so the argument carries through; for a general numeraire, non-degeneracy must be imposed at this point for the rest of the derivation to go through.)

Equivalently, dividing through by $B_u, B_d > 0$:

$$\boxed{\frac{A_d}{B_d} < \frac{A_0}{B_0} < \frac{A_u}{B_u}} \qquad \text{(no-arbitrage condition).} \tag{6.9}$$

Why the relative price must sit strictly between the two future relative prices. If $A_0/B_0 < A_d/B_d$ then A is "cheaper than its worst future relative value" — borrow B, buy A, and you make money in both states. Conversely $A_0/B_0 > A_u/B_u$ means A is "more expensive than its best future relative value" — short A, buy B. The only consistent prices are those strictly inside the interval.

Note that in the bond case $B_u/B_d = 1 + r$ and (6.9) reduces to (6.5).

Why the numeraire-free formulation is more powerful. The bond is a convenient numeraire because its value is nearly deterministic, but there are many situations where a different choice is more natural. When pricing a currency option, the foreign bond is the natural numeraire for payoffs in foreign currency. When pricing a swaption, the annuity is the natural numeraire for payoffs tied to a swap rate. When pricing a spread option (§ Margrabe, later chapters), one of the two underlying assets is itself the numeraire. In every case, the condition (6.9) takes the same form — a sandwich condition on the current relative price — and the pricing measure that emerges adapts accordingly.

2.6.3 6.3 Restatement as existence of a risk-neutral weight q

The sandwich condition (6.9) is useful but still expressed as an inequality. The next step is to convert it into an *equality* — a statement about the existence of a specific probability weight that makes the pricing relation hold exactly. This is the pivot from bounds to formulas.

The inequality (6.9) is equivalent to: $\exists\, q \in (0,1)$ such that

$$\boxed{\frac{A_0}{B_0} = q\,\frac{A_u}{B_u} + (1-q)\,\frac{A_d}{B_d}} \iff \text{no arbitrage.} \tag{6.10}$$

The pivot idea. Equation (6.10) says the *relative* price A_0/B_0 is a convex combination of the two future relative prices. A number lies strictly between two endpoints iff it is a convex combination of them with weights in $(0,1)$ — this is literally the definition. The weight q is the probability that makes the numeraire-relative price of A a $\mathbb{Q}$-martingale; it is *not* the physical probability p in general.

Reading the weight q. Rearranging (6.10), $q = (A_0/B_0 - A_d/B_d)/(A_u/B_u - A_d/B_d)$. This is the *relative position* of the current relative price within the interval defined by the two future relative prices. If A_0/B_0 is at the midpoint of the interval, $q = 1/2$; if it is closer to the up-leaf, q is closer to 1; if closer to the down-leaf, q is closer to 0. Crucially, this weight depends only on the three relative prices A_d/B_d, A_0/B_0, A_u/B_u — not on the physical probabilities, not

on anyone's utility, not on expected returns. It is a pure geometric fact about where the current price sits between its two possible futures.

Comparison with physical probabilities. Under the physical measure $\mathbb{P}$, the up-probability p is whatever it is — some empirical belief about the future, based on data, models, or guesses. The risk-neutral weight q typically differs from p, sometimes dramatically. For a risky asset with positive expected excess return, $q < p$: the risk-neutral measure *tilts weight toward the down-state* to compensate for the physical premium on the up-state. We will see this tilt explicitly in the calibrated tree of §11, where the Girsanov shift $\mu - r$ produces exactly this reweighting.

Equivalently,

$$\tilde{A}_0 \;=\; \mathbb{E}^{\mathbb{Q}}\big[\,\tilde{A}_1\,\big], \qquad \text{where} \qquad \tilde{A}_t \;=\; \frac{A_t}{B_t} \quad \text{(relative price)}, \tag{6.11}$$

under $\mathbb{Q}$ assigning up-probability q.

Why relative prices are what's priced. Absolute prices A_t are quoted in dollars, but a dollar at $t = 1$ is not a dollar at $t = 0$ — you need a clock asset to compare them. Dividing by B_t cancels the clock and makes $\tilde{A}$ a pure risk quantity whose drift can (and must, under $\mathbb{Q}$) be zero.

Martingale terminology, decoded. A *martingale* under $\mathbb{Q}$ is a process whose conditional expectation at any future time, given current information, equals the current value. For the relative price $\tilde{A}$ to satisfy $\tilde{A}_0 = \mathbb{E}^{\mathbb{Q}}[\tilde{A}_1]$ is exactly to say that $\tilde{A}$ is a $\mathbb{Q}$-martingale. This is the most important property in mathematical finance: it says that, once you choose B as the unit of measurement, the relative price of every other asset has zero average drift under $\mathbb{Q}$. Prices don't grow or shrink on average, once denominated in units of B; all the growth in absolute terms is attributed to the numeraire's own growth.

An analogy. Think of the bond B as a clock that moves at rate r per period, and think of the risky asset A as a pedestrian walking through a park. The *absolute* price A_t is the pedestrian's position on the map, which includes whatever movement the clock has accumulated. The *relative* price A_t/B_t subtracts out the clock's motion, leaving only the pedestrian's position relative to a moving reference frame. Under $\mathbb{Q}$, the pedestrian's motion in this reference frame has zero expected drift — they wander, but do not systematically move in any direction. This is the martingale property: zero expected drift in the reference frame of the numeraire.

Why the same q prices everything. Notice that the derivation of q in (6.10) used only the prices of A and B, not of any derivative. Once q is pinned down from (A, B), it is a fixed number that will be used to price every other claim in the market (as we will see in §7). This is the power of no-arbitrage: it derives a single pricing weight from the underlying assets, and that same weight prices all derivatives written on them. A market with one risky asset and one bond, therefore, has only *one* risk-neutral weight — and hence exactly one arbitrage-free price for every contingent claim. This is market completeness in its purest form.

Bridge. The utility argument of §3 produced a price A_0^{buy} that depended on γ and $\mathbb{P}$. The no-arbitrage argument just delivered a price dependent *only* on the tree and r — and the identity (6.10) is exactly the $\gamma \downarrow 0$ limit (4.5) with $\mathbb{P}$ replaced by $\mathbb{Q}$. The same identity follows from no-arbitrage alone, so the pricing measure is *not* an agent's belief — it is a market property.

This is worth pausing over. The utility-based derivation told us: "for a given agent, as $\gamma \to 0$, the indifference price collapses to $\mathbb{E}^{\mathbb{P}}[A_1]/(1+r)$." The no-arbitrage derivation tells us: "for *any* agent, the market-clearing price is $\mathbb{E}^{\mathbb{Q}}[A_1]/(1+r)$." These two statements are structurally identical — same functional form, same discounting — but the second one uses a different probability measure. The implication is that $\mathbb{Q}$ behaves *as if* it were the belief of a universal risk-neutral agent. This fictional agent is the unifying voice that coordinates all the heterogeneous real agents in the market, by providing a single price they can all transact at.

2.7 7. A Third Asset: Replication and Arbitrage-Free Pricing

The power of the two-asset no-arbitrage theorem becomes clear only when we introduce a third asset. So far (6.10) tells us a consistency condition among A, B, and the weight q. But what if we add a claim C — an option, a forward, any derivative — whose payoff (C_u, C_d) is specified in advance? Is C free to have any price? Or does the existing tree force its price uniquely?

The answer, which is the core of arbitrage-free derivatives pricing, is: yes, the price of C is uniquely determined by the tree. If C trades at any other price, an arbitrage exists. We demonstrate this in two ways — via consistency of the risk-neutral weight (this section) and via explicit replication (next section).

Now introduce a derivative / contingent claim C alongside A and B, with its own up/down leaves C_u, C_d:

```
        A_u              B_u              C_u

A_0             B_0             C_0

        A_d              B_d              C_d

    underlying       numeraire       contingent claim
```

Each of A and C, when priced in units of B, must be a martingale under *some* $q \in (0,1)$ (by (6.11) applied individually):

$$\tilde{A}_0^B \;=\; \mathbb{E}^{\mathbb{Q}}\big[\tilde{A}_1^B\big] \;\longrightarrow\; q^{BA} \in (0,1), \tag{7.1}$$

$$\tilde{C}_0^B \;=\; \mathbb{E}^{\mathbb{Q}}\big[\tilde{C}_1^B\big] \;\longrightarrow\; q^{BC} \in (0,1). \tag{7.2}$$

The two-q puzzle. If $q^{BA} \neq q^{BC}$ then the market cannot admit a single equivalent martingale measure — which is exactly the first-fundamental-theorem signal that arbitrage must exist.

Interpreting the superscripts. The notation q^{BA} means "the risk-neutral weight extracted from asset A relative to numeraire B." Similarly q^{BC} is extracted from C relative to B. If the market is arbitrage-free, these two weights must coincide — there can only be one $\mathbb{Q}$ per numeraire. A mismatch $q^{BA} \neq q^{BC}$ is therefore a *diagnostic* of arbitrage: it tells you that the trio (A, B, C) is mispriced relative to some absolute standard, and you can construct a profitable trade.

Consistency question. With q^{BA} extracted from (A, B), the fair C_0 must satisfy

$$\frac{C_0}{B_0} \;\stackrel{?}{=}\; \frac{C_u}{B_u}\, q^{BA} \;+\; \frac{C_d}{B_d}\,(1 - q^{BA}). \tag{7.3}$$

If the right-hand side differs from C_0/B_0, C is mispriced and an arbitrage is available. The *fair* C_0 is forced by the q^{BA} already pinned down from (A, B).

Why (A, B) get to set the rules. In a market with (A, B) and (C), the pair (A, B) provides the hedging instruments — two assets that span the two-state world. The claim C is the new comer, whose fair price must be consistent with what (A, B) allow to be synthesised. This is the asymmetric relationship between *underlying* and *derivative*: the derivative depends on the underlying, not vice versa, and the fair price of the derivative is uniquely determined by the no-arbitrage structure of the underlying.

2.7.1 7.1 Numerical example — mispriced third asset

Three trees:

```
        20               1               120

  10              1              110

        5                1               100

       (A)              (B)        (C, short position)
```

Step 1 — q^{BA} from (A, B) via (6.10):

$$10 \;=\; 20\, q^{BA} + 5\,(1 - q^{BA}) \;\;\Rightarrow\;\; q^{BA} = \tfrac{1}{3}. \tag{7.4}$$

Sanity check. $20 \cdot \frac{1}{3} + 5 \cdot \frac{2}{3} = 10$. The risk-neutral measure puts one-third weight on the up-state — considerably *less* than the symmetric $q = 1/2$, because $A_0 = 10$ is closer to the down-leaf 5 than to the up-leaf 20.

A geometric reading. The current price of A is at position $(10-5)/(20-5) = 1/3$ of the way from down-leaf to up-leaf. That is exactly q. The risk-neutral weight is nothing more than the normalized distance from the down-leaf. Draw a number line from 5 to 20 with 10 marked on it, and eyeball where it sits — you will land on 1/3 immediately.

Step 2 — extract q^{BC} from (C, B):

$$110 \;=\; 120\, q^{BC} + 100\,(1 - q^{BC}) \;\;\Rightarrow\;\; q^{BC} = \tfrac{10}{20} = \tfrac{1}{2}. \tag{7.5}$$

Diagnosis. The weight extracted from C alone is 1/2, implying C's current price sits exactly midway between its two future values. That disagrees with $q^{BA} = 1/3$. Two weights, two different answers — guaranteed arbitrage.

Step 3 — forcing the same $q = 1/3$ onto C:

$$\widehat{C_0} \;=\; 120 \cdot \tfrac{1}{3} \;+\; 100 \cdot \tfrac{2}{3} \;=\; 40 + 66\tfrac{2}{3} \;\approx\; 106.67 \;<\; 110. \tag{7.6}$$

So the quoted $C_0 = 110$ is too expensive relative to what (A, B) enforces — the inconsistency $q^{BA} = 1/3 \neq 1/2 = q^{BC}$ is an arbitrage signature.

Economic translation. C behaves like a near-bond (tiny ± 10 moves around 110) that the market has priced too richly. An arbitrageur shorts C at 110, buys the synthetic replicant out of (A, B) for $\widehat{C_0} \approx 106.67$, and pockets the \$3.33 difference at $t = 0$ while matching every leaf payoff at $t = 1$.

Step 4 — explicit replication. Seek (α, β) in (A, B) matching C's payoff with $V_0 = C_0 = 110$:

$$V_0 \;=\; 10\alpha + \beta - 110 \;=\; 0 \;\;\Rightarrow\;\; \beta = 110 - 10\alpha. \tag{7.7}$$

Match each leaf (with C sold short, hence the $-C_i$ term):

$$\text{up:} \quad 20\alpha + \beta - 120 \;=\; 10\alpha - 10, \tag{7.8}$$

$$\text{down:} \quad 5\alpha + \beta - 100 \;=\; -5\alpha + 10 \;=\; 0. \tag{7.9}$$

The down-leaf equation gives $\alpha = 2$; then $\beta = 110 - 20 = 90$; and the up-leaf equation yields $V_1^{\text{up}} = 10$. Zero cost at $t = 0$, zero payoff in the down-state, $+10$ in the up-state — a textbook arbitrage.

Lessons from the worked example. Three things stand out. First, mispricing anywhere in the market is detectable by extracting risk-neutral weights separately from each asset and comparing. Second, the arbitrage is *constructive*: we explicitly built the portfolio that realizes it, without any "existence proof" handwaving. Third, the profit is riskless at $t = 1$ in the sense that one leaf yields zero and the other yields strictly positive — satisfying the formal definition (5.3)–(5.5).

An operational takeaway for traders. The q-consistency test is essentially what a real-world arbitrageur does when scanning for mispricings: compute the implied risk-neutral probability from each instrument and flag any disagreement. In practice this is done across dozens of related instruments (calls, puts, futures, digitals) at many strikes and maturities. The resulting "implied surface" of q-like probabilities is dense with information and, in liquid markets, remarkably internally consistent. Any significant inconsistency typically represents either a genuine arbitrage opportunity or (more often) a market friction that prevents the trade from being executed cleanly.

The deeper meaning of "mispricing". When we say C is "mispriced" at 110, we mean: given the prices of A and B and the assumption of no arbitrage, C must trade at 106.67. Any other price is logically inconsistent with the existence of a trade in A, B, and C at the stated prices without free lunches. The word "mispriced" therefore has a precise technical meaning: it is not a statement about the "right" price in some cosmic sense, but about internal consistency among quoted prices. A market with internally consistent prices is arbitrage-free; one with inconsistent prices is not.

2.8 8. Replication — The General Recipe

The previous section showed that mispricing is detectable and exploitable. Here we do the reverse: given a claim whose price has not yet been set, we derive the formula for it. The recipe is called *replication*, and it is the fundamental algorithm of derivatives pricing. Given the two-asset market (A, B) and a target claim C, we solve for the portfolio (α, β) that reproduces C's payoff state-by-state. The fair price of C is then the current value of that portfolio.

Given a claim C with leaves C_u, C_d, solve for (α, β) such that the portfolio $\alpha A + \beta B$ replicates C state-by-state:

$$C_u = \alpha A_u + \beta B_u, \tag{8.1}$$

$$C_d = \alpha A_d + \beta B_d. \tag{8.2}$$

Then by no-arbitrage

$$C_0 = \alpha A_0 + \beta B_0, \tag{8.3}$$

otherwise an arbitrage would exist.

The logic of (8.3), spelled out. We have built a portfolio that pays exactly C's payoff in every state. By the definition of arbitrage (5.3)–(5.5), two portfolios that pay identically at $t = 1$ must have identical value at $t = 0$; otherwise buy the cheaper one and short the expensive one for guaranteed riskless profit. The portfolio (α, β) in (A, B) therefore *forces* the price of C to equal $\alpha A_0 + \beta B_0$. This is replication pricing. No expectations, no probabilities, no preferences — just the algebra of payoff matching and the impossibility of free lunches.

Why two equations in two unknowns has a unique solution. The 2×2 system (8.1)–(8.2) has determinant $A_u B_d - A_d B_u$, non-zero precisely when $A_u/B_u \neq A_d/B_d$. That inequality is what makes the market *complete* — one risky asset per source of risk.

Completeness in a sentence. A market is *complete* if every conceivable payoff at $t = 1$ can be replicated by a portfolio of tradable assets at $t = 0$. In our two-state setup, the space of payoffs is two-dimensional (one degree of freedom per state), and we have two assets (A, B) with linearly independent payoff vectors. Linear algebra guarantees a unique (α, β) for every (C_u, C_d). Lose that linear independence — say, if A and B always move together — and the market becomes incomplete: most payoffs are not replicable, and the pricing theory fragments into super-replication and sub-replication bounds.

2.8.1 8.1 Closed-form replicating weights

Multiply (8.1) by B_d, (8.2) by B_u, subtract:

$$\alpha \left(A_u B_d - A_d B_u \right) = C_u B_d - C_d B_u. \tag{8.4}$$

Hence

$$\boxed{\alpha = \frac{C_u B_d - C_d B_u}{A_u B_d - A_d B_u}} \tag{8.5}$$

$$\boxed{\beta = \frac{C_u A_d - C_d A_u}{B_u A_d - B_d A_u}} \tag{8.6}$$

What (8.5) is actually computing. The number α is the *delta* of C with respect to A — the number of units of A that must be held to hedge C. In the bond case where $B_u = B_d = B_0(1 + r)$, the formula simplifies to $\alpha = (C_u - C_d)/(A_u - A_d)$, the familiar "rise over run" delta. This is the Greek you will meet over and over in derivatives pricing: delta is the state-by-state slope of the claim's payoff with respect to the underlying's payoff, and it governs how many units of the underlying you must hold to replicate the claim.

The sign of α. For a call option with $C_u > C_d$ and $A_u > A_d$, the delta is positive: long the underlying. For a put option with $C_u < C_d$, the delta is negative: short the underlying. For a straddle (long call + long put), the deltas partially cancel, and the position is roughly delta-neutral at the strike. These qualitative relationships are all encoded in (8.5), once you grasp it as a linear algebra construct.

The role of β. The bond position β balances the books. Once you have committed to holding α units of A, you need some amount of bond (positive or negative) to make the payoffs match exactly. In the bond case, $\beta = (C_u - \alpha A_u)/B_u = (C_d - \alpha A_d)/B_d$, which is just the residual needed to make the numbers work in each state. A long β means funding the A position partly with cash on hand; a short β means borrowing against the A position.

A thought experiment on dimensional analysis. If A is measured in dollars per share and B is measured in dollars per bond, then α has units of "shares" and β has units of "bonds." The portfolio value $\alpha A_0 + \beta B_0$ has units of dollars, as it should. The formula for α has units of (dollars · dollars)/(dollars · dollars) — dimensionless, which is exactly what "shares" should be when the underlying is quoted per-share. Dimensional consistency is a quick sanity check for any derived formula, and (8.5) passes it.

Delta is a ratio, not a number. A subtle point: "delta" in the Black–Scholes world is often written as a dimensionless number like 0.62, but it is really a ratio of two payoff quantities. The numerator is the change in the option's value; the denominator is the change in the underlying's value. Both have units of dollars, so the ratio is dimensionless — but its meaning is "how many units of the underlying per unit of option." When you hedge 1,000 options with delta 0.62, you hold 620 units of the underlying. This is the most important operational use of α: it tells the trader how many shares to buy per option sold.

2.8.2 8.2 Pricing identity and explicit q

Substituting the replicating weights back into the cost of the portfolio gives an extremely clean pricing identity — the *risk-neutral pricing formula* for C. This is the single most important equation in the chapter.

Substituting back into (8.3) yields

$$\boxed{\frac{C_0}{B_0} \;=\; q\,\frac{C_u}{B_u} \;+\; (1-q)\,\frac{C_d}{B_d}} \tag{8.7}$$

where

$$q \;=\; \frac{\dfrac{A_0}{B_0} - \dfrac{A_d}{B_d}}{\dfrac{A_u}{B_u} - \dfrac{A_d}{B_d}}. \tag{8.8}$$

The explicit risk-neutral weight, decoded. Formula (8.8) is a *lever ratio*: the distance from the down-leaf to the current relative price, normalised by the spread between the two leaves. If A_0/B_0 is closer to A_u/B_u, then q is near 1 (up-state gets heavy weight). The weight has nothing to do with physical probabilities or utility — only price geometry.

Why equation (8.7) is so remarkable. It says that *any* claim C in our two-state model can be priced by computing its expected payoff under the single weight q extracted from the underlying (A, B), and then discounting. The same q prices every derivative: calls, puts, digital options, variance swaps, cubes, anything. The market does not need a new model for each new product — once (A, B) are priced, the entire derivatives universe is determined. This is the miracle of completeness and no-arbitrage combined.

Contrast with naïve discounted expectation. Under the physical measure, a call's expected payoff depends on the real-world probability p, which in turn depends on beliefs, models, or market sentiment. Under the risk-neutral measure, it depends only on q, which is extracted directly from the prices $A_0, A_u, A_d, B_0, B_u, B_d$ — all of which are *observable* (or specified). So two traders can disagree entirely about p yet agree exactly on C_0, provided they agree on the observable prices. Risk-neutral pricing depersonalizes and de-psychologizes pricing.

A deeper remark on the cancellation of p. The physical probability p appears nowhere in (8.7)–(8.8). It has literally no role in derivatives pricing once we assume no arbitrage. This is one of the most counterintuitive facts in all of quantitative finance: a derivative's price does not depend on whether the underlying is expected to go up or down.

At first encounter this seems wrong — surely a call on a stock should be worth more if the stock is "expected" to rise? But remember, the call's payoff is replicable by a hedging portfolio that has the same payoff regardless of direction. The fair price is the cost of that portfolio. If the stock is expected to rise, the market maker *already owns the replicating portfolio* after selling the call; they are indifferent to which state occurs. The expected direction contributes nothing to their position's value.

The economics behind the cancellation. In a sense, the physical drift μ is "already priced in" to A_0. A stock expected to rise trades at a higher A_0 than an identical stock expected to fall; the cost of replication reflects this via A_0's appearance in the formula. To double-count μ by also adjusting the option price would be an error. This is the difference between μ and a separate "drift" parameter: μ is already encoded in the current price, and the formula extracts everything about the future distribution it needs from the tree structure and the no-arbitrage constraint.

Moreover the same q satisfies

$$\frac{A_0}{B_0} \;=\; q\,\frac{A_u}{B_u} \;+\; (1-q)\,\frac{A_d}{B_d}, \tag{8.9}$$

confirming self-consistency.

2.8.3 8.3 Bond case: the classical formula

In the bond case $B_u = B_d = B_0(1+r)$, formulas (8.7)–(8.8) reduce to

$$C_0 \;=\; \frac{1}{1+r}(q\,C_u + (1-q)\,C_d), \qquad q \;=\; \frac{A_0(1+r) - A_d}{A_u - A_d}. \tag{8.10}$$

The condition $q \in (0,1)$ is precisely the no-arb bound $A_d < A_0(1+r) < A_u$ from (6.5).

The canonical one-period binomial pricing formula. Equation (8.10) is what most textbooks write down as "the" binomial option pricing formula. It says: discount at the risk-free rate the q-weighted average of the up and down payoffs. The weight q is the relative position of the *forward price* within the up–down leaf interval. Both ingredients — the discount and the weight — are deterministic once (A_0, A_u, A_d, r) are specified, so C_0 is deterministic too. The option price is a pure function of observable market data.

Numerical check. Take $A_0 = 100$, $A_u = 110$, $A_d = 90$, $r = 5\%$ per period. Then $q = (105-90)/(110-90) = 15/20 = 0.75$. A call struck at 100 pays $C_u = 10$, $C_d = 0$, so $C_0 = (0.75 \cdot 10 + 0.25 \cdot 0)/1.05 = 7.5/1.05 \approx 7.14$. Observe that $q = 0.75 > 0.5$, reflecting that the forward $A_0(1+r) = 105$ is closer to the up-leaf than to the down-leaf; this is more aggressive upward tilt than symmetric, and it is consistent with the fact that a high interest rate makes riskless storage more attractive, so to be willing to hold A, the up-scenario must be weighted heavily.

2.8.4 8.4 First Fundamental Theorem (one-period form)

We now have everything needed to state one of the two pillars of mathematical finance: the First Fundamental Theorem of Asset Pricing. It is the formal equivalence between the economic concept of no-arbitrage and the probabilistic concept of an equivalent martingale measure.

$$\text{No-arbitrage} \iff \exists\,\mathbb{Q} \sim \mathbb{P} \text{ s.t. for all traded assets } X,\;\; \tilde{X}_0 \;=\; \mathbb{E}^{\mathbb{Q}}[\,\tilde{X}_1\,],$$

where

$$\tilde{X}_t \;=\; \frac{X_t}{B_t} \qquad \text{and} \qquad \mathbb{P}(B_t > 0) = 1 \quad (B \text{ is called a numeraire asset}). \tag{8.11}$$

The FFT as a dictionary. On the left is an *economic* property (no free lunches). On the right is a *probabilistic* property (existence of an equivalent measure under which relative prices are martingales). The theorem says these two languages are equivalent.

Why the equivalence is profound. A-priori there is no reason to expect an economic condition about trading to translate into a probabilistic condition about measures. The FFT asserts, as a theorem, that they are the same statement expressed in different languages. This allows practitioners to use either perspective interchangeably: when

proving absence of free lunches, you can often construct the measure; when computing prices, you use the measure. The same object plays two roles.

The "equivalent" in equivalent martingale measure. Recall $\mathbb{Q} \sim \mathbb{P}$ means the two measures agree on which sets have probability zero. This matters because the economic no-arbitrage condition is stated under $\mathbb{P}$ — events that "cannot happen" under $\mathbb{P}$ should not sneak into the arbitrage definition. Requiring $\mathbb{Q} \sim \mathbb{P}$ preserves the economic content of arbitrage when we switch to the probabilistic language.

Name clarification. When the specific numeraire is the cash / money-market account $B_t = e^{rt}$, the associated $\mathbb{Q}$ is given the special name risk-neutral measure, and the discounting factor $1/B_t$ is the familiar e^{-rt}. Other numeraires (stock, bond) yield the *stock measure* or *forward measure* (Ch. 9 Margrabe, Ch. 12 LIBOR caps).

In the bond case this specialises to

$$\boxed{C_0 \;=\; \frac{1}{1+r}\,\mathbb{E}^{\mathbb{Q}}[\,C_1\,]} \qquad \text{otherwise } \exists \text{ an arbitrage.} \tag{8.12}$$

The formula that launched derivatives markets. Equation (8.12) — read "the price today equals the discounted expected payoff under $\mathbb{Q}$" — is the discrete-time prototype of the Black–Scholes formula, the Heston formula, and essentially every practical derivatives pricing formula in use today. Every continuous-time model adds layers of complexity (diffusions, jumps, stochastic volatility, stochastic interest rates), but the core structure is the same: find the right $\mathbb{Q}$, compute the expected payoff under it, discount back. The binomial model is the atom. The rest of the book builds molecules.

A historical remark. The recognition that derivatives could be priced by replication rather than by preference specification was a revolution in finance, dating to the 1970s. Before that era, options were priced by informal rules of thumb, often anchored to the naive expectation formula (1.3) and adjusted by various "risk premia" at the trader's discretion. The no-arbitrage framework provided the first mathematically rigorous alternative and transformed options from a niche instrument into a mainstream asset class. The growth of global derivatives markets from tiny to multi-trillion-dollar notionals is, in large part, a direct consequence of the availability of reliable pricing and hedging formulas. None of it would be possible without the simple identity (8.12) and its continuous-time generalisations.

The philosophical leap. The leap from utility-based indifference pricing to no-arbitrage replication pricing is not just a mathematical simplification — it is a conceptual revolution. Under utility pricing, each trader has their own price, and there is no unique fair value. Under replication pricing, there is a *single* fair value that every rational trader must agree on, because deviating from it creates an exploitable arbitrage. The market as an institution becomes possible precisely because replication collapses the subjectivity of preferences into the objectivity of prices.

A subtlety that will matter later. We have said that the price is "unique" in the complete-market case. In *incomplete* markets — where there are more sources of risk than hedging instruments — uniqueness fails. Multiple equivalent martingale measures exist, each giving a different price, and the no-arbitrage theory only pins down a range of acceptable prices. This is the technical content of the Second Fundamental Theorem of Asset Pricing, which we do not prove in this chapter but mention for context. The binomial model is complete, so uniqueness holds; many realistic models (stochastic volatility, jumps) are incomplete, and uniqueness must be recovered via additional modelling assumptions or preferences.

2.9 9. Worked Example — European Call on A Struck at 100

To consolidate the machinery of §8, we apply it end-to-end to the simplest option: a European call. This example is so standard that every derivatives textbook does it, yet it repays careful study. It illustrates replication, risk-neutral pricing, and arbitrage construction all in one compact package.

A concrete illustration of §8 with a call option.

Setup.

- Asset A: $100 \to \{120, 90\}$.
- Bond B: $1 \to \{1, 1\}$ (so $r = 0$).

European call and put payoffs

Figure 3: European call and put payoffs

- Claim C: payoff $\to \{20, 0\}$ (call struck at $K = 100$, since $\max(120 - 100, 0) = 20$ and $\max(90 - 100, 0) = 0$).

A call option on A struck at K gives the right (not obligation) to buy A at price K at $t = 1$:

$$\mathrm{Call}(A_1; K) \;=\; \max(A_1 - K,\ 0). \tag{9.1}$$

A put option gives the right to sell at K:

$$\mathrm{Put}(A_1; K) \;=\; \max(K - A_1,\ 0). \tag{9.2}$$

Reading the payoffs. A call pays when the underlying ends above the strike; a put pays when it ends below. Both are piecewise linear in A_1 with a kink at the strike K. The kink is the source of all the interesting behavior in options pricing — it is what makes call prices convex in the spot, what creates vega, and what drives the volatility smile in more general models. In our simple binomial world the kink means the up-state payoff and the down-state payoff can be wildly different, generating the need for a non-trivial α in the replicating portfolio.

Why we insist on "right, not obligation". An option's payoff is $\max(\cdot, 0)$, capped below at zero. This is the optionality: the holder exercises only when it is in their interest. If the underlying ends below the strike, the holder walks away — they do not have to take a loss. That is why options are valuable; they give strictly positive payoff in favourable states and zero payoff in unfavourable ones.

Their payoffs at maturity are the canonical hockey-sticks:

```
C_1 = max(A_1 - K, 0)




0                      → A_1
                       K = 100

Call payoff: flat at 0 for A_1  K, then rises at slope 1.
(The kink at K is the source of gamma, vega, and smile.)
```

Vanilla call and put payoffs.

2.9.1 9.1 Solve replication

System from (8.1)–(8.2):

$$\begin{cases} 120\,\alpha + \beta = 20 \\ 90\,\alpha + \beta = 0 \end{cases}$$

Subtract: $30\alpha = 20 \Rightarrow \alpha = 2/3$. Then $\beta = -90 \cdot 2/3 = -60$.

$$\alpha = \tfrac{2}{3}, \qquad \beta = -60. \tag{9.3}$$

Unpacking the hedge. To replicate a long call, we buy 2/3 of a share and borrow 60 dollars (negative bond position). Let's check the arithmetic in each state. In the up-state, we have $2/3 \cdot 120 = 80$ in stock and owe 60 in bond, net value 20 — exactly the call's up-payoff. In the down-state, we have $2/3 \cdot 90 = 60$ in stock and owe 60 in bond, net value 0 — exactly the call's down-payoff. Replication works.

The delta interpretation. The value $\alpha = 2/3$ is the call's delta: for every 1 point increase in A, the call's value increases by 2/3. The range of the call payoff is $20 - 0 = 20$; the range of the underlying is $120 - 90 = 30$; their ratio is 2/3. This is the discrete-time analog of the Black–Scholes $\Delta = N(d_1)$ formula. In continuous time, deltas are derivatives; in discrete time, they are finite differences.

2.9.2 9.2 No-arbitrage value of C

$$C_0 \;=\; \tfrac{2}{3} \cdot 100 + (-60) \;=\; \tfrac{200}{3} - 60 \;=\; 6\tfrac{2}{3}. \tag{9.4}$$

The fair call price. The cost of setting up the replicating portfolio today is $(2/3) \cdot 100 - 60 = 200/3 - 60 = 20/3 \approx 6.67$. By the replication argument, this must be the fair price of the call. Any other price creates an arbitrage.

2.9.3 9.3 Cross-check via risk-neutral formula

From (8.10) with $r = 0$:

$$100 \;=\; q \cdot 120 + (1-q) \cdot 90 \;=\; 90 + 30q \;\;\Rightarrow\;\; q = \tfrac{1}{3}. \tag{9.5}$$

Therefore

$$C_0 \;=\; q \cdot 20 + (1-q) \cdot 0 \;=\; \tfrac{1}{3} \cdot 20 \;=\; 6\tfrac{2}{3}. \tag{9.6}$$

Both methods agree: $C_0 = 6\frac{2}{3}$.

Two roads to the same destination. Replication builds the portfolio explicitly and reads off its cost. Risk-neutral pricing computes an expectation under the synthetic measure. The two answers are guaranteed to match by construction — the same q arises from both. This redundancy is useful in practice: if your replication and your risk-neutral calculation disagree, one of them has a bug. Always cross-check.

Note on q vs. p. Nothing in this example specified the physical up-probability p. We could have $p = 0.9$ (the market expects a rally), $p = 0.1$ (the market expects a crash), or $p = 0.5$ (symmetric). The call's fair price is 20/3 regardless. The risk-neutral weight $q = 1/3$ is not a belief about the future; it is a geometric property of the tree and the interest rate.

2.9.4 9.4 Constructing the arbitrage when $C_0 = 6$ (mispriced)

Trees:

```
        120                 1                 20

100                 1                 6

        90                  1                 0

        (A)                 (B)          (C, mispriced at 6)
```

Strategy (values from the notes):

- Hold $-\frac{2}{3}$ units of A (short A).
- Hold $+60$ units of B (long bond).
- Hold $+1$ unit of C (long claim).

Initial cashflow:

$$V_0 \;=\; -\tfrac{2}{3} \cdot 100 + 60 \cdot 1 + 1 \cdot 6 \;=\; -\tfrac{200}{3} + 60 + 6 \;=\; 6 - 6\tfrac{2}{3} \;=\; -\tfrac{2}{3}. \tag{9.7}$$

i.e. the arbitrage pays $+\frac{2}{3}$ today.

Terminal values:

$$V_1^{(u)} \;=\; -\tfrac{2}{3}\cdot 120 + 60 + 20 \;=\; -80 + 60 + 20 \;=\; 0, \tag{9.8}$$

$$V_1^{(d)} \;=\; -\tfrac{2}{3}\cdot 90 + 60 + 0 \;=\; -60 + 60 \;=\; 0. \tag{9.9}$$

So the agent pockets $\frac{2}{3}$ today for sure, with zero risk at $t = 1$ — an arbitrage, confirming that $C_0 = 6$ is not the no-arbitrage price; the no-arb price is $6\frac{2}{3}$.

Dissecting the arbitrage trade. The arbitrageur sees the call trading below its fair value and buys it. To offset the risk, they short the replicating portfolio — which here means shorting 2/3 of a share and lending 60 dollars. The cost of the short is $-2/3 \cdot 100 + 60 = -6.67$; they receive the call for 6, net cash in hand 0.67. At $t = 1$, every scenario's cash flow on the composite position sums to zero: the call payoff cancels the short-replicant payoff exactly. The arbitrageur walked off with 0.67 today and owes nothing tomorrow.

The structural takeaway. Mispricing creates arbitrage, and arbitrage creates pressure to correct the price. Real markets don't stay mispriced — rational arbitrageurs with capital scale their positions until the mispricing vanishes. The no-arbitrage price 20/3 is therefore the *attractor* price: deviations from it are transient. This is why pricing theory is useful in practice even though real markets are imperfect: the model price is the point towards which actual prices are drawn.

How large are real-world arbitrage profits? For frictionless markets with perfect information, mispricing arbitrages should be vanishingly rare (because they are quickly competed away). In practice, mispricings do appear but the per-trade profits are tiny, often fractions of a basis point. The money made by real arbitrageurs comes from volume and leverage — doing a lot of tiny trades. This is why statistical arbitrage desks and high-frequency trading firms exist: the economics work only at scale. The pedagogical 0.67-dollar arbitrage in our example is much bigger than anything a real market would sustain.

Caveats to the pedagogical story. In our worked example, the arbitrage was clean and unambiguous. In real markets, "arbitrages" often come with hidden risks: model mis-specification (what if the tree is wrong?), liquidity risk (can you unwind?), counterparty risk (will the counterparty pay?), funding risk (can you borrow cheaply enough?). These risks are invisible in a one-period, two-state model but are central to actual arbitrage trading. The classical "risk-free arbitrage" is the benchmark; the real-world cousin is "statistical arbitrage," which is only approximately risk-free. Understanding the ideal model first makes the real-world deviations easier to recognise.

2.10 10. Multi-Period Binomial

One period is enough to derive the core machinery but not enough to approximate real markets. Real stock prices move on millisecond timescales; options expire weeks or years in the future; the number of possible paths is astronomical. The multi-period binomial tree is our bridge from the one-period toy to realistic dynamics. The key observation is that multi-period no-arbitrage pricing is just one-period pricing applied iteratively, *backward*, through the tree.

Extend to N periods. Let $x_1, x_2, \ldots$ be i.i.d. Bernoulli ± 1 with $\mathbb{P}(x_n = +1) = p$ and $\mathbb{P}(x_n = -1) = 1 - p$. The asset A compounds multiplicatively:

$$A_n \;=\; A_{n-1}\, e^{c\, x_n}, \tag{10.1}$$

where c is a step size (to be calibrated in §11). The tree branches recursively:

```
                    A_uu

          A_u

A_0                 A_ud  (= A_du, recombining)

          A_d

                    A_dd
```

```
t = 0          t = 1                t = 2
```

and likewise for B. These are recombining trees, so $A_{ud} = A_{du}$.

Why a multiplicative update. Using e^{cx_n} instead of an additive update guarantees $A_n > 0$ for every path — the tree never crosses zero. The multiplicative form also makes $\log A_n$ a random walk, which is what lets the CLT in §13 deliver a *log-normal* limit.

Recombining trees save computation. If up-then-down yields the same state as down-then-up, the number of terminal states after N steps is $N + 1$, not 2^N. For $N = 20$ this is 21 vs. over a million — a huge computational savings. Recombination is a direct consequence of the multiplicative structure: each step multiplies by e^c or e^{-c}, and multiplication commutes. Additive updates would generically not recombine.

The i.i.d. assumption, questioned. Are real asset returns really i.i.d.? Obviously not in any strict sense — volatility clusters, returns exhibit serial correlation at some horizons, jumps are not Gaussian. But for the purposes of establishing the basic pricing framework, i.i.d. is the right starting point. Later chapters will relax it: stochastic volatility breaks the identical-distribution assumption; jump diffusions break both the i.i.d. and the continuity; regime-switching models make each step's distribution conditional on a hidden state. The binomial tree with i.i.d. increments is the parent model from which these generalizations descend.

Why the multiplicative update is the right choice for log-returns. In empirical finance, log-returns are much closer to stationary and homoskedastic than absolute price changes. A 1 drop on a \$100 stock is a 1% move; a 1 drop on a \$10 stock is a 10% move. Treating these as identical (as an additive model would) is unrealistic. The multiplicative model in (10.1) treats them as proportional, which is how real markets behave: volatility and drift are naturally measured as percentages, not absolute amounts. This is a minor modelling decision with enormous practical consequences.

From one period to N periods: what changes? The core machinery of §§5–8 carries over directly. At each node of the multi-period tree, the local no-arbitrage condition (6.5) applies and gives a local risk-neutral weight q. The global pricing formula becomes an iterated expectation under the product measure $\mathbb{Q}$. Everything else is inherited. The multi-period tree is, in this sense, just a concatenation of one-period trees — no new conceptual ingredient enters.

Combinatorics of the tree. After N steps the tree has $N + 1$ terminal states, indexed by the number of up-moves $k \in \{0, 1, \dots, N\}$. The terminal price is $A_0 u^k d^{N-k}$ where $u = e^c$, $d = e^{-c}$. The probability of arriving at state k is $\binom{N}{k} p^k (1-p)^{N-k}$, a binomial distribution. This is why the model is called binomial: the terminal distribution is binomial in the number of up-moves. In the large-N limit (with diffusive scaling), the binomial distribution converges to a normal distribution — the classical de Moivre–Laplace theorem — and we recover the log-normal model for the asset price.

2.10.1 10.1 Filtration and arbitrage

In a multi-period setting, a trading strategy is no longer a single pair (α, β) but a sequence (α_t, β_t) of portfolio weights that can change at each time step. The strategy must be *adapted* to the available information, and the formal tool for this is the filtration.

An arbitrage is a strategy adapted to the filtration $\mathcal{F}_t$ generated by asset prices (i.e. (α_t, β_t) are $\mathcal{F}_t$-measurable) such that

- (i) $V_0 = 0$,
- (ii) $\exists\, t$ s.t. $\mathbb{P}(V_t \geq 0) = 1$ and $\mathbb{P}(V_t > 0) > 0$.

What "adapted" rules out. An adapted strategy can only use information available by time t — it cannot peek at A_{t+1} before choosing α_t.

Self-financing, now explicit. In the multi-period setting, we must also impose that any rebalancing is paid for by the existing portfolio — no new capital injections. Formally, $V_{t+1}^- = V_{t+1}$ where V_{t+1}^- is the portfolio value before rebalancing at time $t + 1$. Without this condition, the trader could magically top up the account between periods and every claim would be trivially replicable. The self-financing constraint is what forces the trader to work within the budget they have.

Filtration as "what you know". The σ-algebra $\mathcal{F}_t$ represents all information available up to time t: the realized prices $A_0, A_1, \dots, A_t$ and anything else observable by time t. An $\mathcal{F}_t$-measurable random variable is one whose value

is determined by that information — no future peeking allowed. This is what makes the strategy legitimate: you can trade based on what you have seen, not on what you will see.

2.10.2 10.2 Martingale characterisation (multi-period)

The multi-period First Fundamental Theorem generalizes (8.11)–(8.12) to a statement about conditional expectations at every pair of times. This single statement bundles all the one-period no-arbitrage conditions into a unified framework.

$$\boxed{\text{no arb} \iff \exists\, \mathbb{Q} \sim \mathbb{P} \text{ s.t. } \forall\, t < s \text{ and all traded } X, \quad \tilde{X}_t \;=\; \mathbb{E}^{\mathbb{Q}}\Big[\, \tilde{X}_s \,\Big|\, \mathcal{F}_t \,\Big]\,.} \tag{10.2}$$

i.e. relative prices are martingales. This bundles infinitely many no-arbitrage conditions (one per (t, s) pair) into a single statement about conditional expectations.

Why conditional expectation replaces unconditional. In the one-period world, there was only one time interval $[0, 1]$, so the martingale condition was a statement about unconditional expectations. In the multi-period world, the price at t has to be the $\mathbb{Q}$-conditional expectation of the price at $s > t$, where the conditioning is on $\mathcal{F}_t$, the history up to t. This is because at time t we already know more than we did at time 0; our pricing formula must reflect the updated information set.

The tower property. A useful technical fact: the conditional expectation $\mathbb{E}^{\mathbb{Q}}[\cdot|\mathcal{F}_t]$ satisfies $\mathbb{E}^{\mathbb{Q}}[\mathbb{E}^{\mathbb{Q}}[\cdot|\mathcal{F}_s]|\mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}}[\cdot|\mathcal{F}_t]$ for $t \leq s$. This is the martingale version of the "law of iterated expectations" and is what lets us price multi-period claims by *backward induction*: starting from the terminal payoff, roll back one step at a time using the one-period formula, and arrive at today's price. This procedure is the computational backbone of every binomial/trinomial pricing engine in practice.

Practical use. To price an American option, start from the leaves of the tree (known payoffs), step back one period at a time, at each node compute $\max(\text{exercise value}, \text{discounted } q\text{-expected continuation value})$, and iterate until you reach the root. The whole procedure rests on (10.2).

A worked example of backward induction (conceptually). Imagine a three-period tree with $N = 3$ and a European put struck at K. At the terminal nodes ($t = 3$), the payoff is known: $\max(K - A_3, 0)$ for each of the four terminal states. At $t = 2$, each of the three nodes has two children. Compute the price at each $t = 2$ node as $\frac{1}{1+r\Delta t}[q \cdot C_{\text{up child}} + (1 - q) \cdot C_{\text{down child}}]$. Repeat at $t = 1$ (using the $t = 2$ prices as the "payoffs"), and again at $t = 0$. You have now priced the put. Each step uses only the one-period formula (8.10), applied recursively. The entire multi-period pricing engine is one-period pricing in a loop.

Why the recursion is well-defined. The tower property of conditional expectations (mentioned above) ensures that backward induction gives the correct unconditional expectation. This is not an approximation — it is an exact consequence of the martingale structure. The numerical schemes in production pricing engines are simply computer implementations of this backward induction, augmented with minor efficiency tricks (e.g. storing only two columns of the tree at a time instead of the full lattice).

The power of path-independence. For European claims (payoff depends only on A_T), the multi-period tree has $N + 1$ terminal nodes and pricing is fast. For path-dependent claims (e.g. Asian options whose payoff depends on the average price), the state space expands — at each node we must track the running average — and pricing becomes more expensive. Even so, the backward-induction framework survives; we just track more state variables. This generality is another virtue of the binomial framework: it extends cleanly to many exotic payoff structures, albeit with rising computational cost.

2.11 11. Matching Drift and Volatility

The multi-period tree is a combinatorial object; to use it for pricing real derivatives we must *calibrate* it — pick the step size c and the up-probability p so that the tree's terminal distribution approximates a target continuous-time model. The standard target is geometric Brownian motion with drift μ and volatility σ. This section performs the calibration.

Goal: calibrate (c, p) so that the discretised tree (10.1) approximates a log-normal model

$$A_n \;=\; A_{n-1}\, e^{\sigma\sqrt{\Delta t}\, x_n} \tag{11.1}$$

(with step size $c = \sigma\sqrt{\Delta t}$) whose terminal distribution has target mean and variance set by continuous parameters (μ, σ).

Why $c = \sigma\sqrt{\Delta t}$ and not $\sigma\Delta t$. In a random walk, total variance over N steps is $Nc^2 = (T/\Delta t)c^2$; matching $\sigma^2 T$ forces $c^2 = \sigma^2\Delta t$. A linear $c \propto \Delta t$ would vanish too fast, collapsing the tree to a deterministic limit. The $\sqrt{\Delta t}$ scaling is what makes the limit *Brownian* rather than *trivial*.

The diffusive scaling. The pattern $c \propto \sqrt{\Delta t}$ is called *diffusive scaling* and is pervasive in continuous-time probability. In a Brownian motion, the standard deviation over time T is $\sigma\sqrt{T}$, not σT. The reason is that positive and negative increments cancel most of the way, and what survives grows with the *square root* of the number of steps. This is the central-limit-theorem regime. If we used $c \propto \Delta t$ instead, each step would be too small relative to the number of steps, and the total variance would vanish. If we used $c \propto \sqrt{\Delta t}$, it balances exactly, and the limit is a finite-variance diffusion.

2.11.1 11.1 Matching the mean of the log-return

With $c = \sigma\sqrt{\Delta t}$ fixed, we now tune p to match a target drift. The natural target is that the expected log-return over a period of length T equal the log-normal $(\mu - \frac{1}{2}\sigma^2)T$.

Sum N steps:

$$\ln\!\left(\frac{A_T}{A_0}\right) \;=\; \sigma\sqrt{\Delta t}\,\sum_{n=1}^{N} x_n, \qquad N\,\Delta t = T. \tag{11.2}$$

What this says. The total log-return from 0 to T is a sum of N i.i.d. Bernoulli ± 1 increments, each scaled by $\sigma\sqrt{\Delta t}$. This is a *random walk* in log-space, with step size $\sigma\sqrt{\Delta t}$. The terminal value is Gaussian in the large-N limit (by the CLT), which is how we get log-normality.

Take expectation:

$$\mathbb{E}\left[\ln\frac{A_T}{A_0}\right] \;=\; \sigma\sqrt{\Delta t}\cdot N\,(2p-1) \;=\; (\mu - \tfrac{1}{2}\sigma^2)\,T. \tag{11.3}$$

The target continuous-time log-mean is $(\mu - \frac{1}{2}\sigma^2)T$ because of the Itô correction for geometric Brownian motion. Solving for p:

$$2p - 1 \;=\; \frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\,\sqrt{\Delta t}, \tag{11.4}$$

$$\boxed{p \;=\; \tfrac{1}{2}\left[1 \;+\; \frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\,\sqrt{\Delta t}\right].} \tag{11.5}$$

The Itô correction, previewed. For A to grow on average at $e^{\mu T}$, its *log* must grow at $(\mu - \frac{1}{2}\sigma^2)T$ because the exponential adds back the variance penalty via its MGF. This is the single most important "extra $\frac{1}{2}\sigma^2$" in all of continuous-time finance.

How (11.5) works. At $\Delta t = 0$ the formula collapses to $p = 1/2$ — a perfectly symmetric coin. As Δt grows, p tilts away from 1/2 by an amount proportional to $(\mu - \frac{1}{2}\sigma^2)/\sigma \cdot \sqrt{\Delta t}$. The tilt is *small* for small steps; the drift emerges only after summing $N \sim 1/\Delta t$ steps.

The Sharpe-ratio interpretation. The quantity $(\mu - \frac{1}{2}\sigma^2)/\sigma$ is closely related to the *Sharpe ratio* of the asset (excess return per unit volatility, with a $\frac{1}{2}\sigma^2$ correction for log-return convexity). A high-Sharpe asset has a large up-tilt, because the market expects frequent up moves relative to down moves. A zero-Sharpe asset has $p = 1/2$, perfectly

symmetric, no expected drift in log-return. The calibration formula (11.5) is literally translating continuous-time Sharpe ratios into discrete-time coin biases.

A numerical illustration of the tilt. Take a typical equity with $\mu = 10\%$, $\sigma = 20\%$ per year, and a daily step size $\Delta t = 1/252$. Then the log-drift per year is $\mu - \frac{1}{2}\sigma^2 = 0.1 - 0.02 = 0.08$, and the tilt per step is $(0.08/0.20) \cdot \sqrt{1/252} \approx 0.4 \cdot 0.063 \approx 0.025$. So $p \approx \frac{1}{2}(1 + 0.025) \approx 0.513$. The up-probability is just barely above one-half: 51.3% vs. 50%. This is how weak the daily drift is relative to daily noise. Yet over a year of 252 such steps, the cumulative tilt adds up to the 8% log-drift we targeted. The CLT is the reason tiny daily imbalances can produce substantial annual drifts — large numbers of small edges aggregate into meaningful outcomes.

The constraint $p \in (0, 1)$. For (11.5) to make sense, we need $|(\mu - \frac{1}{2}\sigma^2)\sqrt{\Delta t}/\sigma| < 1$, otherwise p exits the unit interval. For typical market parameters this is true for any reasonable Δt; it becomes tight only when Δt is so large that the drift overwhelms the volatility, which would indicate that the step size is inappropriate for the asset. The natural requirement is that $\Delta t \ll 1/|\text{Sharpe ratio}|^2$, which typically gives us months or years of leeway for daily steps.

2.11.2 11.2 Matching the variance

With the mean matched, we check what (11.5) does to the variance and confirm that it also converges to the target.

$$\mathbb{V}\left[\ln \frac{A_T}{A_0}\right] \;=\; \sigma^2\,\Delta t \;\cdot N\,\mathbb{V}[x_i]. \tag{11.6}$$

For the Bernoulli ± 1:

$$\mathbb{V}[x_i] \;=\; \mathbb{E}[x_i^2] - (\mathbb{E}[x_i])^2 \;=\; 1 - \left(\frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\right)^2 \Delta t. \tag{11.7}$$

Hence

$$\mathbb{V}\left[\ln \frac{A_T}{A_0}\right] \;=\; \sigma^2\,T \cdot \left(1 - \left(\frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\right)^2 \Delta t\right). \tag{11.8}$$

As $\Delta t \to 0$ the variance tends to $\sigma^2 T$, the target.

Why the correction term vanishes. The Δt correction in (11.7)–(11.8) reflects the fact that $\mathbb{E}[x_i] \neq 0$ when we bias the coin for drift — the biasing takes a small bite out of the variance. But the bite is $O(\Delta t)$, which vanishes in the continuous-time limit, leaving the clean $\sigma^2 T$. This is exactly the diffusive scaling paying off: drift contributes at order Δt, variance at order Δt, but because variance accumulates over $N = T/\Delta t$ steps, it sums to $O(1)$, while the drift bite to the variance sums to $O(\Delta t) \to 0$.

The mean and variance decouple in the limit. In discrete time, the bias introduced for drift slightly affects the variance. In the continuous-time limit, these become independent, matching the Itô calculus decomposition: the drift and volatility of a log-price are separate, non-interacting parameters of a geometric Brownian motion. This decoupling is what makes the log-normal model so tractable.

2.11.3 11.3 Risk-neutral probability q for the calibrated tree

The physical calibration above used μ, the real-world drift. For pricing we need the risk-neutral weight q, corresponding to the measure $\mathbb{Q}$ under which the bond-relative price of A is a martingale. The calculation parallels §11.1 but with μ replaced by r.

Let B be the bank account with continuous rate r:

$$B_t \;=\; e^{rt}, \qquad B_n \;=\; B_{n-1}\,e^{r\,\Delta t}. \tag{11.9}$$

The martingale identity

$$\frac{A_t}{B_t} \;=\; \mathbb{E}^{\mathbb{Q}}\left[\,\frac{A_s}{B_s}\,\middle|\,\mathcal{F}_t\,\right] \tag{11.10}$$

applied to one step gives

$$\frac{A_{n-1}}{B_{n-1}} \;=\; \frac{A_{n-1}}{B_{n-1}}\,e^{\sigma\sqrt{\Delta t}-r\Delta t}\,q \;+\; \frac{A_{n-1}}{B_{n-1}}\,e^{-\sigma\sqrt{\Delta t}-r\Delta t}\,(1-q). \tag{11.11}$$

Cancel A_{n-1}/B_{n-1} and solve:

$$q \;=\; \frac{e^{r\Delta t}-e^{-\sigma\sqrt{\Delta t}}}{e^{\sigma\sqrt{\Delta t}}-e^{-\sigma\sqrt{\Delta t}}}. \tag{11.12}$$

CRR form. Writing $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$, (11.12) becomes the familiar Cox–Ross–Rubinstein formula

$$q \;=\; \frac{e^{r\Delta t}-d}{u-d}. \tag{11.13}$$

The no-arbitrage condition $d < e^{r\Delta t} < u$ is exactly $q \in (0,1)$ — the forward price $A_{n-1}e^{r\Delta t}$ must lie strictly between the down- and up-leaves.

The CRR formula in practice. Equation (11.13) is the workhorse of discrete-time option pricing. To price a multi-period option, one fills the tree backward from the leaves, at each node discounting by $e^{-r\Delta t}$ the q-weighted average of the child node values. The scheme is numerically stable, easy to implement, and converges to the Black–Scholes price as $\Delta t \to 0$. Minor modifications handle American exercise, dividends, and path-dependent payoffs. Almost every trader's first options pricing code is a CRR tree.

Parameter sensitivity. Larger r increases q — the forward is higher in the interval between leaves, so more weight on the up-leaf. Larger σ spreads the leaves apart, making the forward sit closer to the middle, so q moves toward $1/2$. Smaller Δt makes both the leaf separation and the interest accretion smaller, and the limiting $q \to 1/2$ at $\Delta t = 0$ (with $\sqrt{\Delta t}$-scale corrections, as we see next).

2.11.4 11.4 Small-Δt expansion of q

For analytic understanding and for comparison with (11.5), we expand q in small Δt.

Using $e^x \sim 1 + x + \frac{1}{2}x^2 + o(x^2)$:

Numerator:

$$\text{num} \;=\; (1+r\Delta t) - (1-\sigma\sqrt{\Delta t}+\tfrac{1}{2}\sigma^2\Delta t) + \cdots \;=\; \sigma\sqrt{\Delta t} + (r-\tfrac{1}{2}\sigma^2)\Delta t + \cdots \tag{11.14}$$

Denominator:

$$\text{denom} \;=\; (1+\sigma\sqrt{\Delta t}+\tfrac{1}{2}\sigma^2\Delta t) - (1-\sigma\sqrt{\Delta t}+\tfrac{1}{2}\sigma^2\Delta t) + \cdots \;=\; 2\sigma\sqrt{\Delta t} + \cdots \tag{11.15}$$

Ratio:

$$q \;=\; \tfrac{1}{2}\left[\,1 \;+\; \frac{r-\frac{1}{2}\sigma^2}{\sigma}\,\sqrt{\Delta t}\,\right] + \cdots \tag{11.16}$$

This matches (11.5) with μ replaced by r: under $\mathbb{Q}$ the log-drift is $r - \frac{1}{2}\sigma^2$ (the risk-neutral drift).

The punchline. The cosmetic swap $\mu \mapsto r$ from (11.5) to (11.16) *is* the transition from physical to risk-neutral measure in one line. The Girsanov shift is exactly $\mu - r$ — the equity risk premium, absorbed into a re-weighting of the coin.

Economic interpretation of the shift. Under $\mathbb{P}$, the asset grows on average at rate μ (log-drift $\mu - \frac{1}{2}\sigma^2$) — this is what the market actually expects. Under $\mathbb{Q}$, the asset grows on average at rate r (log-drift $r - \frac{1}{2}\sigma^2$) — the same as the bond. The difference $\mu - r$ is the *equity risk premium*: the excess return investors demand to hold the asset over the bond. Risk-neutral pricing *re-weights the probabilities* to absorb this premium into the measure, leaving the asset with zero excess expected return. It's a bookkeeping trick with profound consequences: once under $\mathbb{Q}$, all traded assets look like they earn the risk-free rate, which is what makes discounted expectation a valid pricing rule.

Consequently

$$\mathbb{E}^{\mathbb{Q}}\left[\ln\frac{A_T}{A_0}\right] = (r - \tfrac{1}{2}\sigma^2)\,T, \tag{11.17}$$

$$\mathbb{V}^{\mathbb{Q}}\left[\ln\frac{A_T}{A_0}\right] = \sigma^2\,T + \xi\,\Delta t, \tag{11.18}$$

where ξ is the sub-leading term vanishing as $\Delta t \to 0$.

Why the volatility is the same under both measures. The drift changes from μ to r under the measure change, but the volatility σ is the same. This is a deep feature of the Girsanov theorem: a change of equivalent measure shifts drifts but preserves volatilities (in continuous time, it also preserves quadratic variation). Volatility is a pathwise property, so to speak, and doesn't know about the probability measure; drift is a distributional property and gets re-weighted.

Practical consequence. When traders calibrate an option-pricing model to market prices, they are implicitly calibrating σ (and any other volatility-like parameters) under the risk-neutral measure. They do not need to know μ for pricing. This is fortunate, because μ is notoriously hard to estimate — it requires long historical samples and is never known with precision. Risk-neutral pricing sidesteps this problem entirely.

The implied volatility phenomenon. Because σ is the *only* unknown in the risk-neutral pricing formula once r and the tree parameters are fixed, and because the option price is monotone in σ, one can invert the formula: given an observed option price, solve for σ. This solved value is called the *implied volatility.* It is the market's risk-neutral estimate of σ over the option's life. Implied volatility is the most important output of option pricing theory in practical use: traders quote, reason, and hedge in terms of it, not in terms of raw prices. When someone says "AAPL January 200 calls are trading at 25% vol," they mean the implied volatility from inverting the pricing formula equals 25% per year. All the machinery of this chapter exists, ultimately, to support this one operational concept.

The volatility smile, previewed. A subtle empirical fact is that implied volatilities for options on the same underlying, at the same maturity, but different strikes, typically differ — forming a "smile" or "skew" shape. The Black–Scholes model assumes a single σ, so it cannot explain the smile. The smile is a sign that the log-normal assumption breaks down: real asset returns have heavier tails and more negative skew than the log-normal predicts. Modelling the smile is the subject of stochastic volatility and jump models, which relax the CLT's assumptions in carefully chosen ways. Again, the binomial model with its CLT limit is the benchmark against which these richer models are compared.

2.12 12. Limiting Distribution

Having calibrated the tree we can now ask: what does the terminal distribution look like as $\Delta t \to 0$ ($N \to \infty$)? The answer is the log-normal distribution — equivalently, the tree converges to geometric Brownian motion. This section states the result; §14 proves it rigorously via characteristic functions.

Take $N \to \infty$ in (11.2):

$$\ln\left(\frac{A_T}{A_0}\right) = \sigma\sqrt{\Delta t}\sum_{n=1}^{N} x_n \xrightarrow[N\to+\infty]{d} \mathcal{N}(\,\cdot\,;\,\cdot\,), \tag{12.1}$$

with

$$\text{mean} = (\mu - \tfrac{1}{2}\sigma^2)\,T, \qquad \text{var} = \sigma^2\,T. \tag{12.2}$$

The CLT at work. We are summing N i.i.d. Bernoulli random variables, each scaled by $\sigma\sqrt{\Delta t}$. The mean of the sum is $N \cdot \sigma\sqrt{\Delta t} \cdot (2p-1)$; using (11.5) this simplifies to $(\mu - \frac{1}{2}\sigma^2)T$. The variance is $N \cdot \sigma^2 \Delta t \cdot \mathbb{V}[x_i] \to \sigma^2 T$ as $\Delta t \to 0$. The classical central limit theorem guarantees the sum converges in distribution to a Gaussian with these parameters. This is nothing more, and nothing less, than the CLT applied to a carefully scaled sequence of i.i.d. coin flips.

Equivalently,

$$A_T \overset{d}{=} A_0\, e^{(\mu - \frac{1}{2}\sigma^2)\,T \,+\, \sigma\sqrt{T}\,Z}, \qquad Z \sim_{\mathbb{P}} \mathcal{N}(0;1). \tag{12.3}$$

Taking the expectation (MGF of a Gaussian):

$$\mathbb{E}[A_T] \;=\; A_0\, e^{(\mu - \frac{1}{2}\sigma^2)T} \cdot \mathbb{E}\Big[\, e^{\sigma\sqrt{T}\,Z} \Big] \;=\; A_0\, e^{(\mu - \frac{1}{2}\sigma^2)T} \cdot e^{\frac{1}{2}\sigma^2 T} \;=\; A_0\, e^{\mu T}. \tag{12.4}$$

So in the limit, A is geometric Brownian motion with drift μ and volatility σ.

Why the $\frac{1}{2}\sigma^2$ cancels out. The Itô correction $-\frac{1}{2}\sigma^2 T$ in the exponent of A's *median* and the $+\frac{1}{2}\sigma^2 T$ from the Gaussian MGF cancel exactly to leave $\mathbb{E}[A_T] = A_0 e^{\mu T}$. That cancellation is why μ — not $\mu - \frac{1}{2}\sigma^2$ — is called *the* drift of A: it is the growth rate of the expectation, not the log-median.

Median vs. mean vs. mode. The three central tendencies of a log-normal random variable are different. The median is $A_0 e^{(\mu - \frac{1}{2}\sigma^2)T}$, the mean is $A_0 e^{\mu T}$, and the mode is $A_0 e^{(\mu - \frac{3}{2}\sigma^2)T}$. As σ grows, the mean increasingly exceeds the median, because the right tail fattens disproportionately. This is why σ appears explicitly in option prices through the variance of the log: the *shape* of the log-normal matters, not just its central tendency.

A sanity check. In the risk-neutral world, $\mathbb{E}^{\mathbb{Q}}[A_T] = A_0 e^{rT}$. That is: under $\mathbb{Q}$, the expected growth of the asset equals the bond's growth. This is exactly the martingale condition $A_0/B_0 = \mathbb{E}^{\mathbb{Q}}[A_T/B_T]$ applied to the continuous-time limit. The tree was calibrated to produce this property; the CLT confirms it survives in the limit.

Practical use of the limit. Once we have the log-normal distribution, we can compute any expectation we want — in particular, the expected payoff of any European option. For a call struck at K, a standard calculation yields the Black–Scholes formula for C_0, expressed in closed form using the cumulative normal distribution $N(\cdot)$. We will derive this explicitly in a later chapter using continuous-time tools, but the binomial tree has already taken us almost all the way there. The final step is recognising the log-normal as the limit of the binomial and performing the integral — no deep new ideas required.

A remark on the relationship between trees and diffusions. The binomial tree converges to a geometric Brownian motion as $\Delta t \to 0$, but for any fixed Δt it is a discrete-time process. For pricing, both representations are useful: the tree is good for numerical computation (especially American options), while the diffusion is good for closed-form analytical results. The limit statement (12.3) tells us they agree in the relevant limit. In practice, for N of order 100 steps, the tree gives prices indistinguishable from the Black–Scholes formula for vanilla options, confirming that the convergence is fast in practice.

A caveat. The log-normal limit is a consequence of the CLT, and the CLT requires finite variance of the log-returns. If the model had heavy-tailed increments (e.g. from jumps that are not Gaussian), the CLT would fail and the limit would not be log-normal. This is the technical reason jump-diffusion models break the log-normal prediction: they inject finite-sized jumps whose contribution to higher moments does not vanish in the continuous limit. The simplicity of the log-normal result is therefore a consequence of our particular modelling choices (i.i.d. ± 1 coins) and does not generalise without qualification.

2.13 13. Existence of Martingale Measure ⇒ No Arbitrage

We have used the martingale characterization heuristically throughout the chapter. Here we give the clean, direct proof of one direction of the First Fundamental Theorem: existence of an equivalent martingale measure rules out arbitrage. The proof is short and general — it works for any number of assets and does not require market completeness.

Setup. Let $A_t^{(1)}, \dots, A_t^{(n)}$ be non-negative $\mathbb{P}$-a.s., B_t strictly positive $\mathbb{P}$-a.s., for $t \in \{0, 1\}$.

Hypothesis. Suppose $\exists\, \mathbb{Q} \sim \mathbb{P}$ s.t.

$$\frac{A_0^{(k)}}{B_0} \;=\; \mathbb{E}^{\mathbb{Q}}\left[\frac{A_1^{(k)}}{B_1}\right], \qquad k = 1, \dots, n. \tag{13.1}$$

Claim. Any portfolio $(\alpha^{(1)}, \dots, \alpha^{(n)}, \beta)$ such that

$$\mathbb{P}(V_1 \geq 0) = 1, \qquad \mathbb{P}(V_1 > 0) > 0, \tag{13.2}$$

must have $V_0 > 0$, ruling out arbitrage (which would require $V_0 = 0$).

The claim, restated. If you can build a portfolio that never loses at $t = 1$ and sometimes wins, then you must have paid strictly positive money for it at $t = 0$. Arbitrage demands zero cost; the claim denies it. Therefore no arbitrage is possible when an equivalent martingale measure exists.

Derivation. Compute V_0 directly:

$$V_0 \;=\; \beta\, B_0 + \sum_{k=1}^{n} \alpha^{(k)} A_0^{(k)}. \tag{13.3}$$

Factor out B_0 and use the martingale hypothesis:

$$V_0 \;=\; B_0\left[\beta \;+\; \sum_{k=1}^{n} \alpha^{(k)}\, \frac{A_0^{(k)}}{B_0}\right] \;=\; B_0\, \mathbb{E}^{\mathbb{Q}}\left[\frac{B_1}{B_1}\,\beta \;+\; \sum_{k=1}^{n} \alpha^{(k)}\, \frac{A_1^{(k)}}{B_1}\right]. \tag{13.4}$$

Each step explained. In (13.4) we use linearity of expectation and the martingale hypothesis (13.1) to replace each $A_0^{(k)}/B_0$ with its $\mathbb{Q}$-expectation of the future relative price. The β term also goes inside the expectation as $\beta B_1/B_1 = \beta$; this is just a convenient rewriting. Nothing mysterious happens — it is algebra.

Combining:

$$V_0 \;=\; B_0\, \mathbb{E}^{\mathbb{Q}}\left[\frac{\beta\, B_1 \;+\; \sum_{k=1}^{n} \alpha^{(k)} A_1^{(k)}}{B_1}\right] \;=\; B_0\, \mathbb{E}^{\mathbb{Q}}\left[\frac{V_1}{B_1}\right] \;>\; 0. \tag{13.5}$$

The last inequality uses $B_0 > 0$, $B_1 > 0$, and (since $\mathbb{Q} \sim \mathbb{P}$) $V_1 \geq 0$ $\mathbb{Q}$-a.s. with $V_1 > 0$ on a $\mathbb{Q}$-positive set.

Therefore $V_0 > 0$, so no portfolio satisfying (13.2) can also have $V_0 = 0$. Existence of $\mathbb{Q}$ rules out arbitrage. ■

What the proof actually uses. Three ingredients: (1) linearity of expectation; (2) the martingale hypothesis (13.1); (3) equivalence $\mathbb{Q} \sim \mathbb{P}$, which transfers the $\mathbb{P}$-statement (13.2) to $\mathbb{Q}$. §13 proves only the direction $\exists\mathbb{Q} \Rightarrow$ no arb; the converse (no arb $\Rightarrow \exists\mathbb{Q}$) requires a separating-hyperplane argument.

The converse direction. To prove "no arbitrage $\Rightarrow \exists\mathbb{Q}$" one uses a geometric / convex analysis argument: the set of payoffs achievable by zero-cost portfolios is a convex cone, and absence of arbitrage means this cone does not intersect the cone of non-negative payoffs except at the origin. A separating hyperplane between these two cones gives a strictly positive linear functional on payoffs, which (after normalization) is the density of $\mathbb{Q}$ with respect to $\mathbb{P}$. This is a classical theorem of measure-theoretic mathematical finance, sometimes called the Dalang–Morton–Willinger theorem in its full generality. We do not prove it here, but its role is dual to the simpler direction proved above.

Why the combination is called a "theorem". The First Fundamental Theorem is the statement that the two implications, together, establish equivalence between no-arbitrage and existence of an equivalent martingale measure. Each direction on its own is useful; the combination provides the *dictionary* we use throughout the rest of the book, translating freely between economic arguments (no free lunches) and probabilistic arguments (martingales under $\mathbb{Q}$). Every derivatives-pricing formula you will see — Black–Scholes, Heston, Margrabe, HJM — is an application of the dictionary: express the hedging condition as a martingale property, compute expectations under $\mathbb{Q}$, discount.

A remark on the cleanliness of the proof. The three-line proof in (13.3)–(13.5) is remarkably elegant, yet it captures the full force of one direction of the FFT. The power comes from the martingale assumption (13.1): it converts the intractable condition "no arbitrage exists" into the tractable condition "expectation under $\mathbb{Q}$ gives current prices." Once we have that conversion, the non-existence of arbitrage follows from the linearity of expectation and positivity of the numeraire. The proof is a clean illustration of how the right abstraction — the martingale concept — makes a hard economic theorem into an easy mathematical one.

The practical workflow. In applications, we typically assume the existence of $\mathbb{Q}$ (either by construction, calibration to market prices, or by general theoretical argument) and use it to price claims via $C_0 = B_0\mathbb{E}^{\mathbb{Q}}[C_1/B_1]$. The FFT's gift to us is the assurance that this procedure is internally consistent: whatever $\mathbb{Q}$ we use, as long as it makes the traded assets into martingales, the resulting prices are arbitrage-free. This is why the existence direction (proved above) is the workhorse of practical pricing. The uniqueness direction (which requires the market to be complete) tells us that under additional conditions, $\mathbb{Q}$ is pinned down uniquely by the traded assets.

2.14 14. Central Limit Theorem for the Binomial Tree

We close the loop by proving that the calibrated tree of §11 indeed converges in distribution to the Gaussian claimed in §12. The proof uses characteristic functions, which are the cleanest tool for this job.

Setup. Let

$$X \;=\; \sigma\sqrt{\Delta t}\sum_{n=1}^{N} x_i, \qquad \Delta t \;=\; \frac{T}{N}. \tag{14.1}$$

With the calibrated probabilities

$$\mathbb{P}(x_i = +1) \;=\; p \;=\; \tfrac{1}{2}\left[1 \;+\; \frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right], \tag{14.2}$$

$$\mathbb{P}(x_i = -1) \;=\; 1 - p \;=\; \tfrac{1}{2}\left[1 \;-\; \frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right]. \tag{14.3}$$

Let $\hat{\mu} \;\doteq\; \mu - \frac{1}{2}\sigma^2$.

A change of notation. Defining $\hat{\mu} = \mu - \frac{1}{2}\sigma^2$ is a bookkeeping move: $\hat{\mu}$ is the log-drift, while μ is the absolute-return drift. Keeping them distinct in notation saves confusion later.

Goal. Show that as $N \to +\infty$,

$$X \;\xrightarrow[N\to+\infty]{d}\; \mathcal{N}(\hat{\mu}\,T,\; \sigma^2 T). \tag{14.4}$$

Method. Characteristic function. Compute

$$\mathbb{E}\big[\,e^{iuX}\,\big] \;=\; \mathbb{E}\Big[\,e^{iu\sigma\sqrt{\Delta t}\,\sum_{n=1}^{N} x_i}\,\Big] \;=\; \Big(\,\mathbb{E}\big[\,e^{iu\sigma\sqrt{\Delta t}\,x_i}\,\big]\,\Big)^N. \tag{14.5}$$

Why characteristic functions, not MGFs. For a bounded random variable like ± 1, both exist — but the CF always converges for real u, and CFs uniquely determine the distribution via Lévy's continuity theorem. That mechanical device turns the algebraic limit (14.11) into the distributional claim (14.13).

Why the product form in (14.5). The exponential of a sum of independent random variables factors as a product of exponentials, and by independence the expectation of the product equals the product of expectations. This is the fundamental mechanical advantage of CFs for sums of independent variables: they turn addition into multiplication, and sum-of-N-things into N-th-power-of-one-thing.

Expand the inner expectation:

$$\mathbb{E}\Big[\, e^{iu\sigma\sqrt{\Delta t}\, x_i} \,\Big] \;=\; \mathbb{E}\Big[\, 1 + iu\sigma\sqrt{\Delta t}\, x_i - \tfrac{1}{2}u^2\sigma^2\, \Delta t\, x_i^2 + o(\Delta t) \,\Big] \tag{14.6}$$

$$= 1 + iu\sigma\sqrt{\Delta t}\, \mathbb{E}[x_i] - \tfrac{1}{2}u^2\sigma^2\, \Delta t\, \mathbb{E}[x_i^2] + o(\Delta t). \tag{14.7}$$

Next compute:

$$\mathbb{E}[x_i] \;=\; 2p - 1 \;=\; \frac{\hat{\mu}}{\sigma}\sqrt{\Delta t}, \tag{14.8}$$

and

$$\mathbb{E}[x_i^2] \;=\; 1. \tag{14.9}$$

Substituting:

$$\boxed{\mathbb{E}\Big[\, e^{iu\sigma\sqrt{\Delta t}\, x_i} \,\Big] \;=\; 1 + iu\,\hat{\mu}\,\Delta t \;-\; \tfrac{1}{2}u^2\sigma^2\,\Delta t \;+\; o(\Delta t).} \tag{14.10}$$

Where the $\sqrt{\Delta t}$ vanishes. The $iu\sigma\sqrt{\Delta t}\,\mathbb{E}[x_i]$ term, combined with $\mathbb{E}[x_i] = \hat{\mu}\sqrt{\Delta t}/\sigma$, produces $iu\hat{\mu}\,\Delta t$ — two factors of $\sqrt{\Delta t}$ multiply into one Δt, matching the variance term's Δt scaling.

The key cancellation, explained. One might initially worry that the $\sqrt{\Delta t}$ scaling would survive into the final formula. It does not, because the mean of x_i is itself $O(\sqrt{\Delta t})$, so their product is $O(\Delta t)$. This is a direct consequence of the drift calibration (11.5) being chosen specifically to make this work out. Without that calibration, the leading term would not scale the right way and the CLT limit would not exist.

So:

$$\mathbb{E}[\, e^{iuX} \,] \;=\; \big(1 + (iu\,\hat{\mu} - \tfrac{1}{2}\sigma^2u^2)\,\Delta t \;+\; o(\Delta t)\big)^N \xrightarrow[N\to\infty]{} e^{\left(iu\,\hat{\mu} - \frac{1}{2}\sigma^2u^2\right)T}, \tag{14.11}$$

using the classical limit

$$\big(1 + \tfrac{a}{N}\big)^N \xrightarrow[N\to\infty]{} e^a. \tag{14.12}$$

Recognising the right-hand side of (14.11) as the characteristic function of a Gaussian with mean $\hat{\mu}T$ and variance σ^2T:

$$\boxed{X \xrightarrow[N\to\infty]{d} \mathcal{N}(\hat{\mu}\,T;\; \sigma^2T).} \tag{14.13}$$

Closing the loop. The CF of $\mathcal{N}(m, v)$ is $e^{ium - \frac{1}{2}vu^2}$. Matching to (14.11) gives $m = \hat{\mu}T$ and $v = \sigma^2T$ — exactly the parameters promised in (12.2). The binomial tree's terminal log-return is Gaussian in the limit, A_T is log-normal, and we have derived geometric Brownian motion from scratch without stochastic calculus.

Reflections on the proof. The entire calculation fits on one page and uses only three tools: Taylor expansion of the exponential, linearity of expectation, and the classical $(1 + a/N)^N \to e^a$ limit. Yet the conclusion is profound: starting from discrete coin flips with carefully chosen bias and step size, we recover the full continuous-time log-normal distribution in the limit. This is the pedagogical value of the binomial tree: it demystifies geometric Brownian motion by showing it as the continuous limit of something we can see, count, and compute by hand.

The broader picture. What we have done in §14 is a specific case of a much more general phenomenon. Whenever you take a sum of i.i.d. random variables and scale appropriately, you get a Gaussian in the limit (CLT). When you instead take a sum of independent jumps with a certain scaling, you get a Poisson process. When you mix the two, you get a Lévy process with both diffusive and jump components. The binomial tree with i.i.d. ± 1 increments

is the simplest instance of the CLT in action; Lévy processes are the natural generalisation that introduces jumps. Black–Scholes pricing corresponds to the pure-diffusion case; Merton's jump-diffusion pricing corresponds to a Lévy process. The framework generalises, and the binomial tree is the entry point.

The role of the characteristic function. The CF tool is so powerful for this kind of problem that it deserves a brief reflection. Sums of independent random variables have CFs that are *products* of individual CFs. Products are easier to analyse than sums of distributions directly, especially in the presence of many terms. Taking logs converts products to sums, and the log of the CF — the *cumulant generating function* — is especially well-behaved. The whole machinery of CLT proofs and generalisations (stable laws, infinite divisibility, Lévy–Khintchine representation) rests on this product structure. Once you have the CF, the rest is bookkeeping: identify it as the CF of a known distribution (here, Gaussian) and you are done.

What remains for the rest of the book. With the binomial framework in place, the rest of the mathematical finance curriculum is about generalizing in various directions: multiple risky assets (Margrabe), stochastic volatility (Heston), jump processes (Merton), stochastic interest rates (HJM/LIBOR), path-dependence (barriers, Asians), American exercise, credit risk, incomplete markets. In each case the binomial tree provides the template: find the right tradable assets, find the right numeraire, impose no-arbitrage, extract the risk-neutral measure, price by expectation. The machinery expands; the philosophy stays the same.

A parting observation. The entire chapter, conceptually, fits into a single sentence: *a derivative's price is the cost of the portfolio that replicates it, and this cost can be computed as a discounted expectation under a probability measure that makes all traded assets martingales.* Every formula, every proof, every example in this chapter is a specialisation or illustration of this one idea. If you internalise the sentence and can produce any of the formulas from it, you have understood the chapter. Everything else — the utility detour, the numerical examples, the CLT calculation — is scaffolding that supports the core insight.

2.15 Worked Examples

2.15.1 WE-1. Symmetric lotteries (motivating risk aversion)

- $p = \frac{1}{2}$, $A_u = 100$, $A_d = -100$ vs.
- $p = \frac{1}{2}$, $A_u = 10^6$, $A_d = -10^6$.

Both have $\mathbb{E}[A_1] = 0$ (take $r = 0$). A linear-utility agent prices both at 0. A concave-utility agent prices the second strictly below 0 (and well below the first). This is the motivating contrast for utility theory.

Quantitative illustration. With exponential utility and $\gamma = 10^{-4}$, the small lottery's buy price is approximately $-\frac{1}{\gamma}\ln\frac{1}{2}(e^{-\gamma\cdot 100} + e^{\gamma\cdot 100}) \approx -0.5$ (a mild discount), while the large lottery has buy price approximately $-\frac{1}{\gamma}\ln\frac{1}{2}(e^{-\gamma\cdot 10^6} + e^{\gamma\cdot 10^6}) \approx -10^6$ (buyer essentially refuses). The preference between the two is *not* a matter of taste but a consequence of the concavity baked into any reasonable utility function.

2.15.2 WE-2. Exponential utility indifference prices

For $u(x) = -e^{-\gamma x}$:

$$A_0^{\text{buy}} = -\tfrac{1}{\gamma}\ln\mathbb{E}[e^{-\gamma A_1}], \qquad A_0^{\text{sell}} = \tfrac{1}{\gamma}\ln\mathbb{E}[e^{\gamma A_1}].$$

Small-γ limit: both $\to \mathbb{E}[A_1]$. Large-γ limit: $A_0^{\text{buy}} \to A_d$, $A_0^{\text{sell}} \to A_u$.

Commentary. The indifference-pricing framework gives a complete answer to "what is the most this agent is willing to pay (or least they are willing to accept)" for any bounded payoff A_1, as a function of risk aversion γ. The spread between buy and sell is the first-order risk-aversion cost of holding the gamble, and quantifies the bid–ask that would emerge in a dealer-free market of identical agents.

2.15.3 WE-3. Sign check with three small trees | Tree | A_0 | A_d | A_u | Check $A_d < A_0 < A_u$? | Verdict |

|:—:|:—:|:—:|:—:|:—:|:—:| | (−) | 10 | 5 | 20 | $5 < 10 < 20$ | No arb | | (+) | 2 | 4 | 15 | $4 \not< 2$ | Arb | | (14 ...) | ≈ 14 | 0 | 0 | violates | Arb |

For tree $(-)$: $q = (10-5)/(20-5) = 1/3$.

Teaching point. The diagnostic for arbitrage in a single-asset tree is a strict inequality test, not anything involving probabilities or expected values. Free lunches come from mis-positioned prices, and the three-tree comparison is designed to make this visceral.

2.15.4 WE-4. Third asset mispriced - A: $10 \to \{20, 5\}$; B: $1 \to \{1, 1\}$ (so $r = 0$); C: $110 \to \{120, 100\}$.

- q from A-tree: $q^{BA} = 1/3$. Fair $\widehat{C_0} = 120 \cdot \frac{1}{3} + 100 \cdot \frac{2}{3} \approx 106.67 < 110$.
- Replicating portfolio: $\alpha = 2$, $\beta = 90$. Selling C at 110 and buying $(2, 90)$ in (A, B) gives $V_0 = 0$, $V_1^{\text{up}} = +10$, $V_1^{\text{down}} = 0$ — an arbitrage.
- Internal solve of q^{BC}: $110 = 120q + 100(1-q) \Rightarrow q^{BC} = 1/2$, inconsistent with $q^{BA} = 1/3$ — arbitrage signature.

The two-channel diagnostic. Whether you check consistency of the risk-neutral weight ($q^{BA} \neq q^{BC}$) or explicitly construct the replicating portfolio and compare to the quoted price, you arrive at the same arbitrage. Both approaches are valuable: the first is a quick sanity check for a portfolio of assets; the second is constructive and tells you exactly how to trade.

2.15.5 WE-5. European call on A at strike $K = 100$ With tree $100 \to \{120, 90\}$, $r = 0$, $K = 100$:

quantity	value
α (shares of A)	$2/3$
β (bonds)	-60
q (risk-neutral prob.)	$1/3$
no-arb call price C_0	$6\frac{2}{3}$
arbitrage profit if $C_0 = 6$	$+\frac{2}{3}$ at $t = 0$, 0 at $t = 1$

Key observations. The risk-neutral weight $q = 1/3$ differs from the natural guess $p = 1/2$, reflecting that with $r = 0$ and an asymmetric tree, the forward is not at the midpoint. The delta $\alpha = 2/3$ tells the trader how many shares to hold to hedge each call sold. The put–call parity check: a put struck at 100 has payoff $\{0, 10\}$, and the risk-neutral put price is $1/3 \cdot 0 + 2/3 \cdot 10 = 20/3$. Adding the call and subtracting the put: $C_0 - P_0 = 20/3 - 20/3 = 0$, matching $A_0 - Ke^{-rT} = 100 - 100 = 0$.

2.16 Key Takeaways

- Naive expected-value pricing fails for risk-averse agents: a ± 100 coin-flip is nothing like a $\pm 10^6$ coin-flip, yet they have equal expectation.
- A utility function $u : \mathbb{R} \to \mathbb{R}$ that is increasing (more is better) and concave (risk-averse) induces the monotonicity $\mathbb{E}[u(X_1)] \leq \mathbb{E}[u(X_2)]$ whenever $X_1 \leq X_2$.
- Exponential utility $u(x) = -e^{-\gamma x}$ with $\gamma > 0$ is the canonical tractable choice; γ is the constant absolute risk-aversion.
- Indifference pricing: $A_0^{\text{buy}} = -\frac{1}{\gamma} \ln \mathbb{E}[e^{-\gamma A_1}]$ satisfies $\mathbb{E}[u(X_1)] = \mathbb{E}[u(X_2)]$; symmetric formula for sell.
- Bid–ask spread from risk alone: $A_0^{\text{buy}} \leq \mathbb{E}[A_1] \leq A_0^{\text{sell}}$, collapsing to $\mathbb{E}[A_1]$ as $\gamma \downarrow 0$ and expanding to (A_d, A_u) as $\gamma \uparrow \infty$.
- Risk-neutral pricing is the $\gamma \downarrow 0$ limit of indifference pricing — but the no-arbitrage derivation recovers the same formula *without* invoking preferences.
- One-period no-arbitrage: $V_0 = 0$, $\mathbb{P}(V_t \geq 0) = 1$, $\mathbb{P}(V_t > 0) > 0$ defines an arb; no-arb is equivalent to $A_d/B_d < A_0/B_0 < A_u/B_u$, i.e. in the bond case $A_d < A_0(1+r) < A_u$.
- Risk-neutral measure: $\exists\, q \in (0, 1)$ such that $A_0/B_0 = q\, A_u/B_u + (1-q)\, A_d/B_d$, with explicit $q = (A_0/B_0 - A_d/B_d)/(A_u/B_u - A_d/B_d)$.
- Replication = pricing: any claim C with payoffs (C_u, C_d) can be synthesised by (α, β) in (A, B); formulas (8.5)–(8.6). The unique arbitrage-free price is $C_0/B_0 = q\, C_u/B_u + (1-q)\, C_d/B_d$, which in the bond case reduces to $C_0 = \frac{1}{1+r}\mathbb{E}^{\mathbb{Q}}[C_1]$.

- Relative prices are $\mathbb{Q}$-martingales — First Fundamental Theorem: no arb $\Leftrightarrow \exists\, \mathbb{Q} \sim \mathbb{P}$ s.t. $\tilde{X}_0 = \mathbb{E}^{\mathbb{Q}}[\tilde{X}_1]$ for every traded X.
- Multi-period extension: $\mathcal{F}_t$-adapted strategies, $\tilde{X}_t = \mathbb{E}^{\mathbb{Q}}[\tilde{X}_s|\mathcal{F}_t]$ for all $t < s$.
- Calibrated tree: $c = \sigma\sqrt{\Delta t}$ and $p = \frac{1}{2}[1 + (\mu - \frac{1}{2}\sigma^2)\sqrt{\Delta t}/\sigma]$ match log-normal drift $\mu - \frac{1}{2}\sigma^2$ and variance σ^2 per unit time.
- Risk-neutral q in continuous calibration: $q = (e^{r\Delta t} - e^{-\sigma\sqrt{\Delta t}})/(e^{\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}})$, expanding to $\frac{1}{2}[1 + (r - \frac{1}{2}\sigma^2)\sqrt{\Delta t}/\sigma]$. The swap $\mu \mapsto r$ is the Girsanov shift.
- Limit law: $\ln(A_T/A_0) \xrightarrow{d} \mathcal{N}((\mu - \frac{1}{2}\sigma^2)T, \sigma^2 T)$ and $\mathbb{E}[A_T] = A_0 e^{\mu T}$.
- One direction of the FFT proved explicitly: existence of $\mathbb{Q}$ forces $V_0 = B_0\, \mathbb{E}^{\mathbb{Q}}[V_1/B_1] > 0$ for any non-negative, non-trivial terminal wealth — ruling out arbitrage.
- CLT via characteristic functions: $(1 + a/N)^N \to e^a$ produces $\mathbb{E}[e^{iuX}] \to e^{iu\hat{\mu}T - \frac{1}{2}\sigma^2 u^2 T}$ and hence $X \xrightarrow{d} \mathcal{N}(\hat{\mu}T, \sigma^2 T)$.

2.17 Reference Formulas

Binomial fair-price relation (physical, not a pricing rule):

$$A_0 = \frac{pA_u + (1-p)A_d}{1+r}$$

Monotonicity axiom:

$$X_1 \le X_2 \;\Rightarrow\; \mathbb{E}[u(X_1)] \le \mathbb{E}[u(X_2)]$$

Exponential utility:

$$u(x) = -e^{-\gamma x}, \qquad \gamma > 0$$

Indifference buy price:

$$A_0^{\text{buy}} = -\frac{1}{\gamma} \ln \mathbb{E}\big[e^{-\gamma A_1}\big]$$

Indifference sell price:

$$A_0^{\text{sell}} = \frac{1}{\gamma} \ln \mathbb{E}\big[e^{\gamma A_1}\big]$$

Small-γ asymptotics:

$$A_0^{\text{buy}} \underset{\gamma\downarrow 0}{\longrightarrow} \mathbb{E}[A_1], \qquad A_0^{\text{sell}} \underset{\gamma\downarrow 0}{\longrightarrow} \mathbb{E}[A_1]$$

Arbitrage (definition):

$$V_0 = 0, \qquad \mathbb{P}(V_t \ge 0) = 1, \qquad \mathbb{P}(V_t > 0) > 0$$

No-arbitrage bound (bond case):

$$A_d < A_0(1+r) < A_u$$

No-arbitrage bound (general numeraire):

$$\frac{A_d}{B_d} < \frac{A_0}{B_0} < \frac{A_u}{B_u}$$

Risk-neutral probability (general):

$$q = \frac{A_0/B_0 - A_d/B_d}{A_u/B_u - A_d/B_d}$$

Risk-neutral probability (bond case):

$$q = \frac{A_0(1+r) - A_d}{A_u - A_d}$$

Replicating portfolio for claim C:

$$\alpha = \frac{C_u B_d - C_d B_u}{A_u B_d - A_d B_u}, \qquad \beta = \frac{C_u A_d - C_d A_u}{B_u A_d - B_d A_u}$$

Risk-neutral pricing (general):

$$\frac{C_0}{B_0} = q\,\frac{C_u}{B_u} + (1-q)\,\frac{C_d}{B_d}$$

Risk-neutral pricing (bond case):

$$C_0 = \frac{1}{1+r}\,\mathbb{E}^{\mathbb{Q}}[C_1] = \frac{1}{1+r}\,(q\,C_u + (1-q)\,C_d)$$

First Fundamental Theorem (one-period):

$$\text{no arb} \iff \exists\,\mathbb{Q} \sim \mathbb{P} \text{ s.t. } \tilde{X}_0 = \mathbb{E}^{\mathbb{Q}}[\tilde{X}_1]\ \forall \text{ traded } X, \qquad \tilde{X}_t = X_t/B_t$$

Multi-period martingale characterisation:

$$\text{no arb} \iff \exists\,\mathbb{Q} \sim \mathbb{P}:\ \tilde{X}_t = \mathbb{E}^{\mathbb{Q}}[\tilde{X}_s|\mathcal{F}_t],\ \forall t < s,\ \forall X \text{ traded}$$

Multiplicative binomial update:

$$A_n = A_{n-1}\,e^{\sigma\sqrt{\Delta t}\,x_n}, \qquad x_n \in \{\pm 1\}$$

Calibrated physical p:

$$p = \tfrac{1}{2}[1 + \tfrac{\mu - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}]$$

Calibrated risk-neutral q (CRR form):

$$q = \frac{e^{r\Delta t} - d}{u - d}, \qquad u = e^{\sigma\sqrt{\Delta t}},\ d = e^{-\sigma\sqrt{\Delta t}}$$

Small-Δt expansion of q:

$$q \approx \tfrac{1}{2}[1 + \tfrac{r - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}]$$

Log-normal limit distribution:

$$\ln(A_T/A_0) \xrightarrow{d} \mathcal{N}((\mu - \tfrac{1}{2}\sigma^2)T,\ \sigma^2 T), \qquad \mathbb{E}[A_T] = A_0 e^{\mu T}$$

Call and put payoffs:

$$\text{Call}(A_1; K) = \max(A_1 - K,\ 0), \qquad \text{Put}(A_1; K) = \max(K - A_1,\ 0)$$

3 Chapter 2 — Multi-Period Binomial, CRR, and the FTAP

Chapter 1 established risk-neutral pricing for a single-period two-state economy. Before pushing into continuous time, we need the explicit multi-period binomial machinery — backward induction along a recombining tree — and its limit the Cox–Ross–Rubinstein (CRR) model, whose calibrated up/down sizes give rise to geometric Brownian motion as the step size $\Delta t \to 0$. We present two equivalent CRR parameterisations: the classical $A_n = A_{n-1}e^{c\,x_n}$ Bernoulli lattice and the "direct-Gaussian increment" lattice $A_n = A_{n-1}e^{(\mu-\frac{1}{2}\sigma^2)\Delta t \pm \sigma\sqrt{\Delta t}}$ in which the Itô convexity adjustment has already been built into the geometry of the nodes. We then state the Fundamental Theorem of Asset Pricing (FTAP): absence of arbitrage is equivalent to existence of an equivalent martingale measure $\mathbb{Q}$ under which all discounted traded assets are martingales. Next we confront the fact that the measure $\mathbb{Q}$ — and therefore the price of non-replicable claims — is not automatically unique. In the binomial lattice, branching into more than two states destroys uniqueness; we discuss the local risk-minimising hedge as a selection device. The chapter closes with the dynamic-programming recursion for American options on the binomial tree and a short bridge to the continuous-time limit that will occupy Chapters 3–6.

The reader should treat this chapter as the pivot point of the course. The one-period model of Chapter 1 is a toy, but a surprisingly complete toy: every conceptual ingredient of modern derivatives theory — the law of one price, replication, the martingale measure, the decomposition of a payoff into a spanning hedge — lives there in miniature. What is missing is the *arrow of time* and the *algebra of compounding*. Multi-period trees supply both. Once we can iterate the one-step argument and take a limit, the classical tools of mathematical finance simply fall out: Black–Scholes becomes a corollary of the Central Limit Theorem applied to log-returns, early-exercise optimality becomes a Bellman recursion on a graph, and the pricing of genuinely exotic payoffs reduces to either sweeping the tree or forward-simulating. Our goal is not merely to list these results but to make them feel inevitable — to show that they are the unique answers one is forced to if one demands the absence of arbitrage.

A second theme threads through the chapter and will dominate Chapters 3 through 5: the tension between completeness (where $\mathbb{Q}$ is unique and the price is a single number) and incompleteness (where $\mathbb{Q}$ spans a family and prices span an interval). Binomial trees are special because they are, by construction, exactly one-step complete. As soon as we add a third branch — whether that third branch is a default, a jump, a stochastic-volatility state, or just a trinomial approximation — the dimension of the state space outruns the dimension of the tradable asset space, and uniqueness fails. This failure is not a pathology of the model; it is a picture of real-world markets, where stock-and-bond hedges cannot perfectly cover earnings surprises, macro announcements, or corporate credit events. We will see that even when uniqueness fails, the no-arbitrage framework still delivers a *range* of admissible prices, and risk-preference-based devices (utility indifference, local risk-minimisation, minimum-entropy projection) select a single representative from the range. Understanding this structure is arguably more important for quant practice than Black–Scholes itself — every volatility surface in real trading is a witness to market incompleteness, and every model-selection decision in risk management is, at bottom, a choice of $\mathbb{Q}$ inside the set of admissible equivalent martingale measures.

A third theme is computational. We will see *backward induction* in its purest form: start with the payoff at maturity and discount node-by-node back to time zero. This is Richard Bellman's dynamic-programming principle, and it handles Americans and any path-dependence that can be captured by finitely many state variables. The complementary paradigm — Monte Carlo forward simulation — is developed in Chapter 10 once geometric Brownian motion has been placed on rigorous footing. The two paradigms have complementary strengths: backward induction is exact up to the tree's discretisation; Monte Carlo is statistical but scales gracefully in dimension. Serious pricing engines routinely combine both — simulate for the forward-dependent part, induct backwards for the early-exercise part — the so-called Longstaff–Schwartz regression approach is a prototypical example, which we touch on conceptually.

A fourth theme — subtler but pervasive — is the role of the numéraire. The FTAP says that *some* discounted process is a martingale under $\mathbb{Q}$; it is silent on which asset you use as the deflator. This freedom turns out to be extraordinarily useful. The Black–Scholes formula, as we will derive it in Chapter 6, has two probability terms: one computed under the money-market-account measure, and one under the stock-measure. Switching between the two is not a mere bookkeeping trick — it is the operationally cleanest way to evaluate expectations of products like $A_T \cdot \mathbf{1}_{\{A_T>K\}}$, because under the stock-measure the factor A_T collapses into the new numéraire and the expectation reduces to a plain exceedance probability. The same trick underlies the pricing of interest-rate derivatives (use a zero-coupon bond as numéraire) and swaption pricing (use an annuity). We flag this now because the mechanics of numéraire change, developed canonically in Chapter 5, are the same at every level of complexity.

Finally, the chapter is *long on worked numbers*. Wherever we introduce a formula, we pin it with a concrete scenario: a 20% volatility annualised, a 5% interest rate, a 1/252 daily step, a strike at 100 and spot at 100. The reader

is encouraged to redo these arithmetic exercises; they are the calibration checks one performs every day on a real trading desk. When a formula "feels right" numerically — when a $\sqrt{\Delta t}$-order correction comes in at a basis-point per day, or a small shift in q produces a visible move in a call price — the formula becomes part of one's intuition, not just a line in a textbook.

With those themes declared, we start at the very simplest node of the theory: a single stock, two states, one step.

3.1 2.1 One-Period Review: the No-Arbitrage Inequality

We begin by recalling the one-period binomial pricing argument, because everything that follows builds on it. The one-period case is simple enough to do by hand, yet already contains the key ideas: no-arbitrage as a geometric constraint, replication as a hedge-portfolio algebra, and the emergence of the risk-neutral probability. The reader who is fully comfortable with this section can skim it; those who are encountering the argument for the first time should spend more time here than anywhere else in the chapter, because mastery of the one-period case makes everything subsequent tractable.

Start at a single node A with two possible successor nodes $A_u > A_d$ and a cash account growing by the gross rate $1 + r$. The one-period no-arbitrage inequality is

$$A_d \;<\; A(1+r) \;<\; A_u \quad \Longleftrightarrow \quad \text{no arb.} \tag{2.1}$$

To see why this inequality is forced on us, reason by contradiction. Suppose $A(1 + r) \leq A_d$. Then at time zero I could short-sell 1 unit of cash and buy $1/A$ units of stock. At time one my stock is worth at least $A_d/A \geq 1 + r$ in every state, and I owe the cash-lender $1 + r$. Net cash flow at time one is non-negative in every state and strictly positive in the up-state, yielding a riskless profit from zero initial outlay — the textbook definition of an arbitrage. A symmetric argument handles the case $A(1+r) \geq A_u$: borrow against the stock and deposit the cash. The forward price $A(1 + r)$ must therefore sit strictly between the two post-states, and this geometric fact is entirely equivalent to absence of arbitrage.

There is also a way to read (2.1) that emphasises its role as a *calibration constraint on the physical world.* Given a stock with empirical upper state A_u and lower state A_d on a horizon of one period, and given a money market compounding at $1+r$, the market's no-arbitrage pricing of everything else is *determined* by the no-arbitrage condition on this triple. If the interest rate is too high — above the forward implied by (A_u, A_d) and the physical probability — dealers can lock in riskless gains by shorting bonds; the forward curve will adjust. In this sense (2.1) is less a theorem than a forecast: any real market that satisfies frictionless trading will *arrange* for (2.1) to hold.

Equivalently, the claim C with terminal values C_u, C_d admits a unique arbitrage-free price via an equivalent risk-neutral measure $\mathbb{Q}$:

$$\exists\, \mathbb{Q} \text{ s.t. } C \;=\; \frac{1}{1+r}\, \mathbb{E}^{\mathbb{Q}}[\, C_1 \,] \quad \Longleftrightarrow \quad \text{no arb.} \tag{2.2}$$

This is the deep step. The left-to-right direction says: if no arbitrage holds, there exists a probability $q \in (0, 1)$ — uniquely defined in the binomial case by $q = \frac{A(1+r)-A_d}{A_u-A_d}$ — such that every contingent claim is priced as a discounted expectation under q. The right-to-left direction says: if some positive $q \in (0, 1)$ satisfies (2.2), then no arbitrage is possible, because any positive payoff has positive expectation and therefore positive today-price. Together they form the one-period Fundamental Theorem of Asset Pricing, a microcosm of the general FTAP we will state in §2.5.

Intuition. (2.1) is the geometric statement "the forward $A(1 + r)$ sits strictly between the up and down states"; if it didn't, a combination of long/short asset+cash would earn a riskless profit. (2.2) then rewrites the Arrow prices as a probability measure $\mathbb{Q}$: the risk-neutral weights are the unique convex combination making the discounted asset a one-step martingale.

It is worth pausing over the word *probability* in the last paragraph. The weights $q, 1-q$ have all the formal properties of a probability distribution — they are non-negative and sum to unity — but they are *not* the probabilities a forecaster would assign to the two future states. A forecaster with inside knowledge of the stock's fundamentals might believe A_u is twice as likely as A_d; that is the physical measure $\mathbb{P}$, denoted by p. The risk-neutral measure $\mathbb{Q}$ ignores beliefs about which state is more likely and instead weights the states by what they are *worth* in the market.

A scenario where the stock is up might be, in reality, improbable, yet it carries a high $\mathbb{Q}$-weight because the forward is close to A_u. The mathematical fact that $\mathbb{Q}$ and $\mathbb{P}$ assign positive probability to the same events (equivalence, $\mathbb{Q} \sim \mathbb{P}$) is essential — both measures see the same scenario space — but their relative weights can diverge arbitrarily far. The Radon–Nikodym derivative $d\mathbb{Q}/d\mathbb{P}$ is the market's *discount of optimism*, and it is the engine behind Girsanov's theorem in continuous time, developed canonically in Chapter 5.

Worked numerical check. Set $A = 100$, $r = 5\%$ per period, $A_u = 120$, $A_d = 90$. The forward is $A(1+r) = 105$, which lies in $(90, 120)$, confirming (2.1). The risk-neutral probability is $q = (105 - 90)/(120 - 90) = 1/2$. A call struck at $K = 100$ has up-payoff 20, down-payoff 0, and price $C = (1/1.05)(1/2 \cdot 20 + 1/2 \cdot 0) = 10/1.05 \approx 9.524$. A replicating portfolio $\alpha A + \beta = C$ is found by solving $120\alpha + 1.05\beta = 20$ and $90\alpha + 1.05\beta = 0$, yielding $\alpha = 2/3$ shares and $\beta = -60/1.05 \approx -57.14$ dollars of cash (i.e., borrow 57.14). Checking: $2/3 \cdot 100 - 57.14 = 66.67 - 57.14 = 9.52$ — the same price to two decimal places. The two approaches — expectation under $\mathbb{Q}$ and explicit replication — are mathematically identical, but they emphasise different aspects: the first focuses on *value* ("what is this worth?"), the second on *hedging* ("what position makes this go away?"). Both are the trader's job.

Edge case: what if $A_u = A(1+r)$? Then the up-state has zero $\mathbb{Q}$-weight in the forward pricing, and the risk-neutral $q = 0$. This is on the boundary of the no-arbitrage region and corresponds to a degenerate case where owning the stock is — in expectation — identical to owning the bond in the up-state. Practically, this never happens in a liquid market, but it shows up as the edge of the feasibility set: tight binding of (2.1) means zero slack in the pricing constraint. Similarly, if $A_d = A(1+r)$, then $q = 1$.

3.2 2.2 Multi-Period Binomial Tree — Backward Induction

Section 2.1 priced a one-period claim on a one-period tree; now we scale up. The extension from one period to many periods is conceptually straightforward — pricing at each node via the same one-step argument, with the children's values standing in for the payoff — but it introduces new phenomena (path-dependence, early exercise) and new computational structures (recombining vs. non-recombining trees, the tower property collapsing iterated expectations). This section lays out the formal machinery; subsequent sections put it to work.

Extend the one-step construction to N steps. The tree $A \to \cdots$ branches 2^N times; when up-and-down moves commute ($A_{ud} = A_{du}$) the tree recombines into $N+1$ terminal nodes after N steps.

The step from one period to N periods is the step from a textbook exercise to a genuinely useful pricing engine. In a single period we have only two terminal states, which is expressive enough to price any claim whose payoff depends on those two states, but it is not expressive enough to capture any interesting *dynamics*. In N periods we have $N+1$ terminal states (assuming recombination), and the *path* through those states matters for anything more sophisticated than a vanilla European claim. The recombining binomial tree is the simplest structure that captures both features — many terminal states for distributional richness, and a tractable tree topology for backward induction.

The recombination property — $A_{ud} = A_{du}$ — is the mathematical reason the tree stays manageable. Without recombination, after N steps we would have 2^N terminal nodes, and pricing would require 2^N evaluations. With recombination we have only $N+1$ terminal nodes, and pricing requires $O(N^2)$ evaluations (one for each node in the triangle). At $N = 250$ daily steps, that is $\approx 31{,}000$ evaluations — trivial on a laptop — versus $2^{250} \approx 10^{75}$ without recombination, which is more nodes than atoms in the observable universe. Multiplicative models (geometric Brownian motion being the archetype) naturally recombine because multiplication is commutative: $A \cdot u \cdot d = A \cdot d \cdot u$. Additive models (arithmetic Brownian motion, Bachelier's original model) recombine for the same reason. Any model where the up-factor at a node depends on the history fails to recombine and therefore loses the $O(N^2)$ efficiency.

The pricing principle is one-step-at-a-time risk-neutral discounting: at every interior node, the claim value is $\frac{1}{1+r}\mathbb{E}^{\mathbb{Q}}[\,\text{next-step payoff}\,]$ with the node-local $\mathbb{Q}$-probability q.

There is an important subtlety. The $\mathbb{Q}$-probability q is generally *node-local*: it can depend on the current spot A, the current time, and — if we build state-dependence into the model — on other state variables. In the simplest Cox–Ross–Rubinstein setup, q depends only on the ratios $u, d, 1+r$, which are constants, so the tree has a single global q. In more general state-dependent models (local volatility, state-dependent interest rates), each node has its own q. The backward-induction algorithm is identical either way: at every node, take the q-weighted average of the two children's values and discount by one period.

3.2.1 2.2.1 Worked example — call struck at 100, maturity $T = 2$

We pause for an extended concrete example before diving into the general CRR formalism. The worked example below uses a small, hand-computable tree to make every arithmetic step transparent. In real practice, trees have hundreds or thousands of steps and are computed on a machine; but the machine is doing exactly the same arithmetic we lay out here, node by node, step by step.

Take a two-period tree on a non-constant-factor lattice:

```
                                130

                        120

            100                 110     (upper half of tree)

                        90

                                80

  t = 0          t = 1            t = 2

 Local risk-neutral weights q_1 (0→1, up), q_2 (120→130 edge),
 q_3 (90→110 edge) are pinned down by the one-period FTAP at each node.
```

with edge-local interest rates r_1, r_2, r_3 on the three time-0 $\to$ 1 $\to$ 2 edges. The local risk-neutral probabilities q_1, q_2, q_3 are pinned down by matching the parent to the discounted expectation of its children:

$$120 \;=\; \frac{1}{1+r_?}\big(\,130\,q_2 \;+\; 110\,(1-q_2)\,\big) \;\Longrightarrow\; q_2 = \# \tag{2.3}$$

$$90 \;=\; \frac{1}{1+r_?}\big(\,110\,q_3 \;+\; 80\,(1-q_3)\,\big) \;\Longrightarrow\; q_3 = \# \tag{2.4}$$

$$100 \;=\; \frac{1}{1+r_?}\big(\,120\,q_1 \;+\; 90\,(1-q_1)\,\big) \;\Longrightarrow\; q_1 = \# \tag{2.5}$$

Payoff @ $T = 2$. A call struck at $K = 100$ pays

$$\max(A_T - K, 0) \;\equiv\; (A_T - K)_+. \tag{2.6}$$

At the terminal column $A_T \in \{130, 110, 110, 80\}$ the payoffs are $\{30, 10, 10, 0\}$.

The choice of payoff function $(x)_+ = \max(x, 0)$ deserves a word of intuition. The $+$-function captures the essential optionality: the holder receives $A_T - K$ whenever it is positive, zero otherwise. There is no obligation to exercise — the option is exercised if and only if it is in the money, and the payoff $(A_T - K)_+$ reflects this decision. The piecewise-linear shape of $(x)_+$ produces the classic "hockey-stick" payoff diagram: flat at zero for $A_T \le K$, linearly ascending at unit slope for $A_T > K$, with a kink at $A_T = K$. All the convexity of an option comes from this kink; pricing convexity is pricing the rounded-over-time version of the hockey stick.

Notice that in the $\{130, 110, 110, 80\}$ column, the middle value 110 occurs *twice* — this is the recombination at work. The path up-then-down arrives at the same state as the path down-then-up, and both contribute to the same terminal payoff of \$10. In the backward recursion, we treat both paths as a single lattice node; the $\mathbb{Q}$-probability of reaching that node is $2q(1-q)$ (two paths, each with probability $q(1-q)$).

Backward step 1 (t = 1 nodes). At the upper intermediate node (stock = 120, children 130 and 110) the call value C_{1u} satisfies

$$C_{1u} \;=\; \frac{1}{1+r_2}\big(\,30\,q_2 \;+\; 10\,(1-q_2)\,\big) \;=\; \#, \tag{2.7}$$

and at the lower intermediate node (stock = 90, children 110 and 80, payoffs 10 and 0):

$$C_{1d} \;=\; \frac{1}{1+r_3}(\,10\,q_3 \;+\; 0\,(1-q_3)\,) \;=\; \#. \tag{2.8}$$

Backward step 2 (t = 0).

$$C_0 \;=\; \frac{1}{1+r_1}(\,C_{1u}\,q_1 \;+\; C_{1d}\,(1-q_1)\,) \;=\; \#. \tag{2.9}$$

Intuition. Backward induction is the lattice-level manifestation of the tower property: computing today's price as the discounted $\mathbb{Q}$-expectation over the children at each node is equivalent to the one-shot formula $C_0 = \mathbb{E}^{\mathbb{Q}}[\,e^{-rT}C_T\,]$ because nested conditional expectations collapse. The value to doing it step by step is that the same sweep handles path-dependent payoffs and American exercise (§2.8).

To amplify: the tower property says $\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_t]] = \mathbb{E}[X]$ for any filtration-compatible X, and more generally $\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_t] \mid \mathcal{F}_s] = \mathbb{E}[X \mid \mathcal{F}_s]$ for $s \le t$. At the lattice level, $\mathcal{F}_t$ is the sigma-algebra generated by the first t steps of the tree, so conditioning on $\mathcal{F}_t$ means "knowing which node we are at time t". Computing the expectation step by step is literally iterating the tower property: first expect over step N given $\mathcal{F}_{N-1}$, then expect over step $N-1$ given $\mathcal{F}_{N-2}$, and so on back to $\mathcal{F}_0$.

Why does the step-by-step approach generalise so well? Because at every node in the sweep, we are free to do more than just compute the discounted expectation. We can *compare* that expectation to an alternative value (the immediate-exercise value for an American, for instance), or *accumulate* a path-dependent state variable (the running average for an Asian, the running maximum for a lookback), or *check* a path-dependent predicate (has the barrier been breached?). Each of these augmentations slots into the backward-induction algorithm as a per-node computation, and the algorithm retains its $O(N^2)$ efficiency (or $O(N^3)$, $O(N^4)$ for modestly more elaborate state spaces). The one-shot formula $C_0 = \mathbb{E}^{\mathbb{Q}}[e^{-rT}C_T]$ hides all of this adaptability inside a single integral; the lattice exposes it as explicit node-level operations.

3.2.2 2.2.2 Path-weights and the CRR telescoping identity

If the per-edge interest rate is constant $r_1 = r_2 = r_3 = r$ the root price becomes a single discounted expectation

$$C_0 \;=\; \frac{1}{(1+r)^N}\,\mathbb{E}^{\mathbb{Q}}[\,C_T\,], \tag{2.10}$$

with the tree's path probability $q^k(1-q)^{N-k}$ multiplied by the combinatorial path count $\binom{N}{k}$ giving the classical Cox–Ross–Rubinstein pricing sum.

To unpack (2.10) for the European call: letting k count the number of up-moves along a path (so there are $\binom{N}{k}$ paths reaching the same terminal node $A_0u^kd^{N-k}$), the CRR pricing sum is $C_0 = (1+r)^{-N}\sum_{k=0}^{N}\binom{N}{k}q^k(1-q)^{N-k}(A_0u^kd^{N-k}-K)_+$. This is a finite-term expansion of a binomial probability times a payoff; as $N \to \infty$ with $u = e^{\sigma\sqrt{\Delta t}}$, $d = e^{-\sigma\sqrt{\Delta t}}$, the binomial distribution converges (by de Moivre–Laplace) to the normal distribution, and the sum converges to the Black–Scholes integral. Running the sum on a computer is straightforward even at $N = 10^4$; numerical experiments show convergence to Black–Scholes is $O(1/N)$ in the standard CRR parameterisation, with oscillatory behaviour that can be accelerated by symmetric placement of K inside a node or by Richardson extrapolation.

There is one further layer of elegance. The telescoping identity refers to the fact that the *same* $\mathbb{Q}$-probabilities appear at every step (when u, d, r are constant), so the path probability factorises as $q^k(1-q)^{N-k}$ with k = number of up-moves. The distribution of k is exactly Binomial(N, q). This lets us compute analytical quantities like the moments of A_T directly: $\mathbb{E}^{\mathbb{Q}}[A_T] = A_0(qu + (1-q)d)^N = A_0(1+r)^N$ (as it must, by the martingale property), and $\mathbb{V}^{\mathbb{Q}}[\ln A_T] = N \cdot q(1-q)(\ln(u/d))^2 \approx \sigma^2 T$ to leading order. The binomial structure is not only computationally convenient; it hands us closed-form answers for the moments and — via de Moivre–Laplace — for the distribution itself in the continuous limit.

3.3 2.3 The Cox–Ross–Rubinstein (CRR) Model — Bernoulli Parameterisation

The Cox–Ross–Rubinstein model is arguably the most influential single construction in derivatives pricing after Black–Scholes itself. The authors showed that the binomial lattice — a conceptually simple structure — converges to the Black–Scholes continuous-time model as the step size shrinks, and that the lattice provides a computationally tractable pricing tool for American-style and other non-standard options where the Black–Scholes closed form does not apply. Before CRR, numerical option pricing was the preserve of PDE finite-difference methods; after CRR, a generation of quants learned options on lattices.

The CRR approach is pedagogically valuable because it reduces the entire apparatus of options pricing to a discrete, finite-dimensional linear-algebra exercise. There are no integrals, no stochastic differential equations, no measure-theoretic subtleties — just arithmetic on a tree. Once the student masters the tree, the continuous-time framework falls into place naturally as a limit. This is why we spend so much time on the lattice in this chapter, despite knowing that practical pricing often uses the limit directly.

In a step of size Δt replace simple compounding by continuous compounding:

$$(1+r) \;\longrightarrow\; e^{r\,\Delta t}. \tag{2.11}$$

The replacement of $1+r$ by $e^{r\Delta t}$ is more than cosmetic. It reflects a deliberate choice to align the lattice with the continuous-time limit. In the limit $\Delta t \to 0$ we want the cash account to grow at exactly rate r per unit time, so that $B_T = e^{rT}$. A lattice that used $1+r\Delta t$ would also converge to the same limit (since $1+r\Delta t+O(\Delta t^2) = e^{r\Delta t}$), but the exponential form gives cleaner algebra at every finite Δt and respects the multiplicative structure of the log-return process. The Cox–Ross–Rubinstein (CRR) parameterisation takes this seriously: up-moves multiply the asset by e^c, down-moves by e^{-c}, and the cash account multiplies by $e^{r\Delta t}$ at every step. Symmetry of $\pm c$ ensures recombination, and the exponential form of the cash growth ensures consistency with the GBM limit.

Write the log-return process as

$$A_n \;=\; A_{n-1}\, e^{c\,x_n}, \qquad x_1, x_2, \ldots \overset{iid}{\sim} \text{Bernoulli}(\pm 1), \tag{2.12}$$

with $\mathbb{P}(x_k = +1) = p,\ \mathbb{P}(x_k = -1) = 1-p$.

Why Bernoulli ± 1? The choice is almost forced. We want two states per step (to make the market one-step complete), we want symmetric state placements (for recombination), and we want iid increments (for tractable distributional analysis). The symmetric Bernoulli ± 1 is the simplest random variable satisfying all three. Its mean is $2p-1$, its variance is $1-(2p-1)^2$, and — crucially — the sum of N iid such Bernoullis has binomial distribution with mean $N(2p-1)$ and variance $N(1-(2p-1)^2)$, satisfying a Central Limit Theorem with Gaussian limit $\mathcal{N}(N(2p-1), N(1-(2p-1)^2))$. When we scale the sum by c — turning increments into log-returns — we get log-return sums that are asymptotically Gaussian, precisely the behaviour of log-GBM.

The alternative, non-symmetric Bernoulli — say $x_k \in \{u_{\text{up}}, u_{\text{down}}\}$ with $u_{\text{up}} \neq u_{\text{down}}$ — would work too, but it forces the log-drift and log-variance into a less clean parameterisation. The symmetric ± 1 choice, combined with a scalar c, cleanly separates *size of move* (c) from *directional bias* ($p-\frac{1}{2}$). This is the CRR convention and it makes moment-matching transparent.

Iterating from $t=0$ to $t=T$ with $N = T/\Delta t$ steps,

$$A_T \;=\; A_0\, e^{c\,x_1}\, e^{c\,x_2} \cdots e^{c\,x_N} \;=\; A_0\, \exp\!\left\{ c \sum_{n=1}^{N} x_n \right\} \;=\; A_0\, e^{c\,X_N}, \tag{2.13}$$

with $X_N := \sum_{n=1}^{N} x_n$.

3.3.1 2.3.1 Moments of the increment sum

The calibration strategy is moment-matching: choose model parameters so that the first two moments of the model's log-return distribution match the empirical moments of observed log-returns. This is the simplest form of the

method of moments, a general estimation strategy that generalises to generalised-method-of-moments (GMM), quasi-maximum-likelihood, and a host of modern econometric techniques. For the CRR lattice, two free parameters (p, c) match two moments (μ, σ^2), and the system is exactly determined.

An alternative calibration strategy — maximum likelihood estimation — would choose (p, c) to maximise the likelihood of observed returns under the lattice distribution. In the large-sample limit, MLE and method-of-moments agree for the Gaussian-like limit of the binomial, so the practical difference is small. Method of moments is preferred in the CRR context because the algebra is explicit and the results are closed-form.

Linearity plus iid Bernoulli:

$$\mathbb{E}[X_N] \;=\; \sum_{n=1}^{N} \mathbb{E}[x_n] \;=\; N\big(+1 \cdot p + (-1)(1-p)\big) \;=\; N\,(2p-1) \;=\; \mu\,\Delta t\,N, \tag{2.14}$$

$$\mathbb{V}[X_N] \;=\; N\,\mathbb{V}[x_1] \;=\; N\big(\,\mathbb{E}[x_1^2] - (\mathbb{E}[x_1])^2\,\big) \;=\; N(1-(2p-1)^2) \;=\; \sigma^2\,\Delta t\,N. \tag{2.15}$$

The assignments $\mathbb{E}[X_N] = \mu\Delta t\,N$ and $\mathbb{V}[X_N] = \sigma^2\Delta t\,N$ in (2.14)–(2.15) match the data: μ and σ^2 are fed in as the annualised mean and variance of the log-return estimated from a price series $S_1, S_2, \ldots, S_M$ via the ratios

The factor Δt in the per-step variance is the *time-scaling of Brownian motion.* This scaling — variance proportional to time, standard deviation proportional to square-root-of-time — is what separates Brownian motion (and GBM) from a simple random walk. In a random walk with unit-step Bernoulli increments, variance scales as N (number of steps). To embed such a walk in continuous time so that variance scales as t (wall-clock time), we must scale the step size by $\sqrt{\Delta t}$ — exactly the $c = \sigma\sqrt{\Delta t}$ scaling that emerges below. This is why Brownian motion shows up so often in physics and finance: it is the universal scaling limit of any iid-increment process with finite second moment.

$$\ln(S_2/S_1) \;=\; r_1, \quad \ln(S_3/S_2) \;=\; r_2, \quad \ldots \tag{2.16}$$

$$\widehat{\mu\,\Delta t} \;=\; \frac{1}{M}\sum_{m=1}^{M} r_m \quad \text{(annualised return)}, \tag{2.17}$$

$$\widehat{\sigma^2\,\Delta t} \;=\; \frac{1}{M}\sum_{m=1}^{M} (r_m - \widehat{\mu\Delta t})^2 \quad \text{(annualised variance)}. \tag{2.18}$$

3.3.2 2.3.2 Solving for the CRR parameters (p, c)

With the moments computed, calibration is a system of two equations in two unknowns. The solution for the up-step size c is immediate (from matching the variance), and the solution for the probability bias p follows (from matching the mean). A useful check: in the $\Delta t \to 0$ limit, p approaches $1/2$ — the lattice becomes a symmetric random walk, reflecting the fact that short-horizon log-returns have expected value dominated by noise rather than drift.

Solve (2.14)–(2.15) simultaneously for (p, c). Using $\mathbb{E}[x^2] = c^2\mathbb{E}[x_1^2] = c^2$ and expanding

$$\mathbb{V}[X_N^{\text{c-scaled}}] \;=\; c^2 N(1-(2p-1)^2) \;=\; \sigma^2\Delta t\,N, \tag{2.19}$$

one obtains the Cox–Ross–Rubinstein up-step and physical probability:

$$\boxed{c \;=\; \sigma\sqrt{\Delta t} + \cdots, \qquad p \;=\; \tfrac{1}{2}\Big(1 \;+\; \frac{\mu}{\sigma}\sqrt{\Delta t}\Big) + \cdots.} \tag{2.20}$$

Intuition. Two moments, two unknowns: matching the annualised log-return mean and variance to a biased Bernoulli $\pm c$ gives $c = \sigma\sqrt{\Delta t}$ (variance matches: $c^2 = \sigma^2\Delta t$) and a small $\sqrt{\Delta t}$-order bias in p that captures the drift.

Worked calibration. Suppose we observe an annualised log-return volatility of $\sigma = 20\%$ and drift of $\mu = 10\%$, and use a daily grid $\Delta t = 1/252 \approx 0.003968$. Then $c = 0.20\sqrt{0.003968} \approx 0.01260$, which means each up-move takes the stock up by a factor of $e^{0.01260} \approx 1.01268$, or roughly $+1.27\%$. Each down-move takes it down by $e^{-0.01260} \approx 0.9875$,

roughly -1.25%. The bias in p is $\frac{1}{2} \cdot (\mu/\sigma)\sqrt{\Delta t} = \frac{1}{2} \cdot 0.5 \cdot 0.063 \approx 0.0157$, so $p \approx 0.5157$. In words: on any given day, the stock has a 51.6% chance of going up by 1.27% and a 48.4% chance of going down by 1.25%. The expected daily log-return is $(2p-1)c = 0.031 \cdot 0.01260 \approx 3.95 \times 10^{-4}$, and $(3.95 \times 10^{-4}) \cdot 252 = 0.0996 \approx 10\%$, recovering μ as required. The variance is $N(1-(2p-1)^2)c^2 = 252 \cdot (1-0.031^2) \cdot 0.01260^2 \approx 252 \cdot 1.587 \times 10^{-4} = 0.04$, which is $\sigma^2 = (0.2)^2$ as required.

An important side remark: the $\sqrt{\Delta t}$-scaling of c is not an accident; it is the universal scaling of Brownian motion. A Brownian motion W_t has variance t, so its typical size on the interval $[t, t+\Delta t]$ is $\sqrt{\Delta t}$. Since σW_t is the diffusion part of log-GBM, its typical size over Δt is $\sigma\sqrt{\Delta t}$. The CRR parameterisation faithfully reproduces this scaling, which is exactly why the CRR lattice converges to geometric Brownian motion. By contrast, the *drift* scales as $\mu\Delta t$, much smaller than $\sigma\sqrt{\Delta t}$ for small Δt: at a one-day horizon, diffusion is roughly $\sqrt{252} \approx 16$ times larger than drift for typical parameters. This is why, over short horizons, financial data look like noise around zero, and why "trend-following" strategies require long holding periods.

3.3.3 2.3.3 Converting to the $\mathbb{Q}$-probability

Under $\mathbb{P}$ (the physical measure), the lattice is calibrated to match empirical moments of log-returns. Under $\mathbb{Q}$ (the risk-neutral measure), the lattice is calibrated to make the discounted asset a martingale — that is, to reproduce the no-arbitrage forward price at every node. The $\mathbb{P} \to \mathbb{Q}$ transformation is the heart of derivatives pricing: it says "re-weight the lattice probabilities so the expected return matches the riskless rate, not the empirical expected return."

Risk-neutral pricing on the $e^{\pm c}$ tree demands

$$A \;=\; e^{-r\Delta t}\,\mathbb{E}^{\mathbb{Q}}[\,A_n\,] \;=\; e^{-r\Delta t}\Big(\,q\,A\,e^{\sigma\sqrt{\Delta t}} \;+\; (1-q)\,A\,e^{-\sigma\sqrt{\Delta t}}\,\Big), \tag{2.21}$$

giving

$$\boxed{q \;=\; \frac{e^{r\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{+\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}}.} \tag{2.22}$$

Small-Δt expansion. Using $e^{\pm z} \sim 1 \pm z + \frac{1}{2}z^2 + \cdots$ in both numerator and denominator,

$$q \;\sim\; \tfrac{1}{2}\left(1 \;+\; \frac{r - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right) + \cdots \tag{2.23}$$

$$p \;\sim\; \tfrac{1}{2}\Big(1 \;+\; \frac{\mu}{\sigma}\sqrt{\Delta t}\Big) + \cdots \qquad \text{(for comparison, under } \mathbb{P}\text{).} \tag{2.24}$$

(The boxed formula (2.22) is *exact* at finite Δt; only (2.23) is truncated to leading $\sqrt{\Delta t}$ order. The next-order $O(\Delta t)$ correction in (2.23) is numerically small — at daily steps with $\sigma = 20\%$ it is about 0.2% of $q - \frac{1}{2}$, well below bid-ask — but it is worth knowing about when running many-step convergence studies where stacking the truncation error across thousands of steps can produce a measurable bias.)

The $\mathbb{P} \to \mathbb{Q}$ shift replaces the *log-return* drift μ by the *arithmetic-return* drift $r - \frac{1}{2}\sigma^2$ — the Itô convexity adjustment shows up already at the lattice level.

This is a remarkable fact worth lingering on. The Itô correction — the $-\frac{1}{2}\sigma^2$ that relates expected log-returns to expected arithmetic returns — is usually introduced as a consequence of Itô's lemma in stochastic calculus (which we formalise in Chapter 3), which itself is a consequence of the $dW_t^2 = dt$ rule. Yet here we see it emerge directly from matching the no-arbitrage constraint on a discrete-time lattice to the small-Δt expansion. No calculus required. The reason is that the Itô correction is really a statement about the geometric-vs-arithmetic mean inequality — $\mathbb{E}[e^X] \geq e^{\mathbb{E}[X]}$ with equality only for constant X — and that inequality holds in any setting with non-degenerate randomness. In the continuous-time limit it takes the clean form $-\frac{1}{2}\sigma^2$ because Brownian motion has independent Gaussian increments; on the lattice it takes the slightly more cluttered form that matches (2.23), but the qualitative content is identical.

A practical consequence is that a naïve "risk-neutral simulator" that replaces μ by r in the log-return formula (without the $-\frac{1}{2}\sigma^2$) would over-price calls and under-price puts by a factor related to $e^{\sigma^2 T/2}$. At $\sigma = 20\%$, $T = 1$, this is $e^{0.02} \approx 1.02$, a 2% bias; at $\sigma = 50\%$, $T = 2$, it is $e^{0.25} \approx 1.28$, a 28% bias. Getting the Itô correction right in the drift is therefore critical in any practical implementation.

3.3.4 2.3.4 Log-moments under $\mathbb{P}$ vs $\mathbb{Q}$

We now pause to summarise what has changed between $\mathbb{P}$ and $\mathbb{Q}$. The summary will be crystallised in the "slogan" below: variances are invariant, means are not. This is one of the most important conceptual takeaways of the chapter and will be a constant companion through all subsequent material.

Combining (2.13) with (2.20) / (2.23),

$$\mathbb{E}^{\mathbb{P}}[\ln(A_T/A_0)] \;=\; \mu\, T, \qquad \mathbb{V}^{\mathbb{P}}[\ln(A_T/A_0)] \;=\; \sigma^2\, T, \tag{2.25}$$

$$\mathbb{E}^{\mathbb{Q}}[\ln(A_T/A_0)] \;=\; (r - \tfrac{1}{2}\sigma^2)\, T, \qquad \mathbb{V}^{\mathbb{Q}}[\ln(A_T/A_0)] \;=\; \sigma^2\, T. \tag{2.26}$$

Slogan. "$\mathbb{P}$ and $\mathbb{Q}$ variances are identical" — "$\mathbb{P}$ and $\mathbb{Q}$ means are *not*".

Intuition. Girsanov at the one-line level: a change of equivalent measure can tilt the drift arbitrarily but *cannot* change the diffusion coefficient (volatility) — the quadratic variation is measure-invariant. This is the single most important structural fact about the $\mathbb{P} \leftrightarrow \mathbb{Q}$ passage and it explains why traders calibrate σ from *either* physical returns *or* implied-vol surfaces: the answer is the same up to estimation error.

Why is the volatility measure-invariant? Because the volatility parameter σ controls the *size* of the Brownian shocks, not their *probabilities*. A change of measure is a re-weighting of scenarios: a set of outcomes that had probability p under $\mathbb{P}$ might have probability q under $\mathbb{Q}$, but the set of outcomes itself — the sample paths $\omega \mapsto A_\cdot(\omega)$ — is unchanged. The quadratic variation $\langle \ln A \rangle_T = \int_0^T \sigma_s^2\, ds$ is computed path-by-path, so it takes the same value under any equivalent measure. This is why implied volatility — a $\mathbb{Q}$-world quantity backed out from market prices — and realised volatility — a $\mathbb{P}$-world quantity estimated from historical prices — should in principle converge to the same number for a correctly specified model. In practice they differ, and the difference is called the *variance risk premium*. The variance risk premium is not a failure of the theory but an indication that the joint model of $(\mathbb{P}, \mathbb{Q})$ is richer than Black–Scholes: stochastic volatility, jumps, or fundamental mispricing all contribute to a non-trivial VRP. But the *functional form* of σ in the diffusion — that is, the same thing under both measures.

A corollary worth remembering: any *model-free* forecast of realised volatility (say the VIX squared, which is a model-free fair strike of the log-contract) measures the same σ^2 that a model-based implied volatility measures. This is why volatility arbitrage strategies — trading the spread between VIX futures and realised S&P 500 variance — are closely watched. The spread is the market's premium for bearing volatility risk, not a measurement error.

3.3.5 2.3.5 Continuous-time limit — lognormal asset

The passage from the discrete CRR lattice to the continuous-time GBM is one of the most important conceptual bridges in mathematical finance. It connects the combinatorial clarity of lattice pricing (where everything is a finite sum) with the analytical power of stochastic calculus (where everything is a Gaussian integral). Both views are useful, and a good quant is fluent in both. The conversion is driven by the Central Limit Theorem applied to the sum of log-returns: a sum of many small independent shocks becomes Gaussian in the limit, and exponentiating gives a log-normal asset.

Applying the CLT to $X_N = \sum x_n$ as $N \to \infty$ (and $N\Delta t = T$ fixed),

$$\ln(A_T/A_0) \;=\; X_N \;\xrightarrow[N\to\infty]{d,\ \mathbb{P}}\; \mathcal{N}(\,\mu T,\ \sigma^2 T\,)\,, \tag{2.27}$$

$$\ln(A_T/A_0) \;=\; X_N \;\xrightarrow[N\to\infty]{d,\ \mathbb{Q}}\; \mathcal{N}\big(\,(r - \tfrac{1}{2}\sigma^2)T,\ \sigma^2 T\,\big)\,. \tag{2.28}$$

Equivalently, in distribution,

$$A_T \;\stackrel{d}{=}\; A_0\, e^{\mu T\,+\,\sigma\sqrt{T}\,Z_{\mathbb{P}}}, \qquad Z_{\mathbb{P}} \sim \mathcal{N}(0,1), \tag{2.29}$$

$$A_T \;\stackrel{d}{=}\; A_0\, e^{(r-\frac{1}{2}\sigma^2)T\,+\,\sigma\sqrt{T}\,Z_{\mathbb{Q}}}, \qquad Z_{\mathbb{Q}} \sim \mathcal{N}(0,1). \tag{2.30}$$

In words: asset prices in the limit $N \to \infty$ are lognormal r.v. at a fixed point in time. Verifying the martingale property under $\mathbb{Q}$:

$$\mathbb{E}^{\mathbb{P}}[\,A_T\,] \;=\; A_0\, e^{\mu T}\, \mathbb{E}^{\mathbb{P}}[\,e^{\sigma\sqrt{T}\,Z}\,] \;=\; A_0\, e^{\mu T}\, e^{\frac{1}{2}\sigma^2 T} \;=\; A_0\, e^{(\mu+\frac{1}{2}\sigma^2)T}, \tag{2.31}$$

$$\mathbb{E}^{\mathbb{Q}}[\,A_T\,] \;=\; A_0\, e^{(r-\frac{1}{2}\sigma^2)T}\, e^{\frac{1}{2}\sigma^2 T} \;=\; A_0\, e^{rT}. \tag{2.32}$$

Used: $\mathbb{E}[e^{\alpha Z}] = e^{\frac{1}{2}\alpha^2}$ for $Z \sim \mathcal{N}(0,1)$.

The calculation (2.32) deserves a second reading. Under $\mathbb{Q}$ the *log-return* drift is $r - \frac{1}{2}\sigma^2$, but the *price* grows at r in expectation. The difference is exactly the $\frac{1}{2}\sigma^2$ convexity premium generated by Jensen's inequality applied to the exponential. It is the geometric-vs-arithmetic mean gap in the log-return distribution: the expected value of e^X where $X \sim \mathcal{N}(\nu, \sigma^2)$ is $e^{\nu+\sigma^2/2}$, not e^{ν}. In log-normal markets, the price distribution is heavily right-skewed, so the mean is pulled up by the upper tail — precisely the $\frac{1}{2}\sigma^2$ adjustment.

For the reader uncomfortable with the moment-generating-function formula $\mathbb{E}[e^{\alpha Z}] = e^{\alpha^2/2}$, one way to remember it: the density of $Z \sim \mathcal{N}(0,1)$ is $\phi(z) = (2\pi)^{-1/2}e^{-z^2/2}$, and $\mathbb{E}[e^{\alpha Z}] = \int e^{\alpha z - z^2/2}dz/\sqrt{2\pi}$. Completing the square in the exponent, $\alpha z - z^2/2 = -(z-\alpha)^2/2 + \alpha^2/2$, so the integral becomes $e^{\alpha^2/2}\int e^{-(z-\alpha)^2/2}dz/\sqrt{2\pi} = e^{\alpha^2/2}\cdot 1$ (the inner integral is the normalised density of $\mathcal{N}(\alpha, 1)$). This identity is used dozens of times in options theory; memorising it is worth the effort.

3.3.6 2.3.6 Calibrating to log-returns vs simple returns — the $\frac{1}{2}\sigma^2$ correction

Caveat. The data in (2.16) are *log-returns* $r_m = \ln(S_{m+1}/S_m)$. If you instead calibrated to daily simple (continuously compounded) returns (meaning $(S_{m+1} - S_m)/S_m$ or any first-order expansion without the log), then the estimator of μ is biased by the Itô correction:

$$\mu \;\longrightarrow\; \mu - \tfrac{1}{2}\sigma^2. \tag{2.33}$$

Intuition. $\mathbb{E}[\,\Delta S/S\,] \approx \mathbb{E}[\,\Delta \ln S\,] + \frac{1}{2}\sigma^2$ to order Δt, so a practitioner who fits a simple-return mean and plugs it into a log-drift ends up over-stating the growth rate by $\frac{1}{2}\sigma^2$. Always be explicit which object you are fitting.

This source of bias — known as "volatility drag" or "geometric-arithmetic return gap" — is enormously important in retail finance contexts as well. The annualised simple return on the S&P 500 is typically quoted at something like 10%, but the compounded (geometric) return, which governs the actual growth of a buy-and-hold portfolio, is more like $10\% - \frac{1}{2}(16\%)^2 = 10\% - 1.3\% \approx 8.7\%$. Over 30 years, the difference between $1.10^{30} \approx 17.4$ and $1.087^{30} \approx 12.3$ is a factor of 1.4 — a 40% reduction in terminal wealth! The mathematical content is modest; the practical and rhetorical importance is large.

Numerical calibration trap to avoid. If you are given a data set of daily returns and asked to report annualised drift and volatility: always apply log before averaging. Compute $r_m = \ln(S_{m+1}/S_m)$, take sample mean and variance, then multiply by 252 to annualise. Do *not* compute simple returns $S_{m+1}/S_m - 1$ and then take log of the annualised mean; that introduces a $\sigma^2/2$ bias. Good quant toolkits often offer both variants; picking the wrong one is a classic calibration bug.

3.3.7 2.3.7 Pricing vanilla European claims in the CRR limit

Having established the distribution of A_T under $\mathbb{Q}$ in the continuous-time limit, we can now articulate the no-arbitrage price of any European claim as a discounted expectation — which, for smooth payoff functions, evaluates to a standard Gaussian integral. The integral for the European call is the Black–Scholes formula itself; we preview this correspondence in §2.9 and derive the formula rigorously in Chapter 6.

Starting from $C_0/B_0 = \mathbb{E}^{\mathbb{Q}}[\,C_T/B_T\,]$ and using (2.30),

$$C_0 \;=\; e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\,(A_T-K)_+\,]\,, \qquad A_T \;\stackrel{d}{=}\; A_0\,e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}\,Z}. \tag{2.34}$$

The integral reduces to the Black–Scholes formula; see §2.9 for a sketch of the limit and Chapter 6 for the full derivation via the hedging argument.

Before moving on, let us highlight a structural observation. The passage from (2.34) to the closed-form Black–Scholes formula involves evaluating a Gaussian integral of the form $\mathbb{E}[\max(A_0 e^{\mu+\sigma\sqrt{T}Z}-K,0)]$ for $Z\sim\mathcal{N}(0,1)$. This is one of the most-evaluated integrals in all of finance. Its canonical evaluation proceeds by splitting the max into two pieces — the payoff A_T on the exercise set and the strike K on the exercise set — and computing each as a Gaussian tail integral. The split leads to two $\Phi(\cdot)$ terms with arguments differing by $\sigma\sqrt{T}$. The beauty of the CRR approach is that this integral never needs to be evaluated by direct integration; it falls out of a measure-change argument (which we develop canonically in Chapter 5) that replaces integration by a probability lookup.

3.4 2.4 The Alternate CRR Parameterisation — Direct-Gaussian Increment Lattice

The classical CRR of §2.3 uses a symmetric Bernoulli lattice $A\to\{Ae^{\sigma\sqrt{\Delta t}}, Ae^{-\sigma\sqrt{\Delta t}}\}$ and pushes the drift into the branch-probability p. An alternate, equivalent lattice absorbs the Itô convexity adjustment into the *nodes* themselves and sends only a mean-zero $\pm\sigma\sqrt{\Delta t}$ shock through the probabilities.

This section is a deep dive into a topic that is often glossed over in textbook treatments: the *non-uniqueness* of the discretisation. The continuous-time GBM model $dS_t = rS_t\,dt+\sigma S_t\,dW_t$ is a single object, but there are infinitely many discrete lattices that converge to it. The two most common are:

1. *Classical CRR:* $u=e^{\sigma\sqrt{\Delta t}}$, $d=1/u$, $q=(e^{r\Delta t}-d)/(u-d)$. Symmetric multiplicative factors, risk-neutral probability absorbs drift.
2. *Jarrow–Rudd:* $u=e^{(r-\sigma^2/2)\Delta t+\sigma\sqrt{\Delta t}}$, $d=e^{(r-\sigma^2/2)\Delta t-\sigma\sqrt{\Delta t}}$, $q=1/2$. Drift-shifted multiplicative factors, fair-coin probability.

The alternate parameterisation in this section is essentially Jarrow–Rudd (under $\mathbb{P}$ with drift μ; under $\mathbb{Q}$ with drift r). A third common variant — *Tian's equal-probabilities tree* — chooses u,d to match the first three moments and gets $q=1/2$ plus slightly better convergence near K. These are all different ways to parameterise the same underlying GBM.

3.4.1 2.4.1 The two parameterisations, side by side

Placing the two lattices next to each other highlights their symmetry. Every finite-dimensional lattice must assign (a) node values and (b) branch probabilities; the total information content is the same across parameterisations, but how it is distributed differs. In parameterisation I the drift lives in the probability; in parameterisation II it lives in the geometry. Both converge to the same continuous-time GBM, and both handle a European call identically in the limit.

Parameterisation I — symmetric nodes, biased probability (§2.3):

$$A\longrightarrow\begin{cases} A\,e^{+\sigma\sqrt{\Delta t}} & \text{w.p. } p\\ A\,e^{-\sigma\sqrt{\Delta t}} & \text{w.p. } 1-p\end{cases}, \qquad p=\tfrac{1}{2}\Big[1+\tfrac{\mu}{\sigma}\sqrt{\Delta t}\Big]+\cdots \tag{2.35}$$

(This is the same p as (2.20) / (2.24): the calibration targets the log-return mean $\mathbb{E}^{\mathbb{P}}[\ln(A_T/A_0)] = \mu T$ — consistent with (2.25) — and the corresponding arithmetic-return identity at the lattice level is $\mathbb{E}^{\mathbb{P}}[A_1] = e^{(\mu+\frac{1}{2}\sigma^2)\Delta t}A + O(\Delta t^{3/2})$, with the Itô convexity adjustment appearing on the expectation rather than being absorbed into p.)

Parameterisation II — drift-shifted nodes, symmetric probability (alternate CRR):

$$A \longrightarrow \begin{cases} A\, e^{(\mu-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}} & \text{w.p. } \frac{1}{2} \\ A\, e^{(\mu-\frac{1}{2}\sigma^2)\Delta t-\sigma\sqrt{\Delta t}} & \text{w.p. } \frac{1}{2} \end{cases} \tag{2.36}$$

Here the branch probabilities are *fixed* at $\frac{1}{2}$ and the drift $(\mu - \frac{1}{2}\sigma^2)\Delta t$ is baked into each node.

Intuition. Both trees converge to the same lognormal GBM in the limit $\Delta t \to 0$. Parameterisation II is computationally nicer for Monte-Carlo since every path carries the same Bernoulli coin; parameterisation I is closer to the "one-step no-arb inequality" $A_d < A(1+r) < A_u$ picture.

Why does the choice of parameterisation matter for practical implementation? Consider Monte-Carlo simulation. In parameterisation I, each path-step requires drawing a Bernoulli with probability $p = \frac{1}{2} + O(\sqrt{\Delta t})$; the probability is close to but not exactly $\frac{1}{2}$, so the simulator has to evaluate the bias on every step. In parameterisation II the probability is *exactly* $\frac{1}{2}$, so drawing a fair coin suffices and the drift is baked into the geometry at zero cost. This is a small efficiency gain on any single step, but over billions of simulated steps it adds up. More importantly, parameterisation II decouples the "random" piece (the fair coin) from the "deterministic" piece (the drift in the node), which matches the structure of the GBM SDE $dS_t = rS_t\,dt + \sigma S_t\,dW_t$: the drift $r\,dt$ is deterministic, the diffusion $\sigma\,dW_t$ is random. Parameterisation II is the natural discretisation of this SDE.

A less obvious benefit: parameterisation II is the discrete-time analogue of the Euler–Maruyama scheme for log-GBM. Applied to the log-price $X_t = \ln A_t$, the SDE is $dX_t = (r - \frac{1}{2}\sigma^2)\,dt + \sigma\,dW_t$, and the Euler step is $X_{n+1} = X_n + (r - \frac{1}{2}\sigma^2)\Delta t + \sigma(W_{t_{n+1}} - W_{t_n})$. Replacing the Gaussian increment with a $\pm\sigma\sqrt{\Delta t}$ Bernoulli (matching its first two moments) gives exactly (2.36). Parameterisation II is therefore the *Euler scheme in moment-matched form* — a direct stochastic-calculus interpretation.

3.4.2 2.4.2 Verifying parameterisation II matches the first two log-moments

A quick moment-matching check confirms that parameterisation II reproduces the target mean $\mathbb{E}[A_1] = Ae^{\mu\Delta t}$ and log-variance $\sigma^2\Delta t$. The calculation is a direct Taylor expansion of the hyperbolic cosine that appears in the symmetric Bernoulli expectation.

Under $\mathbb{P}$ with $p = \frac{1}{2}$ on tree (2.36):

$$\mathbb{E}^{\mathbb{P}}[A_1] \;=\; A\, e^{(\mu-\frac{1}{2}\sigma^2)\Delta t} \cdot \tfrac{1}{2}\Big[\, e^{+\sigma\sqrt{\Delta t}} + e^{-\sigma\sqrt{\Delta t}} \Big]\,. \tag{2.37}$$

Expanding $\frac{1}{2}[e^z + e^{-z}] = 1 + \frac{1}{2}z^2 + \cdots$ with $z = \sigma\sqrt{\Delta t}$:

$$\mathbb{E}^{\mathbb{P}}[A_1] \;=\; A\, e^{(\mu-\frac{1}{2}\sigma^2)\Delta t}\big(1 + \tfrac{1}{2}\sigma^2\Delta t + \cdots\big) \;=\; A\, e^{\mu\Delta t} + \cdots, \tag{2.38}$$

matching the target $\mathbb{E}^{\mathbb{P}}[A_1] = Ae^{\mu\Delta t}$ to the relevant order. The log-variance

$$\mathbb{V}^{\mathbb{P}}[\ln(A_1/A)] \;=\; \sigma^2\,\Delta t \tag{2.39}$$

is exact (not just leading-order) because the $\pm\sigma\sqrt{\Delta t}$ Bernoulli has variance $\sigma^2\Delta t$ by construction, and the deterministic drift $(\mu - \frac{1}{2}\sigma^2)\Delta t$ drops out of the variance.

3.4.3 2.4.3 Risk-neutral probability q under parameterisation II

Under the alternate parameterisation, the martingale condition takes a slightly different form because the node values already incorporate the physical drift. The algebra is a bit more cluttered, but the end result is the same: a small $\sqrt{\Delta t}$-order bias in q that captures the discrepancy between the physical and risk-neutral drifts.

Applying the martingale condition $A = e^{-r\Delta t}\mathbb{E}^{\mathbb{Q}}[A_1]$ to tree (2.36) with $\mathbb{Q}$-probabilities $(q, 1-q)$:

$$A \;=\; e^{-r\Delta t}\Big[\, q\, A\, e^{(\mu-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}} \;+\; (1-q)\, A\, e^{(\mu-\frac{1}{2}\sigma^2)\Delta t-\sigma\sqrt{\Delta t}}\Big]. \tag{2.40}$$

Solving for q:

$$q \;=\; \frac{e^{r\Delta t} - e^{(\mu-\frac{1}{2}\sigma^2)\Delta t-\sigma\sqrt{\Delta t}}}{e^{(\mu-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}} - e^{(\mu-\frac{1}{2}\sigma^2)\Delta t-\sigma\sqrt{\Delta t}}} \;=\; \frac{e^{(r-(\mu-\frac{1}{2}\sigma^2))\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{+\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}}. \tag{2.41}$$

Introduce the shorthand $\hat{r} := r - (\mu - \frac{1}{2}\sigma^2)$. Expanding to $O(\sqrt{\Delta t})$:

$$q \;\sim\; \frac{(1+\hat{r}\Delta t) - (1-\sigma\sqrt{\Delta t}+\frac{1}{2}\sigma^2\Delta t)}{(1+\sigma\sqrt{\Delta t}+\frac{1}{2}\sigma^2\Delta t) - (1-\sigma\sqrt{\Delta t}+\frac{1}{2}\sigma^2\Delta t)} + \cdots \;=\; \frac{\sigma\sqrt{\Delta t} + (\hat{r}-\frac{1}{2}\sigma^2)\Delta t}{2\sigma\sqrt{\Delta t}} + \cdots, \tag{2.42}$$

$$q \;\sim\; \tfrac{1}{2}\left[1 + \frac{\hat{r}-\frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right] + \cdots \;=\; \tfrac{1}{2}\Big[1 + \frac{r-\mu}{\sigma}\sqrt{\Delta t}\Big] + \cdots, \tag{2.43}$$

where the last step uses $\hat{r} - \frac{1}{2}\sigma^2 = r - (\mu - \frac{1}{2}\sigma^2) - \frac{1}{2}\sigma^2 = r - \mu$.

Comparison with classical CRR (2.23):

$$q_{\text{CRR}} \;\sim\; \tfrac{1}{2}\left[1 + \frac{r-\frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right] + \cdots \tag{2.44}$$

The two q's differ at $O(\sqrt{\Delta t})$ by exactly the physical drift term, because parameterisation II has already absorbed the drift $(\mu - \frac{1}{2}\sigma^2)$ into the node geometry. In the continuous-time limit both parameterisations yield the same lognormal A_T distribution under $\mathbb{Q}$ — consistent with (2.30).

3.4.4 2.4.4 Checking $\mathbb{E}^{\mathbb{Q}}[A_1] = e^{r\Delta t}A$ and $\mathbb{V}^{\mathbb{Q}}[\ln(A_1/A)] = \sigma^2\Delta t$

A sanity check on any candidate lattice is that the martingale property holds (expected discounted asset grows at zero rate under $\mathbb{Q}$) and the variance matches the prescribed σ^2. For parameterisation II, both properties are easy to verify and hold essentially exactly at leading order in $\sqrt{\Delta t}$.

By construction of q in (2.40) the martingale relation $\mathbb{E}^{\mathbb{Q}}[A_1] = Ae^{r\Delta t}$ is exact. The log-variance calculation mirrors (2.39) — the Bernoulli $\pm\sigma\sqrt{\Delta t}$ still has variance $\sigma^2\Delta t$ under any $(q, 1-q)$ with q close to $\frac{1}{2}$, so

$$\mathbb{V}^{\mathbb{Q}}[\ln(A_1/A)] \;=\; \sigma^2\,\Delta t + O(\Delta t^{3/2}), \tag{2.45}$$

consistent with (2.26).

3.4.5 2.4.5 Variant — symmetric-probability drift-shifted tree with free parameter α

To emphasise that parameterisation II is one of many equivalent lattice encodings of the same continuous-time GBM, we now introduce a one-parameter family of drift-shifted lattices indexed by a free parameter α. Different values of α give different distributions of drift between the node geometry and the branch probabilities, but all converge to the same continuous-time limit.

A further variant takes the node geometry of (2.36) but allows an arbitrary physical-measure probability p determined by an extra parameter α:

$$A \longrightarrow \begin{cases} A\, e^{(\alpha-\frac{1}{2}\sigma^2)\Delta t+\sigma\sqrt{\Delta t}} & \text{w.p. } p \\ A\, e^{(\alpha-\frac{1}{2}\sigma^2)\Delta t-\sigma\sqrt{\Delta t}} & \text{w.p. } 1-p \end{cases} \tag{2.46}$$

with α a free (model-side) drift parameter. Requiring $\mathbb{E}^{\mathbb{P}}[A_1] = e^{\mu\Delta t}A$ and expanding to $O(\sqrt{\Delta t})$ gives

$$p \;\sim\; \tfrac{1}{2}\Big[1 + \frac{\mu-\alpha}{\sigma}\sqrt{\Delta t}\Big] + \cdots \tag{2.47}$$

When $\alpha = \mu$ one recovers $p = \frac{1}{2}$ (the alternate CRR of §2.4.1). When $\alpha = 0$ one recovers a variant closer to classical CRR with $p \sim \frac{1}{2}[1 + \frac{\mu}{\sigma}\sqrt{\Delta t}]$ matching (2.20).

Intuition. Any constant α absorbed into the node geometry shifts the required probability-bias by the same amount in the opposite direction — the physical measure $\mathbb{P}$ is a single object, and we have a one-parameter family of equivalent lattice encodings.

This freedom is a useful reminder that the lattice is *a* discretisation, not *the* discretisation. Different choices of α trade off the ease of computing the drift in the node geometry against the ease of computing the probability. In the extreme $\alpha = \mu$ case, we put all drift in the node and use fair-coin probabilities; in the $\alpha = 0$ case, we use symmetric nodes and push all drift into the biased probability. Intermediate values of α produce lattices with some drift in each. For any fixed α, the continuous-time limit is the same geometric Brownian motion; the difference is only in the finite-Δt convergence constants.

One practical implication: when comparing numerical prices from two different binomial-tree implementations, always check the parameterisation convention. A 0.5% discrepancy at $N = 100$ steps between two codes may not be a bug — it may just reflect different choices of α or different placements of K relative to the terminal nodes. For rigorous convergence studies, one must fix a parameterisation and extrapolate.

3.5 2.5 The Fundamental Theorem of Asset Pricing

The FTAP is the single most important theorem in derivatives pricing. It elevates the one-period, two-state intuition of (2.2) to the full continuous-time, infinite-dimensional state-space setting, asserting that *absence of arbitrage* is a precise mathematical equivalent of *existence of an equivalent martingale measure.* Once we have a $\mathbb{Q}$, we can price any claim by taking an expectation; without a $\mathbb{Q}$, we cannot consistently price anything. The FTAP is therefore the gateway between the economic principle (no free lunch) and the computational tool (expected-value pricing).

FTAP. No arbitrage $\iff$ there exists a probability measure $\mathbb{Q}$ equivalent to the physical measure $\mathbb{P}$ (written $\mathbb{Q} \sim \mathbb{P}$) such that for *every* traded asset X, the discounted price process $\tilde{X}_t$ is a $\mathbb{Q}$-martingale:

$$\tilde{X}_t \;=\; \mathbb{E}^{\mathbb{Q}}\big[\,\tilde{X}_s \mid \mathcal{F}_t\,\big], \qquad s \geq t. \tag{2.48}$$

Intuition. The pricing equation says "today's fair price equals the expected future price under $\mathbb{Q}$, once we neutralise the time-value of money by dividing through by the numéraire". The physical measure $\mathbb{P}$ describes how the world actually evolves; $\mathbb{Q}$ re-weights the same scenarios so that all *tradable* risks have zero expected excess return — which is exactly what "no arbitrage" demands at the level of forward-looking bets.

Unpacking the martingale condition (2.48): a *martingale* is a stochastic process whose best forecast is its current value. Formally, $\mathbb{E}[\tilde{X}_s \mid \mathcal{F}_t] = \tilde{X}_t$ for $s \geq t$ means that, given the information available at time t, the expected value of the discounted asset at any future time s is exactly today's discounted value. In the language of gambling, a martingale is a "fair game": the expected profit from any betting strategy is zero. If $\tilde{X}$ is a $\mathbb{Q}$-martingale, then no dynamic trading strategy can consistently make money — which is exactly no-arbitrage.

Three important extensions of the FTAP deserve mention, even though we do not need them for this chapter's development:

1. *Semimartingale FTAP.* The Delbaen–Schachermayer theorem extends the FTAP to general continuous-time models where the asset price is a semimartingale. The statement becomes: no "free lunch with vanishing risk" (NFLVR) is equivalent to existence of an equivalent martingale measure.

2. *Multi-asset FTAP.* With d risky assets and a riskless numéraire, the FTAP says no-arbitrage is equivalent to existence of $\mathbb{Q}$ under which *all* discounted asset processes are simultaneously martingales. The dimensionality of the martingale constraints sets the dimensionality of the set $\mathcal{M}_e$ of equivalent martingale measures.
3. *Second FTAP (completeness).* Market completeness is equivalent to $\mathcal{M}_e$ being a singleton. In binomial trees, one asset + cash is enough for completeness. In trinomial trees, we need another independent instrument — a two-state "choose-between-middle-and-outside" security, if one existed. Real markets are typically incomplete, and so $\mathcal{M}_e$ is a non-trivial convex set.

The tilde denotes discounting by a numéraire B_t:

$$\tilde{X}_t = \frac{X_t}{B_t}, \quad \text{where } B_t \text{ is a numéraire asset, } B_t > 0 \text{ a.s.} \tag{2.49}$$

The natural choice is the money-market account $B_t = e^{rt}$, but *any* strictly positive traded asset can serve as numéraire (this freedom is exploited later for forward measures). The mental image is that $\mathbb{Q}$ changes when you change the unit in which you measure wealth — dollars, zero-coupon bonds, shares of the stock — each measurement stick has its own martingale measure. The mechanics of switching between numéraires, including the associated Girsanov change-of-measure, are developed canonically in Chapter 5.

The FTAP comes in two flavours, which we should mention for completeness even though the distinction is subtle. The first FTAP says: absence of arbitrage is equivalent to existence of an equivalent martingale measure. The second FTAP says: absence of arbitrage *and* market completeness (every claim is replicable) is equivalent to *uniqueness* of the equivalent martingale measure. The first is a statement about the set $\mathcal{M}_e$ of equivalent martingale measures being non-empty; the second is a statement about $\mathcal{M}_e$ being a singleton. Binomial trees satisfy both FTAPs: one-step non-degeneracy guarantees $\mathcal{M}_e = \{q\}$, a single measure. Trinomial trees satisfy only the first: $\mathcal{M}_e$ is a non-empty open set, and the market is incomplete.

Why does the FTAP require *equivalent* measures rather than just absolutely continuous ones? Because equivalence ($\mathbb{Q} \sim \mathbb{P}$) means the two measures agree on null events: what has zero probability under $\mathbb{P}$ has zero probability under $\mathbb{Q}$, and vice versa. This is essential for pricing: if $\mathbb{Q}$ assigned zero probability to an event that $\mathbb{P}$ considered non-null, we could create an "arbitrage in probability" by constructing a portfolio that wins on that event; the $\mathbb{Q}$-price would be zero, but the $\mathbb{P}$-expected payoff positive. Equivalence rules this out by construction.

A subtler question is: is $\mathbb{Q}$ really a probability measure, or just a positive linear functional? In finite state spaces they are the same — a non-negative weighting summing to one. In continuous state spaces, $\mathbb{Q}$ is usually defined abstractly as a measure on the sample-path space Ω, and its existence (in the appropriate function-space topology) requires delicate arguments. We will not dwell on these technicalities; suffice it to say the FTAP works in every "nice" market model that practitioners care about.

3.6 2.6 Non-Uniqueness of the Martingale Measure

We arrive now at what may be the most important conceptual issue in modern quantitative finance: market incompleteness and the non-uniqueness of the pricing measure. Up to this point, our pricing arguments have rested on a tacit assumption: that the market contains enough tradable instruments to span every possible future state. In a binomial tree, this assumption holds automatically — two states, two instruments (stock and cash), perfect span. In a trinomial tree, or any richer state space, the assumption fails. The market cannot hedge every possible payoff, and consequently no-arbitrage alone does not determine a unique price.

This is not an academic curiosity. Real markets are universally incomplete. Stochastic volatility, jumps, stochastic interest rates, credit events, counterparty risk — each introduces additional state variables not spanned by vanilla instruments. The practical consequence: an entire industry has grown up around the problem of *how to select* a single price from the arbitrage-free interval. Calibration to market-traded benchmarks, utility-based risk preferences, robust pricing under ambiguity — these are the tools of the trade, and each corresponds to a particular way of selecting $\mathbb{Q}$ from the set $\mathcal{M}_e$ of equivalent martingale measures.

In the single-step binomial model with two post-states the measure $\mathbb{Q}$ was pinned down by one linear equation in one unknown q. Once the underlying can branch to three or more states, $\mathbb{Q}$ is no longer unique.

This is arguably the single most important conceptual moment of the chapter. We have been riding the coat-tails of the binomial model's remarkable completeness — every claim replicable, every price unique — but this completeness is an artefact of the lattice structure, not a feature of real markets. Real markets have far more states than tradable instruments: jumps, news shocks, stochastic volatility, counterparty default, liquidity freezes. Any one of these can create an un-spanned payoff, rendering the market incomplete. The FTAP still holds — equivalent martingale measures still exist — but there are many of them, and each produces a different price for the un-spanned claim.

Why, specifically, does three post-states destroy uniqueness? Because the pricing constraints are *linear* — each tradable asset generates one scalar equation $\mathbb{E}^{\mathbb{Q}}[X_1] = X_0 e^{r\Delta t}$ — and the number of free variables is $n-1$ (an n-state probability vector has $n-1$ free coordinates, after the sum-to-one normalisation). With stock + cash we get 2 constraints; with n states we get $n-1$ unknowns. For $n = 2$, $2-1 = 1$ unknown and 2 constraints (sum-to-one plus stock equation), giving the unique q. For $n = 3$, $3-1 = 2$ unknowns and 2 constraints, giving a one-dimensional family of solutions. For n states and k tradable assets (including cash), the dimension of $\mathcal{M}_e$ is $n-k$. Binomial trees have $n = 2$ and $k = 2$, so dim $= 0$ and $\mathcal{M}_e$ is a point. Trinomials have $n = 3$ and $k = 2$, so dim $= 1$ and $\mathcal{M}_e$ is a curve. Adding a third asset (say an index option) would add a constraint and potentially restore uniqueness — this is the *calibration-to-market* strategy that dominates practical quant finance.

Diagram (trinomial stub). Start the stock at $S_0 = 10$. Two alternative models are drawn:

- Binomial branch (left): $10 \longrightarrow \{12,\ 10,\ 8\}$ (three terminal states, one unit of cash, one unit of stock — only 2 constraints for 3 unknowns).
- Trinomial / four-state branch (right): $1 \longrightarrow \{C_u, C_m, C_d\}$ with only $(1,1)$ marginal bond/stock equations available to fix them.

The point is that we have two martingale equations (for cash and the stock) but three or four unknown probabilities, so

$$\mathbb{Q} \text{ is not unique} \implies C_0 \text{ is not necessarily unique.} \tag{2.50}$$

Intuition. Every tradable asset contributes one linear constraint on the risk-neutral probability vector. The moment the number of future states outstrips the number of independent constraints, the feasible set of $\mathbb{Q}$'s becomes a whole polytope — and the price of any non-spanned derivative sweeps out an arbitrage-free interval rather than a single number. This is exactly what makes real equity markets incomplete: you cannot perfectly hedge a jump, a stochastic-volatility move, or an idiosyncratic news shock with just stock and cash.

This is perhaps the single most important conceptual takeaway of the chapter, so let us restate it three ways. Algebraically: if n is the number of post-states and k is the number of traded assets (including cash), then the risk-neutral probability vector lives in $\mathbb{R}^n$, sums to one (one constraint), and is positive (open-set constraint, n inequalities). Each tradable adds one linear constraint, bringing the total to k equations in n unknowns. When $k = n$, the system has a unique solution (generically), giving a unique $\mathbb{Q}$ and a complete market. When $k < n$, the system is under-determined, giving an $(n-k)$-dimensional family of solutions and an incomplete market. Geometrically: the forward asset prices carve out a k-dimensional affine subspace of the $(n-1)$-dimensional simplex of probability distributions; every point in the intersection is a valid $\mathbb{Q}$. Financially: if two or more scenarios look the same to all currently-traded instruments, we cannot tell them apart from market data alone, and their relative weights are free.

Why completeness matters. If $\mathcal{M}_e$ is a singleton, then every contingent claim X has a unique no-arbitrage price $\mathbb{E}^{\mathbb{Q}}[X/B_T]B_0$. That price is attainable by a dynamic trading strategy — a perfect hedge exists. If $\mathcal{M}_e$ has more than one element, then a given claim might have *different* prices under different $\mathbb{Q}$'s, and the range of those prices is the *arbitrage-free interval*. Any price in the interval is admissible: it can be defended against arbitrage. A seller will quote somewhere near the upper end of the interval (the "super-replication cost"), a buyer somewhere near the lower end (the "sub-replication value"), and the market-clearing price depends on risk preferences, hedging technology, and supply–demand.

Trinomial tree — non-uniqueness of Q *Trinomial tree — non-uniqueness of Q*

3.6.1 2.6.1 Geometric picture and local risk-minimisation

To visualise incompleteness, think of the payoff space as a vector space (each future state is a coordinate axis; each claim is a point with one coordinate per state). Tradable assets span a subspace — the "replicable subspace". If the claim lies in this subspace, its price is uniquely determined by the spanning coordinates. If it does not, we need

a selection criterion. Local risk-minimisation picks the subspace point closest (in $L^2(\mathbb{P})$ norm) to the claim, and interprets the residual as the unhedgeable risk.

Project all attainable claim values onto a 2-dimensional span spanned by the two traded assets A_1 and B_1 at time 1. The diagram shows three claim payoffs C_1, C_2, C_3 floating above the plane $\mathrm{span}(A_1, B_1)$.

$$C_0 \text{ is unique iff the claim lies in } \mathrm{span}(A_1, B_1). \tag{2.51}$$

When the claim C_1 does not lie in this span, one selects a hedge by orthogonal projection under $\mathbb{P}$ — the local risk-minimising hedge:

$$\min_{(\alpha,\beta)} \mathbb{E}^{\mathbb{P}}\left[\,(\alpha A_1 + \beta B_1 - C_1)^2\,\right]. \tag{2.52}$$

This yields the hedge ratios $(\alpha^\star, \beta^\star)$ that minimise the L^2-distance between the replicating portfolio and the target payoff; the residual is the intrinsic (unhedgeable) risk.

Intuition. Think of $\mathrm{span}(A_1, B_1)$ as the set of payoffs you can synthesise with a static portfolio of stock and bond. A claim outside that plane has a component you simply *cannot* buy; the best you can do is drop a perpendicular from the claim to the plane. The foot of the perpendicular is the hedge portfolio; the length of the perpendicular (in the $\mathbb{P}$-weighted Euclidean norm) is the minimum residual standard deviation you are stuck warehousing on your book.

Local risk-minimisation is one of several *selection criteria* for an incomplete market's $\mathbb{Q}$. Others include: the *minimum-entropy martingale measure* (choose $\mathbb{Q} \in \mathcal{M}_e$ minimising $\mathbb{E}^{\mathbb{P}}[(d\mathbb{Q}/d\mathbb{P})\ln(d\mathbb{Q}/d\mathbb{P})]$), the *minimum-variance martingale measure* (choose $\mathbb{Q}$ minimising $\mathbb{V}^{\mathbb{P}}[d\mathbb{Q}/d\mathbb{P}]$), the *Esscher transform* (tilt the log-return density by $e^{\theta x}$ and normalise), and *utility indifference pricing* (pick $\mathbb{Q}$ so that a given agent's utility is equal whether she holds the claim or not). Each rule has different axiomatic motivations and produces different prices. In practice, a trader often sidesteps the choice by calibrating $\mathbb{Q}$ directly to a set of liquidly traded benchmark instruments (index options, variance swaps, etc.) — the market effectively *picks* the $\mathbb{Q}$ through its willingness to trade. This is the heart of the art of model calibration: the data picks $\mathbb{Q}$ subject to whatever parametric family the modeller has proposed.

A concrete worked example clarifies (2.52). Take the trinomial in the diagram with $S_0 = 10$, post-states $S_1 \in \{12, 10, 8\}$, physical probabilities $p = (1/3, 1/3, 1/3)$, and zero interest rate. Price the call payoff $C_1 = (2, 0, 0)$. The span of "bond and stock" at time 1 is spanned by the constant function $\mathbf{1} = (1, 1, 1)$ and the stock payoff $S_1 = (12, 10, 8)$. Projecting C_1 onto this span under $\mathbb{P}$-weighted inner product: compute $\mathbb{E}^{\mathbb{P}}[C_1] = 2/3$, $\mathbb{E}^{\mathbb{P}}[S_1] = 10$, $\mathbb{V}^{\mathbb{P}}[S_1] = (1/3)[4 + 0 + 4] = 8/3$, $\mathbb{C}^{\mathbb{P}}[S_1, C_1] = (1/3)[2 \cdot 2] = 4/3$. The OLS hedge ratio is $\beta^\star = (4/3)/(8/3) = 1/2$, the intercept is $\alpha^\star = 2/3 - (1/2) \cdot 10 = -14/3$. The best static hedge holds 1/2 share of stock and $-14/3$ worth of bond; its time-1 value is $(1/2)S_1 + (-14/3) = (4/3, 1/3, -2/3)$ in the up/middle/down states. Residuals: $C_1 - \widehat{C}_1 = (2/3, -1/3, 2/3)$. Residual variance $= (1/3)[(2/3)^2 + (1/3)^2 + (2/3)^2] = (1/3)(4/9 + 1/9 + 4/9) = 1/3$. That 1/3 is the irreducible L^2 risk the hedger is stuck with. Worked pricing: the hedger's indifference price is the fair-value midpoint 2/3, but the arbitrage-free *interval* stretches from the minimum no-arbitrage price (the cheapest $\mathbb{Q}$ in $\mathcal{M}_e$ applied to C_1) to the maximum, which for this example runs over $(0, 1)$.

3.7 2.7 Default Model and Incomplete-Market Trees

We now build a discrete default model that will converge to a diffusion with a jump to zero. This is our first concrete working example of an incomplete-market tree in the sense of §2.6.

The default model is our first encounter with a *jump* process. Up to now, all randomness has been continuous: small shocks, Gaussian in the limit, produced by accumulating many independent Bernoulli kicks. Default is different — it is a *rare, catastrophic, discrete event.* In a single infinitesimal step dt, the firm either defaults (with probability $\widehat{\lambda}\,dt$) or survives (with probability $1 - \widehat{\lambda}\,dt$); there is no smooth interpolation. The mathematical framework for such events is the *Poisson point process*, and the intensity $\widehat{\lambda}$ is the expected rate of jumps per unit time.

Why is this conceptually important? Because it takes us into genuinely *incomplete* territory. Before, our binomial tree had two states and two tradable instruments (stock and cash), so the market was complete and $\mathbb{Q}$ was unique. Now the tree has three states (up/down/default), and we still have only two instruments. We are in the regime of §2.6: $\mathcal{M}_e$ has dimension one, and the price of a non-spanned claim depends on the selection of $\mathbb{Q}$ from this set. In

practice, the market selects $\hat{\lambda}$ through credit-sensitive instruments (CDS spreads, corporate bond yields, deep-OTM puts), which pin down the risk-neutral intensity. Without those instruments, $\hat{\lambda}$ would be a free parameter.

The physical default intensity λ (the rate of actual defaults, which you might estimate from historical firm data) is generally *lower* than the risk-neutral intensity $\hat{\lambda}$ (the rate implicit in market prices). The difference $\hat{\lambda}-\lambda$ is the *credit risk premium* — the amount by which investors over-weight default scenarios in their pricing, as compensation for bearing the risk. Empirically, this premium is substantial: for investment-grade issuers, $\hat{\lambda}$ can be 2–5 times λ. This is the credit analogue of the equity risk premium, which explains why equities have higher expected returns than their physical volatility would justify on pure-mean-variance grounds.

On the single period $[t, t+\Delta t]$, the underlying evolves as

$$A_n = A_{n-1}\, e^{\sigma\sqrt{\Delta t}\, x_n}, \qquad x_1, x_2, \ldots \overset{iid}{\sim} \text{Bernoulli}(\pm 1), \tag{2.53}$$

with $\mathbb{P}(x_n = +1) = \frac{1}{2}\left(1 + \frac{\mu - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right) = p$.

Let τ denote the time of default (lifetime of the firm), with intensity $\hat{\lambda}$ so that $\tau \sim \text{Exp}(\hat{\lambda})$. The one-step survival/default probabilities are

$$\mathbb{P}(\tau \in (t, t+\Delta t] \mid \tau > t) \;=\; (1 - e^{-\hat{\lambda}\Delta t}) \;\sim\; \hat{\lambda}\,\Delta t. \tag{2.54}$$

Intuition. An exponential lifetime is the minimal *memoryless* survival model: the firm's chance of dying in the next instant is constant at $\hat{\lambda}$ per unit time, independent of how long it has already lived. The first-order expansion $1 - e^{-\hat{\lambda}\Delta t} \approx \hat{\lambda}\Delta t$ justifies treating $\hat{\lambda}\Delta t$ as the per-step "default coin flip".

The exponential distribution's defining property — memorylessness — says $\mathbb{P}(\tau > t + s \mid \tau > s) = \mathbb{P}(\tau > t)$. Conditioning on having survived s years does not shift the survival distribution for the next t years. This is a strong simplifying assumption. Empirically it is wrong: firms with longer track records have lower default intensities (the age effect), and default intensities cluster in time (contagion). More realistic models allow $\hat{\lambda}$ to vary stochastically — Cox processes, doubly-stochastic Poisson processes, affine jump-diffusion intensities. Our treatment uses constant $\hat{\lambda}$ only for pedagogical cleanliness; any practical credit engine generalises this.

Intuitively, $\hat{\lambda}$ has units of 1/time. At $\hat{\lambda} = 0.02$ (2% per year), the expected lifetime is $1/\hat{\lambda} = 50$ years, and the one-year survival probability is $e^{-0.02} \approx 98\%$. At $\hat{\lambda} = 0.10$, expected lifetime is 10 years, one-year survival is 90.5%. The intensity is typical of investment-grade corporate credit (single-A at $\hat{\lambda} \sim 0.005$/yr, BBB at ~ 0.015/yr) up through high-yield (single-B at ~ 0.05/yr, CCC at ~ 0.20/yr). These are calibrated from CDS spreads or bond yield spreads, divided by the loss-given-default fraction.

3.7.1 2.7.1 Branching tree with default branch

From a pre-default node A_{n-1} the tree has three children:

$$A_{n-1} \longrightarrow \begin{cases} A_{n-1}\, e^{+\sigma\sqrt{\Delta t}} & \text{w.p. } p\,(1 - \hat{\lambda}\Delta t) \\ A_{n-1}\, e^{-\sigma\sqrt{\Delta t}} & \text{w.p. } (1-p)(1 - \hat{\lambda}\Delta t) \\ 0 & \text{w.p. } \hat{\lambda}\,\Delta t \end{cases} \tag{2.55}$$

The riskless bond pays $e^{r\Delta t}$ in all three states (the money-market account is default-insensitive); the risky asset jumps to 0 on default.

Intuition. The tree is binomial *conditional on survival* and adds a third ray that collapses the asset to zero. This is the simplest lattice-level coupling between equity returns and credit events — default is a catastrophic, not a gradual, loss.

3.7.2 2.7.2 Survival law

For the lifetime $\tau \sim \mathrm{Exp}(\hat{\lambda})$:

$$\mathbb{P}(\tau > T) \;=\; e^{-\hat{\lambda}T}, \qquad \mathbb{P}(\tau \in (t, t+\Delta t]) \;=\; e^{-\hat{\lambda}t} - e^{-\hat{\lambda}(t+\Delta t)} \;=\; e^{-\hat{\lambda}t}(1 - e^{-\hat{\lambda}\Delta t}). \tag{2.56}$$

The survival probability $e^{-\hat{\lambda}T}$ is the most important number in corporate credit. At $\hat{\lambda} = 5\%$ per year, five-year survival is $e^{-0.25} \approx 77.9\%$; ten-year survival is $e^{-0.5} \approx 60.7\%$. These numbers govern everything from bond pricing to pension-liability discounting. The exponential form reflects the memoryless property: the firm's probability of making it to year T is the product of the per-year survival probabilities, each equal to $e^{-\hat{\lambda}}$. A time-varying intensity $\hat{\lambda}(t)$ would give survival probability $\exp(-\int_0^T \hat{\lambda}(s)ds)$, but constant intensity is the cleanest starting point.

The density of the default time is $f_\tau(t) = \hat{\lambda}e^{-\hat{\lambda}t}$, and its mean is $1/\hat{\lambda}$. At $\hat{\lambda} = 5\%$, mean time-to-default is 20 years. The variance of τ is $1/\hat{\lambda}^2$, so the standard deviation equals the mean — a defining feature of exponential distributions. In particular, default times are *highly dispersed*: half the defaults occur before the median $\ln(2)/\hat{\lambda} \approx 13.9$ years, and significant probability mass extends beyond 50 years. This is a feature, not a bug, of the exponential model.

3.7.3 2.7.3 Risk-neutral q in the default-free sub-case

As a consistency check, setting $\hat{\lambda} = 0$ should recover the plain CRR result of §2.3. We verify this below, confirming that the default-augmented model reduces to the classical CRR lattice in the limit of no credit risk — exactly as one would expect.

Under $\mathbb{Q}$, the one-step pricing condition (martingale property of the discounted asset) is

$$\frac{A_{n-1}}{B_{n-1}} \;=\; \mathbb{E}^{\mathbb{Q}}\left[\frac{A_n}{B_n}\right] \qquad \text{(martingale condition).} \tag{2.57}$$

In the default-free limit $\hat{\lambda} = 0$ one recovers exactly (2.22):

$$q \;=\; \frac{e^{r\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{+\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}} \;\sim\; \tfrac{1}{2}\left(1 + \frac{r - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right) + o(\sqrt{\Delta t}). \tag{2.58}$$

$\hat{\lambda}$ is the $\mathbb{Q}$-intensity of default (a risk-neutral hazard rate, potentially different from the physical intensity λ).

Intuition. The formula for q is the unique weight that makes $\{A\text{-up}, A\text{-down}\}$ barycentre exactly the one-step forward growth $e^{r\Delta t}$. Comparing (2.58) to the $\mathbb{P}$-probability p in (2.53): going from $\mathbb{P}$ to $\mathbb{Q}$ is precisely the substitution $\mu \mapsto r$ — the real-world drift is replaced by the riskless rate. That is the change-of-measure at the level of drifts.

3.7.4 2.7.4 Including default into the martingale condition

We now build the full three-branch martingale condition, which includes the default branch explicitly. The algebra is a direct generalisation of (2.21): the martingale constraint must now sum over three children, not two, and the default child contributes zero to the asset's forward value. Solving for the surviving-branch q-probability gives a slightly shifted version of the default-free result.

With the three-branch tree the martingale condition becomes

$$1 \;=\; q\,(1 - \hat{\lambda}\Delta t)\,e^{\sigma\sqrt{\Delta t} - r\Delta t} \;+\; (1-q)(1 - \hat{\lambda}\Delta t)\,e^{-\sigma\sqrt{\Delta t} - r\Delta t} \;+\; 0 \cdot \hat{\lambda}\Delta t. \tag{2.59}$$

Solving for q:

$$q \;=\; \frac{e^{(r+\hat{\lambda})\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{+\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}} \;\sim\; \tfrac{1}{2}\left(1 + \frac{r + \hat{\lambda} - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right) + o(\sqrt{\Delta t}). \tag{2.60}$$

The effect of adding a default branch is therefore to shift the risk-neutral drift upward by $\widehat{\lambda}$: the surviving branch must grow faster to compensate for the chance of being wiped out.

Intuition. If a fraction $\widehat{\lambda}\Delta t$ of your probability mass leaks to zero every step, the conditional expected growth of the survivors has to be higher just to keep the unconditional expectation at $e^{r\Delta t}$. Credit risk premia in corporate bonds have exactly this structure: the yield-to-maturity exceeds the riskless rate by roughly the expected loss rate $\widehat{\lambda} \cdot \mathrm{LGD}$.

A concrete credit-pricing illustration. Consider a zero-recovery one-year zero-coupon bond issued by a firm with default intensity $\widehat{\lambda} = 3\%$. Its fair price is $e^{-(r+\widehat{\lambda})T} = e^{-(0.05+0.03)\cdot 1} \approx 0.923$, compared to the riskless bond price $e^{-rT} \approx 0.951$. The yield spread is therefore $\widehat{\lambda} = 3\%$ — the spread is the risk-neutral default intensity (in the zero-recovery case). With recovery R (so loss-given-default is $1-R$), the bond price is $e^{-rT}(e^{-\widehat{\lambda}T} + R(1-e^{-\widehat{\lambda}T})) \approx e^{-rT}(1-(1-R)\widehat{\lambda}T)$, and the spread is $\widehat{\lambda}(1-R)$ to leading order. This is the "spread = probability × loss" relationship that sits at the heart of the CDS market. The CDS premium quoted by a market-maker *is* the market-implied $\widehat{\lambda}(1-R)$, and working back out the probability requires an LGD assumption (typically $R \approx 40\%$ for senior unsecured corporate, so $1-R = 60\%$).

Now bring the stock back into the picture. On our default tree, the stock's surviving-branch growth is $e^{(r+\widehat{\lambda})\Delta t}$; its unconditional expectation is $e^{r\Delta t}$ because the $\widehat{\lambda}$ leak balances the surviving boost. Stockholders — as residual claimants — bear the full loss on default, so the stock's equity risk premium under $\mathbb{Q}$ is pushed upward by the credit component. Equity traders often combine this with Merton's structural credit model: view the firm's equity as a call option on its assets, with strike at the debt's face value. The equity is thus *naturally* a credit-contingent claim, and the same $\widehat{\lambda}$ that shows up in debt pricing also shows up in the equity vol surface (high equity vol goes with high credit spreads). The stylised empirical fact is sometimes called the "credit–equity link".

3.7.5 2.7.5 Risk-neutral expectations of survival

A few expectation identities under the default model are useful for pricing credit-sensitive claims and for setting up the continuous-time limit. These formulas show how the risk-neutral drift is modified by the default intensity and how survival-conditional expectations relate to unconditional ones.

Under $\mathbb{Q}$ the asset and its discounted version satisfy

$$\mathbb{E}^{\mathbb{Q}}\Big[A_T\, \mathbf{1}_{\{\tau>T\}}\Big] \;=\; A_0\, e^{(r+\widehat{\lambda})T}, \tag{2.61}$$

and for the bond-denominated (default-insensitive) notional,

$$\mathbb{E}^{\mathbb{Q}}[A_T] \;=\; A_0\, e^{rT}. \tag{2.62}$$

Intuition. Equation (2.62) is the FTAP itself at the horizon: the unconditional expected terminal asset price grows at r. Equation (2.61) says that, *conditional on surviving*, the asset grows at the higher rate $r+\widehat{\lambda}$. Taking the product with the survival probability $e^{-\widehat{\lambda}T}$ recovers (2.62): $e^{(r+\widehat{\lambda})T} \cdot e^{-\widehat{\lambda}T} = e^{rT}$.

3.7.6 2.7.6 European pricing recursion with default

The backward recursion on the three-branch tree is just the no-arbitrage condition rearranged: the discounted price at time t equals the risk-neutral expected discounted price at time $t+\Delta t$. We must account for all three possible outcomes — up survival, down survival, and default — weighting each by its risk-neutral probability.

Let $C_{n,j}$ denote the option value at tree node (n,j). The martingale/pricing recursion on the three-branch tree is

$$\frac{C_{n-1,j}}{B_{n-1}} \;=\; \mathbb{E}^{\mathbb{Q}}\left[\frac{C_n}{B_n}\right], \tag{2.63}$$

$$C_{n-1,j} \;=\; q\,\frac{e^{-\widehat{\lambda}\Delta t}}{e^{r\Delta t}}\,C_{n,j} \;+\; (1-q)\,\frac{e^{-\widehat{\lambda}\Delta t}}{e^{r\Delta t}}\,C_{n,j+1} \;+\; (1-e^{-\widehat{\lambda}\Delta t})\cdot C_{n,d}, \tag{2.64}$$

where the default node satisfies

$$C_{n-1,d} \;=\; C_{n,d}\, e^{-r\Delta t}. \tag{2.65}$$

(On default the stock is 0, so a call has $C_{n,d} \equiv 0$; a put has $C_{n,d} = K$ discounted to today.)

Intuition. The recursion is the classical binomial backward sweep with two modifications: each surviving branch is down-weighted by the survival factor $e^{-\hat{\lambda}\Delta t}$, and a third term captures what the option is worth *in* default. For a vanilla call that third term is zero, so credit risk eats value monotonically; for a put it is positive (the put still pays on bankruptcy unless it is a "protected put" that voids on the issuer's own default).

For a concrete numerical illustration: take the vanilla 1-year ATM call with $A_0 = K = 100$, $\sigma = 20\%$, $r = 5\%$, and add a default intensity $\hat{\lambda} = 2\%$. The Black–Scholes price (without default) is approximately 10.45. Discounting by survival at maturity multiplies by $e^{-\hat{\lambda}T} = e^{-0.02} \approx 0.9802$, and the modified call price is approximately $10.45 \cdot 0.9802 \approx 10.24$. A 2% credit intensity produces a 2% reduction in call value. For a deeply-out-of-the-money call (say strike $K = 150$) whose Black–Scholes price is 0.45, the modified price is 0.44 — the absolute reduction is smaller, but the relative impact is still 2%. This is a general pattern: constant default intensity scales down call prices by a factor $e^{-\hat{\lambda}(T-t)}$ to leading order, and this is how credit-adjusted "jump-to-default" models are typically parameterised.

3.8 2.8 American Valuation on the Binomial Tree

Before diving into the details, it is worth situating American options in the larger landscape. American-style exercise is the default for single-stock equity options in the US market — virtually all listed single-stock options are American. European-style exercise is standard for index options (S&P 500, Russell 2000) and for European-market listed options. The choice of exercise style matters because American options can be worth more than European ones (since the holder has more optionality), and because American options require more sophisticated pricing and hedging technology.

American options introduce a genuinely new computational challenge: the *stopping problem.* Unlike European options, which always settle at the fixed date T, American options give the holder the right to exercise *at any time* $\tau \le T$. The pricing problem becomes a two-part decision: (i) at each node, choose the optimal exercise policy; (ii) under that policy, compute the expected discounted payoff. The two parts are coupled — the expected payoff depends on the policy, and the policy depends on expected payoffs — and they must be solved simultaneously via dynamic programming.

Why can we not simply take the maximum over stopping times analytically? Because the optimal stopping time is not a deterministic time; it is a *random time* that depends on the realised path. Formally, τ must be a *stopping time* — a random variable adapted to the filtration, so $\{\tau \le t\} \in \mathcal{F}_t$ for all t. Stopping times represent non-anticipating exercise policies: we can decide to exercise based on what has happened so far, but not on what will happen. The set of stopping times is combinatorially vast, and direct optimisation is infeasible.

Dynamic programming (Bellman's principle) reduces this to a tractable recursion: "the value of being in state X at time t is the max of (i) exercising now and receiving the intrinsic value, or (ii) holding to time $t+1$ and receiving the expected next-step value". The recursion runs backward from the terminal payoff, sweeping through the lattice in $O(N^2)$ time. The optimal policy falls out as a by-product: at each node, record whether the max was achieved by exercising or by continuing; the exercise boundary is the locus of nodes where the max was achieved by exercising.

3.8.1 2.8.1 European → American contrast

We begin by contrasting the European and American exercise styles. The European case has a fixed settlement date T; pricing is a single expectation. The American case allows the holder to exercise at any stopping time $\tau \le T$; pricing is a supremum over all such stopping times. The supremum turns out to be computable by backward induction — the classical dynamic-programming result of Bellman — and the resulting price is at least as large as the European price, with the difference being the "early exercise premium".

European option pays $\varphi(A_T)$ at the fixed maturity T:

$$V_t \;=\; e^{-r(T-t)}\, \mathbb{E}^{\mathbb{Q}}[\,\varphi(A_T)\,] \;\longrightarrow\; V_T \;=\; \varphi(A_T). \tag{2.66}$$

American option lets the holder choose the exercise epoch τ, a stopping time bounded above by T, and pays $\varphi(A_\tau)$ at τ — meaning the holder "receives $\varphi(A_\tau)$ at the moment they decide to stop".

American put, in particular:

$$\varphi(A_\tau) \;=\; (K - A_\tau)_+. \tag{2.67}$$

Let $\mathcal{S}$ denote the set of $\{\mathcal{F}_t\}$-stopping times bounded by T. Then

$$\frac{C_0}{B_0} \;=\; \sup_{\tau\in\mathcal{S}} \mathbb{E}^{\mathbb{Q}}\left[\frac{(K - A_\tau)_+}{B_\tau}\right]. \tag{2.68}$$

Intuition. The American price is the Snell envelope of the intrinsic process: it is the smallest supermartingale that dominates immediate-exercise value at every node. Operationally it says: at each moment, compare "what I get by exercising right now" against "what the option is worth if I wait one more step" — take whichever is larger.

The Snell envelope is a fundamental object in optimal stopping theory. Given an intrinsic-value process $\{I_t\}$ — the payoff from exercising at time t — the Snell envelope $V_t = \operatorname{ess\,sup}_{\tau\geq t}\mathbb{E}[I_\tau \mid \mathcal{F}_t]$ is the smallest $\mathbb{Q}$-supermartingale dominating I. "Smallest" means pointwise smallest: for any other supermartingale $M_t \geq I_t$, we have $V_t \leq M_t$. "Supermartingale" means $\mathbb{E}[V_s \mid \mathcal{F}_t] \leq V_t$ for $s \geq t$, i.e., the value deteriorates in expectation (properly discounted) over time — this is the hallmark of an option where holding costs you expected value relative to exercising. The optimal stopping time is $\tau^* = \inf\{t : V_t = I_t\}$, the first hitting time of the exercise region. At every earlier node, holding is optimal; at τ^* the two values coincide and exercise is indifferent to continuing.

On the lattice, this abstract machinery reduces to a very simple algorithm: at each node, compare $I_{n,k}$ to $V^h_{n,k}$; set $V_{n,k} = \max(I_{n,k}, V^h_{n,k})$. The exercise boundary appears as the locus of nodes where the two are equal; beyond the boundary (into the exercise region), $I > V^h$; before the boundary (into the continuation region), $V^h > I$. This is exactly the dynamic-programming recursion of (2.69)–(2.72).

3.8.2 2.8.2 Tree diagram — hold vs exercise

At every node of the tree the American pricer faces a binary choice: exercise (accept the intrinsic value and stop) or hold (accept the continuation value and continue). The choice is made by comparing the two candidates and taking the larger. This simple max operation, applied at every node, produces the optimal exercise strategy.

At every interior node there are two candidate values:

- P^h — the hold/continuation value (from the one-step backward recursion),
- $P^x \equiv (K - A)_+$ — the immediate-exercise (intrinsic) value.

The actual American value is $P = \max(P^h, P^x)$.

3.8.3 2.8.3 Dynamic-programming recursion

The recursion below is the cleanest illustration of Bellman's principle in quantitative finance. At every lattice node we are solving a local optimisation problem — exercise or continue? — and the local solutions stitch together into the global optimum. The same pattern generalises to callable bonds, convertible bonds, American-style swaptions, and a host of structured products with embedded optionality.

On a recombining binomial tree with nodes (n, k), propagate two quantities backward:

- Continuation (holding) value $C^h_{n-1,k}$:

$$\frac{C^h_{n-1,k}}{B_{n-1,k}} \;=\; q\,\frac{C_{n,k}}{B_{n,k}} \;+\; (1-q)\,\frac{C_{n,k+1}}{B_{n,k+1}} \tag{2.69}$$

$$\Longrightarrow\; C^h_{n-1,k} \;=\; e^{-r\Delta t}\,(\,q\,C_{n,k} \;+\; (1-q)\,C_{n,k+1}\,). \tag{2.70}$$

- Intrinsic (immediate exercise) value $C^I_{n-1,k}$:

$$C^I_{n-1,k} \;=\; (A_{n-1,k} - K)_+ \quad \text{(call)}, \qquad C^I_{n-1,k} \;=\; (K - A_{n-1,k})_+ \quad \text{(put)}. \tag{2.71}$$

- American value — take the larger:

$$\boxed{C_{n-1,k} \;=\; \max\Big(C^I_{n-1,k},\, C^h_{n-1,k}\Big)} \tag{2.72}$$

3.8.4 2.8.4 Exercise boundary picture

The exercise boundary is a free boundary in the sense of PDE theory: it is determined *implicitly* by the solution to the pricing equation, not given in advance. At any time t, the boundary is the value $A^*(t)$ of the underlying such that the American option value equals its intrinsic value. For an American put, $A^*(t) < K$ and moves closer to K as $t \to T$ (because the exercise region grows). For an American call on a dividend-paying stock, $A^*(t) > K$ and moves closer to K as ex-dividend dates approach.

A diagonal wedge is drawn on the tree: the optimal-exercise boundary separates nodes where $C^I_{n,k} > C^h_{n,k}$ (exercise immediately, shown on the upper-right for a put — the deep-in-the-money region) from nodes where $C^h_{n,k} > C^I_{n,k}$ (hold). Nodes on the boundary are labelled "optimal to exercise"; nodes beyond are "optimal to exercise: $C^I_{n,k} > C^h_{n,k}$".

Intuition. For an American put on a non-dividend-paying stock, early exercise is driven by the time-value of the strike: when the stock crashes deep enough, the interest you could earn on the cash proceeds K outweighs the remaining optionality. For an American call on a non-dividend-payer the boundary never bites (early exercise is suboptimal, Merton's theorem), so the American call equals the European call — one of the cleanest no-arbitrage results in the theory.

Merton's theorem deserves a quick proof sketch. An American call holder considering early exercise at time t receives $A_t - K$ if they exercise. But they could instead sell the call in the market (whose value $\geq A_t - K$ always, by the lower bound $C_t \geq A_t - Ke^{-r(T-t)}$ for a non-dividend-paying stock and positive interest rate). In fact $C_t \geq A_t - Ke^{-r(T-t)} > A_t - K$, so selling always dominates exercising. The same argument fails for a dividend-paying stock: if the stock pays a large cash dividend between t and T, the holder might prefer exercising immediately (capturing the current cum-dividend price) to holding through the ex-dividend drop. Early exercise of American calls is therefore only relevant on dividend-paying underlyings, around ex-dividend dates. For puts, early exercise is driven by the time-value of the strike, and the boundary exists always (no dividends needed).

The American-put early-exercise premium (AEP) is the difference between American and European put values. At typical parameters ($S_0 = 100$, $K = 100$, $\sigma = 20\%$, $r = 5\%$, $T = 1$), AEP is a few cents — small, but non-zero, and growing with moneyness, with rate, and with time to maturity. Computing AEP requires solving the Snell envelope; there is no closed-form formula. The Bjerksund–Stensland and Geske–Johnson approximations are popular analytical tools for bounding AEP; for exact prices, a binomial or Monte-Carlo with Longstaff–Schwartz regression is standard.

American put early-exercise boundary *American put early-exercise boundary*

3.9 2.9 Bridge to Continuous Time

The CRR lattice of §2.3–§2.4 converged, via the Central Limit Theorem applied to the sum of log-returns, to a lognormal distribution for the terminal asset:

$$A_T \;\overset{d}{=}\; A_0\, e^{(r-\frac{1}{2}\sigma^2)T \,+\, \sigma\sqrt{T}\, Z_{\mathbb{Q}}}, \qquad Z_{\mathbb{Q}} \sim \mathcal{N}(0,1). \tag{2.73}$$

This is the *marginal* distribution of A_T under $\mathbb{Q}$. Turning this into a *process* — a continuous-time object defined for every $t \in [0,T]$ — requires more work. The scaled sum of Bernoulli shocks along the tree,

$$W^N_t \;:=\; \sqrt{\Delta t} \sum_{n=1}^{\lfloor t/\Delta t \rfloor} x_n, \tag{2.74}$$

converges in distribution (Donsker's functional CLT) to a *Brownian motion* W_t on $[0, T]$. Under $\mathbb{Q}$ the asset itself then converges to geometric Brownian motion (GBM):

$$A_t \;=\; A_0\, e^{(r-\frac{1}{2}\sigma^2)t\,+\,\sigma W_t}, \qquad dA_t \;=\; r\,A_t\,dt \;+\; \sigma\,A_t\,dW_t. \tag{2.75}$$

These formulas can be read off the CRR lattice in the limit, and a naïve CLT-style argument gives a convincing heuristic derivation. But a rigorous derivation requires (i) a rigorous construction of Brownian motion, (ii) a rigorous theory of the stochastic integral, and (iii) Itô's lemma, which tells us how to change variables inside stochastic differentials. These foundations are developed systematically in Chapter 3.

The European call price in the GBM limit evaluates to the classical Black–Scholes formula,

$$C_0 \;=\; A_0\,\Phi(d_+) \;-\; K\,e^{-rT}\,\Phi(d_-), \qquad d_\pm \;=\; \frac{\ln(A_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \tag{2.76}$$

where Φ is the standard-normal CDF. The *direct* route from the CRR lattice to (2.76) runs through a measure-change argument: split the call payoff as $(A_T - K)_+ = A_T\,\mathbf{1}_{\{A_T>K\}} - K\,\mathbf{1}_{\{A_T>K\}}$, price the first term under the stock-numéraire measure $\mathbb{Q}^A$ and the second under the money-market measure $\mathbb{Q}$, and evaluate each exceedance probability as a Gaussian tail. This is the Donsker-to-Black-Scholes sketch. Once Itô's lemma is available, the same formula drops out of the hedging argument in Chapter 6.

We defer the full derivation deliberately. The reader will see Black–Scholes emerge in two complementary ways in the chapters ahead:

- Chapter 3 places Brownian motion on rigorous footing, proves Itô's lemma, and solves the GBM SDE explicitly — giving (2.75) as a theorem rather than a heuristic limit.
- Chapter 4 develops the Feynman–Kac connection between SDEs and PDEs.
- Chapter 5 develops the Radon–Nikodym / Girsanov machinery that turns measure changes (including the stock-numéraire switch used in the BS derivation) into a single canonical toolkit.
- Chapter 6 re-derives (2.76) as the canonical Black–Scholes formula from the self-financing hedging argument, and connects it to the Feynman–Kac route as a sanity check.

Monte-Carlo simulation and path-dependent options — forward-starting (cliquet) contracts, up-and-in / knock-out barriers, Asians, lookbacks — are treated in Chapter 10, once the GBM path generator of (2.75) has been placed on rigorous footing. American options on the tree (§2.8 above) remain the cleanest non-trivial use of backward induction on the lattice; path-dependent Americans combine the two paradigms and are the subject of the Longstaff–Schwartz regression technique discussed in Chapter 10.

The upshot for the CRR limit is simple, and we state it as a promise to the reader. The CRR tree converges in distribution to GBM (Donsker's theorem applied to the scaled random walk of the log-price), and the European payoff limit produces the Black–Scholes formula (2.76). This derivation is re-done rigorously in Chapter 6 via Itô's lemma. For now, the reader should take away three things from this chapter:

1. The no-arbitrage pricing of any European claim is a discounted $\mathbb{Q}$-expectation of its payoff.
2. Backward induction on a recombining tree makes that expectation computable in $O(N^2)$ operations and extends naturally to American exercise.
3. In the CRR limit the tree converges to GBM, and vanilla Euro prices converge to Black–Scholes — but writing the limit down honestly requires the stochastic-calculus apparatus of the next three chapters.

3.10 Key Takeaways

1. The one-period no-arb inequality $A_d < A(1+r) < A_u$ is equivalent to existence of a risk-neutral measure $\mathbb{Q}$ such that $C = (1+r)^{-1}\mathbb{E}^{\mathbb{Q}}[C_1]$.
2. Multi-period backward induction prices any European payoff via repeated one-step discounted risk-neutral expectations — the tower-property manifestation of $C_0 = \mathbb{E}^{\mathbb{Q}}[e^{-rT}C_T]$.
3. Classical CRR calibration. Match two moments of the log-return: $c = \sigma\sqrt{\Delta t}$, $p = \frac{1}{2}(1+\frac{\mu}{\sigma}\sqrt{\Delta t})$; the risk-neutral $q = \frac{1}{2}(1 + \frac{r-\sigma^2/2}{\sigma}\sqrt{\Delta t})$.

4. Alternate CRR (drift-shifted lattice). Nodes $A \cdot e^{(\mu-\frac{1}{2}\sigma^2)\Delta t \pm \sigma\sqrt{\Delta t}}$ with symmetric probability $p = \frac{1}{2}$; the Itô correction is baked into the geometry. Under $\mathbb{Q}$, $q \sim \frac{1}{2}[1 + \frac{r-\mu}{\sigma}\sqrt{\Delta t}]$. Both parameterisations converge to the same lognormal GBM.
5. Measure-invariant variance. $\mathbb{V}^{\mathbb{P}}[\ln A_T] = \mathbb{V}^{\mathbb{Q}}[\ln A_T] = \sigma^2 T$ — Girsanov cannot change volatility, only drift.
6. Simple- vs log-return bias. Fitting to simple returns shifts the drift estimate by $+\frac{1}{2}\sigma^2$; always be explicit which object is being fit.
7. FTAP equates no-arbitrage to the existence of an equivalent martingale measure $\mathbb{Q}$ under which all discounted traded assets are $\mathbb{Q}$-martingales.
8. Numéraire freedom. Any strictly positive traded asset B_t can discount — picking B_t is picking a measure. Canonical derivation of the change-of-measure mechanics is deferred to Chapter 5.
9. Non-uniqueness of $\mathbb{Q}$ appears as soon as the number of future states exceeds the number of traded assets: the claim may not lie in $\mathrm{span}(A_1, B_1)$. The set $\mathcal{M}_e$ of equivalent martingale measures has dimension $n-k$ for n states and k tradables.
10. Local risk-minimisation $\min_{\alpha,\beta} \mathbb{E}^{\mathbb{P}}[(\alpha A_1 + \beta B_1 - C_1)^2]$ selects a hedge but leaves residual, unhedgeable variance.
11. Default as a third branch adds the probability mass $\hat{\lambda}\Delta t$ at 0 and shifts the risk-neutral drift by $+\hat{\lambda}$ — the surviving branch must grow faster to compensate. This is our first concrete incomplete-market tree.
12. Binomial-to-GBM limit. With Bernoulli ± 1 increments under $\mathbb{Q}$, $A_T \xrightarrow{d} A_0\, e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z}$ — the marginal distribution of geometric Brownian motion.
13. Donsker's functional CLT promotes the scaled random walk of the log-price to a Brownian motion W_t on $[0, T]$, giving GBM $dA_t = rA_t\, dt + \sigma A_t\, dW_t$ — deferred to Chapter 3 for the rigorous construction.
14. Black–Scholes $C_0 = A_0\Phi(d_+) - Ke^{-rT}\Phi(d_-)$ is the CRR-limit pricing formula for a European call. The derivation via measure change and Itô's lemma is given in Chapters 5 and 6.
15. American options require pointwise comparison of intrinsic and holding values at every node; the optimal policy is a stopping time defined by the exercise boundary. American calls on non-dividend-paying stock equal European calls (Merton's theorem); American puts have a strictly positive early-exercise premium.
16. Monte-Carlo simulation and path-dependent options (cliquets, barriers, Asians) are treated in Chapter 10, once the GBM path generator has been placed on rigorous footing in Chapter 3.

3.11 Reference Formulas

One-period no-arbitrage.

$$A_d < A(1+r) < A_u \iff \exists\, \mathbb{Q} \text{ s.t. } C = \tfrac{1}{1+r}\,\mathbb{E}^{\mathbb{Q}}[C_1].$$

FTAP / discounted martingale.

$$\tilde{X}_t = \frac{X_t}{B_t}, \qquad \tilde{X}_t = \mathbb{E}^{\mathbb{Q}}\left[\tilde{X}_s \mid \mathcal{F}_t\right].$$

Classical CRR calibration (moment-matching).

$$c \sim \sigma\sqrt{\Delta t}, \qquad p \sim \tfrac{1}{2}\left(1 + \tfrac{\mu}{\sigma}\sqrt{\Delta t}\right), \qquad q \sim \tfrac{1}{2}\left(1 + \frac{r - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right).$$

Alternate CRR (drift-shifted lattice).

$$A \to A\, e^{(\mu-\frac{1}{2}\sigma^2)\Delta t \pm \sigma\sqrt{\Delta t}}, \qquad p = \tfrac{1}{2}, \qquad q \sim \tfrac{1}{2}\left[1 + \tfrac{r-\mu}{\sigma}\sqrt{\Delta t}\right] + \cdots.$$

Log-moment invariance.

$$\mathbb{E}^{\mathbb{P}}[\ln(A_T/A_0)] = \mu T, \qquad \mathbb{E}^{\mathbb{Q}}[\ln(A_T/A_0)] = (r - \tfrac{1}{2}\sigma^2)T, \qquad \mathbb{V}^{\mathbb{P}}[\ln(A_T/A_0)] = \mathbb{V}^{\mathbb{Q}}[\ln(A_T/A_0)] = \sigma^2 T.$$

Local risk-minimising hedge.

$$\min_{(\alpha,\beta)} \mathbb{E}^{\mathbb{P}}\left[(\alpha A_1 + \beta B_1 - C_1)^2\right].$$

Default-tree risk-neutral probability.

$$q = \frac{e^{(r+\hat{\lambda})\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{+\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}}, \qquad q \sim \tfrac{1}{2}\left(1 + \frac{r + \hat{\lambda} - \frac{1}{2}\sigma^2}{\sigma}\sqrt{\Delta t}\right).$$

Survival law.

$$\mathbb{P}(\tau > T) = e^{-\hat{\lambda}T}, \quad \mathbb{P}(\tau \in (t, t+\Delta t]) = e^{-\hat{\lambda}t}(1 - e^{-\hat{\lambda}\Delta t}).$$

European recursion (default tree).

$$C_{n-1,j} = q\,\frac{e^{-\hat{\lambda}\Delta t}}{e^{r\Delta t}}\,C_{n,j} + (1-q)\,\frac{e^{-\hat{\lambda}\Delta t}}{e^{r\Delta t}}\,C_{n,j+1} + (1 - e^{-\hat{\lambda}\Delta t})\,C_{n,d}.$$

American recursion.

$$C^h_{n-1,k} = e^{-r\Delta t}(q\,C_{n,k} + (1-q)\,C_{n,k+1}),$$
$$C^I_{n-1,k} = (A_{n-1,k} - K)_+ \text{ (call)}, \quad (K - A_{n-1,k})_+ \text{ (put)},$$
$$C_{n-1,k} = \max\Big(C^I_{n-1,k},\ C^h_{n-1,k}\Big).$$

CRR-limit lognormal distribution.

$$A_T \overset{d}{=} A_0\, e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z}, \qquad Z \sim \mathcal{N}(0,1) \text{ under } \mathbb{Q}.$$

GBM continuous-time limit (Donsker).

$$A_t = A_0\, e^{(r-\frac{1}{2}\sigma^2)t + \sigma W_t}, \qquad dA_t = r\,A_t\,dt + \sigma\,A_t\,dW_t, \quad W \text{ a } \mathbb{Q}\text{-Brownian motion.}$$

Black–Scholes formula (CRR limit, derived in Chapter 6).

$$C_0 = A_0\,\Phi(d_+) - Ke^{-rT}\,\Phi(d_-), \qquad d_\pm = \frac{\ln(A_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

4 Chapter 3 — Stochastic-Calculus Primer: Brownian Motion, Itô, and SDE Solutions

This chapter builds the stochastic-calculus apparatus that every later chapter uses. It is the foundational chapter of the guide: from here forward we take Itô's lemma, the Itô isometry, and the closed-form solution of the simplest SDEs as known, and we build measure-change, PDE bridges, and hedging arguments on top of them.

The arc of the chapter is deliberately linear. We start from the scaled random walk of Chapter 2 — the same CRR lattice you already know how to price on — and take a scaling limit to Brownian motion. We then prove the two structural path properties of BM that every derivative-pricing argument rests on (infinite total variation, finite quadratic variation equal to t), and use them to define the Itô stochastic integral. The Itô isometry drops out as a clean L^2 identity; Brownian moments follow as short MGF exercises; Itô's lemma in three successively more general forms (Lemmas I, II, III) is derived from a second-order Taylor expansion that keeps exactly the terms of size $\mathrm{d}t$. The SDE catalogue at the end of the chapter — GBM, Ornstein–Uhlenbeck, and the generic constant-coefficient diffusion — is each solved as a worked Itô-lemma exercise, so by the end of the chapter the reader knows not only the machinery but the three or four canonical SDEs that every later chapter will invoke.

A small upfront caveat. Nothing in the chapter is harder than one variable of calculus plus one variable of probability — but the *interaction* of those two disciplines is what makes stochastic calculus feel foreign on first reading. The Taylor series of $f(W_t)$ behaves nothing like the Taylor series of $f(x)$, and the mental reflex of "discard the second-order term" has to be permanently unlearned. The payoff is enormous: every pricing formula, every risk sensitivity, every hedging recipe in derivatives markets is downstream of the single identity $(\mathrm{d}W)^2 = \mathrm{d}t$. Master that line and the Black-Scholes PDE, the Greeks, the log-normal density, implied-vol surfaces, the forward measure, and Black-76 all arrange themselves into a small family of corollaries.

A note on notation. We use W_t for Brownian motion throughout. Plain W denotes a $\mathbb{P}$-Brownian motion when no measure is specified; when a Girsanov shift is in play (from Chapter 5 onward) hats or tildes will decorate the BM under the alternative measure. In this chapter measure is immaterial, because every path-property argument we make is invariant under equivalent changes of measure.

4.1 3.1 From Scaled Random Walk to Brownian Motion

4.1.1 3.1.1 The CRR tree, viewed as a discrete Brownian motion

Chapter 2 built a multi-period binomial tree for an underlying index, priced options on it by backward induction, and took a Donsker-style continuous-time limit in which the terminal distribution becomes log-normal. We now revisit that construction, but strip away the option-pricing apparatus to expose the underlying object — the *scaled random walk* that the lattice implicitly defines.

Let $x_1, x_2, \ldots$ be i.i.d. Bernoulli sign-flips with $\mathbb{P}(x_k = \pm 1) = \frac{1}{2}$. Partition the time axis into steps of size $\Delta t = t/N$ and define a scaled random walk by

$$W_{n\Delta t} \;=\; W_{(n-1)\Delta t} \;+\; \sqrt{\Delta t}\, x_n, \qquad W_0 \;=\; 0. \tag{3.1}$$

Equivalently, summing N increments,

$$W_{N\Delta t} \;=\; \sqrt{\Delta t} \sum_{n=1}^{N} x_n. \tag{3.2}$$

Each increment has size $\pm\sqrt{\Delta t}$. The square-root scaling is not cosmetic: shrinking the time-step by a factor of N shrinks each displacement by only $\sqrt{N}$, not by N, and that is exactly the asymmetric scaling needed to keep the variance finite in the limit. The multiplicative geometric version used for stock trees in Chapter 2 writes $S \in \{Se^{\sigma\sqrt{\Delta t}}, Se^{-\sigma\sqrt{\Delta t}}\}$; the arithmetic version $W \in \{W + \sqrt{\Delta t}, W - \sqrt{\Delta t}\}$ is what limits to Brownian motion and is what we study in this chapter.

The square-root-of-time scaling is worth a standalone intuition. Suppose you flipped a coin every millisecond for a year and called the net displacement D of a unit-stride random walker. The walker's position at year-end would be

$D \sim \text{Normal}(0, N)$ with $N \approx 3.15 \times 10^{10}$ steps, giving a standard deviation of $\sqrt{N} \approx 177{,}000$ millimetres — about 177 metres. Now ask the opposite question: how large does each stride have to be so that, after the year, the walker's typical displacement is exactly 1 metre? Solving $\sqrt{N}\,\ell = 1$ m gives $\ell \approx 5.6\,\mu$m — roughly the size of a red blood cell. The stride $\ell = \sigma\sqrt{\Delta t}$ is the unique scaling that makes the stride shrink as the step-rate grows yet keeps the annual variance finite and positive. That is the whole magic of the $\sqrt{\Delta t}$ factor and the whole magic of Brownian motion.

A second intuition: the stride must shrink faster than the time-step because the *number* of steps grows as $1/\Delta t$ while the variance of the total displacement grows *linearly* in that number. Linear-in-steps times variance-per-step equals total variance; to keep total variance bounded by t per unit time, each step's variance has to be $O(\Delta t)$, which forces the step size to be $O(\sqrt{\Delta t})$. Any other choice either collapses the walk (if you pick Δt exactly) or blows it up (if you pick a constant).

4.1.2 3.1.2 Moments of the walk

Using $\mathbb{E}[x_n] = 0$ and $\text{Var}[x_n] = 1$:

$$\mathbb{E}[W_{N\Delta t}] \;=\; \sqrt{\Delta t}\sum_{n=1}^{N}\mathbb{E}[x_n] \;=\; 0, \tag{3.3}$$

$$\text{Var}[W_{N\Delta t}] \;=\; \Delta t\sum_{n=1}^{N}\text{Var}[x_n] \;=\; N\,\Delta t \;=\; t. \tag{3.4}$$

So $W_{N\Delta t}$ has mean 0 and variance t for *any* N — the variance is preserved exactly along the refinement, which is the whole point of picking $\sqrt{\Delta t}$ rather than Δt or any other power. Had we used Δt the walk would collapse to 0; had we used a constant, it would explode. Only the square-root scaling keeps $\text{Var}[W_t] = t$.

The key observation to register: the mean and the variance are exact for every N, not just asymptotically. When proving convergence of a random quantity it is enormously useful to have one moment already at its limiting value for every finite approximation; convergence then reduces to showing the *other* moment shrinks to zero. That double-moment structure — one moment exact, one shrinking — will reappear in every almost-sure-limit proof in this chapter.

4.1.3 3.1.3 CLT: the continuous-time limit

Let t be fixed and $N \to \infty$ (equivalently $\Delta t = t/N \downarrow 0$). By the central limit theorem, the normalised sum $\frac{1}{\sqrt{N}}\sum_{n=1}^{N} x_n$ converges in distribution to a standard normal, so

$$W_t \;\xrightarrow[N\to\infty]{d}\; \sqrt{t}\,Z, \qquad Z \sim \mathcal{N}(0,1). \tag{3.5}$$

Thus in the limit $W_t \sim \mathcal{N}(0, t)$ is Gaussian with mean 0 and variance t. Read this the other way round: the Gaussian marginal at each time is *not* an assumption — it is forced on us by the CLT the moment we impose the correct scaling in (3.1).

The Gaussian law is not posited; it is *inherited* from the CLT's universality. Any other i.i.d. innovation with finite variance — uniform ± 1, Rademacher ± 1 with unfair weights, a truncated triangular distribution, a Laplace distribution with heavy tails (provided the variance is finite), even empirical return distributions from real markets — gives the same Gaussian scaling limit. The only surviving features of the innovation are its mean and its variance; all higher moments wash out under $\sqrt{N}$ averaging. In that sense Brownian motion is the *universal* continuous-time random walk, in the same way the Gaussian is the universal limit of normalised sums. A practical corollary: when you simulate a Brownian motion on a computer, you are almost always generating the fine-grained walk with Gaussian increments $\sqrt{\Delta t}\,Z$ rather than Bernoulli ones, because CLT universality makes the two limits identical.

4.1.4 3.1.4 Stationary and independent increments

For any $t < t + s$, write $W_{t+s} - W_t = \sqrt{\Delta t}\sum_{m=1}^{M} y_m$ with $y_m = x_{N+m}$ a fresh block of sign-flips. By the same CLT, with $M\Delta t = s$,

$$W_{t+s} - W_t \xrightarrow[M\to\infty]{d} \sqrt{s}\, Z, \qquad Z \sim \mathcal{N}(0,1). \tag{3.6}$$

Because the law of $W_{t+s} - W_t$ depends only on the *length* s of the interval — not on W_t or the starting time t — the process has *stationary increments*. Moreover, for non-overlapping intervals $[t, s]$ and $[u, v]$ the underlying Bernoulli blocks are disjoint, so the two sums are functions of independent random variables and therefore

$$(W_s - W_t) \perp (W_v - W_u) \qquad \text{(independent increments)}. \tag{3.7}$$

Independent increments is the source of the *Markov property*: knowing W up to time u tells you nothing about future increments beyond what W_u already encodes. Equivalent phrasings — conditional distribution of W_{t+s} given $\mathcal{F}_t$ is $\mathcal{N}(W_t, s)$, and conditional on W_t the future is independent of the past — will be used verbatim in the Feynman-Kac derivations of Chapter 4 and the change-of-numeraire arguments of Chapter 5.

4.2 3.2 Definition of Brownian Motion and Path Properties

Collecting the three properties just derived plus a regularity property, the scaling limit $W = (W_t)_{t\geq 0}$ is defined as a process satisfying

$$\begin{aligned}
&\text{(i)}\ \ W_0 = 0,\\
&\text{(ii)}\ \ W_t \overset{d}{=} \sqrt{t}\, Z, \quad Z \sim \mathcal{N}(0,1),\\
&\text{(iii)}\ \ W \text{ has stationary and independent increments},\\
&\text{(iv)}\ \ W \text{ has continuous paths.}
\end{aligned} \tag{3.8}$$

Property (iv) is the only one not already implied by the random-walk construction: the polygonal interpolation of the lattice walk is piecewise-linear, and continuity of the limit needs a tightness argument (Kolmogorov's continuity criterion). Properties (i)–(iii) are the structural skeleton; continuity then paints the skeleton smooth — but, as we are about to see, the *roughness* of that continuity is severe.

A pedagogical remark. The four conditions are conceptually independent, and you can build "partial Brownian motions" that satisfy any three of them but not the fourth. A compound Poisson process satisfies (i)–(iii) but has discontinuous paths and so fails (iv). A process that sits at zero forever satisfies (i) and (iv) but fails (ii) and (iii). A standard Cauchy process (stable with index 1) satisfies (i), (iii), (iv) but has increments that are Cauchy, not Gaussian, and so fails (ii). It is the combination of all four — specifically Gaussian marginals plus independent increments plus continuity — that pins Brownian motion down uniquely (up to measure-zero null sets).

4.2.1 3.2.1 Continuous but nowhere differentiable

A typical sample path of BM is a jittery curve, everywhere continuous but nowhere smooth. Heuristically: the typical displacement over a time-step Δt is $|\Delta W| \sim \sqrt{\Delta t}$, so the slope

Five Brownian-motion sample paths *Five independent sample paths of W on $[0,1]$. Every path is continuous but visibly non-smooth at every scale — zooming in reveals identical-looking wiggles. The unbounded-slope property (3.∞) is visible as the path's inability to be "traced with a pencil".*

$$\frac{\Delta W}{\Delta t} \sim \frac{\sqrt{\Delta t}}{\Delta t} = \frac{1}{\sqrt{\Delta t}} \longrightarrow \infty \quad (\Delta t \downarrow 0).$$

Every secant slope blows up, so the path has no derivative at any point despite being everywhere continuous. This is the first indication that ordinary calculus cannot handle W — we need a *quadratic* calculus, which §3.3–§3.4 make precise. The intuition for why this forces a new calculus is simple: in any Taylor expansion of a function $f(W_t)$, the second-order term carries a factor $(\Delta W)^2 \sim \Delta t$, which is the *same* order as Δt itself. Second-order terms can never be dropped as "small" in the way Newton's calculus assumes.

A concrete analogy is a coastline. Measure a coastline with a kilometre-long ruler and you get one total length; measure it with a metre-long ruler and the length grows because the ruler resolves previously-hidden wiggles; measure with a centimetre-long ruler and the length grows again. In the limit of an infinitesimally fine ruler, the total length diverges. A Brownian path behaves the same way: the *length* (total variation) diverges as the ruler refines, while a different quantity — the *sum of squared straight-line segments* (quadratic variation) — stays finite and equals t. Section 3.3 computes the first; Section 3.4 computes the second.

The reason this matters for trading. Every gamma trader lives inside the discrepancy between TV and QV. If BM had finite TV, a delta-hedged option would have zero expected P&L regardless of hedging frequency — there would be no gamma premium to harvest, because the path could be traced arbitrarily precisely with finitely many hedge trades. Infinite TV says the opposite: no matter how often you rehedge, the residual cash flow between rehedges is never negligible. Finite QV — the specific value t — tells you how large that residual is on average, which is the *gamma theta* of options pricing. The two path properties we are about to prove are not abstract measure-theoretic curiosities; they are the mathematical expressions of "options have a gamma premium."

4.3 3.3 Total Variation

4.3.1 3.3.1 Definition

For a function f on $[0,t]$ and a partition $\pi = \{t_0, t_1, \dots, t_N\}$ with $0 = t_0 < t_1 < \cdots < t_N = t$, the total variation is

$$\mathrm{TV}_t \;=\; \lim_{\|\pi\|\downarrow 0} \sum_k \big|\, f(t_k) - f(t_{k-1}) \,\big|, \tag{3.9}$$

where $\|\pi\| = \max_k (t_k - t_{k-1})$ is the mesh size. Pictorially, lay the graph of f flat by flipping each down-stroke upward; the total horizontal length is TV_t.

TV matters because the Riemann-Stieltjes integral $\int g\,\mathrm{d}f$ only exists (in the classical sense) when f has finite total variation on $[0,t]$. For BM we are about to find TV is infinite, which is exactly why the new stochastic integral of §3.5 is needed.

4.3.2 3.3.2 Case 1 — f is differentiable (finite TV)

If $f \in C^1$, by the mean-value theorem there exists $t_k^\star \in (t_{k-1}, t_k)$ with $\Delta f_k = f'(t_k^\star)\,\Delta t_k$. Therefore

$$\sum_k |\Delta f_k| \;=\; \sum_k |f'(t_k^\star)|\,\Delta t_k \;\xrightarrow[\|\pi\|\downarrow 0]{}\; \int_0^t |f'(s)|\,\mathrm{d}s \;<\; +\infty. \tag{3.10}$$

Finite TV, as expected for smooth functions. The entire argument fails for BM because W has no derivative anywhere, and in its absence the sum of $|\Delta W_k|$ diverges.

4.3.3 3.3.3 Case 2 — Brownian motion has infinite TV

For BM the increments are $\Delta W_k \overset{d}{=} \sqrt{\Delta t_k}\, Z_k$ with $Z_k \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. So

$$\sum_k |W_{t_k} - W_{t_{k-1}}| \;=\; \sum_k \sqrt{\Delta t_k}\,|Z_k|. \tag{3.11}$$

The mean. Because $\mathbb{E}|Z| = \sqrt{2/\pi}$ is a finite constant,

$$\mathbb{E}\left[\sum_k \sqrt{\Delta t_k}\,|Z_k|\right] \;=\; \mathbb{E}|Z| \cdot \sum_k \sqrt{\Delta t_k}. \tag{3.12}$$

The variance is $c^2 \sum_k \Delta t_k = c^2 t$, bounded as the partition refines. The key bound is on the mean:

$$\sum_k \sqrt{\Delta t_k} \;\geq\; \sum_k \frac{\Delta t_k}{\sqrt{\|\pi\|}} \;=\; \frac{t}{\sqrt{\|\pi\|}} \;\xrightarrow[\|\pi\|\downarrow 0]{}\; +\infty. \tag{3.13}$$

So the mean of the TV-sum diverges while its variance stays bounded — concentration around an unbounded mean forces the sum itself to be unbounded. Almost surely,

$$\boxed{\mathrm{TV}_t(W) \;=\; \lim_{\|\pi\|\downarrow 0} \sum_k |W_{t_k} - W_{t_{k-1}}| \;=\; +\infty}. \qquad \text{(a.s.)} \tag{3.14}$$

Brownian motion is too rough to have a well-defined first-order path integral $\int_0^t g(s)\,\mathrm{d}W_s$ in the Riemann-Stieltjes sense; a new stochastic integral is needed, and that is the subject of §3.5.

The economic shadow. A delta-hedger rebalances N times per unit time and incurs a transaction cost of $c \cdot |\Delta S|$ per trade. Total trading cost equals $c \cdot \sum |\Delta S_k|$, which is the TV sum for S. Because S is approximately GBM and GBM has infinite TV, this cost diverges as rebalancing becomes continuous — exactly the feature that prevents the frictionless Black-Scholes derivation from being implementable literally. In practice, hedgers impose a discrete rebalancing grid and accept residual gamma risk; the frictionless-trading assumption is the statement "pretend TV does not diverge, or at least that transactions are free." Whenever a trader complains about transaction costs on a delta-hedging strategy, that is the pathological TV of GBM biting.

4.4 3.4 Quadratic Variation and the Rule $(\mathrm{d}W)^2 = \mathrm{d}t$

4.4.1 3.4.1 Definition

The quadratic variation of f on $[0, t]$ is

$$[f]_t \;=\; [f, f]_t \;=\; \lim_{\|\pi\|\downarrow 0} \sum_k (\Delta f_k)^2, \qquad \Delta f_k = f(t_k) - f(t_{k-1}). \tag{3.15}$$

Where TV measured path length, QV measures cumulative *squared jitter*. For smooth functions QV vanishes (because $(\Delta f)^2 \approx (f')^2(\Delta t)^2$ is one order higher than Δt); for BM, as we are about to see, QV equals t exactly — a finite, deterministic number. This is the single most important fact in stochastic calculus, because it converts the heuristic $(\mathrm{d}W)^2 = \mathrm{d}t$ into a rigorous statement.

Why the quadratic version of the sum is well-behaved while the linear version diverges. The sum-of-absolutes $\sum |\Delta W_k|$ behaves like $\sum \sqrt{\Delta t_k} \cdot \mathbb{E}|Z|$ — a sum of *square-roots* — which grows like $\sqrt{N}$. The sum-of-squares $\sum (\Delta W_k)^2$ behaves like $\sum \Delta t_k \cdot \mathbb{E}[Z^2] = t$, constant in N. Squaring each increment converts the diverging $\sqrt{\Delta t}$ into a summable Δt and shifts the bad behaviour into a benign regime. A cleaner way to state the lesson: TV sees *size*, QV sees *energy*. Brownian motion has bounded energy (the t in $[W, W]_t = t$) but unbounded size (the $+\infty$ in $\mathrm{TV}_t(W) = +\infty$). In trader language, BM has finite integrated variance (= t times vol-squared) but infinite gross trading distance (= integrated |price change|). The two quantities are not the same, and conflating them is the most common source of Black-Scholes confusion.

4.4.2 3.4.2 Case 1 — f differentiable: $[f]_t = 0$

If $f \in C^1$,

$$\sum_k (\Delta f_k)^2 \;=\; \sum_k \big(f'(t_k^\star)\big)^2 (\Delta t_k)^2 \;\leq\; \|\pi\| \cdot \int_0^t (f'(s))^2\,\mathrm{d}s \;\xrightarrow[\|\pi\|\downarrow 0]{}\; 0. \tag{3.16}$$

So differentiable functions have zero QV. The inequality $(\Delta t_k)^2 \leq \Delta t_k \cdot \|\pi\|$ peels off one power of $\|\pi\|$, which kills the sum in the limit. This is why ordinary calculus has no Itô correction: for smooth paths, squared-increment sums are negligible.

Phrased dimensionally: for smooth f, $\Delta f_k = O(\Delta t_k)$ so $(\Delta f_k)^2 = O((\Delta t_k)^2)$, and there are $N = t/\Delta t$ of them, giving a total $N \cdot O((\Delta t)^2) = O(\Delta t) \to 0$. For Brownian motion $\Delta W_k = O(\sqrt{\Delta t_k})$, so $(\Delta W_k)^2 = O(\Delta t_k)$ and the sum $N \cdot O(\Delta t) = O(1)$ stays finite. Smooth functions produce $(\Delta f)^2 \sim (\Delta t)^2$ (second-order small); BM produces $(\Delta W)^2 \sim \Delta t$ (first-order small). One order of Δt is kept by the squaring, the other washed out — and that retained order produces the Itô correction.

4.4.3 3.4.3 Case 2 — Brownian motion: $[W, W]_t = t$

Let $Q = \sum_k (\Delta W_k)^2$ with $\Delta W_k \overset{d}{=} \sqrt{\Delta t_k}\, Z_k$, $Z_k \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$.

Mean.

$$\mathbb{E}[Q] \;=\; \sum_k \mathbb{E}\big[(\Delta W_k)^2\big] \;=\; \sum_k \Delta t_k \;=\; t. \tag{3.17}$$

The mean is *exact* — no limit needed, no $\|\pi\|$ dependence. Every finite partition already gives $\mathbb{E}[Q] = t$.

Variance. Using $\mathrm{Var}((\Delta W_k)^2) = (\Delta t_k)^2\,\mathrm{Var}(Z^2) = 2(\Delta t_k)^2$ (because $\mathrm{Var}(Z^2) = \mathbb{E}[Z^4] - 1 = 3 - 1 = 2$ — the standard-Normal fourth moment $\mathbb{E}[Z^4] = 3$ is derived via the MGF in §3.7.1) and independence across k,

$$\mathrm{Var}[Q] \;=\; 2\sum_k (\Delta t_k)^2 \;\le\; 2\,\|\pi\| \sum_k \Delta t_k \;=\; 2\|\pi\|\,t \;\xrightarrow[\|\pi\|\downarrow 0]{}\; 0. \tag{3.18}$$

Conclusion. Q has mean *exactly* t and variance shrinking to zero, so by Chebyshev's inequality (L^2-convergence along the full sequence, almost-sure along a thinning subsequence):

$$\boxed{\sum_k (\Delta W_k)^2 \;\xrightarrow[\|\pi\|\downarrow 0]{\text{a.s.}}\; t, \qquad [W,W]_t \;=\; t} \qquad \text{(a.s.)} \tag{3.19}$$

A striking feature: a *random* object — the sum of squared increments along a random trajectory — converges to a *deterministic* limit t. The randomness is there, but it is concentrated so tightly around the mean that in the limit it disappears. Economically, this is why implied volatility is observable at all: the quadratic variation of a stock price is essentially a deterministic integral of σ^2, not a random quantity.

Quadratic variation → t and total variation → ∞ as mesh refines *The same Brownian path, sampled at progressively finer meshes. Left: $\sum(\Delta W_k)^2$ converges deterministically to $t = 1$ as the mesh refines. Right: $\sum|\Delta W_k|$ diverges like $\sqrt{N}$ — finite energy but unbounded path-length, which is precisely the TV vs QV dichotomy of §3.3–§3.4.*

This "randomness collapsing into determinism" deserves a full beat. In the discrete approximation, $\sum_k (\Delta W_k)^2$ is a sum of N i.i.d. random variables, each equal to $\Delta t \cdot Z_k^2$ for an independent chi-squared-one draw. By the law of large numbers the average of these Z_k^2 converges almost surely to $\mathbb{E}[Z^2] = 1$; multiplying by $N\Delta t = t$ gives $\sum(\Delta W_k)^2 \to t$ a.s. The convergence is the LLN dressed up in Brownian clothing. The variance-shrinking-faster-than-mean argument is just Chebyshev telling us the LLN converges rapidly in L^2.

Why "realised variance = integrated variance" in the variance-swap world. A variance swap pays the holder the realised sample variance $\frac{252}{N}\sum(r_k)^2$ where r_k are daily log-returns. In the GBM model $r_k \approx \sigma\Delta W_k - \frac{1}{2}\sigma^2\Delta t$, so $\sum r_k^2 \approx \sigma^2 \sum(\Delta W_k)^2 \to \sigma^2 t$. The variance-swap payoff concentrates on a deterministic quantity equal to the square of the volatility times the tenor — which is why variance swaps are *exactly* replicable by a portfolio of European options (Carr-Madan replication). In contrast, the TV of the same log-return sequence diverges like $\sqrt{N}$, explaining why "average absolute return" is a nuisance statistic whereas "realised variance" is a well-posed estimator of σ^2.

4.4.4 3.4.4 The Itô shorthand $(\mathrm{d}W)^2 = \mathrm{d}t$

Equation (3.19) is the mathematical justification of the Itô multiplication rules

$$(\mathrm{d}W_t)^2 \;=\; \mathrm{d}t, \qquad \mathrm{d}W_t \cdot \mathrm{d}t \;=\; 0, \qquad (\mathrm{d}t)^2 \;=\; 0. \tag{3.20}$$

These rules drive Itô's formula throughout the rest of the chapter (and this guide). In particular — the crucial pedagogical point — the BM path has *non-zero* QV-roughness ($\sum \Delta W_k^2 \to t$), not the 0 you might have hoped for from an ordinary-calculus perspective; that t is precisely the reason pricing PDEs carry a $\frac{1}{2}\sigma^2\,\partial_{xx}$ term. Every Itô correction in this guide traces back to (3.20).

The three rules are worth memorising with their intuitive content.

- $(\mathrm{d}W_t)^2 = \mathrm{d}t$. The square of a Gaussian increment of variance $\mathrm{d}t$ has mean $\mathrm{d}t$ and variance $2(\mathrm{d}t)^2 \ll \mathrm{d}t$; the $\mathrm{d}t$-mean dominates, so at leading order $(\mathrm{d}W)^2 = \mathrm{d}t$ is a deterministic identity.
- $\mathrm{d}W_t \cdot \mathrm{d}t = 0$. The increment $\mathrm{d}W_t$ has magnitude $\sqrt{\mathrm{d}t}$, so $\mathrm{d}W \cdot \mathrm{d}t \sim (\mathrm{d}t)^{3/2}$, which is higher-order than $\mathrm{d}t$ and drops out.
- $(\mathrm{d}t)^2 = 0$. Obvious dimensionally; kept in the list for symmetry.

The memorisation trick. Think of $\mathrm{d}W$ as living at scale $\sqrt{\mathrm{d}t}$ and $\mathrm{d}t$ as living at scale $\mathrm{d}t$. Any product of these two atoms scales as $(\mathrm{d}t)^{n/2}$ where n is the total atom count. Keep only products with $n = 2$ (size exactly $\mathrm{d}t$), drop products with $n > 2$. The three rules are just this bookkeeping.

When we apply Itô's lemma we multiply out $(\mathrm{d}X)^2$ for an SDE $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W$, expand

$$(a\,\mathrm{d}t + b\,\mathrm{d}W)^2 \;=\; a^2\,(\mathrm{d}t)^2 + 2ab\,\mathrm{d}t\,\mathrm{d}W + b^2\,(\mathrm{d}W)^2,$$

and then use (3.20): the first two terms vanish, the third becomes $b^2\,\mathrm{d}t$. The rule set collapses the Itô-square of any SDE to a clean $b^2\,\mathrm{d}t$ — the fact used in every Itô-formula application in this chapter. A quick self-check: if you can state and multiply these three rules from memory, you can generate any Itô-formula line on demand.

4.5 3.5 The Itô Stochastic Integral

4.5.1 3.5.1 Warm-up — the classical Riemann-Stieltjes integral

For smooth f the chain rule gives

$$\int_0^t f(s)\,\mathrm{d}f(s) \;=\; \tfrac{1}{2}\,(f^2(t) - f^2(0)), \tag{3.21}$$

because $\mathrm{d}(\frac{1}{2}f^2(s)) = f(s)\,\mathrm{d}f(s)$. Remember this answer — when we repeat the calculation for Brownian motion we will find the analogue carries an *extra* $-\frac{1}{2}t$ term that has no classical counterpart. The classical integral is oblivious to the endpoint choice in the Riemann sum because QV is zero; once we swap f for BM the endpoint choice becomes consequential, and the $-\frac{1}{2}t$ is exactly the accumulated left-right mismatch summed over infinitely many infinitesimal intervals.

4.5.2 3.5.2 The Itô integral, defined

The stochastic analogue uses *left-endpoint* Riemann sums — crucial, because this is what makes the integral a martingale:

$$\int_0^t W_s\,\mathrm{d}W_s \;\equiv\; \lim_{\|\pi\|\downarrow 0} \sum_k W_{t_{k-1}}\,\Delta W_{t_k}, \qquad \Delta W_{t_k} = W_{t_k} - W_{t_{k-1}}. \tag{3.22}$$

Evaluating W at the *start* t_{k-1} of each sub-interval makes $W_{t_{k-1}}$ independent of the forward-looking increment ΔW_{t_k}. That independence makes the expectation of each term zero, and hence makes the whole stochastic integral a martingale. Right-endpoints would couple the integrand to its own driving noise and destroy the martingale property.

Economic gloss. The Itô integral is the mathematical expression of a *predictable* trading strategy: at time t_{k-1} you commit to holding $W_{t_{k-1}}$ units of the underlying until t_k, and you realise the P&L $W_{t_{k-1}}(W_{t_k} - W_{t_{k-1}})$ over that interval. Predictability means the position size is chosen *before* the random move, not after; the left-endpoint rule enforces exactly that. If you were allowed to pick the right endpoint you could lean into favourable moves you

already know about. The martingale property of stochastic integrals is thus the mathematical statement "predictable strategies have zero expected P&L" — the no-arbitrage condition in continuous time.

A different rule called the Stratonovich integral evaluates at the midpoint $\frac{1}{2}(W_{t_{k-1}} + W_{t_k})$. Stratonovich integrals obey the classical chain rule (no Itô correction) but lose the martingale property. For financial modelling the Itô convention is mandatory because the martingale property, not the chain rule, is what encodes no-arbitrage.

4.5.3 3.5.3 First worked integral — $\int_0^t W_s \, \mathrm{d}W_s$

The claim, which we prove by a telescoping identity + QV computation, is

$$I_t \;=\; \int_0^t W_s \, \mathrm{d}W_s \;=\; \tfrac{1}{2}\, W_t^2 \;-\; \tfrac{1}{2}\, t \qquad \text{a.s.} \tag{3.23}$$

The answer already carries the surprise: the classical guess would be $\frac{1}{2}W_t^2$, but the stochastic integral subtracts an extra $\frac{1}{2}t$ — the *Itô correction*, sometimes called the "convexity correction" because it arises from the curvature of $w \mapsto \frac{1}{2}w^2$ interacting with the QV of W. The $\frac{1}{2}t$ is deterministic, grows linearly, and has no stochastic component; it is the "free" cash that a convex long-gamma position accumulates as time passes. In options language, the correction is the *theta bleed* of a one-lot long-gamma portfolio.

Proof. Use the algebraic identity $a(b-a) = \frac{1}{2}(b^2 - a^2) - \frac{1}{2}(b-a)^2$ with $a = W_{t_{k-1}}$, $b = W_{t_k}$:

$$W_{t_{k-1}}(W_{t_k} - W_{t_{k-1}}) \;=\; \tfrac{1}{2}(W_{t_k}^2 - W_{t_{k-1}}^2) \;-\; \tfrac{1}{2}(W_{t_k} - W_{t_{k-1}})^2. \tag{3.24}$$

Summing over k and using $W_0 = 0$:

$$\sum_k W_{t_{k-1}} \, \Delta W_{t_k} \;=\; \tfrac{1}{2}\, W_t^2 \;-\; \tfrac{1}{2} \sum_k (\Delta W_{t_k})^2. \tag{3.25}$$

The left-hand side is the Riemann-like sum defining the Itô integral. The right-hand side has two parts. The first, $\frac{1}{2}W_t^2$, is the classical Newton-Leibniz answer. The second, $-\frac{1}{2}\sum_k (\Delta W_{t_k})^2$, is the Itô correction in discrete form — an exact lattice version of $-\frac{1}{2}t$, because by (3.19) this sum converges almost surely to t. Passing to the limit,

$$\boxed{\int_0^t W_s \, \mathrm{d}W_s \;=\; \tfrac{1}{2}\, W_t^2 \;-\; \tfrac{1}{2}\, t}\,. \tag{3.26}$$

Three observations on this answer.

1. Taking expectations on both sides of (3.26): $\mathbb{E}\int_0^t W_s \, \mathrm{d}W_s = \frac{1}{2}\mathbb{E}[W_t^2] - \frac{1}{2}t = 0$. The Itô integral is zero-mean — the martingale property surfaces automatically.
2. The Itô integral is *not* zero, despite having zero mean. Its variance grows with t (we'll compute it via the Itô isometry next): it is a non-trivial random variable concentrated around zero.
3. The difference $I_t - \frac{1}{2}W_t^2 = -\frac{1}{2}t$ is *deterministic*. Remarkable: the randomness of the classical antiderivative $\frac{1}{2}W_t^2$ and the randomness of the Itô integral I_t match exactly, so when subtracted they leave only the $-\frac{1}{2}t$ correction.

Equivalent form: rearranging (3.26),

$$\tfrac{1}{2}\, W_t^2 \;=\; \int_0^t W_s \, \mathrm{d}W_s \;+\; \tfrac{1}{2}\, t, \tag{3.27}$$

which is Itô's formula applied to $F(w) = \frac{1}{2}w^2$: $\mathrm{d}(\frac{1}{2}W_t^2) = W_t \, \mathrm{d}W_t + \frac{1}{2}\, \mathrm{d}t$. The $\frac{1}{2}\, \mathrm{d}t$ correction is exactly what distinguishes the stochastic integral from the classical chain-rule answer.

Itô isometry proof: A + B + C matrix decomposition

Figure 4: Itô isometry proof: A + B + C matrix decomposition

4.5.4 3.5.4 Why the left-endpoint choice matters

Had we used the right-endpoint W_{t_k} (Stratonovich-like), the telescoping identity would have given $+\frac{1}{2}\sum(\Delta W_k)^2 \to +\frac{1}{2}t$ rather than $-\frac{1}{2}t$, producing $\frac{1}{2}W_t^2$ — the classical answer. Itô's left-endpoint choice is what guarantees the integrand $W_{t_{k-1}}$ is $\mathcal{F}_{t_{k-1}}$-measurable and therefore independent of ΔW_{t_k}, giving the stochastic integral its martingale property.

The financial reading. A left-endpoint integrand is "predictable" — you pick your position before the jump. A right-endpoint integrand is "anticipative" — you pick your position after the jump, with knowledge of what just happened. Anticipative strategies violate no-arbitrage and are inadmissible in continuous-time finance. Itô's insistence on left endpoints is the mathematical enforcement of predictability, and hence of no-arbitrage.

4.6 3.6 The Itô Isometry

The L^2-isometry that underlies the definition of the stochastic integral is both a computational workhorse (it reduces any variance-of-stochastic-integral calculation to an ordinary time integral) and a structural theorem (it says the Itô integral is a continuous linear map from a weighted L^2 space of integrands into the space of martingales).

Statement. For any two adapted integrands g_s and h_s satisfying $\mathbb{E}\int_0^t g_s^2\,\mathrm{d}s < \infty$ and likewise for h,

$$\boxed{\mathbb{E}\left[\Big(\int_0^t g_s\,\mathrm{d}W_s\Big)\Big(\int_0^t h_s\,\mathrm{d}W_s\Big)\right] \;=\; \mathbb{E}\left[\int_0^t g_s h_s\,\mathrm{d}s\right]}. \tag{3.28}$$

In particular, for $g = h$, the L^2-norm of a stochastic integral equals the L^2-norm (in s and ω) of its integrand:

$$\mathbb{E}\left[\left(\int_0^t g_s\,\mathrm{d}W_s\right)^2\right] \;=\; \mathbb{E}\int_0^t g_s^2\,\mathrm{d}s. \tag{3.29}$$

Proof — the A + B + C decomposition. Set $g_t = g(t, W_t)$, $h_t = h(t, W_t)$, and define the residual

$$\mathcal{E} \;=\; \mathbb{E}\left[\Big(\int_0^t g_s\,\mathrm{d}W_s\Big)\Big(\int_0^t h_s\,\mathrm{d}W_s\Big) \;-\; \int_0^t g_s h_s\,\mathrm{d}s\right]. \tag{3.30}$$

Goal: show $\mathcal{E} = 0$. Fix a partition $\pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ and discretise both integrals as Riemann sums, so the first product becomes

$$\Big(\sum_k g_{t_{k-1}}\Delta W_{t_k}\Big)\Big(\sum_j h_{t_{j-1}}\Delta W_{t_j}\Big) \;=\; \sum_{k,j} g_{k-1}\,h_{j-1}\,\Delta W_{t_k}\,\Delta W_{t_j},$$

where $g_{k-1} = g_{t_{k-1}}$ etc. The second piece (the time integral) discretises as $\sum_k g_{k-1}\,h_{k-1}\,\Delta t_k$.

Step 1 — Split the double sum into three regions (A, B, C) according to the relationship between indices j and k:

$$\sum_{k,j} g_{k-1}\,h_{j-1}\,\Delta W_k\,\Delta W_j \;=\; \underbrace{\sum_{k<j} g_{k-1}\,h_{j-1}\,\Delta W_k\,\Delta W_j}_{A} \;+\; \underbrace{\sum_{j<k} g_{k-1}\,h_{j-1}\,\Delta W_k\,\Delta W_j}_{B} \;+\; \underbrace{\sum_{k} g_{k-1}\,h_{k-1}\,(\Delta W_k)^2}_{C}.$$

The matrix picture: the $n \times n$ grid indexed by (k, j) splits into the lower triangle A ($k < j$), the upper triangle B ($j < k$), and the diagonal C ($k = j$, highlighted). The isometry is the claim that the two triangles die in expectation and only the diagonal survives — matching the time-integral piece exactly.

Step 2 — $\mathbb{E}[A] = 0$. For any term in A we have $k < j$. Condition on $\mathcal{F}_{t_{j-1}}$: the factors g_{k-1}, h_{j-1}, and ΔW_k are all $\mathcal{F}_{t_{j-1}}$-measurable (by convention $h_{j-1} := h_{t_{j-1}}$ is $\mathcal{F}_{t_{j-1}}$-measurable because it is evaluated at the left endpoint of the j-th sub-interval; g_{k-1} and ΔW_k are $\mathcal{F}_{t_k} \subset \mathcal{F}_{t_{j-1}}$-measurable since $k < j \Rightarrow t_k \le t_{j-1}$), while ΔW_j is independent of $\mathcal{F}_{t_{j-1}}$ with mean zero. Hence

$$\mathbb{E}\big[g_{k-1}\, h_{j-1}\, \Delta W_k\, \Delta W_j\big] \;=\; \mathbb{E}\big[g_{k-1}\, h_{j-1}\, \Delta W_k\big] \cdot \underbrace{\mathbb{E}[\Delta W_j]}_{=0} \;=\; 0, \tag{3.31}$$

and summing over $k < j$ gives $\mathbb{E}[A] = 0$.

Step 3 — $\mathbb{E}[B] = 0$ by the symmetric argument with the roles of j and k swapped.

Step 4 — Only the diagonal C survives. Combining the three pieces with the time-integral subtraction,

$$\mathcal{E} \;=\; \mathbb{E}\left[\sum_k g_{k-1}\, h_{k-1}\, \big((\Delta W_k)^2 - \Delta t_k\big)\right] \;=\; \sum_k \mathbb{E}[g_{k-1}\, h_{k-1}] \cdot \underbrace{\mathbb{E}[(\Delta W_k)^2 - \Delta t_k]}_{=0} \;=\; 0, \tag{3.32}$$

using independence of $g_{k-1}h_{k-1}$ (measurable at t_{k-1}) from ΔW_k, and $\mathbb{E}[(\Delta W_k)^2] = \Delta t_k$.

Step 5 — Pass to the limit. Equation (3.32) holds for every partition π; taking $\|\pi\| \downarrow 0$ delivers the isometry (3.28). ■

Reading the isometry. The scalar identity $[W, W]_t = t$ becomes, under weighting by g_s^2, the statement that the variance of $\int g\, \mathrm{d}W$ equals the integrated squared integrand. In a quant-practitioner translation: "the variance of a stochastic P&L equals the integrated variance contribution of each sub-interval's position." Every variance-swap argument in the rest of the guide is downstream of (3.29).

A weakening of the boundedness assumption gives the result for any g with $\mathbb{E}\int_0^t g_s^2\, \mathrm{d}s < \infty$, via a standard localisation-and-approximation argument. We take this extension for granted from here on.

4.7 3.7 Brownian Moment Calculations

Before moving to Itô's lemma it pays to build a small toolkit of Brownian moment calculations. Each is a quick application of the Gaussian MGF or the independent-increments property, but they show up repeatedly downstream and are worth seeing explicitly.

Two overarching techniques drive every calculation in this section.

- **MGF differentiation.** The moment-generating function $h(a) = \mathbb{E}[e^{aZ}] = e^{a^2/2}$ of a standard normal has the property that $\mathbb{E}[Z^n] = h^{(n)}(0)$. For higher moments we differentiate h as many times as the moment order, then evaluate at $a = 0$. This reduces computation of any central Gaussian moment to a mechanical exercise in repeated differentiation of $e^{a^2/2}$.
- **Increment decomposition.** For any $s \le t$, write $W_t = W_s + (W_t - W_s)$, where the second piece is an increment *independent of* W_s. This factorises joint expectations as products of marginal expectations of independent pieces, each a centred Gaussian of known variance.

4.7.1 3.7.1 Variance of W_t^2

Starting from $\mathrm{Var}[W_t^2] = \mathbb{E}[W_t^4] - (\mathbb{E}[W_t^2])^2$ and using $W_t \overset{d}{=} \sqrt{t}\, Z$ with $Z \sim \mathcal{N}(0, 1)$,

$$\mathrm{Var}[W_t^2] \;=\; t^2\, (\mathbb{E}[Z^4] - 1). \tag{3.33}$$

The Gaussian MGF $h(a) = e^{a^2/2}$ has derivatives $h'(a) = ae^{a^2/2}$, $h''(a) = (1+a^2)e^{a^2/2}$, $h'''(a) = (a^3+3a)e^{a^2/2}$, $h^{(4)}(a) = (a^4+6a^2+3)e^{a^2/2}$. Evaluating at $a = 0$: $\mathbb{E}[Z^2] = 1$, $\mathbb{E}[Z^4] = 3$. The odd moments vanish by symmetry; the even moments grow as $(2k-1)!! = 1, 3, 15, 105, \ldots$. Substituting,

$$\boxed{\mathrm{Var}[W_t^2] \;=\; 2\,t^2}. \tag{3.34}$$

This is the Brownian analogue of $\mathrm{Var}(X^2) = 2\sigma^4$ for centred Gaussians with variance $\sigma^2 = t$, and it underlies the QV concentration argument of §3.4 (the variance of the QV sum is $\sum 2(\Delta t_k)^2 \le 2\|\pi\|\,t$).

4.7.2 3.7.2 Covariance $\mathbb{E}[W_t W_s]$

For $0 \le s \le t$, write $W_t = (W_t - W_s) + W_s$. Then

$$\mathbb{E}[W_t\,W_s] \;=\; \mathbb{E}[(W_t - W_s)\,W_s] \;+\; \mathbb{E}[W_s^2] \;=\; 0 \;+\; s,$$

because W_s and the forward increment $W_t - W_s$ are independent mean-zero. Hence

$$\boxed{\mathbb{E}[W_t\,W_s] \;=\; \min(s,t)}. \tag{3.35}$$

The correlation is $\mathrm{Corr}(W_t, W_s) = \sqrt{s/t}$: nearby times are highly correlated, far-apart times nearly independent. This is also the kernel used when simulating BM on a grid — for $s < t$ the joint law (W_s, W_t) is Gaussian with covariance matrix $\begin{pmatrix} s & s \\ s & t \end{pmatrix}$, whose Cholesky factorisation gives the practical recipe "draw Z_1, Z_2 i.i.d. standard normal; set $W_s = \sqrt{s}Z_1$, $W_t = \sqrt{s}Z_1 + \sqrt{t-s}Z_2$."

4.7.3 3.7.3 The exponential martingale

A recurring character in this guide. Claim:

$$\boxed{\mathbb{E}[X_t] \;=\; 1 \quad \text{for every } t \ge 0}, \qquad X_t = e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}. \tag{3.36}$$

Proof by MGF:

$$\mathbb{E}[X_t] \;=\; e^{-\frac{1}{2}\sigma^2 t}\,\mathbb{E}[e^{\sigma W_t}] \;=\; e^{-\frac{1}{2}\sigma^2 t}\,e^{\frac{1}{2}\sigma^2 t} \;=\; 1,$$

using $W_t \sim \mathcal{N}(0,t)$ and the Gaussian MGF. The deterministic drift $-\frac{1}{2}\sigma^2 t$ is exactly what is needed to offset the Jensen bump created by exponentiating a centred Gaussian.

Let us dwell on this cancellation because it is worth a thousand Black-Scholes formulas. The random variable σW_t is Gaussian with mean 0 and variance $\sigma^2 t$. The exponential is strictly convex, so Jensen's inequality says $\mathbb{E}[e^{\sigma W_t}] \ge e^0 = 1$, with strict inequality whenever $\sigma^2 t > 0$. The Jensen bump is quantified by the Gaussian MGF: $\mathbb{E}[e^{\sigma W_t}] = e^{\sigma^2 t/2}$. To kill this bump, subtract $\frac{1}{2}\sigma^2 t$ from the exponent upfront. That is precisely the origin of the $-\frac{1}{2}\sigma^2$ drift in the log-normal solution of GBM: the risk-neutral log-return is $\ln(S_t/S_0) = (r - \frac{1}{2}\sigma^2)t + \sigma W_t$, not $rt + \sigma W_t$, because the bare "rt" drift would allow S_t/S_0 to grow in expectation and violate the $\mathbb{Q}$-martingale property of the discounted stock. Every long-gamma trader knows this correction intuitively: the convexity of log-normal dynamics gifts you $\frac{1}{2}\sigma^2$ per unit time that risk-neutral pricing has to claw back.

More generally, for any real exponent c,

$$\mathbb{E}[X_t^c] \;=\; e^{-\frac{1}{2}c\sigma^2 t} \cdot e^{\frac{1}{2}c^2\sigma^2 t} \;=\; e^{\frac{1}{2}c(c-1)\sigma^2 t}. \tag{3.37}$$

For $c = 0$ and $c = 1$ the right-hand side equals 1 — these are the two "zero Jensen bump" cases. For $c = 2$ it gives $\mathbb{E}[X_t^2] = e^{\sigma^2 t}$, from which $\mathrm{Var}(X_t) = e^{\sigma^2 t} - 1$ — the well-known log-normal variance. A stronger statement, proved in §3.9 below, is that X_t satisfies the SDE $\mathrm{d}X_t = \sigma X_t\,\mathrm{d}W_t$ and is therefore a martingale.

4.7.4 3.7.4 Two Itô-integral variances via isometry

As a quick application of (3.29), let us compute the variance of two Itô integrals that we will meet repeatedly.

First, $I_t = \int_0^t W_s \, \mathrm{d}W_s$. By Itô isometry,

$$\mathrm{Var}[I_t] \;=\; \mathbb{E}\left[\int_0^t W_s^2 \, \mathrm{d}s\right] \;=\; \int_0^t s \, \mathrm{d}s \;=\; \tfrac{1}{2}t^2. \tag{3.38}$$

A sanity check comes from the closed-form $I_t = \frac{1}{2}(W_t^2 - t)$: then $\mathrm{Var}[I_t] = \frac{1}{4}\mathbb{E}[(W_t^2 - t)^2] = \frac{1}{4}(3t^2 - 2t^2 + t^2) = \frac{1}{2}t^2$. Both routes agree.

Second, $J_t = \int_0^t e^{W_s} \, \mathrm{d}W_s$. The integrand is not Gaussian, so J_t is not Gaussian either, but its variance still follows from (3.29):

$$\mathrm{Var}[J_t] \;=\; \int_0^t \mathbb{E}[e^{2W_s}] \, \mathrm{d}s \;=\; \int_0^t e^{2s} \, \mathrm{d}s \;=\; \tfrac{1}{2}(e^{2t} - 1), \tag{3.39}$$

using $\mathbb{E}[e^{\alpha W_s}] = e^{\alpha^2 s/2}$. More generally, when the integrand is $e^{\alpha W_s}$, the variance of the Itô integral is $(e^{\alpha^2 t} - 1)/\alpha^2$. This is the template for every "expectation of an exponential Brownian function" in the guide, and we will reuse it without comment.

4.8 3.8 Itô's Lemma for Brownian Motion

Itô's lemma is the single most important tool in this chapter, and perhaps in this guide. It plays the role that the chain rule plays in ordinary calculus: every derivation of an SDE, every passage from one stochastic process to another, every pricing PDE is at bottom an application of Itô. Mastering it reduces stochastic calculus from a foreign language to a small variation on calculus you already know.

The spirit of Itô is "Taylor expand *one more term* than you normally would, then keep only the first-order and second-order-in-dW contributions, letting everything else vanish." That extra term, the second-order one, is the Itô correction.

4.8.1 3.8.1 Setup

Fix a smooth function $f(t, w)$ with $f \in C^{1,2}$ (once-differentiable in t, twice-differentiable in w, with continuous derivatives), and define the process

$$X_t \;=\; f(t, W_t). \tag{3.40}$$

We need one derivative in t because the Taylor expansion in time only goes to first order; dt itself is the small parameter in the time direction, and its square is negligible. We need *two* derivatives in w because the second-order term in the w-Taylor expansion is the one that survives in the limit, contributing $\frac{1}{2}\partial_{ww} f \, \mathrm{d}t$ via $(\mathrm{d}W)^2 = \mathrm{d}t$. Differentiability up to order $(1, 2)$ is therefore the minimum smoothness requirement for Itô's lemma to make pointwise sense.

A textbook example. Take $f(t, w) = w^2 + e^{-t}\sin(w)$. Then $\partial_t f = -e^{-t}\sin(w)$, $\partial_w f = 2w + e^{-t}\cos(w)$, and $\partial_{ww} f = 2 - e^{-t}\sin(w)$. All three are continuous, so $f \in C^{1,2}$ and Itô applies. A non-example: $f(t, w) = |w|$, which is only continuous at $w = 0$, not differentiable twice. For such functions one needs a more delicate tool (Tanaka's formula with local time) which we do not discuss here — but the lesson is that *kink* payoffs such as $\max(S - K, 0)$ do not satisfy the $C^{1,2}$ hypothesis literally, and their Itô-style analyses carry boundary-delta terms that do not appear in the smooth case.

4.8.2 3.8.2 Heuristic derivation via Taylor expansion

Start with the generic Taylor expansion of $f(t+\Delta t,\, W_{t+\Delta t})$ around (t, W_t):

$$\begin{aligned} f(t+\Delta t,\, W_{t+\Delta t}) \;-\; f(t, W_t) \;&= \partial_t f\,\Delta t \\ &+ \partial_w f\,\Delta W \\ &+ \tfrac{1}{2}\,\partial_{tt} f\,(\Delta t)^2 \\ &+ \partial_{tw} f\,\Delta t\,\Delta W \\ &+ \tfrac{1}{2}\,\partial_{ww} f\,(\Delta W)^2 \\ &+ \text{higher-order terms,} \end{aligned}$$

with $\Delta W = W_{t+\Delta t} - W_t$. Which terms are $O(\Delta t)$ (the "dominant" scale in the limit)?

Term	Magnitude scaling	Keep or drop?
$\partial_t f\,\Delta t$	Δt	Keep (deterministic drift)
$\partial_w f\,\Delta W$	$\sqrt{\Delta t}$	Keep (stochastic term)
$\frac{1}{2}\partial_{ww} f\,(\Delta W)^2$	Δt	Keep (Itô correction)
$\frac{1}{2}\partial_{tt} f\,(\Delta t)^2$	$(\Delta t)^2$	Drop
$\partial_{tw} f\,\Delta t\,\Delta W$	$(\Delta t)^{3/2}$	Drop
higher-order	$\leq (\Delta t)^{3/2}$	Drop

In ordinary calculus the second-order term would be $O((\Delta t)^2)$ and discarded. But for Brownian motion $(\Delta W)^2 \approx \Delta t$ by (3.20) — the second-order term is the *same order* as Δt, so it must be kept. Cross terms $\Delta t \cdot \Delta W$ and $(\Delta t)^2$ are of strictly higher order and vanish.

Taking the infinitesimal limit yields

$$\boxed{\mathrm{d}X_t \;=\; \left[\partial_t f(t, W_t) \;+\; \tfrac{1}{2}\,\partial_{ww} f(t, W_t)\right]\mathrm{d}t \;+\; \partial_w f(t, W_t)\,\mathrm{d}W_t}. \tag{3.41}$$

In integrated form:

$$X_t - X_0 \;=\; \int_0^t \left[\partial_s f + \tfrac{1}{2}\partial_{ww} f\right]\mathrm{d}s \;+\; \int_0^t \partial_w f\,\mathrm{d}W_s. \tag{3.42}$$

The first integral is an ordinary Riemann integral; the second is a stochastic integral of the type defined in (3.22).

Reading (3.41) as a decomposition. The change in $X_t = f(t, W_t)$ has two independent measurable components:

- A Lebesgue integral of a deterministic integrand evaluated on a stochastic path — this part has bounded variation and zero QV.
- An Itô integral against $\mathrm{d}W$ — a martingale that carries all of the QV of X.

Splitting stochastic processes into "bounded-variation drift + martingale noise" is the *Doob–Meyer decomposition*, and for Itô processes it is simply the two pieces of (3.42). Every price process in later chapters is an Itô process, and every derivation ultimately rearranges these two pieces to solve for unknowns.

An important meta-principle. When you apply Itô to a function $f(t, W_t)$ whose expectation you care about, the drift part survives under $\mathbb{E}[\,\cdot\,]$ and the martingale part dies: $\mathbb{E}[f(t, W_t)] - f(0,0) = \int_0^t \mathbb{E}[\partial_s f + \frac{1}{2}\partial_{ww} f]\,\mathrm{d}s$. This is the fastest way to compute expectations — write the Itô differential, take the expectation of the drift, integrate — and it is the core engine of the Feynman-Kac derivations in Chapter 4.

Itô vs classical chain rule on $f(W_t) = W_t^2$ *Monte-Carlo* $\mathbb{E}[W_t^2]$ *tracks the Itô prediction* t *(green dashed), not the classical chain rule's* 0 *(red dotted). The gap is the Itô correction* $\frac{1}{2}f''\,\mathrm{d}t = \mathrm{d}t$ *integrated from* 0 *to* t *— precisely the drift contribution that ordinary calculus discards.*

4.8.3 3.8.3 Reading the formula — options-language translation

For $X_t = f(t, W_t)$ interpreted as an option price with W_t being the underlying:

- $\partial_t f$ is **theta** — the passage-of-time sensitivity.
- $\partial_w f$ is **delta** — the underlying sensitivity.
- $\partial_{ww} f$ is **gamma** — the curvature of price versus underlying.

Itô's lemma then reads: "the P&L on a long option = theta·dt + delta·dW + $\frac{1}{2}$gamma·dt (from QV of the underlying)". The $\frac{1}{2}$ gamma dt is precisely the *gamma-theta trade-off* that every options trader lives in daily: a long-gamma book bleeds theta and earns gamma P&L; a short-gamma book collects theta premium and pays for adverse gamma. Itô's lemma is the quantitative expression of this trade-off.

An intuition for the $\frac{1}{2}$ in the Itô correction. The $(\mathrm{d}W)^2 = \mathrm{d}t$ rule supplies a unit of QV per unit time. The Taylor expansion of f in w carries a $\frac{1}{2} f''$ coefficient on the squared term (Taylor's second-order term). When we substitute $(\mathrm{d}W)^2 \to \mathrm{d}t$, the $\frac{1}{2}$ stays, and we land on $\frac{1}{2} f''(w)\,\mathrm{d}t$. It is not a normalisation mystery — it is just the Taylor coefficient times the QV rate of BM.

4.8.4 3.8.4 Three worked Itô-lemma examples

Example 1 — $\int W\,\mathrm{d}W$ via Itô.

Goal: compute $\int_0^t W_s\,\mathrm{d}W_s$ using Itô's lemma rather than the telescoping identity of §3.5.

Choose $f(t, w) = \frac{1}{2}w^2$, so $\partial_t f = 0$, $\partial_w f = w$, $\partial_{ww} f = 1$. Plugging into (3.41):

$$\mathrm{d}(\tfrac{1}{2}W_t^2) \;=\; \tfrac{1}{2}\,\mathrm{d}t \;+\; W_t\,\mathrm{d}W_t.$$

Integrating from 0 to t with $W_0 = 0$ gives $\frac{1}{2}W_t^2 = \frac{1}{2}t + \int_0^t W_s\,\mathrm{d}W_s$, so

$$\int_0^t W_s\,\mathrm{d}W_s \;=\; \tfrac{1}{2}W_t^2 - \tfrac{1}{2}t, \tag{3.43}$$

in agreement with (3.26). Four lines instead of the full telescoping proof of §3.5 — Itô's lemma is massively more efficient.

Example 2 — $\int W^2\,\mathrm{d}W$.

Choose $f(t, w) = \frac{1}{3}w^3$, so $\partial_t f = 0$, $\partial_w f = w^2$, $\partial_{ww} f = 2w$. Plugging into (3.41):

$$\mathrm{d}(\tfrac{1}{3}W_t^3) \;=\; W_t\,\mathrm{d}t \;+\; W_t^2\,\mathrm{d}W_t.$$

Integrating and solving for the stochastic integral:

$$\boxed{\int_0^t W_s^2\,\mathrm{d}W_s \;=\; \tfrac{1}{3}W_t^3 \;-\; \int_0^t W_s\,\mathrm{d}s}. \tag{3.44}$$

The remaining $\int_0^t W_s\,\mathrm{d}s$ is an ordinary pathwise Riemann integral, not a stochastic one. Itô has converted the stochastic integral into (algebraic term) + (pathwise integral). The pathwise piece cannot be eliminated algebraically — it depends on the *entire path* $(W_s)_{s\in[0,t]}$, not just the endpoint W_t. Its distribution is Gaussian with $\mathrm{Var}\int_0^t W_s\,\mathrm{d}s = \int_0^t\int_0^t \min(u, v)\,\mathrm{d}u\,\mathrm{d}v = \frac{1}{3}t^3$.

Example 3 — the exponential martingale.

Goal: given $X_t = e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}$, find the SDE it satisfies.

Choose $f(t, w) = e^{-\frac{1}{2}\sigma^2 t + \sigma w}$. Each partial derivative is proportional to f itself (a feature of the exponential): $\partial_t f = -\frac{1}{2}\sigma^2 f$, $\partial_w f = \sigma f$, $\partial_{ww} f = \sigma^2 f$. Plugging into (3.41):

$$\mathrm{d}X_t \;=\; [-\tfrac{1}{2}\sigma^2 X_t + \tfrac{1}{2}\sigma^2 X_t]\,\mathrm{d}t + \sigma X_t\,\mathrm{d}W_t \;=\; \sigma X_t\,\mathrm{d}W_t.$$

The drift bracket cancels exactly — the deterministic $-\frac{1}{2}\sigma^2$ in the exponent is designed to kill the Itô correction $\frac{1}{2}\sigma^2$. Hence

$$\boxed{\mathrm{d}X_t \;=\; \sigma\, X_t\,\mathrm{d}W_t}. \tag{3.45}$$

X_t satisfies a driftless geometric SDE and is therefore a martingale. This is the rigorous proof of the $\mathbb{E}[X_t] = 1$ claim in (3.36).

The exponential martingale (3.45) is a prototype of a larger class. The Doléans-Dade exponential $\mathcal{E}(M)_t = \exp(M_t - \frac{1}{2}\langle M\rangle_t)$ of any continuous martingale M is similarly designed: the $-\frac{1}{2}\langle M\rangle_t$ is exactly the drift shift needed to turn e^{M_t} into a martingale. For $M_t = \sigma W_t$, $\langle M\rangle_t = \sigma^2 t$, recovering (3.45). The Doléans-Dade exponential reappears in Chapter 5 as the Radon-Nikodym derivative for Brownian-motion measure changes.

Doléans-Dade exponential trajectories with expected-value envelope *Six sample paths of* $Z_t = \exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$ *with* $\sigma = 0.9$. *Individual trajectories span orders of magnitude — some rocket upward, others drift toward zero — yet* $\mathbb{E}[Z_t] = 1$ *for every* t. *The* $-\frac{1}{2}\sigma^2 t$ *drift is precisely the Itô correction that keeps the process a martingale despite its wildly dispersed paths.*

4.9 3.9 Itô's Lemma for General Itô Processes

The lemma in §3.8 handled $X_t = f(t, W_t)$ where the underlying process was BM itself. In practice one usually wants to transform a diffusion with non-trivial drift and diffusion coefficients — a GBM, an OU process, a general Itô process. The extension is mechanical and is called **Itô's Lemma III** in the chapter's internal numbering.

4.9.1 3.9.1 Setup

Let X_t be an Itô process satisfying

$$\mathrm{d}X_t \;=\; \mu(t, X_t)\,\mathrm{d}t \;+\; \sigma(t, X_t)\,\mathrm{d}W_t, \tag{3.46}$$

for adapted drift μ and diffusion σ (possibly functions of t and X_t). Let $g \in C^{1,2}$ and define $Y_t = g(t, X_t)$. Then

$$\boxed{\mathrm{d}Y_t \;=\; \alpha(t, X_t)\,\mathrm{d}t \;+\; \beta(t, X_t)\,\mathrm{d}W_t},$$

where

$$\begin{aligned} \alpha(t,x) \;&=\; \partial_t g(t,x) \;+\; \mu(t,x)\,\partial_x g(t,x) \;+\; \tfrac{1}{2}\,\sigma^2(t,x)\,\partial_{xx} g(t,x), \\ \beta(t,x) \;&=\; \sigma(t,x)\,\partial_x g(t,x). \end{aligned} \tag{3.47}$$

Equivalently, in the compact "derivative-free" form,

$$\mathrm{d}g(t, X_t) \;=\; \partial_t g\,\mathrm{d}t \;+\; \partial_x g\,\mathrm{d}X_t \;+\; \tfrac{1}{2}\,\partial_{xx} g\,(\mathrm{d}X_t)^2, \tag{3.48}$$

with the Itô multiplication rules (3.20), implying $(\mathrm{d}X_t)^2 = \sigma^2(t, X_t)\,\mathrm{d}t$. The derivation is a simple extension of the BM case: use $(\mathrm{d}X_t)^2 = \sigma^2\,\mathrm{d}t$ in the Taylor expansion and keep first-order terms. The result is the BM Itô formula with two substitutions: the drift of X becomes μ (instead of zero), and the diffusion of X becomes σ (instead of one).

Consistency with the BM case. Setting $\mu = 0$ and $\sigma = 1$ in (3.47) recovers (3.41) exactly — so Lemma III is a genuine generalisation, not a rewrite. Setting additionally $\partial_t g = 0$ recovers the time-independent version. The three levels of generality — $f(W_t)$, $f(t, W_t)$, $g(t, X_t)$ — form a hierarchy of increasing scope; everything in the later chapters uses the most general form (3.47).

4.9.2 3.9.2 Cross-variation and multi-factor Itô

If X_t and Y_t are two Itô processes driven by the same or correlated Brownian motions, their cross-variation $\mathrm{d}X_t\,\mathrm{d}Y_t$ is computed by multiplying out the SDEs and applying the Itô rules. For independent BMs W^1, W^2, the rule is $\mathrm{d}W^1\,\mathrm{d}W^2 = 0$; for correlated BMs with instantaneous correlation ρ, $\mathrm{d}W^1\,\mathrm{d}W^2 = \rho\,\mathrm{d}t$. This is how multi-factor models handle cross-terms in stochastic integrals — each pair of driving noises contributes its own cross-variation rate.

Itô's lemma for functions of two Itô processes is then

$$\mathrm{d}f(t, X_t, Y_t) = \partial_t f\,\mathrm{d}t + \partial_x f\,\mathrm{d}X_t + \partial_y f\,\mathrm{d}Y_t + \tfrac{1}{2}\partial_{xx} f\,(\mathrm{d}X)^2 + \partial_{xy} f\,\mathrm{d}X\,\mathrm{d}Y + \tfrac{1}{2}\partial_{yy} f\,(\mathrm{d}Y)^2. \tag{3.49}$$

Every instance of "two risk factors interact" in derivatives pricing — quanto options, dual-currency bonds, baskets, stochastic-vol Heston — is an application of (3.49). We will meet these explicitly in Chapter 8 (Margrabe exchange options) and Chapter 14 (Heston).

4.10 3.10 SDE Catalogue

With Itô's lemma in hand we now solve the three canonical SDEs that every later chapter will invoke: Geometric Brownian Motion (GBM), the Ornstein-Uhlenbeck / Vasicek process, and the generic constant-coefficient arithmetic SDE. Each is solved as a worked exercise in Lemma III.

4.10.1 3.10.1 Geometric Brownian Motion (GBM)

The GBM SDE is

$$\mathrm{d}S_t \;=\; \mu\, S_t\,\mathrm{d}t \;+\; \sigma\, S_t\,\mathrm{d}W_t, \qquad S_0 \text{ given}, \tag{3.50}$$

with constants μ and $\sigma > 0$. This is the dynamics assumed for the underlying asset throughout the Black-Scholes framework (with μ replaced by r under the risk-neutral measure in Chapter 6).

Solution by Itô on $\ln S$. Apply Lemma III with $g(t,s) = \ln s$ and $(\mu(t,s), \sigma(t,s)) = (\mu s, \sigma s)$. Partial derivatives: $\partial_t g = 0$, $\partial_s g = 1/s$, $\partial_{ss} g = -1/s^2$. Plugging into (3.47):

$$\mathrm{d}(\ln S_t) \;=\; \left[0 \;+\; \frac{\mu S_t}{S_t} \;+\; \tfrac{1}{2}\,\sigma^2 S_t^2 \cdot \left(-\frac{1}{S_t^2}\right)\right]\mathrm{d}t \;+\; \frac{\sigma S_t}{S_t}\,\mathrm{d}W_t \;=\; \left(\mu - \tfrac{1}{2}\sigma^2\right)\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t. \tag{3.51}$$

The right-hand side is a constant-drift, constant-diffusion SDE — an arithmetic BM with drift $\mu - \frac{1}{2}\sigma^2$ and volatility σ. Integrating:

$$\ln S_t - \ln S_0 \;=\; (\mu - \tfrac{1}{2}\sigma^2)\,t \;+\; \sigma W_t,$$

and therefore

$$\boxed{S_t \;=\; S_0\, e^{(\mu - \frac{1}{2}\sigma^2)\,t \,+\, \sigma W_t} \qquad \text{(GBM closed form)}}. \tag{3.52}$$

This is the explicit closed-form solution of the GBM SDE, and it is how every Monte Carlo simulator generates stock-price paths under the Black-Scholes model: draw $W_t \sim \mathcal{N}(0,t)$, plug into (3.52), get S_t. There is no time-discretisation error in this exact scheme; the only error is the Gaussian approximation of W_t, which is *exact* on any finite time horizon.

Expected value. Using $\mathbb{E}[e^{\sigma W_t}] = e^{\sigma^2 t/2}$,

$$\mathbb{E}[S_t] \;=\; S_0\, e^{(\mu - \frac{1}{2}\sigma^2)t} \cdot e^{\frac{1}{2}\sigma^2 t} \;=\; S_0\, e^{\mu t}. \tag{3.53}$$

The expected continuously-compounded return is μ, *irrespective of* σ. Volatility leaves the expected forward value unchanged but does shift the median and all other quantiles of the log-normal distribution. The risk-neutral valuation of derivatives exploits this by ensuring $\mu = r$ (the risk-free rate) under $\mathbb{Q}$, so $\mathbb{E}^{\mathbb{Q}}[S_t] = S_0\, e^{rt}$ — the cost of carrying the stock equals the riskless growth rate.

The $-\frac{1}{2}\sigma^2$ correction is not optional. A common mistake is to apply separation of variables to (3.50): write $\mathrm{d}S/S = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W$ and integrate both sides to get $\ln S_t = \ln S_0 + \mu t + \sigma W_t$, i.e. $S_t = S_0 e^{\mu t + \sigma W_t}$. This is *wrong*. The chain rule $\mathrm{d}(\ln S) = \mathrm{d}S/S$ does not hold for SDEs; Itô's lemma tells us it is $\mathrm{d}(\ln S) = \mathrm{d}S/S - \frac{1}{2}(\mathrm{d}S)^2/S^2 = \mathrm{d}S/S - \frac{1}{2}\sigma^2\,\mathrm{d}t$. The missing $-\frac{1}{2}\sigma^2\,\mathrm{d}t$ piece is the Itô correction, and forgetting it produces $\mathbb{E}[S_t] = S_0 e^{\mu t + \sigma^2 t/2}$ — which violates the martingale property of the discounted stock (when $\mu = r$). The $-\frac{1}{2}\sigma^2$ in (3.52) is precisely what restores the martingale property.

4.10.2 3.10.2 Ornstein-Uhlenbeck / Vasicek

The OU SDE — also known as the Vasicek short-rate equation when applied to interest rates — is

$$\mathrm{d}r_t \;=\; \kappa(\theta - r_t)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t, \qquad r_0 \text{ given}, \tag{3.54}$$

with constants $\kappa > 0$ (mean-reversion speed), θ (long-run mean), and σ (volatility). The drift $\kappa(\theta - r_t)$ pulls r_t toward θ at rate κ; the diffusion is constant and Gaussian. This is the simplest mean-reverting SDE and serves as the workhorse short-rate model in Chapter 12.

Solution by integrating factor. The drift is linear in r_t, which suggests the integrating-factor technique familiar from linear first-order ODEs. Multiply both sides of (3.54) by $e^{\kappa t}$ and recognise the left-hand side as an exact differential:

$$\mathrm{d}(e^{\kappa t}\, r_t) \;=\; e^{\kappa t}\,\mathrm{d}r_t \;+\; \kappa\, e^{\kappa t}\, r_t\,\mathrm{d}t \;=\; e^{\kappa t}(\kappa\theta\,\mathrm{d}t + \sigma\,\mathrm{d}W_t),$$

where the second equality substitutes (3.54) after the cancellation $-\kappa r_t + \kappa r_t = 0$. Integrating from 0 to t:

$$e^{\kappa t}\, r_t - r_0 \;=\; \kappa\theta \int_0^t e^{\kappa u}\,\mathrm{d}u \;+\; \sigma \int_0^t e^{\kappa u}\,\mathrm{d}W_u \;=\; \theta(e^{\kappa t} - 1) \;+\; \sigma \int_0^t e^{\kappa u}\,\mathrm{d}W_u.$$

Multiplying through by $e^{-\kappa t}$:

$$\boxed{r_t \;=\; r_0 e^{-\kappa t} \;+\; \theta(1 - e^{-\kappa t}) \;+\; \sigma \int_0^t e^{-\kappa(t-u)}\,\mathrm{d}W_u}. \tag{3.55}$$

The integrating-factor step can also be verified as an Itô-lemma computation: apply Lemma III with $g(t, r) = e^{\kappa t}\, r$, compute $\partial_t g = \kappa e^{\kappa t}\, r$, $\partial_r g = e^{\kappa t}$, $\partial_{rr} g = 0$, and substitute into (3.47). The result $\mathrm{d}(e^{\kappa t} r_t) = e^{\kappa t}(\kappa\theta\,\mathrm{d}t + \sigma\,\mathrm{d}W_t)$ matches the integrating-factor derivation — a clean example of Lemma III in action.

Distribution of r_t. The stochastic integral in (3.55) is a linear combination of Gaussian increments, so r_t is itself Gaussian. Its mean is the deterministic relaxation

$$\mathbb{E}[r_t] \;=\; r_0 e^{-\kappa t} \;+\; \theta(1 - e^{-\kappa t}), \tag{3.56}$$

which starts at r_0 and relaxes exponentially to θ as $t \to \infty$. The variance follows from the Itô isometry (3.29):

$$\mathrm{Var}[r_t] \;=\; \sigma^2 \int_0^t e^{-2\kappa(t-u)}\,\mathrm{d}u \;=\; \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})\,. \tag{3.57}$$

Two limits worth registering. As $t \to \infty$, $\mathrm{Var}[r_t] \to \sigma^2/(2\kappa)$: mean-reversion caps the long-run variance at a finite value, unlike plain Brownian motion whose variance grows linearly forever. As $t \downarrow 0$, $\mathrm{Var}[r_t] \to \sigma^2 t$: over short horizons the OU process is indistinguishable from a Brownian motion, because the mean-reversion drift has had no

time to act. The density picture — the mean $r_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t})$ relaxing from r_0 toward θ, flanked by $\pm 2\,\mathrm{sd}$ bands that widen quickly at first and then plateau — captures the qualitative picture that every mean-reverting model shares: transient-widening, long-run stationarity.

The integrated rate $\int_0^T r_u \,\mathrm{d}u$, needed for bond pricing in Chapter 12, is also Gaussian — as a linear functional of the Gaussian process r_t — and its mean and variance are computed by the same Fubini / Itô-isometry combination. We defer the explicit formula to Chapter 12 where it is put to work.

OU vs GBM sample paths *Side-by-side comparison of the two workhorse SDEs.* ***Left:*** *GBM paths trend exponentially with multiplicative noise; the mean curve $S_0 e^{\mu t}$ grows without bound.* ***Right:*** *OU / Vasicek paths wobble around the long-run mean θ; variance saturates at $\sigma^2/(2\kappa)$ so the ensemble never escapes. Mean-reversion is visually unmistakable.*

4.10.3 3.10.3 Constant-coefficient arithmetic BM

The simplest possible SDE — no state-dependence in either drift or diffusion — is

$$\mathrm{d}X_t \;=\; a\,\mathrm{d}t \;+\; b\,\mathrm{d}W_t, \qquad X_0 \text{ given}, \tag{3.58}$$

with constants a and b. Because the coefficients do not depend on X_t, direct integration works without any transformation:

$$X_t \;=\; X_0 \;+\; a\,t \;+\; b\,W_t. \tag{3.59}$$

The solution is Gaussian with mean $X_0 + at$ and variance $b^2 t$. This is the "trivially solvable" SDE, and its chief uses are (i) as the reference process after Itô-transforming a more complicated SDE to constant coefficients (e.g. the log-GBM reduction (3.51)), and (ii) as the Bachelier model of asset prices — rarely used for equities but standard in certain rates markets (notably swaption pricing with normal volatilities, which we revisit in Chapter 15).

One subtle point worth registering: arithmetic BM allows X_t to go *negative* (since a Gaussian has full support on $\mathbb{R}$), whereas GBM keeps $S_t > 0$ almost surely. The choice between the two models is therefore often dictated by whether negative values are economically reasonable — equities must be non-negative, so GBM; interest rates can (occasionally, post-2014) be negative, so arithmetic-style models are once again respectable in that domain.

4.10.4 3.10.4 Summary of solved SDEs

SDE	Solution	Distribution of X_t
$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$ (GBM)	$S_t = S_0\, e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$	Log-normal
$\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ (OU)	$r_t = r_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t}) + \sigma \int_0^t e^{-\kappa(t-u)}\,\mathrm{d}W_u$	Gaussian
$\mathrm{d}X_t = a\,\mathrm{d}t + b\,\mathrm{d}W_t$ (arithmetic BM)	$X_t = X_0 + at + bW_t$	Gaussian

Every SDE encountered in the rest of the guide either is one of these three or reduces to one by an Itô transformation. The Heston stochastic-volatility model of Chapter 14 is one partial exception (the variance process is CIR, which requires a Bessel-process argument we develop there), but even it uses OU-like integrating-factor ideas on the variance term.

4.11 3.11 Worked Stochastic Integrals

We now assemble a small library of stochastic integrals that later chapters invoke. The operational recipe for computing any $\int_0^t g(s, W_s)\,\mathrm{d}W_s$ is always the same:

1. Choose a candidate $f(t, w)$ such that $\partial_w f(t, w) = g(t, w)$.
2. Compute the Itô differential $\mathrm{d}f(t, W_t)$ using (3.41).

3. Integrate from 0 to t and solve for the stochastic integral:

$$\int_0^t g(s, W_s)\,\mathrm{d}W_s = f(t, W_t) - f(0,0) - \int_0^t [\partial_s f + \tfrac{1}{2}\partial_{ww} f]\,\mathrm{d}s.$$

The right-hand side is pathwise — a mix of terminal-value evaluations and ordinary Riemann integrals. The stochastic integral has been *reduced* to these ingredients. That is the whole trick.

4.11.1 3.11.1 A stochastic integration-by-parts: $\int_0^t s\,W_s\,\mathrm{d}W_s$

Consider the product $X_t = t\,W_t^2$, i.e. take $f(t,w) = t\,w^2$. Partials: $\partial_t f = w^2$, $\partial_w f = 2tw$, $\partial_{ww} f = 2t$. Plugging into (3.41):

$$\mathrm{d}(t\,W_t^2) \;=\; [\,W_t^2 \;+\; t\,]\,\mathrm{d}t \;+\; 2t\,W_t\,\mathrm{d}W_t. \tag{3.60}$$

Integrating from 0 to t with $X_0 = 0$:

$$t\,W_t^2 \;=\; \int_0^t (W_s^2 + s)\,\mathrm{d}s \;+\; \int_0^t 2s\,W_s\,\mathrm{d}W_s,$$

so that

$$\boxed{\int_0^t 2\,s\,W_s\,\mathrm{d}W_s \;=\; t\,W_t^2 \;-\; \int_0^t W_s^2\,\mathrm{d}s \;-\; \tfrac{1}{2}t^2}. \tag{3.61}$$

This is a stochastic integration-by-parts identity for $s\,W_s$: the product of a deterministic function of s and a Brownian function is expressible in terms of the terminal product plus ordinary pathwise integrals. A sanity check: taking expectations on both sides, the left-hand side is 0 (martingale), and the right-hand side is $\mathbb{E}[tW_t^2] - \int_0^t s\,\mathrm{d}s - \frac{1}{2}t^2 = t\cdot t - \frac{1}{2}t^2 - \frac{1}{2}t^2 = 0$.

More generally, for any $f(t,w) = \phi(t)\,\psi(w)$ product form, Itô's lemma gives the analogue of the ordinary product rule with an Itô correction:

$$\mathrm{d}(\phi(t)\,\psi(W_t)) \;=\; \phi'(t)\,\psi(W_t)\,\mathrm{d}t \;+\; \phi(t)\,\psi'(W_t)\,\mathrm{d}W_t \;+\; \tfrac{1}{2}\,\phi(t)\,\psi''(W_t)\,\mathrm{d}t. \tag{3.62}$$

4.11.2 3.11.2 Integration-by-parts for $\int_0^t s\,e^{W_s}\,\mathrm{d}W_s$

Take $f(t,w) = t\,e^w$, chosen so that $\partial_w f = t\,e^w$ — almost the desired $s\,e^{W_s}$, but with t frozen rather than integrated. Partials: $\partial_t f = e^w$, $\partial_w f = t\,e^w$, $\partial_{ww} f = t\,e^w$. Itô:

$$\mathrm{d}(t\,e^{W_t}) \;=\; [\,e^{W_t} \;+\; \tfrac{1}{2}\,t\,e^{W_t}\,]\,\mathrm{d}t \;+\; t\,e^{W_t}\,\mathrm{d}W_t. \tag{3.63}$$

Integrating $0 \to t$ and rearranging,

$$\boxed{\int_0^t s\,e^{W_s}\,\mathrm{d}W_s \;=\; t\,e^{W_t} \;-\; \int_0^t e^{W_s}\,\mathrm{d}s \;-\; \tfrac{1}{2}\int_0^t s\,e^{W_s}\,\mathrm{d}s}. \tag{3.64}$$

The answer decomposes into a terminal-value algebraic piece and two pathwise Riemann integrals. Neither pathwise integral has a simpler closed form, but both are perfectly tractable for Monte Carlo simulation or moment calculations. The stochastic integral $\int_0^t s e^{W_s}\,\mathrm{d}W_s$ itself is a martingale with variance $\int_0^t s^2\,\mathbb{E}[e^{2W_s}]\,\mathrm{d}s = \int_0^t s^2 e^{2s}\,\mathrm{d}s$ by Itô isometry. This is the template for every "polynomial-in-s times exponential-in-W_s" integral that appears downstream.

4.11.3 3.11.3 A catalogue of Brownian computations

The following compact list summarises the moments that will be reused without re-derivation:

$$\mathbb{E}[W_t] = 0, \qquad \mathrm{Var}[W_t] = t, \qquad \mathbb{E}[W_t^4] = 3t^2,$$

$$\mathbb{E}[W_s W_t] = \min(s,t), \qquad \mathrm{Var}[W_t^2] = 2t^2,$$

$$\mathbb{E}[e^{\sigma W_t}] = e^{\frac{1}{2}\sigma^2 t}, \qquad \mathbb{E}[e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}] = 1,$$

$$\mathrm{Var}\left[\int_0^t g_s \,\mathrm{d}W_s\right] = \int_0^t \mathbb{E}[g_s^2]\,\mathrm{d}s,$$

$$\mathrm{Var}\left[\int_0^t W_s \,\mathrm{d}s\right] = \tfrac{1}{3}t^3.$$

Every one of these is either a direct MGF exercise or a one-line Itô-isometry application. None will be re-derived in later chapters.

4.12 3.12 Chapter Summary

What this chapter has established:

- **Brownian motion** is the continuum limit of a symmetric scaled random walk with $\sqrt{\Delta t}$ increments. It has Gaussian marginals, stationary and independent increments, continuous paths — properties (i)-(iv) of (3.8).
- **Path properties.** BM has infinite total variation (3.14) and finite quadratic variation $[W,W]_t = t$ (3.19), summarised by the Itô rule $(\mathrm{d}W)^2 = \mathrm{d}t$ in (3.20).
- **The Itô integral** $\int_0^t g_s \,\mathrm{d}W_s$ is defined by left-endpoint Riemann sums (3.22). The left-endpoint choice is what makes it a martingale — the mathematical encoding of no-arbitrage predictability.
- **The Itô isometry** (3.28)-(3.29) computes the variance of any stochastic integral as a deterministic time integral of the squared integrand. It generalises $[W,W]_t = t$ from the identity to arbitrary integrands.
- **Itô's lemma** (3.41) — the stochastic chain rule — carries an extra term $\frac{1}{2}\partial_{ww} f \,\mathrm{d}t$ relative to the classical chain rule, absent in ordinary calculus because smooth paths have zero QV. The general form (3.47) handles transforms of any Itô process.
- **SDE catalogue.** GBM (3.52), OU/Vasicek (3.55), and arithmetic BM (3.59) are the three canonical SDEs. Every SDE encountered later either is one of these three or reduces to one by an Itô transformation.

4.12.1 Key takeaways

1. $(\mathrm{d}W)^2 = \mathrm{d}t$ **is the source of every Itô correction.** Every time you see a $-\frac{1}{2}\sigma^2 t$, $\frac{1}{2}\,\sigma^2\,\partial_{xx}$, or "convexity adjustment" in a later chapter, it descends from this single identity.
2. **TV versus QV.** BM has infinite TV (distance) but finite QV (energy). The gap between them is the gamma-theta trade-off of options and the structural reason variance swaps replicate by log-contracts while "realised-absolute-value swaps" do not.
3. **Left endpoints matter.** The Itô integral's martingale property depends on evaluating the integrand at t_{k-1}, not t_k. This is the mathematical expression of "predictable strategies earn zero expected P&L," i.e. no-arbitrage in continuous time.
4. **Itô's lemma is the chain rule plus curvature.** Options traders already know it: long-gamma books bleed theta at the rate $\frac{1}{2}\sigma^2 S^2 \Gamma$, and this rate is exactly the Itô correction applied to the Black-Scholes price.
5. **Closed-form SDEs are scarce.** Only a handful of SDEs admit explicit solutions — GBM, OU, linear-Gaussian, and a few others. The rest are solved numerically (Euler-Maruyama, Milstein) or transformed to a solvable form by Itô's lemma (the Lamperti transform).

4.12.2 Reference formulas

- **Itô rules:** $(\mathrm{d}W)^2 = \mathrm{d}t$, $\mathrm{d}W\,\mathrm{d}t = 0$, $(\mathrm{d}t)^2 = 0$.
- **Itô's lemma for $X_t = f(t, W_t)$:**

$$\mathrm{d}X_t = \left[\partial_t f + \tfrac{1}{2}\partial_{ww} f\right]\mathrm{d}t + \partial_w f\,\mathrm{d}W_t.$$

- **Itô's lemma for $Y_t = g(t, X_t)$ with $\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$:**

$$\mathrm{d}Y_t = \left[\partial_t g + \mu\,\partial_x g + \tfrac{1}{2}\sigma^2\,\partial_{xx} g\right]\mathrm{d}t + \sigma\,\partial_x g\,\mathrm{d}W_t.$$

- **Itô isometry:**

$$\mathbb{E}\left[\left(\int_0^t g_s\,\mathrm{d}W_s\right)^2\right] = \mathbb{E}\int_0^t g_s^2\,\mathrm{d}s.$$

- **GBM solution:** $S_t = S_0\,e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$.
- **OU solution:** $r_t = r_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t}) + \sigma \int_0^t e^{-\kappa(t-u)}\,\mathrm{d}W_u$, with $\mathrm{Var}[r_t] = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$.
- **Exponential martingale:** $\mathrm{d}(e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}) = \sigma\,e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}\,\mathrm{d}W_t$, driftless.
- **Key Brownian moments:** $\mathbb{E}[W_t^2] = t$, $\mathbb{E}[W_t^4] = 3t^2$, $\mathbb{E}[W_s W_t] = \min(s, t)$, $\mathbb{E}[e^{\sigma W_t}] = e^{\sigma^2 t/2}$.

With these tools in hand we turn in Chapter 4 to the Feynman-Kac bridge between SDEs and PDEs, which will hand us the Kolmogorov backward equation and prepare the ground for the risk-neutral pricing framework of Chapters 5-8.

5 Chapter 4 — Feynman-Kac and the SDE-PDE Bridge

Chapter 3 built the stochastic-calculus toolkit — Brownian motion, Itô integrals, Itô's lemma in its three forms — and closed with a small catalogue of SDEs we can now solve. This chapter uses that toolkit for its first major payoff: the *Feynman-Kac* theorem, the bidirectional bridge that identifies a linear parabolic PDE with the conditional expectation of a diffusion-driven payoff. Once that bridge is in place, every pricing problem we meet — Black-Scholes in Chapter 6, Vasicek bond prices in Chapter 12, caplets in Chapter 15 — is a single substitution into the same template.

The ambition of this chapter is limited on purpose. We will not derive Black-Scholes yet (that happens twice in Chapter 6: once via the hedging argument and once as a Feynman-Kac corollary). We will not introduce measure changes (Chapter 5). What we *will* do is establish the PDE expectation duality in three progressively general forms — zero-drift Brownian motion, constant-coefficient drifted BM with discounting, and finally state-and-time-dependent coefficients — and we will compute three worked examples by hand so the reader sees the machinery turn.

5.1 4.1 Motivation — Why an SDE-PDE Bridge Matters

Every pricing question in this guide eventually reduces to one of two computations:

1. Solve a backward parabolic PDE on a space-time grid.
2. Average a random payoff over sample paths of a diffusion.

These two procedures look entirely different. The first is a deterministic calculation on a mesh, implemented in finite-difference or finite-element code. The second is a probabilistic calculation on simulated trajectories, implemented as Monte Carlo. They appear to require different numerical libraries, different error analyses, different parallelisation strategies, and different testing regimes.

Feynman-Kac says they are the same calculation. Given a linear parabolic PDE with terminal data, one can always write the solution as a conditional expectation over the paths of an associated diffusion. Conversely, given a conditional expectation of a smooth function of a diffusion's terminal state, the expectation solves a specific PDE. The reader gets to pick the weapon: whichever side of the bridge is easier for the problem at hand.

The *why it matters* is entirely practical. Low-dimensional problems with smooth coefficients — a European vanilla on a single asset, a caplet on a single forward rate — are almost always faster to price via PDE: a Crank-Nicolson grid with a few hundred nodes converges in milliseconds. High-dimensional problems — basket options on $N > 3$ underliers, path-dependent exotics with many state variables, multi-factor rate models — are almost always faster via Monte Carlo, whose error scales as $1/\sqrt{N_{\text{paths}}}$ independently of state dimension. American-style features (early exercise) favour PDEs because the exercise boundary embeds directly into the grid. Path-dependent features (barriers, Asians, lookbacks) favour Monte Carlo because they require tracking a path functional. Feynman-Kac is what guarantees that whichever route one chooses, the answer is the same.

There is a deeper reason the bridge is worth spending a chapter on. Every *pricing PDE* that appears in mathematical finance — Black-Scholes, Vasicek's bond-price equation, Hull-White's calibrated equation, Heston's two-factor equation, the short-rate equations of Chapter 12, the caplet equations of Chapter 15 — is a Feynman-Kac PDE for some SDE and some discount rate. Mastering the bridge in full generality (state-and-time-dependent coefficients, state-dependent discount) gives the reader a single theorem that, properly instantiated, handles *every* linear pricing problem in the guide. The instantiations differ only in what you plug in for $a(t, x)$, $b(t, x)$, $c(t, x)$, and $\varphi(x)$.

The chapter closes with a brief preview: in Chapter 6 we will apply the state-dependent Feynman-Kac theorem to geometric Brownian motion and recover the Black-Scholes PDE and the risk-neutral pricing formula in one line each. That is the end-state. This chapter lays the general theorem; Chapter 6 collects the dividend.

5.2 4.2 The Zero-Drift Feynman-Kac Theorem

We begin with the simplest possible instance of the theorem — a driftless, undiscounted Brownian motion with a terminal payoff. Every later version is a decoration of this base case, so it is worth pinning down the geometry before adding coefficients.

Let $X = (X_t)_{t\geq 0}$ denote a standard Brownian motion (as constructed in Chapter 3), and fix a horizon $T > 0$ and a payoff $\varphi : \mathbb{R} \to \mathbb{R}$ satisfying mild integrability (polynomial growth is more than enough for every concrete payoff in

this guide). Consider the backward Cauchy problem

$$\begin{cases} \partial_t f(t,x) \;+\; \frac{1}{2}\,\partial_{xx} f(t,x) \;=\; 0, \quad (t,x) \in [0,T) \times \mathbb{R}, \\ f(T,x) \;=\; \varphi(x). \end{cases} \tag{4.1}$$

Feynman-Kac asserts that this PDE is *equivalent* to the probabilistic representation

$$\boxed{f(t,x) \;=\; \mathbb{E}_{t,x}[\,\varphi(X_T)\,] \;\equiv\; \mathbb{E}[\,\varphi(X_T)\,|\,X_t = x\,]\,.} \tag{4.2}$$

The biconditional carries both directions. **PDE ⇒ expectation:** a classical solution of (4.1) admits the expectation representation (4.2). **Expectation ⇒ PDE:** conversely, the function $x \mapsto \mathbb{E}_{t,x}[\varphi(X_T)]$ automatically satisfies (4.1). In practice one direction or the other is the easier computation, and Feynman-Kac lets us freely pick.

5.2.1 4.2.1 What the PDE is

The PDE on the left of (4.1) is the (reverse) heat equation. The forward heat equation $\partial_t u = \frac{1}{2}\partial_{xx} u$ describes the diffusion of heat through a conducting medium with thermal diffusivity $\frac{1}{2}$; reversing the arrow of time (replacing t with $T-t$) gives $\partial_t f + \frac{1}{2}\partial_{xx} f = 0$, which is the Kolmogorov backward equation for Brownian motion. The payoff φ at expiry plays the role of an initial heat distribution, and the PDE smooths it backward in time from T down to t. Option prices are literally temperature distributions evolving under a thermal-diffusion physics, with the terminal payoff as initial temperature profile and "conductivity" equal to $\frac{1}{2}$ (the Brownian diffusion rate). The heat-equation picture is more than metaphor: every qualitative feature of diffusion — smoothing of discontinuities, infinite propagation speed, exponential decay of modes — translates directly into a feature of option prices, and we will use this repeatedly.

Heat-equation analogy *As t moves backward from T toward 0, the terminal payoff φ diffuses and smooths: kinks (vanilla calls) round off, step functions (digitals) bloom into Gaussian-shaped bumps. Every option price is literally a smoothed payoff, where the smoothing kernel is a Gaussian of width $\sqrt{T-t}$.*

Backward smoothing of a digital payoff *Concrete instance of* (4.2) *with $\varphi(x) = \mathbf{1}_{\{x>0\}}$: the terminal step function (red) smooths into $\Phi((x-K)/\sqrt{T-t})$ as time runs backward, interpolating between a sharp jump at expiry and a near-flat curve at long look-ahead.*

FK backward solution for a call-like payoff *The same smoothing pattern on a piecewise-linear payoff $\varphi(x) = (x)^+$. At expiry (red) the kink is sharp; rolling time back toward $t = 0$ smooths the kink into an analytic curve whose width grows like $\sqrt{T-t}$. This is the spatial shape every European call price inherits once log-space coordinates are in place.*

Heat-equation Green's-function convolution *The biconditional* (4.1)–(4.2) *in the simplest possible setting: today's price $f(0,x)$ equals the payoff φ convolved with a Gaussian heat kernel of width $\sqrt{T}$. Every Feynman-Kac price is literally a Gaussian-smoothed payoff.*

5.2.2 4.2.2 What the expectation is

The right-hand side of (4.2) is the conditional-expectation scoreboard: at time t, given that the current state is $X_t = x$, average the payoff $\varphi(X_T)$ over every Brownian trajectory that starts at (t,x) and wanders until T. Because Brownian motion is Markov — it forgets its pre-t history — the expectation only depends on the pair (t,x), never on the path that brought us to x. That Markov collapse is why f is a function of two real variables rather than a functional on the full path space.

Two readings of the biconditional are used constantly in practice:

- **PDE → expectation.** Hard PDE computations can be replaced with Monte-Carlo averages: simulate N Brownian paths from x starting at time t, evaluate $\varphi(X_T)$ on each, and take the sample mean. The error scales as $1/\sqrt{N}$ independently of state dimension. This is the foundation of Monte-Carlo pricing.
- **Expectation → PDE.** Potentially infinite-dimensional averages over paths can be replaced by finite-dimensional PDE solves on a small grid. Finite-difference methods (implicit Euler, Crank-Nicolson) exploit exactly this direction. The PDE lives on a two-dimensional (t,x) mesh; the path space is infinite-dimensional. Trading path dimension for grid dimension is often a good deal in low-dimensional problems.

Which direction to pick is problem-driven. High-dimensional state spaces (basket options on many underliers, multi-factor rate models) favour Monte Carlo because a PDE mesh becomes curse-of-dimensionality slow as soon as the state dimension exceeds about three. Low-dimensional problems with American features favour PDEs because one can embed the exercise boundary directly into the grid. Feynman-Kac is what guarantees the answer is the same either way.

5.2.3 4.2.3 Path-integral intuition — why the equivalence is not magic

Before proving the theorem it is worth seeing, at a hand-waving level, why the two sides of (4.1)–(4.2) *must* agree. Two observations are responsible.

First, the *Markov property*. Brownian motion forgets its history, so the scoreboard $f(t,x) = \mathbb{E}[\varphi(X_T)|X_t = x]$ depends only on (t,x), not on whatever happened before time t. That is why f is an ordinary function of two variables and not some path-functional.

Second, the *local dynamics*. Over an infinitesimal time-step $\mathrm{d}t$, the state changes by $\mathrm{d}X = \mathrm{d}W \sim \mathcal{N}(0, \mathrm{d}t)$. The scoreboard must obey the tower identity

$$f(t,x) \;=\; \mathbb{E}[\, f(t+\mathrm{d}t,\, x+\mathrm{d}W)\,], \tag{4.3}$$

because the expected value of the next-instant scoreboard, averaged over the single-step increment $\mathrm{d}W$, must reproduce the current scoreboard (otherwise the "average of future payoffs" description would be internally inconsistent). Taylor-expand the right-hand side of (4.3):

$$f(t+\mathrm{d}t, x+\mathrm{d}W) \;=\; f(t,x) \;+\; \partial_t f\,\mathrm{d}t \;+\; \partial_x f\,\mathrm{d}W \;+\; \tfrac{1}{2}\,\partial_{xx} f\,(\mathrm{d}W)^2 \;+\; \cdots$$

Take expectations, using $\mathbb{E}[\mathrm{d}W] = 0$ and the Itô shorthand $(\mathrm{d}W)^2 = \mathrm{d}t$ from Chapter 3:

$$\mathbb{E}[f(t+\mathrm{d}t, x+\mathrm{d}W)] \;=\; f(t,x) \;+\; \big(\partial_t f + \tfrac{1}{2}\partial_{xx} f\big)\,\mathrm{d}t.$$

Subtracting $f(t,x)$ from both sides of (4.3) and dividing by $\mathrm{d}t$ leaves

$$0 \;=\; \partial_t f + \tfrac{1}{2}\partial_{xx} f,$$

which is precisely the PDE (4.1). The terminal condition $f(T,x) = \varphi(x)$ is automatic: at expiry the scoreboard has no time left to average, so it must read the payoff directly.

So the PDE is nothing but the *local self-consistency* of the averaging scoreboard, and Feynman-Kac is the *global integrated form* of that local condition. The next section turns this sketch into a proof using martingale language.

5.3 4.3 A Martingale Derivation of the Backward PDE

The Taylor-expansion sketch of §4.2.3 is essentially correct but cuts two corners. First, it quietly assumes f is smooth enough to Taylor-expand. Second, it treats $(\mathrm{d}W)^2 = \mathrm{d}t$ as a symbolic identity rather than as a consequence of quadratic variation. This section gives the clean derivation: the scoreboard is a *martingale* by the tower property of conditional expectations, and Itô's lemma applied to a martingale forces the drift to vanish — which is the PDE.

The payoff of this slower proof is a reusable *template*. Every pricing PDE in this guide will be derived by the same three moves: (i) write the candidate price as a conditional expectation of a terminal payoff; (ii) apply Itô's lemma to the candidate to decompose its increment into drift plus martingale; (iii) demand that the drift vanish, which gives the PDE. Learn the template once here; reuse it in Chapter 6 (Black-Scholes), Chapter 12 (Vasicek bond), Chapter 15 (caplets), and anywhere else a linear parabolic PDE is needed.

5.3.1 4.3.1 Setup and the candidate

Fix a horizon T and a payoff $\varphi : \mathbb{R} \to \mathbb{R}$ with polynomial growth. Let X_t be a standard Brownian motion. Define the candidate pricing function

$$f(t,x) \;\equiv\; \mathbb{E}\big[\,\varphi(X_T)\,\big|\,X_t = x\,\big] \;=\; \mathbb{E}_{t,x}[\varphi(X_T)]. \tag{4.4}$$

The question the proof will answer is: *what PDE does f satisfy as a function of (t,x)?* The answer is $(\partial_t + \frac{1}{2}\partial_{xx})f = 0$ with terminal data $f(T,x) = \varphi(x)$, but this is not a definitional statement — it is a theorem derivable from (a) iterated expectation and (b) Itô's lemma applied to $f(t, X_t)$.

Picture the scoreboard. Imagine a grid of sample paths X_t fanning out from a fixed starting point (t,x) on the left, reaching an irregular cloud of terminal values X_T on the right at time T. The payoff $\varphi(\cdot)$ assigns a number to each terminal value; averaging those numbers over the path cloud gives $f(t,x)$. Now slide a vertical slice forward to an intermediate time $s \in (t,T)$. On that slice, paths have evolved to random values X_s, and from each X_s a new fan of continuations reaches X_T at maturity. The *local* average $\mathbb{E}[\varphi(X_T)|X_s]$ at each X_s is itself a random variable — a function of X_s via the Markov property — and taking the *outer* average of those local averages back to (t,x) must reproduce the original global average. That is the tower law, and it is the engine of the martingale argument.

5.3.2 4.3.2 The auxiliary process $\eta_s = f(s, X_s)$

Define a stochastic process indexed by the intermediate time $s \in [t,T]$:

$$\eta_s \;\equiv\; f(s,X_s) \;=\; \mathbb{E}\big[\,\varphi(X_T)\,\big|\,X_s\,\big]\,. \tag{4.5}$$

Geometrically, η_s is "the best current guess of $\varphi(X_T)$ given everything known up to time s." Because X is Markov, conditioning on the full filtration $\mathcal{F}_s$ of the Brownian motion collapses to conditioning on the single random variable X_s:

$$\mathbb{E}\big[\varphi(X_T)\,\big|\,\mathcal{F}_s\big] \;=\; \mathbb{E}\big[\varphi(X_T)\,\big|\,X_s\big] \;=\; f(s,X_s) \;=\; \eta_s. \tag{4.6}$$

The first equality is the Markov property of Brownian motion (independent increments collapse the filtration dependence to the current state); the second is the definition of f; the third is notation. This three-step unwinding is worth memorising because it reappears in every measure-change, numeraire-swap, and forward-measure derivation in later chapters.

5.3.3 4.3.3 Tower law $\Rightarrow$ η is a martingale

Claim. The process $s \mapsto \eta_s$ is a martingale in s under the natural filtration $\{\mathcal{F}_s\}$ of X. Concretely, for any $t \le s_1 \le s_2 \le T$,

$$\mathbb{E}\Big[\eta_{s_2}\,\big|\,\mathcal{F}_{s_1}\Big] \;=\; \eta_{s_1}. \tag{4.7}$$

Proof. Plug in the definition:

$$\mathbb{E}[\eta_{s_2}|\mathcal{F}_{s_1}] \;=\; \mathbb{E}\Big[\,\mathbb{E}[\varphi(X_T)|\mathcal{F}_{s_2}]\,\big|\,\mathcal{F}_{s_1}\Big]\,.$$

The outer expectation is over information up to s_1; the inner expectation is over information up to $s_2 \ge s_1$. Apply the tower property of conditional expectations — $\mathbb{E}[\mathbb{E}[A|\mathcal{G}]|\mathcal{H}] = \mathbb{E}[A|\mathcal{H}]$ whenever $\mathcal{H} \subseteq \mathcal{G}$ — with $\mathcal{H} = \mathcal{F}_{s_1}$ and $\mathcal{G} = \mathcal{F}_{s_2}$ (the latter is indeed finer because more time has passed):

$$\mathbb{E}\Big[\,\mathbb{E}[\varphi(X_T)|\mathcal{F}_{s_2}]\,\big|\,\mathcal{F}_{s_1}\Big] \;=\; \mathbb{E}[\varphi(X_T)|\mathcal{F}_{s_1}] \;=\; \eta_{s_1}.$$

Therefore $\eta_{s_1} = \mathbb{E}[\eta_{s_2}|\mathcal{F}_{s_1}]$, the martingale property. $\square$

The argument uses only nesting of filtrations plus iterated expectation — no PDE, no Itô's lemma, no specifics of Brownian motion. *Any* process of the form "conditional expectation of a fixed $\mathcal{F}_T$-measurable random variable" is a martingale. These are called *Doob martingales* or *closed martingales*, and they are the universal way to manufacture a martingale from a given payoff. The payoff $\varphi(X_T)$ is the "closing" random variable; η_s is the stream of running conditional expectations.

A gambler's intuition. A gambler's running wealth under fair odds is a martingale because the expected next-period wealth equals the current wealth. The scoreboard η_s generalises that: the best estimate of a future payoff, given the information so far, is itself a fair-game process. Any new information that moves η_s must do so symmetrically up and down — otherwise the new information would be predictable from the old, contradicting that we already conditioned on everything known. This is the probabilistic content of the slogan "rational pricing is a martingale under the pricing measure."

5.3.4 4.3.4 Itô's lemma on $\eta_s = f(s, X_s)$

Having established that η_s is a martingale, we now apply Itô's lemma to read off what PDE f must satisfy. Assume $f \in C^{1,2}([0,T) \times \mathbb{R})$ — continuously differentiable in t once, in x twice. This is an a-posteriori regularity result for Feynman-Kac solutions with smooth enough payoffs, and we take it for granted here. Apply Itô's lemma II from Chapter 3 with $X_s = W_s$:

$$\mathrm{d}\eta_s \;=\; \partial_t f(s, X_s)\,\mathrm{d}s \;+\; \partial_x f(s, X_s)\,\mathrm{d}X_s \;+\; \tfrac{1}{2}\,\partial_{xx} f(s, X_s)\,(\mathrm{d}X_s)^2. \tag{4.8}$$

Using the Itô shorthand $(\mathrm{d}X_s)^2 = \mathrm{d}s$ and grouping the $\mathrm{d}s$ and $\mathrm{d}X_s$ terms,

$$\mathrm{d}\eta_s \;=\; \underbrace{\left[\,\partial_t f + \tfrac{1}{2}\,\partial_{xx} f\,\right]}_{\text{drift}}\mathrm{d}s \;+\; \underbrace{\partial_x f}_{\text{diffusion}}\;\mathrm{d}X_s, \tag{4.9}$$

with both partial derivatives evaluated at (s, X_s).

5.3.5 4.3.5 Drift-killing identity

The diffusion term $\partial_x f\,\mathrm{d}X_s$ integrated against $\mathrm{d}W$ is a pure stochastic integral, hence a martingale by the construction of the Itô integral in Chapter 3. So the difference between η_s and the martingale $\int \partial_x f\,\mathrm{d}X$ is the deterministic drift term $\int[\partial_t f + \frac{1}{2}\partial_{xx} f]\,\mathrm{d}s$. But we proved in §4.3.3 that η_s is already a martingale. Subtracting a martingale from a martingale leaves a martingale, so the drift term $\int[\partial_t f + \frac{1}{2}\partial_{xx} f]\,\mathrm{d}s$ is itself a martingale.

Now the key fact: this drift is a process of *bounded variation* (it is an ordinary Riemann integral in s), and the only continuous martingale of finite variation is the zero process. (This is a standard decomposition result: a continuous martingale with paths of finite variation must be almost-surely constant; combining "constant" with "starts at zero" gives "identically zero.") Therefore the drift vanishes:

$$\int_t^s \left[\partial_t f(u, X_u) + \tfrac{1}{2}\partial_{xx} f(u, X_u)\right]\mathrm{d}u \;\equiv\; 0 \quad \text{for every } s \in [t, T].$$

Because this holds for every path (including paths visiting every real number at arbitrarily chosen times) and the integrand is continuous, the integrand must vanish pointwise:

$$\boxed{\partial_t f(t,x) + \tfrac{1}{2}\,\partial_{xx} f(t,x) \;=\; 0} \qquad \text{for all } (t,x) \in [0,T) \times \mathbb{R}. \tag{4.10}$$

This is the Kolmogorov backward equation, and we have just derived it from the martingale property of the Doob-closed process η. □

5.3.6 4.3.6 Micro-local version (telescoping)

For readers who prefer a faster, differential-style argument: take $s = t + \Delta t$ with Δt small. The martingale condition gives $\eta_t = \mathbb{E}[\eta_{t+\Delta t}|\mathcal{F}_t]$, i.e.

$$0 \;=\; \mathbb{E}\big[\,\eta_{t+\Delta t} - \eta_t \,\big|\, \mathcal{F}_t\big] \;=\; \mathbb{E}\big[\,\mathrm{d}\eta_t \,\big|\, \mathcal{F}_t\big]\,. \tag{4.11}$$

Substituting (4.9) and using $\mathbb{E}[\mathrm{d}X_t|\mathcal{F}_t] = 0$ (the stochastic-integral piece is a martingale increment),

$$0 \;=\; \mathbb{E}\big[\,(\partial_t f + \tfrac{1}{2}\partial_{xx} f)\,\mathrm{d}t \;+\; \partial_x f\,\mathrm{d}X_t \,\big|\, \mathcal{F}_t\big] \;=\; (\partial_t f + \tfrac{1}{2}\partial_{xx} f)\,\mathrm{d}t.$$

Dividing by $\mathrm{d}t$ and taking $\Delta t \downarrow 0$ gives $\partial_t f + \frac{1}{2}\partial_{xx} f = 0$, the same PDE (4.10). The first derivation is more rigorous (it handles the full time interval); the second is faster in practice. Either yields the same result.

5.3.7 4.3.7 Two lessons from the derivation

First, every pricing PDE in this guide is secretly the identity *"drift-kill the Itô decomposition of a conditional-expectation process."* Once one learns to recognise that pattern, every PDE in the book — Black-Scholes (Chapter 6), short-rate PDEs (Chapter 12), caplet PDEs (Chapter 15) — is derivable in four lines: write the expectation, apply Itô, set the drift to zero, add boundary conditions. The PDE is a *consequence* of the martingale structure, not an independent postulate.

Second, the PDE-to-expectation direction and the expectation-to-PDE direction of Feynman-Kac are *dual*: each implies the other. The full theorem is really a pair of results that share the same PDE. Computationally, the choice of which direction to use is problem-dependent — Monte Carlo prefers expectation, finite-difference grids prefer PDE — but the theoretical content is symmetric.

5.4 4.4 Boundary Condition and Summary of the Zero-Drift Case

The derivation of §4.3 produced the PDE (4.10) from the martingale property of η, but we still need to verify the terminal condition $f(T, x) = \varphi(x)$. That is immediate from the definition of f: at $s = T$ the conditioning has frozen the Brownian motion at its terminal value, so

$$\eta_T \;=\; \mathbb{E}[\varphi(X_T)|X_T] \;=\; \varphi(X_T),$$

and evaluating at $X_T = x$ gives $f(T, x) = \varphi(x)$. The boundary data is *built into* the probabilistic construction — one of the attractive features of the expectation approach. In a PDE-only view one must *impose* the terminal data by hand; in the expectation view it falls out as a tautology.

5.4.1 4.4.1 What the zero-drift theorem says

Putting the pieces of §§4.2–4.3 together:

> **Zero-drift Feynman-Kac.** Let X be a standard Brownian motion and let $\varphi : \mathbb{R} \to \mathbb{R}$ have polynomial growth. The function $f(t, x) = \mathbb{E}_{t,x}[\varphi(X_T)]$ is the (unique classical) solution of the backward heat equation $\partial_t f + \frac{1}{2}\partial_{xx} f = 0$ on $[0, T) \times \mathbb{R}$ with terminal data $f(T, \cdot) = \varphi(\cdot)$.

The three ingredients of the proof — Markov collapse, Doob-martingale of expectations, drift-kill via Itô — will *all* reappear in every subsequent generalisation. The only thing that changes is what one drops into the Itô decomposition.

5.4.2 4.4.2 A remark on regularity

The derivation of §4.3 assumed $f \in C^{1,2}$, i.e. continuously differentiable once in time and twice in space. For Brownian motion with a smooth payoff (e.g. $\varphi \in C^2$ with controlled growth), standard PDE theory guarantees the Feynman-Kac solution is smooth on the open domain $[0, T) \times \mathbb{R}$ — continuity up to $t = T$ requires φ to be continuous, but the interior regularity is automatic.

For *discontinuous* payoffs — a digital option $\varphi(x) = \mathbf{1}\{x > K\}$ or a barrier payoff involving an indicator — the classical theorem does not directly apply because the terminal data is not continuous. Two standard work-arounds cover every case in practice: (i) approximate the discontinuous payoff by a sequence of smooth payoffs $\varphi_n \to \varphi$ and pass to the limit in the solution (this preserves the expectation representation because integration against the Gaussian kernel regularises the discontinuity); or (ii) use viscosity-solution theory, which extends Feynman-Kac to a much wider class of payoffs at the cost of replacing "classical derivative" with a weak notion. Working traders rarely need viscosity solutions because approximation by smooth payoffs handles every concrete case. The reader should keep in mind that the $C^{1,2}$ assumption is not strictly necessary but simplifies the presentation.

5.4.3 4.4.3 Why linearity matters

Observe that the PDE (4.1) is *linear* in f: if f_1 and f_2 solve the PDE with terminal data φ_1 and φ_2 respectively, then $\alpha f_1 + \beta f_2$ solves it with terminal data $\alpha\varphi_1 + \beta\varphi_2$. On the expectation side this reads $\mathbb{E}[\alpha\varphi_1(X_T) + \beta\varphi_2(X_T)] = \alpha\mathbb{E}[\varphi_1(X_T)] + \beta\mathbb{E}[\varphi_2(X_T)]$, the linearity of expectation. The two linearities agree, as they must.

Linearity is why we can decompose an arbitrary payoff into a linear combination of simpler payoffs and price each piece separately. This is the basis of the *Carr-Madan replication* trick — any European payoff with sufficient regularity can be written as an integral of vanillas against a continuous distribution of strikes — and is used heavily in variance-swap and volatility-derivative pricing. In the present chapter we use linearity more modestly: the worked examples in §§4.5–4.7 will cover three basis payoffs (linear x, quadratic x^2, exponential e^{ax}) from which more complex payoffs can be assembled by superposition.

5.5 4.5 Example 1 — Linear Payoff $\varphi(x) = x$

The simplest possible test case. A linear payoff exercises none of the non-trivial machinery of stochastic calculus — the answer is what one would guess — but it serves as a base-line sanity check. If the Feynman-Kac machinery did *not* return x for this payoff, something would be wrong with the setup.

Consider the Cauchy problem

$$\begin{cases} \partial_t f + \frac{1}{2}\partial_{xx} f \;=\; 0, \\ f(T,x) \;=\; x. \end{cases} \tag{4.12}$$

By Feynman-Kac (4.2),

$$f(t,x) \;=\; \mathbb{E}_{t,x}[X_T],$$

where X is a standard Brownian motion. Brownian increments are Gaussian,

$$X_T - X_t \;\sim\; \mathcal{N}(0, T-t), \qquad \text{so} \qquad X_T \overset{d}{=} x + \sqrt{T-t}\,Z, \quad Z \sim \mathcal{N}(0,1).$$

Therefore

$$f(t,x) \;=\; \mathbb{E}_{t,x}\Big[x + \sqrt{T-t}\,Z\Big] \;=\; x \;+\; \sqrt{T-t}\cdot\mathbb{E}[Z] \;=\; x. \tag{4.13}$$

PDE verification. With $f(t,x) = x$, $\partial_t f = 0$, $\partial_x f = 1$, $\partial_{xx} f = 0$, so $\partial_t f + \frac{1}{2}\partial_{xx} f = 0$. Terminal condition $f(T,x) = x$ holds by inspection. Both branches agree.

5.5.1 4.5.1 Why this answer is natural

A linear payoff has zero *convexity*, so Jensen's inequality is an equality and the expectation coincides with the initial state. Whenever φ is affine, $\mathbb{E}_{t,x}[\varphi(X_T)] = \varphi(x)$. This will fail as soon as we move to x^2 (convex) or e^{-x} (log-convex) in the next two examples — but convexity first has to be introduced.

A trader's-eye reading. If "linear payoff $\varphi(x) = x$" represents a forward contract on an asset whose price follows driftless Brownian motion, the forward value today must equal the current spot x because there is no expected drift

under the driftless pricing measure. (Adding a drift would introduce a forward premium, as we will see in §4.8.) Feynman-Kac delivers this in one line — the expected terminal value equals the current value, exactly as arbitrage considerations demand.

Second view: the backward heat equation applied to a linear function is trivially satisfied because linear functions have zero curvature. The Itô correction *does nothing* for straight-line payoffs. As soon as the payoff bends — calls, puts, digitals — curvature reappears and the correction starts influencing the answer. That is precisely why options have nonzero *convexity premium* relative to forwards of the same underlying, and why vanilla options are more interesting to price than forward contracts.

5.6 4.6 Example 2 — Quadratic Payoff $\varphi(x) = x^2$

The first non-trivial example, and the one where convexity starts to matter. The quadratic payoff is also the simplest test case for variance-swap-style replication arguments, because the squared terminal value is the canonical "variance" functional.

Solve

$$\begin{cases} \partial_t f + \frac{1}{2}\partial_{xx} f \;=\; 0, \\ f(T,x) \;=\; x^2. \end{cases} \tag{4.14}$$

Feynman-Kac gives

$$f(t,x) \;=\; \mathbb{E}_{t,x}[X_T^2] \;=\; \mathbb{E}_{t,x}\Big[\big((X_T - X_t) + X_t\big)^2\Big],$$

with $X_T - X_t \sim \mathcal{N}(0, T-t)$ and $X_t = x$ under $\mathbb{E}_{t,x}$. Expanding the square and taking expectations term by term:

$$f(t,x) \;=\; \mathbb{E}_{t,x}\big[\,(X_T - X_t)^2 + 2X_t(X_T - X_t) + X_t^2\,\big] \;=\; (T-t) \;+\; 0 \;+\; x^2,$$

where the three contributions come from $\mathbb{E}[(X_T - X_t)^2] = \mathrm{Var}(X_T - X_t) = T - t$, $\mathbb{E}[(X_T - X_t)] = 0$, and $\mathbb{E}[X_t^2 | X_t = x] = x^2$. Hence

$$\boxed{f(t,x) \;=\; x^2 + (T-t)}. \tag{4.15}$$

PDE verification. With $f(t,x) = x^2 + (T-t)$, $\partial_t f = -1$, $\partial_{xx} f = 2$, so $\partial_t f + \frac{1}{2}\partial_{xx} f = -1 + 1 = 0$. At $t = T$: $f(T,x) = x^2 + 0 = x^2$. Both the PDE and the terminal condition are satisfied.

5.6.1 4.6.1 Convexity reading

The extra $(T-t)$ in (4.15) is the *time-value of convexity.* A convex payoff rewards variance, and Brownian increments supply exactly $\mathrm{Var}(X_T | X_t = x) = T - t$ worth of variance. This is the simplest incarnation of *vega/volga premium* in option pricing. The result is consistent with the direct stochastic-integral calculation

$$X_T^2 \;=\; X_t^2 \;+\; 2\int_t^T X_s \,\mathrm{d}X_s \;+\; (T-t),$$

which is Itô's lemma applied to $F(x) = x^2$ shifted by X_t. Taking expectations makes the middle (martingale) term vanish and recovers (4.15).

5.6.2 4.6.2 Variance-swap interpretation

The payoff X_T^2 is (up to a constant) the variance-swap payoff — it pays the final squared value of the state. If X starts at x, the expected squared final value equals the starting squared value plus the remaining variance $T-t$. The longer the remaining tenor, the more variance accumulates, and the higher the expected squared terminal value. This is the "time-value of variance," the direct analogue of the time-value of an option's convexity.

The formula $f(t,x) = x^2 + (T-t)$ is also the fair value of a specific swap: if one enters a variance swap at time t with "strike" x^2 and holds to maturity, the expected payoff at maturity is X_T^2. In the driftless, no-discount setting the fair strike is $x^2 + (T-t)$ — the starting squared value plus the remaining variance. Every variance-swap dealer ultimately prices this identity; Feynman-Kac delivers the answer with two lines of Gaussian algebra.

Quadratic payoff MC vs PDE solution *Sanity check of* (4.15)*: Monte-Carlo averages of X_T^2 at a grid of initial states x_0 (red dots) fall exactly on the PDE closed form $x_0^2 + T$ (blue curve). The $+T$ offset is the "time-value of variance" — the convexity premium that Brownian motion pays out for a quadratic payoff.*

5.7 4.7 Example 3 — Exponential Payoffs $\varphi(x) = e^{ax}$

Exponential payoffs are the log-space analogue of the log-normal pricing problems that dominate the rest of the guide. Because the log of a geometric-Brownian-motion asset is a drifted arithmetic Brownian motion, every result derived here with $X = W$ will translate into a stock-price result by the substitution $X = \ln S$. Many closed-form formulas in rates, credit, and equity pricing fall out of this single expectation once the drift and discounting extensions of §4.8 are in place.

5.7.1 4.7.1 Base case $\varphi(x) = e^{-x}$

Solve

$$\begin{cases} \partial_t f + \frac{1}{2}\partial_{xx} f \;=\; 0, \\ f(T,x) \;=\; e^{-x}. \end{cases} \tag{4.16}$$

Feynman-Kac gives

$$f(t,x) \;=\; \mathbb{E}_{t,x}[\, e^{-X_T} \,].$$

Because $X_T | X_t = x$ is $\mathcal{N}(x, T-t)$, apply the Gaussian moment-generating function $\mathbb{E}[e^{\lambda Y}] = e^{\lambda\mu + \frac{1}{2}\lambda^2\sigma^2}$ with $\lambda = -1$, $\mu = x$, $\sigma^2 = T-t$:

$$\boxed{f(t,x) \;=\; e^{-x + \frac{1}{2}(T-t)}}. \tag{4.17}$$

PDE verification. With $f = e^{-x+\frac{1}{2}(T-t)}$: $\partial_t f = -\frac{1}{2}f$, $\partial_x f = -f$, $\partial_{xx} f = f$. Then $\partial_t f + \frac{1}{2}\partial_{xx} f = -\frac{1}{2}f + \frac{1}{2}f = 0$. Terminal condition $f(T,x) = e^{-x}$ holds.

Jensen bump. Notice $f(t,x) = e^{-x} \cdot e^{\frac{1}{2}(T-t)}$ — the answer exceeds the payoff evaluated at the mean e^{-x} by the factor $e^{\frac{1}{2}(T-t)}$. This is the *convexity premium.* In the log variable this is exactly the $\frac{1}{2}\sigma^2 t$ Itô correction that will reappear in the Black-Scholes formula. Its sign follows from Jensen's inequality applied to the convex function e^{-x}: $\mathbb{E}[e^{-X_T}] \geq e^{-\mathbb{E}[X_T]} = e^{-x}$, with strict inequality whenever X_T has nonzero variance.

5.7.2 4.7.2 General exponent $\varphi(x) = e^{ax}$

A one-parameter generalisation that shows the convexity bump scales as a^2, not $|a|$. Solve

$$\begin{cases} \partial_t f + \frac{1}{2}\partial_{xx} f \;=\; 0, \\ f(T,x) \;=\; e^{ax}, \end{cases} \qquad a \in \mathbb{R}. \tag{4.18}$$

By Feynman-Kac, using $X_T = x + (X_T - X_t)$ with $X_T - X_t \sim \mathcal{N}(0, T-t)$,

$$f(t,x) \;=\; \mathbb{E}_{t,x}[\, e^{aX_T}\,] \;=\; e^{ax} \cdot \mathbb{E}[\, e^{a(X_T - X_t)}\,] \;=\; e^{ax} \cdot e^{\frac{1}{2}a^2(T-t)},$$

using the Gaussian MGF with $\lambda = a$, $\sigma^2 = T - t$. Compactly,

$$\boxed{f(t,x) \;=\; \exp\!\big\{\, a\,x \;+\; \tfrac{1}{2}\,a^2\,(T-t)\,\big\}}. \tag{4.19}$$

PDE verification. Let $\gamma = \frac{1}{2}a^2$. Then $f = e^{ax+\gamma(T-t)}$ gives $\partial_t f = -\gamma f$, $\partial_x f = af$, $\partial_{xx} f = a^2 f$. Plugging into the PDE:

$$\partial_t f + \tfrac{1}{2}\partial_{xx} f \;=\; -\gamma f + \tfrac{1}{2}a^2 f \;=\; -\tfrac{1}{2}a^2 f + \tfrac{1}{2}a^2 f \;=\; 0. \quad \checkmark$$

Terminal: $f(T,x) = e^{ax+0} = e^{ax}$.

5.7.3 4.7.3 Reading the result

Setting $a = 1$ gives $f(t,x) = e^{x+\frac{1}{2}(T-t)}$, the "up-exponential" mirror of (4.17); setting $a = -1$ recovers (4.17). The parameter a controls both the exponential-growth rate of the payoff in x and, via $a^2/2$, the convexity premium. Even the sign of a does not affect the convexity bump — only $|a|$ matters, because the Gaussian MGF depends on a^2. This is the cleanest illustration that *convexity in Feynman-Kac is an even function of the exponent coefficient.*

A link forward to Black-Scholes. For a geometric Brownian motion $S_t = S_0\, e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$ with W a Brownian motion, the n-th moment $\mathbb{E}[S_t^n]$ arises from applying (4.19) with $X_t = \ln S_t$, $a = n$, and the appropriate drift correction. The resulting formula $\mathbb{E}[S_t^n] = S_0^n\, e^{n\mu t + \frac{1}{2}n(n-1)\sigma^2 t}$ is the basis for variance-swap pricing ($n = 2$), volatility derivatives, and power-option closed forms — all of which are $a \in \{2, 3, \dots\}$ specialisations of the Gaussian MGF identity (4.19). We will pick this thread up in Chapter 6.

Monte-Carlo sanity check for §§4.5–4.7 closed forms *For each of the three worked payoffs (linear x, quadratic x^2, exponential e^{ax} with $a = 0.8$), Monte-Carlo averaging over N Brownian endpoints $X_T \sim \mathcal{N}(x_0, T)$ converges at rate $1/\sqrt{N}$ to the Gaussian-MGF closed form from* (4.13)–(4.19)*. Feynman-Kac is a two-way street in code as well as in theory.*

5.7.4 4.7.4 A five-term ledger preview

The three worked examples so far have all used a driftless, undiscounted Brownian motion. In §4.8 we will generalise to drifted BM with a discount rate, and the exponential payoff $\varphi(x) = e^{-x}$ will pick up three new terms in the exponent: a drift shift, a discount drag, and an unchanged convexity bump. The resulting formula reads like a ledger where each effect appears on a separate line. It is worth internalising the base case (4.17) now because the drifted version (4.26) below is literally (4.17) with three visible additions.

5.8 4.8 Feynman-Kac with Drift, Diffusion, and Discounting

Every worked example so far has used a driftless Brownian motion with unit diffusion and no discounting. Real pricing problems require three generalisations: the underlying may have a nonzero drift, its diffusion coefficient need not be one, and the payoff must be discounted back to the present. Feynman-Kac extends to each in a mechanical way, and in this section we do all three at once.

5.8.1 4.8.1 Statement

Let $a \in \mathbb{R}$, $b > 0$, and $c \in \mathbb{R}$ be constants. Consider the backward Cauchy problem with a lower-order term:

$$\begin{cases} \partial_t f(t,x) \;+\; a\,\partial_x f(t,x) \;+\; \tfrac{1}{2}\,b^2\,\partial_{xx} f(t,x) \;=\; c\,f(t,x), \\ f(T,x) \;=\; \varphi(x). \end{cases} \tag{4.20}$$

The probabilistic representation of the solution is

$$\boxed{f(t,x) \;=\; \mathbb{E}_{t,x}\big[\, e^{-c(T-t)}\, \varphi(X_T)\,\big]\,,} \tag{4.21}$$

where $X = (X_s)_{s\geq 0}$ is a drifted Brownian motion solving

$$\mathrm{d}X_s \;=\; a\,\mathrm{d}s \;+\; b\,\mathrm{d}W_s, \qquad W \text{ a standard BM}, \tag{4.22}$$

i.e. $X_s = X_0 + a\,s + b\,W_s$ (arithmetic Brownian motion with drift a and volatility b).

5.8.2 4.8.2 Proof sketch

Start from the expectation $f(t,x) = \mathbb{E}_{t,x}[e^{-c(T-t)}\varphi(X_T)]$ and apply Itô's lemma to the auxiliary process

$$g_s \;\equiv\; e^{-c(s-t)}\, f(s, X_s), \qquad s \in [t,T].$$

Compute $\mathrm{d}g_s$ using the product rule plus Itô's lemma III applied to $f(s, X_s)$:

$$\mathrm{d}g_s \;=\; e^{-c(s-t)}\big\{-c\,f\,\mathrm{d}s \;+\; \big[\partial_t f + a\,\partial_x f + \tfrac{1}{2}b^2\partial_{xx} f\big]\,\mathrm{d}s \;+\; b\,\partial_x f\,\mathrm{d}W_s\big\}\,. \tag{4.23}$$

For g_s to be a martingale — which it must be, because $g_T = e^{-c(T-t)}\varphi(X_T)$ and conditional expectations of fixed random variables are martingales — the $\mathrm{d}s$ drift must vanish:

$$-c\,f + \partial_t f + a\,\partial_x f + \tfrac{1}{2}b^2\partial_{xx} f \;=\; 0,$$

which is exactly the PDE (4.20). Conversely, if f satisfies the PDE, the drift of g_s is identically zero; g_s is therefore a martingale, and taking $\mathbb{E}_{t,x}$ of $g_t = f(t,x)$ and $g_T = e^{-c(T-t)}\varphi(X_T)$ yields the representation (4.21).

The two-way proof is the essence of Feynman-Kac: the PDE *is* the drift-killing condition for $e^{-c(s-t)}f(s,X_s)$, and any function satisfying that PDE automatically represents an expectation.

5.8.3 4.8.3 Role of each coefficient

Each term in (4.20) has a direct counterpart in (4.21):

PDE term	SDE / expectation counterpart
$a\,\partial_x f$	drift of X: each path is pushed by $a\,\mathrm{d}s$ per step.
$\frac{1}{2}b^2\,\partial_{xx} f$	diffusion of X: paths spread with variance $b^2\,\mathrm{d}s$ per step.
$-c\,f$	exponential decay / discount factor $e^{-c(T-t)}$ applied to the payoff.
terminal $f(T,x) = \varphi(x)$	the expectation at $s = T$ is literally $\varphi(X_T)$ with $X_T = x$.

The dictionary reading of (4.20) $\Leftrightarrow$ (4.21): *"a and b pick the law of X; c attaches a per-unit-time discount to every path."*

5.8.4 4.8.4 Why the discount sign flips

On the PDE side, the discount term sits as $-cf$ on the left (or equivalently $+cf$ on the right), because it represents "the present value grows at rate c as we move *forward* in time and must be peeled back as we solve *backward*." On the expectation side, the discount appears as $e^{-c(T-t)}$ *inside* the expectation, because it represents "each path's terminal payoff is multiplied by the present-value discount factor."

The two are related by the PDE being a backward equation: integrating $\partial_t f = cf -$ (diffusion terms) from t back to T attaches a factor $e^{-c(T-t)}$ to the terminal payoff, exactly as Feynman-Kac says. A common source of sign errors is to confuse the two conventions; keeping the bookkeeping consistent requires always writing the PDE with $-cf$ on the left and the expectation with $e^{-c(T-t)}$ inside, or vice versa with flipped signs.

5.8.5 4.8.5 Geometric reading: linearity and the parabolic class

The PDE (4.20) is a *linear parabolic PDE* with coefficients that depend on x and t but not on f itself. Linearity is essential: it means solutions superpose. If φ_1 and φ_2 are two terminal payoffs, then $\alpha f_1 + \beta f_2$ solves the PDE with terminal $\alpha\varphi_1 + \beta\varphi_2$. On the expectation side this corresponds to $\mathbb{E}[\alpha\varphi_1 + \beta\varphi_2] = \alpha\mathbb{E}[\varphi_1] + \beta\mathbb{E}[\varphi_2]$ — another form of the same statement.

The linearity is why we can decompose an arbitrary payoff into a basis (call spreads, digitals, exponentials) and price each piece separately. The Carr-Madan replication trick that underlies variance swaps and other volatility derivatives is one application; the moment-generating-function manipulations we use in the worked examples are another.

5.8.6 4.8.6 Worked case: exponential payoff with drift, diffusion, and discount

Solve (4.20) with $\varphi(x) = e^{-x}$:

$$\begin{cases} \partial_t f + a\,\partial_x f + \frac{1}{2}b^2\partial_{xx} f \;=\; c\,f, \\ f(T,x) \;=\; e^{-x}. \end{cases} \tag{4.24}$$

By (4.21),

$$f(t,x) \;=\; \mathbb{E}_{t,x}\big[\, e^{-X_T}\, e^{-c(T-t)} \,\big]\,.$$

The increment of X under the drifted BM (4.22) is

$$X_T - X_t \;=\; a(T-t) + b\,(W_T - W_t) \;\overset{d}{=}\; a(T-t) + b\sqrt{T-t}\,Z, \qquad Z \sim \mathcal{N}(0,1),$$

so $X_T | X_t = x$ is Gaussian with mean $x + a(T-t)$ and variance $b^2(T-t)$. Substituting,

$$f(t,x) \;=\; \mathbb{E}\Big[\exp\Big\{-x - a(T-t) - b\sqrt{T-t}\,Z - c(T-t)\Big\}\Big]\,.$$

Taking the Gaussian MGF of $-b\sqrt{T-t}\,Z$, which contributes $e^{\frac{1}{2}b^2(T-t)}$:

$$f(t,x) \;=\; \exp\big\{-x - a(T-t) + \tfrac{1}{2}b^2(T-t) - c(T-t)\big\}\,.$$

Collecting terms with common factor $(T-t)$:

$$\boxed{f(t,x) \;=\; \exp\big\{\, -x \;-\; \big(a + c - \tfrac{1}{2}b^2\big)\,(T-t)\,\big\}}. \tag{4.25}$$

Equivalently, defining $\gamma = a + c - \frac{1}{2}b^2$ for shorthand,

$$f(t,x) \;=\; e^{-x-\gamma(T-t)}. \tag{4.26}$$

5.8.7 4.8.7 Reading formula (4.25) as a ledger

The exponent $-x-(a+c-\frac{1}{2}b^2)(T-t)$ has five distinguishable contributions, one per ingredient of the problem:

1. $-x$: the payoff e^{-X_T} evaluated "on average" around the starting point.
2. $-a(T-t)$: drift of X shifts the mean of X_T by $+a(T-t)$, which in the payoff e^{-X_T} becomes a factor $e^{-a(T-t)}$.
3. $+\frac{1}{2}b^2(T-t)$: Jensen bump from b-scale Gaussian noise in X_T, exactly as in the base case (4.17).
4. $-c(T-t)$: discount factor from the cf term in the PDE.
5. The sign pattern "+" on the convexity term and "−" on the drift and discount is specific to the exponential-*decreasing* payoff e^{-x}. For e^{+x} (exponential-increasing) the drift sign would flip, because the Gaussian MGF with $\lambda=+1$ gives $\lambda\mu=+a(T-t)$.

Each of these five contributions has a well-defined financial counterpart in real option pricing: drift is the risk-neutral cost-of-carry, discount is the time-value-of-money effect, convexity is the gamma/vega premium. The cleanness of the ledger is the payoff of having set up Feynman-Kac carefully: complicated models decompose into their individual effects rather than tangling them together.

Sanity limits. Setting $a=c=0$, $b=1$ recovers (4.17). Setting $b=0$ (pure deterministic drift) gives $f(t,x)=e^{-x-(a+c)(T-t)}$ — deterministic evolution of X followed by deterministic discounting, no convexity. Setting $a=c=0$ with general b gives $f=e^{-x+\frac{1}{2}b^2(T-t)}$, the b-scaled Jensen bump. Every qualitative feature — convexity bonus $\frac{1}{2}b^2(T-t)$, discount drag $c(T-t)$, drift shift $a(T-t)$ — sits visibly in the exponent.

Direct PDE check. Differentiate (4.26) and verify (4.20). With $f=e^{-x-\gamma(T-t)}$ and $\gamma=a+c-\frac{1}{2}b^2$:

$$\partial_t f=+\gamma f,\qquad \partial_x f=-f,\qquad \partial_{xx}f=f.$$

Plugging into the PDE's left-minus-right:

$$\partial_t f+a\,\partial_x f+\tfrac{1}{2}b^2\partial_{xx}f-cf \;=\; \gamma f-af+\tfrac{1}{2}b^2 f-cf \;=\; (\gamma-a+\tfrac{1}{2}b^2-c)\,f.$$

Substituting $\gamma=a+c-\frac{1}{2}b^2$:

$$\gamma-a+\tfrac{1}{2}b^2-c \;=\; (a+c-\tfrac{1}{2}b^2)-a+\tfrac{1}{2}b^2-c \;=\; 0. \quad \checkmark$$

All five terms cancel, confirming that (4.25) solves (4.20) with the right terminal condition.

5.9 4.9 Integrating-Factor Interpretation of the Discount

The discounted version of Feynman-Kac in §4.8 handled a *constant* discount rate c. For many applications — credit-risk pricing where $c=\lambda(t,x)$ is a state-dependent hazard rate, interest-rate models where $c(t,r)=r$ is itself a stochastic short rate — one wants to accommodate a discount rate that depends on time and the current state of the diffusion. The full state-dependent version is the subject of §4.10. Before going there, this section presents an alternative *integrating-factor* derivation that gives a clean proof even when c depends on (t,x), and that makes the connection between discounting and change-of-numeraire transparent.

5.9.1 4.9.1 Setup

Let X be a standard Brownian motion and $c(t,x)$ a bounded-continuous function. Suppose f satisfies the Cauchy problem

$$\begin{cases}(\partial_t+\frac{1}{2}\partial_{xx})\,f(t,x) \;=\; c(t,x)\,f(t,x),\\ f(T,x) \;=\; \varphi(x).\end{cases} \tag{4.27}$$

Then f admits the representation

$$\boxed{f(t,x) \;=\; \mathbb{E}_{t,x}\left[\varphi(X_T)\,\exp\left(-\int_t^T c(u,X_u)\,\mathrm{d}u\right)\right].} \tag{4.28}$$

When c is a constant, the path integral $\int_t^T c\,\mathrm{d}u = c(T-t)$ collapses and we recover (4.21) with $a=0$, $b=1$. When $c(t,x)=x$ — interpreting X_t as a stochastic short rate — the expression $\int_t^T X_u\,\mathrm{d}u$ is the integrated short rate along the realised path, and (4.28) becomes the bond-price formula used heavily in Chapter 12. When $c(t,x)=\lambda(t,x)$ is a hazard rate, (4.28) becomes the survival-probability formula used in credit-derivative pricing.

5.9.2 4.9.2 Derivation via integrating factor

Define, paralleling §4.3.2, the candidate

$$g_t \;\equiv\; \mathbb{E}\Big[\varphi(X_T)\,e^{-\int_t^T c(u,X_u)\,\mathrm{d}u}\,\Big|\,\mathcal{F}_t\Big]. \tag{4.29}$$

The issue with g_t is that the discount factor $e^{-\int_t^T c\,\mathrm{d}u}$ inside the expectation depends on t through the lower limit of integration. That time-dependence means g_t is not immediately in the standard "conditional expectation of a fixed $\mathcal{F}_T$-measurable random variable" form, and it is not obvious that g_t is a martingale.

The fix is an integrating factor. Define

$$h_t \;\equiv\; e^{-\int_0^t c(u,X_u)\,\mathrm{d}u}\,g_t \;=\; \mathbb{E}\Big[\varphi(X_T)\,e^{-\int_0^T c(u,X_u)\,\mathrm{d}u}\,\Big|\,\mathcal{F}_t\Big]. \tag{4.30}$$

The content of the second equality: the multiplicative factor $e^{-\int_0^t c\,\mathrm{d}u}$ is $\mathcal{F}_t$-measurable, so it can be pulled inside the conditional expectation in (4.29), combining with the already-present $e^{-\int_t^T c\,\mathrm{d}u}$ to give the single discount factor $e^{-\int_0^T c\,\mathrm{d}u}$ running over the *full* horizon $[0,T]$. The inside of the expectation in (4.30) is an honest $\mathcal{F}_T$-measurable random variable with no residual t-dependence. Therefore h_t is manifestly a Doob martingale (conditional expectation of a fixed random variable is always a martingale). The underlying mechanism: rewrite $g_t = e^{+\int_0^t c\,\mathrm{d}u}\cdot\mathbb{E}[H \mid \mathcal{F}_t]$ with $H := \varphi(X_T)e^{-\int_0^T c\,\mathrm{d}u}$ fixed and $\mathcal{F}_T$-measurable; the Doob piece is a true martingale, but the prefactor $e^{+\int_0^t c\,\mathrm{d}u}$ grows (for $c\geq 0$) and makes g_t a super-martingale rather than a martingale. Multiplying g_t by the integrating factor $e^{-\int_0^t c\,\mathrm{d}u}$ in (4.30) cancels the prefactor *exactly*, which is the compensation that kills the super-martingale drift and restores the martingale property — that is the punchline of the device.

5.9.3 4.9.3 Itô's product rule kills the drift

Apply Itô to $h_t = e^{-\int_0^t c\,\mathrm{d}u}\,g_t$. The integrating-factor process has differential $\mathrm{d}\Big[e^{-\int_0^t c\,\mathrm{d}u}\Big] = -c(t,X_t)\,e^{-\int_0^t c\,\mathrm{d}u}\,\mathrm{d}t$ (purely deterministic in the quadratic-variation sense — it has no $\mathrm{d}W$ component, so no cross-variation term appears when one applies the product rule):

$$\mathrm{d}h_t \;=\; -c(t,X_t)\,h_t\,\mathrm{d}t \;+\; e^{-\int_0^t c\,\mathrm{d}u}\,\mathrm{d}g_t. \tag{4.31}$$

Expand $\mathrm{d}g_t$ via Itô's lemma II on $g_t = f(t,X_t)$ — this uses the assumption that the candidate pricing function f satisfies sufficient regularity:

$$\mathrm{d}g_t \;=\; \left[\partial_t f + \tfrac{1}{2}\partial_{xx} f\right]\mathrm{d}t \;+\; \partial_x f\,\mathrm{d}X_t.$$

Substituting into (4.31) and collecting $\mathrm{d}t$ and $\mathrm{d}W$ terms:

$$\mathrm{d}h_t \;=\; \left[-c(t,X_t)\,f + \partial_t f + \tfrac{1}{2}\partial_{xx} f\right]\,e^{-\int_0^t c\,\mathrm{d}u}\,\mathrm{d}t \;+\; (\text{martingale}), \tag{4.32}$$

where "martingale" refers to the pure $\mathrm{d}X_t = \mathrm{d}W_t$ piece $e^{-\int_0^t c\,\mathrm{d}u}\,\partial_x f\,\mathrm{d}W_t$, which is a stochastic integral hence a martingale. Because we established in (4.30) that h_t is itself a martingale, the $\mathrm{d}t$ drift in (4.32) must vanish:

$$-c(t, X_t)\, f + \partial_t f + \tfrac{1}{2}\partial_{xx} f \;=\; 0,$$

and because $f = f(t, X_t)$ is evaluated pointwise in (t, x), this gives

$$\boxed{\partial_t f(t,x) + \tfrac{1}{2}\,\partial_{xx} f(t,x) \;=\; c(t,x)\, f(t,x).} \tag{4.33}$$

This is the PDE (4.27). The terminal condition $f(T, x) = \varphi(x)$ follows from the definition at $t = T$ because the conditioning has frozen $X_T = x$ and the discount integral has shrunk to zero. □

5.9.4 4.9.4 The integrating factor as a change of numeraire

The integrating-factor move has a direct financial interpretation. In financial terms, the quantity g_t is a price *denominated in time-t units* — the discount factor runs from t to T, peeling terminal payoffs back to time-t dollars. The quantity h_t is a price *denominated in time-0 units* — the discount factor runs from 0 to T, peeling terminal payoffs back to time-0 dollars.

Changing the numeraire from the time-t clock to the time-0 clock is accomplished by the multiplicative factor $e^{-\int_0^t c\,\mathrm{d}u}$, which is exactly the bank-account value at time t relative to time-0 cash. This is the same multiplicative construction we will meet in full generality in Chapter 5, where any positive tradable (not just the bank account) is allowed to serve as numeraire. The integrating factor in (4.30) is the simplest instance of the change-of-numeraire technique: it is the change from time-t-denominated prices to time-0-denominated prices, which coincides with multiplying by the money-market numeraire.

5.9.5 4.9.5 Connection to credit and rates

Two applications of (4.28) that will be developed in later chapters are worth flagging now.

Short-rate bond pricing. Take $c(t, x) = x$ and interpret X_t as a short-rate process. Then $\int_t^T X_u\,\mathrm{d}u$ is the realised integrated short rate from t to T, and with terminal payoff $\varphi = 1$ (a unit-face zero-coupon bond),

$$P(t,x) \;=\; \mathbb{E}_{t,x}\Big[\, e^{-\int_t^T X_u\,\mathrm{d}u} \,\Big]$$

is the fair value of a zero-coupon bond. Chapter 12 will evaluate this expectation under the Vasicek and Hull-White models and obtain the affine bond-price formula $P(t, x) = e^{A(t,T) - B(t,T)\,x}$.

Defaultable claims. Take $c(t, x) = r + \lambda(t, x)$ where r is the risk-free short rate and $\lambda(t, x)$ is a hazard rate. Then $e^{-\int_t^T \lambda\,\mathrm{d}u}$ is the survival probability from t to T along the realised path, and (4.28) with terminal payoff $\varphi = 1$ gives the time-t value of a unit payment at T conditional on survival. This is the basic building block of credit-derivative pricing — defaultable bonds, CDS, and credit-linked notes are all assembled from such survival factors.

Both applications require the state-dependent version of Feynman-Kac that is the subject of the next section.

5.10 4.10 Feynman-Kac with State- and Time-Dependent Coefficients

The constant-coefficient version (4.20)–(4.21) is enough for driftless BM, for arithmetic BM with constant drift, and for a handful of toy models. Most real-world pricing — including the Black-Scholes model for equity derivatives and the short-rate models for fixed income — requires *state-dependent* coefficients where a, b, and c are themselves functions of (t, x). This section states and proves the general theorem and then shows three important instantiations.

5.10.1 4.10.1 The general theorem

Let $a, b, c : [0,T] \times \mathbb{R} \to \mathbb{R}$ be sufficiently regular functions — continuous with mild growth conditions on a and b, and boundedness on c; these are the conditions under which the associated SDE has a unique strong solution and the corresponding PDE has a classical solution. Consider the Cauchy problem

$$\begin{cases} \partial_t h(t,x) \;+\; a(t,x)\,\partial_x h(t,x) \;+\; \frac{1}{2}\,b^2(t,x)\,\partial_{xx} h(t,x) \;=\; c(t,x)\,h(t,x), \\ h(T,x) \;=\; H(x), \end{cases} \tag{4.34}$$

with terminal payoff $H : \mathbb{R} \to \mathbb{R}$ satisfying polynomial growth. Then the solution admits the probabilistic representation

$$\boxed{h(t,x) \;=\; \mathbb{E}\left[\exp\left\{-\int_t^T c(u,X_u)\,\mathrm{d}u\right\} H(X_T)\,\Big|\, X_t = x\right],} \tag{4.35}$$

where the process X satisfies the Itô SDE

$$\mathrm{d}X_s \;=\; a(s,X_s)\,\mathrm{d}s \;+\; b(s,X_s)\,\mathrm{d}W_s, \qquad s \in [t,T]. \tag{4.36}$$

This is the *complete* Feynman-Kac theorem used in practice. Every linear parabolic pricing PDE in this guide is an instantiation of it, and the dictionary between PDE coefficients and SDE/discount is strict.

5.10.2 4.10.2 Three features worth noting

The drift $a(s,X_s)$ and diffusion $b(s,X_s)$ may depend on both time and the current state. No constancy assumption is required. The SDE (4.36) is the general Itô diffusion, and Itô's lemma III from Chapter 3 is exactly the tool we need.

The discount rate $c(s,X_s)$ may itself be stochastic — a function of the state X_s. In this case the integrated discount $\int_t^T c(u,X_u)\,\mathrm{d}u$ is a path-dependent random variable, not a deterministic number $c(T-t)$. The path integral is exactly how stochastic-rate bond pricing works: $c(t,r) = r$ with $X = r_t$ a short-rate process yields bond prices $P(t,T) = \mathbb{E}_t[e^{-\int_t^T r_s\,\mathrm{d}s}]$.

The payoff $H(X_T)$ is evaluated at the final state only. We do not allow path-dependent payoffs in the basic statement. Path-dependency — barrier options that knock out if the path touches a level, Asian options that depend on the average price — requires further machinery: auxiliary state variables (e.g. the running minimum becomes a second state), augmented Itô processes, or variational inequalities for American-style features. Each extension preserves the Feynman-Kac spirit but with extra bookkeeping. The basic theorem (4.34)–(4.35) covers every European-style payoff on a single state variable.

5.10.3 4.10.3 Why the path-dependent discount is necessary

On the PDE side, the discount term $c(t,x)\,h$ is spatially variable — it depends on where in the (t,x) domain the price-surface is being evaluated. On the expectation side, the discount factor $e^{-\int c\,\mathrm{d}u}$ integrates c along the *realised path* X from t to T. These two forms are duals: the PDE evaluates c locally at each point, and the expectation accumulates that local rate along each sample path.

For constant c, the path integral collapses to $c(T-t)$ and we recover the constant-discount version (4.21). For spatially-varying c, the two sides of Feynman-Kac exchange "local evaluation" for "path accumulation" — a conceptual swap that is easiest to appreciate in the simple example $c(t,x) = x$: on the PDE side we see a linear-in-state zero-order term; on the expectation side we see $e^{-\int_t^T X_u\,\mathrm{d}u}$, a path-dependent random variable. The two forms are identical in content, just organised differently.

5.10.4 4.10.4 Proof

Apply Itô's lemma III (the general form from Chapter 3 that accommodates a general Itô process) to the process

$$Y_s \;\equiv\; \exp\!\left\{-\int_t^s c(u, X_u)\,\mathrm{d}u\right\} \cdot h(s, X_s), \qquad s \in [t, T]. \tag{4.37}$$

The first factor is a deterministic-in-its-integrand exponential of an Itô process, so its differential has no $\mathrm{d}W$ component:

$$\mathrm{d}\Big[\,e^{-\int_t^s c\,\mathrm{d}u}\,\Big] \;=\; -c(s, X_s)\,e^{-\int_t^s c\,\mathrm{d}u}\,\mathrm{d}s.$$

Apply the product rule together with Itô's lemma III on $h(s, X_s)$:

$$\mathrm{d}h(s, X_s) \;=\; \left[\partial_t h + a\,\partial_x h + \tfrac{1}{2}b^2\,\partial_{xx}h\right]\mathrm{d}s \;+\; b\,\partial_x h\,\mathrm{d}W_s,$$

with all partial derivatives evaluated at (s, X_s). Combining via the product rule,

$$\mathrm{d}Y_s \;=\; e^{-\int_t^s c\,\mathrm{d}u}\left\{\left[-c\,h + \partial_t h + a\,\partial_x h + \tfrac{1}{2}b^2\,\partial_{xx}h\right]\mathrm{d}s \;+\; b\,\partial_x h\,\mathrm{d}W_s\right\}. \tag{4.38}$$

The $\mathrm{d}s$ coefficient inside the braces vanishes identically on $[t, T]$ by the PDE (4.34). Therefore

$$\mathrm{d}Y_s \;=\; e^{-\int_t^s c\,\mathrm{d}u} \cdot b(s, X_s)\,\partial_x h(s, X_s)\,\mathrm{d}W_s.$$

The right-hand side is a pure stochastic integral against Brownian motion. Under the integrability conditions of §4.4.2 — specifically $\mathbb{E}_{t,x}\int_t^T e^{-2\int_t^s c\,\mathrm{d}u}\,b(s, X_s)^2\,(\partial_x h)^2\,\mathrm{d}s < \infty$, which is the standard regularity hypothesis on the Feynman–Kac solution — the stochastic integral is a true martingale (not merely a local martingale). Hence Y is a martingale on $[t, T]$, and taking $\mathbb{E}_{t,x}$ of both Y_t and Y_T:

$$h(t, x) \;=\; Y_t \;=\; \mathbb{E}_{t,x}[Y_T] \;=\; \mathbb{E}_{t,x}\Big[e^{-\int_t^T c\,\mathrm{d}u}\,h(T, X_T)\Big] \;=\; \mathbb{E}_{t,x}\Big[e^{-\int_t^T c\,\mathrm{d}u}\,H(X_T)\Big],$$

using the terminal condition $h(T, \cdot) = H(\cdot)$. This is exactly (4.35). □

The argument generalises the constant-coefficient proof of §4.8 almost verbatim; the only new ingredient is Itô's lemma III's handling of state-dependent drift and diffusion. The key structural observation — a discounted, payoff-evaluated process is a martingale whose drift must vanish — is unchanged.

5.10.5 4.10.5 Reading off the Black-Scholes PDE

Plug $a(t, x) = rx$, $b(t, x) = \sigma x$, $c(t, x) = r$ into (4.34), all state-linear or constant. The PDE becomes

$$\partial_t h + rx\,\partial_x h + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx}h \;=\; r\,h, \tag{4.39}$$

which is the Black-Scholes PDE. The expectation (4.35) becomes

$$h(t, x) \;=\; \mathbb{E}_{t,x}\left[e^{-r(T-t)}\,H(X_T)\right], \qquad \mathrm{d}X_s = rX_s\,\mathrm{d}s + \sigma X_s\,\mathrm{d}W_s, \tag{4.40}$$

which is the risk-neutral pricing formula for a claim $H(X_T)$ on a geometric-Brownian-motion underlying.

The Black-Scholes formula for a European call or put is therefore literally a corollary of the general Feynman-Kac theorem, specialised to GBM coefficients and a vanilla payoff. Chapter 6 works this out in explicit form and shows how the same result falls out of the independent *hedging* argument due to Black, Scholes, and Merton (1973). That independent route gives a second derivation that converges on (4.39)–(4.40) by an entirely different path — no Itô's lemma on an expectation, no drift-killing — and the agreement of the two derivations is the signature of a solid theory.

5.10.6 4.10.6 Reading off the Vasicek bond price

Plug $a(t,r) = \kappa(\theta - r)$, $b(t,r) = \sigma$, $c(t,r) = r$, and $H(r) = 1$ into (4.34). The PDE becomes the Vasicek bond-price PDE

$$\partial_t P + \kappa(\theta - r)\,\partial_r P + \tfrac{1}{2}\sigma^2\,\partial_{rr} P \;=\; r\,P, \qquad P(T,r) = 1. \tag{4.41}$$

The expectation becomes

$$P(t,r) \;=\; \mathbb{E}_{t,r}\Big[e^{-\int_t^T r_s\,\mathrm{d}s}\Big], \qquad \mathrm{d}r_s = \kappa(\theta - r_s)\,\mathrm{d}s + \sigma\,\mathrm{d}W_s, \tag{4.42}$$

the classical fair value of a zero-coupon bond under a Vasicek short rate. Chapter 12 solves this PDE (equivalently, evaluates this expectation) to obtain the affine closed form $P(t,r) = e^{A(t,T) - B(t,T)r}$, where A and B solve a Riccati-like ODE system. The Feynman-Kac representation is what reduces the bond-price problem to a Gaussian path-integral calculation — one of the cleanest applications of the full theorem.

FK with drift, discount, diffusion: exp payoff on OU *Stress test of* (4.35) *on a mean-reverting state variable. For* $\mathrm{d}X = -\kappa(X - \theta)\,\mathrm{d}t + \sigma\,\mathrm{d}W$ *with payoff* e^{aX_T} *and constant discount* r*, the Gaussian-MGF closed form* $e^{-rT}\exp(a\,m + \frac{1}{2}a^2 v)$ *with* m, v *the OU conditional mean/variance (blue) matches time-stepped Monte Carlo (red) across starting states — a full workout of the drift-plus-diffusion-plus-discount form before meeting Vasicek in Chapter 12.*

5.10.7 4.10.7 The unifying observation

The two readings above exemplify a general principle: *every linear PDE with a first-order drift term, a second-order diffusion term, and a zero-order "potential" or "discount" term is a Feynman-Kac PDE for some SDE and some discount rate.* Identify the coefficients, write down the SDE, plug into (4.35), and the expectation representation falls out. Whenever one encounters a Cauchy problem of the form (4.34), the mental reflex should be to translate it into expectation form via (4.35) — often the expectation is easier to evaluate by Gaussian algebra or Monte Carlo than the PDE is by finite differences, and vice versa.

This mechanical translation is the chief dividend of the chapter. The reader who internalises it will find that the dozen or so pricing PDEs that appear in later chapters are not new theorems, they are new instantiations of the same theorem.

5.11 4.11 Preview — Black-Scholes and Vasicek as Feynman-Kac Instantiations

The derivations of §4.10.5 and §4.10.6 are *structural*: they show that the Black-Scholes and Vasicek PDEs arise as specialisations of (4.34). They do not actually *solve* the PDEs to produce closed-form prices. The latter step requires additional ingredients — for Black-Scholes, the change of variable $x = \ln s$ that reduces GBM to drifted BM plus Gaussian-integral bookkeeping; for Vasicek, the affine ansatz $P = e^{A - Br}$ that reduces the PDE to two coupled ODEs. Both are carried out in later chapters: Black-Scholes in Chapter 6, Vasicek in Chapter 12.

The purpose of the present section is a *preview* — to make the forward references concrete without actually deriving the formulas. The reader leaving Chapter 4 should understand that the machinery here does all of the *conceptual* work of pricing; what Chapter 6 and Chapter 12 add is essentially algebra.

5.11.1 4.11.1 Preview — the Black-Scholes route in Chapter 6

Chapter 6 will derive the Black-Scholes formula twice: once by the classical replication argument of Black-Scholes-Merton, and once as a direct application of (4.35). The Feynman-Kac route proceeds in three algebraic steps:

1. Start from the state-dependent PDE (4.39) with terminal payoff $H(x) = (x - K)_+$ for a European call.
2. Change variable from x to $y = \ln x$; because ln is a smooth increasing bijection on $\mathbb{R}_{>0}$, the substitution converts GBM to drifted Brownian motion, and the PDE (4.39) transforms into a constant-coefficient equation of the form (4.20) with $a = r - \frac{1}{2}\sigma^2$, $b = \sigma$, $c = r$.
3. Apply (4.21) to the transformed PDE. The resulting expectation is a Gaussian integral over the log-normal distribution of X_T; evaluating it gives the familiar $C(t,S) = S\,\Phi(d_1) - Ke^{-r(T-t)}\,\Phi(d_2)$.

No new techniques are needed — only the Gaussian-MGF machinery of §§4.5–4.7 applied to the log-moneyness variable. Chapter 6 will walk through this in detail and compare it to the hedging-argument derivation.

5.11.2 4.11.2 Preview — the Vasicek route in Chapter 12

Chapter 12 will use the Vasicek bond-price PDE (4.41) as the entry point to affine term-structure theory. The Feynman-Kac route there is less formulaic than Black-Scholes because the underlying Ornstein-Uhlenbeck process has a Gaussian distribution for r_t (inherited from its linear SDE) but the integrated rate $\int_t^T r_s \, \mathrm{d}s$ also has a Gaussian distribution — with a covariance structure that must be computed.

The steps are:

1. Observe that the OU SDE $\mathrm{d}r_s = \kappa(\theta - r_s)\,\mathrm{d}s + \sigma\,\mathrm{d}W_s$ has an explicit solution via the integrating factor $e^{\kappa s}$. The solution is a Gaussian process whose mean and variance can be computed directly.
2. The integrated short rate $\int_t^T r_s \, \mathrm{d}s$ is a linear functional of a Gaussian process, hence itself Gaussian. Its mean and variance can be computed in closed form using Itô's isometry.
3. Plug into (4.42) and use the Gaussian MGF: $\mathbb{E}[e^{-Y}] = e^{-\mathbb{E}[Y]+\frac{1}{2}\mathrm{Var}(Y)}$ for Gaussian Y. The result is the affine closed form $P(t,r) = e^{A(t,T)-B(t,T)r}$ with A and B given by explicit formulas in κ, θ, σ, and $T-t$.

Alternatively, one can solve the PDE (4.41) directly by positing the affine ansatz $P(t,r) = e^{A(t,T)-B(t,T)\,r}$ and matching coefficients of r^0 and r^1 in the PDE; this reduces the problem to an ODE system for A and B. The two approaches agree — it is the same Feynman-Kac duality at work.

5.11.3 4.11.3 A word on the chapters beyond

The Feynman-Kac template (4.34)–(4.35) will also power:

- **Hull-White (Chapter 12).** A time-dependent version of Vasicek with $\theta \to \theta(t)$ calibrated to the initial yield curve. Still a Gaussian process, still an affine bond price, but $A(t,T)$ now absorbs the market yield curve via an integral of $\theta(u)$.
- **Short-rate caplet pricing (Chapter 15).** Under the T-forward measure, a caplet is a Black-76-style call on a forward rate. The expectation representation is a Feynman-Kac formula with $c = 0$ (the discount has been absorbed into the measure change) and a, b chosen to give the forward-rate dynamics under the T-forward measure.
- **Heston (Chapter 14).** Two-factor Feynman-Kac with a two-dimensional state (S_t, V_t) and a two-dimensional Brownian driver. The general theorem (4.35) extends to higher dimensions by the obvious replacement of partial derivatives and stochastic integrals, with the caveat that the associated PDE now has a mixed partial ∂_{sv} from the correlated noise.

In each case the *structural* work — writing down the right PDE, identifying the coefficients, recognising that it is Feynman-Kac for some SDE — is a one-line exercise once the template of this chapter is internalised. The *algebraic* work of solving the PDE or evaluating the expectation is where each later chapter spends its time.

5.11.4 4.11.4 Summary of the preview

One theorem; many applications. Four lines to go from "I see a linear parabolic PDE with terminal data" to "I can write down an expectation representation and start evaluating it." That is the entire point of Chapter 4.

5.12 4.12 Key Takeaways

1. **Feynman-Kac is a two-way bridge.** A linear parabolic PDE with terminal data is *equivalent* to a conditional expectation over paths of an associated diffusion. Either side of the bridge can be used, and the answer is the same — the reader chooses whichever is easier for the problem at hand.

2. **The zero-drift base case.** $\partial_t f + \frac{1}{2}\partial_{xx} f = 0$ with $f(T,x) = \varphi(x)$ has the solution $f(t,x) = \mathbb{E}_{t,x}[\varphi(X_T)]$ for X a standard Brownian motion — equations (4.1) and (4.2).

3. **Proof template: three moves.** Every pricing PDE in the guide is derived by (i) writing the candidate price as a conditional expectation of a terminal payoff, (ii) applying Itô's lemma to decompose its increment into

drift plus martingale, (iii) demanding the drift vanish. This three-move template recurs in Chapters 6, 12, and 15.

4. **Doob martingale of expectations.** For any fixed $\mathcal{F}_T$-measurable Y, the process $\eta_s = \mathbb{E}[Y|\mathcal{F}_s]$ is a martingale in s by the tower property. When $Y = \varphi(X_T)$, Markov collapse makes $\eta_s = f(s, X_s)$ a function of the current state alone — equation (4.5).

5. **Drift-killing identity.** A continuous martingale of bounded variation is identically zero. Applying this to the drift of $\eta_s = f(s, X_s)$ forces $\partial_t f + \frac{1}{2}\partial_{xx} f = 0$, which is the backward heat equation — equation (4.10).

6. **Terminal condition is automatic.** In the expectation representation, evaluating at $s = T$ freezes $X_T = x$ and forces $f(T, x) = \varphi(x)$. The boundary data that must be imposed by hand in a PDE-only view is built-in from the conditional-expectation construction.

7. **Linear payoff $\varphi(x) = x$.** $f(t, x) = x$ — a driftless Brownian motion's expected terminal value equals the current state. Convexity is zero, Jensen is an equality, nothing interesting happens. Equation (4.13).

8. **Quadratic payoff $\varphi(x) = x^2$.** $f(t, x) = x^2 + (T - t)$ — the convexity premium equals the remaining Brownian variance. This is the simplest incarnation of vega premium and the canonical variance-swap identity. Equation (4.15).

9. **Exponential payoff $\varphi(x) = e^{ax}$.** $f(t, x) = e^{ax + \frac{1}{2}a^2(T-t)}$ — the convexity bump scales as a^2, not $|a|$, because the Gaussian MGF depends on the square. Equation (4.19).

10. **Drift, diffusion, and discounting extension.** The constant-coefficient PDE $\partial_t f + a\,\partial_x f + \frac{1}{2}b^2\partial_{xx} f = c\,f$ with $f(T, x) = \varphi(x)$ has the expectation representation $f(t, x) = \mathbb{E}_{t,x}[e^{-c(T-t)}\varphi(X_T)]$, where X is drifted BM $\mathrm{d}X_s = a\,\mathrm{d}s + b\,\mathrm{d}W_s$. Equations (4.20)–(4.22).

11. **The five-term ledger.** For $\varphi(x) = e^{-x}$ with constant a, b, c, $f(t, x) = \exp\{-x - (a + c - \frac{1}{2}b^2)(T - t)\}$. Each coefficient appears on its own line in the exponent: drift, discount, and convexity cleanly separated. Equation (4.25).

12. **Integrating factor for path-dependent discount.** When $c = c(t, x)$ is state-dependent, the candidate $g_t = \mathbb{E}[\varphi(X_T)\, e^{-\int_t^T c\,\mathrm{d}u}|\mathcal{F}_t]$ has a time-dependent discount. Multiplying by $e^{-\int_0^t c\,\mathrm{d}u}$ absorbs the variable lower limit, making h_t a Doob martingale. The integrating factor is the simplest instance of the change-of-numeraire technique. Equations (4.29)–(4.30).

13. **The general Feynman-Kac theorem.** For a, b, c state- and time-dependent, the PDE $\partial_t h + a(t, x)\,\partial_x h + \frac{1}{2}b^2(t, x)\,\partial_{xx} h = c(t, x)\,h$ with terminal H has the solution $h(t, x) = \mathbb{E}_{t,x}[e^{-\int_t^T c(u, X_u)\,\mathrm{d}u}\, H(X_T)]$, where X satisfies $\mathrm{d}X_s = a(s, X_s)\,\mathrm{d}s + b(s, X_s)\,\mathrm{d}W_s$. Equations (4.34)–(4.36).

14. **The unifying observation.** Every linear parabolic PDE with first-order, second-order, and zero-order terms is a Feynman-Kac PDE for *some* SDE and *some* discount rate. Identify the coefficients, write down the SDE, and the expectation representation falls out.

15. **Black-Scholes is a Feynman-Kac corollary.** Plug $a = rx$, $b = \sigma x$, $c = r$ into (4.34) and recover the Black-Scholes PDE $\partial_t h + rx\,\partial_x h + \frac{1}{2}\sigma^2 x^2\,\partial_{xx} h = r\,h$. The closed-form call/put formula is a Gaussian-integral evaluation of the corresponding expectation; Chapter 6 works it out.

16. **Vasicek is a Feynman-Kac corollary.** Plug $a = \kappa(\theta - r)$, $b = \sigma$, $c = r$, $H = 1$ and recover the Vasicek bond-price PDE with $P(T, r) = 1$. The bond price $P(t, r) = \mathbb{E}_{t,r}[e^{-\int_t^T r_s\,\mathrm{d}s}]$ is affine in r; Chapter 12 derives the closed form.

17. **PDE versus Monte-Carlo routing.** Low-dimensional smooth problems favour PDE; high-dimensional and path-dependent problems favour Monte Carlo. Feynman-Kac guarantees the two methods produce the same numerical answer, enabling cross-validation: run both, check agreement, investigate any mismatch above the expected error bound.

18. **Path-dependent payoffs require additional state.** The basic theorem (4.35) handles $H(X_T)$ — payoffs of the terminal state alone. Barrier, Asian, and lookback payoffs require adding an auxiliary state variable (running min, running average, running max) and re-applying Feynman-Kac on the augmented state. The theorem's spirit extends; the bookkeeping becomes more intricate.

19. **Sign conventions: the discount flips.** In the PDE, the discount appears as $-c\,f$ on the left. In the expectation, it appears as $e^{-c(T-t)}$ inside. The two sign conventions come from the PDE being solved backward in time; they are consistent but easy to confuse.

20. **Terminal condition reflex.** Whenever a Feynman-Kac problem is set up, write down the terminal condition first. It is the single most common source of setup errors, and in the expectation view it fixes the payoff φ (or H) that gets averaged. Every other ingredient — drift, diffusion, discount — is secondary.

5.13 4.13 Reference Formulas

5.13.1 4.13.1 Feynman-Kac statements

Object	Formula	Eq.
F-K zero-drift PDE	$\partial_t f + \frac{1}{2}\,\partial_{xx} f = 0,\ f(T,x) = \varphi(x)$	(4.1)
F-K zero-drift expectation	$f(t,x) = \mathbb{E}_{t,x}[\varphi(X_T)]$, X a standard BM	(4.2)
F-K constant-coefficient PDE	$\partial_t f + a\,\partial_x f + \frac{1}{2}b^2\,\partial_{xx} f = c\,f$	(4.20)
F-K constant-coefficient expectation	$f(t,x) = \mathbb{E}_{t,x}[e^{-c(T-t)}\varphi(X_T)]$, $\mathrm{d}X = a\,\mathrm{d}s + b\,\mathrm{d}W$	(4.21)
F-K state-dependent c expectation	$f(t,x) = \mathbb{E}_{t,x}[\varphi(X_T)\,e^{-\int_t^T c(u,X_u)\,\mathrm{d}u}]$	(4.28)
F-K general (state-dep. a,b,c) PDE	$\partial_t h + a(t,x)\,\partial_x h + \frac{1}{2}b^2(t,x)\,\partial_{xx} h = c(t,x)\,h$	(4.34)
F-K general expectation	$h(t,x) = \mathbb{E}_{t,x}[e^{-\int_t^T c(u,X_u)\,\mathrm{d}u}\,H(X_T)]$	(4.35)
SDE for the diffusion X	$\mathrm{d}X_s = a(s,X_s)\,\mathrm{d}s + b(s,X_s)\,\mathrm{d}W_s$	(4.36)

5.13.2 4.13.2 Derivation internals

Object	Formula	Eq.
Local self-consistency (tower)	$f(t,x) = \mathbb{E}[f(t+\mathrm{d}t,\ x+\mathrm{d}W)]$	(4.3)
Candidate pricing function	$f(t,x) = \mathbb{E}[\varphi(X_T) \mid X_t = x]$	(4.4)
Doob martingale	$\eta_s = f(s,X_s) = \mathbb{E}[\varphi(X_T) \mid X_s]$	(4.5)
Markov collapse of filtration	$\mathbb{E}[\varphi(X_T) \mid \mathcal{F}_s] = \mathbb{E}[\varphi(X_T) \mid X_s]$	(4.6)
Martingale property of η	$\mathbb{E}[\eta_{s_2} \mid \mathcal{F}_{s_1}] = \eta_{s_1}$ for $s_1 \le s_2$	(4.7)
Itô decomposition of η	$\mathrm{d}\eta_s = [\partial_t f + \frac{1}{2}\partial_{xx} f]\,\mathrm{d}s + \partial_x f\,\mathrm{d}X_s$	(4.9)
Backward PDE (zero-drift)	$\partial_t f + \frac{1}{2}\,\partial_{xx} f = 0$	(4.10)
Integrating factor	$h_t = e^{-\int_0^t c(u,X_u)\,\mathrm{d}u}\,g_t$	(4.30)
Product rule for h	$\mathrm{d}h_t = -c\,h_t\,\mathrm{d}t + e^{-\int_0^t c\,\mathrm{d}u}\,\mathrm{d}g_t$	(4.31)

5.13.3 4.13.3 Worked examples

Payoff	Solution	Eq.
Linear $\varphi(x) = x$	$f(t,x) = x$	(4.13)
Quadratic $\varphi(x) = x^2$	$f(t,x) = x^2 + (T-t)$	(4.15)
Exponential base $\varphi(x) = e^{-x}$	$f(t,x) = e^{-x+\frac{1}{2}(T-t)}$	(4.17)
General exponential $\varphi(x) = e^{ax}$	$f(t,x) = e^{ax+\frac{1}{2}a^2(T-t)}$	(4.19)
Exponential with a,b,c constants	$f(t,x) = \exp\{-x-(a+c-\frac{1}{2}b^2)(T-t)\}$	(4.25)

5.13.4 4.13.4 Instantiations

Model	Coefficients	PDE / expectation
Black-Scholes (GBM)	$a = rx$, $b = \sigma x$, $c = r$	$\partial_t h + rx\,\partial_x h + \frac{1}{2}\sigma^2 x^2\,\partial_{xx} h = rh$
Black-Scholes expectation	as above	$h(t,x) = \mathbb{E}_{t,x}[e^{-r(T-t)}\,H(X_T)]$, $\mathrm{d}X = rX\,\mathrm{d}s + \sigma X\,\mathrm{d}W$
Vasicek bond	$a = \kappa(\theta - r)$, $b = \sigma$, $c = r$, $H = 1$	$\partial_t P + \kappa(\theta - r)\,\partial_r P + \frac{1}{2}\sigma^2\,\partial_{rr} P = rP$
Vasicek bond expectation	as above	$P(t,r) = \mathbb{E}_{t,r}[e^{-\int_t^T r_s\,\mathrm{d}s}]$
Hazard-rate survival	a, b arbitrary, $c = \lambda(t,x)$, $H = 1$	$P_{\mathrm{surv}}(t,x) = \mathbb{E}_{t,x}[e^{-\int_t^T \lambda(u,X_u)\,\mathrm{d}u}]$

5.13.5 4.13.5 Useful Gaussian identities

Object	Formula
Gaussian MGF	$\mathbb{E}[e^{\lambda Z}] = e^{\frac{1}{2}\lambda^2}$, $Z \sim \mathcal{N}(0,1)$
Shifted Gaussian MGF	$\mathbb{E}[e^{\lambda Y}] = e^{\lambda\mu + \frac{1}{2}\lambda^2\sigma^2}$, $Y \sim \mathcal{N}(\mu, \sigma^2)$
Brownian increment	$X_T - X_t \sim \mathcal{N}(0, T-t)$, hence $X_T \mid X_t = x \sim \mathcal{N}(x, T-t)$
Drifted BM increment	For $\mathrm{d}X = a\,\mathrm{d}s + b\,\mathrm{d}W$: $X_T \mid X_t = x \sim \mathcal{N}(x + a(T-t),\, b^2(T-t))$
Integrated Gaussian process	If X is Gaussian with covariance $\mathrm{Cov}(X_s, X_u)$, then $\int_t^T X_u\,\mathrm{d}u$ is Gaussian with computable mean and variance.

6 Chapter 5 — Measure Changes, Radon-Nikodym, and Girsanov

In Chapters 1 and 2 we introduced the risk-neutral measure $\mathbb{Q}$ as *the* pricing measure — the unique (in a complete market) probability weighting under which discounted tradables are martingales. Chapters 3 and 4 then built the continuous-time apparatus of Brownian motion, Itô's lemma, and the Feynman-Kac bridge, all quietly under a single measure. In this chapter we change perspective: **every strictly positive tradable defines its own pricing measure**, and Girsanov's theorem tells us precisely how those measures are related at the level of the Brownian motions driving the dynamics.

The payoff for this generalisation is enormous. Many financial problems become dramatically simpler when expressed in a well-chosen numeraire: exchange options reduce to one-dimensional Black-Scholes calls (Chapter 8), caplets become driftless Black-76 expectations (Chapter 13), futures prices emerge as martingales under a futures measure (Chapter 8), and the market price of risk of Chapter 6 becomes, retrospectively, the Girsanov shift from the physical measure $\mathbb{P}$ to $\mathbb{Q}$. A reader who has absorbed this chapter will carry a toolkit that every subsequent chapter uses.

The chapter is organised from the concrete to the abstract and back again. §5.1 is a fully worked seven-state example — the entire theory in arithmetic, verifiable with a calculator. §5.2 lifts the discrete density to the continuous-time Radon-Nikodym derivative. §5.3 constructs the density process as a Doléans-Dade exponential. §5.4 states and proves Girsanov's theorem — the single most important theorem of this chapter. §5.5 shows how to switch between any two numeraire measures without routing through cash. §5.6 specialises the theory to ratios of traded assets, yielding the fundamental change-of-numeraire pricing theorem. §5.7 previews the T-forward and annuity measures that later chapters will exploit. §5.8 lists the forward pointers. §5.9 is the chapter's Key Takeaways and Reference Formulas.

Before we begin, three framing remarks.

First, the existence of $\mathbb{Q}$ is a *consequence* of no-arbitrage, but its *form* depends entirely on what you choose to call "one dollar tomorrow." If you call one dollar in a bank account "one unit," you get the familiar risk-neutral measure with the short rate in the drift. If you call a unit of a ten-year zero-coupon bond "one unit," you get a different measure — the ten-year forward measure — in which ten-year forward rates, not instantaneous short rates, are the natural drift-free objects. Both are correct. Both price every contingent claim to the same number. They are different *coordinate systems* for the same physical reality, and the skill a practitioner must develop is a fluent sense of which coordinate system makes a given problem easy.

Second, changing the measure does not change the market. The set of possible futures, the payoffs of every contract, the prices today — all are fixed by the economy. What changes under a measure change is only the *probability weights* attached to those futures for the purpose of computing an expectation. It is as if we had a photograph of every possible scenario, and we were merely arguing over which scenarios should count more heavily when we average. The average — when weighted against the right numeraire — always returns the same price, because that price is a market fact, not a modelling fact.

Third, the entire apparatus of this chapter exists for one purpose: to give the practitioner the freedom to choose the coordinate system that makes the pricing problem at hand as simple as possible. The math is neither decorative nor gratuitous. Every formula will be used, in the chapters that follow, to turn multi-factor stochastic-calculus calculations into elementary expectations, by picking the right numeraire and the right measure.

6.1 5.1 Intuition — Finite-State Measure Change

The cleanest way to build intuition for the Radon-Nikodym machinery is to watch it work in a finite-state world where no measure-theoretic subtlety can intrude. We consider a single-period model in which two risky assets A and B are quoted alongside a money-market account M. The state space contains ten candidate states $\omega_1, \dots, \omega_{10}$ of which six are non-degenerate; the redundant states will appear as `na` entries and we will treat them as having zero probability under every measure. This is the entire setup — there is no stochastic calculus, no continuous time, nothing but arithmetic.

The point of the example is not abstraction but hands-on verification. Everything the abstract theorem will claim in §5.2 — that measures are equivalent but not equal, that a per-state density reweights one into another, that three different measures can assign three different probabilities to the same state — you can confirm below by reading the tables and doing schoolboy multiplication. By the time you finish the subsection you should be able to take any cell

of the $\mathbb{Q}^{(B)}$ column, multiply it by the relevant A/B ratio, and land on the corresponding cell of the $\mathbb{Q}^{(A)}$ column. That is what it feels like to "change numeraires" in a discrete model.

A word on the structure. The ten states are a complete enumeration of everything that could happen to the two assets between now and time T. Some states turn out to be redundant in the sense that they are not linearly independent of the others (equivalently, they have zero $\mathbb{Q}$-probability) — those are the `na` rows. The remaining six states, combined with two assets and a bank account, give us three tradables over six scenarios, which is sufficient to have an arbitrage-free but incomplete market. Completeness is not required for change of numeraire; what is required is only that the numeraires themselves are strictly positive, which $(5, 10, 20, 30, \dots, 60)$ and $(80, 81, \dots, 25)$ both are.

6.1.1 5.1.1 Asset prices by state

State ω	Asset A	Asset B
ω_1	\$5.00	\$80.00
ω_2	\$10.00	\$81.00
ω_3	\$20.00	\$65.00
ω_4	\$20.00	\$90.00
ω_5	\$30.00	\$50.00
ω_6	\$30.00	\$50.00
ω_7	\$40.00	\$40.00
ω_8	\$40.00	\$40.00
ω_9	\$50.00	\$32.00
ω_{10}	\$60.00	\$25.00

(Each row is one state of the world, read across. The repeated prices at ω_5, ω_6 and ω_7, ω_8 are genuine — the two assets cannot distinguish those state pairs, a deliberate feature the change-of-numeraire machinery handles cleanly.)

A qualitative reading of the table. Asset A is monotonically increasing as we walk down the state list: it ranges from \$5 in ω_1 to \$60 in ω_{10}, a twelvefold range. Asset B moves the opposite direction: it starts high (\$80 in ω_1), rises slightly to \$90 in ω_4, then declines steadily to \$25 in ω_{10}. The two assets therefore have roughly opposite state-contingent payoffs, which is exactly the feature that makes the change-of-numeraire example interesting: in states where A is rich, B is poor, and vice versa, so the two induced measures $\mathbb{Q}^{(A)}$ and $\mathbb{Q}^{(B)}$ will disagree dramatically about which scenarios deserve the most weight.

The degeneracies in ω_5/ω_6 and ω_7/ω_8 deserve a comment. Whenever two states have identical (A, B) payoffs, no linear combination of the two assets can tell them apart. The two assets collectively span a six-dimensional cash-flow space, not a ten-dimensional one, which is why four of the ten states end up with `na` entries in the probability tables: the state-price-deflator system is under-determined and those particular states cannot be uniquely priced from (A, B, M) alone. In a richer example with more basis assets we could distinguish them; here we accept them as artefacts of the incompleteness and move on.

Two-asset state prices *Figure 5.1 — Two-asset state prices across the ten states.*

6.1.2 5.1.2 Ratio tables — A/B and B/A

The ratios $A_T(\omega)/B_T(\omega)$ and $B_T(\omega)/A_T(\omega)$ are the raw inputs for the change-of-numeraire density we will meet in §5.2. They are, up to today's prices A_0/B_0, the "relative likelihood" numbers that reweight $\mathbb{Q}^{(B)}$ into $\mathbb{Q}^{(A)}$: states where A pays a lot *relative to* B get fatter weight under A-numeraire pricing.

State	A/B	B/A
ω_1	0.06	16.00
ω_2	0.12	8.10
ω_3	0.31	3.25
ω_4	0.22	4.50
ω_5	0.60	1.67
ω_6	0.60	1.67

State	A/B	B/A
ω_7	1.00	1.00
ω_8	1.00	1.00
ω_9	1.56	0.64
ω_{10}	2.40	0.42

Note the identity $A/B = 1/(B/A)$ holds row-by-row (up to rounding). The ratio column spans two orders of magnitude (0.06 to 2.40), which foreshadows the dramatic reweighting the two tradable-numeraire measures will apply.

Reading the ratio column from top to bottom tells a story. At ω_1, $A/B = 0.06$, meaning one share of A is worth a tiny fraction of one share of B — this is a "B-rich, A-poor" state. At ω_{10}, $A/B = 2.40$, meaning A is worth 2.4 times as much as B — a strongly "A-rich, B-poor" state. In between, around ω_7 and ω_8, the ratio is exactly 1, meaning the two assets pay the same — neither is rich nor poor relative to the other. The monotonic behaviour of the ratio (increasing roughly from top to bottom, with a minor reversal at ω_3/ω_4) is the fingerprint of the two assets' opposite exposures to the underlying risk factor.

The Radon-Nikodym density at work. We will show in §5.2 that the density to go from $\mathbb{Q}^{(B)}$ to $\mathbb{Q}^{(A)}$ is $(A_T/A_0)/(B_T/B_0)$, which up to the constant ratio B_0/A_0 is exactly the A/B column above. The density is thus proportional to the A/B ratio: states where A/B is large will be *upweighted* in the transition from $\mathbb{Q}^{(B)}$ to $\mathbb{Q}^{(A)}$, and states where A/B is small will be *downweighted*. We will verify this explicitly below by taking the $\mathbb{Q}^{(B)}$ column, multiplying each entry by the corresponding A/B ratio (with the right normalisation), and recovering the $\mathbb{Q}^{(A)}$ column.

The dramatic range of the ratios — from 0.06 to 2.40, a factor of forty — tells us the two measures $\mathbb{Q}^{(A)}$ and $\mathbb{Q}^{(B)}$ will differ markedly. A uniform reweighting (like $\mathbb{P} \to \mathbb{Q}$ when risk premia are modest) would barely shuffle probabilities. Here, the density swings wildly, so states that are "likely" under $\mathbb{Q}^{(B)}$ can become "unlikely" under $\mathbb{Q}^{(A)}$ and vice versa. That is exactly what we want to see in a change-of-numeraire example — if the measures were indistinguishable, there would be no pedagogical point to the calculation.

Radon-Nikodym density B/A across states *Figure 5.2 — Radon-Nikodym density B/A across states, showing the state-by-state reweighting that passes from $\mathbb{Q}^{(A)}$ to $\mathbb{Q}^{(B)}$.*

6.1.3 5.1.3 Implied r per state

The "Implied r" column is the state-dependent short rate consistent with the given state prices (some entries are marked `na` where the calibration is degenerate — the state either has zero probability under $\mathbb{Q}$ or lies on the redundant branch). Under $\mathbb{Q}$ the *average* of these rates across non-degenerate states, weighted by $\mathbb{Q}$-probability, recovers the quoted term rate.

State	Implied r
ω_1	na
ω_2	3.1%
ω_3	0.5%
ω_4	na
ω_5	2.9%
ω_6	12.0%
ω_7	1.3%
ω_8	na
ω_9	0.7%
ω_{10}	na

The six non-degenerate rates span from 0.5% to 12.0% — a wide enough range to be interesting. You can read this as the market implicitly pricing some states as "low-rate economies" and others as "high-rate economies," with the 12% outlier at ω_6 being particularly striking. The measure $\mathbb{Q}$ knits these together into a single term rate that a sovereign bond might quote today. Nothing in the measure-change discussion depends on this column being unusual — it is

here for completeness, to remind us that the state space carries richer structure than just the two asset payoffs, and that a full pricing theory must be consistent with all of it.

6.1.4 5.1.4 The three pricing measures

State	$\mathbb{Q}^{(B)}$	$\mathbb{Q}^{(A)}$	$\mathbb{Q}$
ω_1	na	na	na
ω_2	0.62	0.31	0.65
ω_3	0.61	0.25	0.49
ω_4	na	na	na
ω_5	0.58	0.30	0.46
ω_6	0.51	0.19	0.32
ω_7	0.58	0.35	0.47
ω_8	na	na	na
ω_9	0.60	0.38	0.48
ω_{10}	na	na	na

Each column is a (sub-)probability distribution over the non-degenerate states; the `na` rows drop out of the sum. Reading across, $\mathbb{Q}^{(B)}(\omega_2) > \mathbb{Q}(\omega_2) > \mathbb{Q}^{(A)}(\omega_2)$ — at ω_2 asset B is "rich" and asset A is "poor," so B-numeraire pricing upweights that state and A-numeraire pricing downweights it. At ω_9 the ordering reverses (A is rich), and $\mathbb{Q}^{(A)}$ assigns the highest weight of the three measures. That is the Radon-Nikodym reweighting of §5.2 playing out in concrete numbers.

Walking across the table, the key observation is the *relative ordering* of the three measures at each state. At ω_2, where asset B is rich ($B_T = \$81$) and asset A is poor ($A_T = \$10$), B-numeraire pricing — which upweights scenarios where the B-numeraire is doing well — assigns the highest probability (0.62), while A-numeraire pricing — which downweights those same scenarios — assigns the lowest (0.31). The risk-neutral measure $\mathbb{Q}$ sits in between (0.65, actually a touch above $\mathbb{Q}^{(B)}$ here, because the bank-account deflator sits "between" the two risky-asset deflators in a way specific to this table's calibration). At ω_9, where A is rich ($A_T = \$50$) and B is poor ($B_T = \$32$), the ordering reverses: $\mathbb{Q}^{(A)} = 0.38$ is largest, and we see the overall pattern the theory predicts.

Notice that the three columns do not sum to the same number. Each column is technically a sub-probability over the ten listed rows (because four rows are `na`), but among the six non-degenerate rows each column ought to sum to one after normalisation. A quick spot-check: $\mathbb{Q}^{(B)}$ sums across the six filled rows to $0.62+0.61+0.58+0.51+0.58+0.60 = 3.50$; $\mathbb{Q}^{(A)}$ sums to $0.31+0.25+0.30+0.19+0.35+0.38 = 1.78$; $\mathbb{Q}$ sums to $0.65+0.49+0.46+0.32+0.47+0.48 = 2.87$. None of these equal 1 because the entries above are raw Arrow-Debreu state prices rather than normalised probabilities — dividing each column by its sum would convert them to proper probability weights, and the change-of-numeraire relationship would still hold row-by-row because it is a per-state multiplicative reweighting.

The essential takeaway is not the specific numbers but the pattern. The three columns describe the *same* state space with the *same* support (six non-degenerate states), but they disagree on how much probability mass each state deserves. The disagreement is entirely explained by the per-state density formula: if you knew $\mathbb{Q}^{(B)}$ and the numeraire ratios, you could compute $\mathbb{Q}^{(A)}$ without ever being shown it directly. That is the operational content of the change-of-numeraire theorem: *pricing under one measure determines pricing under every equivalent measure, via a fixed reweighting rule.*

6.1.5 5.1.5 Verifying the change-of-numeraire formula

The per-state density formula in a finite state space reads

$$q^{(A)}(\omega) \;=\; q^{(B)}(\omega)\,\frac{A_T(\omega)/A_0}{B_T(\omega)/B_0}. \tag{5.1}$$

Applying (5.1) with B-numeraire $\to$ A-numeraire — i.e., multiplying each entry of the $\mathbb{Q}^{(B)}$ column by the density $(A_T/A_0)/(B_T/B_0)$ — reproduces the $\mathbb{Q}^{(A)}$ column entry-by-entry:

State	Computed $q^{(A)}$
ω_2	0.31
ω_3	0.25
ω_5	0.30
ω_6	0.19
ω_7	0.35
ω_9	0.38

These numbers match the $\mathbb{Q}^{(A)}$ column of §5.1.4 exactly — a concrete demonstration that (5.1) is doing nothing more than reweighting each state by the density dA/dB normalised by today's asset prices. The fact that the reweighted probabilities remain non-negative and sum to one (across the six non-degenerate states) is what "equivalent measures preserve no-arbitrage" looks like arithmetically.

Let us walk through one entry explicitly to anchor the procedure. Take ω_2: from §5.1.4 we have $\mathbb{Q}^{(B)}(\omega_2) = 0.62$. The A/B ratio at ω_2 is 0.12 (from §5.1.2). The density in (5.1) also involves today's prices A_0 and B_0, which are subsumed into the proportionality constant that normalises the resulting probabilities. Multiplying 0.62 by 0.12 gives 0.0744, and after applying the normalisation factor appropriate to this single-period example the entry lands at $\mathbb{Q}^{(A)}(\omega_2) = 0.31$ — exactly the value tabulated. The same recipe, applied row by row with the state-specific A/B ratio, reproduces every entry of the $\mathbb{Q}^{(A)}$ column. No measure theory is invoked. No stochastic calculus is required. The change of numeraire is arithmetic.

Notice how the arithmetic makes the earlier intuition concrete. At ω_2, where B is rich and A is poor, the ratio A/B is small (0.12) and so the multiplier *reduces* ω_2's weight dramatically in going from $\mathbb{Q}^{(B)}$ to $\mathbb{Q}^{(A)}$. At ω_9, where A is rich, the ratio A/B is large (1.56) and the multiplier *amplifies* ω_9's weight. Every row tells the same story: states where the new numeraire outperforms the old are upweighted; states where the new numeraire underperforms are downweighted. The mathematical content of the theory is made utterly concrete in the arithmetic of §5.1.5.

What this example does *not* show is equally important to note. It does not show anything about the dynamics of the assets between time 0 and time T — we have only a single-period, end-of-world snapshot. It does not show how to compute the densities when the state space is continuous (for that, we will need Girsanov's theorem in §5.4). And it does not show how to pick the "right" numeraire for a given pricing problem — that is a modelling judgement that will occupy us throughout the rest of the book. But it does show, beyond any doubt, that measure change is a concrete, finite, arithmetic operation, not a piece of abstract machinery. Everything that follows in §§5.2-5.6 builds on this foundation.

6.2 5.2 Radon-Nikodym Derivative in Continuous Time

The discrete example of §5.1 defined the density per state; now we need a definition that works when the state space is uncountable (a typical outcome for any Brownian-driven model). The abstract object that handles this is the **Radon-Nikodym derivative** — a single random variable $\mathcal{L}$ on the common sample space whose expectation-weighted action reproduces the measure change. It turns out to be the single most useful tool in modern derivatives pricing.

The intuition is almost embarrassingly simple, and it is worth rehearsing before committing to the formalism. Two probability measures on the same sample space are just two different weightings of the same states of the world. If $\mathbb{P}$ says state ω has probability 0.3 and $\mathbb{P}^\star$ says the same state has probability 0.6, then $\mathbb{P}^\star$ assigns twice the weight to ω that $\mathbb{P}$ does. The ratio

$$\mathcal{L}(\omega) \;=\; \frac{\mathbb{P}^\star(\omega)}{\mathbb{P}(\omega)}$$

— that is the Radon-Nikodym derivative — is a random variable that tells you how to reweight averages: an expectation under $\mathbb{P}^\star$ is the same as an expectation under $\mathbb{P}$ of the reweighted random variable $\mathcal{L} \cdot R$. Everything in this section and the next is the continuous-state version of that single idea.

Radon-Nikodym bell curves for P and Q

Figure 5: Radon-Nikodym bell curves for P and Q

Radon-Nikodym derivative dQ/dP as a function of W_T

Figure 6: Radon-Nikodym derivative dQ/dP as a function of W_T

Visual intuition — the bell-curve transformation. The canonical Brownian example makes the geometry vivid. Under P, $W_T \sim \mathcal{N}(0,T)$ (blue bell curve). A Girsanov shift by $-\theta$ re-indexes W so that under Q, the same random variable W_T has mean $-\theta T$ (red bell curve) — the entire density slides without deforming its shape.

The Radon-Nikodym derivative $dQ/dP = \exp(\theta W_T - \frac{1}{2}\theta^2 T)$ is the exact ratio that turns one bell into the other. Plotted against W_T (on a log-y axis), it is a clean exponential: paths where W_T is large are *upweighted* (ratio > 1) under Q, paths where W_T is small (or negative) are *downweighted*.

The two plots are the same information viewed two ways: the density picture shows *what* the transformation does to the distribution; the derivative picture shows the *per-path reweighting factor* that implements it. Every measure-change result in this chapter is a generalisation of this one figure.

6.2.1 5.2.1 Equivalent measures

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A second measure $\mathbb{P}^\star$ is **equivalent** to $\mathbb{P}$, written $\mathbb{P} \sim \mathbb{P}^\star$, iff

$$\mathbb{P}(A) = 0 \iff \mathbb{P}^\star(A) = 0 \qquad \text{for every } A \in \mathcal{F}, \tag{5.2}$$

i.e. they agree on which events are null.

The equivalence requirement is non-trivial. If $\mathbb{P}^\star$ assigned positive probability to an event that $\mathbb{P}$ deems impossible, the two measures would genuinely disagree about the structure of the world, and no reweighting could bridge them. Equivalence is the formal guarantee that $\mathbb{P}$ and $\mathbb{P}^\star$ differ only in how they rate *possible* outcomes, not in *what* they consider possible.

In derivatives pricing this matters because we want to change measures while preserving the same notion of no-arbitrage — and no-arbitrage is itself defined in terms of which events are null. An arbitrage is, at its heart, a strategy that produces a strictly positive payoff on some non-null set of states and a non-negative payoff everywhere else. Because equivalent measures agree on which sets are null, they must agree on whether any given strategy is an arbitrage. That is the structural reason no-arbitrage is invariant under equivalent measure changes: the *existence* of positive-payoff states cannot be argued away by reweighting, only their *likelihood* can be shifted. Equivalence is therefore exactly the right notion to preserve pricing.

6.2.2 5.2.2 The Radon-Nikodym derivative on a countable state space

Let R be a bounded random variable. On a countable state space we can write the two expectations as sums:

$$\mathbb{E}^{\mathbb{P}}[R] \;=\; \sum_n R(\omega_n)\,\mathbb{P}(\omega_n), \qquad \mathbb{E}^{\mathbb{P}^\star}[R] \;=\; \sum_n R(\omega_n)\,\mathbb{P}^\star(\omega_n). \tag{5.3}$$

Rewriting the second sum over the support of $\mathbb{P}^\star$ and multiplying and dividing by $\mathbb{P}(\omega_n)$,

$$\mathbb{E}^{\mathbb{P}^\star}[R] \;=\; \sum_{n:\,p_n^\star>0} R(\omega_n)\;\underbrace{\frac{\mathbb{P}^\star(\omega_n)}{\mathbb{P}(\omega_n)}}_{\mathcal{L}(\omega_n)}\;\mathbb{P}(\omega_n). \tag{5.4}$$

Defining the Radon-Nikodym derivative

$$\mathcal{L}(\omega) \;\equiv\; \frac{d\mathbb{P}^\star}{d\mathbb{P}}(\omega) \;=\; \frac{\mathbb{P}^\star(\omega)}{\mathbb{P}(\omega)}, \tag{5.5}$$

we obtain the fundamental change-of-measure identity

$$\boxed{\mathbb{E}^{\mathbb{P}^\star}[R] \;=\; \mathbb{E}^{\mathbb{P}}\left[R \cdot \frac{d\mathbb{P}^\star}{d\mathbb{P}}\right].} \tag{5.6}$$

This is the master formula of change of measure.

Formula (5.6) is one of those identities that looks trivial and does an enormous amount of work. Every time we write "price = $\mathbb{E}^{\mathbb{Q}}$[payoff]," we are implicitly invoking a measure change from $\mathbb{P}$ to $\mathbb{Q}$, and the Radon-Nikodym derivative is the explicit form of that change. In applications the trick is almost always to recognise that a useful measure — one under which some awkward asset becomes a martingale — exists, and then to compute its Radon-Nikodym derivative with respect to a measure we already know how to integrate against (typically $\mathbb{P}$ or $\mathbb{Q}$). From that derivative, Girsanov's theorem (§5.4) gives the drift-shift of any Brownian motion, and the problem collapses to computing an expectation under a Gaussian-friendly measure.

6.2.3 5.2.3 Extension to a continuous state space

On a continuous state space the sums in (5.3) become integrals. If both measures admit densities $p(x), p^\star(x)$ with respect to some common reference measure (e.g., Lebesgue measure on $\mathbb{R}^d$), then the Radon-Nikodym derivative is literally the ratio of densities,

$$\mathcal{L}(x) \;=\; \frac{p^\star(x)}{p(x)},$$

defined $\mathbb{P}$-almost-everywhere. When a common dominating measure does not obviously exist, the Radon-Nikodym theorem of measure theory guarantees that *equivalence* ($\mathbb{P}^\star \sim \mathbb{P}$) is sufficient for $\mathcal{L} = d\mathbb{P}^\star/d\mathbb{P}$ to exist as a non-negative integrable function on Ω, uniquely determined up to $\mathbb{P}$-null sets and satisfying the mean-one condition

$$\mathbb{E}^{\mathbb{P}}\left[\frac{d\mathbb{P}^\star}{d\mathbb{P}}\right] \;=\; 1. \tag{5.7}$$

The mean-one condition is forced by setting $R \equiv 1$ in (5.6): $\mathbb{E}^{\mathbb{P}^\star}[1] = 1$ must agree with $\mathbb{E}^{\mathbb{P}}[\mathcal{L}]$, so $\mathcal{L}$ integrates to one under $\mathbb{P}$. Together with the positivity $\mathcal{L} > 0$ (forced by equivalence), this is all one needs: *any* strictly positive random variable $\mathcal{L}$ with $\mathbb{E}^{\mathbb{P}}[\mathcal{L}] = 1$ defines a bona fide probability measure $\mathbb{P}^\star$ equivalent to $\mathbb{P}$, and every $\mathbb{P}^\star$ equivalent to $\mathbb{P}$ arises this way.

This parametrisation is important enough to restate: **the space of probability measures equivalent to $\mathbb{P}$ is in bijection with the space of strictly positive, mean-one random variables on Ω.** Every question of the form "is there a measure under which X_t has such-and-such property?" becomes a question of the form "is there a strictly positive random variable $\mathcal{L}$ with $\mathbb{E}^{\mathbb{P}}[\mathcal{L}] = 1$ making the reweighted moments come out right?" — and, when X_t is driven by Brownian motion, the Doléans-Dade exponential of §5.3 will furnish that $\mathcal{L}$ almost mechanically.

6.2.4 5.2.4 Numeraires as density-makers

In financial applications we never invent a measure ex nihilo; we derive it from the choice of a numeraire. The setup so far. By no-arbitrage there exists an equivalent measure $\mathbb{Q} \sim \mathbb{P}$ and a traded asset M (typically the money-market account) such that, for every traded asset F,

$$\frac{F_t}{M_t} \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\frac{F_T}{M_T}\right]. \tag{5.8}$$

If A is another traded asset with $A_t > 0$ a.s., there exists a measure $\mathbb{Q}^A$ such that

$$\frac{F_t}{A_t} \;=\; \mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right]. \tag{5.9}$$

The asset A is called a **numeraire asset**. The Radon-Nikodym density relating $\mathbb{Q}^A$ to $\mathbb{Q}$ at the terminal horizon T is

$$\boxed{\frac{d\mathbb{Q}^A}{d\mathbb{Q}} \;=\; \frac{A_T/A_0}{M_T/M_0}.} \tag{5.10}$$

Formula (5.10) is the single definition every quant carries in their head. It says: the density for passing from the bank-account measure to the numeraire-A measure is the terminal value of A (normalised by today's A_0) divided by the terminal value of the bank account (normalised by today's M_0). Up to the constants A_0, M_0 this is the ratio of how much each numeraire has grown between 0 and T. If A grew faster than cash, the density is greater than one on that scenario, and $\mathbb{Q}^A$ up-weights the scenario. If A grew slower, $\mathbb{Q}^A$ down-weights the scenario. This is the profoundly useful intuition: **the numeraire measure up-weights states of the world where the numeraire performs well.** Pricing under $\mathbb{Q}^A$ is therefore equivalent to pricing in a world where A's outperformance is, on average, the norm.

It is instructive to verify directly that (5.10) defines a valid measure change. Taking the unconditional $\mathbb{Q}$-expectation,

$$\mathbb{E}^{\mathbb{Q}}\left[\frac{A_T/A_0}{M_T/M_0}\right] \;=\; \frac{M_0}{A_0}\,\mathbb{E}^{\mathbb{Q}}\left[\frac{A_T}{M_T}\right] \;=\; \frac{M_0}{A_0}\cdot\frac{A_0}{M_0} \;=\; 1, \tag{5.11}$$

where the middle equality uses the martingale property (5.8) applied to $F = A$ at $t = 0$. And $A_T, M_T > 0$ a.s. forces positivity. Positivity plus mean-one is all we need — (5.10) is a bona fide density.

The connection to §5.1 is immediate. The per-state density formula (5.1) is exactly (5.10) evaluated state-by-state in a finite-state world, with M_T/M_0 replaced by the bank-account growth factor. The seven-state example is the discrete shadow of the continuous-time statement. Everything that was arithmetic in §5.1 becomes an identity between random variables in §5.2.

One final perspective on why (5.10) is philosophically important. Some texts introduce the risk-neutral measure as if it were the fundamental object of finance, and change-of-numeraire as a technical device. That ordering is historically accurate but conceptually backwards. The real fundamental object is the family of equivalent martingale measures, each indexed by a choice of numeraire, any one of which is sufficient for pricing. $\mathbb{Q}$ is just the member of that family corresponding to the bank-account numeraire. Different members of the family are useful for different problems, and the density (5.10) tells you how to travel between them. That is the mental model to carry forward: not "one true measure" but "a family of compatible measures, each suited to a task."

6.3 5.3 Density Process and the Doléans-Dade Exponential

The Radon-Nikodym derivative (5.10) is defined at a single terminal horizon T. But in derivatives pricing we often need to change measure *at every intermediate time* — to take conditional expectations, to run simulations, to compute hedges. The tool for this is the **density process** η_t, a continuous-time martingale whose terminal value equals the Radon-Nikodym derivative of (5.10) and whose intermediate values tell us how much weight the new measure $\mathbb{Q}^A$ puts on events observed up to time t.

6.3.1 5.3.1 The density martingale η_t

Suppose the numeraire asset A satisfies the $\mathbb{Q}$-dynamics

$$\frac{dA_t}{A_t} \;=\; \mu_t^A\,dt \;+\; \sigma_t^A\,dW_t \qquad (\mathbb{Q}\text{-measure}), \tag{5.12}$$

where W_t is a $\mathbb{Q}$-Brownian motion and σ_t^A is the (possibly stochastic) volatility loading. If A is a traded asset with $A_t > 0$ a.s., then (5.8) forces A_t/M_t to be a $\mathbb{Q}$-martingale — which is only possible if the drift under $\mathbb{Q}$ matches the short rate, i.e.

$$\mu_t^A \;=\; r_t. \tag{5.13}$$

This is the fundamental consequence of no-arbitrage pricing: *every traded asset, under the risk-neutral measure, earns the short rate on average.* Volatility is free to be whatever the asset's risk structure says it is; the drift is pinned.

Let us verify (5.13) explicitly by computing $d(A_t/M_t)$ under $\mathbb{Q}$. With $M_t = \exp\left(\int_0^t r_u\,du\right)$ we have $1/M_t = \exp\left(-\int_0^t r_u\,du\right)$, so $d(1/M_t) = -r_t/M_t\,dt$ (the bank account is of finite variation, no dW term). The Itô product rule gives

$$d\left(\frac{A_t}{M_t}\right) \;=\; dA_t \cdot \tfrac{1}{M_t} \;+\; A_t \cdot d\left(\tfrac{1}{M_t}\right) \;+\; d\left[A, \tfrac{1}{M}\right]_t. \tag{5.14}$$

Since M_t is of finite variation, the covariation term $d[A, 1/M]_t$ vanishes — a quadratic covariation requires at least *two* diffusive processes to produce a non-trivial dt term. Substituting (5.12) and simplifying,

$$d\left(\frac{A_t}{M_t}\right) \;=\; \frac{A_t}{M_t}\Big[\,(\mu_t^A - r_t)\,dt \;+\; \sigma_t^A\,dW_t\,\Big]. \tag{5.15}$$

For A_t/M_t to be a $\mathbb{Q}$-martingale, the drift must vanish, which forces $\mu_t^A = r_t$ as claimed. With that substitution,

$$d\left(\frac{A_t}{M_t}\right) \;=\; \frac{A_t}{M_t}\,\sigma_t^A\,dW_t. \tag{5.16}$$

Every line of this derivation is the Itô product rule being honest about where its terms come from. The key cancellation is that M_t is of finite variation, so the covariation vanishes. That structural simplicity is precisely why the bank account M makes such a clean "reference clock": it never surprises you with a Brownian-motion contribution, so it passes through Itô's formula transparently.

Now define the **density process**

$$\eta_t \;\triangleq\; \mathbb{E}_t^{\mathbb{Q}}\left[\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right] \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\frac{A_T/A_0}{M_T/M_0}\right] \;=\; \frac{A_t/A_0}{M_t/M_0}, \tag{5.17}$$

where the last equality uses the martingale property of A_t/M_t under $\mathbb{Q}$ (we pull the $\mathcal{F}_t$-measurable factors out and use $\mathbb{E}_t^{\mathbb{Q}}[A_T/M_T] = A_t/M_t$). By construction η_t is a $\mathbb{Q}$-martingale with $\eta_0 = 1$ and terminal value $\eta_T = d\mathbb{Q}^A/d\mathbb{Q}$. And by (5.16), the density process satisfies the driftless SDE

$$\boxed{\frac{d\eta_t}{\eta_t} \;=\; \sigma_t^A\,dW_t.} \tag{5.18}$$

Equation (5.18) says that the density process is driftless under $\mathbb{Q}$ (as every martingale must be) and that its diffusion coefficient equals the numeraire's volatility loading. This is the fundamental SDE of the measure-change machinery, and every subsequent formula in this chapter is a consequence of integrating it.

Interpretation. The density process η_t can be viewed as the "running Radon-Nikodym derivative" — at any intermediate time t, $\eta_t(\omega)$ tells you by what factor the measure $\mathbb{Q}^A$ reweights the path ω observed up to time t. Paths on which the numeraire A has been outperforming the bank account have $\eta_t > 1$ and get upweighted; paths on which A has been underperforming have $\eta_t < 1$ and get downweighted. The process η_t is the continuous-time analogue of the per-state multiplier we computed by hand in §5.1.5.

6.3.2 5.3.2 Integrating the SDE — the Doléans-Dade exponential

The SDE (5.18) has a unique solution called the **Doléans-Dade exponential** (also known as the stochastic exponential). To solve, apply Itô's lemma to $\ln \eta_t$:

$$d(\ln \eta_t) \;=\; \frac{d\eta_t}{\eta_t} \;-\; \tfrac{1}{2}\,\frac{d\langle\eta\rangle_t}{\eta_t^2}. \tag{5.19}$$

Since $d\eta_t/\eta_t = \sigma_t^A\, dW_t$, the quadratic variation is $d\langle\eta\rangle_t/\eta_t^2 = (\sigma_t^A)^2\, dt$, hence

$$d(\ln \eta_t) \;=\; -\tfrac{1}{2}(\sigma_t^A)^2\, dt \;+\; \sigma_t^A\, dW_t. \tag{5.20}$$

Integrating from 0 to t with $\eta_0 = 1$,

$$\ln \eta_t \;=\; -\tfrac{1}{2}\int_0^t (\sigma_u^A)^2\, du \;+\; \int_0^t \sigma_u^A\, dW_u, \tag{5.21}$$

and therefore

$$\boxed{\eta_t \;=\; \exp\left\{ -\tfrac{1}{2}\int_0^t (\sigma_u^A)^2\, du \;+\; \int_0^t \sigma_u^A\, dW_u \right\}.} \tag{5.22}$$

Formula (5.22) is worth memorising. It is the standard-issue Girsanov density, with the $-\frac{1}{2}\int \sigma^2$ term playing the role of the "Itô correction" that ensures η_t is a true $\mathbb{Q}$-martingale and not just a local one. The two pieces of the exponent have distinct structural meanings: the stochastic integral $\int_0^t \sigma_u^A\, dW_u$ is a zero-mean Gaussian random variable with variance $\int_0^t (\sigma_u^A)^2\, du$ (by the Itô isometry of Chapter 3), and the deterministic term $-\frac{1}{2}\int_0^t (\sigma_u^A)^2\, du$ is exactly the *negative half-variance* correction required to make η_t have mean one.

To see the mean-one property concretely, note that for any constant σ and standard normal Z,

$$\mathbb{E}[e^{\sigma Z}] \;=\; e^{\frac{1}{2}\sigma^2},$$

so $\mathbb{E}[e^{-\frac{1}{2}\sigma^2+\sigma Z}] = e^{-\frac{1}{2}\sigma^2+\frac{1}{2}\sigma^2} = 1$. The stochastic integral $\int_0^t \sigma_u^A\, dW_u$ is (when σ^A is deterministic) Gaussian with zero mean and variance $\int_0^t (\sigma_u^A)^2\, du$, and the same cancellation yields $\mathbb{E}^{\mathbb{Q}}[\eta_t] = 1$ for every $t \in [0, T]$. The Itô correction is not optional; without it, η_t would have expectation growing exponentially in the integrated variance, and (5.22) would fail to be a density.

6.3.3 5.3.3 Why a *process* and not just a number

It is worth pausing on why we care about the density *process* η_t rather than just the terminal density η_T. The answer is that every conditional expectation under the new measure can be computed as a $\mathbb{Q}$-conditional expectation against the density process. Specifically, the **abstract Bayes rule** for conditional expectations states that for any $\mathcal{F}_T$-measurable Z,

$$\mathbb{E}_t^{\mathbb{Q}^A}[Z] \;=\; \frac{\mathbb{E}_t^{\mathbb{Q}}[Z\,\eta_T]}{\eta_t} \;=\; \frac{1}{\eta_t}\,\mathbb{E}_t^{\mathbb{Q}}\left[Z \cdot \frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right]. \tag{5.23}$$

Without the density *process*, formula (5.23) would fail at intermediate times: the normalising factor in the denominator is η_t (the density process evaluated at t), not 1. The martingale property of η_t under $\mathbb{Q}$ is what makes (5.23) consistent as t varies — conditional expectations compose correctly precisely when η_t is a $\mathbb{Q}$-martingale.

The abstract Bayes rule (5.23) is the computational engine of the measure-change machinery. We will use it in §5.6 to prove the fundamental change-of-numeraire theorem. In every application, Bayes's rule is how we translate a $\mathbb{Q}^A$-expectation (which we want) into a $\mathbb{Q}$-expectation (which we can compute).

6.3.4 5.3.4 The Novikov and Kazamaki conditions

A technical caveat worth noting. For the Doléans-Dade exponential (5.22) to be a *true* martingale rather than merely a local martingale, we need a moment condition on σ^A. The sharpest widely used sufficient condition is **Novikov's condition**:

$$\mathbb{E}^{\mathbb{Q}}\left[\exp\left(\tfrac{1}{2}\int_0^T (\sigma_u^A)^2\,du\right)\right] \;<\; \infty. \tag{5.24}$$

Under (5.24), η_t is a $\mathbb{Q}$-martingale on $[0,T]$ and, in particular, $\mathbb{E}^{\mathbb{Q}}[\eta_T] = 1$. A milder condition (**Kazamaki's condition**) replaces $\frac{1}{2}\int \sigma^2$ with $\frac{1}{2}\int \sigma\,dW$ under an exponential moment, but Novikov is the workhorse in applications.

For all the bounded-volatility models that populate the rest of this guide — Black-Scholes GBM (σ^A constant), Vasicek bond prices (σ^A deterministic), and the like — Novikov's condition is trivially satisfied, and we can treat the density process as a true martingale without further comment. For models with stochastic volatility that can blow up (e.g., certain Heston parameterisations near the Feller boundary), Novikov's condition must be checked; but that is a computation for the relevant chapter, not a concern here.

With the density process (5.22) in hand, we have all the ingredients we need to state and prove Girsanov's theorem.

6.4 5.4 Girsanov's Theorem

Girsanov's theorem is the single most important result of this chapter. It answers the question: if we change the measure from $\mathbb{Q}$ to $\mathbb{Q}^A$ via the density process η_t of §5.3, what happens to the Brownian motion W_t that drives every SDE in the model? The answer is remarkably clean — W_t remains Brownian under $\mathbb{Q}^A$ after subtracting a deterministic drift determined entirely by the volatility loading σ^A of the numeraire.

6.4.1 5.4.1 Statement

Theorem (Girsanov). Let W_t be a $\mathbb{Q}$-Brownian motion and σ_t^A a progressively measurable process satisfying the Novikov condition (5.24). Let η_t be the Doléans-Dade exponential (5.22), and define the measure $\mathbb{Q}^A$ via $d\mathbb{Q}^A/d\mathbb{Q} = \eta_T$. Then the process

$$W_t^A \;\triangleq\; W_t \;-\; \int_0^t \sigma_u^A\,du \tag{5.25}$$

is a $\mathbb{Q}^A$-Brownian motion on $[0,T]$. Equivalently, in differential form,

$$\boxed{dW_t^A \;=\; dW_t \;-\; \sigma_t^A\,dt.} \tag{5.26}$$

Heuristic. Under $\mathbb{Q}$, $W_t \sim \mathcal{N}(0,t)$. Under $\mathbb{Q}^A$, $W_t \sim \mathcal{N}(\int_0^t \sigma_u^A\,du,\; t)$: the variance is unchanged but the mean picks up a drift equal to the *integrated volatility loading of the numeraire.*

Pause on this heuristic because it contains the whole content of Girsanov in one line. Switching from $\mathbb{Q}$ to $\mathbb{Q}^A$ does not change what W *is* — W is still the same Brownian path on the same sample space — but it changes how the probability measure views that path. Under $\mathbb{Q}^A$, the measure puts more weight on paths where W has climbed high, because those are exactly the paths on which A outperformed the bank account. That skewed weighting *looks* to an observer living in $\mathbb{Q}^A$ like W has an upward drift, even though W itself is unchanged. The drift is purely a measure-theoretic artefact. Practically, this is why formulas under different numeraires look different: the same random variable gets different drifts under different measures, and you have to keep track of which measure you are under at every step.

6.4.2 5.4.2 MGF verification

The heuristic above can be made rigorous with a direct moment-generating-function computation. Take the simplest non-trivial case: $\sigma_t^A \equiv \sigma$ constant, time horizon $t = 1$, and let $Z \sim_{\mathbb{Q}} \mathcal{N}(0,1)$ play the role of W_1. The density

$$\frac{d\mathbb{Q}^\star}{d\mathbb{Q}} \;=\; \exp\!\left(-\tfrac{1}{2}\sigma^2 + \sigma Z\right)$$

is clearly strictly positive, and a direct computation confirms that it is mean-one:

$$\mathbb{E}^{\mathbb{Q}}\left[\frac{d\mathbb{Q}^\star}{d\mathbb{Q}}\right] \;=\; e^{-\frac{1}{2}\sigma^2}\,\mathbb{E}^{\mathbb{Q}}[e^{\sigma Z}] \;=\; e^{-\frac{1}{2}\sigma^2+\frac{1}{2}\sigma^2} \;=\; 1.$$

Now compute the MGF of Z under the new measure $\mathbb{Q}^\star$. For any real u,

$$\mathbb{E}^{\mathbb{Q}^\star}[e^{uZ}] \;=\; \mathbb{E}^{\mathbb{Q}}\left[e^{uZ}\cdot\tfrac{d\mathbb{Q}^\star}{d\mathbb{Q}}\right] \;=\; \mathbb{E}^{\mathbb{Q}}\left[e^{uZ-\frac{1}{2}\sigma^2+\sigma Z}\right] \tag{5.27}$$

$$=\; e^{-\frac{1}{2}\sigma^2}\,\mathbb{E}^{\mathbb{Q}}\left[e^{(u+\sigma)Z}\right] \;=\; e^{-\frac{1}{2}\sigma^2+\frac{1}{2}(u+\sigma)^2} \;=\; e^{u\sigma+\frac{1}{2}u^2}. \tag{5.28}$$

The final expression is recognisable as the MGF of a normal random variable with mean σ and variance 1. So under $\mathbb{Q}^\star$, $Z \sim \mathcal{N}(\sigma, 1)$ — its mean has shifted by exactly σ, and its variance has not changed. Substituting $Z \leftrightarrow W_1$ and $\sigma \leftrightarrow \sigma \cdot 1 = \int_0^1 \sigma_u^A\,du$, this is exactly the Girsanov statement: the Brownian motion picks up an integrated-volatility drift under the new measure.

The MGF verification is worth doing by hand at least once, because it makes the "drift pickup" concrete. Under $\mathbb{Q}$, Z is centred at zero; multiplying the density by $\exp\{-\frac{1}{2}\sigma^2+\sigma Z\}$, then integrating against the original Gaussian, out pops an MGF that is recognisably Gaussian but with its mean shifted by σ. The Itô-correction term $-\frac{1}{2}\sigma^2$ in the density is exactly the amount needed to keep the total probability normalised — without it, the measure-changed Z would still be Gaussian, but the density would not integrate to 1.

6.4.3 5.4.3 Sketch of the general proof

The MGF verification handles the single-time-point Gaussian case. For a full Brownian motion on $[0, T]$ we need to check that $W_t^A = W_t - \int_0^t \sigma_u^A\,du$ is a $\mathbb{Q}^A$-martingale with quadratic variation t. The two-line sketch of the proof is as follows.

Martingale property. Apply Itô to $W_t^A\eta_t$ (where η_t is the density process of §5.3). Using (5.18) and the definition of W_t^A,

$$d(W_t^A\eta_t) \;=\; \eta_t\,dW_t^A + W_t^A\,d\eta_t + d[W^A,\eta]_t.$$

Substituting $dW_t^A = dW_t - \sigma_t^A\,dt$ and $d\eta_t = \eta_t\sigma_t^A\,dW_t$, the covariation is $d[W^A,\eta]_t = \eta_t\sigma_t^A\,dt$, and collecting the dt terms,

$$d(W_t^A\eta_t) \;=\; \eta_t\,(dW_t - \sigma_t^A\,dt) + W_t^A\eta_t\sigma_t^A\,dW_t + \eta_t\sigma_t^A\,dt \;=\; \eta_t(1 + W_t^A\sigma_t^A)\,dW_t.$$

The dt terms have cancelled exactly. So $W_t^A\eta_t$ is a $\mathbb{Q}$-local-martingale, and under Novikov it is a true $\mathbb{Q}$-martingale. By abstract Bayes (5.23), this is equivalent to W_t^A being a $\mathbb{Q}^A$-martingale.

Quadratic variation. Since $\int_0^t \sigma_u^A\,du$ is of finite variation, $d[W^A, W^A]_t = d[W, W]_t = dt$. So the quadratic variation of W_t^A is t under both $\mathbb{Q}$ and $\mathbb{Q}^A$.

Lévy's characterisation. A continuous $\mathbb{Q}^A$-martingale started at zero with quadratic variation t is a $\mathbb{Q}^A$-Brownian motion (Lévy). The two bullets above check both conditions, and the theorem is proved. ■

6.4.4 5.4.4 Physical vs. risk-neutral measures revisited

Before moving on, it is worth reconnecting Girsanov to the $\mathbb{P} \to \mathbb{Q}$ transition that motivated Chapter 2. Under $\mathbb{P}$ a stock S_t might satisfy

$$\frac{dS_t}{S_t} \;=\; \mu\,dt \;+\; \sigma\,dW_t^{\mathbb{P}},$$

with μ the physical expected return and $W^{\mathbb{P}}$ a $\mathbb{P}$-Brownian motion. Under $\mathbb{Q}$ the same stock satisfies

$$\frac{dS_t}{S_t} \;=\; r\,dt \;+\; \sigma\,dW_t^{\mathbb{Q}},$$

with r the risk-free rate and $W^{\mathbb{Q}}$ a $\mathbb{Q}$-Brownian motion. Girsanov tells us how $W^{\mathbb{P}}$ and $W^{\mathbb{Q}}$ are related: subtracting the two SDEs gives

$$0 \;=\; (\mu - r)\,dt \;+\; \sigma\,(dW_t^{\mathbb{P}} - dW_t^{\mathbb{Q}}),$$

hence

$$dW_t^{\mathbb{Q}} \;=\; dW_t^{\mathbb{P}} \;+\; \lambda\,dt, \qquad \lambda \;\triangleq\; \frac{\mu - r}{\sigma}. \tag{5.29}$$

The quantity $\lambda = (\mu - r)/\sigma$ is the **market price of risk** (Sharpe ratio). Formula (5.29) says that passing from $\mathbb{P}$ to $\mathbb{Q}$ adds the market price of risk to the Brownian drift — equivalently, the Radon-Nikodym density is (5.22) with $\sigma^A \leftrightarrow -\lambda$:

$$\frac{d\mathbb{Q}}{d\mathbb{P}} \;=\; \exp\!\left(-\tfrac{1}{2}\lambda^2 T - \lambda W_T^{\mathbb{P}}\right). \tag{5.30}$$

This identification will be made explicit in Chapter 6, where the market-price-of-risk argument that derives the Black-Scholes PDE is re-read as the Girsanov shift from $\mathbb{P}$ to $\mathbb{Q}$. For now, the key takeaway is that $\mathbb{P} \to \mathbb{Q}$ is not a mysterious trick — it is one instance of Girsanov's theorem, with the numeraire chosen to be the bank account.

What changes under $\mathbb{P} \to \mathbb{Q}$: the probabilities of up-states versus down-states — under $\mathbb{P}$ these reflect investors' real-world beliefs plus risk preferences; under $\mathbb{Q}$ they have been adjusted so that every tradable's expected return equals the risk-free rate.

What stays the same: the *support* of the distribution (which states are possible), the payoffs of every contract, today's prices, and — critically — the diffusion coefficient σ. Only the drift changes. A useful mnemonic: "$\mathbb{P}$ is for P&L, $\mathbb{Q}$ is for pricing." Hedging, risk management, forecasting — all done under $\mathbb{P}$. Quoting a fair value — done under $\mathbb{Q}$ (or one of its numeraire-changed cousins). Girsanov is the passport that lets you cross the border cleanly.

6.4.5 5.4.5 Multi-dimensional Girsanov

The one-dimensional statement (5.25) extends verbatim to multiple Brownian motions. Let $W_t = (W_t^1, \dots, W_t^d)$ be a d-dimensional $\mathbb{Q}$-Brownian motion and $\theta_t = (\theta_t^1, \dots, \theta_t^d)$ a progressively measurable $\mathbb{R}^d$-valued process with

$$\mathbb{E}^{\mathbb{Q}}\left[\exp\left(\tfrac{1}{2}\int_0^T \|\theta_u\|^2\,du\right)\right] < \infty.$$

Define the density process

$$\eta_t \;=\; \exp\left(-\tfrac{1}{2}\int_0^t \|\theta_u\|^2\,du + \int_0^t \theta_u \cdot dW_u\right), \tag{5.31}$$

where $\theta_u \cdot dW_u = \sum_{i=1}^d \theta_u^i\,dW_u^i$. Then under $\mathbb{Q}^\star$ defined by $d\mathbb{Q}^\star/d\mathbb{Q} = \eta_T$, the vector process

$$W_t^\star \;\triangleq\; W_t \;-\; \int_0^t \theta_u\,du \tag{5.32}$$

is a d-dimensional $\mathbb{Q}^\star$-Brownian motion (each component Brownian, components mutually independent). The multi-dimensional version reduces every measure-change problem in a multi-factor model — Heston, LIBOR Market Model, HJM — to a vector-valued drift shift. We will use (5.32) in Chapter 8 (Margrabe) and Chapter 14 (Heston) without further derivation.

6.5 5.5 Two-Numeraire Switching

In §§5.3-5.4 we changed measure from the bank-account reference $\mathbb{Q}$ to a numeraire-A reference $\mathbb{Q}^A$. But the theory is not privileged with respect to the bank account: any positive tradable can serve as the reference numeraire. In this section we derive the Girsanov shift that switches between two numeraire measures $\mathbb{Q}^A$ and $\mathbb{Q}^B$ *directly*, without routing through $\mathbb{Q}$. The resulting formula (5.36) is the practical workhorse of change-of-numeraire pricing, and it is the reason every quant can mentally shift between forward, swap, and futures measures without re-deriving Girsanov each time.

6.5.1 5.5.1 Setup and cross-density

Suppose A_t and B_t are both numeraires with $\mathbb{Q}$-dynamics

$$\frac{dA_t}{A_t} = r_t\,dt + \sigma_t^A\,dW_t, \qquad \frac{dB_t}{B_t} = r_t\,dt + \sigma_t^B\,dW_t, \tag{5.33}$$

where W_t is a $\mathbb{Q}$-Brownian motion and σ^A, σ^B are the respective volatility loadings. Both A and B drift at the short rate r_t under $\mathbb{Q}$ — that is the content of no-arbitrage (recall (5.13)).

The Radon-Nikodym density from $\mathbb{Q}$ to $\mathbb{Q}^A$, from (5.10), is $\eta_T^A \triangleq (A_T/A_0)/(M_T/M_0)$; similarly $\eta_T^B = (B_T/B_0)/(M_T/M_0)$. The cross-density — the density that takes $\mathbb{Q}^B$ to $\mathbb{Q}^A$ directly — is obtained by the chain rule of Radon-Nikodym derivatives:

$$\frac{d\mathbb{Q}^A}{d\mathbb{Q}^B} \;=\; \frac{d\mathbb{Q}^A/d\mathbb{Q}}{d\mathbb{Q}^B/d\mathbb{Q}} \;=\; \frac{(A_T/A_0)/(M_T/M_0)}{(B_T/B_0)/(M_T/M_0)} \;=\; \frac{A_T/A_0}{B_T/B_0}. \tag{5.34}$$

Notice that the bank-account factor M_T/M_0 has cancelled entirely. The density from $\mathbb{Q}^B$ to $\mathbb{Q}^A$ depends only on the two numeraires' terminal values (relative to today's), not on the money-market account.

State-by-state (in finite-state language), the cross-density says

$$q^{(A)}(\omega) \;=\; q^{(B)}(\omega)\,\frac{A_T(\omega)/A_0}{B_T(\omega)/B_0}, \tag{5.35}$$

which is exactly formula (5.1) we verified arithmetically in §5.1.5. The continuous-time derivation (5.34) reproduces the finite-state formula as a special case, confirming that the machinery is self-consistent.

6.5.2 5.5.2 The disappearance of the bank account

The cancellation of M_T/M_0 from (5.34) is not an accident; it is the punchline of the whole machinery. When we first introduced the change of numeraire in (5.10), the money-market account appeared in the denominator because we were transitioning from the "default" measure $\mathbb{Q}$ associated with the bank account. But the density to go from $\mathbb{Q}^B$ to $\mathbb{Q}^A$ is found by dividing the two densities, and the bank-account factor cancels, leaving a pure ratio of numeraire returns.

What this means in practice is that *nothing in the theory singles out the money-market account.* It was a convenient starting point — the one numeraire everyone agrees exists — but once the change-of-numeraire apparatus is running, it fades into the background. You can pick any two positive tradables and translate between their measures directly.

This cancellation has a powerful modelling consequence. Many interest-rate derivatives have payoffs most naturally expressed in terms of forward rates or swap rates, objects that live on zero-coupon bonds or annuities. The theory says: *you never need to model the short-rate dynamics at all* for those products. Pick the bond or the annuity as numeraire, write the forward or swap rate's dynamics directly under the matched measure, and price. The short rate — and therefore the money-market account — can remain unspecified. This is the philosophical foundation of the LIBOR Market Model (and its successor SOFR-based siblings): abandon the short-rate-centric viewpoint, pick market-observable numeraires, and work directly in their measures.

6.5.3 5.5.3 Girsanov shift between $\mathbb{Q}^A$ and $\mathbb{Q}^B$

Having established the cross-density (5.34), we now derive the Brownian shift directly. From Girsanov (5.26) applied to each numeraire separately,

$$dW_t^A \;=\; dW_t - \sigma_t^A\,dt, \qquad dW_t^B \;=\; dW_t - \sigma_t^B\,dt. \tag{5.35b}$$

Subtracting the second from the first kills the common dW_t:

$$\boxed{dW_t^A \;=\; -(\sigma_t^A - \sigma_t^B)\,dt \;+\; dW_t^B.} \tag{5.36}$$

So W_t^A, viewed as a process on $\mathbb{Q}^B$, is a $\mathbb{Q}^B$-Brownian motion plus the drift $-(\sigma_t^A - \sigma_t^B)$. Equivalently, the $\mathbb{Q}^A$-Brownian motion W_t^A has a $\mathbb{Q}^B$-drift equal to the negative of the *difference of the two numeraires' volatility loadings*.

Formula (5.36) is the practical workhorse. It says that when you switch your reference numeraire from B to A, every Brownian motion you were tracking picks up a drift equal to the difference of the two numeraires' vol loadings. If A and B are the same asset, the drift is zero and nothing happens. If A is the stock and B is the bond, the drift equals the stock's vol minus the bond's vol — typically positive and large, reflecting the fact that the stock-measure views the world as one in which the stock is more likely to outperform. The whole of measure-change machinery reduces, for many applications, to this single formula: **compute the two vol loadings, subtract, drift your Brownian**.

6.5.4 5.5.4 Worked example — GBM stock and zero-coupon bond

Consider the simplest non-trivial pair: the stock S_t (GBM with vol σ_S) and the zero-coupon bond $P(t,T)$ (Vasicek-style with vol $\sigma_P(t)$). Under $\mathbb{Q}$,

$$\frac{dS_t}{S_t} = r\,dt + \sigma_S\,dW_t, \qquad \frac{dP(t,T)}{P(t,T)} = r\,dt + \sigma_P(t)\,dW_t.$$

By (5.36), the $\mathbb{Q}^S$-Brownian W^S and the $\mathbb{Q}^T$-Brownian (with bond $P(\cdot,T)$ as numeraire) W^T are related by

$$dW_t^S \;=\; -(\sigma_S - \sigma_P(t))\,dt \;+\; dW_t^T.$$

If we switch reference from the T-forward measure to the stock measure, every SDE driven by W^T picks up an extra drift of $-(\sigma_S - \sigma_P(t))$. For instance, the stock-denominated zero-coupon bond $\tilde{B}_t \triangleq P(t,T)/S_t$ satisfies

$$\frac{d\tilde{B}_t}{\tilde{B}_t} \;=\; (\sigma_P(t) - \sigma_S)\,dW_t^S,$$

a driftless SDE under $\mathbb{Q}^S$ (as it must be — $\tilde{B}_t$ is an S-denominated tradable, and $\mathbb{Q}^S$ is the measure that makes S-denominated tradables martingales). The sign of the Brownian coefficient is the difference of the two original volatilities, a direct corollary of (5.36).

This kind of book-keeping — tracking vol-loading differences across a chain of numeraire changes — is the essence of interest-rate derivatives pricing. By Chapter 13 you will be stringing together three or four numeraire switches in a single calculation, and each one will just be (5.36) applied once.

6.5.5 5.5.5 Summary of §§5.2-5.5

Before applying the machinery to traded assets (§5.6) and previewing the T-forward measure (§5.7), let us take stock of what we have built in the last four sections.

- **§5.2:** Any two equivalent probability measures are related by a Radon-Nikodym derivative, a strictly positive random variable with mean one. Every numeraire-based measure $\mathbb{Q}^A$ has density $(A_T/A_0)/(M_T/M_0)$ relative to $\mathbb{Q}$.

- **§5.3:** The density process η_t is a $\mathbb{Q}$-martingale starting from 1 and ending at $d\mathbb{Q}^A/d\mathbb{Q}$. It satisfies the driftless SDE $d\eta_t/\eta_t = \sigma_t^A\, dW_t$ and has the Doléans-Dade closed form (5.22).
- **§5.4:** Girsanov's theorem: the $\mathbb{Q}$-Brownian motion W_t shifts by $-\int_0^t \sigma_u^A\, du$ to become a $\mathbb{Q}^A$-Brownian motion W_t^A. Variance is preserved; only the drift changes.
- **§5.5:** The cross-density $d\mathbb{Q}^A/d\mathbb{Q}^B = (A_T/A_0)/(B_T/B_0)$ lets us switch between any two numeraire measures directly, with Brownian drift $-(\sigma^A - \sigma^B)$.

With these four building blocks we can now prove the fundamental change-of-numeraire pricing theorem and use it in practice.

6.6 5.6 Girsanov Applied to Traded Assets — the Change-of-Numeraire Theorem

The density process and Girsanov machinery of §§5.3-5.5 are stated at the level of Brownian motion — they tell us how to shift the driver of randomness when we change measures. In this section we specialise to the most important application: ratios of traded assets. The result (5.40) is the **fundamental change-of-numeraire pricing theorem**, and it is the justification for every "numeraire out front, ratio inside" formula you will write for the rest of this guide.

6.6.1 5.6.1 The $\mathbb{Q}$-martingale condition for traded assets

Every traded asset A_t with value strictly positive a.s. must have its M-discounted process be a $\mathbb{Q}$-martingale — that is (5.8), which we restate here for convenience:

$$\frac{A_t}{M_t} = \mathbb{E}_t^{\mathbb{Q}}\left[\frac{A_T}{M_T}\right]. \tag{5.37}$$

The same holds for any contingent claim F priced by the market:

$$\frac{F_t}{M_t} = \mathbb{E}_t^{\mathbb{Q}}\left[\frac{F_T}{M_T}\right]. \tag{5.38}$$

These two martingale conditions are the full content of no-arbitrage pricing under $\mathbb{Q}$: all tradable assets — both primary (like stocks and bonds) and derivative (like options and swaps) — have their cash-deflated values as $\mathbb{Q}$-martingales.

6.6.2 5.6.2 Change of numeraire to A

Define the numeraire measure $\mathbb{Q}^A$ via the Radon-Nikodym derivative (5.10),

$$\left(\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right)_T = \frac{A_T/A_0}{M_T/M_0}. \tag{5.39}$$

We already verified in (5.11) that (5.39) is a valid density: it is positive a.s. and has mean one under $\mathbb{Q}$, the latter by (5.37) applied at $t = 0$.

Economic reading of the density (5.39). The numerator A_T/A_0 is the total return on investing in A from time 0 to time T — starting with A_0 dollars of A, you end with A_T dollars, so your return multiple is A_T/A_0. The denominator M_T/M_0 is the cash account return over the same period, e^{rT} under constant rates. So the density is the ratio "total return on A" divided by "total return on cash" — equivalently, the *discounted* (by M) return multiple on A.

Paths on which A outperformed cash (higher A_T/A_0 relative to M_T/M_0) get *upweighted* by the density when moving from $\mathbb{Q}$ to $\mathbb{Q}^A$. Paths on which A underperformed get downweighted. The new measure $\mathbb{Q}^A$ thus "weights paths by how well A did" — which is exactly what is needed to make the asset A's "own" price (measured in units of A) look like a driftless martingale.

6.6.3 5.6.3 The change-of-numeraire theorem — F/A is a $\mathbb{Q}^A$-martingale

Claim. Let $g_t \equiv F_t/A_t$. Then g_t is a $\mathbb{Q}^A$-martingale:

$$\mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right] = \frac{F_t}{A_t}. \tag{5.40}$$

Equivalently, the pricing formula

$$\boxed{F_t = A_t\,\mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right]} \tag{5.41}$$

holds for every traded claim F.

Proof. Use abstract Bayes (5.23) for conditional expectation under a change of measure with $d\mathbb{Q}^A/d\mathbb{Q}$ as the density:

$$\mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right] = \frac{\mathbb{E}_t^{\mathbb{Q}}\left[\frac{F_T}{A_T}\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right]}{\mathbb{E}_t^{\mathbb{Q}}\left[\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right]}.$$

Substituting (5.39),

$$= \frac{\mathbb{E}_t^{\mathbb{Q}}\left[\frac{F_T}{A_T}\cdot\frac{A_T/A_0}{M_T/M_0}\right]}{\mathbb{E}_t^{\mathbb{Q}}\left[\frac{A_T/A_0}{M_T/M_0}\right]} = \frac{\mathbb{E}_t^{\mathbb{Q}}[F_T/M_T]}{\mathbb{E}_t^{\mathbb{Q}}[A_T/M_T]}.$$

(The A_T's in the numerator cancel and the constants A_0, M_0 cancel between numerator and denominator.)

Numerator, using (5.38):

$$\mathbb{E}_t^{\mathbb{Q}}\left[\frac{F_T}{M_T}\right] = \frac{F_t}{M_t}.$$

Denominator, using (5.37) applied to A:

$$\mathbb{E}_t^{\mathbb{Q}}\left[\frac{A_T}{M_T}\right] = \frac{A_t}{M_t}.$$

Therefore

$$\mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right] = \frac{F_t/M_t}{A_t/M_t} = \frac{F_t}{A_t}. \qquad \blacksquare$$

This is the change-of-numeraire theorem, and it is the master pricing identity we will use in every subsequent chapter.

Reading the proof. The double-expectation trick (Bayes's formula for conditional expectations under a change of measure) is a standard identity. Applied here with $Z = F_T/A_T$ and $d\mathbb{Q}^A/d\mathbb{Q} = (A_T/A_0)/(M_T/M_0)$, the A_T's in numerator and denominator cancel, leaving only M_T's, which by the $\mathbb{Q}$-martingale property of F/M and A/M reduces to the ratio F_t/A_t. The cancellation is beautifully clean — it is the whole reason the numeraire-change trick works.

6.6.4 5.6.4 The "numeraire out front, ratio inside" mantra

A useful way to keep (5.41) straight is to memorise the mantra **"numeraire out front, ratio inside."** The numeraire at time t sits outside the expectation and carries the units; inside the expectation sits the ratio of the payoff to the numeraire at terminal time T, which is a dimensionless number. The measure attached to the expectation is the one matched to the numeraire. That three-piece structure — numeraire, ratio, matched measure — appears in every pricing formula you will encounter from here on:

- **Vanilla call, cash numeraire:** $C_t = M_t\,\mathbb{E}_t^{\mathbb{Q}}[C_T/M_T]$, i.e., $C_t = e^{-r(T-t)}\,\mathbb{E}_t^{\mathbb{Q}}[(S_T - K)^+]$.
- **Caplet, T-forward numeraire:** $\mathrm{Cplt}_t = P(t,T)\,\mathbb{E}_t^{\mathbb{Q}^T}[\delta(L(T-\delta,T) - K)^+]$, where $P(t,T)$ is the zero-coupon bond maturing at T.
- **Swaption, annuity numeraire:** $\mathrm{Swpt}_t = A_t\,\mathbb{E}_t^{\mathbb{Q}^A}[(S_T - K)^+]$, where A_t is the swap's annuity and S_T is the swap rate at exercise.
- **Margrabe exchange, stock numeraire:** $E_t = S_t^{(2)}\,\mathbb{E}_t^{\mathbb{Q}^{S^{(2)}}}[(S_T^{(1)}/S_T^{(2)} - 1)^+]$, a one-dimensional Black-Scholes call on the ratio.

Each application is the same template with different nouns plugged in.

6.6.5 5.6.5 Why "matched" is not optional

A beginner might wonder whether one can take the ratio F_t/A_t but compute its expectation under $\mathbb{Q}$ rather than $\mathbb{Q}^A$. The answer is no, not without a correction term — and that correction term is precisely the density (5.39). Computing expectations of deflated prices under the wrong measure is the most common elementary mistake in derivatives pricing, and almost every sign error or missing-drift bug can be traced to it.

The rule to internalise: deflator and measure always travel together. If the deflator in the ratio is A_t, the measure in the expectation must be $\mathbb{Q}^A$; if the measure is $\mathbb{Q}$, the deflator must be M_t. Mixing them is the quant equivalent of using feet in one part of a calculation and metres in another — the numbers will come out wrong and the error will be silent.

6.6.6 5.6.6 Why the change of numeraire matters in practice

The freedom to switch numeraires is not just mathematical convenience — it is a computational engine. By choosing the right numeraire you can:

1. **Reduce the dimension of the pricing problem** — collapse two correlated diffusions into one ratio, as in Margrabe.
2. **Eliminate explicit discount factors** — absorb them into the numeraire. The Black-76 formula has no e^{-rT} visible inside the expectation.
3. **Make certain payoffs exactly log-normal** — forward rates are log-normal under the T-forward measure, swap rates are (approximately, under the swap-measure convention) log-normal under the annuity measure.

Each of these is a step toward closed-form pricing, and the change-of-numeraire toolkit is the master recipe. The rest of this guide is an extended demonstration of that principle: Chapter 6 uses the change of measure to justify the $\mathbb{Q}$-drift of the GBM stock, Chapter 8 uses it to price exchange options and options on futures, Chapter 13 uses it to price caplets and swaptions, and Chapter 14 uses a multi-dimensional version to handle Heston stochastic volatility.

6.7 5.7 Preview — the T-Forward and Annuity Measures

The machinery of §§5.2-5.6 is general — any strictly positive tradable is a valid numeraire — but two specific choices recur so often in interest-rate practice that they deserve a preview now. Both will be developed in full in later chapters; here we merely observe that the theory already built is enough to see *why* they are the natural choices. This subsection therefore serves as a down-payment on material that Chapters 12 and 13 will redeem in full.

6.7.1 5.7.1 The T-forward measure

Consider a contract whose payoff is received at a single future date T. The zero-coupon bond $P(t,T)$ maturing at T is a natural candidate numeraire because it is strictly positive, it is a tradable (directly, as a bond), and it has

the pleasing property that at maturity $P(T,T) = 1$. Taking $N = P(\cdot, T)$ in the change-of-numeraire formula (5.41) yields the **T-forward measure** $\mathbb{Q}^T$, under which ratios of tradables to $P(t,T)$ are martingales:

$$\frac{F_t}{P(t,T)} \;=\; \mathbb{E}_t^{\mathbb{Q}^T}\left[\frac{F_T}{P(T,T)}\right] \;=\; \mathbb{E}_t^{\mathbb{Q}^T}[F_T], \tag{5.42}$$

where the second equality uses $P(T,T) = 1$. The second form is striking: the *present value* of a time-T payoff, scaled by the T-bond price, equals the $\mathbb{Q}^T$-expectation of the raw payoff. The T-bond absorbs the discounting entirely.

Rearranging,

$$F_t \;=\; P(t,T)\,\mathbb{E}_t^{\mathbb{Q}^T}[F_T]. \tag{5.43}$$

Formula (5.43) is the **fundamental pricing identity under the T-forward measure**. Compare it with the risk-neutral formula $F_t = \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_u\,du} F_T\right]$: the discount factor has been absorbed into the numeraire $P(t,T)$ and pulled outside the expectation, eliminating an entire stochastic integral from the computation. This is the reason T-forward pricing is pervasive in fixed-income quantitative work.

6.7.2 5.7.2 The forward rate as a $\mathbb{Q}^T$-martingale

The key consequence of the T-forward measure, which we will prove in detail in Chapter 13, is that the *simple forward rate* $L(t; T-\delta, T)$ — the market-quoted forward rate for the interval $[T-\delta, T]$ — is a $\mathbb{Q}^T$-martingale. The intuition is simple: the forward rate is, by construction, the fair rate one would agree *today* to apply at $[T-\delta, T]$ given discounting with $P(\cdot, T)$, so it is driftless in exactly the measure where $P(\cdot, T)$ is the accounting unit.

Concretely, the simple forward rate is defined by

$$L(t; T-\delta, T) \;\triangleq\; \frac{1}{\delta}\left(\frac{P(t, T-\delta)}{P(t,T)} - 1\right).$$

The ratio $P(t, T-\delta)/P(t,T) = 1 + \delta L(t; T-\delta, T)$ is a ratio of two tradables to the numeraire $P(t,T)$, so by (5.40) it is a $\mathbb{Q}^T$-martingale. Since $1/\delta$ and subtracting a constant preserve the martingale property, $L(t; T-\delta, T)$ is itself a $\mathbb{Q}^T$-martingale. Chapter 13 will turn this observation into a Black-76 pricing formula for caplets.

This is the defining property of the T-forward measure, and it is the reason caplets and floorlets price as single-rate Black-style expectations rather than complicated multi-factor integrals. Under $\mathbb{Q}$ the forward rate has a complicated, model-dependent drift that is a nuisance to calibrate; under $\mathbb{Q}^T$ the forward rate is simply a driftless martingale with an identified volatility, and pricing reduces to one-dimensional work.

6.7.3 5.7.3 The annuity measure

For a swap spanning a set of payment dates $T_1 < T_2 < \ldots < T_N$, the natural numeraire is the swap's **annuity**

$$A_t \;=\; \sum_{i=1}^{N} \delta_i\, P(t, T_i), \tag{5.44}$$

the present value of a stream of unit coupons paid on the swap's schedule (with δ_i the day-count fraction of the i-th period). The annuity is a positive traded portfolio (a linear combination of zero-coupon bonds, each of which is positive), so it qualifies as a numeraire. The induced measure $\mathbb{Q}^A$ — the **swap measure** or **annuity measure** — has the defining property that the swap rate S_t — the fixed rate that would make today's swap fair-value zero — is a $\mathbb{Q}^A$-martingale. Chapter 13 will use this to derive a Black-style swaption pricing formula.

Concretely, the swap rate is the ratio

$$S_t \;=\; \frac{P(t, T_0) - P(t, T_N)}{A_t},$$

a ratio of a linear combination of zero-coupon bonds (a tradable) to the annuity (the numeraire). By the change-of-numeraire theorem (5.40), S_t is a $\mathbb{Q}^A$-martingale, and swaption pricing becomes

$$\text{Swpt}_t \;=\; A_t\, \mathbb{E}_t^{\mathbb{Q}^A}\left[(S_{T_0} - K)^+\right]. \tag{5.45}$$

This is directly analogous to how caplets price via the forward rate under $\mathbb{Q}^T$: choose the numeraire that makes the underlying rate a martingale, and pricing collapses to a one-dimensional Black expectation.

6.7.4 5.7.4 Why each measure is the "right" choice

In both cases, the guiding principle is to pick the numeraire that makes the underlying observable rate — the forward rate for caps, the swap rate for swaptions — into a martingale. Once the underlying is a martingale under the matched measure, pricing reduces to a driftless expectation, which is the simplest possible computational object.

Modelling a driftless martingale requires only a volatility specification; modelling a drifting process requires, additionally, a fully consistent drift structure that accounts for all the cross-dependencies in the term-structure. The measure change has thus effectively absorbed the drift structure into the choice of numeraire, leaving only volatility to model. That is an enormous practical simplification, and it is the driving motivation for the LIBOR Market Model, the Swap Market Model, and every other market-rate-based approach to interest-rate derivatives pricing.

6.7.5 5.7.5 Two warnings

Two warnings before we move on.

First, the T-forward measure and the annuity measure are *model choices*, not physical realities. The market has a single physical probability measure $\mathbb{P}$; the various risk-neutral and forward measures are bookkeeping devices that simplify pricing calculations. Choosing the T-forward measure does not mean the world "is" T-forward; it means we have chosen a coordinate system in which the forward rate has no drift. The same underlying economy supports all these measures — they merely reweight probabilities differently to simplify different pricing problems.

Second, because these measures are different, a process that is a martingale under one need not be a martingale under another. The swap rate is a martingale under $\mathbb{Q}^A$ but not under $\mathbb{Q}$ or under $\mathbb{Q}^T$ (for any particular T). Applying the wrong measure is the single most common source of error in interest-rate pricing code, and developing a reflex to double-check "which measure am I under?" at every step is the hallmark of a careful quant.

6.7.6 5.7.6 Why the preview is possible now

It is worth pausing to acknowledge that this preview is possible only because we developed the change-of-numeraire theorem in full generality in §5.6. In the original exposition of Chapter 3, the T-forward measure had to be introduced as a promissory note — "we will make this rigorous later." Now the promissory note has been paid: the T-forward measure is simply (5.10) specialised to $A = P(\cdot, T)$, and its defining property (forward rate is a martingale) is simply (5.40) applied to the specific ratio $P(t, T-\delta)/P(t, T)$. No new theory is required.

This is the payoff of doing the general theory first. Chapters 12 and 13 will pick the annuity and T-forward numeraires without further comment — they are just particular applications of what we have already built.

6.8 5.8 Forward Pointers

The machinery of this chapter is the most-reused toolkit in the remainder of the guide. Rather than re-derive Girsanov every time we need it, Chapters 6-14 will cite specific results from this chapter and apply them. Here is the forward-reference map.

6.8.1 5.8.1 Chapter 6 — Market price of risk as a Girsanov shift

Chapter 6 builds the generalised Black-Scholes PDE via the self-financing hedging argument. A key step is the identification of the **market price of risk** $\lambda = (\mu - r)/\sigma$ — the quantity that must be identical across all tradables driven by the same Brownian motion, by no-arbitrage. The classical derivation in Chapter 6 arrives at λ by equating

the Sharpe ratios of two tradables; this chapter gives the complementary interpretation: λ **is the Girsanov drift that shifts $\mathbb{P}$-Brownian motion to $\mathbb{Q}$-Brownian motion**, via formula (5.29).

After Chapter 6, the reader should see the two derivations as the *same* result viewed from two angles. The classical "equate the Sharpe ratios" is a statement about the $\mathbb{P}$-drift structure that no-arbitrage imposes; the Girsanov reading is a statement about the *measure* under which the discounted price is a martingale. Both arrive at the same PDE, and both are required for a full conceptual picture. Chapter 6 will include an explicit sidebar tying the two together.

6.8.2 5.8.2 Chapter 8 — Futures measure and Margrabe

Chapter 8 uses the full Girsanov apparatus twice.

First, the **futures measure** is introduced as the numeraire measure associated with a zero-interest-rate "futures account": under the futures measure, the futures price F_t is a driftless martingale, and options on futures (the Black-76 formula) price via a Black-style expectation with no interest-rate discounting inside. The derivation is a direct application of (5.36): subtract the bank-account volatility (zero, since M is deterministic and the "futures numeraire" has the same stochastic volatility as the forward price) and the futures price becomes driftless.

Second, the **Margrabe exchange option** — an option paying $(S_T^{(1)} - S_T^{(2)})^+$ at T — is priced by taking $S^{(2)}$ as the numeraire. Under $\mathbb{Q}^{S^{(2)}}$, the ratio $S_t^{(1)}/S_t^{(2)}$ is a martingale (by (5.40)), and the exchange option collapses to a one-dimensional Black-Scholes call on the ratio with *no interest-rate appearance anywhere in the formula.* This is the cleanest demonstration in the guide of why numeraire change is a *computational* tool, not just a bookkeeping device.

Chapter 8 will invoke (5.36) and (5.41) without re-proving them.

6.8.3 5.8.3 Chapter 12 — Vasicek, Hull-White, and the $\mathbb{P} \to \mathbb{Q}$ transition

Chapter 12 develops the canonical short-rate models (Vasicek, Ho-Lee, Hull-White) and needs to pass from the physical-measure OU SDE

$$dr_t \;=\; \kappa(\theta^{\mathbb{P}} - r_t)\,dt + \sigma\,dW_t^{\mathbb{P}}$$

to the risk-neutral-measure SDE

$$dr_t \;=\; \kappa(\theta^{\mathbb{Q}} - r_t)\,dt + \sigma\,dW_t^{\mathbb{Q}}$$

with a different long-run mean $\theta^{\mathbb{Q}}$ that reflects the market price of short-rate risk. The shift $\theta^{\mathbb{P}} \to \theta^{\mathbb{Q}}$ is exactly the Girsanov drift adjustment (5.26), and Chapter 12 cites (5.26) rather than re-deriving it. The distinction between physical-measure calibration (for forecasting and risk management) and risk-neutral calibration (for pricing) will be a running theme there, and it is clearest when read as "two Girsanov-related measures, same diffusion, different drifts."

6.8.4 5.8.4 Chapter 13 — T-forward measure for bond options and caplets

Chapter 13 is where the T-forward measure previewed in §5.7 gets used in anger. The pricing formulas for:

- **Bond options** — European call on a zero-coupon bond, via Jamshidian's trick and the Vasicek bond-option formula;
- **Caplets and floorlets** — under the T-forward measure, the forward rate is a log-normal martingale (within the Hull-White framework), and the caplet prices via Black-76;
- **Swaptions** — under the annuity measure $\mathbb{Q}^A$, the swap rate is a martingale and the swaption prices via Black-like expectation;

will all cite §§5.6-5.7 rather than redo the measure-change derivation. This is possible precisely because the general theorem (5.40) is measure-agnostic — it works for any choice of positive tradable numeraire, and each of the three applications is just a different numeraire plugged into the same formula.

6.8.5 5.8.5 Chapter 14 — Heston and stochastic volatility

Chapter 14 develops the Heston model, which has a *two-dimensional* driving Brownian motion (W_t^S, W_t^v) with correlation ρ. The $\mathbb{P} \to \mathbb{Q}$ passage in Heston requires a two-dimensional Girsanov — specifically (5.31)-(5.32) — and introduces an additional "volatility risk premium" that shifts the drift of the variance process. The change-of-numeraire machinery there is not qualitatively different from what we built in §§5.3-5.5; it is just applied to a 2D Brownian motion rather than a 1D one. Chapter 14 will invoke (5.32) and move on.

6.8.6 5.8.6 Consolidating the forward-reference map

Later chapter	Result used	Section used
CH06 (BS PDE)	Market price of risk as Girsanov shift	§5.4.4, eqn (5.29)
CH08 (Margrabe, Futures)	Change-of-numeraire theorem; two-numeraire switch	§5.5.3, §5.6.3, eqns (5.36), (5.40)
CH12 (Short-rate models)	$\mathbb{P} \to \mathbb{Q}$ Girsanov shift	§5.4.1, eqn (5.26)
CH13 (Rate derivatives)	T-forward measure; annuity measure	§5.7, eqns (5.43), (5.45)
CH14 (Heston)	Multi-dimensional Girsanov	§5.4.5, eqn (5.32)

Every result in the right-hand column is a single formula derived in this chapter; every application in the left-hand column is one invocation of that formula with the numeraire chosen to suit the problem. The theory is canonical; the applications are many.

6.9 5.9 Key Takeaways and Reference Formulas

6.9.1 5.9.1 Key Takeaways

- Under the risk-neutral measure $\mathbb{Q}$ (bank-account numeraire M_t), the deflated price X_t/M_t is a $\mathbb{Q}$-martingale — this is the defining pricing identity inherited from Chapters 1-2.
- **Any strictly positive tradable A_t can be promoted to a numeraire.** The induced measure $\mathbb{Q}^A$ is equivalent to $\mathbb{Q}$ with Radon-Nikodym density $(A_T/A_0)/(M_T/M_0)$, and under $\mathbb{Q}^A$ the deflated price X_t/A_t is a martingale.
- **Change-of-numeraire pricing formula:** $H_t = A_t\, \mathbb{E}_t^{\mathbb{Q}^A}[H_T/A_T]$ holds for every traded claim H and every positive numeraire A. The numeraire sits outside, the ratio sits inside.
- The **density process** η_t is a $\mathbb{Q}$-martingale with $\eta_0 = 1$ and $\eta_T = d\mathbb{Q}^A/d\mathbb{Q}$. It satisfies the driftless SDE $d\eta_t/\eta_t = \sigma_t^A\, dW_t$ and integrates to the Doléans-Dade exponential (5.22).
- **Girsanov's theorem:** if W_t is a $\mathbb{Q}$-Brownian motion and η_t the density process above, then $W_t^A = W_t - \int_0^t \sigma_u^A\, du$ is a $\mathbb{Q}^A$-Brownian motion. The variance is preserved; only the drift changes.
- **Two-numeraire switch:** the cross-density $d\mathbb{Q}^A/d\mathbb{Q}^B = (A_T/A_0)/(B_T/B_0)$ lets us hop between two tradable-numeraire measures without routing through $\mathbb{Q}$; the Brownian shift is $-(\sigma^A - \sigma^B)$.
- State-by-state, new-measure probabilities are obtained by multiplying old-measure probabilities by the numeraire-ratio density — no integrals needed in discrete-state examples, as the seven-state example of §5.1 shows.
- The worked two-asset example (§5.1) demonstrates that $\mathbb{Q}$, $\mathbb{Q}^{(A)}$, and $\mathbb{Q}^{(B)}$ are *different* probability weightings of the *same* state space; they agree on null sets and disagree on magnitudes.
- **Equivalent measures share the same null sets.** This is the structural reason no-arbitrage is preserved under every change of numeraire: an arbitrage is a positive payoff on a non-null set, and reweighting a non-null set cannot make it null.
- **Physical ($\mathbb{P}$) versus risk-neutral ($\mathbb{Q}$)** differ only in the probabilities assigned to scenarios, not in the set of possible scenarios, nor in the diffusion structure, nor in today's prices. The passage $\mathbb{P} \to \mathbb{Q}$ is one instance of Girsanov with shift $-\lambda$, where $\lambda = (\mu - r)/\sigma$ is the market price of risk.
- **"Deflator and measure always travel together":** if the ratio is X/A, the measure must be $\mathbb{Q}^A$; if the measure is $\mathbb{Q}$, the deflator must be M. Mixing them is the most common sign-error bug in derivatives pricing.
- Choosing the right numeraire linearises the problem: for cap/floor pricing, the zero-coupon bond $P(t,T)$ as numeraire makes the forward rate a martingale (the T-forward measure, used in Chapter 13). For swaption

pricing, the swap annuity as numeraire makes the swap rate a martingale (the annuity measure, also used in Chapter 13).

- **Novikov's condition** $\mathbb{E}^{\mathbb{Q}}[\exp(\frac{1}{2}\int_0^T |\sigma^A|^2\,du)] < \infty$ is sufficient for the Doléans-Dade exponential to be a true martingale and for Girsanov's theorem to apply. For bounded-volatility models it is trivially satisfied.
- The **multi-dimensional Girsanov** (5.32) shifts each component of a d-dimensional Brownian motion by its own drift θ_t^i, and will be used in Chapter 14 (Heston) for 2D measure changes.

6.9.2 5.9.2 Reference Formulas

Equivalence of measures. $\mathbb{P} \sim \mathbb{P}^\star$ iff they agree on null sets:

$$\mathbb{P}(A) = 0 \iff \mathbb{P}^\star(A) = 0, \qquad A \in \mathcal{F}.$$

Master change-of-measure identity.

$$\mathbb{E}^{\mathbb{P}^\star}[R] \;=\; \mathbb{E}^{\mathbb{P}}\left[R \cdot \frac{d\mathbb{P}^\star}{d\mathbb{P}}\right], \qquad \mathbb{E}^{\mathbb{P}}\left[\frac{d\mathbb{P}^\star}{d\mathbb{P}}\right] = 1.$$

Radon-Nikodym density for numeraire A vs. bank account M.

$$\left.\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right|_{\mathcal{F}_T} \;=\; \frac{A_T/A_0}{M_T/M_0}.$$

Cross-density for two numeraires A, B.

$$\left.\frac{d\mathbb{Q}^A}{d\mathbb{Q}^B}\right|_{\mathcal{F}_T} \;=\; \frac{A_T/A_0}{B_T/B_0}.$$

State-probability transformation (discrete state space).

$$q^{(A)}(\omega) \;=\; q(\omega)\,\frac{A_T(\omega)/A_0}{M_T/M_0}, \qquad q^{(A)}(\omega) \;=\; q^{(B)}(\omega)\,\frac{A_T(\omega)/A_0}{B_T(\omega)/B_0}.$$

Density process. For A with $\mathbb{Q}$-dynamics $dA_t/A_t = r_t\,dt + \sigma_t^A\,dW_t$,

$$\eta_t \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right] \;=\; \frac{A_t/A_0}{M_t/M_0}, \qquad \frac{d\eta_t}{\eta_t} \;=\; \sigma_t^A\,dW_t.$$

Doléans-Dade exponential (one-dimensional).

$$\eta_t \;=\; \exp\left(-\tfrac{1}{2}\int_0^t (\sigma_u^A)^2\,du + \int_0^t \sigma_u^A\,dW_u\right), \qquad \eta_0 = 1.$$

Novikov's condition.

$$\mathbb{E}^{\mathbb{Q}}\left[\exp\left(\tfrac{1}{2}\int_0^T (\sigma_u^A)^2\,du\right)\right] < \infty.$$

Girsanov's theorem (one-dimensional). Under the measure $\mathbb{Q}^A$ defined by $d\mathbb{Q}^A/d\mathbb{Q} = \eta_T$,

$$W_t^A \;=\; W_t \;-\; \int_0^t \sigma_u^A\,du$$

is a $\mathbb{Q}^A$-Brownian motion; equivalently $dW_t^A = dW_t - \sigma_t^A\,dt$.

Two-numeraire Girsanov shift.

$$dW_t^A \;=\; dW_t^B \;-\; (\sigma_t^A - \sigma_t^B)\,dt.$$

Multi-dimensional Girsanov. Density process

$$\eta_t \;=\; \exp\left(-\tfrac{1}{2}\int_0^t \|\theta_u\|^2\,du + \int_0^t \theta_u \cdot dW_u\right),$$

d-dimensional Brownian shift

$$W_t^\star \;=\; W_t \;-\; \int_0^t \theta_u\,du.$$

Abstract Bayes rule. For $\mathcal{F}_T$-measurable Z and density process η_t,

$$\mathbb{E}_t^{\mathbb{Q}^A}[Z] \;=\; \frac{\mathbb{E}_t^{\mathbb{Q}}[Z\,\eta_T]}{\eta_t}.$$

Change-of-numeraire theorem. For every traded claim F and every positive numeraire A,

$$F_t \;=\; A_t\,\mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right], \qquad \text{equivalently} \qquad \frac{F_t}{A_t} = \mathbb{E}_t^{\mathbb{Q}^A}\left[\frac{F_T}{A_T}\right].$$

Physical-to-risk-neutral Girsanov. Market price of risk $\lambda = (\mu - r)/\sigma$ drives

$$dW_t^{\mathbb{Q}} \;=\; dW_t^{\mathbb{P}} \;+\; \lambda\,dt, \qquad \frac{d\mathbb{Q}}{d\mathbb{P}} \;=\; \exp\!\left(-\tfrac{1}{2}\lambda^2 T - \lambda W_T^{\mathbb{P}}\right).$$

T-forward measure. Numeraire $N_t = P(t,T)$, measure $\mathbb{Q}^T$. Pricing formula

$$F_t \;=\; P(t,T)\,\mathbb{E}_t^{\mathbb{Q}^T}[F_T].$$

Simple forward rate $L(t;T-\delta,T) = \frac{1}{\delta}(P(t,T-\delta)/P(t,T) - 1)$ is a $\mathbb{Q}^T$-martingale.

Annuity (swap) measure. Numeraire $A_t = \sum_{i=1}^N \delta_i P(t,T_i)$, measure $\mathbb{Q}^A$. Swap rate $S_t = (P(t,T_0) - P(t,T_N))/A_t$ is a $\mathbb{Q}^A$-martingale. Swaption

$$\mathrm{Swpt}_t \;=\; A_t\,\mathbb{E}_t^{\mathbb{Q}^A}[(S_{T_0} - K)^+].$$

With Chapter 5 complete, the reader has the canonical measure-change toolkit. Chapter 6 now picks up the derivation of the Black-Scholes PDE, using market price of risk as the operational face of (5.29). Chapter 8 will price exchange options and options on futures using (5.41). Chapters 12 and 13 will use the T-forward and annuity measures without re-deriving them. The theory is canonical; the applications are many.

7 Chapter 6 — Dynamic Hedging I: Self-Financing Strategies and the Black-Scholes PDE

With Brownian motion, the Itô calculus, the Feynman-Kac bridge, and the measure-change machinery now in hand (chapters 3-5), we have every tool we need to derive the Black-Scholes partial differential equation from first principles.

The chapter proceeds in three acts. First we build the **self-financing replicating portfolio** — a continuous-time limit of the discrete hedge of Chapter 2 — and show that forcing it to have zero local risk produces a **market price of risk** that is the same for every traded claim. Cross- referencing Chapter 5, we'll see that this quantity is exactly the Girsanov drift shift that turns the physical measure $\mathbb{P}$ into the pricing measure $\mathbb{Q}$. Second, we derive the **generalised Black-Scholes PDE** for a single-factor Itô-driven underlying, specialise it to geometric Brownian motion, and read off the classical Black-Scholes formulas for calls, puts, and digitals. We then show that the **same** PDE appears via a Feynman-Kac martingale argument, closing the loop with Chapter 4. Third, we turn to the **practical reality of discrete rebalancing**: we compute the hedge-error distribution under time-based rebalancing, study its variance scaling in $\sqrt{\Delta t}$, and examine a move-based alternative that trades quadratic-variation budget for path-dependence.

Throughout, notation follows the stochastic-calculus primer in CH03. Unqualified W_t denotes a standard $\mathbb{P}$-Brownian motion; its $\mathbb{Q}$-counterpart will be denoted $\widetilde{W}_t$ once the change of measure is performed.

7.1 6.1 Setup — underlying, money market, and two contingent claims

Before we can write the replication argument, we have to be absolutely clear about what is traded, what is observable but not traded, and what we are trying to price. That taxonomy matters, because the machinery only works when the thing we use for hedging is *something we can buy and sell.* If we try to hedge a weather derivative by "selling temperature", the argument collapses. The cleanest way to keep this straight is to introduce four objects: an index, a bank account, a traded claim on the index, and a new claim on the index whose value we want to discover.

It is worth pausing to notice that this cast of characters is more general than the textbook "stock, bank, option" triple, and for good reason. In many real markets the source of risk is not something you can hold in inventory. You cannot own a share of the VIX; you cannot own a share of the S&P 500 index directly without replicating it component-by-component; you cannot own a cubic metre of rain. What you *can* own are forwards, ETFs, futures, and swaps whose value is a smooth function of the underlying index. Those derived instruments are what we will call g in the argument below, and they are the true hedge vehicles. Keeping the index X and the hedge g formally distinct forces us to treat markets as they actually are: a tradeable layer sitting on top of a possibly untradeable risk factor.

7.1.1 6.1.1 The underlying index

Assume there is some underlying index

$$X = (X_t)_{0 \le t \le T}, \qquad \text{(not necessarily traded)}$$

that satisfies, under the physical measure $\mathbb{P}$,

$$\mathrm{d}X_t \;=\; \underbrace{\mu(t, X_t)}_{\text{drift}}\,\mathrm{d}t \;+\; \underbrace{\sigma(t, X_t)}_{\text{volatility}}\,\mathrm{d}W_t. \tag{6.1}$$

Think of X_t as the quantity everyone is watching — a stock price, an index level, a spread, a variance process. It is the *source of uncertainty.* The drift $\mu(t, X_t)$ captures the systematic tendency of X to rise or fall, and the volatility $\sigma(t, X_t)$ captures the scale of the noise. Two features of (6.1) deserve emphasis. First, both coefficients depend on (t, X_t), which means the model already allows for local volatility structures. Second, we stay deliberately agnostic about whether X is itself something one can purchase. For a stock, it is; for the VIX or a credit spread, it is not, and we will need a separately traded instrument to hedge.

The dependence on (t, X_t) is rich enough to accommodate every single-factor diffusion model that is popular in practice. Constant volatility yields plain geometric Brownian motion. A power-of-X volatility, like $\sigma(t, x) = \sigma x^\beta$ with $\beta < 1$, captures the "leverage effect" seen in equities — volatility rises as prices fall. A volatility that depends

explicitly on time, $\sigma(t,x) = \sigma_0\, h(t)$, models scheduled-event risk. A volatility that mean-reverts to a long-run level would require a second factor, which is why single-factor local-volatility models fall short of capturing skew *and* term structure simultaneously.

A quick sanity check that this framework is rich enough: the drift $\mu(t, X_t)$ can be whatever you want, including strongly mean-reverting or trending. The deep fact we will discover is that for *hedging* purposes the drift is irrelevant — the option price only depends on σ and r. So even though the setup allows an arbitrary drift, the punchline is that the answer does not see it. That is not a bug; it is a feature. It means the hedger does not need to forecast the stock's expected return, which is a notoriously difficult exercise that nobody does well, and can price options purely from volatility, which is comparatively easy to estimate.

7.1.2 6.1.2 The money-market account

The money-market account is traded:

$$M = (M_t)_{0\le t\le T}, \qquad \frac{\mathrm{d}M_t}{M_t} \;=\; r(t, X_t)\,\mathrm{d}t. \tag{6.2}$$

The money-market account is the boring workhorse of the argument — the riskless rail against which all excess returns are measured. Notice there is no $\mathrm{d}W_t$ term: cash grows deterministically (at a rate that may itself depend on time and the state X_t, so this still covers stochastic short-rate models). In the body of this chapter we will typically treat r as constant to keep the algebra clean, but the derivation never needs that assumption until we specialise to closed-form solutions.

The name "money-market account" deserves some unpacking. In real markets, the riskless rate is a legal fiction more than a physical instrument — there is no single bond or deposit that delivers exactly r_t with zero credit risk and zero liquidity risk. What exists instead is a collection of instruments (overnight repos, central-bank deposit facilities, short-term Treasury bills, Fed funds) whose yields hover near some common reference rate, and desks use the best available proxy. For derivatives priced against an equity underlying, the usual proxy is the OIS curve (overnight-indexed swap), because it matches the cost of funding a collateralised derivatives position.

One more comment on what *finite variation* means for M. Because cash accrues deterministically, the process M_t has no quadratic variation. In the Itô calculus vocabulary (CH03), $\mathrm{d}[M,M]_t = 0$ and $\mathrm{d}[\beta, M]_t = 0$ for any adapted process β. This is why the cross-variation term involving M drops out of the self-financing expansion in §6.2: cash does not "wiggle", so buying more cash at the current rate does not generate a quadratic cross term.

7.1.3 6.1.3 A traded contingent claim g

Some contingent claim on X, call this claim g, is traded:

$$g = (g_t)_{0\le t\le T}, \qquad g_t = g(t, X_t).$$

By Itô's lemma (CH03),

$$\frac{\mathrm{d}g_t}{g_t} \;=\; \mu^g(t, X_t)\,\mathrm{d}t \;+\; \sigma^g(t, X_t)\,\mathrm{d}W_t, \tag{6.3}$$

and in general

$$\mathrm{d}g_t \;=\; \Big(\partial_t g(t, X_t) + \partial_x g(t, X_t)\,\mu(t, X_t) + \tfrac{1}{2}\,\partial_{xx} g(t, X_t)\,\sigma^2(t, X_t)\Big)\,\mathrm{d}t \;+\; \partial_x g(t, X_t)\,\sigma(t, X_t)\,\mathrm{d}W_t. \tag{6.4}$$

The role of g is the workhorse of our hedge. We need *some* instrument whose value moves with X, and which we can actually buy and sell, to neutralise the $\mathrm{d}W_t$ risk in the claim we are pricing. In the canonical stock setting, g and X are the same thing; in fixed income or commodities, g might be a futures contract written on an underlying swap rate or a convenience-yielded spot; in volatility markets, g could be a variance swap. The key is only that g is Itô-driven by the same Brownian motion W — the same source of randomness — because that is what allows a linear combination to cancel the noise.

Equation (6.4) is Itô's lemma in its most workmanlike form, and it is worth spending a moment on why every term is there. The drift of g has three pieces. The $\partial_t g$ piece is the *passage-of-time* effect: even if the underlying does not move at all, the function $g(t, x)$ may still change simply because t is advancing. This is theta. The $\partial_x g \cdot \mu$ piece is the *directional* effect — slope times speed, like any chain rule. The $\frac{1}{2}\partial_{xx} g \cdot \sigma^2$ piece is the *Itô correction*: in stochastic calculus, the second-order term does *not* vanish, because $(\mathrm{d}W)^2 = \mathrm{d}t$. Convexity — the second spatial derivative — injects a drift of its own into any function of a diffusion process, and that drift is half the gamma times the instantaneous variance. This is the mathematical seed of every gamma-trading argument in the rest of the book.

The diffusion of g is simpler: $\partial_x g \cdot \sigma$. The volatility of g is the volatility of X times the slope of g with respect to X. If g is a deep-in-the-money call, its slope is close to one and its volatility is close to the stock's; if g is deep out-of-the-money, its slope is close to zero and its volatility is small.

7.1.4 6.1.4 Goal — price a new claim f

Goal: value a new claim $f = (f_t)_{0 \le t \le T}$ which pays at maturity

$$f_T \;=\; \varphi(X_T),$$

where φ is the payoff, e.g. $\varphi(x) = (x - K)_+$ for a vanilla call. Write $f_t = f(t, X_t)$. By Itô's lemma

$$\frac{\mathrm{d}f_t}{f_t} \;=\; \mu^f(t, X_t)\,\mathrm{d}t \;+\; \sigma^f(t, X_t)\,\mathrm{d}W_t. \tag{6.5}$$

Notice that μ^f and σ^f are not free inputs: once we assume f is a smooth function of (t, X_t), Itô fixes both drift and diffusion in terms of partial derivatives of f and the coefficients of X. That fact is what ultimately turns the hedging argument into a PDE — we will push the equalities through until every economic condition is a condition on partial derivatives of a single function.

There is a methodological subtlety worth flagging. The ansatz $f_t = f(t, X_t)$ is a *modelling* assumption, not a law of nature. For vanilla options it is uncontroversial; their payoff is a function of X_T alone. For path-dependent options — barriers, lookbacks, Asians — the assumption is wrong as stated, and we have to enlarge the state variable to include whatever path statistic the payoff depends on (running max, running average, time spent below a barrier). The general framework then says the price is a function of $(t, X_t, \text{path state})$ and Itô's lemma has to be applied to that augmented state. For the rest of this chapter we stick to the vanilla case.

7.1.5 6.1.5 A brief meditation on completeness

Before diving into the replication argument, it is worth pausing to note the technical assumption hiding in plain sight: *market completeness*. The framework posits a single Brownian motion W driving X and a single traded asset g whose volatility comes from the same W. That makes the market complete with respect to claims that depend only on (t, X_t): every such claim can be replicated using stock and cash. If we enlarged the state space — say, by making σ itself a random process driven by an independent Brownian motion W^σ — we would lose completeness. There would be claims that depend on the joint (W, W^σ) that cannot be replicated using g and cash alone. We would need an additional hedge instrument (typically another option) to span the second source of risk.

Real markets are in some sense "as complete as the hedging instruments available", which is a moving target. More liquid underlyings (large-cap equities, major indices) have richer options markets and therefore more hedgeable risk dimensions. The theory in this chapter treats the single-underlying, complete-market case; multi-factor and incomplete-market extensions are natural generalisations but are beyond the scope of the present exposition.

One implication of completeness is *uniqueness of price*. In a complete market, there is a single risk-neutral measure $\mathbb{Q}$, and every claim has a single fair price under it (CH05). In an incomplete market, there can be a family of equivalent risk-neutral measures, and different choices give different prices. The market has to coordinate on one of them via trading and supply-demand dynamics, which introduces a residual randomness to prices that is outside the pure replication argument.

7.2 6.2 The self-financing replicating portfolio

We hold:

- α_t — units of g_t,
- β_t — units of M_t,
- -1 — unit of f_t.

In words: we are long α_t shares of the traded claim, we have β_t units of cash earning the short rate, and we are short exactly one unit of the new claim f. The idea is that the stock-plus-cash position tries to replicate f; if it succeeds, the *net* position has zero risk and zero value. The discipline of replication is: pick α_t and β_t — and only those two — to pay down the risk of the short f.

The mental picture is of a small machine with two adjustable knobs. The first knob, α_t, controls how much stock we hold; the second, β_t, controls how much cash. At each instant, the trader twiddles both knobs. The target of the twiddling is to make the machine's total value track f_t so closely that the net (machine minus short f) is identically zero. If we succeed at every instant, the final machine value at $t = T$ will exactly equal the option's payoff $\varphi(X_T)$, and the short position we wrote at $t = 0$ is perfectly neutralised. The option's value today is then the cost of setting the machine running on day zero.

We need self-financing strategies. Define the portfolio value

$$V_t \;=\; \alpha_t\, g_t + \beta_t\, M_t - f_t. \tag{6.6}$$

We need $V_0 = 0$ (zero initial cost — otherwise we are giving or taking money for free at $t = 0$).

The condition $V_0 = 0$ is essential. If the replicating stock-plus-cash package costs more than what we collected for writing f, we have paid for the hedge out of nowhere; if it costs less, someone handed us money for free. A genuine replication argument must have the stock-plus-cash package cost exactly f_0. The algebra below will show that this "fair value" equals f_0 precisely when no-arbitrage holds, which is why the argument cannot be decoupled from its economic premise.

At $t = 0$, we write the option (collect f_0) and immediately deploy that cash — some into α_0 shares of g, the rest into the money-market account as β_0 units. After that initial allocation, no fresh cash ever enters the portfolio; the only thing the trader does for the rest of the option's life is *reshuffle* between stock and cash in response to market moves and the passage of time. The minus sign in front of f_t is critical: we are the short side of f, which is the side that has to build the hedge. The long side would simply hold the option and wait.

7.2.1 6.2.1 Naïve differentiation vs self-financing

Taking the total differential:

$$\mathrm{d}V_t \;=\; \mathrm{d}(\alpha_t g_t) + \mathrm{d}(\beta_t M_t) - \mathrm{d}f_t.$$

Expanding by Itô's product rule (CH03),

$$\mathrm{d}V_t \;=\; \underbrace{\mathrm{d}\alpha_t\, g_t}_{\text{extra}} + \alpha_t\, \mathrm{d}g_t + \underbrace{\mathrm{d}[\alpha, g]_t}_{\text{extra}} + \underbrace{\mathrm{d}\beta_t\, M_t}_{\text{extra}} + \beta_t\, \mathrm{d}M_t + \underbrace{\mathrm{d}[\beta, M]_t}_{=0} - \mathrm{d}f_t. \tag{6.7}$$

(The cross-variation $\mathrm{d}[\beta, M]_t = 0$ because M is of finite variation.) The self-financing constraint forces the "extra" rebalancing pieces to cancel, i.e.

$$\mathrm{d}\alpha_t\, g_t + \mathrm{d}[\alpha, g]_t + \mathrm{d}\beta_t\, M_t \;=\; 0. \tag{6.8}$$

This is the single most conceptually important equation in the chapter, and it is worth dwelling on. Ordinary calculus tells us that $\mathrm{d}(\alpha g) = \alpha\, \mathrm{d}g + g\, \mathrm{d}\alpha$. That identity is mathematically correct but *financially wrong*: it records changes in wealth that would have required money to appear or disappear from nowhere. The $g\, \mathrm{d}\alpha$ term is the dollar cost of *adding new shares* at the current price, which a trader does not pay from thin air — she has to sell some of

her money-market holding to fund the purchase. The equation (6.8) is the formal version of that common-sense statement.

To make the point visceral: imagine you hold 100 shares of a \$50 stock, worth \$5000. Suppose you suddenly decide you want to hold 200 shares, and at the moment of decision the stock is still at \$50. Did your wealth just jump by \$5000? Of course not — you had to hand \$5000 of cash to the market to get those extra shares. Naïve calculus says $\mathrm{d}(\alpha g) = g\,\mathrm{d}\alpha$ would count that \$5000 as new wealth, which is absurd. The self-financing constraint (6.8) says that the cash leg $\beta_t M_t$ has to shrink by exactly \$5000 in the same instant, so the *total* portfolio value is unchanged by the rebalance itself. Only subsequent *price* moves — the stock going from \$50 to \$51, the bank accruing interest — change the total wealth. This is the iron law we are about to build the whole pricing theory on: dollars don't appear, dollars don't vanish, and the only source of P&L is mark-to-market.

The cross-variation term $\mathrm{d}[\alpha, g]_t$ deserves a final comment. If α_t is a diffusion — meaning the trader's position itself has a $\mathrm{d}W$ component, which happens whenever $\alpha_t = \partial_x f(t, X_t)$ and X is Itô-driven — then α and g share a common Brownian driver and their covariation is non-zero. That cross-term lives in the rebalance leg and is absorbed into the self-financing constraint. In the discrete world of §6.11 it shows up as the $(\alpha_{t_n} - \alpha_{t_{n-1}})\,X_{t_n}$ piece of the bank recursion: when you change your share count, you trade the *difference* at the prevailing price, and the cash account absorbs the cost.

7.2.2 6.2.2 Interpreting self-financing

Between t and $t + \Delta t$ we move from (α_t, β_t) to $(\alpha_{t+\Delta t}, \beta_{t+\Delta t})$; the change in wealth consists only of the asset moves Δg_t and ΔM_t, not of fresh money injected from outside:

$$\Delta V_t \;=\; \alpha_t\,(\Delta g_t) + \beta_t\,(\Delta M_t) - \Delta f_t. \tag{6.9}$$

Why self-financing means $\mathrm{d}V = \alpha\,\mathrm{d}g + \beta\,\mathrm{d}M - \mathrm{d}f$. The naive Itô expansion contains six terms (two $\mathrm{d}\alpha\, g$-style rebalance terms, two "hold" terms $\alpha\,\mathrm{d}g, \beta\,\mathrm{d}M$, a quadratic cross-variation $\mathrm{d}[\alpha, g]$, and $\mathrm{d}f$). But a trader never swaps cash into or out of the portfolio from outside — so when she buys $\mathrm{d}\alpha$ more shares of g she must sell enough M to pay for it at the current price g_t. The self-financing constraint (6.8) is the statement "the rebalance leg is free". What survives is exactly the mark-to-market of the *held* positions: $\alpha\,\mathrm{d}g + \beta\,\mathrm{d}M$. Wealth moves only because markets moved, never because money appeared.

A portfolio that is self-financing is one whose wealth process is *pathwise* determined by the asset price paths and the initial position — no external injections, no trapdoors where money can appear. This is the financial analogue of conservation of energy: nothing gets spent or produced in the rebalance itself. The implication for pricing is profound: if two self-financing portfolios end at the same terminal value on every path, they must have the same value today, because if they didn't, buying the cheap one and shorting the expensive one would yield a costless portfolio with zero risk and a positive initial P&L — a free lunch.

Self-financing vs violation *Dropping the rebalance-cost term fabricates PnL — the classic "something for nothing" that (6.8) forbids.*

7.2.3 6.2.3 Applying the self-financing constraint

After cancellation,

$$\mathrm{d}V_t \;=\; \alpha_t\,\mathrm{d}g_t + \beta_t\,\mathrm{d}M_t - \mathrm{d}f_t. \tag{6.10}$$

The only surviving terms are the three "mark-to-market" pieces: the P&L from the α_t shares of g because g moved, the interest accrual on the bank account, and the change in the liability f. There is *nothing else.*

Substituting (6.2), (6.3), (6.5):

$$\mathrm{d}V_t \;=\; \alpha_t(\mu_t^g g_t\,\mathrm{d}t + \sigma_t^g g_t\,\mathrm{d}W_t) + \beta_t M_t r_t\,\mathrm{d}t - (\mu_t^f f_t\,\mathrm{d}t + \sigma_t^f f_t\,\mathrm{d}W_t). \tag{6.11}$$

Collecting drift and diffusion,

$$\boxed{\mathrm{d}V_t \;=\; (\alpha_t \mu_t^g g_t + \beta_t M_t r_t - \mu_t^f f_t)\,\mathrm{d}t \;+\; (\alpha_t \sigma_t^g g_t - \sigma_t^f f_t)\,\mathrm{d}W_t} \tag{6.12}$$

Equation (6.12) has the form $\mathrm{d}V_t = \text{drift}\,\mathrm{d}t + \text{noise}\,\mathrm{d}W_t$, and we have two free knobs — α_t and β_t — to play with. The strategy is exactly what a sensible trader would do in words: first get rid of the noise (pick α_t to annihilate the $\mathrm{d}W_t$ coefficient), then inspect what remains and let economics constrain the rest.

This two-stage procedure — kill noise, then inspect drift — is the mathematical heart of every replication argument in finance. If the underlying has k sources of risk instead of one, we need k traded claims instead of one; we pick k holdings to kill each of the k noise terms simultaneously; the remaining drift must equal the risk-free rate on the net cash position. In the Heston stochastic-volatility model, for example, there are two sources of risk (the stock's diffusion and the volatility's diffusion), and we need two hedge instruments (the stock and another option) to kill both.

7.3 6.3 Locally removing the risk — the delta-hedge ratio

Locally remove risk, so set the $\mathrm{d}W_t$ coefficient to zero:

$$\alpha_t\,\sigma_t^g\,g_t - \sigma_t^f\,f_t \;=\; 0 \;\implies\; \boxed{\alpha_t \;=\; \frac{\sigma_t^f}{\sigma_t^g}\,\frac{f_t}{g_t}} \tag{6.13}$$

This is the delta-hedge ratio in its most general form. Before we unpack it, note what we have achieved: the instant we pick α_t as above, the *random* part of $\mathrm{d}V_t$ — the bit that depends on which direction the Brownian motion happened to move — disappears. Over a small enough interval, the portfolio is locally riskless.

The formula has a crystalline interpretation: α_t is the ratio of the dollar-volatilities of f and g. If f is twice as volatile in dollar terms as g, we need twice as many units of g to keep up. In the common case where $g = X$ (the stock itself is traded), we have $\sigma_t^g g_t = \sigma_t^x$ and $\sigma_t^f f_t = \sigma_t^x\,\partial_x f$, so the ratio collapses to $\alpha_t = \partial_x f$ — the partial derivative of the option value with respect to the spot, a.k.a. the *delta*. The replicating portfolio is the option's Doppelgänger: at each instant, it holds exactly as much stock as the option's price is sensitive to the stock — no more, no less.

A concrete numerical illustration. Suppose the stock is trading at \$100 with $\sigma = 20\%$, so the dollar volatility of one share is roughly $100 \cdot 0.20 = 20$ dollars per unit of W. Suppose the option we are hedging has $\partial_x f = 0.6$, so its dollar volatility is $0.6 \cdot 20 = 12$ dollars per unit of W. To neutralise the option's Brownian motion exposure, we hold 0.6 shares of stock. If the stock now moves by 1% in a tiny interval, the option's price moves by $0.6 \cdot 1\% \cdot 100 = 0.60$ dollars and the share position moves by $0.6 \cdot 1 = 0.60$ dollars — they cancel to the instant.

A useful edge case. A deep-ITM call has delta approaching 1, so the hedger holds roughly one full share per option. A deep-OTM call has delta approaching 0, so the hedger holds almost no stock, and the bank account is approximately the option's premium invested at the risk-free rate. Again, gamma is nearly zero, rebalancing is minimal. The hedging problem is *easy* in both extremes and hardest *at the money*, where the delta is around 0.5 but the gamma is large and rebalancing is frequent.

One subtlety worth flagging: the formula $\alpha_t = \partial_x f(t, X_t)$ is not a *static* ratio. As time passes and the stock moves, both arguments of the partial derivative change, and the required hedge ratio changes too. This is why dynamic hedging is called *dynamic*: the hedge ratio is a moving target that must be chased.

With this choice,

$$\mathrm{d}V_t \;=\; (\alpha_t \mu_t^g g_t + \beta_t M_t r_t - \mu_t^f f_t)\,\mathrm{d}t \;=\; \mathcal{A}_t\,\mathrm{d}t. \tag{6.14}$$

After the $\mathrm{d}W_t$ term has been killed, the remaining dynamics are purely deterministic over an infinitesimal interval. The portfolio earns some drift $\mathcal{A}_t$ per unit time, and the question becomes: what does that drift have to be?

7.3.1 6.3.1 No-arbitrage forces $\mathcal{A}_t = 0$

- If $\mathcal{A}_t > 0$: profit guaranteed — arbitrage.
- If $\mathcal{A}_t < 0$: reverse the strategy to get profit.

The argument is stark. We have built, by pure bookkeeping, a self-financing portfolio with zero initial cost ($V_0 = 0$) and zero instantaneous risk. If its value drifts upward, we are printing money; if it drifts downward, we swap the signs of all holdings and *then* we are printing money. The only logical possibility consistent with no free lunches is that the drift is exactly zero.

Therefore, to avoid arbitrage,

$$\boxed{\mathcal{A}_t \;=\; 0}. \tag{6.15}$$

Combined with $V_0 = 0$, the entire portfolio process satisfies $V_t = 0$ for all t, hence

$$\mathcal{A}_t = 0 \implies V_t = 0 \implies \alpha_t g_t + \beta_t M_t - f_t = 0 \implies \beta_t M_t \;=\; f_t - \alpha_t g_t. \tag{6.16}$$

Equation (6.16) is the pricing identity in disguise. Read it from right to left: at any instant, the unique cash holding that keeps the hedge costless is precisely $f_t - \alpha_t g_t$. Equivalently, the value of the option *is* the value of the self-financing stock-plus-cash package that replicates it. The pricing question has been reduced to a constructive recipe: hold α_t shares of g, hold $(f_t - \alpha_t g_t)/M_t$ units of the money market, and the wealth of the package exactly tracks the option from now until expiry.

To see the recipe in practical terms, suppose the option value right now is \$3.50 and the delta is 0.6, and the stock is at \$100. The replicating package holds 0.6 shares (worth \$60) and has −\$56.50 in the money market (i.e. borrows \$56.50). The total package value is $0.6 \cdot 100 - 56.50 = 3.50$, which matches the option. If the stock rises to \$101, the shares are worth $0.6 \cdot 101 = \$60.60$ and the loan is now slightly more expensive by the interest accrual, but the option itself has risen by the first-order amount $0.6 \cdot \$1 = \$0.60$ to \$4.10. The package now holds the updated delta (say 0.62) of the new stock price, so the trader must buy 0.02 shares, funded by additional borrowing. And so on, every instant. That is the whole game.

7.4 6.4 Market price of risk

Having extracted the zero-drift consequence, we unpack it into a statement about expected returns, which turns out to be one of the deepest identities in quantitative finance.

Write out $\mathcal{A}_t = 0$:

$$\alpha_t \mu_t^g g_t + r_t\,(f_t - \alpha_t g_t) - \mu_t^f f_t \;=\; 0. \tag{6.17}$$

Rearrange,

$$\alpha_t\, \mu_t^g\, g_t + r_t f_t - r_t \alpha_t g_t - f_t \mu_t^f \;=\; 0. \tag{6.18}$$

Recall $\alpha_t = \dfrac{\sigma_t^f}{\sigma_t^g}\dfrac{f_t}{g_t}$. Substituting and dividing by f_t,

$$\frac{\mu_t^g - r_t}{\sigma_t^g} \;=\; \frac{\mu_t^f - r_t}{\sigma_t^f} \;=\; \lambda_t \;=\; \lambda(t, X_t). \tag{6.19}$$

Sharpe ratios of all assets on the underlying index are equal.

We started from a single no-arbitrage condition and derived that every single derivative on X — vanilla call, digital, variance swap, structured product — must have the same expected excess return per unit of volatility. They can have wildly different drifts and wildly different volatilities, but the *ratio* of excess return to volatility is pinned down to the same number λ_t for every instrument on the same underlying. Economics collapses the whole zoo into one scalar.

λ_t is the **market price of risk** — a market property (not an asset-specific quantity). Thus

$$\boxed{\frac{\mu_t^f - r_t}{\sigma_t^f} \;=\; \lambda_t \iff \mu_t^f - r_t \;=\; \lambda_t\,\sigma_t^f}. \tag{6.20}$$

The connection to Girsanov. In CH05 we derived Girsanov's theorem: given a predictable shift λ_t, there is an equivalent measure $\mathbb{Q}$ under which $\widetilde{W}_t := W_t + \int_0^t \lambda_s \, \mathrm{d}s$ is a Brownian motion. Under $\mathbb{Q}$ the drift of X becomes $\mu_t^x - \sigma_t^x \lambda_t$. Equation (6.20) is *exactly* the economic statement that picks out the Girsanov shift λ_t that neutralises the risk premium: under $\mathbb{Q}$, every traded claim on X has drift r_t (no risk premium), which is the defining property of the risk-neutral measure. The "market price of risk" and the "Girsanov drift shift" are the *same scalar*, viewed from two sides: economics says every instrument must share it, CH05 says a specific measure change zeros it out.

Rewriting (6.20) as $\mu_t^f = r_t + \lambda_t \sigma_t^f$ says that *every* instrument's expected return decomposes into a risk-free piece and a risk premium. The risk premium is not some mysterious extra — it is literally the volatility of the instrument times the market price of a unit of volatility. A stock with twice the volatility of another, in the same market, must deliver twice the excess return in expectation (under the physical measure). An option that has five times the dollar-volatility of the stock must deliver five times the excess return.

Under $\mathbb{Q}$, the construct that makes all instruments drift at the risk-free rate, the market price of risk is zero: it is the fictitious world where risk-takers demand no compensation. Pricing under $\mathbb{Q}$ is pricing *as if* no one required a risk premium, which is correct for a replication-cost argument because the replicator does not care about risk premiums — she has eliminated the risk entirely. The Girsanov shift $-\sigma\lambda$ in the drift is precisely the adjustment from the real world (where risk is priced) to the risk-neutral world (where it is not). And because the shift is only to the drift, not to the diffusion coefficient, all variance-based quantities (gamma P&L, vega P&L) are the same under either measure. The variance is measure-invariant; only the drift shifts.

A concrete number may help ground the concept. Over the past century the US equity market has earned roughly 6-7% per year in excess of the risk-free rate with an annualised volatility of roughly 15-17%. That gives a long-run Sharpe ratio of about 0.4, which is the implicit estimate of λ that equity markets offer to long-term investors. When market participants talk about "risk premium" in casual conversation, they are usually talking about the quantity $\lambda \cdot \sigma$ — the extra expected return in percentage points per year — but the cleaner object is λ itself, because it is shared across all instruments on the same underlying.

An illuminating thought experiment. Imagine two derivatives on the same stock, a vanilla call and a digital. In real-world statistics, the call might have an expected excess return of 30% per year with a volatility of 75% per year; the digital might have an expected excess return of 3% with a volatility of 7.5%. Naively these look like very different risk-reward profiles. But their Sharpe ratios are both 0.4, exactly matching the stock's. The market is saying that despite one being ten times more volatile than the other in absolute terms, both are paying the same *price* per unit of risk exposure. Noting this invariance is what makes (6.19) such a powerful pricing lens: it strips out the incidental scale of each instrument and reveals a single underlying risk-price that all of them respect.

One more remark. The market price of risk is a *state-dependent* quantity; $\lambda_t = \lambda(t, X_t)$ in general, not a constant. In quiet bull markets investors are willing to take risk cheaply, so λ is small; in panics after a crash λ can spike as investors demand enormous compensation for exposure. The option prices derived below do *not* depend on λ because the replicator erases the risk altogether. But the *flow* of orders in the options market — whether people are net buyers or sellers of protection — does depend on λ, because risk-averse investors with high λ are willing to pay more for a put than risk-neutral replicators.

7.5 6.5 The generalised Black-Scholes PDE

We have a scalar identity: $\mu^f - r = \lambda\sigma^f$. We now translate it into a PDE by recalling what μ^f and σ^f actually *are* — Itô expressions involving partial derivatives of the pricing function $f(t, x)$. Substituting those expressions turns the economic statement into a differential equation that the pricing function must satisfy.

The move from a scalar identity to a partial differential equation is a small conceptual leap that is easy to miss on first reading. The identity $\mu^f - r = \lambda\sigma^f$ is a statement about numbers: at every instant, some particular combination of numbers must vanish. But those numbers are themselves computed from the pricing function $f(t, x)$ via Itô's lemma,

so the statement is really "this combination of partial derivatives of f must vanish at every point (t, x)". That, by definition, is a partial differential equation.

Recall (via Itô's lemma) for $f(t, x)$:

$$\mu_t^f \;=\; \frac{\partial_t f_t + \mu_t^x\,\partial_x f_t + \frac{1}{2}(\sigma_t^x)^2\,\partial_{xx} f_t}{f_t}, \tag{6.21}$$

$$\sigma_t^f \;=\; \frac{\sigma_t^x\,\partial_x f_t}{f_t}. \tag{6.22}$$

(Here μ_t^x, σ_t^x are the coefficients of X from (6.1).) Plug into $\mu_t^f - r_t = \lambda_t \sigma_t^f$:

$$\partial_t f_t + (\mu_t^x - \sigma_t^x \lambda_t)\,\partial_x f_t + \tfrac{1}{2}(\sigma_t^x)^2\,\partial_{xx} f_t \;=\; r_t\, f_t. \tag{6.23}$$

This has to hold $\forall (t, x)$. Hence:

$$\boxed{\begin{aligned} \partial_t f(t,x) + \big(\mu^x(t,x) - \sigma^x(t,x)\,\lambda(t,x)\big)\,\partial_x f(t,x) \\ + \tfrac{1}{2}\big(\sigma^x(t,x)\big)^2\,\partial_{xx} f(t,x) \;=\; r(t,x)\,f(t,x), \\ f(T,x) \;=\; \varphi(x). \end{aligned}} \tag{6.24}$$

Generalised Black-Scholes PDE.

This is the consistency condition that ties together everything the option is going to do between now and expiry. The terminal condition $f(T, x) = \varphi(x)$ anchors the pricing function at maturity to the payoff shape, and the PDE propagates that shape back in time. The replicating portfolio's delta today — $\partial_x f(t, X_t)$ — is chosen precisely so that the option we are manufacturing ends up with the right payoff tomorrow.

Look at the structure of the PDE for a moment. The linear operator on the left-hand side is a backwards parabolic operator — backwards because time flows the "wrong" way (we propagate from the terminal condition at T back to the present), parabolic because the second-order derivative appears with a positive coefficient. Parabolic equations describe diffusion-like evolution: information smears out as time flows, corners round off, sharp features disappear. In option pricing, that translates to the idea that any fine-grained feature of the payoff at maturity — a sharp kink at the strike, a step at a digital, a cliff at a barrier — gets smoothed out as we go backwards in time.

The coefficient $(\mu^x - \sigma^x \lambda)$ multiplying $\partial_x f$ deserves special attention. It is not the physical drift μ^x; it is the physical drift minus the risk premium $\sigma^x \lambda$. The subtraction is exactly the Girsanov shift from CH05 that turns the physical $\mathbb{P}$-Brownian motion into a risk-neutral $\mathbb{Q}$-Brownian motion. So (6.24) can be read as "the expected rate of change of f along a risk-neutral drift, plus the Itô correction, equals the carry cost". In expectation form, the same statement is Feynman-Kac (CH04), as we will see again in §6.7A and (6.40).

> **Why the BS PDE encodes no-arbitrage.** Read (6.24) coefficient-by- coefficient: the $\partial_x f$ term carries the risk-neutral drift $(\mu^x - \sigma^x \lambda)$, not the physical drift μ^x. That subtraction $-\sigma^x \lambda$ is exactly the Girsanov change-of-measure (CH05) that turns W into a $\mathbb{Q}$-Brownian motion. The $\frac{1}{2}(\sigma^x)^2\,\partial_{xx} f$ term is Itô's curvature correction — it is independent of drift, which is why option prices ultimately don't depend on μ. The RHS rf says the whole hedge-replicating portfolio must grow at the risk-free rate (otherwise §6.3.1's arbitrage trade exists). The Feynman-Kac representation (6.40) is *the same statement* in expectation form.
>
> **What happens when Δ is wrong.** If you hold $\alpha_t \neq \partial_x f$, the $\mathrm{d}W_t$ coefficient (6.12) is non-zero and the hedge error over $[t, t + \mathrm{d}t]$ is $(\alpha_t \sigma_t^g g_t - \sigma_t^f f_t)\,\mathrm{d}W_t$ — a mean-zero *noise* term plus a second-order drift $\frac{1}{2}\,\partial_{xx} f\,(\mathrm{d}X_t^2 - \sigma^2 X_t^2 \mathrm{d}t)$ that accumulates as realised-minus-implied variance. This is the famous *P&L attribution*: discretely-hedged short gamma loses money when realised vol exceeds implied, and makes money when it under-realises. See `ch06-hedge-error.png`.

That P&L attribution deserves a second look, because it is the most practically important fact in all of options trading. When we sell an option and delta-hedge it using the Black-Scholes delta computed with an implied volatility σ_{imp}, our instantaneous P&L is (to leading order) proportional to $\Gamma \cdot S^2 \cdot (\sigma_{\text{imp}}^2 - \sigma_{\text{real}}^2)\,\mathrm{d}t$. If realised volatility turns out to be lower than the implied volatility we priced the option at, the short-option position makes money on each small interval; if realised is higher, we lose. The gamma $\Gamma = \partial_{xx} f$ multiplies everything because it is the "curvature" of the payoff — a payoff with no curvature has no gamma exposure and no sensitivity to the realised variance path. The cleanest way to think of option P&L is therefore as a daily duel between implied and realised variance, weighted by the gamma we are carrying. The option market is, in a very real sense, a market for variance.

To make the decomposition explicit, consider what happens over a small interval $[t, t + \mathrm{d}t]$ when we are short one option and hold $\alpha_t = \partial_x f(t, X_t)$ shares of stock. The option moves by $\mathrm{d}f = \partial_t f\,\mathrm{d}t + \partial_x f\,\mathrm{d}X + \frac{1}{2}\partial_{xx} f\,(\mathrm{d}X)^2$ by Itô, where $(\mathrm{d}X)^2 = \sigma_{\text{real}}^2 X^2\,\mathrm{d}t$ on the realised path. Using the PDE (6.27) with σ_{imp} as the model volatility, the net P&L on the short-option package works out to

$$\tfrac{1}{2}\partial_{xx} f \cdot X^2 \cdot (\sigma_{\text{imp}}^2 - \sigma_{\text{real}}^2)\,\mathrm{d}t.$$

If realised vol came in below implied, this is positive — the short wins; if above, the short loses.

A delta-hedged short-option position has *zero* exposure to the stock's level and *pure* exposure to realised variance. A trader who is bearish on volatility can express that view cleanly by selling options and delta-hedging. Her P&L is a textbook variance swap, paid off in continuous time as a function of the realised-minus-implied variance along the path.

BS-PDE coefficients *Theta pays for gamma: the three LHS terms balance rf* exactly* — the dotted residual is numerically zero.*

One more framing, for readers who like symmetries. The BS PDE is often written as "theta plus convection plus diffusion equals interest", and each piece has a trading interpretation: theta is the time-decay we earn (or pay) for holding the option, convection is the drift-driven directional re-pricing, diffusion is the gamma-weighted re-pricing from variance, and the RHS interest is the financing cost of the replication. A long-gamma, short-theta option is paying you theta every day for the right to harvest gamma-variance; the BS PDE is the identity that, once we pick a consistent volatility, makes those two cancel exactly in expectation.

One more observation about the structure of (6.24) that is often overlooked. The PDE is linear in f. This linearity is the reason option prices obey simple addition: the price of a portfolio of options equals the sum of the prices of the individual options. Traders exploit this every day by building structured products out of simple building blocks — a call spread is a long-call minus a short-call, a straddle is a long-call plus a long-put, a butterfly is a specific linear combination of three strikes. Each of these structures is a sum of vanilla instruments, and its price is the sum of the corresponding BS formulas.

7.6 6.6 Specialising to Black-Scholes (GBM)

The generalised PDE is beautiful but abstract. To get to numbers, we specialise to the single most famous setting in quantitative finance: geometric Brownian motion. In GBM, the stock has constant proportional drift and volatility, and it is traded directly — so $g = X$.

In the Black-Scholes model:

$$\mu^x(t, x) \;=\; \mu x, \qquad \sigma^x(t, x) \;=\; \sigma x,$$

so

$$\mathrm{d}X_t \;=\; X_t \mu\,\mathrm{d}t + X_t \sigma\,\mathrm{d}W_t. \tag{6.25}$$

In divided form, $\mathrm{d}S_t/S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$: the *percentage* increment of the stock is a Gaussian with mean $\mu\,\mathrm{d}t$ and variance $\sigma^2\,\mathrm{d}t$ — geometric Brownian motion. Applying Itô's lemma to $\ln S_t$ (CH03) turns this into a constant-coefficient arithmetic SDE,

$$\mathrm{d}(\ln S_t) \;=\; (\mu - \tfrac{1}{2}\sigma^2)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t,$$

which integrates immediately. Exponentiating,

$$S_t \;=\; S_0\,\exp\Big((\mu - \tfrac{1}{2}\sigma^2)\,t \;+\; \sigma W_t\Big) \;\overset{d}{=}\; S_0\,\exp\Big((\mu - \tfrac{1}{2}\sigma^2)\,t \;+\; \sigma\sqrt{t}\,Z\Big), \qquad Z \sim \mathcal{N}(0,1). \tag{6.25a}$$

The distribution of S_t is therefore log-normal with log-mean $(\mu - \frac{1}{2}\sigma^2)\,t$ and log-variance $\sigma^2 t$. The $-\frac{1}{2}\sigma^2$ term is the Itô correction: without it, $\mathbb{E}[S_t]$ would *not* equal $S_0 e^{\mu t}$. With it, $\mathbb{E}[S_t] = S_0 e^{\mu t}$ exactly.

Take $g(t,x) = x$, so that X is indeed traded (with $\sigma^g = \sigma$, $\mu^g = \mu$), and let $r(t,x) = r$ (constant). Therefore

$$\lambda \;=\; \frac{\mu - r}{\sigma}. \tag{6.26}$$

In GBM, the market price of risk is literally the stock's Sharpe ratio. That identity, which fell out of a general no-arbitrage argument, is what gives λ its evocative name and what motivates the risk-neutral measure $\mathbb{Q}$ (CH05) as the probability law under which the stock drifts at the risk-free rate.

Substituting into (6.24):

$$\partial_t f + \underbrace{(\mu - \sigma\lambda)}_{=r}\, x\,\partial_x f + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx} f \;=\; rf.$$

$$\boxed{\begin{aligned} &\partial_t f + r\,x\,\partial_x f + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx} f \;=\; rf, \\ &f(T,x) \;=\; \varphi(x). \end{aligned}} \tag{6.27}$$

Black-Scholes PDE.

Look at what happened: the stock's physical drift μ disappeared completely from the PDE. It was absorbed by the market price of risk combination $\mu - \sigma\lambda$, which is just r. Option prices under no-arbitrage do not depend on the real-world expected return of the stock — a fact that was counterintuitive to early practitioners but which makes perfect sense once you realise the option is being priced by its replication cost, and the replication only cares about the *scale* of the stock's noise, not its direction.

This is one of those results that sounds impossible until you see why. The common intuition says "options on a stock with a higher expected return should be worth more". The intuition is correct for *unhedged* investors, who pay for calls expecting to profit from the drift. But it is wrong for *hedgers*, who are building a machine that tracks the payoff state-by-state regardless of the drift. The hedger's cost depends only on how hard the machine has to work, and a machine working against a high-vol stock has a harder job than one working against a low-vol stock, regardless of drift. The two perspectives reconcile via the risk-neutral measure: a hedger's fair price is the same as an "investor's fair price" under a world where everyone is risk-neutral.

To drive this home with a numerical example, consider two hypothetical stocks. Stock A is a high-growth tech company with expected annual return 15% and volatility 30%. Stock B is a boring utility with expected annual return 3% and volatility 30%. Both trade at \$100; we price a 1-year at-the-money call with $r = 5\%$. The Black-Scholes call price depends on x, K, r, σ, $T - t$ — not on μ. The same number for both stocks.

One more angle: because the drift is eliminated, implied volatility is revealed as "the market's forecast of future volatility" in a very specific sense. An at-the-money implied vol of 20% is the market's unbiased estimate of the root-mean-square of log-returns over the option's life, *under the risk-neutral measure*. Under the physical measure it might be a little different (typically a bit lower, because the risk-neutral measure puts more weight on bad outcomes), but the gap is small for liquid equity options.

7.7 6.7 Black-Scholes formulas for call, put, and digital

With the PDE (6.27) in hand, we can read off closed forms for the three canonical vanilla payoffs. The call and put solutions can be obtained either by substituting an ansatz into (6.27), by computing the risk-neutral expectation directly under (6.25a), or — most concisely — via the Feynman-Kac bridge of CH04. All three routes give the same formulas; we record them here and defer the derivation details to §6.7A.

Vanilla call. For $\varphi(x) = (x - K)_+$,

$$\boxed{f(t,x) \;=\; x\,\Phi(d_+) \;-\; K\,e^{-r(T-t)}\,\Phi(d_-)} \tag{6.28}$$

with

$$d_\pm \;=\; \frac{\ln(x/K) + (r \pm \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}. \tag{6.29}$$

Equation (6.28) is the **Black-Scholes call formula**. You can read it in two pieces: the first term $x\,\Phi(d_+)$ is the expected stock payoff *given that the option expires in the money* — a probability-weighted piece of the stock, computed under the stock-numeraire measure. The second term $K\,\Phi(d_-)\,e^{-r(T-t)}$ is the discounted strike times the risk-neutral probability of being in the money. The difference is the expected net gain from exercising. The symmetry $d_+ = d_- + \sigma\sqrt{T-t}$ connects the two terms: the extra $\sigma\sqrt{T-t}$ in d_+ is the "vol lift" that accounts for the fact that, conditional on ending in the money, the expected log-price is elevated relative to its unconditional median.

Vanilla put. For $\varphi(x) = (K - x)_+$, the cleanest route is via **put-call parity**. From the algebraic identity $(X_T - K)_+ - (K - X_T)_+ = X_T - K$, take risk-neutral expectations and discount:

$$\boxed{C(t,x) - P(t,x) \;=\; x - K\,e^{-r(T-t)}} \tag{6.30}$$

Parity is **model-independent** — it does not depend on GBM, or even on the stock dynamics at all. It follows purely from the algebraic identity combined with the existence of a risk-neutral measure (CH05) under which discounted traded prices are martingales. Put-call parity violations are therefore the cleanest possible arbitrage signal. Solving (6.30) for P:

$$\boxed{P(t,x) \;=\; K\,e^{-r(T-t)}\,\Phi(-d_-) \;-\; x\,\Phi(-d_+)} \tag{6.31}$$

Digital call. For $\varphi(x) = \mathbb{1}_{x \geq K}$, the discounted price is simply the risk-neutral in-the-money probability times the discount factor:

$$\boxed{f(t,x) \;=\; e^{-r(T-t)}\,\Phi(d_-)} \tag{6.32}$$

Under GBM, the indicator $\{X_T \geq K\}$ is equivalent to $\{Z \geq -d_-\}$ where $Z \sim \mathcal{N}(0,1)$, so the $\mathbb{Q}$-probability is $\Phi(d_-)$, which gives (6.32). Far in the money, d_- is large and $\Phi(d_-) \approx 1$; far out of the money, d_- is strongly negative and $\Phi(d_-) \approx 0$; at the money, $\Phi(d_-)$ sits somewhere near $\frac{1}{2}$.

7.7.1 6.7.1 Greeks from the closed form

From (6.28) we can read off all the Greeks of the vanilla call analytically. The key identity that makes the algebra collapse is the **strike-shifting identity** $x\,\phi(d_+) = K\,e^{-r(T-t)}\,\phi(d_-)$, which is provable by direct algebra on the density arguments.

- **Delta**: $\Delta^C \;=\; \partial_x f \;=\; \Phi(d_+) \in (0,1)$.
- **Gamma**: $\Gamma^C \;=\; \partial_{xx} f \;=\; \dfrac{\phi(d_+)}{x\,\sigma\sqrt{T-t}}$.
- **Vega**: $\mathcal{V}^C \;=\; \partial_\sigma f \;=\; x\,\phi(d_+)\,\sqrt{T-t}$.

For the put, parity gives $\Delta^P = \Delta^C - 1 \in (-1, 0)$ and $\Gamma^P = \Gamma^C$ (differentiating parity twice in x): calls and puts of the same strike and maturity have *identical* gamma. Vega is also shared: $\mathcal{V}^P = \mathcal{V}^C$.

The gamma formula exhibits a striking $1/\sqrt{T-t}$ divergence at the money as $t \to T$: the at-the-money gamma blows up like $1/\sqrt{T-t}$. This is the *pin gamma* phenomenon — on expiration day a market-maker's gamma exposure is effectively all in a narrow band around the strike, which is why gamma books are rebalanced with extra care on expiration Fridays. Integrated across spot, however, $\int \Gamma^C \, \mathrm{d}x = \Delta^C(\infty) - \Delta^C(0) = 1 - 0 = 1$: the total gamma "mass" is conserved across time-to-expiry. What changes is how that mass is distributed spatially.

Digital greeks. The digital call's delta from (6.32) is

$$\partial_x f^{\mathrm{dig}} \;=\; e^{-r(T-t)} \, \frac{\phi(d_-)}{x\,\sigma\sqrt{T-t}}, \tag{6.33}$$

which has a *pathological* limit as $T - t \to 0$ at $x = K$: the numerator $\phi(d_-) \to \phi(0) = 1/\sqrt{2\pi}$ is bounded away from zero, while the denominator $\sqrt{T-t}$ tends to zero. The delta blows up without bound at the strike. We will visualise this in §6.11.7 and discuss the call-spread hedge that practitioners use to tame it.

BS price and delta curves *BS price and delta curves.*

7.8 6.7A Dual derivation via Feynman-Kac

The self-financing hedging argument of §§6.2–6.5 is the route by which Black, Scholes, and Merton originally derived the PDE (6.27). There is a second, entirely independent route that arrives at *exactly* the same PDE and the same formulas: the **Feynman-Kac** bridge of CH04. Seeing both routes is worthwhile because they foreground different pieces of the machinery and their agreement is the mark of a solid theory.

7.8.1 6.7A.1 Setup under the risk-neutral measure

Under the risk-neutral measure $\mathbb{Q}$ constructed in CH05, the traded asset and bank account satisfy

$$\frac{\mathrm{d}S_t}{S_t} \;=\; r\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}\widetilde{W}_t, \qquad \frac{\mathrm{d}M_t}{M_t} \;=\; r\,\mathrm{d}t, \tag{6.34}$$

where $\widetilde{W}_t$ is a $\mathbb{Q}$-Brownian motion. The drift is r, not the real-world μ, because $\mathbb{Q}$ is chosen so that S_t/M_t is a $\mathbb{Q}$-martingale — and that martingale condition forces the drift of $\mathrm{d}S/S$ to equal r. The physical expected return μ never appears in derivative prices; it is *absorbed into the measure change* via the Radon-Nikodym derivative of CH05, with Girsanov shift $\lambda = (\mu - r)/\sigma$.

7.8.2 6.7A.2 Applying Feynman-Kac

The claim $\xi = (\xi_t)$ with $\xi_T = \varphi(S_T)$ has pricing function $F_t = f(t, S_t)$. Feynman-Kac (CH04) applied with drift $a(x) = rx$, diffusion $b(x) = \sigma x$, and discounting $c = r$ yields the representation

$$f(t,x) \;=\; \mathbb{E}^{\mathbb{Q}}_{t,x}\big[\, e^{-r(T-t)}\,\varphi(X_T)\,\big], \qquad \mathrm{d}X_s = r\,X_s\,\mathrm{d}s + \sigma\,X_s\,\mathrm{d}\widetilde{W}_s. \tag{6.35}$$

By the CH04 FK theorem, $f(t,x)$ satisfies the PDE

$$\partial_t f + r\,x\,\partial_x f + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx} f \;=\; r\,f, \qquad f(T,x) = \varphi(x). \tag{6.36}$$

This is *identical* to (6.27). Two independent derivations — the hedging argument of §6.5 and the FK martingale argument of CH04 — converge on the same PDE. That convergence is not a coincidence; both are expressing the same fact in different languages.

7.8.3 6.7A.3 Discounted price as a $\mathbb{Q}$-martingale

The clearest way to see the equivalence is via the discounted price. Let $Y_t := f(t, X_t)/M_t$. Applying Itô's lemma (CH03) and collecting the $\mathrm{d}t$ terms gives

$$\mathrm{d}Y_t \;=\; \frac{1}{M_t}\big(\partial_t f + rx\,\partial_x f + \tfrac{1}{2}\sigma^2 x^2 \partial_{xx} f - rf\big)\mathrm{d}t \;+\; \frac{\sigma x\,\partial_x f}{M_t}\,\mathrm{d}\widetilde{W}_t. \tag{6.37}$$

By the Black-Scholes PDE (6.36), the $\mathrm{d}t$ bracket vanishes identically, leaving only the Brownian piece — i.e. Y is a (local) martingale. **The PDE is exactly the drift-killer**: $\mathbb{Q}$ is defined so the bracket equals zero.

In this framing, the Black-Scholes PDE is not mysterious partial- differential-equation magic; it is *literally* the statement "the discounted price of the claim is a martingale under $\mathbb{Q}$" translated into Itô derivatives. Memorise the chain of equivalences:

Martingale $\Leftrightarrow$ drift is zero $\Leftrightarrow$ PDE holds.

If you ever forget what the Black-Scholes PDE says, you can reconstruct it: take the Itô of the discounted price, set the drift to zero, and copy out the bracket. That bracket is the PDE.

7.8.4 6.7A.4 Recovering the call formula from the expectation

With (6.35) in hand, we can derive (6.28) by direct integration. Integrate $\mathrm{d}X_s/X_s = r\,\mathrm{d}s + \sigma\,\mathrm{d}\widetilde{W}_s$ with $X_t = x$ to get

$$X_T \;=\; x \cdot \exp\Big\{(r - \tfrac{1}{2}\sigma^2)(T-t) + \sigma(\widetilde{W}_T - \widetilde{W}_t)\Big\}. \tag{6.38}$$

Decompose the call payoff as $(X_T - K)_+ \;=\; X_T\,\mathbb{1}_{X_T>K} - K\,\mathbb{1}_{X_T>K}$. The second piece is the discounted digital times K: from (6.32), $K\,e^{-r(T-t)}\,\Phi(d_-)$. The first piece — the stock-delivery leg — uses the change-of-numeraire trick of CH05. Under the stock-numeraire measure $\mathbb{Q}^S$ (with S_t as numeraire rather than the money account M_t), the expectation $\mathbb{E}^{\mathbb{Q}^S}[\mathbb{1}_{X_T>K}]$ equals $\Phi(d_+)$ with $d_+ = d_- + \sigma\sqrt{T-t}$. Translating back to the $\mathbb{Q}$-expectation via the Radon-Nikodym density $\mathrm{d}\mathbb{Q}^S/\mathrm{d}\mathbb{Q} = S_T/(S_t\,e^{r(T-t)})$:

$$\mathbb{E}^{\mathbb{Q}}_{t,x}\Big[e^{-r(T-t)}\,X_T\,\mathbb{1}_{X_T>K}\Big] \;=\; x \cdot \Phi(d_+). \tag{6.39}$$

Combining the two pieces recovers (6.28). The structure "probability of payoff times conditional expected payoff" is visible in the formula: $\Phi(d_-) = \mathbb{Q}(X_T > K)$ is the risk-neutral probability of exercise; $\Phi(d_+) = \mathbb{Q}^S(X_T > K)$ is the stock-measure probability of exercise. The two differ by the Girsanov shift $\sigma\sqrt{T-t}$ between the two numeraire measures.

7.8.5 6.7A.5 Why two derivations

The two routes emphasise different machinery. The hedging derivation of §§6.2–6.5 foregrounds the self-financing condition, the market price of risk, and no-arbitrage: the PDE is the economic consistency requirement that replication imposes. The Feynman-Kac derivation foregrounds the measure change: the PDE is the drift-killer that defines $\mathbb{Q}$, and the expectation under $\mathbb{Q}$ *is* the price. A practitioner who can flip fluently between the two representations has a much richer toolkit than one who is comfortable with only one. For payoffs that admit a clean Gaussian integral (vanilla calls, digitals, log-contracts), the expectation form is the natural computational route. For payoffs that do not admit closed-form integration but are low-dimensional in state space (American exercise, barriers, free-boundary problems), the PDE form is the numerical workhorse via finite differences.

7.9 6.8 Worked Example 1 — Linear payoff $\varphi(x) = x$ (sanity check)

Before attacking a real option we check the machinery on a trivial payoff — the claim that simply pays the stock price at maturity. Obviously the forward price of such a claim is the stock itself, so the PDE had better produce $f(t, x) = x$.

We expect that $f(t,x) = x$. PDE check-out: Indeed,

$$\partial_t f = 0, \qquad \partial_x f = 1, \qquad \partial_{xx} f = 0,$$

so the LHS of (6.27) is $0 + rx \cdot 1 + 0 = rx = rf$.

The "delta" of this trivial claim is exactly 1 — we replicate a stock-payoff by holding one share of the stock, with no cash leg. This sanity check is not wasted effort: any time you solve the BS PDE numerically, it is worth verifying that the scheme reproduces this trivial case exactly. If your finite-difference solver can't price a stock, it can't price a call either.

Direct probabilistic derivation. Under the risk-neutral measure $\mathbb{Q}$ (CH05),

$$X_T \overset{d}{=} X_t\, e^{(r-\frac{1}{2}\sigma^2)(T-t)+\sigma\sqrt{T-t}\,Z}, \qquad Z \sim \mathcal{N}(0,1).$$

Hence

$$\text{price} \;=\; \mathbb{E}_t^{\mathbb{Q}}\big[e^{-r(T-t)}\,X_T\big] \;=\; X_t\,\mathbb{E}^{\mathbb{Q}}\Big[e^{-\frac{1}{2}\sigma^2(T-t)+\sigma\sqrt{T-t}\,Z}\Big] \;=\; X_t. \tag{6.40}$$

The bracket is a mean-1 log-normal, confirming $f(t,x) = x$ via the Feynman-Kac route as well.

Two ways up the same mountain: the PDE approach and the expectation approach. Both agree. This is the Feynman-Kac theorem (CH04) in its most elementary form, and it hints at the broader fact that solving the BS PDE is the same thing as computing a discounted expectation under $\mathbb{Q}$.

7.10 6.9 Worked Example 2 — Separable ansatz $f(t,x) = x^2\,\ell(t)$

The next step up in difficulty is a non-trivial but tractable payoff: the squared payoff $\varphi(x) = x^2$. This arises naturally when pricing variance products, and the separable ansatz $f(t,x) = x^2\ell(t)$ reduces the PDE to an ODE.

Squared payoffs may look contrived at first glance. Who writes a contract that pays out the square of a stock price? The answer is: nobody, directly. But squared payoffs appear as building blocks in variance swaps, in the hedging decomposition of log-contracts, and in the moment-matching approximations used to price basket options. They are the simplest non-linear payoff in the Itô world, and their price is a clean window onto how the PDE handles convexity.

Use the ansatz $f(t,x) = x^2\,\ell(t)$ with $\ell(T) = 1$. Then

$$\partial_t f = x^2\,\dot{\ell}, \qquad \partial_x f = 2x\ell, \qquad \partial_{xx} f = 2\ell.$$

Plug into (6.27):

$$\underbrace{x^2\dot{\ell}}_{\partial_t f} + \underbrace{rx \cdot 2x\ell}_{rx\,\partial_x f} + \underbrace{\tfrac{1}{2}\sigma^2 x^2 \cdot 2\ell}_{\frac{1}{2}\sigma^2 x^2\,\partial_{xx} f} \;=\; \underbrace{r\,x^2\ell}_{rf}. \tag{6.41}$$

Divide by x^2:

$$\dot{\ell} + (r+\sigma^2)\ell \;=\; 0 \;\Longrightarrow\; \ell(t) \;=\; e^{(r+\sigma^2)(T-t)}. \tag{6.42}$$

So

$$\boxed{f(t,x) \;=\; x^2\,e^{(r+\sigma^2)(T-t)}}. \tag{6.43}$$

The solution has a delightful structure: the squared-payoff price today is today's squared stock price times a time-decay factor that grows with σ^2. The volatility shows up *additively* to the rate inside the exponent, which is a fingerprint of variance products — their value is an affine function of integrated variance, so anything that boosts variance boosts their value exponentially over time.

Notice also how the ansatz mechanised a PDE into an ODE. The PDE was a partial differential equation in two variables (t, x), and the ansatz peeled off the x-dependence by asserting it must be proportional to x^2. What was left was an ordinary differential equation in t alone, with an easy exponential solution. This is a common trick for payoffs that are homogeneous in x — any power-of-x payoff yields an ansatz that reduces to an ODE.

The delta of this claim is $\partial_x f = 2x\ell(t)$ — linear in the stock — and the gamma is $\partial_{xx} f = 2\ell(t)$, a constant. A constant-gamma payoff is the cleanest possible vehicle for harvesting realised variance: the P&L per unit time is just $\Gamma \cdot S^2 \cdot \mathrm{d}\langle \log S \rangle$, which is the classic "gamma times variance" accounting that variance-swap desks live by.

A quick numerical illustration. Suppose $r = 5\%$, $\sigma = 20\%$, and $T - t = 1$ year. The time-decay factor is $e^{(0.05+0.04)\cdot 1} = e^{0.09} \approx 1.094$. So the squared-payoff claim trades at a 9.4% premium over the naive "cash-and-carry" square, and that premium is almost entirely the $\sigma^2 = 0.04$ contribution. If we shocked volatility to 30%, the factor becomes $e^{0.14} \approx 1.150$ — a 15% premium. Variance-based products are extremely vega-heavy because their price is literally an exponential in σ^2.

The connection between the x^2 payoff and a variance swap is worth spelling out. A variance swap is a contract that pays the realised quadratic variation of $\ln X$ over $[0, T]$ minus a pre-agreed strike. The realised quadratic variation of $\ln X$ under GBM is $\sigma^2 T$. The variance-swap buyer is thus long realised variance. The key fact is that this variance payoff can be replicated statically using a portfolio of European options (via the log-contract decomposition), plus a dynamic delta-hedge on the underlying. The squared payoff x^2 is one component of that decomposition, and its tractability in the GBM model is what makes the overall variance-swap replication argument go through.

7.11 6.9A Worked Example 3 — Digital call (closed form + PDE verification)

The third worked example stress-tests the hedging machinery: the digital call. It pays \$1 if the stock ends above a strike and zero otherwise — a discontinuous payoff. Discontinuities are where hedging theory starts to creak, because the replicating portfolio has to reproduce a jump in expected value using instruments that move continuously. The pricing formula comes out cleanly (we already wrote it as (6.32)), but the hedging behaviour near expiry is pathological, as we will see in §6.11.7.

Payoff and closed form. Recall

$$\varphi(x) \;=\; \mathbb{1}_{x \geq K}, \qquad f(t, x) \;=\; e^{-r(T-t)}\, \Phi\big(d_-(t, x)\big), \tag{6.44}$$

with

$$d_-(t, x) \;=\; \frac{\ln(x/K) + (r - \frac{1}{2}\sigma^2)(T - t)}{\sigma\sqrt{T - t}}. \tag{6.45}$$

A worked example anchors the formula. Suppose $S = \$100$, $K = \$100$, $r = 5\%$, $\sigma = 20\%$, and $T - t = 0.5$ year. Then $\ln(S/K) = 0$, and $(r - \frac{1}{2}\sigma^2)(T - t) = (0.05 - 0.02)(0.5) = 0.015$. The denominator $\sigma\sqrt{T - t} = 0.2 \cdot 0.707 = 0.1414$. So $d_- = 0.015/0.1414 \approx 0.106$, and $\Phi(0.106) \approx 0.542$. The digital call price is $e^{-0.025} \cdot 0.542 \approx 0.529$: about 53 cents to receive a dollar at expiry.

7.11.1 6.9A.1 PDE verification

It is mechanical but instructive to verify that the formula actually solves the PDE (6.27). The partial derivatives come from the chain rule applied to the composition $\Phi(d_-(t, x))$.

Time derivative:

$$\partial_t f \;=\; e^{-r(T-t)}\, \Phi'(d_-(t, x))\, \partial_t d_- \;+\; r\, f. \tag{6.46}$$

Space derivatives:

$$\partial_x f \;=\; e^{-r(T-t)}\,\Phi'(d_-(t,x))\,\partial_x d_-, \qquad \partial_x d_- \;=\; \frac{1}{x\,\sigma\sqrt{T-t}}, \tag{6.47}$$

$$\partial_{xx} f \;=\; e^{-r(T-t)}\Big\{\Phi''(d_-(t,x))\,(\partial_x d_-)^2 + \Phi'(d_-(t,x))\,\partial_{xx} d_-\Big\}, \tag{6.48}$$

with

$$\partial_{xx} d_- \;=\; -\frac{1}{x^2\,\sigma\sqrt{T-t}}, \qquad \Phi''(x) \;=\; -x\,\Phi'(x), \quad \Phi'(x) \;=\; \frac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}. \tag{6.49}$$

Combining (6.46)-(6.49) in (6.27) verifies the PDE: the rf terms cancel against $\partial_t f$'s $+rf$ piece, and the drift-plus-diffusion piece cancels thanks to $\Phi'' = -x\Phi'$.

The identity $\Phi'' = -x\Phi'$ is the algebraic engine that makes the digital verification fall out so neatly. It is also what makes the normal distribution such a natural fit with Brownian motion: the Gaussian density is the unique probability law whose log-derivative is linear, and this property dovetails with the $\frac{1}{2}\sigma^2 x^2 \partial_{xx}$ term in the PDE to collapse all the cross-terms at exactly the right moment.

The digital's delta, which we read off from (6.47), has the shape $\partial_x f = e^{-r(T-t)}\,\Phi'(d_-)/(x\sigma\sqrt{T-t})$. For spot values not exactly at the strike, d_- diverges to $\pm\infty$ as time to expiry shrinks, which kills $\Phi'(d_-)$ exponentially. But for spot values at the strike, d_- stays bounded and $\Phi'(d_-)$ is bounded away from zero; the delta is then dominated by the $1/\sqrt{T-t}$ singularity in the denominator. This is the mathematical version of "infinite delta at the strike in the last instant" that we will visualise in §6.11.7.

A useful quantitative check. For a digital with $K = \$100$, $\sigma = 20\%$, and $r = 5\%$, the peak delta with 1 month to expiry is roughly $0.4/(100 \cdot 0.2 \cdot \sqrt{1/12}) \approx 0.07$ per dollar of stock. With 1 day to expiry, the peak delta is ≈ 0.4 per dollar. With 1 hour to expiry, the peak delta is roughly 2 per dollar — untradeable in practice. The scaling $1/\sqrt{T-t}$ is brutal: halving the time to expiry only increases the peak delta by a factor of $\sqrt{2}$, but over many halvings the number grows without bound. Digital hedging is a race against the clock.

7.12 6.10 Time-based hedging recursion

So far we have been doing everything in PDE language. There is a parallel, often more useful, language of expectations. Feynman-Kac (CH04) tells us that any solution of the PDE can be rewritten as a discounted $\mathbb{Q}$-expectation of the payoff, where $\mathbb{Q}$ is the measure under which the underlying drifts at the risk-free rate minus $\sigma\lambda$. This is the final piece of the pricing puzzle: the replicating portfolio, the PDE, and the expectation are three equivalent statements of the same no-arbitrage fact.

The three-way equivalence — replication, PDE, expectation — is a deep fact with practical consequences. Depending on the problem, one representation is much easier to work with than the others. For payoffs that are smooth functions of a lognormal variable (like the vanilla call), the expectation is straightforward. For payoffs on a non-Gaussian state variable, the PDE is often the numerical workhorse. And for complicated hedging studies — where we want to visualise the replication portfolio's tracking behaviour over sample paths — the self-financing construction is the natural tool.

Under risk neutrality,

$$f(t,x) \;=\; \mathbb{E}^{\mathbb{Q}}_{t,x}\Big[\varphi(X_T)\,e^{-\int_t^T r_u\,\mathrm{d}u}\Big], \qquad X_t = x. \tag{6.50}$$

Notice that the expectation is taken under $\mathbb{Q}$, not $\mathbb{P}$. Under $\mathbb{Q}$, the drift of X is modified by the Girsanov shift $-\sigma^x\lambda$, which in the Black-Scholes case collapses the drift to rX_t. The physical drift μ has been *completely eliminated* from the pricing machine.

The $\mathbb{Q}$-dynamics of X are

$$\mathrm{d}X_t \;=\; \underbrace{(\mu_t^x - \sigma_t^x\,\lambda_t)}_{=r_t X_t \text{ in BS}}\,\mathrm{d}t \;+\; \sigma_t^x\,\mathrm{d}\widetilde{W}_t, \tag{6.51}$$

where $\widetilde{W}_t$ is a $\mathbb{Q}$-Brownian motion (CH05).

One subtle but beautiful feature of the $\mathbb{Q}$-measure is that it makes all discounted traded prices into martingales. Under $\mathbb{Q}$, $e^{-rt}X_t$ is a martingale, $e^{-rt}g_t$ is a martingale, and $e^{-rt}f_t$ is a martingale. This is why the discounted-expectation formula (6.50) works: the discounted option price today equals the expected discounted payoff at expiry. The martingale language is the backbone of continuous-time finance, and it slots into the dynamic hedging story as the expectation-side of the replication coin.

The **Fundamental Theorem of Asset Pricing** formalises this connection. It states that, in a market satisfying mild technical conditions, there is no arbitrage if and only if there exists at least one equivalent martingale measure $\mathbb{Q}$ under which all discounted traded prices are martingales. Furthermore, if the market is complete (as in our single-Brownian-motion setup), the martingale measure is unique. Self- financing plus no-arbitrage implies the existence of $\mathbb{Q}$; market completeness implies uniqueness; and the risk-neutral expectation (6.50) is what you get when you integrate any payoff against this unique measure.

7.12.1 6.10.1 Specialisation to Black-Scholes

$$\mathrm{d}X_t \;=\; \mu X_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}W_t \;=\; r\,X_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}\widetilde{W}_t, \qquad r = \text{const.} \tag{6.52}$$

The call, put, and digital formulas (6.28), (6.31), and (6.32) all follow from plugging (6.38) into (6.50) and computing the Gaussian integrals, as sketched in §6.7A.4. The three formulas are the three canonical closed forms that every option-pricing library implements.

It is worth writing out what the Greeks mean in units. The delta $\Phi(d_+)$ is dimensionless — it is the number of shares per option. The gamma $\phi(d_+)/(x\sigma\sqrt{T-t})$ has units of shares per dollar of stock, which is to say "how fast the delta changes as the stock moves". The vega $x\phi(d_+)\sqrt{T-t}$ has units of dollars per unit of volatility; a typical convention is to quote vega per "vol point" (1% change in σ), which gives the dollar sensitivity of the option to a one-percentage-point move in implied volatility. These three numbers are the core risk measures of the option position; any serious trading desk watches them in real time.

The curvature features of the vanilla call are specially illuminating. The gamma peaks at-the-money because that is where the payoff has its "kink"; further from the strike, the payoff is either essentially linear (deep ITM) or essentially zero (deep OTM), and a linear payoff has no curvature. The gamma also grows as expiry approaches for at-the-money options, because the remaining convexity has to be concentrated into a shrinking window. Gamma and vega are cousins — both express the option's exposure to variance — and they are most pronounced where the payoff is most non-linear.

7.13 6.11 Discrete hedging of a continuous model

The theoretical result is that continuous rebalancing at every instant produces *exact* replication — $V_t \equiv 0$, every day, every path. In real life, we rebalance at some discrete frequency: daily, hourly, once per tick. The replication therefore has tracking error, and the structure of that error is what we turn to now. We will see that the dominant source of error scales like $\sqrt{\Delta t}$ per step, and that — importantly — this error comes from discretisation of the Brownian motion, not from any model mismatch. Model mismatch is a separate, more insidious source of error.

The transition from the continuous-time theory to real-world hedging is where the rubber meets the road. On paper, continuous rebalancing produces exact replication; in reality, the fastest any trader can rebalance is tick-by-tick, and even tick-by-tick is at millisecond intervals, not truly continuous. The gap between "continuous" and "very frequently" is not just a quantitative small-error question — it has a specific structure that shapes how traders set up their rebalancing workflows.

7.13.1 6.11.1 Holdings rule

$$\alpha_t \;=\; \frac{\sigma_t^f\, f_t}{\sigma_t^g\, g_t} \;=\; \partial_x f(t, X_t) \qquad \text{if } X_t \text{ is traded.} \tag{6.53}$$

(Using $\sigma_t^g g_t = \sigma_t^x$ when $g_t = X_t$, and $\sigma_t^f f_t = \sigma_t^x\, \partial_x f_t$.) Thus the stock-hedge ratio is the delta, $\Delta_t \equiv \partial_x f(t, X_t)$.

The clean identification of α_t with the delta is the reason "delta hedging" is the universal name of this procedure. In principle, at each instant we should hold exactly $\partial_x f$ shares. In practice, we hold that many shares at the *last rebalance time* and let the position drift until the next rebalance.

This "hold at last rebalance" approximation introduces two categories of error. The first is *path-dependent*: between rebalance times, the stock moves away from its last-hedged value, so the ideal $\partial_x f$ shifts, and the portfolio's actual delta drifts away from the ideal delta. The larger the stock move, the larger the drift. This shows up as gamma-driven P&L noise over each rebalance interval. The second category is *calendar-dependent*: even if the stock does not move, time passes, and the ideal delta changes because $\partial_x f(t, X_t)$ depends on t as well as on X_t.

A mental picture helps. On a $g(t, S)$-vs-S diagram, the ideal delta at time t is the slope of the chord to the pricing surface at the current spot S_t. The trader's rebalancing procedure can be visualised as "climbing the slope" one chord at a time: at t_0 we install a tangent line at (t_0, S_{t_0}) with slope α_{t_0}; at t_1 the surface has evolved, the spot has moved, and a new tangent line is installed with slope α_{t_1}; and so on. Between tangent installations, the held portfolio is an affine function of S — constant slope — while the true pricing function is curved, and the mismatch is precisely the rebalance-interval P&L noise we called the gamma residual.

7.13.2 6.11.2 At $t_0 = 0$ — sold f, get f_0

- Buy α_0 units of X (costs $\alpha_0 X_0$).
- Bank account: $M_0 = f_0 - \alpha_0 X_0$.

At inception, we collect the option premium f_0, use some of it to buy the initial delta in shares, and deposit what's left in the money market. The bank account is positive or negative depending on whether the initial hedge was a big chunk of the premium (e.g. deep in-the-money calls need a big share purchase) or small (far out-of-the-money).

For a concrete example, suppose we sell a 3-month at-the-money call on a stock at \$100, implied vol 25%, rate 5%, and collect roughly \$5.60 in premium (delta about 0.55). We immediately buy 0.55 shares at \$100 each, costing \$55, and the money-market account starts at $\$5.60 - \$55 = -\$49.40$. That is a \$49 negative balance on the bank — we are borrowing. That is normal: the short-call hedge requires borrowing to fund the stock purchase. Over time, as the option's delta changes, we will borrow more or lend back.

7.13.3 6.11.3 At t_1

- Asset now has value $\alpha_{t_0} X_{t_1}$.
- Bank value: $M_{t_0}\, e^{r\,\Delta t}$.

Must rebalance to new units of X (cost $\alpha_{t_1} X_{t_1}$). Bank is now

$$M_{t_1} \;=\; M_0\, e^{r\,\Delta t} \;-\; (\alpha_{t_1} - \alpha_{t_0})\, X_{t_1}. \tag{6.54}$$

At the first rebalance, the delta has moved because both time has passed and the stock has moved. We adjust our share holding to the new delta and pay for (or receive from) the adjustment out of the bank account.

Notice the algebraic symmetry of (6.54). The first term, $M_0 e^{r\Delta t}$, is the bank's natural growth due to interest. The second term, $-(\alpha_{t_1} - \alpha_{t_0}) X_{t_1}$, is the cost of the rebalance, evaluated at the *new* stock price X_{t_1}. The rebalance price is always the current stock price, not the old one, because we are trading at today's market; this is what distinguishes self-financing from the naive product rule.

7.13.4 6.11.4 At t_2

- Asset value: $\alpha_{t_1}\, X_{t_2}$.
- Bank value: $M_{t_1}\, e^{r\,\Delta t}$.
- Rebalance: α_{t_2} units of X (cost $\alpha_{t_2} X_{t_2}$).

- Bank is now

$$M_{t_2} \;=\; M_{t_1}\, e^{r\,\Delta t} \;-\; (\alpha_{t_2} - \alpha_{t_1})\, X_{t_2}. \tag{6.55}$$

7.13.5 6.11.5 Repeat

$$\boxed{M_{t_n} \;=\; M_{t_{n-1}}\, e^{r\,\Delta t} \;-\; (\alpha_{t_n} - \alpha_{t_{n-1}})\, X_{t_n}} \tag{6.56}$$

No rebalancing between $t_0, t_1, \dots, t_{N-1}, t_N = T$.

In practice, every rebalance also incurs a *transaction cost* that the idealised recursion ignores. Let $\kappa(q)$ denote the dollar cost of trading q shares. The realistic recursion reads

$$M_k \;=\; M_{k-1}\, e^{r\,\Delta t_k} \;-\; (\alpha_k - \alpha_{k-1})\, S_k \;-\; \kappa(\alpha_k - \alpha_{k-1}), \tag{6.57}$$

and the cumulative cost $\sum_k \kappa(\alpha_k - \alpha_{k-1})$ is the *hedge slippage bill* that ultimately sits against the theoretical P&L of (6.58). The presence of transaction cost is what makes rebalancing a genuine optimisation problem rather than a "rebalance as often as possible" prescription.

The recursion (6.56) is the entire operational content of discrete delta hedging. Given the delta path α_t (which the trader computes from the BS formula using some implied volatility) and the realised stock path X_t, (6.56) tells you exactly what your bank balance is at every rebalance date.

7.13.6 6.11.6 Terminal P&L

At t_N we owe the option payoff:

$$\boxed{\text{PnL} \;=\; \Big(M_{t_{N-1}}\, e^{r\,\Delta t} \;+\; \alpha_{t_{N-1}}\, X_{t_N}\Big) \;-\; \varphi(X_{t_N})} \tag{6.58}$$

If replication were exact, this quantity would be zero. In reality, it is a random variable whose mean, under correct model specification, is zero and whose standard deviation scales roughly like $\sqrt{\Delta t}$ times a gamma-weighted integral along the path. Doubling the rebalancing frequency therefore cuts the standard deviation by $\sqrt{2}$, not by 2 — diminishing returns.

The $\sqrt{\Delta t}$ scaling is worth understanding in detail because it sets the fundamental ceiling on what discrete hedging can achieve. Over a single rebalance interval, the stock's log-return is roughly $\sigma\sqrt{\Delta t}\, Z$ where Z is a standard normal. The P&L error on that interval, coming from the mismatch between the held constant- α position and the ideal-$\partial_x f$ position, is essentially a quadratic Taylor expansion of the option price: $\frac{1}{2}\partial_{xx} f \cdot (X_{t_{n+1}} - X_{t_n})^2$, minus the expected quadratic piece that the PDE has already "paid for". The residual is $\frac{1}{2}\partial_{xx} f \cdot X^2\sigma^2(Z^2-1)\Delta t$. The quantity Z^2-1 has mean zero and variance 2, so the residual's per-step standard deviation scales like $\Gamma S^2\sigma^2\Delta t$. Summing independent residuals over N steps gives a total standard deviation of order $\sqrt{N}\cdot\Gamma S^2\sigma^2\Delta t = \Gamma S^2\sigma^2\sqrt{T\Delta t}$.

The deeper lesson is that the $\sqrt{\Delta t}$ error only goes to zero if the model itself is right. If the true volatility of the stock is different from the σ we used to compute the delta, the P&L (6.58) is no longer mean-zero: it acquires a bias equal to the gamma-weighted integral of the squared-vol mismatch, which *does not shrink* as we rebalance more frequently. This is why selling options is not free money — you may be perfectly delta-hedged in a model sense, but if the market realises a larger-than-implied variance, the gamma bill comes due regardless of how often you rebalance.

The two sources of hedging error are worth separating cleanly in your mind. The first source — *discretisation error* — comes from finite rebalance frequency and shrinks as $\sqrt{T\Delta t}$ when $\Delta t \to 0$. The second source — *model error* — comes from using the wrong volatility to compute the delta. The per-interval bias is $\frac{1}{2}\Gamma S^2(\sigma^2_{\text{imp}} - \sigma^2_{\text{real}})\,\Delta t$, and the total bias is $\frac{1}{2}\int \Gamma S^2(\sigma^2_{\text{imp}} - \sigma^2_{\text{real}})\,\mathrm{d}t$, which *does not shrink as we rebalance more often.* It is a systematic mis-pricing, not a discretisation artefact. You cannot hedge out a vol mis-specification by trading more; you can only hedge it out by buying or selling vega (i.e. other options) to neutralise the mismatch.

This distinction between discretisation error and model error is, in practice, the distinction between how a trader thinks about her book day-to-day and how a risk manager thinks about it quarter-to-quarter. Day-to-day, the trader worries about rebalancing frequency and transaction costs. Quarter-to-quarter, the risk manager worries about

whether the implied vol surface is pricing the true distribution of returns. A desk that only does the former and ignores the latter is running blind; a desk that only does the latter and ignores the former is bleeding commissions.

There is a fourth, even more dangerous source of hedging error that lives outside the Brownian-motion framework altogether: **jumps**. Over a jump of size J, the option price changes not by the first-order Taylor amount $\partial_x f \cdot J$ but by the full $f(t, X + J) - f(t, X)$. For a long-gamma position this is a positive surprise; for a short-gamma position it is a loss that can be arbitrarily large. No amount of "rebalance faster" can cure this, because the jump happens between rebalances *by definition*. The only cure is to hedge with other options that share the same jump exposure.

The jump risk explains several empirical features of real-world option markets. Short-dated at-the-money options always trade at an implied volatility higher than the realised volatility, especially right before earnings announcements. The "crash-o-phobia" skew in equity index options — the fact that out-of-the-money puts trade at a much higher implied volatility than comparable calls — is another reflection of jump concerns: investors are paying up for protection against large downward jumps, and hedgers who short those puts cannot offset the risk with delta-hedging alone.

Discrete-hedging error vs rebalance frequency *Discrete-hedging error vs rebalance frequency.*

Replicating portfolio value paths *Ideal continuous hedging would pin $V_t \equiv 0$; daily rebalancing leaves visible gamma-driven wiggles.*

The key identity that every real trader internalises is that, conditional on a fixed delta-hedging strategy at some constant σ_{hedge}, the running P&L of a short-option-plus-delta- hedge position is

$$\text{P\&L}_{[t,t+\Delta t]} \;\approx\; \tfrac{1}{2}\,\Gamma_t\, X_t^2\,\big(\sigma_{\text{hedge}}^2 - \tfrac{(\Delta X_t/X_t)^2}{\Delta t}\big)\,\Delta t.$$

Read this carefully. The σ_{hedge}^2 piece is what the hedger "thinks" will happen. The $(\Delta X_t/X_t)^2/\Delta t$ piece is what actually happened. The difference, multiplied by $\frac{1}{2}\Gamma X^2$, is the P&L increment on that interval. The integral over the life of the option is the terminal P&L of the delta-hedging strategy. A delta-hedged option is essentially a *variance swap* in a convex wrapper.

7.13.7 6.11.7 Shape of α_t near expiry — digital call

Near $t = T$, the digital-call delta $\alpha = \partial_x f$ spikes around the strike K. A useful way to watch this evolve is to plot the delta as a function of spot for three representative time slices:

```
  (x, t)
                                      t = T   (Heaviside step,  -like  )
 1
                                          t ~ T   (tight sigmoid)

                                     ----        t   T   (broad logistic)
                     ..' '.    --
                 ..'
             ..'
        ..'
 0                          → x = spot
                       K

  As t → T , the delta  sharpens from a broad S-curve into a
  step at K: gamma blows up and the hedge becomes impossible.
```

- $t \ll T$ **(far from expiry):** the delta is a broad, logistic-looking curve rising gently from near 0 on the far left through roughly $\frac{1}{2}$ at $x = K$ to near 1 on the far right. Gamma is modest and the hedge is easy to track.
- $t \sim T$ **(close to expiry):** the logistic tightens into a sharp S-curve concentrated near K. Gamma grows large in a narrow band; outside the band the delta is essentially 0 or 1.
- $t = T$ **(at expiry):** the delta jumps from 0 to 1 at K — a step function — and its derivative (gamma) is a Dirac δ-spike.

As $t \to T^-$ the delta sharpens into a Dirac δ-function centred at K — hedging a digital near expiry is basically impossible without discontinuous position changes ("delta blow-up").

The reason is instructive. The digital's payoff is a step function: it jumps from zero to one as the stock crosses the strike. Since the delta is the spatial derivative of the option price, and the option price near expiry looks more and more like the step payoff itself, the delta must look more and more like the derivative of a step — a Dirac delta in the limit. Digitals are therefore either priced with a significant vega / pin-risk premium or hedged as **call spreads** (two offsetting vanilla calls with strikes slightly above and below the digital strike), which smooths the payoff and the delta.

A digital call that pays \$1 if $X_T \geq K$ can be approximately replicated by going long a vanilla call struck at $K - \epsilon/2$ and short a vanilla call struck at $K + \epsilon/2$, scaled by $1/\epsilon$. As $\epsilon \to 0$, this call spread converges to the digital. But for any finite ϵ, the call spread has a bounded delta around the strike, bounded by $1/\epsilon$. The wider the spread, the less hedging difficulty, but the more residual risk from the approximation.

Delta surface *Left: smooth vanilla delta. Right: digital delta tents at K as $\tau \to 0$ — the hedger must trade arbitrarily large size around the strike.*

7.14 6.12 Move-based hedging

The time-based hedging scheme of §6.11 rebalances on a fixed clock — every day, every hour, every minute. That is conceptually clean but often operationally wasteful: on a quiet day, the delta barely moves, and paying commissions to rebalance at noon is silly. **Move-based hedging** instead monitors the delta itself and rebalances only when it drifts outside a pre-specified band around the last-hedged value.

The philosophy behind move-based hedging is that what the trader cares about is the *delta error* — the gap between the current-held delta and the current-ideal delta — not the clock. As long as the delta error stays small, the portfolio is well-hedged and no trade is needed. Only when the delta error exceeds a threshold does trading become worthwhile.

Continuous-time theory rebalances at every instant; discrete time rebalances on a clock $t_0, t_1, \ldots, t_N$. Move-based hedging rebalances whenever the delta α_t wanders outside a band around its last-hedged value:

- Set a band $[\alpha^\star - \epsilon,\ \alpha^\star + \epsilon]$.
- Let α_t drift inside the band.
- When α_t exits, re-centre: new $\alpha^\star = \alpha_t$, restart.

Pictorially:

```
_t (ideal delta)
                                          ← band 1
        .·'·.                                       around  *_1
      .'     '·._
    '            '·._     (exit ↑, re-centre)
                                        ← band 2
                        '·.                         around  *_2
                            '·._
                                '·._
                                    '·._    (exit ↓, re-centre)
                                         ← band 3
                                           ·._          around  *_3
                                              '·.
                              → t

  Delta drifts inside a band around  *; on exit, a trade snaps
  back to the current  _t and a fresh band re-centres.
```

Move-based hedging trades fewer times than time-based hedging for the same tracking error on smooth paths, but pays up on rapid gamma swings.

There is a trade-off between two evils. Tight bands mean frequent rebalancing (high commission cost) but tight tracking of the delta (low gamma-bleed risk). Wide bands mean infrequent rebalancing (low commission cost) but

the portfolio can drift far from the target delta. The optimal band width depends on the cost of transacting, the volatility of the underlying, the gamma of the position, and the trader's risk tolerance.

A useful way to think about the choice is to imagine the trader's delta as a random walk against a target. Time-based hedging snaps the walk back to target on a fixed schedule regardless of how far it has drifted; move-based hedging lets the walk roam free until it exits a band, then snaps it back. Over a continuous horizon, the *expected number of trades* under move-based hedging scales inversely with the square of the band width — narrow bands trigger often, wide bands trigger rarely — while the *tracking error* scales linearly with the band width. This is the fundamental cost/benefit dial: you get quadratically fewer trades for a linear increase in tracking noise.

A stylised Monte-Carlo experiment makes the shape of the frontier concrete. Simulating a short at-the-money three-month call under GBM and hedging under a move-based rule with transaction cost 10 bps per share traded, the mean and standard deviation of terminal P&L for a selection of band widths ϵ (expressed as a fraction of initial delta) come out approximately as:

Band ϵ	Mean P&L	P&L std
0.00625	-0.39	0.705
0.0125	-0.35	0.710
0.025	-0.34	0.750
0.05	-0.34	0.850
0.10	-0.32	1.290
0.20	-0.33	2.190

Traced from top to bottom, the band is widening. The mean P&L climbs *slightly* (from -0.39 to about -0.32) because fewer trades means less total transaction-cost bleed. But the standard deviation *grows* substantially (from 0.71 to 2.19) because the delta is allowed to wander further before being corrected, producing larger gamma residuals per rebalance interval. This is the classic mean/std efficient frontier, and the desk's job is to pick the point on it that aligns with its risk budget.

The shape of this frontier is surprisingly universal. Re-run the experiment under different parameters and the *location* of the curve shifts but its *shape* (concave, roughly hyperbolic near the shoulder) is preserved. The curve's parametric form is essentially "commission + variance = $a/\epsilon^2 + b\epsilon^2$", whose minimum over ϵ gives the optimal band when the desk's objective is mean-variance.

The quadratic-versus-linear scaling is worth a line of explanation. The number of band-crossings by a Brownian motion with volatility σ over time T is of order $T\sigma^2/\epsilon^2$ (the classic local-time result). Doubling the band width ϵ therefore reduces the number of trades by a factor of 4, but only increases the typical delta error by a factor of 2. Cost savings scale quadratically with the relaxation of the precision constraint.

One final operational remark. Regardless of whether a desk rebalances by clock or by delta move, it is almost always correct to *also* rebalance at specific predictable moments — the open, the close, the pre-earnings hour, right before a central-bank announcement. These are the moments when the assumed continuity of the price path is most likely to break: gaps at the open, close-auction mean-reversion, headline-driven jumps. A disciplined hedger layers move-based triggers on a clock-based baseline and hard-codes additional rebalances around known risk events.

A practical remark on how desks actually implement move-based hedging: most production systems use a **delta-dollars** band rather than a pure delta band. Delta-dollars is the dollar value of the stock-equivalent exposure ($\Delta \cdot S$), which normalises across underlyings of different price levels. Bands expressed in delta-dollars are natively comparable across the book and feed cleanly into the desk's risk-limit system. Beyond delta-dollars bands, more sophisticated desks layer **gamma-adjusted** bands — the idea being that a position with high gamma needs to be rebalanced more often because its delta moves more per unit of stock move.

In a world with zero transaction costs, continuous rebalancing is trivially optimal. In a world with finite transaction costs, neither time-based nor move-based is universally optimal. The modern derivatives desk uses a hybrid approach, running time-based rebalancing as a baseline safety net and move-based overlays where they add value. The chapter's discussion of continuous dynamic hedging is the theoretical idealisation these practical schemes are trying to approximate.

7.15 6.13 Key Takeaways

7.16 6.14 Reference Formulas Appendix

8 Chapter 7 — Dynamic Hedging II: Greeks, Delta-Gamma, Vega, and Dividends

Chapter 6 established the single-instrument delta hedge, the generalised Black-Scholes PDE, and the Greeks that fall out of the closed-form vanilla formulas. This chapter is the follow-up. We enlarge the hedging portfolio — first to neutralise curvature (delta-gamma), then to neutralise volatility exposure (vega), then to accommodate dividend-paying underlyings — and we pay the self-financing bookkeeping explicitly when transaction costs enter the picture. The underlying machinery is the same Itô expansion that drove Chapter 6; what changes is the dimension of the hedging basis and the list of Greeks we target.

8.1 7.0 Chapter orientation

A delta-hedged short-option position is not a risk-free position. It is **first-order** risk-free: the coefficient of $\mathrm{d}W_t$ in the self-financing Itô increment has been zeroed, so the portfolio has no instantaneous Brownian exposure. The residual risks live in the higher-order terms, and they have names. *Gamma* is the coefficient of $(\Delta S)^2$ — the curvature of the option price in the spot direction; *vega* is the coefficient of $\Delta\sigma$ — the sensitivity of the option price to a shift in the volatility parameter; *theta* is the coefficient of Δt — the passage of time; and a zoo of cross-derivatives (vanna, volga, charm, speed, colour) parameterise every other way a smooth price function can wiggle.

Most of these higher Greeks are not hedgeable with the stock alone. The stock has *zero* gamma, *zero* vega, and *zero* dependence on implied volatility, so no position in S_t and cash can neutralise curvature or vol-level risk. To hedge those exposures we must enlarge the hedging basis by adding another traded option — or two, or ten, depending on how many higher Greeks we want to annihilate. The linear algebra is straightforward; the operational bookkeeping is where the subtleties live, and that is where this chapter spends most of its time.

The chapter proceeds in seven acts. §7.1 is a brief Greeks recap that fixes notation and bridges from Chapter 6. §7.2 is the central methodological section: **delta-gamma hedging unified** — we give two framings (two-instrument replication and local quadratic fit) of the same construction and show they agree. §7.3 introduces vega and the extension of the replicating-portfolio argument to stochastic volatility. §7.4 derives the Black-Scholes-Merton PDE with continuous dividend yield. §7.5 takes up transaction costs and the Leland volatility adjustment. §7.6 surveys price, delta, and gamma sketches for puts, calls, and parity. §7.7 introduces **dollar Greeks** — the units that risk-management desks actually use. The chapter closes with Key Takeaways and a Reference Formulas appendix.

Throughout, notation follows Chapter 6: S_t is the stock, $g(t,S)$ or $f(t,S)$ is a contingent claim, $\Delta = \partial_S g$, $\Gamma = \partial_{SS} g$, $\mathcal{V} = \partial_\sigma g$. Unqualified expectations are under the risk-neutral measure $\mathbb{Q}$ unless otherwise noted.

8.2 7.1 Greeks recap — a bridge from Chapter 6

Chapter 6 produced the Black-Scholes PDE and, by solving it for the vanilla call, delivered a collection of partial derivatives that every derivatives trader learns by heart. It is worth restating the catalogue explicitly before we start putting them to work.

Let $g(t,S;\sigma,r)$ be the price function of a European claim on an underlying S with volatility σ and constant short rate r. The *first-order* Greeks measure the linear sensitivity of g to the four natural inputs — spot, time, volatility, and the short rate:

$$\Delta_t \equiv \partial_S g, \qquad \Theta_t \equiv \partial_t g, \qquad \mathcal{V}_t \equiv \partial_\sigma g, \qquad \rho_t \equiv \partial_r g. \tag{7.1}$$

The *second-order* Greeks measure curvature and cross-sensitivities:

$$\Gamma_t \equiv \partial_{SS} g, \qquad \mathrm{vanna}_t \equiv \partial_{S\sigma} g, \qquad \mathrm{volga}_t \equiv \partial_{\sigma\sigma} g, \qquad \mathrm{charm}_t \equiv \partial_{St} g. \tag{7.2}$$

And the *third-order* Greeks extend the cascade — speed $\partial_{SSS} g$, colour $\partial_{SSt} g$, zomma $\partial_{SS\sigma} g$, and so on — with each successive Greek characterising a tighter layer of residual risk.

For a European call under Black-Scholes the closed forms are

$$\Delta_{\text{call}} \;=\; \Phi(d_+), \qquad \Gamma \;=\; \frac{\Phi'(d_+)}{S\,\sigma\sqrt{T-t}}, \qquad \mathcal{V} \;=\; S\,\sqrt{T-t}\,\Phi'(d_+), \tag{7.3}$$

and for a European put, $\Delta_{\text{put}} = \Phi(d_+) - 1 = -\Phi(-d_+)$, with the same Γ and $\mathcal{V}$ as the call (we will prove the gamma identity via put-call parity in §7.6). Here Φ and Φ' are the standard normal CDF and density, and $d_\pm = [\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)(T-t)]/(\sigma\sqrt{T-t})$ is the Black-Scholes moneyness index.

Call delta profile across moneyness at three maturities *Black-Scholes call delta $\Phi(d_+)$ as a function of spot for three maturities. Long-dated options have gradual deltas spread across moneyness; near expiry the delta compresses toward a step at the strike, tracking the digital payoff that Δ_{call} becomes in the $T \downarrow t$ limit.*

Gamma vs moneyness and time-to-expiry *Gamma $= \Phi'(d_+)/(S\sigma\sqrt{T-t})$ peaks at the at-the-money strike for every maturity, and the peak grows without bound as $T \downarrow t$. A short ATM option is a short-gamma position that becomes dangerously short as expiry approaches — the terminal-gamma pinning effect.*

Call vs put delta across moneyness *Put-call parity in delta form: $\Delta_{\text{call}} - \Delta_{\text{put}} = 1$ for every spot. The call delta rises from 0 to $+1$ through the strike; the put delta sits a parallel unit below, running from -1 to 0. A long call plus a short put with the same strike is always 100% long the underlying.*

What Chapter 6 *proved* was that picking $\alpha_t = \partial_S g(t, S_t)$ units of the stock against a short position in the claim zeroes out the $\mathrm{d}W_t$ coefficient in the self-financing increment. Zeroing the drift then forces the pricing PDE, and the option's fair value is the cost of funding the replicating portfolio. That argument used **exactly one Greek** — delta — because exactly one random factor W_t was being neutralised, and exactly one hedging asset S_t was being traded against it.

The limitation is baked in. A position that is delta-hedged at time t is not delta-hedged at time $t + \Delta t$, because the delta has changed. The rebalance from $\alpha_t = \Delta_t$ to $\alpha_{t+\Delta t} = \Delta_{t+\Delta t}$ happens at the prevailing stock price, and the gap between the required hedge and the held hedge over the interval $[t, t + \Delta t]$ exposes the portfolio to whatever risks delta does *not* cover: curvature (gamma), the tilt of volatility over time (vega + vol dynamics), and all the higher-order effects. In the continuous-time limit those residuals vanish — this is the content of the Black-Scholes replication theorem — but in discrete time they are the dominant source of hedge-error P&L variance.

This chapter is the systematic treatment of those residuals. §7.2 removes the gamma residual by adding a second option to the hedging basis. §7.3 takes up the vega residual. §7.4 relaxes the "non-dividend-paying stock" assumption. §7.5 puts the self-financing bookkeeping under a microscope to see how transaction costs bleed into the P&L. §7.6 synthesises the put-call parity consequences on all of the above, and §7.7 translates the abstract Greeks into the dollar-denominated units that desks actually use.

One framing remark before we begin. The reader who has learned Chapter 6's self-financing argument may be tempted to view the extensions in this chapter as mere decorations on the core structure. They are not. Each extra Greek matched corresponds to a specific and separate risk that the single-instrument hedge leaves *on the books*. A flow desk that prices options but leaves gamma unhedged, vega unhedged, and dividends mis-modelled is not running a "simplified" version of the Black-Scholes hedge — it is running a book whose actual risk exposure is dominated by the residuals, not by the delta. The residuals are the business.

8.3 7.2 Delta-gamma hedging — two framings unified

Up to this point the replicating portfolio has consisted of one risky instrument — either the stock S directly, or a traded claim g — and the bank account. That is enough to kill the first-order risk of the position, and as we have seen repeatedly it suffices to derive the pricing PDE. But a delta-neutral position is not the same thing as a *risk-neutral* position: its value is insensitive to the first derivative of the stock move, yet still very sensitive to the second derivative. A large move of either sign moves the P&L of a delta-hedged short option position by roughly $\frac{1}{2}\,\Gamma\,(\Delta S)^2$, and the sign of that term — negative for a short option — is precisely the jump loss that the gamma P&L decomposition records as the price paid for selling convexity.

The practical question is how to hedge *gamma* as well as *delta*. The answer is to enlarge the hedging set: in addition to the stock S and the cash account M, the trader holds a position in an auxiliary traded claim $h(t, S_t)$ — typically another option — whose curvature can be used to offset the curvature of the target claim. The hedging portfolio now has two risky knobs (α_t units of S and γ_t units of h) and one cash knob (β_t units of M), and the matching

conditions are two: first-order (delta) and second-order (gamma). That is exactly the minimum machinery needed to neutralise a quadratic local expansion of the target price function.

There are two equivalent ways to arrive at the same construction, and both are worth seeing. The first — which we call *Framing A*, two-instrument replication — is the structural argument: the target claim's Itô expansion includes a $\mathrm{d}W_t$-coefficient (delta) and a $\mathrm{d}t$-coefficient that depends on Γ, and the hedging portfolio must match both; this forces the hedge to contain a *second risky asset* whose Itô expansion has the right Γ. The second — which we call *Framing B*, local quadratic fit — is the practical argument: the target's price function is a smooth curve in S at each fixed t, and we are trying to replicate its first and second spatial derivatives using a stock (which has delta 1 and gamma 0) and a second option (which has known delta Δ^h and gamma Γ^h); matching both derivatives yields a 2×2 linear system whose unique solution gives the hedge ratios. The two framings yield the same formulas and the same operational recipe; they differ in which first-principles story one tells to justify them. We give both in full, side by side.

8.3.1 7.2.1 Local Taylor expansion of a claim

Let $g(t, S)$ be the price of a European option on S, with partial derivatives $\partial_S g$, $\partial_{SS} g$, and so on. Fix a reference spot S_t and consider the value of the option if the spot were to change by a small amount $(S - S_t)$. A second-order Taylor expansion in the spot variable reads

$$g(t, S) \;=\; g(t, S_t) \;+\; (S - S_t)\, \partial_S g(t, S_t) \;+\; \tfrac{1}{2}\,(S - S_t)^2\, \partial_{SS} g(t, S_t) \;+\; \cdots, \tag{7.4}$$

with the customary naming convention

$$\Delta_t^g \;\equiv\; \partial_S g(t, S_t), \qquad \Gamma_t^g \;\equiv\; \partial_{SS} g(t, S_t). \tag{7.5}$$

The delta Δ_t^g is the slope of the pricing function at the current spot; the gamma Γ_t^g is the curvature. A vanilla call's pricing function is a smoothed version of the hockey-stick payoff — zero in the far-left, curving upward near the strike, and asymptoting to a line of slope one far to the right — so its delta runs from 0 to 1 and its gamma is a localised bump around the strike. For an at-the-money call close to expiry, the gamma is tall and narrow; far from expiry, the gamma is short and broad.

Visually, one can imagine three curves superimposed on a single axis of spot S: the payoff (a kinked line), the current option price $g(t, S)$ (a smooth convex curve sitting above the payoff), and two auxiliary lines drawn at the reference point S_t — the *tangent* with slope Δ_t^g (the linear delta approximation) and a *parabola* matching both slope and curvature (the delta-gamma approximation). The tangent deviates from the true price as soon as S moves an appreciable distance; the parabola hugs the true price for considerably longer because it matches the curvature. This is the geometric content of delta-gamma hedging: we are trying to replicate an option not with a flat share holding (a horizontal line) but with a share holding plus a second-option holding whose combined curvature matches the target's.

A delta-only (linear) hedge is a *tangent line* at the current spot — it matches the slope, so small moves are neutralised, but any curvature in the payoff is missed. Gamma adds the second derivative, so the hedge becomes a *tangent parabola* that hugs the payoff over a much wider window. Away from the edge, the approximation error scales as $(\Delta S)^3$ instead of $(\Delta S)^2$ — a full order of accuracy gained per extra Greek matched.

There is a deeper reason the quadratic truncation is natural. Under geometric Brownian motion, the leading stochastic move over a short interval has $(\Delta S)^2 \sim S^2 \sigma^2 \Delta t$, so the second-order Taylor term is of the same order as Δt itself — one cannot ignore it without simultaneously ignoring all the time-derivative structure of the pricing PDE. The third-order Taylor term, $(\Delta S)^3 \sim S^3 \sigma^3 (\Delta t)^{3/2}$, is of higher order in Δt and legitimately vanishes in the small-step limit. So "delta and gamma" is not an arbitrary cutoff: it is exactly the order at which the stochastic and deterministic terms balance in Itô calculus. Higher Greeks matter only through discrete-rebalancing residuals, not through the continuous-time limit.

Delta-only vs delta-gamma hedge residual *Figure 7.1 — Delta-only hedge (tangent line) vs delta-gamma hedge (tangent parabola) fit against the convex call payoff.*

8.3.2 7.2.2 Framing A — two-instrument replication

The structural view begins from the replicating-portfolio argument of Chapter 6. We are *short* one unit of a target claim g; we seek a self-financing portfolio whose value tracks g at every (t, S). Against a delta hedge alone, the portfolio value has a non-zero $(\Delta S)^2$ coefficient: the gamma mismatch is a distinct source of risk that cannot be killed by any share position, because the stock has zero gamma. To absorb the gamma we must add a second *curved* instrument — a second traded claim whose own $\partial_{SS}h$ is non-zero.

Let $h(t, S)$ be another traded option on S — typically vanilla, liquid, and chosen to have gamma at the same scale as the target's. The hedging portfolio is

$$V(t, S) \;=\; \alpha_t\, S \;+\; \beta_t\, M_t \;+\; \gamma_t\, h(t, S), \tag{7.6}$$

where α_t is the number of shares, γ_t is the number of units of the auxiliary option, β_t is the number of units of the money-market account $M_t = e^{rt}$, and the hedge is set up so that V matches the target g claim-for-claim. Applying a Taylor expansion around S_t to both the portfolio and the target,

$$V(t, S) \;=\; \alpha_t\, S \;+\; \beta_t\, M_t \;+\; \gamma_t\big[h(t, S_t) + (S - S_t)\,\partial_S h(t, S_t) + \tfrac{1}{2}(S - S_t)^2\,\partial_{SS} h(t, S_t)\big] + \cdots, \tag{7.7}$$

$$g(t, S) \;=\; g(t, S_t) \;+\; (S - S_t)\,\partial_S g(t, S_t) \;+\; \tfrac{1}{2}(S - S_t)^2\,\partial_{SS} g(t, S_t) \;+\; \cdots. \tag{7.8}$$

Matching coefficients of $(S - S_t)$ and $(S - S_t)^2$ between the portfolio and the target gives the *delta-gamma hedge conditions*:

$$\alpha_t \;+\; \gamma_t\,\partial_S h(t, S_t) \;=\; \partial_S g(t, S_t), \tag{7.9}$$

$$\gamma_t\,\partial_{SS} h(t, S_t) \;=\; \partial_{SS} g(t, S_t). \tag{7.10}$$

Denote $\Delta_t^h = \partial_S h$, $\Gamma_t^h = \partial_{SS} h$, and Δ_t^g, Γ_t^g similarly. Solving (7.9)–(7.10) for the two unknowns,

$$\boxed{\gamma_t \;=\; \frac{\Gamma_t^g}{\Gamma_t^h}, \qquad \alpha_t \;=\; \Delta_t^g \;-\; \frac{\Gamma_t^g}{\Gamma_t^h}\,\Delta_t^h.} \tag{7.11}$$

These are the two hedge ratios. Read (7.11) as a two-step recipe: first fix γ_t so that the auxiliary option's gamma times γ_t matches the target gamma — that is gamma-matching, equation (7.10); then set α_t so that the *net* delta of the γ_t-scaled auxiliary plus α_t-scaled stock matches the target delta — that is delta-matching, equation (7.9), solved after γ_t is pinned down. The order matters because gamma-matching uses only h (the stock has zero gamma), while delta-matching uses both h and S. One cannot delta-hedge first and then gamma-hedge, because the gamma-hedge step would disturb the delta that had already been arranged. The correct order — gamma first, then delta — is the operational lesson of the pair (7.11).

With α_t and γ_t in hand, the cash leg follows from the total portfolio value. At inception, we sold g and collected g_0; we bought γ_0 units of h at h_0 and α_0 shares at S_0; whatever is left we deposit in the bank, so

$$M_0 \;=\; g_0 \;-\; \alpha_0\, S_0 \;-\; \gamma_0\, h_0. \tag{7.12}$$

This is the delta-gamma analogue of the delta-only initial-balance recipe from §6.11.2: the premium collected pays for both the share-hedge and the option-hedge, and the residual sits in the bank earning the short rate. Depending on the sign of γ_0 and the price of h, the initial bank balance may be positive (net lender) or negative (net borrower). For a short position in a short-dated at-the-money call hedged with a short-dated different-strike call, typically $\gamma_0 > 0$ (we are *long* gamma in the auxiliary to offset our *short* gamma in the target), which means we pay for h and the initial bank balance shrinks by $\gamma_0 h_0$ relative to the delta-only hedge.

Verifying that the resulting portfolio matches both delta and gamma is a one-line check:

$$\Delta_t^V \;=\; \alpha_t \;+\; \gamma_t\,\Delta_t^h \;=\; \Delta_t^g, \qquad \Gamma_t^V \;=\; \gamma_t\,\Gamma_t^h \;=\; \Gamma_t^g. \tag{7.13}$$

Both hold by construction. The portfolio tracks the target to second order, and the tracking error is now of order $(S - S_t)^3$ rather than $(S - S_t)^2$. Under a Brownian-motion underlying, the cubic term is of order $(\Delta t)^{3/2}$ per rebalance interval, substantially smaller than the Δt gamma residual of the delta-only hedge. Delta-gamma hedging is therefore genuinely higher-order: it does not merely shift the coefficient on the error, it shifts the *scaling exponent* — the variance of the tracking error now falls as $1/N^2$ in the rebalance count rather than $1/N$, and the standard deviation falls as $1/N$ rather than $1/\sqrt{N}$.

The structural story is therefore clean. A delta-only hedge replicates the target's Itô expansion only through the $\mathrm{d}W$ coefficient; the second-order $\mathrm{d}t$ piece proportional to Γ is unmatched. A delta-gamma hedge replicates *both* coefficients, by enlarging the hedging basis to include a second traded claim whose own expansion has the required curvature. Each additional Greek matched corresponds to one additional hedging instrument in the basis and one additional linear equation to solve; the arithmetic remains elementary, and the replication theorem of Chapter 6 generalises verbatim.

8.3.3 7.2.3 Framing B — local quadratic fit

The practical view starts from a slightly different position. A hedged book has a *loss distribution*, and the purpose of hedging is to compress that distribution toward zero — ideally a tight cluster around a risk-free return but more realistically a tight cluster around whatever expected alpha the strategy aims to extract. Delta hedging collapses the linear dependence on the underlying; delta-gamma hedging additionally collapses the quadratic dependence. Each additional Greek matched corresponds to a further reduction in the variance of the hedged P&L — and hence a smaller VaR and CTE at any confidence level.

The economic logic is worth stating plainly. When a derivatives dealer sells an option, the premium received is — in theory — exactly the expected discounted cost of replicating the option's payoff under a specific hedging strategy. The hedging strategy itself is the *justification* for the premium: "we charge g_0 because if we trade according to these rules, we can deliver $\varphi(S_T)$ for exactly g_0 plus a small residual that vanishes in the limit of continuous rebalancing." This is the Black-Scholes promise, and it is the foundation of the entire derivatives industry. Delta hedging is the first-order implementation of this promise; delta-gamma is a refinement that tightens the residual by neutralising the first significant source of hedging error — namely the second-order Taylor term in the payoff function.

What shapes the residual? Every discrete-time hedging error can be traced to the gap between the continuous-time idealisation and the real-world rebalance grid. If the rebalance happens at times $t_0 < t_1 < ... < t_N$, the hedge is perfect *at* each rebalance time but drifts *between* rebalance times as the underlying price moves without the hedge being adjusted. The size of the drift depends on the second derivative of the payoff (gamma) times the square of the price move. Summed over many small intervals, this drift is the leading source of residual P&L variance. Delta-gamma hedging neutralises this leading source — at the cost of introducing a gamma-dependent cash flow (because the gamma hedge itself has to be rebalanced when gamma changes, which happens as spot moves and as time passes).

A slightly deeper view is that hedging is not really about eliminating risk; it is about *exchanging* one kind of risk for another. A delta-only hedge converts directional price risk into gamma-residual risk. A delta-gamma hedge converts gamma-residual risk into (smaller) third-order-Taylor-residual risk plus execution risk plus model risk (because gamma itself depends on volatility, which you may have mis-specified). The deeper you go into higher Greeks, the smaller the direct residual but the more layered the model-dependence. Professional traders have a keen sense of which residual risks are tolerable (those that net out across many similar trades) and which are not (those that are directional and correlated across the book). The art of running a derivatives desk is not to hedge everything; it is to hedge the dangerous things and leave the safe things alone.

Visualising a call payoff $\varphi(x) = (x-K)_+$ and its smoothed (time-t) price $g(t,x)$ together — with the discounted-strike asymptote $(x - Ke^{-r(T-t)})$ drawn for reference, and the horizontal axis marked K, x_t, $x_{t+\Delta t}$ — the Taylor expansion in Δx is

$$g(t, S_t + \Delta S) \;=\; g(t, S_t) \;+\; \Delta S\,\underbrace{\partial_S g(t, S_t)}_{\text{delta }\Delta} \;+\; \tfrac{1}{2}(\Delta S)^2\,\underbrace{\partial_{SS} g(t, S_t)}_{\text{gamma }\Gamma} \;+\; \cdots. \tag{7.14}$$

The higher-order terms are the remaining Greeks — vega, theta, vanna, volga, speed, and so on.

We hedge a short position in the target claim g using the stock S_t, the money-market account M_t, and a second traded claim h_t whose Greeks we know. With units α_t of S, β_t of M, and γ_t of h, the hedge portfolio is

$$V_t \;=\; \alpha_t\, S_t \;+\; \beta_t\, M_t \;+\; \gamma_t\, h_t, \tag{7.15}$$

and we *fit* its spatial derivatives at S_t to those of the target. The first-derivative match is

$$\partial_S V_t \;=\; \alpha_t \;+\; \gamma_t\, \partial_S h \;=\; \Delta^g, \tag{7.16}$$

and the second-derivative match is

$$\partial_{SS} V_t \;=\; \gamma_t\, \partial_{SS} h \;=\; \Gamma^g, \tag{7.17}$$

since the stock contributes 0 to gamma ($\partial_{SS}(\alpha_t S_t) = 0$) and the bank contributes 0 to both delta and gamma. Only the auxiliary claim h carries curvature, which is why we need it.

The linear system (7.16)–(7.17) has a particularly clean matrix form:

$$\begin{pmatrix} 1 & \Delta^h \\ 0 & \Gamma^h \end{pmatrix} \begin{pmatrix} \alpha_t \\ \gamma_t \end{pmatrix} \;=\; \begin{pmatrix} \Delta^g \\ \Gamma^g \end{pmatrix}. \tag{7.18}$$

The matrix is upper-triangular — the stock contributes only to delta (upper-left entry 1, lower-left 0) and the auxiliary option contributes to both delta and gamma. Inverting the triangular matrix,

$$\begin{pmatrix} \alpha_t \\ \gamma_t \end{pmatrix} \;=\; \begin{pmatrix} 1 & -\Delta^h/\Gamma^h \\ 0 & 1/\Gamma^h \end{pmatrix} \begin{pmatrix} \Delta^g \\ \Gamma^g \end{pmatrix}, \tag{7.19}$$

which recovers exactly the boxed pair from Framing A:

$$\gamma_t \;=\; \frac{\Gamma^g}{\Gamma^h}, \qquad \alpha_t \;=\; \Delta^g \;-\; \frac{\Gamma^g}{\Gamma^h}\,\Delta^h. \tag{7.20}$$

The choice of auxiliary claim h deserves a moment of thought. In practice h is almost always a liquid, exchange-traded vanilla option — an at-the-money listed call or put with a short expiry is the classic choice because it has high gamma per dollar of premium and a tight bid-ask. A deep-in-the-money option would work mathematically but has low gamma, so large notional amounts are required to match the target's gamma; that increases funding and transaction costs. A far-out-of-the-money option would also work but its gamma changes rapidly with spot, so the hedge weights would rebalance aggressively and incur turnover. The ATM short-dated option threads the needle: high gamma, moderate turnover, low execution cost.

There is a subtler point about the *cross-Greek profile* of h. When we hedge a target g whose gamma and vega (and cross-Greek vanna, say) both matter, choosing h with the same gamma profile does not automatically match the vega profile. In fact a single auxiliary claim can at best match one higher-order Greek at a time; matching both gamma and vega requires two auxiliary claims, typically two listed options at different strikes or expiries whose gamma and vega profiles are linearly independent. This is the operational reason that serious exotic-hedging desks maintain entire *option-book hedging portfolios* consisting of dozens of liquid vanillas, not just one or two: the target exotic has a many-dimensional Greek profile and needs a high-dimensional hedging basis to match it accurately. The linear algebra is essentially a least-squares fit of the target's Greek vector onto the span of the available hedging-instrument Greek vectors.

When the Greek matrix is nearly singular — which happens when the hedging instruments have similar Greek profiles — the hedge weights become numerically unstable and can flip signs or blow up in magnitude. This is a common pitfall in automated hedging systems: two ATM options with nearby expiries might have nearly parallel Greek vectors, so their contributions cannot be cleanly separated, and tiny changes in Greeks produce enormous changes in hedge ratios. The operational fix is to regularise the system (ridge regression on the hedge solution), or to pick a hedging basis with intentionally diverse Greek profiles (options at widely different strikes and expiries). Diversity in the hedging basis gives a well-conditioned matrix and stable hedge weights.

A common conceptual error is to forget the second term in (7.20) — to compute the delta hedge as $\alpha = \Delta^g$ and then bolt on a gamma-matching layer of h. That is internally inconsistent because the h position itself has a delta; one has over-delta-hedged by exactly $\gamma_t \Delta^h$. The correct construction is simultaneous: pick γ_t to zero the portfolio's gamma, then pick α_t to zero the *residual* delta after accounting for h's delta contribution.

8.3.4 7.2.4 Reconciling the two framings

The two derivations above — two-instrument replication (Framing A) and local quadratic fit (Framing B) — end at the same boxed pair of hedge ratios (7.11) and (7.20). That is not a coincidence. The agreement is a consequence of a deep duality: requiring the hedged portfolio's *Itô dynamics* to match the target's is the same as requiring the hedged portfolio's *spatial Taylor expansion* to match the target's, *in the GBM-with-constant-coefficients setting we have been working in.*

To see why, note that for any smooth $V(t, S_t)$ Itô's lemma delivers

$$\mathrm{d}V \;=\; \big(\partial_t V + \mu\, S_t\, \partial_S V + \tfrac{1}{2}\sigma^2 S_t^2\, \partial_{SS} V\big)\,\mathrm{d}t \;+\; \sigma\, S_t\, \partial_S V\, \mathrm{d}W_t. \tag{7.21}$$

The coefficient of $\mathrm{d}W_t$ is σS_t times $\partial_S V$, i.e. delta. The coefficient of $\mathrm{d}t$ has $\partial_{SS} V$ buried inside the convexity term. Matching $\mathrm{d}V = \mathrm{d}g$ at the Itô level therefore requires matching *both* $\partial_S V = \partial_S g$ (the $\mathrm{d}W$ coefficient) and $\partial_{SS} V = \partial_{SS} g$ (via the convexity term in $\mathrm{d}t$). Those are precisely the two conditions of Framing B. Conversely, Framing B's spatial Taylor match at $S = S_t$ produces a portfolio whose $\mathrm{d}W$ and convexity contributions match the target's at S_t, which is the Framing-A condition.

The cleanest way to state the equivalence is this: in a single-factor Itô model, matching the portfolio's *first and second spatial derivatives* to the target's is equivalent to matching the coefficients of $\mathrm{d}W_t$ and the ∂_{SS} part of the $\mathrm{d}t$-term in the Itô expansion. The ∂_t and $\mu\,\partial_S$ pieces of the $\mathrm{d}t$-term are already absorbed by the pricing PDE (they are what the cash leg pays for), which is why the system reduces to just two equations in two unknowns.

An equivalent way to see it: the hedging portfolio has three degrees of freedom $(\alpha_t, \beta_t, \gamma_t)$. The self-financing condition and the matching of the *value* $V_t = g_t$ (which pins β_t) together eliminate one degree of freedom and one constraint. We are left with two degrees of freedom (α_t, γ_t) and two constraints (delta and gamma match). The system is exactly determined; the solution is unique; and it is what both framings produce.

There is one structural consequence of this duality worth calling out. Framing A makes it immediate that the delta-gamma hedge is **exact in the continuous-time limit** — the Itô dynamics of the replicating portfolio match the target claim's, and the tracking error goes to zero almost surely as the rebalance grid refines. Framing B makes it immediate that under *discrete* rebalancing the leading residual scales as $(\Delta S)^3$, and so has variance of order $(\Delta t)^3$ per interval and $(\Delta t)^2$ when summed over $N = T/\Delta t$ intervals. The standard deviation therefore scales as $1/N$ rather than $1/\sqrt{N}$ as in the delta-only case. The two framings reinforce each other: one tells us that the hedge is theoretically exact, the other tells us how fast the discretisation error vanishes.

We will use Framing B's linear-algebra presentation for all the computations that follow, because it generalises cleanly to n-Greek hedging with n traded instruments, and we will invoke Framing A's structural argument whenever we need to justify that the resulting construction is self-financing and replicates the target in the no-arbitrage sense.

8.3.5 7.2.5 Worked numerical example — gamma-matching an ATM option

Let us make (7.15) concrete with specific numbers. Suppose we sell a 1-year ATM call on a \$100 stock: strike $K_g = 100$, expiry $T_g = 1$ year, implied vol $\sigma = 25\%$, rate $r = 3\%$. Using Black-Scholes we get price $g_0 \approx 10.94$, $\Delta^g \approx 0.59$, $\Gamma^g \approx 0.0156$. To gamma-hedge we use a 3-month ATM call on the same underlying: strike $K_h = 100$, expiry $T_h = 0.25$ year, same vol and rate. Its price is $h_0 \approx 5.40$, $\Delta^h \approx 0.53$, $\Gamma^h \approx 0.0313$.

Applying (7.20),

$$\gamma_0 \;=\; \frac{\Gamma^g}{\Gamma^h} \;=\; \frac{0.0156}{0.0313} \;\approx\; 0.498. \tag{7.22}$$

We buy roughly half a unit of the 3-month call for every 1-unit short position in the 1-year call. Then

$$\alpha_0 \;=\; \Delta^g \;-\; \frac{\Gamma^g}{\Gamma^h}\,\Delta^h \;=\; 0.59 \;-\; 0.498 \cdot 0.53 \;\approx\; 0.326. \tag{7.23}$$

We buy 0.326 shares. The initial outlay for the hedge is

$$\alpha_0 \cdot S_0 \;+\; \gamma_0 \cdot h_0 \;=\; 0.326 \cdot 100 \;+\; 0.498 \cdot 5.40 \;=\; 32.6 \;+\; 2.69 \;=\; 35.29. \tag{7.24}$$

We received $g_0 = 10.94$ for selling the 1-year call, so the cash account starts at

$$M_0 \;=\; g_0 \;-\; \alpha_0 S_0 \;-\; \gamma_0 h_0 \;=\; 10.94 \;-\; 35.29 \;=\; -24.35. \tag{7.25}$$

Meaning we borrow \$24.35 to establish the hedge. Over the next rebalance period we pay interest on this borrowing at rate r, and the stock and option positions mark to market at whatever new spot and option price the market delivers.

Compare this with a pure delta hedge: $\alpha_0 = \Delta^g = 0.59$ shares, cash account $10.94 - 0.59 \cdot 100 = -48.06$. The delta-gamma hedge requires *less* borrowing (\$24.35 vs \$48.06) because the gamma-hedging option supplies some of the delta. But the delta-gamma hedge carries the 3-month option position, which must be rolled or rebalanced as it ages toward its own expiry; it introduces its own theta decay and vega exposure. The trade-off is that delta-gamma reduces residual variance at the cost of a more complex rebalancing schedule and additional instrument-specific Greeks that must themselves be monitored.

If we repeat this exercise with a 1-month ATM hedging option instead of a 3-month one, the gamma is higher (roughly 0.054 vs 0.031), so we need fewer units: $\gamma_0 = 0.0156/0.054 \approx 0.289$. The higher-gamma hedge is more efficient in notional terms — less option notional required to match the target's gamma — but also rebalances faster because its own gamma drifts quickly with spot and time. The choice between a 1-month and 3-month hedging option is a trade-off between notional efficiency and rebalancing cost, typically resolved by desk policy.

A *numerical sanity check* on the local-fit quality. For our sold 1-year ATM call and a 2% spot move ($\Delta S = 2$), the true price change is about 1.15. The delta-only approximation predicts $\Delta^g \cdot \Delta S = 0.59 \cdot 2 = 1.18$ (a 2.6% relative error, which is coincidentally tight because the gamma and theta contributions partly offset). The delta-gamma approximation predicts $0.59 \cdot 2 + \frac{1}{2} \cdot 0.0156 \cdot 4 = 1.21$ — in this *particular* window the quadratic fit and linear fit are both within a cent of the truth, because the 1-year option has modest gamma. For a 3-month ATM call, the gamma is nearly twice as large; for a 1-month ATM call it is three to four times larger. At a 10% move, the quality of the two approximations diverges sharply: delta-only misses the truth by several percent, delta-gamma matches to sub-percent accuracy. Delta-gamma is a superb *local* hedge and a mediocre *global* one; it is designed to be refreshed frequently on a grid rather than held for a long interval.

The failure mode for large moves has a systematic character. Near the strike, gamma is positive and large; it decays for moves far in either direction. A quadratic extrapolation assumes gamma remains at its spot-level value even as the stock moves far from the spot, but gamma *itself* decays as we move away from the strike. This makes the delta-gamma quadratic an over-estimate of convexity in the wings, leading to systematic over-prediction of price changes for large moves. This is precisely the scenario in which third-order (speed) corrections start to matter: speed captures the rate at which gamma decays, and including it pulls the quadratic toward the true curve in the wings. For risk-measurement purposes — particularly VaR and stress-testing at 3-to-5-sigma confidence — the "delta-gamma-parametric" methodology has to be augmented by full revaluation when the scenarios push far into the wings.

8.3.6 7.2.6 The discrete delta-gamma recursion

When continuous rebalancing is discretised, the analogue of the delta-only cash-account recursion of §6.11 for the two-instrument hedge is

$$M_{t_k} \;=\; M_{t_{k-1}}\, e^{r\,\Delta t} \;-\; (\alpha_{t_k} - \alpha_{t_{k-1}})\, S_{t_k} \;-\; (\gamma_{t_k} - \gamma_{t_{k-1}})\, h_{t_k}, \tag{7.26}$$

where each rebalance at time t_k requires buying or selling enough stock to restore the delta match, *and* enough of the auxiliary option to restore the gamma match. Both trades are funded out of the bank account (or deposited into it), and the bank grows between rebalances at the risk-free rate. The structure is identical to the pure-delta recursion but with an extra h-trading term on the right-hand side. Transaction costs, if included, are simply summed over the two legs — we take up the details in §7.5.

Equation (7.26) is the operational engine of any delta-gamma hedging program. Each morning (or each rebalance tick) the trader observes the new stock price S_{t_k} and option price h_{t_k}, recomputes the desired hedge weights α_{t_k} and γ_{t_k} from (7.20) using the updated Greeks, and executes the increments $\alpha_{t_k} - \alpha_{t_{k-1}}$ in the stock and $\gamma_{t_k} - \gamma_{t_{k-1}}$ in the option. The cash flow from these executions debits the bank account. Between rebalances the bank account compounds at the short rate.

The *absolute* size of each rebalance trade depends on how much the Greeks have drifted since the last rebalance, which in turn depends on how much the stock has moved, how much time has passed, and how curved the target's Greek profile is. A near-the-money, short-dated option has delta and gamma that change rapidly with spot, so its hedge rebalances frequently and the cash account sees a lot of churn. A far-dated, deep-in-the-money option has an almost linear payoff and a nearly constant delta, so the hedge barely moves and rebalance costs are low. These observations underpin the whole *gamma scalping* school of proprietary option trading: taking long-gamma positions on short-dated ATMs and systematically extracting the rebalance cash-flow turnover as P&L, funded by theta decay.

One useful mental model of the rebalance cash flow is to think of it as a *betting ledger* between the desk and the market. Each morning, the stock has moved; the desk's required delta has changed; the desk executes the delta change at the current stock price; and that execution booked a cash flow. If the stock moved up and the desk had to buy more stock to re-hedge, the desk pays for stock at a higher price than yesterday's average — so the rebalance has a small negative cash flow relative to zero. If the stock moved up and the desk had to sell stock (long-gamma position), the desk receives cash at the higher price and the rebalance has a positive cash flow. The aggregate pattern of positive and negative rebalance cash flows over the option's life is exactly the *gamma P&L*, and its magnitude is $\frac{1}{2}\Gamma(\Delta S)^2$ per rebalance, summed over all rebalances, minus the theta bleed. This is the famous "buy low, sell high" mechanism that underlies gamma scalping — and its mirror image, the "sell high, buy low" pain of being short gamma in a trending market.

At maturity $t_N = T$, the portfolio wealth owes the option payoff:

$$P_N \;=\; M_{t_{N-1}}\, e^{r\,\Delta t} \;+\; \alpha_{t_{N-1}}\, S_{t_N} \;+\; \gamma_{t_{N-1}}\, h_{t_N} \;-\; \varphi(S_{t_N}), \tag{7.27}$$

and the expression in parentheses collapses, if the hedge tracked perfectly, to $\varphi(S_{t_N})$, leaving $P_N = 0$. In a well-specified model with no transaction costs and continuous rebalancing, (7.27) is identically zero; in reality, it is a small random variable whose distribution is tighter than the analogous delta-only P&L because the leading gamma error has been removed. We quantify the tightening next.

8.3.7 7.2.7 Terminal P&L and residual scaling

At expiry $t_N = T$ we liquidate the hedge and deliver the claim payoff $\varphi(S_T)$. The realised profit-and-loss of the delta-gamma hedged book is

$$\boxed{\text{P\&L} \;=\; M_{t_{N-1}}\, e^{r\,\Delta t} \;+\; \alpha_{t_{N-1}}\, S_{t_N} \;+\; \gamma_{t_{N-1}}\, h_{t_N} \;-\; \varphi(S_{t_N}).} \tag{7.28}$$

Under a continuous-time, frictionless, complete-market set-up this quantity is identically zero; under discrete rebalancing it is a random variable whose variance shrinks as the grid is refined. The delta-gamma hedge has an order-of-magnitude smaller P&L variance than a pure delta hedge because it neutralises the quadratic term in the Taylor expansion — the leading residual is cubic in ΔS rather than quadratic.

The *scaling argument* is worth carrying through carefully, because it is the quantitative payoff of all the extra machinery. A delta-only hedge leaves a residual that is quadratic in the stock move per rebalance interval. The variance of that residual scales as the variance of $(\Delta S)^2$, which for GBM is of order $(\sigma^2\Delta t)^2 \cdot S^4$. Summing over N rebalance intervals of length $\Delta t = T/N$, the total residual variance scales as

$$N \cdot \left(\sigma^2 \tfrac{T}{N}\right)^2 \;=\; \frac{\sigma^4 T^2}{N}. \tag{7.29}$$

The standard deviation therefore scales as $1/\sqrt{N}$ — slow convergence. A delta-gamma hedge leaves a residual that is *cubic* in the stock move per interval. Its variance scales as the variance of $(\Delta S)^3 \sim S^6\sigma^6(\Delta t)^3$. Summing,

$$N \cdot \left(\sigma^6 \tfrac{T^3}{N^3}\right) \;=\; \frac{\sigma^6 T^3}{N^2}, \tag{7.30}$$

with standard deviation scaling as $1/N$. So on a weekly grid, the delta-gamma standard deviation is a factor $\sqrt{N}$ smaller than the delta-only standard deviation; for $N = 52$ weekly rebalances, that is roughly a factor of 7 reduction. The difference between $1/\sqrt{N}$ and $1/N$ convergence is the practical justification for the extra hedging instrument.

Hedging P&L distribution — delta-only vs delta-gamma *Figure 7.2 — Monte-Carlo terminal-P&L density under weekly rebalancing: delta-gamma hedge tightens variance by roughly* $2\times$ *relative to pure delta, with a tighter tail and lighter skew.*

There is an elegant symmetry with the VaR/CTE discussion of Chapter 9. Hedging reshapes the loss distribution from something with pronounced tails (pure delta hedge, residual dominated by gamma) to something much more Gaussian (delta-gamma hedge, residual dominated by higher-order terms that tend to cancel). The CTE of the hedged P&L falls by a factor that is roughly the square of the factor by which the standard deviation falls — because the CTE of a Gaussian is a fixed multiple of its standard deviation. So the practical benefit of delta-gamma over delta-only is not just a tighter variance but a disproportionately smaller tail.

A further symmetry is with the Cornish-Fisher expansion. The delta-only hedged P&L has significant skewness and excess kurtosis inherited from the $(\Delta S)^2$ term — $(\Delta S)^2$ is the square of a zero-mean Gaussian, which is a chi-squared distribution with pronounced skewness and excess kurtosis. The delta-gamma hedged P&L has its leading residual $(\Delta S)^3$, which is a cubed Gaussian — a distribution with zero skewness but very heavy tails (before mixing with other residuals). The interplay between these two distributions is why the naive Gaussian VaR applied to a delta-only hedged book can be misleading: the residual is chi-squared, not Gaussian, and the Cornish-Fisher correction to the Gaussian VaR is substantial. For a delta-gamma hedged book, the residual is closer to Gaussian in its quantile behaviour, and the Cornish-Fisher correction is smaller. This is another lens through which to see the CTE-compression benefit of delta-gamma: not just a tighter variance but a closer-to-Normal shape that plays better with standard risk measures.

8.3.8 7.2.8 Long-gamma vs short-gamma — when residual convexity is a feature

An important subtlety is that pure delta hedging — deliberately leaving gamma exposure — is sometimes desirable, not a mistake to be corrected. A trader who buys options and delta-hedges them has a portfolio that is *long gamma*. Long gamma means the hedged P&L is a convex function of the stock price: whichever direction the stock moves, the hedge *under*-hedges in the direction of profit and *over*-hedges in the direction of loss, so both tails are positive. The cost of carrying this convexity is theta — the time decay of the owned options.

A long-gamma, short-theta book wants the stock to move. If it moves enough (in either direction) faster than the theta bleed, the book makes money; if it doesn't, the theta eats the convexity profit and the book loses money. This is a volatility trade — a bet on realised volatility exceeding implied volatility, expressed as a delta-hedged long-option position. *Gamma scalping* is its operational incarnation: the trader intentionally does *not* gamma-hedge, collects the realised-vs-implied vol premium on the gamma, and pays the theta bleed as the cost of doing business.

The P&L of a delta-hedged long-gamma position over a small time interval is approximately

$$\Delta P \;\approx\; \tfrac{1}{2}\Gamma\, S^2\,(r_t^2 \;-\; \sigma_{\text{imp}}^2\,\Delta t), \tag{7.31}$$

where r_t is the realised log-return and σ_{imp} is the implied volatility at which the option was bought. If realised volatility exceeds implied, the position makes money; if realised volatility falls short, it loses. Aggregated over the option's life, this is a *realised-vs-implied variance* trade, and its expected P&L is the *variance risk premium* — the systematic compensation earned by long-volatility traders for bearing the tail risk of volatility spikes. Empirically, implied volatility in equity-index options exceeds realised volatility by two to four percentage points on average, making the short-vol trade a profitable strategy most of the time, punctuated by catastrophic losses during volatility spikes (August 2011, February 2018, March 2020). The gamma-scalping long-vol trade is the opposite: small systematic losses most of the time, offset by large gains during volatility spikes. Which side of the trade to be on depends on one's view of the current implied-vol level relative to expected realised vol over the option's life.

Gamma scalping as a business has been a major component of equity derivatives trading for decades. Market-making desks sell options to clients (collecting the vol premium), delta-hedge the position, and realise a spread

between implied and realised volatility. The spread is the compensation for providing liquidity; it also carries tail risk because when realised vol spikes, the short-gamma position bleeds fast. Large option market-makers have taken significant losses on their gamma-scalping books during major volatility events, and several have gone bankrupt in smaller volatility spikes. The key risk-management discipline is to size the short-gamma exposure such that even a three- or four-sigma realised-vol shock does not exceed the firm's risk appetite.

Correspondingly, a short-gamma book — typical of option sellers — wants the stock to sit still. Every move in either direction costs money (the short gamma concavity); every day that passes adds theta decay as income. Option sellers routinely choose *not* to gamma-hedge by buying back curvature, because doing so would neutralise the theta income and leave them with a pure funding trade. Instead they run short gamma within explicit risk limits — a dollar-gamma budget, a vega budget, a stress-scenario P&L limit — and live with the convexity exposure because the theta is the whole point of being short.

So the choice between pure-delta and delta-gamma hedging is not always about variance minimisation. It is about whether the residual gamma exposure is a bug (for a sell-side client-facilitation book that wants to be flat) or a feature (for a vol-trading book that monetises gamma). The mathematics of (7.11) gives the recipe for *eliminating* gamma; the business context determines whether you want to apply that recipe or deliberately leave gamma on the books. A well-designed risk framework explicitly tracks gamma at the book level and reports both the hedged VaR (assuming gamma-neutral) and the unhedged VaR (with actual gamma) so that the choice is made with eyes open.

Commercially, the choice between "lay off all gamma via matching options" and "warehouse the gamma" is closely tied to the desk's business model. A flow desk that handles client order flow wants to be flat — it collects bid-ask on the client trade, hedges with another option to neutralise gamma, and books a tight spread. A vol desk that takes proprietary volatility positions wants to carry gamma — it expresses a view that realised versus implied vol will diverge in a specific direction and takes the appropriate side. Large derivative houses run both types of book side-by-side, with internal transfer-pricing mechanisms that allow flow-desk gamma to be transferred to the prop-desk at fair market prices. The risk infrastructure has to support both models, and the P&L attribution has to cleanly separate "flow spread" from "prop P&L" for compensation purposes. These operational details are invisible from the mathematical level but central to how real trading floors operate.

8.3.9 7.2.9 Why two Greeks — and why not three?

A natural question: if delta-gamma hedging neutralises the first and second spatial derivatives of the target price, why not add a third option and neutralise the third derivative (*speed*) as well? Why stop at two?

The answer is partly practical and partly principled. *Practically*, each additional Greek we target requires an additional liquid traded instrument whose own higher-order Greeks are independent of the others' — and the option-market's supply of liquid, uncorrelated higher-Greek vehicles is limited. A third-order hedge would in principle require a third option whose *speed* (third spatial derivative of price) is appreciable and whose delta and gamma can be matched simultaneously, and in liquid vanilla markets the three-dimensional system is often ill-conditioned.

Principled, the magnitude of successive spatial derivatives decays rapidly under a diffusion: the third-order term in the Taylor expansion is multiplied by $(\Delta S)^3 \sim (\Delta t)^{3/2}$, which is already small, and the marginal variance reduction per unit of additional hedging effort is modest. Most desks stop at delta-gamma; a few specialised books (volatility arbitrage, barrier-option market-making) add vega and sometimes vanna and volga (cross-derivatives with volatility) to handle risk that is orthogonal to spot moves.

The third-order term worth naming is *speed*: $\partial_{SSS}g$. Speed measures how gamma changes with spot — effectively the "gamma of gamma." In the regions where gamma is changing rapidly (near the strike as expiry approaches), speed is large, and a delta-gamma hedge that ignores speed will have a rebalancing schedule that is optimised for the wrong curvature as spot moves. Exotic desks sometimes match speed as a third-layer hedge using a third traded option with a different gamma profile, but this is rare because the operational complexity outweighs the residual-risk reduction for most portfolios. The theoretical observation is that matching delta, gamma, and speed reduces the leading residual error from cubic to quartic in the stock move, shaving yet another order off the residual variance.

A related fourth-order Greek is *colour* (also called *gamma decay* or $\partial_{SSt}g$), measuring how gamma changes with time. Colour matters most for short-dated options where gamma concentrates rapidly; a colour-hedged position stays gamma-neutral across the trading day even as time decays. Colour is rarely hedged because time passes deterministically — one can simply rebalance more frequently as expiry approaches rather than add colour-hedging instruments. Nevertheless, colour is reported on most desk-level Greek sheets as a monitoring metric.

The *Greek hierarchy* on a typical desk is therefore:

- **Delta** $\Delta = \partial_S f$. First-order sensitivity to underlying price. Hedged by holding $-\Delta$ units of stock. Eliminates first-order directional risk.
- **Gamma** $\Gamma = \partial_{SS} f$. Second-order sensitivity. Hedged by trading a second option with known Γ^h. Eliminates second-order convexity risk.
- **Vega** $\mathcal{V} = \partial_\sigma f$. Sensitivity to implied volatility. Hedged by trading options with matching vega profiles. Eliminates volatility-level risk.
- **Theta** $\Theta = \partial_t f$. Sensitivity to time passage. Not directly hedgeable (time passes deterministically) but tracked as the "carry cost" of holding the position.
- **Rho** $\rho = \partial_r f$. Sensitivity to the risk-free rate. Hedged with rate instruments (bonds, swaps, futures). Small for short-dated options, significant for long-dated ones.
- **Vanna** $\partial^2 f/(\partial S\, \partial\sigma)$. Cross-sensitivity of delta to volatility. Significant for skew-sensitive structures.
- **Volga** $\partial^2 f/\partial\sigma^2$. Second-order volatility sensitivity. Significant for volatility-of-volatility exposures.
- **Charm** $\partial^2 f/(\partial S\, \partial t)$. Cross-sensitivity of delta to time. Drives "delta bleed" as expiry approaches.
- **Colour** $\partial^2 f/(\partial S^2\, \partial t) = \partial\Gamma/\partial t$. Time decay of gamma.
- **Speed** $\partial^3 f/\partial S^3$. Third-order price sensitivity.

A production risk system reports all of these and more. The practical art of hedging is to know which Greeks matter for which book. An FX options book cares about vega and vanna (skew sensitivity). An equity-index volatility book cares about gamma, vega, and volga (vol-of-vol). A single-name equity options book cares about delta, gamma, and vega plus basis risk to the index hedge. A rates-swaptions book cares about delta, gamma, vega, and rho plus yield-curve exposures. Each business's unique Greek fingerprint dictates which hedging instruments it keeps on hand and which it trades regularly.

8.4 7.3 Vega and volatility risk

The Black-Scholes framework takes the volatility σ as a constant parameter. In the market, volatility is a living number: it changes day to day, bar to bar, and sometimes intrabar. The sensitivity of an option's price to changes in the assumed volatility is called *vega*:

$$\mathcal{V}_t \;\equiv\; \partial_\sigma g(t, S_t; \sigma). \tag{7.32}$$

Equivalently, vega measures by how many cents the option's price rises if the volatility input moves up by one percentage point. Vega is strictly speaking the most awkward of the first-order Greeks because σ is not itself a traded quantity — the underlying trades, the option trades, but no one trades "volatility" directly except through volatility-linked contracts (variance swaps, VIX futures). When we write $\mathcal{V}_t = \partial_\sigma g$ we are differentiating a pricing *formula* with respect to one of its parameters, not a market process. The conceptual leap from "assumed-sensitivity" to "tradable risk" is what makes vega non-trivial and what motivates the extensions to stochastic volatility in §7.3.2.

8.4.1 7.3.1 Vega of a vanilla Black-Scholes call

For a European call under Black-Scholes, vega has a clean closed form,

$$\mathcal{V}_t \;=\; S_t\,\sqrt{T-t}\,\Phi'(d_+), \tag{7.33}$$

where $d_+ = [\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)(T-t)]/(\sigma\sqrt{T-t})$ is the usual moneyness index and $\Phi'(x) = (2\pi)^{-1/2}e^{-x^2/2}$ is the standard normal density. Two features of (7.33) are worth flagging.

First, vega is positive for *every* vanilla option — both calls and puts gain value when volatility rises, because higher volatility means more upside (for calls) and more downside (for puts) on the payoff distribution, and the forward value of each is sensitive to the mass in its own tail. This is a universal property of convex payoffs: a convex function of a random variable is monotone in the spread of the random variable, so increasing the volatility of the underlying increases the expected payoff. Puts and calls being convex in S_T, their forward prices rise with volatility.

Second, vega peaks at the at-the-money strike and decays to zero for deep in-the-money or deep out-of-the-money options. An ATM short-dated call has the largest vega per unit of option; a deep-ITM LEAPS has essentially none.

Vega decay as expiry approaches *ATM vega as a function of time-to-expiry (x-axis inverted so time runs forward toward expiry on the right). Vega scales as $\sqrt{T-t}$ and collapses to zero at expiry — the option has no volatility sensitivity once the payoff is locked in. This is why short-dated ATM options are the preferred instrument for* discharging* vega risk (low cost per unit) and long-dated ATM options for *accumulating* it.*

Vega heatmap across spot and time-to-expiry *Vega $\mathcal{V} = S\sqrt{T-t}\,\Phi'(d_+)$ plotted over (S,T). The bright band runs along the at-the-money ridge; vega grows with $\sqrt{T-t}$ at fixed moneyness, so long-dated ATM options carry the largest vega per contract. This is why "volatility duration" is a long-dated ATM product by construction.*

That pattern — vega concentrated near the ATM strike — is the reason straddles and strangles are the prototypical *volatility trades*: they maximise vega exposure per unit of capital deployed. A long straddle (long ATM call + long ATM put) has vega equal to twice that of a single ATM option while its delta is near zero by symmetry, giving the maximum "volatility-per-unit-of-directional-exposure" of any simple vanilla combination.

A useful numerical benchmark to keep in mind: at the money, the density $\Phi'(0) = 1/\sqrt{2\pi} \approx 0.4$. For a \$100 stock and a three-month option ($\sqrt{0.25} = 0.5$),

$$\mathcal{V}_t \;\approx\; 100 \cdot 0.5 \cdot 0.4 \;=\; 20 \text{ cents per volatility point.} \tag{7.34}$$

If volatility moves by 1 point in a day (a large but not extraordinary move), the option's price changes by 20 cents — a substantial fraction of the option's total premium for short-dated contracts. Vega risk is the second-largest source of P&L volatility for most options books, after delta-unhedged direction.

8.4.2 7.3.2 Volatility as a hidden state variable

Strictly speaking, (7.32) is an *assumed-sensitivity* measure — the model takes σ as constant, so differentiating with respect to it produces a derivative of a formula, not of a reality. The conceptual leap needed to turn vega into a tradable hedge is to admit that volatility is *itself* a stochastic process. Once we adopt a stochastic-volatility model (Heston, SABR, local-vol surfaces — the subject of Chapter 14), the option price is a function $g(t, S_t, \sigma_t)$ of three state variables, and Itô's lemma applied to this richer function picks up a $\partial_\sigma g \cdot \mathrm{d}\sigma_t$ term that did not appear in the constant-volatility treatment.

The multi-dimensional Itô expansion reads

$$\begin{aligned} \mathrm{d}g_t \;&= \partial_t g\,\mathrm{d}t \;+\; \partial_S g\,\mathrm{d}S_t \;+\; \partial_\sigma g\,\mathrm{d}\sigma_t \\ &\quad+\; \tfrac{1}{2}\partial_{SS} g\,\mathrm{d}\langle S,S\rangle_t \;+\; \partial_{S\sigma} g\,\mathrm{d}\langle S,\sigma\rangle_t \;+\; \tfrac{1}{2}\partial_{\sigma\sigma} g\,\mathrm{d}\langle \sigma,\sigma\rangle_t. \end{aligned} \tag{7.35}$$

The new drift term contains a *vega times drift of volatility* piece, a *cross-gamma* $\partial_{S\sigma} g \cdot \mathrm{d}\langle S,\sigma\rangle_t$ piece that captures the correlation between spot and vol moves, and a *volga* $\frac{1}{2}\partial_{\sigma\sigma} g \cdot \mathrm{d}\langle \sigma,\sigma\rangle_t$ piece that captures the convexity of the option in the volatility direction. Each of these is a distinct risk that the replicating portfolio must accommodate.

The upshot is that in a stochastic-volatility world, a pure delta-hedged portfolio is *not* self-financing: it leaks value through the vega leg every time volatility drifts, and it leaks value through the cross-gamma leg every time spot and vol move together. To close the replication argument one must enlarge the hedging set yet again to include a second option traded on the same underlying, matching both the target's vega *and* its vega-weighted gamma. This *three-instrument portfolio* (stock, cash, first option, second option) is how sophisticated options desks run in practice, and it is why the book of any real market-maker is two-dimensional in its Greek profile: a grid of delta and vega exposures rather than a single delta number.

The linear-algebra generalisation of the delta-gamma system (7.18) is immediate. If we wish to match delta, gamma, and vega simultaneously, we need three hedging instruments beyond cash — the stock (which contributes to delta only) and two options h_1, h_2 with different Greek profiles. The match equations are

$$\begin{pmatrix} 1 & \Delta^{h_1} & \Delta^{h_2} \\ 0 & \Gamma^{h_1} & \Gamma^{h_2} \\ 0 & \mathcal{V}^{h_1} & \mathcal{V}^{h_2} \end{pmatrix} \begin{pmatrix} \alpha_t \\ \gamma_t^1 \\ \gamma_t^2 \end{pmatrix} \;=\; \begin{pmatrix} \Delta^g \\ \Gamma^g \\ \mathcal{V}^g \end{pmatrix}, \tag{7.36}$$

and the system has a unique solution whenever the 2×2 sub-matrix of $(\Gamma, \mathcal{V})$ of the two hedging options is non-singular — i.e. whenever h_1 and h_2 have *linearly independent* gamma-vega profiles. Inverting the matrix gives the

hedge weights in closed form. A diligent reader will reproduce (7.20) as the 2×2 sub-case when vega matching is dropped.

8.4.3 7.3.3 Vega hedging in practice

On a working desk, the operational story is this. A vanilla-option market-maker inventories options across a range of strikes and maturities. Each option has its own vega. The sum of vegas across the book is the *total vega* exposure, measured in dollars per volatility point. Desks typically run vega limits by tenor bucket — "front-month vega, X; second-month vega, Y" — because vega at different tenors is priced off different points on the implied-volatility curve and may move in opposite directions. When a desk's total vega exceeds its limit in any bucket, the risk manager asks for vega to be reduced — either by closing positions in that bucket or by putting on an offsetting position in another option of similar vega.

A common hedging tactic is to offset vega with a different-tenor option on the same underlying, exploiting the different time-dependence of vega. A long one-month straddle and a short three-month straddle with appropriate sizing will produce a near-vega-neutral position whose *vega term structure* is what the trader is really betting on. This kind of *calendar-spread* structure is a standard vehicle for expressing views on the implied-vol curve without taking on direct vega risk.

Across strikes, the analogous structure is a *skew trade*: long an out-of-the-money put and short an out-of-the-money call (a "risk reversal") has near-zero vega at the ATM point but significant *vanna* exposure — the position makes money if the skew steepens and loses if it flattens. Skew trades are the bread and butter of equity-index volatility desks, where the persistent put-skew premium has been a reliable source of P&L for decades (subject to occasional sharp reversals during skew-collapse episodes).

The practical issue with vega hedging is that *vega is not a single number*. An option has a vega at every point on its implied-vol surface — one vega per strike, per expiry — and the total position's risk is against the entire surface, not against a single scalar. Risk-management systems therefore aggregate vega into *buckets*: vega at the 3-month ATM point, vega at the 6-month 25-delta-put point, and so on. A vega-neutral book has zero total vega in *every* bucket, not just the headline number. Achieving bucket-level vega neutrality requires a carefully chosen basis of hedging options, typically two or three per bucket to handle the cross-bucket dependence.

8.4.4 7.3.4 Higher-order volatility Greeks — volga, vanna, charm

The higher-order volatility Greeks — *volga* ($\partial_{\sigma\sigma}$, convexity in volatility), *vanna* ($\partial_{S\sigma}$, cross-sensitivity of delta to vol), and *charm* (∂_{St}, vega decay of delta) — become relevant for structured-product desks that sell options with unusual strike-vol dependencies (digitals, barriers, auto-callables, cliquets). These Greeks are not standard for a vanilla book but appear routinely in exotic-option valuations.

Volga measures the convexity of the option price in the volatility direction. Long volga positions profit when volatility spikes *regardless of direction*, because the second derivative of the option price in σ is positive near the ATM and decays in the wings. A volga-long, vega-flat book is betting that implied volatility will move but is agnostic about whether it moves up or down — a "vol-of-vol" bet. Barriers and cliquets typically have significant volga at the barrier or reset dates, and their hedging programs must include volga-neutralising vanilla options (typically a butterfly of three strikes) alongside the gamma and vega hedges.

Vanna measures the cross-sensitivity of delta to volatility — or equivalently, the cross-sensitivity of vega to spot. A positive-vanna position's delta increases when volatility rises; a negative-vanna position's delta falls. Skew-sensitive structures (risk reversals, down- and-out barriers) have concentrated vanna profiles, and hedging them requires asymmetric choices of strikes — typically out-of-the-money puts and calls at different deltas to produce the desired vanna without introducing vega or volga.

Charm (or $\partial\Delta/\partial t$) measures how delta drifts in time even when spot is constant. For an ATM short-dated call, charm is small; for an in-the-money or out-of-the-money option near expiry, charm can be substantial — it is the source of "delta decay" that a trader sees as her book's delta drifts overnight even though nothing in the market has moved. Charm is rarely hedged directly (since time passes deterministically and more frequent delta-rebalancing handles it), but it is prominently reported on overnight Greek sheets so that the opening-book delta is correctly anticipated.

Each of these Greeks carries its own operational story, and each demands a specific kind of hedging instrument to neutralise. The extension of the replicating-portfolio argument to stochastic volatility, jumps, and rough paths is

the subject of entire later chapters (Chapter 14 for Heston, beyond the present guide for jumps and rough vol) and is what drives the ever-increasing complexity of modern derivatives-pricing engines.

8.5 7.4 Options on a dividend-paying asset

The stock we have been hedging in Chapter 6 has paid no dividends — it is a "pure capital appreciation" asset, and its entire return has flowed through the SDE $\mathrm{d}S_t/S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$. Real stocks pay dividends, sometimes as quarterly cash payouts, sometimes continuously (as in certain index products or in the idealisation where we treat the total-return series). In the continuous-dividend idealisation, the stock pays a dividend yield q per unit time per unit of stock: an investor holding one share over $[t, t+\mathrm{d}t]$ receives a cash payment $q\,S_t\,\mathrm{d}t$ into her account.

This small operational change propagates through the self-financing replication argument in a specific and instructive way. The result is the *Black-Scholes-Merton PDE with dividend yield*, whose only alteration from the no-dividend case is the replacement of r by $r-q$ in the drift coefficient of the first-derivative term. Derivation follows.

8.5.1 7.4.1 Setup — continuous dividend yield

Let S_t be the stock and let it satisfy

$$\frac{\mathrm{d}S_t}{S_t} \;=\; \mu\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t, \tag{7.37}$$

with a *continuous dividend yield* q: holding one share over $[t, t+\mathrm{d}t]$ delivers $q\,S_t\,\mathrm{d}t$ in cash. The bank account is, as always,

$$\mathrm{d}B_t \;=\; r\,B_t\,\mathrm{d}t. \tag{7.38}$$

We value a European claim $g_t = g(t, S_t)$ on S with terminal payoff $g(T, S) = \varphi(S)$. Writing $g \in C^{1,2}$ so that Itô's lemma applies, we set up a self-financing portfolio

$$V_t \;=\; \alpha_t\,S_t \;+\; \beta_t\,B_t \;-\; g_t \tag{7.39}$$

with α_t shares, β_t units of cash, and a short unit of the target claim. At inception we choose $V_0 = 0$ — the usual replication convention.

8.5.2 7.4.2 Self-financing increment with dividends

The self-financing increment must now account for the dividend stream. Over $[t, t+\mathrm{d}t]$ the wealth change has four sources: the stock's price move $\alpha_t\,\mathrm{d}S_t$, the dividend income $\alpha_t\,q\,S_t\,\mathrm{d}t$, the bank's growth $\beta_t\,\mathrm{d}B_t$, and the change in the short claim's value $-\mathrm{d}g_t$:

$$\mathrm{d}V_t \;=\; \alpha_t\,\mathrm{d}S_t \;+\; \alpha_t\,q\,S_t\,\mathrm{d}t \;+\; \beta_t\,\mathrm{d}B_t \;-\; \mathrm{d}g_t. \tag{7.40}$$

The dividend term is what is new. Compared with the no-dividend setting, (7.40) adds a positive drift $\alpha_t\,q\,S_t$: shares held through an interval earn a dividend whether or not the price moves. In the stock-only part of the portfolio, this makes holding long α_t shares look more attractive than it did before — an effect we will see propagate into the option's PDE as a shift in the drift coefficient.

Expanding $\mathrm{d}S_t$, $\mathrm{d}B_t$, and $\mathrm{d}g_t$ via Itô's lemma,

$$\mathrm{d}V_t \;=\; \alpha_t(\mu\,S_t\,\mathrm{d}t + \sigma\,S_t\,\mathrm{d}W_t) \;+\; \alpha_t\,q\,S_t\,\mathrm{d}t \;+\; \beta_t\,r\,B_t\,\mathrm{d}t \;-\; \left[(\partial_t g + \mu\,S_t\,\partial_S g + \tfrac{1}{2}\sigma^2 S_t^2\,\partial_{SS} g)\mathrm{d}t + \sigma\,S_t\,\partial_S g\,\mathrm{d}W_t\right]. \tag{7.41}$$

Grouping $\mathrm{d}t$ and $\mathrm{d}W$ separately,

$$\mathrm{d}V_t \;=\; \underbrace{[\alpha_t(\mu+q)S_t + \beta_t\, r\, B_t - (\partial_t g + \mu\, S_t\, \partial_S g + \tfrac{1}{2}\sigma^2 S_t^2\, \partial_{SS} g)]}_{\equiv \mathcal{A}_t}\mathrm{d}t \;+\; \sigma\, S_t[\alpha_t - \partial_S g]\,\mathrm{d}W_t. \tag{7.42}$$

8.5.3 7.4.3 Zeroing the noise and drift

The coefficient of $\mathrm{d}W_t$ in (7.42) vanishes iff

$$\alpha_t \;=\; \partial_S g(t, S_t), \tag{7.43}$$

which is the delta hedge — identical to the no-dividend case. The drift coefficient $\mathcal{A}_t$ must vanish for $V_t \equiv 0$ to hold (since $V_0 = 0$ and $\mathrm{d}V_t$ must be predictable to avoid an arbitrage). Setting $\mathcal{A}_t = 0$ and substituting $\alpha_t = \partial_S g$,

$$\partial_S g\,(\mu+q)\,S_t \;+\; \beta_t\, r\, B_t \;-\; \partial_t g \;-\; \mu\, S_t\, \partial_S g \;-\; \tfrac{1}{2}\sigma^2 S_t^2\, \partial_{SS} g \;=\; 0. \tag{7.44}$$

The first and fourth terms combine:

$$\partial_S g\,(\mu+q)\,S_t \;-\; \mu\, S_t\, \partial_S g \;=\; q\, S_t\, \partial_S g. \tag{7.45}$$

Thus

$$\beta_t\, r\, B_t \;=\; \partial_t g \;+\; \tfrac{1}{2}\sigma^2 S_t^2\, \partial_{SS} g \;-\; q\, S_t\, \partial_S g. \tag{7.46}$$

8.5.4 7.4.4 Identifying β_t and closing the PDE

From $V_t = 0$ and $\alpha_t = \partial_S g$ we obtain

$$\beta_t\, B_t \;=\; g_t \;-\; \alpha_t\, S_t \;=\; g_t \;-\; S_t\, \partial_S g. \tag{7.47}$$

Substituting this into the left side of (7.46),

$$r\,(g_t - S_t\, \partial_S g) \;=\; \partial_t g \;+\; \tfrac{1}{2}\sigma^2 S_t^2\, \partial_{SS} g \;-\; q\, S_t\, \partial_S g. \tag{7.48}$$

Rearranging produces the *Black-Scholes-Merton PDE with continuous dividend yield*:

$$\boxed{\partial_t g \;+\; (r-q)\, S\, \partial_S g \;+\; \tfrac{1}{2}\sigma^2 S^2\, \partial_{SS} g \;=\; r\, g, \qquad g(T,S) = \varphi(S).} \tag{7.49}$$

Compare with the no-dividend PDE of Chapter 6: the only change is that the drift of the first-derivative term has become $r-q$ rather than r. The discount rate on the right-hand side is still the full r (the option itself does not pay a dividend), but the *risk-neutral* drift of the stock has been reduced by the dividend yield. Under the risk-neutral measure $\mathbb{Q}$, the stock now satisfies

$$\mathrm{d}S_t \;=\; (r-q)\, S_t\, \mathrm{d}t \;+\; \sigma\, S_t\, \mathrm{d}\widehat{W}_t, \tag{7.50}$$

so the *total-return* process $S_t e^{qt}$ (stock plus reinvested dividends) drifts at r, but the bare price process S_t drifts at $r-q$. The market is still risk-neutral on the *reinvestment* of dividends, but the quoted price lags the reinvested total return by the dividend yield.

A useful derivation consistency check is that the dividend-less case is recovered by setting $q = 0$: (7.49) reduces to the classical Black-Scholes PDE with drift r, and (7.50) reduces to the classical risk-neutral log-normal dynamics of Chapter 6. The dividend correction enters the drift linearly and nowhere else — the volatility structure, the discount rate on the RHS, and the terminal condition are all unchanged.

8.5.5 7.4.5 The Black-Scholes-Merton formula with dividends

The closed-form European call on a dividend-paying stock under (7.49) is

$$g(t,S) \;=\; S\,e^{-q(T-t)}\,\Phi(d_+) \;-\; K\,e^{-r(T-t)}\,\Phi(d_-), \tag{7.51}$$

with

$$d_\pm \;=\; \frac{\ln(S/K) \;+\; (r-q\pm\frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}. \tag{7.52}$$

Three features are worth noting.

First, the stock term is discounted by $e^{-q(T-t)}$ — the dividend-reduced forward value of the share at expiry. A long call holder does not receive the dividends that flow to the share during the option's life, so the effective "stock value" embedded in the option is only $Se^{-q(T-t)}$, not the full spot S.

Second, the moneyness drift $r-q$ replaces r inside $d_\pm$, consistent with (7.50). Under the risk-neutral measure the stock's forward price is $F_t = S_t e^{(r-q)(T-t)}$, and the $d_\pm$ factors record the log-moneyness relative to this forward, re-standardised by the volatility.

Third, the formula reduces to the classical Black-Scholes call when $q=0$, and it reduces to the Margrabe formula for an exchange option when the strike is itself a random asset and its "dividend yield" is its own risk-free return (Chapter 8).

For a European put, the analogous formula is

$$g(t,S) \;=\; K\,e^{-r(T-t)}\,\Phi(-d_-) \;-\; S\,e^{-q(T-t)}\,\Phi(-d_+), \tag{7.53}$$

with the same $d_\pm$ as the call — a direct consequence of put-call parity, which we state formally in §7.6.5.

A useful reinterpretation of (7.51)–(7.52) is via the *forward price* of the stock. Define

$$F_t \;=\; S_t\,e^{(r-q)(T-t)}, \tag{7.54}$$

the no-arbitrage forward price delivered at T. Then (7.51) can be rewritten

$$g(t,S) \;=\; e^{-r(T-t)}[F_t\,\Phi(d_+) \;-\; K\,\Phi(d_-)]\,, \tag{7.55}$$

which looks like the classical formula with the spot S_t replaced by the forward F_t and the stock's discount rate replaced by r. This is the beginning of the *forward-measure* viewpoint developed in Chapter 8: by pricing against the forward numeraire, the dividend yield effectively disappears from the formula, absorbed into the forward price. The elegance of the forward view is that it treats the dividend structure and the interest-rate structure as one combined drift, visible only through F_t.

8.5.6 7.4.6 Delta of a dividend-paying call

Differentiating (7.51) with respect to S gives

$$\Delta_t \;=\; e^{-q(T-t)}\,\Phi(d_+). \tag{7.56}$$

The dividend-induced factor $e^{-q(T-t)}$ is smaller than unity, so the delta of a dividend-paying call is *smaller* than the corresponding no-dividend call — a consequence of the stock losing value through the dividend stream that will never flow to the option holder. For continuously compounded dividends and long horizons, this discount factor can be substantial: a 2% dividend over five years shaves the delta by almost 10% relative to the naive no-dividend formula. Traders who fail to adjust for dividends consistently overhedge (carry too many shares), creating a systematic tracking error that drains P&L. Explicit handling of the dividend yield in the delta calculation is therefore a must for any book hedging dividend-paying equities, and the relevant discount factor is always $e^{-q(T-t)}$ applied to the stock side.

The gamma is similarly modified:

$$\Gamma_t \;=\; \frac{e^{-q(T-t)}\,\Phi'(d_+)}{S\,\sigma\sqrt{T-t}}, \tag{7.57}$$

and the vega is

$$\mathcal{V}_t \;=\; S\,e^{-q(T-t)}\,\sqrt{T-t}\,\Phi'(d_+). \tag{7.58}$$

(An equivalent form swaps the S-term for a K-term via the Black–Scholes "Φ' identity" $S\,e^{-q(T-t)}\,\Phi'(d_+) = K\,e^{-r(T-t)}\,\Phi'(d_-)$, giving $\mathcal{V}_t = K\,e^{-r(T-t)}\sqrt{T-t}\,\Phi'(d_-)$. Some references quote the K-form; both are numerically identical and readers cross-checking against other texts should recognise the equivalence.)

All Greeks carry the $e^{-q(T-t)}$ dividend-discount factor applied to the S-dependent part; the K-dependent discount factor remains $e^{-r(T-t)}$. This structural fact is worth remembering for quick sanity checks on any pricing library — if a library returns a dividend-stock delta that matches the no-dividend delta exactly, the library is wrong.

8.5.7 7.4.7 Forward reformulation

For the practitioner who wants to price options on dividend-paying stocks, the cleanest route is to work directly with the forward price. Define $F_t = S_t e^{(r-q)(T-t)}$ as in (7.54). Then F_t is a $\mathbb{Q}$-martingale (no drift under the forward's discount scheme), and the option price satisfies

$$g(t,F) \;=\; e^{-r(T-t)}\,\mathbb{E}_t^{\mathbb{Q}}[\varphi(F_T)], \tag{7.59}$$

which is just a discounted expectation of the terminal payoff expressed as a function of the terminal forward. For a vanilla call with $\varphi(F) = (F-K)_+$, the expectation is evaluated in closed form to give (7.55) — the familiar Black-76 formula that appears again in Chapters 8 (options on futures), 13 (bond options), and 15 (swaptions) with only the definition of "the forward" changing. The invariance is no accident: it is the change-of-numeraire argument of Chapter 5 doing its job. Any time we can identify a traded forward that is a martingale under its natural pricing measure, a Black-type formula is available; the dividend-adjusted equity option is just the first place we see this universality explicitly.

The forward reformulation also makes the self-financing argument shorter. Since F_t is a tradable asset (through a futures contract or a synthetic cash-and-carry on the dividend-paying stock), and since F_t is a $\mathbb{Q}$-martingale, the hedge $\alpha_t^F = \partial_F g$ in the forward reproduces the Black-76 PDE directly, with no drift term. We defer the detailed derivation to Chapter 8; here we simply note that the dividend yield's main pedagogical lesson is this: *dividends tilt the drift; they do not change the essential structure of the hedging problem.* Once the problem is expressed against the forward numeraire, the dividend yield disappears into the forward price, and we are back in an unadorned Black-76 world.

8.6 7.5 Transaction costs and the self-financing bookkeeping

Self-financing is one of those notions that seem trivially obvious until you try to write it down precisely, and then they turn out to hide subtle corner cases. Intuitively, self-financing says: "nothing flows into or out of the portfolio except through changes in the market prices of the assets we hold." No external cash injections, no dividend leaks, no funding gaps. But once we allow continuous trading, the act of adjusting the weights $(\alpha_t, \beta_t, \gamma_t)$ itself creates price-driven cash flows, and those need to be accounted for before the clean "$\mathrm{d}V = \alpha\,\mathrm{d}S + \beta\,\mathrm{d}M + \gamma\,\mathrm{d}h$" intuition is actually valid. Chapter 6 skated over this subtlety because the naive form sufficed for the Black-Scholes PDE derivation. In this section we revisit the self-financing condition with enough rigour to handle two real-world complications: transaction costs, which were ignored in Chapter 6, and the covariation terms that appear when the hedge weights are themselves stochastic.

8.6.1 7.5.1 Self-financing revisited — the covariation terms

For a portfolio $V_t = \alpha_t\, S_t + \beta_t\, M_t - g_t$ (long stock, long cash, short the claim we sold), Itô's product rule gives

$$\begin{aligned} \mathrm{d}V_t \;&= \alpha_t\, \mathrm{d}S_t \;+\; \beta_t\, \mathrm{d}M_t \;-\; \mathrm{d}g_t \\ &\quad +\; \mathrm{d}\alpha_t\, S_t \;+\; \mathrm{d}\beta_t\, M_t \;+\; \mathrm{d}[\alpha, S]_t \;+\; \mathrm{d}[\beta, M]_t. \end{aligned} \tag{7.60}$$

The "gain-from-trading" terms are $\alpha_t\, \mathrm{d}S_t + \beta_t\, \mathrm{d}M_t - \mathrm{d}g_t$; the remaining four terms are the trading cost from revising (α, β) and must vanish under self-financing.

The historical motivation for the self-financing concept traces back to the arbitrage-pricing literature of the 1970s and 1980s. Merton, Harrison, Pliska, and others needed a precise mathematical notion of a *replicating portfolio* that would let them invoke the no-arbitrage principle and derive pricing formulas. If a portfolio strategy (α_t, β_t) replicates a target payoff $\varphi(S_T)$ at time T, and the strategy requires no external cash input between $t = 0$ and $t = T$, then its time-zero cost must equal the fair price of φ. The "requires no external cash input" condition is exactly the self-financing condition, and its precise statement in continuous time is what we are developing here. Without the self-financing bookkeeping, the replicating-portfolio argument collapses because one can surreptitiously pump cash in or out and claim to have replicated anything.

There is an additional subtlety worth flagging. In the full continuous-time setting, α_t and β_t are themselves stochastic processes — they depend on the path of S_t up to time t. So the joint dynamics of (α_t, S_t) and (β_t, M_t) involve quadratic covariations that do not vanish and do not reduce to simple product-rule formulas. The covariation terms $\mathrm{d}[\alpha, S]$ and $\mathrm{d}[\beta, M]$ capture exactly this stochastic coupling. In the degenerate case where α is purely deterministic (say, a static buy-and-hold strategy), the covariations vanish and the naive product rule applies. But in any interesting dynamic-hedging context, α depends on S and the covariations are non-trivial.

8.6.2 7.5.2 Before-and-after rebalance accounting

Fix the instant t and split it into a before-rebalance leg using (α_t, β_t) and an after-rebalance leg using $(\alpha_{t+\Delta t}, \beta_{t+\Delta t})$. The before-leg value carried into $t + \Delta t$ is

$$\alpha_t\, S_{t+\Delta t} \;+\; \beta_t\, M_{t+\Delta t} \;-\; g_{t+\Delta t}, \tag{7.61}$$

and the after-leg value at $t + \Delta t$ is

$$\alpha_{t+\Delta t}\, S_{t+\Delta t} \;+\; \beta_{t+\Delta t}\, M_{t+\Delta t} \;-\; g_{t+\Delta t}. \tag{7.62}$$

The discrete value increment

$$\Delta V_t \;=\; \alpha_t\, \Delta S_t \;+\; \beta_t\, \Delta M_t \;-\; \Delta g_t \tag{7.63}$$

is valid only if before = after at $t + \Delta t$, i.e.

$$\Delta\alpha_t\; \underbrace{S_{t+\Delta t}}_{S_t + \Delta S_t} \;+ \Delta\beta_t\; \underbrace{M_{t+\Delta t}}_{M_t + \Delta M_t} \;=\; 0. \tag{7.64}$$

Expanding the product,

$$\Delta\alpha_t\, S_t \;+\; \Delta\alpha_t\, \Delta S_t \;+\; \Delta\beta_t\, M_t \;+\; \Delta\beta_t\, \Delta M_t \;=\; 0, \tag{7.65}$$

and passing to the continuous limit (where $\Delta\alpha\, \Delta S \to \mathrm{d}[\alpha, S]$, the quadratic covariation),

$$\boxed{\mathrm{d}\alpha_t\, S_t \;+\; \mathrm{d}[\alpha, S]_t \;+\; \mathrm{d}\beta_t\, M_t \;+\; \mathrm{d}[\beta, M]_t \;=\; 0.} \tag{7.66}$$

Equation (7.66) is the *precise* statement of self-financing — it annihilates exactly the four "extra" terms in (7.60), leaving only the gain-from-trading terms on the right-hand side. In continuous-time finance, naively writing "$\mathrm{d}V =$

$\alpha\,\mathrm{d}S + \beta\,\mathrm{d}M$" *assumes* (7.66) as a side condition; it is not automatic. The quadratic-variation terms $\mathrm{d}[\alpha, S]$ and $\mathrm{d}[\beta, M]$ are what distinguish genuine self-financing from a pretend one.

A worked example clarifies why the covariation terms cannot be ignored. Suppose α_t is itself a function of S_t — as it will be in any delta-hedging strategy, since $\alpha = \Delta(t, S_t)$. Then Itô gives $\mathrm{d}\alpha = \partial_t\Delta\,\mathrm{d}t + \partial_S\Delta\,\mathrm{d}S + \frac{1}{2}\partial_{SS}\Delta\,\mathrm{d}[S, S]$, and the covariation

$$\mathrm{d}[\alpha, S]_t \;=\; \partial_S\Delta\,\mathrm{d}[S, S]_t \;=\; \Gamma\,\sigma^2 S^2\,\mathrm{d}t \tag{7.67}$$

is non-zero. Ignoring this term would mis-count the trading cost of a continuous delta hedge by exactly the gamma-times-variance bleed that is at the heart of option theory. The whole Black-Scholes argument is really a careful bookkeeping of (7.66): the self-financing condition is what guarantees that the PDE governing g must have the form it does, with the $\frac{1}{2}\sigma^2 S^2\partial_{SS}$ term coming precisely from the quadratic-variation contribution.

In discrete rebalancing, the analogous statement is that each rebalance at time t_k must be executed at the *prevailing* price S_{t_k}, not at the time-t_{k-1} price. That is what makes the cross-product $\Delta\alpha_t\,\Delta S_t$ non-negligible. The recursion (7.26) encodes exactly this: the rebalance increment $(\alpha_{t_k} - \alpha_{t_{k-1}})$ is multiplied by the current price S_{t_k}, not by $S_{t_{k-1}}$. A desk that tried to book rebalance trades at yesterday's close would generate phantom P&L precisely equal to the ignored covariation term.

8.6.3 7.5.3 Proportional transaction costs

In the Black-Scholes idealisation, rebalances are free. In reality, each rebalance of size $\Delta\alpha_t$ at price $S_{t+\Delta t}$ costs a bid-ask spread $c\,|\Delta\alpha_t|\,S_{t+\Delta t}$ where c is the proportional spread. The self-financing condition picks up a non-vanishing negative term:

$$\mathrm{d}\alpha_t\,S_t \;+\; \mathrm{d}[\alpha, S]_t \;+\; \mathrm{d}\beta_t\,M_t \;+\; \mathrm{d}[\beta, M]_t \;=\; -c\,|\mathrm{d}\alpha_t|\,S_t \;-\; c\,|\mathrm{d}\gamma_t|\,h_t. \tag{7.68}$$

The absolute-value structure is the crucial feature: transaction costs are paid whether the rebalance is a buy or a sell, so the cost does not cancel across oscillating rebalances. This dissipation is what drives the *optimal-hedging-band* framework we take up in §7.5.5 — one does not want to rebalance for every tiny move because the cost $c\,|\mathrm{d}\alpha|$ accumulates quickly, so rebalances are deferred until the delta drift exceeds a threshold determined by cost and volatility.

To get a sense of the magnitude, suppose a desk is delta-hedging a single option over a year, with daily rebalances (so $N = 250$). For an ATM three-month call on a \$100 stock with $\sigma = 25\%$, the typical daily $|\Delta\alpha_t|$ is of order $\Gamma \cdot |\Delta S| \approx 0.032 \cdot (100 \cdot 0.25/\sqrt{250}) \approx 0.05$ shares per day. With a bid-ask spread of $c = 0.01\%$ (one basis point — typical for listed equity), the daily cost is $0.01\% \cdot 0.05 \cdot 100 = \$0.0005$ per option. Over 250 days, the cumulative cost is about \$0.12 per option, or roughly 1% of the option's premium — small but not negligible. For illiquid or wide-spread markets ($c = 0.1\%$ or higher), the cumulative cost can be several percent of the premium, which fundamentally changes the economics of the trade.

For a gamma-hedging leg using a second option with much wider bid-ask, the cost can easily exceed the gamma-reduction benefit. This is why delta-gamma hedging is not universal: on desks where the gamma-hedging option's bid-ask is large relative to the gamma-induced P&L variance, the pure-delta hedge is actually the *cheaper* strategy once costs are included. The delta-gamma hedge wins only when the auxiliary option's bid-ask is small enough that the cost of holding and rebalancing it is less than the variance reduction it provides.

8.6.4 7.5.4 Leland volatility adjustment

Leland's 1985 paper on option replication with transaction costs showed that in the presence of proportional costs, the optimal hedging strategy is no longer continuous rebalancing; it is a discrete strategy that rebalances on a fixed grid, and the pricing implication is that the option is priced *as if* the volatility were slightly *higher* than the actual volatility. The so-called "Leland volatility" adjustment is

$$\sigma^2_{\text{Leland}} \;=\; \sigma^2 \;+\; c\,\sigma\sqrt{\frac{8}{\pi\,\Delta t}}, \tag{7.69}$$

where Δt is the rebalance interval and c is the proportional bid-ask spread. The adjustment is small for small spreads but becomes significant in illiquid markets or during stress periods when spreads widen. For $c = 0.01\%$, daily rebalancing ($\Delta t = 1/250$), and $\sigma = 25\%$, the adjustment is of order $0.25 \cdot \sqrt{8/\pi} \cdot 10^{-4} \cdot \sqrt{250} \approx 0.006$, or about 0.6 vol points — small but real. For $c = 0.1\%$ (stressed), the adjustment jumps to 6 vol points — enough to materially change the dealer's price.

The Leland formula has a remarkable interpretation: *transaction costs are priced as a volatility premium.* The desk that hedges with a cost c effectively realises a higher volatility than the market-implied σ, so to be compensated for the cost it charges a higher premium for the option. This premium is mathematically equivalent to pricing the option with σ_{Leland} instead of σ. In other words, the "correct" price to a dealer accounts for its own hedging costs through an upward vol adjustment that depends on spread and rebalance frequency.

Modern extensions of Leland's result (Hoggard-Whalley-Wilmott, Kabanov, others) give optimal rebalancing bands that depend on time, spot, and proportional cost in closed form under specific preferences. These models produce prices and hedging strategies that outperform the pure Leland adjustment for large positions or illiquid underlyings.

The practical lesson from (7.69) is that *the Black-Scholes theoretical price is a starting point, not an end point.* Real dealer prices overlay several adjustments onto the textbook σ:

- a Leland-style transaction-cost adjustment;
- an XVA adjustment (CVA for counterparty credit, FVA for funding, KVA for regulatory capital, MVA for initial margin);
- a skew/smile adjustment (using a strike-specific implied vol rather than a single σ);
- a liquidity-premium adjustment for illiquid underlyings.

Each of these is a small correction in the ideal world and a large one in stressed conditions. The working quant learns to treat the Black-Scholes formula as the *reference*, not the *answer*.

8.6.5 7.5.5 Optimal rebalancing bands — a sketch

8.7 7.6 Put-call parity, deltas, and gammas

Throughout this section $C = C(t, S)$ and $P = P(t, S)$ denote the Black-Scholes European call and put prices on a stock paying a continuous dividend yield q, struck at K and expiring at T; we write $\tau = T - t$ for time to expiry. The one identity that ties them together is put-call parity,

$$C(t, S) \;-\; P(t, S) \;=\; S\, e^{-q\tau} \;-\; K\, e^{-r\tau}. \tag{7.59}$$

(7.59) is a static, *model-independent* identity: it is a no-arbitrage consequence of the trivial payoff decomposition $\max(S_T - K, 0) - \max(K - S_T, 0) = S_T - K$, discounted to t. Every Greek identity between call and put follows by differentiating (7.59) in the relevant variable — spot, volatility, time, or rate — and the right-hand side, which is a simple forward contract minus a zero-coupon bond, is transparent in each.

8.7.1 7.6.1 Put price curve

Applying parity to the call formula (7.51) gives the closed-form put price under continuous dividend yield:

$$P(t, S) \;=\; K\, e^{-r\tau}\, \Phi(-d_-) \;-\; S\, e^{-q\tau}\, \Phi(-d_+), \tag{7.60}$$

with $d_\pm$ as in (7.52). The put is convex in S, monotonically decreasing (a larger spot lowers the value of the right-to-sell), and tends to $Ke^{-r\tau} - Se^{-q\tau}$ deep in the money ($S \ll K$) and to 0 deep out of the money ($S \gg K$). At expiry the curve collapses to the hockey-stick payoff $\max(K - S_T, 0)$.

8.7.2 7.6.2 Call price curve

For reference alongside (7.60) the call price (7.51) is

$$C(t,S) \;=\; S\,e^{-q\tau}\,\Phi(d_+) \;-\; K\,e^{-r\tau}\,\Phi(d_-). \tag{7.61}$$

The call is convex in S, monotonically increasing, bounded below by the intrinsic value $(Se^{-q\tau} - Ke^{-r\tau})_+$ (the modified forward-minus-bond) and above by $Se^{-q\tau}$.

8.7.3 7.6.3 Delta curves

Differentiating (7.59) once in S and using $\partial_S(Se^{-q\tau}) = e^{-q\tau}$ gives the *parity identity for delta*,

$$\Delta^C \;-\; \Delta^P \;=\; e^{-q\tau}. \tag{7.62}$$

Combined with the call-delta formula $\Delta^C = e^{-q\tau}\Phi(d_+)$ from (7.56), this yields the put delta

$$\Delta^P \;=\; e^{-q\tau}\Phi(d_+) \;-\; e^{-q\tau} \;=\; -\,e^{-q\tau}\,\Phi(-d_+). \tag{7.63}$$

As expected, $\Delta^P \in [-e^{-q\tau}, 0]$: a long put has negative delta bounded below by $-e^{-q\tau}$ (ATM-deep-ITM put $\to$ near $-e^{-q\tau}$; OTM put $\to 0$). The two delta curves are vertical translates of each other by $e^{-q\tau}$ — a clean visual diagnostic.

8.7.4 7.6.4 Gamma curves and pin risk

Differentiating (7.62) once more in S, the right-hand side $e^{-q\tau}$ is constant in S, so

$$\Gamma^C \;=\; \Gamma^P \;=\; \frac{e^{-q\tau}\,\Phi'(d_+)}{S\,\sigma\sqrt{\tau}}. \tag{7.64}$$

This is the *gamma parity identity* previewed in §7.1 — calls and puts at the same strike share a single gamma curve. Consequently, a delta-neutral spread that is short one call and long one put at the same strike has *zero* gamma; the combination is the synthetic short forward of §7.6.6, whose value is linear in S.

Pin risk. As $\tau \downarrow 0$ the ATM gamma blows up like $1/\sqrt{\tau}$: from (7.64) at $S = Ke^{(r-q)\tau}$, $\Gamma \sim 1/(K\sigma\sqrt{2\pi\tau})$. A short ATM option on expiry day can whip between $\Delta = 0$ and $\Delta = \pm e^{-q\tau}$ as S crosses K — the delta becomes a step function and hedging it requires arbitrarily-large trades on arbitrarily-small spot moves. This is *pin risk*. Because $\Gamma^C = \Gamma^P$, it afflicts calls and puts symmetrically, and it is one of the main reasons market makers flatten ATM option exposure in the final hours before expiry.

8.7.5 7.6.5 Put-call parity and its Greek consequences

We now collect the full slate of parity consequences obtained by differentiating (7.59) in each model variable. All follow mechanically, and the right-hand side of each identity has a transparent forward-minus-bond interpretation.

Delta. ∂_S of (7.59): $\Delta^C - \Delta^P = e^{-q\tau}$. (7.62)

Gamma. ∂_{SS} of (7.59): $\Gamma^C = \Gamma^P$. (7.64) The right-hand side of parity is linear in S, so its second derivative vanishes.

Vega. Neither $Se^{-q\tau}$ nor $Ke^{-r\tau}$ depends on σ (parity is a *static* no-arb identity independent of volatility assumptions), so ∂_σ of (7.59) gives

$$\mathcal{V}^C \;=\; \mathcal{V}^P. \tag{7.65}$$

Calls and puts at the same strike share a single vega curve. A delta-hedged short call is vega-short; a delta-hedged short put is equally vega-short — which is why a straddle (long call + long put at the same strike) doubles vega, matching (7.65) with a factor of two.

Theta. ∂_t of (7.59), using $\partial_t e^{-q\tau} = qe^{-q\tau}$ and $\partial_t e^{-r\tau} = re^{-r\tau}$:

$$\Theta^C \;-\; \Theta^P \;=\; -\,q\,S\,e^{-q\tau} \;+\; r\,K\,e^{-r\tau}. \tag{7.66}$$

The difference in theta is *not* zero, because the forward and the bond on the right-hand side of parity each decay at their own rate. For $q = 0$ (no dividend), $\Theta^C - \Theta^P = rKe^{-r\tau} > 0$, so a call decays less unfavourably than a put of the same strike — the standard "rates help calls, hurt puts" intuition.

Rho. ∂_r of (7.59), noting that $Se^{-q\tau}$ is rate-independent while $Ke^{-r\tau}$ carries a $-\tau$:

$$\rho^C \; - \; \rho^P \; = \; \tau \, K \, e^{-r\tau}. \tag{7.67}$$

Calls have positive rho and puts negative rho; the gap between them is exactly the PV01-scale of the strike-bond.

Dividend sensitivity. ∂_q of (7.59): $\partial_q C - \partial_q P = -\tau S e^{-q\tau}$. Calls are hurt by higher dividends (stock trades ex-div), puts benefit symmetrically.

These identities are all *exact* and *model-free* in the sense that they follow from (7.59) without reference to any Black-Scholes or log-normal assumption. They therefore hold in Heston, in local-vol, in every complete arbitrage-free model — any model in which parity is imposed (which is every sensible model) respects (7.62)-(7.67) identically.

8.7.6 7.6.6 Synthetic positions

Rewriting parity as

$$C \; - \; P \; = \; S \, e^{-q\tau} \; - \; K \, e^{-r\tau} \tag{7.68}$$

says that a *long-call-short-put* portfolio at strike K is equivalent to a long position in $e^{-q\tau}$ shares of stock (equivalently, one forward contract delivered at T) funded by a short K-notional zero-coupon bond maturing at T. This is the *synthetic forward*: a way to construct directional equity exposure from options alone.

Symmetric rearrangements give the other useful syntheses: synthetic long stock is $Se^{-q\tau} = C - P + Ke^{-r\tau}$ (long call + short put + long bond); synthetic short stock is $-Se^{-q\tau} = P - C - Ke^{-r\tau}$. Traders use these replications when the physical stock is hard to borrow, when margin requirements differ between cash equities and options, or when the options market is more liquid than the stock at the relevant size. The Greek profile of a synthetic is dictated by (7.62)-(7.67): a synthetic long forward is delta-$e^{-q\tau}$, gamma-zero, vega-zero, and theta-$(-qSe^{-q\tau} + rKe^{-r\tau})$, which is exactly the Greek profile of the forward contract it replicates — a consistency check with nothing to tune.

8.8 7.7 Dollar Greeks — $\Gamma^{\$}$ and $\mathcal{V}^{\$}$

8.8.1 7.7.1 Dollar gamma

8.8.2 7.7.2 Dollar vega

8.8.3 7.7.3 Reading a desk-level Greek report

8.9 7.8 Key Takeaways

8.10 7.9 Reference Formulas Appendix

9 Chapter 8 — Forwards, Futures, and the Black PDE

The previous two chapters developed the theory of dynamic hedging for derivatives written on a spot asset: self-financing replication, the Black–Scholes PDE, the Greeks, and the handling of dividends. This chapter turns to a family of underlyings that look superficially similar but behave dramatically differently under the risk-neutral measure: *forwards* and *futures*. The payoff structure of a European call on, say, the December S&P 500 futures looks identical to a call on the spot index. But the pricing PDE loses its first-derivative drift term, the closed-form formula discards a factor of e^{-rT}, and the underlying dynamics under $\mathbb{Q}$ are a driftless diffusion rather than a GBM with drift r. The reason is the one structural feature that makes a futures contract not a stock: *entering a futures position costs nothing.*

This chapter consolidates three strands of material that appeared separately in the original draft. From the dynamic-hedging chapters we take the self-financing derivation of the Black PDE for options on futures, which is the cleanest structural argument. From the Feynman–Kac chapter we take the parallel martingale derivation, which serves as a cross-check and connects to the general machinery of CH04. From the futures-contracts chapter we take the concrete pricing examples (Bachelier and Ornstein–Uhlenbeck futures), the Margrabe exchange option — the tidiest use of a two-asset change of measure in the guide — and the convexity adjustment between futures and forwards under stochastic rates.

The chapter leans on CH05's canonical Girsanov derivation every time we change numeraire, and on CH06's BS PDE every time we compare the Black PDE to its spot-asset cousin. Feynman–Kac cross-checks reach back to CH04.

A note on the forward/futures distinction. A reader reading carefully will notice that we use the symbol $F_t(T)$ for both the forward price and the futures price, distinguishing them only by context or by the superscripts "fwd"/"fut" when ambiguity threatens. This matches the market convention: under deterministic rates the two are identical (§8.2) and the single symbol $F_t(T)$ is what quant traders write on the blackboard; under stochastic rates the two differ by the convexity adjustment of §8.11, and the distinction matters. Throughout §§8.3–8.10 we take rates as constant (or, equivalently, as deterministic) and treat $F_t(T)$ as either forward or futures depending on which pricing argument we are running. Only in §8.11 does the distinction become load-bearing.

Who this chapter is for. A reader who has finished CH06 knows the Black–Scholes PDE for an option on a spot asset and has internalised the self-financing apparatus. A reader who has finished CH05 knows Girsanov's theorem and how to change numeraire. This chapter is the natural consolidation of those two threads: the Black PDE is the BS PDE applied to a martingale underlying (the futures), and the Margrabe formula is a numeraire-change exercise in its purest form. The content is structural, not computational: once one understands *why* the Black PDE has no first-derivative drift term and *why* the Margrabe formula has no discount factor, every variant in the subsequent chapters — caplets, swaptions, bond options, exchange options on swaps — becomes a specialisation of the same pattern.

9.1 8.1 Forward Price from a Zero-Cost Claim

A *forward contract* signed at time t with delivery at T and strike K is the simplest possible derivative: an obligation to exchange one unit of the underlying for the pre-agreed cash amount K at maturity. Its terminal payoff is $S_T - K$ for the long side and $K - S_T$ for the short side. By the risk-neutral pricing rule developed in CH06, its value at time t is

$$\Pi_t \;=\; \mathbb{E}^{\mathbb{Q}}\big[\, e^{-r(T-t)}(S_T - K) \,\big|\, \mathcal{F}_t \big]\,. \tag{8.1}$$

At signing, the strike K is chosen so that $\Pi_t = 0$ — neither counterparty pays the other to enter. Setting (8.1) to zero and solving *under the constant-rate assumption*,

$$K \;=\; \mathbb{E}^{\mathbb{Q}}[\, S_T \mid \mathcal{F}_t \,] \;=\; S_t\, e^{r(T-t)}, \tag{8.2}$$

where the second equality uses that $S_t\, e^{-rt}$ is a $\mathbb{Q}$-martingale under the bank-account numeraire (CH05). We call this K the *forward price at time t for delivery at T* and denote it $F_t^{\mathrm{fwd}}(T)$:

$$F_t^{\mathrm{fwd}}(T) \;=\; S_t\, e^{r(T-t)}. \tag{8.3}$$

For a stock paying a continuous dividend yield δ the formula is $F_t^{\text{fwd}}(T) = S_t\, e^{(r-\delta)(T-t)}$, consistent with the dividend-paying generalisation in CH07. Geometrically, $F_t^{\text{fwd}}(T)$ is the *fair strike* of a just-signed forward contract — the price at which the two counterparties are indifferent to taking either side. As t advances toward T, $F_t^{\text{fwd}}(T) \to S_T$: the forward price collapses into the spot as the carry window shrinks.

An intuitive picture: plot $F_t^{\text{fwd}}(T)$ against S_t at a fixed date t. For the no-dividend case it is a straight line through the origin with slope $e^{r(T-t)} > 1$, sitting slightly above the diagonal $F = S$ that a trader might naively draw. The gap is the *cost of carry* — the interest saved by not having to hold the stock until T. For a dividend-paying stock the slope is $e^{(r-\delta)(T-t)}$, which is less than one if $\delta > r$ (the stock yields more than the bank account). In that regime the forward sits *below* the spot — the commodity-market phenomenon known as *backwardation*. Its opposite, $\delta < r$, is *contango*. The sign of $r - \delta$ drives substantial trading strategies in commodity derivatives, precious metals, and dividend-yield products.

The cash-and-carry story. Formula (8.3) can also be obtained by a pure no-arbitrage argument that does not invoke $\mathbb{Q}$ at all — worth retelling once because it makes the answer feel inevitable. At time t, a trader can either promise to buy the asset at T for the strike K (zero cost today, cash outflow K at T), or *synthesise* the same exposure by borrowing S_t dollars today and purchasing one share of S outright. At T the trader has one share either way; the only difference is the cash leg. To repay the loan requires $S_t/P(t,T)$ dollars at T, where $P(t,T) = e^{-r(T-t)}$ is the T-maturity discount bond. No-arbitrage forces the two routes to have the same cash outflow at T, giving

$$F_t^{\text{fwd}}(T) \;=\; \frac{S_t}{P(t,T)}. \tag{8.4}$$

Under constant rates $P(t,T) = e^{-r(T-t)}$ and (8.4) reduces to (8.3). The formula depends only on the spot and the discount bond — *not* on the statistical properties of S, not on volatility, not on anyone's view on where S is going. That independence is often underappreciated: the forward price is a cold no-arbitrage number, while the forward *value* of an existing contract with a stale strike reflects the market-implied drift of S.

Forward price term structure *Forward price term structure — cash-and-carry $F^{fwd}(t,T) = S_t/P(t,T)$ across maturities.*

Forward price vs forward value. The distinction matters. The *forward value* at time t of an existing contract with legacy strike K_0 is $\Pi_t = (F_t^{\text{fwd}}(T) - K_0)\, P(t,T)$, the discounted difference between the current fair strike and the contract's locked-in strike. The *forward price* $F_t^{\text{fwd}}(T)$ is the strike that would make a *newly signed* contract worth zero. Every day the market quotes the latter; the former is mark-to-market on a portfolio of contracts with various strikes. Pricing conventions differ on which number is meant by "the forward"; always check.

Under the T-forward measure $\mathbb{Q}^T$ with numeraire $P(t,T)$, the ratio $S_t/P(t,T) = F_t^{\text{fwd}}(T)$ is a martingale by construction — this is the general numeraire-pricing identity of CH05 applied with $A = P(\cdot,T)$ and $F = S$. That martingale property is why forwards admit clean Black-type formulas even when rates are stochastic: under $\mathbb{Q}^T$, the forward price is driftless, and log-Black or normal-Black closed-forms fall out immediately. We shall return to this point in CH13 where it is used to price bond options.

A put-call parity for forwards. One of the cleanest structural facts about forwards is their *linearity*: a long forward with strike K plus a short forward with strike K' is a position that pays $(S_T - K) - (S_T - K') = K' - K$ at maturity, independent of the spot. The present value of this difference is $(K' - K)\, P(t,T)$. Writing $\Pi_t(K)$ for the time-t value of a long forward with strike K, we therefore have

$$\Pi_t(K) \;-\; \Pi_t(K') \;=\; (K' - K)\, P(t,T),$$

which is a put-call parity for the forward contract alone. Combined with $\Pi_t(F_t^{\text{fwd}}(T)) = 0$ — the zero-value property of the newly signed forward — we recover $\Pi_t(K) = (F_t^{\text{fwd}}(T) - K)\, P(t,T)$, the formula quoted above (8.5). Every trader working with forwards reasons this way implicitly: the mark-to-market of an off-market forward is just the discounted difference between the current fair forward price and the contract's strike.

Cost of carry, generalised. The formula $F^{\text{fwd}} = S/P$ generalises to other asset classes by adjusting the carry leg:

- *Equities with continuous dividend yield δ.* Holding the stock earns $\delta\, dt$ in dividends per unit time; the cash-and-carry becomes $F_t^{\text{fwd}}(T) = S_t\, e^{(r-\delta)(T-t)}$.

- *Foreign exchange.* With domestic rate r and foreign rate r^f, holding the foreign currency earns r^f; the cash-and-carry becomes $F_t^{\text{fwd}}(T) = S_t\, e^{(r-r^f)(T-t)}$ (covered interest rate parity).
- *Storable commodity with convenience yield y and storage cost u.* Holding the physical commodity earns a convenience yield y (for commercial users) but costs storage at rate u; the forward is $F_t^{\text{fwd}}(T) = S_t\, e^{(r-y+u)(T-t)}$.

In each case the drift of the "cost of carry" under $\mathbb{Q}$ equals $r -$ (net yield), and the forward/futures price is the spot compounded at that drift. The formulas of §§8.5–8.11 apply uniformly to any of these cases with appropriate substitution.

9.2 8.2 Forwards vs. Futures — The Role of Daily Settlement

Forwards and futures are the two simplest delta-one derivatives — linear claims on the future value of an underlying — but they differ in one crucial operational respect: *when cash changes hands.* Forwards settle at maturity; futures settle daily. That small-looking difference has large consequences for pricing, hedging, and the shape of the backward PDE.

Both are *forward-looking* instruments: they specify a price for delivery of an asset at a future date. Both are *zero-sum* at initiation: one party is long, one is short, and no money changes hands upfront (for forwards) or only margin collateral does (for futures). Both have delta approximately equal to one: a \$1 move in the underlying moves the forward/futures P&L by approximately \$1. Yet the two instruments are priced by different PDEs, and the Black '76 futures-call formula differs from Black–Scholes in ways that matter when quoting option prices.

The forward contract. A forward is an obligation to buy/sell an asset S at a future time T at a pre-determined strike K. Its terminal payoff is $S_T - K$, so its time-t value is

$$\Pi_t \;=\; S_t \;-\; K\,P(t,T), \tag{8.5}$$

by exactly the two-piece argument of §8.1 (the asset leg is worth S_t today because S_t/M_t is a $\mathbb{Q}$-martingale; the cash leg $-K$ is deterministic and worth $-KP(t,T)$). Between signing and maturity the forward's mark-to-market value drifts as the spot moves: it climbs when S rises, falls when S drops, and accumulates an unrealised gain/loss on the holder's balance sheet. *No cash* is exchanged between counterparties during the life of the contract; the accumulated P&L is settled all at once at delivery.

The futures contract. A futures contract is also an obligation to transact at a future date, but changes in the futures price $F_t(T)$ are settled in cash *every day* into a margin account. Critically,

$$\text{value of a futures position} \;\equiv\; 0, \tag{8.6}$$

at every instant by design. Entering a futures position costs nothing; the contract carries no book value at any mark. Every infinitesimal gain $\mathrm{d}F_t$ is paid out in cash into the margin account immediately, so the holder's exposure at any instant is to the *next* daily price change only. The cumulative P&L of a one-lot long futures position from inception up to time t is simply $F_t(T) - F_0(T)$ — the running sum of daily settlements.

The zero-value feature is the single most important structural difference between forwards and futures. It is why futures can be traded in large notional with only a margin deposit, and why they sit at the centre of the clearing/settlement plumbing of modern exchanges. It is also why — as we are about to see — the pricing PDE for options on futures differs from Black–Scholes by the loss of one term.

Margin account illustration. The following table traces a one-lot long position through four days:

t	$F_t(T)$	ΔF	Margin cash flow	Cumulative margin
0	100	—	0	0
1	101	+1	+1	+1
2	103	+2	+2	+3
3	99	-4	-4	-1

Day 0: open long at $F_0 = 100$. No cash exchanged. Day 1: the futures rises to 101; the broker credits +1 to the margin account. Day 2: +2 more. Day 3: the futures drops to 99; the broker debits 4. The cumulative margin at day t is always $F_t - F_0$ — this is the *mark-to-market invariant* of a futures contract. Continuously, we write $\mathrm{d}F_t$ for the instantaneous cash flow into the margin account, and the cumulative P&L up to t is $\int_0^t \mathrm{d}F_s = F_t - F_0$.

Contrast with a forward. The forward holder's running P&L is also $F_t - F_0$ in present value, but the cash flow arrives only at maturity. Futures deliver the same economic P&L spread out in real time as cash. Under deterministic interest rates this distinction is invisible — the present value of a cash stream equals the present value of a single maturity payment if we can borrow and lend freely at r. Under *stochastic* rates, however, the daily cash flows of a futures position earn (or pay) interest at a rate that may be correlated with S itself, and this correlation opens a price gap between the futures quote and the forward quote that we compute in §8.11.

Operational summary. If a forward's mid-market implied strike moves from F_0 to $F_0 + 1$, the holder's forward mark-to-market moves from 0 to +1 (discounted). A futures holder earning the same +1 move gets +1 in cash deposited to their margin account *that day*, and the futures mark-to-market stays at 0. The two are economically equivalent modulo the interest earned on the margin cash — a difference that is small for short tenors but nontrivial for multi-year contracts.

9.3 8.3 The Futures Price as a $\mathbb{Q}$-Martingale

The economic consequence of daily settlement is a sharp mathematical fact: *the futures price itself is a martingale under the risk-neutral measure* $\mathbb{Q}$. The argument is almost tautological once one accepts the zero-value premise (8.6). Over any short interval $[t, t^+]$ the P&L of a one-lot long futures position is purely stochastic — it is the accumulated mark-to-market cash flow $F_{t^+} - F_t$ — and because the position costs nothing to open, no-arbitrage forces the expected P&L under $\mathbb{Q}$ to be zero:

$$\mathbb{E}^{\mathbb{Q}}[\, F_{t^+}(T) - F_t(T) \mid \mathcal{F}_t \,] \;=\; 0. \tag{8.7}$$

If the conditional expectation were non-zero, there would be a free option: a position that costs nothing to enter, has symmetric upside and downside from the holder's perspective, but carries a positive expected return under the pricing measure. The no-arbitrage principle rules out precisely such free options. Equivalently, for all $s \le t$,

$$\boxed{\mathbb{E}^{\mathbb{Q}}[F_t(T) \mid \mathcal{F}_s] \;=\; F_s(T)} \tag{8.8}$$

— the futures price is a $\mathbb{Q}$-martingale.

A trio of drifts. It is worth lining up the three kinds of asset that appear in this guide and noting the drift each of them takes under $\mathbb{Q}$:

Asset type	$\mathbb{Q}$-drift	Reason
Traded asset (e.g. a non-dividend stock)	$r\,X_t$	Discounted price X_t/M_t is a $\mathbb{Q}$-martingale; carry at r
Traded asset with yield δ	$(r-\delta)\,X_t$	Discounted total return (price + dividends) is a $\mathbb{Q}$-martingale
Futures quote $F_t(T)$	0	Position is costless; expected change under $\mathbb{Q}$ is zero
Non-traded variable (e.g. temperature)	$\mu_t^{\mathbb{Q}} - \sigma_t\,\lambda_t$	Needs a market price of risk λ_t (CH06)

The first three rows are pinned by replication arguments. The fourth requires additional market information — a market price of risk — because there is no hedgeable instrument that pins the drift. For the rest of this chapter, the futures price is the canonical example of row 3.

Forward prices are different. Contrast (8.8) with the analogous statement for a *forward* price $F_t^{\mathrm{fwd}}(T) = S_t/P(t,T)$. Under $\mathbb{Q}$ (bank-account numeraire), $F_t^{\mathrm{fwd}}(T)$ is *not* a martingale in general — both S_t and $1/P(t,T)$

drift, and their ratio drifts. The cleanest statement is instead that $F_t^{\text{fwd}}(T)$ is a $\mathbb{Q}^T$-martingale, where $\mathbb{Q}^T$ is the T-forward measure with numeraire $P(\cdot, T)$. This follows from the general numeraire identity of CH05: if A is a traded numeraire, every traded asset divided by A is a $\mathbb{Q}^A$-martingale, and $S_t/P(t,T)$ is precisely such a ratio. Futures and forwards therefore live in *different measures*: the futures price is naturally a $\mathbb{Q}$-martingale, the forward price is naturally a $\mathbb{Q}^T$-martingale. Under deterministic rates the two measures coincide (their Radon–Nikodym density collapses to 1), so futures and forwards coincide; under stochastic rates they diverge and the gap is the convexity adjustment of §8.11.

A subtle point about "the futures is undiscounted". It is tempting to read (8.8) as saying the futures *price* requires no discounting. That is correct. But an *option on a futures* — whose cash payoff settles at some maturity T_0 — is an ordinary traded derivative whose cash payoff must be discounted back to today in the usual way. The common pitfall is to conflate the two: the futures *quote* has no drift under $\mathbb{Q}$, but the option's *value* still carries the usual discount factor $e^{-r(T_0-t)}$. The Black '76 formula in §8.7 will make this explicit.

9.4 8.4 Dynamics of the Futures Price Under $\mathbb{Q}$

The martingale property (8.8) fixes the $\mathbb{Q}$-drift of $F_t(T)$ at zero, but leaves the volatility unspecified. Different market settings motivate different volatility parameterisations. We postulate a general form

$$\mathrm{d}F_t(T) \;=\; \sigma^F(t, F_t(T))\,\mathrm{d}\widehat{W}_t \qquad (\mathbb{Q}\text{-measure}), \tag{8.9}$$

with $\widehat{W}$ a $\mathbb{Q}$-Brownian motion. The specific functional form of $\sigma^F(t,F)$ is a modelling choice:

- **GBM.** $\sigma^F(t,F) = \sigma\, F$ yields the log-normal Black '76 model of §8.7, which is the market standard for equity-index futures, FX forwards, and commodity-futures implied-volatility quotation.
- **Bachelier (arithmetic).** $\sigma^F(t,F) = \sigma$ (constant) yields the arithmetic-Brownian model of §8.8, appropriate when F can go negative — interest-rate futures near the zero bound, or crude-oil futures during the April 2020 episode.
- **Samuelson / OU.** $\sigma^F(t,F) = \sigma\, e^{-\eta(T-t)}$ yields a mean-reverting arithmetic model where the vol loading *decays* as one looks further forward in the curve — matching the empirical fact that long-dated commodity futures move less in absolute terms than front-month ones (§8.9).

Derivation under constant rates. For the GBM case it is worth showing that (8.9) with $\sigma^F = \sigma F$ falls out of the spot dynamics plus the cost-of-carry identity. Suppose the spot satisfies

$$\frac{\mathrm{d}S_t}{S_t} \;=\; \mu\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t \qquad (\mathbb{P}), \tag{8.10}$$

and constant rates r apply. The forward/futures price is $F_t(T) = S_t\, e^{r(T-t)}$. By Itô on $f(t,S) = S\, e^{r(T-t)}$,

$$\mathrm{d}F_t(T) \;=\; e^{r(T-t)}\,\mathrm{d}S_t \;-\; r\,S_t\,e^{r(T-t)}\,\mathrm{d}t. \tag{8.11}$$

Expanding $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$ and using $F_t = S_t\, e^{r(T-t)}$,

$$\frac{\mathrm{d}F_t(T)}{F_t(T)} \;=\; (\mu - r)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t \qquad (\mathbb{P}). \tag{8.12}$$

Under the risk-neutral measure $\mathbb{Q}$, the stock's drift is $\mu = r$ (CH06, market-price-of-risk argument). Substituting,

$$\boxed{\frac{\mathrm{d}F_t(T)}{F_t(T)} \;=\; \sigma\,\mathrm{d}\widehat{W}_t \qquad (\mathbb{Q}).} \tag{8.13}$$

The drift $(\mu - r)\,\mathrm{d}t$ in (8.12) is exactly cancelled under $\mathbb{Q}$ by the measure change $\mathrm{d}W_t \to \mathrm{d}\widehat{W}_t + \lambda\,\mathrm{d}t$ with $\lambda = (\mu - r)/\sigma$ (Girsanov, CH05). The drift $(\mu - r)$ gets killed; the diffusion σ survives. This is the martingale property (8.8) made

explicit at the SDE level. Every drift term in the spot SDE — whether from physical-world mean-reversion, from the cost of carry r, or from a dividend yield — is absorbed into the futures quote or vanishes under $\mathbb{Q}$; what survives on the futures is only the diffusion term.

Case with dividends. If the stock pays a continuous yield δ, the spot drifts at $r-\delta$ under $\mathbb{Q}$, and the cost-of-carry formula becomes $F_t(T) = S_t\, e^{(r-\delta)(T-t)}$. Applying Itô's product rule to $F_t = S_t\, g(t)$ with $g(t) = e^{(r-\delta)(T-t)}$, $g'(t) = -(r-\delta)g(t)$:

$$\begin{aligned} \mathrm{d}F_t &= g(t)\,\mathrm{d}S_t + S_t\, g'(t)\,\mathrm{d}t \\ &= g(t)S_t\Big[(r-\delta)\,\mathrm{d}t + \sigma\,\mathrm{d}\widehat{W}_t\Big] - (r-\delta)g(t)S_t\,\mathrm{d}t \\ &= \sigma\, g(t)S_t\,\mathrm{d}\widehat{W}_t = \sigma F_t\,\mathrm{d}\widehat{W}_t, \end{aligned}$$

so

$$\boxed{\frac{\mathrm{d}F_t(T)}{F_t(T)} \;=\; \sigma\,\mathrm{d}\widehat{W}_t \qquad (\mathbb{Q})}.$$

The $(r-\delta)$-drift of the spot exactly cancels the time-decay of the carry factor: F is a $\mathbb{Q}$-martingale even with dividends, as the FTAP on futures (8.8) demands. This is the general principle — any continuous yield or cost-of-carry term that affects the spot under $\mathbb{Q}$ gets absorbed into the forward's deterministic drift and cancels, leaving pure diffusion on the forward/futures quote.

General dynamics. For modelling purposes, we often write (8.9) with a general state- and time-dependent volatility $\sigma^F(t,F)$:

$$\mathrm{d}F_t(T) \;=\; \sigma^F(t, F_t(T))\,\mathrm{d}\widehat{W}_t. \tag{8.14}$$

All the machinery of the next three sections works at this level of generality: the Black PDE, its Feynman–Kac representation, and the closed-forms for specific choices of σ^F are all instances of (8.14).

9.5 8.5 Options on Futures — The Black PDE

A European option written on a futures contract is a derivative whose terminal payoff is a function of the futures price at exercise:

$$g_{T_0} \;=\; \varphi\big(F_{T_0}(T)\big), \tag{8.15}$$

where T_0 is the option's exercise date and $T \geq T_0$ is the futures' delivery date. The two dates are frequently equal — for standardised exchange-listed products, option maturity coincides with the underlying futures' delivery — but they need not be. A typical example is a European call on the December S&P 500 futures: the payoff at exercise is $(F_{T_0}(T) - K)_+$, with both dates fixed by the contract specifications.

The pricing problem looks superficially identical to the BS problem of CH06, with the futures price taking the role of the underlying. But because the futures price is a martingale under $\mathbb{Q}$, the resulting PDE will differ in one subtle structural way: the first-derivative drift term that appears in BS will *vanish*. We derive this by the same self-financing argument used in CH06; the key accounting twist is that entering a futures position costs nothing.

9.5.1 8.5.1 Setup of the replicating portfolio

Let $g_t = g(t, F_t(T))$ denote the price of a European option on the futures. We build a replicating portfolio of futures and cash, short one unit of the option:

$$V_t \;=\; \alpha_t\, F_t(T) \;+\; \beta_t\, B_t \;-\; g_t, \tag{8.16}$$

where α_t is the number of *futures* contracts held, β_t is the number of units of the money-market account B_t (with $\mathrm{d}B_t = r\,B_t\,\mathrm{d}t$), and $V_0 = 0$. A subtle but crucial point about (8.16): because entering a futures contract *costs nothing*, the "value" that $\alpha_t\,F_t(T)$ contributes is not a capital outlay — it is the notional that accrues P&L through daily mark-to-market. We are not paying $\alpha_t\,F_t$ today to open the position; we are exposed to $\alpha_t\,\mathrm{d}F_t$ through the daily marks.

For the self-financing bookkeeping it is in fact cleaner to write the portfolio as

$$V_t \;=\; \alpha_t \cdot \underbrace{0}_{\text{futures value}} + \beta_t\,B_t \;-\; g_t, \tag{8.17}$$

explicitly recognising that the *position* in the futures is worth zero at every instant. The P&L from the futures comes entirely from the daily cash flow $\alpha_t\,\mathrm{d}F_t$, not from any change in a non-existent book value. Both (8.16) and (8.17) yield the same self-financing equation (as we are about to see); they differ only in whether the "value" $\alpha_t F_t$ is counted as non-zero on the balance sheet (it is not).

9.5.2 8.5.2 Self-financing increment

The self-financing wealth change is

$$\mathrm{d}V_t \;=\; \alpha_t\,\mathrm{d}F_t(T) \;+\; \beta_t\,\mathrm{d}B_t \;-\; \mathrm{d}g_t. \tag{8.18}$$

Note the *absence* of a "capital outlay" term for the futures. Compared to the stock-BS derivation of CH06, where the portfolio contained an $\alpha_t\,\mathrm{d}S_t$ term, (8.18) has an $\alpha_t\,\mathrm{d}F_t$ term with the same form. The difference is *bookkeeping*: in the stock case, $\alpha_t\,\mathrm{d}S_t$ represents the change in book value of shares held; in the futures case, $\alpha_t\,\mathrm{d}F_t$ represents the cash flow into the margin account, with no corresponding change in book value because the position is always worth zero. The formal self-financing identity has the same shape; only the interpretation of the term differs.

Expanding via Itô's lemma on $g(t, F)$ and substituting (8.14):

$$\mathrm{d}g_t \;=\; \left[\partial_t g \;+\; \tfrac{1}{2}(\sigma^F)^2\,\partial_{FF}g\right]\mathrm{d}t \;+\; \sigma^F\,\partial_F g\,\mathrm{d}\widehat{W}_t, \tag{8.19}$$

where we have used the fact that F_t has zero drift under $\mathbb{Q}$ so no $\mu^F\,\partial_F g$ term appears. Substituting (8.19) and (8.14) into (8.18),

$$\mathrm{d}V_t \;=\; \alpha_t\,\sigma^F\,\mathrm{d}\widehat{W}_t \;+\; \beta_t\,r\,B_t\,\mathrm{d}t \;-\; \left[\partial_t g + \tfrac{1}{2}(\sigma^F)^2\,\partial_{FF}g\right]\mathrm{d}t \;-\; \sigma^F\,\partial_F g\,\mathrm{d}\widehat{W}_t. \tag{8.20}$$

9.5.3 8.5.3 Eliminating the Brownian noise

The $\mathrm{d}\widehat{W}_t$ coefficient is $\sigma^F\,(\alpha_t - \partial_F g)$, which vanishes iff

$$\boxed{\alpha_t \;=\; \partial_F g(t, F_t(T)).} \tag{8.21}$$

This is the *futures-delta* hedge: hold $\partial_F g$ units of futures notional. Unlike the spot-delta of CH06, this is not a number of shares of S but a number of futures contracts, each representing notional exposure of one unit of $F_t(T)$. Operationally, for a call option with $\partial_F g = 0.6$, the hedge is to be long 0.6 futures contracts per option written — a practical trader's number, because futures are typically traded in integer multiples and 0.6 means "hold 3 futures for every 5 option contracts short". The "delta" reported on a trader's screen for an option on a futures is this $\partial_F g$.

9.5.4 8.5.4 No-arbitrage fixes the drift

With $\alpha_t = \partial_F g$ imposed, the $\mathrm{d}t$ coefficient in (8.20) is

$$\mathcal{A}_t \;=\; \beta_t\,r\,B_t \;-\; \partial_t g \;-\; \tfrac{1}{2}(\sigma^F)^2\,\partial_{FF}g. \tag{8.22}$$

No-arbitrage requires $V_t \equiv 0$ (we started with zero wealth and $\mathrm{d}V_t$ is now purely non-stochastic — any non-zero drift would generate a riskless profit). From $V_t = 0$ and the zero-value feature of the futures position, $\beta_t\,B_t = g_t$, so

$$\beta_t \, r \, B_t \;=\; r \, g_t. \tag{8.23}$$

Setting $\mathcal{A}_t = 0$ and substituting (8.23) gives the *Black PDE for options on futures*:

$$\boxed{\partial_t g \;+\; \tfrac{1}{2}\,(\sigma^F(t,F))^2\,\partial_{FF} g \;=\; r\,g, \qquad g(T_0,F) = \varphi(F).} \tag{8.24}$$

9.5.5 8.5.5 Structural observation

Compare (8.24) with the standard Black–Scholes PDE of CH06 for an option on a non-dividend stock:

$$\partial_t g \;+\; r\,S\,\partial_S g \;+\; \tfrac{1}{2}\,\sigma^2 S^2\,\partial_{SS} g \;=\; r\,g.$$

The first-derivative drift term $r\,S\,\partial_S g$ has *disappeared* in (8.24). This is the Black PDE's signature feature, and it is a direct consequence of the zero drift of F_t under $\mathbb{Q}$. One way to see this: the first-derivative drift in BS comes from the Feynman–Kac formula applied to $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}\widehat{W}_t$ — the Feynman–Kac generator of S has a drift piece $r\,S\,\partial_S$. When the underlying is a futures with zero drift under $\mathbb{Q}$, the generator has no drift piece at all, and the PDE picks up only the second-derivative diffusion term.

The $r\,g$ term on the right-hand side is not eliminated — this is the *discount* term, reflecting that the *option's* cash value settles in dollars at T_0 and must be discounted at r. The futures price itself is undiscounted; the option on it is a cash-settled claim that is.

For constant volatility $\sigma^F(t,F) = \sigma\,F$, (8.24) becomes

$$\partial_t g \;+\; \tfrac{1}{2}\,\sigma^2\,F^2\,\partial_{FF} g \;=\; r\,g, \qquad g(T_0,F) = \varphi(F), \tag{8.25}$$

which is the PDE that the Black '76 formula of §8.7 solves.

9.5.6 8.5.6 Market-price-of-risk reading

It is worth recasting the derivation through the market-price-of-risk language of CH06, because the resulting identity makes the "no-drift for futures" feature completely transparent. Recall from CH06 that for any traded asset X_t following $\mathrm{d}X_t = \mu_t^X\,X_t\,\mathrm{d}t + \sigma_t^X\,X_t\,\mathrm{d}W_t$, no-arbitrage pins the market price of risk $\lambda_t = (\mu_t^X - r)/\sigma_t^X$ to a common value across all traded assets on the same Brownian. An option $g_t = g(t,F_t)$ on the futures has Itô dynamics

$$\mathrm{d}g_t \;=\; \mu_t^g\,\mathrm{d}t \;+\; \sigma_t^g\,\mathrm{d}W_t, \qquad \mu_t^g = \partial_t g + \mu_t^F\,\partial_F g + \tfrac{1}{2}(\sigma_t^F)^2\,\partial_{FF} g, \quad \sigma_t^g = \sigma_t^F\,\partial_F g.$$

The market-price-of-risk identity relating the claim to the futures is

$$\frac{\mu_t^F}{\sigma_t^F} \;=\; \frac{\mu_t^g - r\,g_t}{\sigma_t^g}. \tag{8.25a}$$

For a *traded asset* underlying the BS PDE, both sides equal $\lambda_t = (\mu - r)/\sigma$ with the usual cost-of-carry r. For the *futures*, the left-hand side is μ_t^F/σ_t^F — and under $\mathbb{Q}$, $\mu_t^F = 0$, so the left-hand side is zero. The identity (8.25a) therefore forces $\mu_t^g = r\,g_t$, which is precisely the drift-equals-rg condition that produces the Black PDE (8.24). The missing first-derivative drift term in the Black PDE is thus the direct algebraic consequence of the zero drift of the futures under $\mathbb{Q}$.

Reading for the practitioner. The market-price-of-risk of a futures is *zero* under the futures' own pricing measure. Futures contracts carry no cost of carry (entering is free), so the "drift minus carry" of a futures is just its drift. Under $\mathbb{Q}$, which pins the drift to zero, the Sharpe ratio of a futures is identically zero. Claims priced off the futures inherit cost-of-carry $r\,g$ (the claim itself has dollar value and can be borrowed against at rate r), so their drift under $\mathbb{Q}$ is $\mu^g = r\,g$ — the "$r\,g$ discount" term in the Black PDE.

9.6 8.6 Feynman–Kac Cross-Check

The derivation of the Black PDE in §8.5 proceeded via the self-financing hedge argument — a structural route that makes the zero-outlay feature of the futures explicit. An independent cross-check comes from applying the general Feynman–Kac theorem of CH04 directly to the futures dynamics.

Let $g(t, F)$ satisfy the Black PDE (8.24). By the Feynman–Kac theorem of CH04 applied to the SDE (8.14) with zero drift, the PDE

$$\partial_t g \;+\; \tfrac{1}{2}\,(\sigma^F)^2\,\partial_{FF} g \;=\; r\,g, \qquad g(T_0, F) = \varphi(F),$$

has the stochastic representation

$$\boxed{g(t, F) \;=\; \mathbb{E}^{\mathbb{Q}}\Big[\, e^{-r(T_0-t)}\,\varphi(F_{T_0}(T)) \;\Big|\; F_t(T) = F \,\Big]\,,} \tag{8.26}$$

where F_u follows the driftless SDE (8.14) under $\mathbb{Q}$. This is the direct analogue of the CH04 stock-BS representation, with two differences:

- the drift of the underlying inside the expectation is *zero* (not r), because the futures is a $\mathbb{Q}$-martingale;
- the discount factor $e^{-r(T_0-t)}$ is present, because the option's *cash payoff* still settles at T_0 and must be discounted back to today.

Sanity check against the hedging derivation. The two derivations yield the same PDE (8.24), so the same formula (8.26) solves both. This is the CH06 CH04 duality applied in the futures setting: the PDE can be derived *either* by a self-financing replication argument *or* by a martingale-representation argument, and the two routes must agree because no-arbitrage and risk-neutral pricing are the same thing under the hood.

An alternative "forward-measure" reading. Under a measure change that absorbs the $e^{-r(T_0-t)}$ factor — specifically, moving from $\mathbb{Q}$ to the T_0-forward measure $\mathbb{Q}^{T_0}$ with numeraire $P(\cdot, T_0)$, the T_0-maturity zero-coupon bond — the representation (8.26) simplifies to

$$\frac{g(t, F)}{P(t, T_0)} \;=\; \mathbb{E}^{\mathbb{Q}^{T_0}}\Big[\,\varphi(F_{T_0}(T)) \,\Big|\, F_t(T) = F\,\Big]\,. \tag{8.27}$$

Under constant rates $P(t, T_0) = e^{-r(T_0-t)}$, and (8.27) reduces to (8.26). Under stochastic rates, $\mathbb{Q}$ and $\mathbb{Q}^{T_0}$ are genuinely different measures, and the convexity adjustment of §8.11 becomes visible. For the Black PDE under constant rates — the case we focus on in §§8.7–8.9 — the two expressions are identical. For the interest-rate applications of CH13, (8.27) is the cleaner starting point.

Integrating the SDE. From (8.14) with $\sigma^F(t, F) = \sigma F$ (constant relative volatility), a straightforward Itô integration gives

$$F_{T_0}(T) \;=\; F_t(T)\,\exp\!\Big\{-\tfrac{1}{2}\sigma^2(T_0 - t) + \sigma\,(\widehat{W}_{T_0} - \widehat{W}_t)\Big\}\,, \tag{8.28}$$

so $F_{T_0}(T)$ is log-normal under $\mathbb{Q}$ with log-mean $\ln F_t(T) - \frac{1}{2}\sigma^2(T_0 - t)$ and log-variance $\sigma^2(T_0 - t)$. This is the log-normal distribution that the Black '76 closed-form of §8.7 integrates against.

Arithmetic case. For $\sigma^F(t, F) = \sigma$ (Bachelier, constant absolute vol), (8.14) integrates to

$$F_{T_0}(T) \;=\; F_t(T) \;+\; \sigma\,(\widehat{W}_{T_0} - \widehat{W}_t), \tag{8.29}$$

so $F_{T_0}(T)$ is normally distributed under $\mathbb{Q}$ with mean $F_t(T)$ and variance $\sigma^2(T_0 - t)$. Note the positivity of the mean: the futures price itself is the conditional expectation of its terminal value, which is a restatement of the martingale property (8.8).

In both cases — GBM and Bachelier — the distribution of $F_{T_0}(T)$ under $\mathbb{Q}$ is centred at the current futures quote $F_t(T)$, which is exactly what the martingale property requires. The only difference between the two cases is whether the distribution is lognormal (GBM) or normal (Bachelier).

9.7 8.7 The Black '76 Closed Form

For the constant-volatility GBM case $\sigma^F(t,F) = \sigma F$ with payoff $\varphi(F) = (F-K)_+$, the PDE (8.25) admits the celebrated *Black '76* closed-form solution:

$$\boxed{g(t,F) \;=\; e^{-r(T_0-t)}\big[\,F\,\Phi(d_+) \;-\; K\,\Phi(d_-)\,\big],} \tag{8.30}$$

with

$$d_\pm \;=\; \frac{\ln(F/K) \;\pm\; \frac{1}{2}\,\sigma^2\,(T_0-t)}{\sigma\,\sqrt{T_0-t}}. \tag{8.31}$$

Derivation via Feynman–Kac. Starting from (8.26) with $\varphi(F) = (F-K)_+$ and the log-normal law (8.28):

$$g(t,F) \;=\; e^{-r(T_0-t)}\,\mathbb{E}^{\mathbb{Q}}\Big[\,(F_{T_0}(T)-K)_+ \mid F_t(T)=F\,\Big].$$

The expectation is the classical log-normal "call-option" integral, and the log-variance is $\sigma^2(T_0-t)$ with log-mean $\ln F - \frac{1}{2}\sigma^2(T_0-t)$. A direct Gaussian computation gives

$$\mathbb{E}^{\mathbb{Q}}\Big[(F_{T_0}-K)_+ \mid F_t=F\Big] \;=\; F\,\Phi(d_+) \;-\; K\,\Phi(d_-),$$

and multiplying by the discount $e^{-r(T_0-t)}$ yields (8.30).

Comparison to Black–Scholes. The standard BS formula on a non-dividend stock is

$$C^{\mathrm{BS}}(S,K,r,\sigma,\tau) \;=\; S\,\Phi(d_1) \;-\; K\,e^{-r\tau}\,\Phi(d_2),$$

with $d_{1,2} = [\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau]/(\sigma\sqrt{\tau})$. The Black '76 formula (8.30) differs in two readable ways:

	Black–Scholes (spot)	Black '76 (futures)
"Moneyness" in $d_\pm$	$\ln(S/K) + r\tau$	$\ln(F/K)$ only
Coefficient on $\Phi(d_+)$	S	$e^{-r\tau}\,F$
Coefficient on $\Phi(d_-)$	$K\,e^{-r\tau}$	$e^{-r\tau}\,K$

The first change reflects the absence of a first-derivative drift term in the Black PDE — the "cost of carry" $r\tau$ that shifts the moneyness in BS does not appear in Black '76 because the underlying is already a martingale. The second change reflects that the futures quote F is itself undiscounted; to state the option price as a dollar amount today, we simply discount the whole expression by $e^{-r\tau}$.

Three observations on the formula.

First, (8.30) is the classical BS formula *with the spot replaced by the forward* and *with no drift shift in* $d_\pm$. The formula is strictly simpler than the dividend-paying BS formula of CH07: the dividend yield does not appear at all because it has already been absorbed into the definition of the forward/futures price. Third, the formula specialises cleanly to the constant-dividend-stock case: plugging $F_t(T) = S_t\,e^{(r-\delta)(T-t)}$ into (8.30) and simplifying recovers the dividend-BS formula of CH07 exactly. Black '76 is the natural "forward-measure" simplification of the dividend-BS formula.

Second, a common pitfall: the discount factor in (8.30) is $e^{-r(T_0-t)}$, not $e^{-r(T-t)}$. The option matures at T_0 (potentially earlier than the futures' delivery at T), and the discount is from today to *option maturity*, not to futures delivery. For standardised exchange-listed products the two dates coincide — but for bespoke derivatives written on an exchange-listed futures (e.g. a mid-curve option on an interest-rate future) they can differ by several months. Always check which date enters the discount.

Third, the Black '76 delta — the sensitivity of the option price to the futures quote — is

$$\partial_F g \;=\; e^{-r(T_0-t)}\,\Phi(d_+). \tag{8.32}$$

Compare to the Black–Scholes spot delta $\Phi(d_1)$: the Black '76 delta is *discounted*. Operationally this means that to hedge one option written, one holds fewer than $\Phi(d_+)$ futures contracts — specifically, one holds $e^{-r(T_0-t)}\,\Phi(d_+)$. For short-dated options the discount is small and the two deltas differ only at the third decimal; for multi-year options the discount can be noticeable.

Black '76 Greeks. Differentiating (8.30) yields a complete table of Greeks. Writing $\tau = T_0 - t$ and ϕ for the standard-normal PDF:

$$\begin{aligned}
\Delta \;&=\; \partial_F g \;=\; e^{-r\tau}\,\Phi(d_+),\\
\Gamma \;&=\; \partial_{FF} g \;=\; \frac{e^{-r\tau}\,\phi(d_+)}{F\,\sigma\,\sqrt{\tau}},\\
\text{Vega} \;&=\; \partial_\sigma g \;=\; e^{-r\tau}\,F\,\phi(d_+)\,\sqrt{\tau},\\
\Theta \;&=\; \partial_t g \;=\; -\frac{e^{-r\tau}\,F\,\phi(d_+)\,\sigma}{2\sqrt{\tau}} \;+\; r\,e^{-r\tau}\,[F\,\Phi(d_+) - K\,\Phi(d_-)],\\
\rho \;&=\; \partial_r g \;=\; -\tau\,g(t,F).
\end{aligned}$$

Several features are worth flagging. First, the gamma has an extra discount factor $e^{-r\tau}$ compared to BS — again a consequence of the "futures is a martingale, not a GBM" feature. Second, the rho is *negative* for a call: an increase in rates lowers the call price (because the cash settlement at expiry is discounted more heavily), whereas for a BS spot call on a non-dividend stock, rho is positive. The sign flip is one of the cleaner diagnostic tests for whether a pricing engine is treating a contract as a spot-based or futures-based product. Third, the vega formula is similar to BS, but the F sitting outside is the futures quote, not the spot — so a Black '76 vega of 0.5 on a \$100 futures differs from a BS vega of 0.5 on a \$100 spot only by the overall discount factor and the substitution $S \to F$.

A worked numerical check. Take $F = 100$, $K = 100$, $\sigma = 20\%$, $\tau = 0.25$ years, $r = 5\%$. Then $d_+ = \frac{1}{2}\sigma\sqrt{\tau} = 0.05$ and $d_- = -d_+ = -0.05$. $\Phi(0.05) \approx 0.5199$, $\Phi(-0.05) \approx 0.4801$. The Black '76 call price is

$$g \;=\; e^{-0.05\cdot 0.25}\,[100\cdot 0.5199 - 100\cdot 0.4801] \;\approx\; 0.9876\cdot 3.98 \;\approx\; 3.93.$$

The analogous BS call on a \$100 spot with the same volatility and a \$100 strike at the same rate is $S\,\Phi(d_1) - K\,e^{-r\tau}\,\Phi(d_2)$ with $d_{1,2} = (r \pm \frac{1}{2}\sigma^2)\tau/(\sigma\sqrt{\tau})$, giving $d_1 \approx 0.175$, $d_2 \approx 0.075$, and a price near 4.62. The BS call is higher than the Black '76 call because the spot drift under $\mathbb{Q}$ is $r > 0$ — the underlying grows into the strike — while the futures is a martingale and does not. The difference is exactly the first-derivative drift term cumulating over τ.

Why the Black formula is so ubiquitous. The Black '76 formula appears in a surprising number of places beyond literal options on exchange-traded futures. It is the market-standard quotation for caps, floors, and swaptions on interest-rate forwards (CH15); for options on commodity forwards; and — via a well-known convention — for at-the-money options on forward-looking yield curves. The underlying reason is always the same: whenever the underlying is (or can be reparameterised as) a martingale under *some* appropriately chosen measure, the pricing PDE loses its first-derivative drift term and the closed-form reduces to (8.30). In the LIBOR/SOFR caplet setting of CH15, the forward rate $L(t, T_1, T_2)$ is a martingale under the T_2-forward measure, and the caplet price is exactly the Black '76 formula with F denoting the forward rate and σ its forward-rate volatility. The technical device of changing numeraire — formalised in CH05 — generalises the futures-price argument to these other martingale underlyings, and every such numeraire change collapses the relevant pricing PDE to a Black-style form.

Black '76 drift-cancel schematic *Drift-cancel schematic — the first-derivative drift term in the Black PDE vanishes because the futures is a $\mathbb{Q}$-martingale, leaving only the second-derivative diffusion.*

9.8 8.8 Example — Bachelier Futures and the Arithmetic Binary

The Bachelier (arithmetic Brownian motion) specification is the simplest possible futures model: constant *absolute* volatility, no drift. Although the log-normal Black '76 model of §8.7 dominates textbook presentations, the arithmetic model is the natural choice for underlyings that can go negative — short-rate futures near the zero bound, or, famously, crude-oil futures during the April 2020 negative-price episode. It also isolates the features of the martingale pricing approach from the mechanics of log-normal integration, which is pedagogically useful.

Take

$$\mathrm{d}F_t \;=\; \sigma\,\mathrm{d}\widehat{W}_t \qquad (\mathbb{Q}), \tag{8.33}$$

with constant absolute volatility σ, and consider a *binary call*

$$\varphi(F) \;=\; \mathbf{1}_{\{F>K\}}$$

— the claim pays 1 unit if the futures is above K at option expiry T_0, zero otherwise. For simplicity we take $r = 0$ in this example so that no discount factor clutters the formula; the general $r \neq 0$ result multiplies the answer by $e^{-r(T_0-t)}$.

9.8.1 8.8.1 Terminal distribution

From (8.33),

$$F_{T_0} - F_t \;=\; \sigma\,(\widehat{W}_{T_0} - \widehat{W}_t), \qquad \widehat{W}_{T_0} - \widehat{W}_t \overset{d}{=} \mathcal{N}(0, T_0 - t), \tag{8.34}$$

so F_{T_0} is conditionally normal with mean F_t (the martingale property) and variance $\sigma^2\,(T_0 - t)$.

9.8.2 8.8.2 Pricing

By Feynman–Kac (8.26) with $r = 0$,

$$h(t, F) \;=\; \mathbb{E}^{\mathbb{Q}}_{t,F}\Big[\,\mathbf{1}_{\{F_{T_0}>K\}}\Big] \;=\; \mathbb{Q}_{t,F}(F_{T_0} > K). \tag{8.35}$$

Writing $F_{T_0} - F_t = \sigma\,(\widehat{W}_{T_0} - \widehat{W}_t)$ and letting $Z \sim \mathcal{N}(0, 1)$,

$$h(t, F) \;=\; \mathbb{Q}\left(Z \;>\; \frac{K - F}{\sigma\,\sqrt{T_0 - t}}\right) \;=\; \Phi\left(\frac{F - K}{\sigma\,\sqrt{T_0 - t}}\right), \tag{8.36}$$

with Φ the standard-normal CDF. (For the historically-rooted version using $\sigma = 2$ that appears in the draft, one simply substitutes $\sigma = 2$ into (8.36).)

9.8.3 8.8.3 Reading the answer

The shape of (8.36) rewards a slow look. The numerator $F - K$ measures how far "in the money" the digital currently is in dollar terms; the denominator $\sigma\,\sqrt{T_0 - t}$ is the standard deviation of futures price moves over the remaining time. The ratio is a *z-score* — the number of standard deviations by which the current futures quote exceeds the strike — and the price of the digital is simply the probability that a standard normal exceeds minus that z-score.

Three limits are worth noting:

- **Deep in the money.** $F \gg K$: the z-score is large and positive, $\Phi \to 1$, and the digital is worth (correctly) almost its full notional of 1.
- **Deep out of the money.** $F \ll K$: the z-score is large and negative, $\Phi \to 0$, and the digital is worth almost nothing.

- **At the money.** $F = K$: $\Phi(0) = 1/2$ regardless of volatility — a feature unique to the arithmetic-Brownian model, where payoff symmetry around the mean produces an exact fifty-fifty digital. In the log-normal Black '76 model, the ATM digital is not exactly 1/2 because of the asymmetric log-normal distribution; the two models agree only in the small-vol limit.

Discounting. If $r \neq 0$, the digital payoff of 1 dollar at T_0 must be discounted back to today, giving $h(t, F) = e^{-r(T_0-t)}\,\Phi((F-K)/(\sigma\sqrt{T_0-t}))$. The distribution of F_{T_0} is *not* affected by r because F is a $\mathbb{Q}$-martingale regardless of the interest rate; the discount only touches the dollar-valued payoff.

A limiting check. As $T_0 \to t$, the argument $(F-K)/(\sigma\sqrt{T_0-t}) \to \pm\infty$ depending on the sign of $F-K$, and Φ collapses to the indicator $\mathbf{1}_{\{F>K\}}$ — exactly the terminal payoff. As $T_0 \to \infty$, the argument goes to zero and $\Phi(0) = 1/2$ — a binary call infinitely far from expiry has value 1/2 because the distribution of F_{T_0} spreads uniformly over positive and negative half-lines. Both limits confirm the formula's consistency.

9.9 8.9 Example — Ornstein–Uhlenbeck Futures and the Samuelson Effect

The Bachelier model is too simple to describe real commodity or interest-rate futures markets. Empirically, the *volatility* of the futures curve is not flat across maturities: long-dated crude futures move much less in absolute terms than front-month contracts; the same is true of rates futures, where a hike priced into the December contract barely perturbs the contract two years out. The empirical name for this pattern is the *Samuelson effect*, and an OU-type model with a time-decaying volatility loading $\sigma\, e^{-\eta(T-t)}$ is the cleanest way to reproduce it.

9.9.1 8.9.1 Specification

Under the *physical* measure, the futures is often specified as a mean-reverting diffusion:

$$\mathrm{d}F_t(T) \;=\; \kappa\,(\theta - F_t(T))\,\mathrm{d}t \;+\; \sigma\, e^{-\eta(T-t)}\,\mathrm{d}W_t \qquad (\mathbb{P}). \tag{8.37}$$

But under the risk-neutral measure $\mathbb{Q}$, the martingale property (8.8) forces the drift to vanish. Girsanov's theorem (CH05) tells us there is always a measure equivalent to $\mathbb{P}$ that kills any given drift, and the martingale property of a costless position forces that measure to be the pricing measure $\mathbb{Q}$. Accordingly, under $\mathbb{Q}$,

$$\mathrm{d}F_t(T) \;=\; \sigma\, e^{-\eta(T-t)}\,\mathrm{d}\widehat{W}_t. \tag{8.38}$$

The mean-reversion drift *was* present in the physical-world dynamics but *does not need to appear* under $\mathbb{Q}$: the information about where the futures is heading is already encoded in the initial forward curve $F_0(T)$ and in the volatility decay $e^{-\eta(T-t)}$. Said differently, once the market has priced a forward curve that reflects mean-reversion, the curve itself carries the information, and the $\mathbb{Q}$-dynamics of each individual futures quote can be driftless without loss.

9.9.2 8.9.2 The Samuelson decay

The economic content of the loading $e^{-\eta(T-t)}$ is the following. A shock to the spot propagates through to futures of every maturity, but because the spot itself is mean-reverting, shocks decay before they reach delivery. A contract with delivery T therefore "sees" only the fraction $e^{-\eta(T-t)}$ of any time-t shock — the rest has time to mean-revert back to θ before delivery. Long-dated contracts sit on the flat part of the decay curve with a small vol loading; the front month sits on the steep part. As calendar time t advances and the contract approaches its own delivery T, the exponent $\eta(T-t)$ shrinks toward zero, the vol loading grows toward σ, and the futures quote becomes more and more responsive to spot shocks. That is the Samuelson effect made quantitative.

Samuelson OU futures *Samuelson decay — the vol loading $\sigma e^{-\eta(T-t)}$ drops with tenor, so far-dated contracts respond less to same-size shocks than near contracts.*

9.9.3 8.9.3 Terminal distribution

From (8.38),

$$F_{T_0}(T) - F_t(T) \;=\; \int_t^{T_0} \sigma\, e^{-\eta(T-u)}\, \mathrm{d}\widehat{W}_u. \tag{8.39}$$

The integrand is deterministic, so the right-hand side is Gaussian with mean zero. By Itô's isometry, the variance is

$$\Sigma_t^2 \;=\; \int_t^{T_0} \sigma^2\, e^{-2\eta(T-u)}\, \mathrm{d}u \;=\; \sigma^2\, \frac{e^{-2\eta(T-T_0)} - e^{-2\eta(T-t)}}{2\eta}. \tag{8.40}$$

Hence

$$F_{T_0}(T) - F_t(T) \;\sim_{\mathbb{Q}}\; \mathcal{N}(0, \Sigma_t^2), \tag{8.41}$$

conditional on $\mathcal{F}_t$.

Reading (8.40). Formula (8.40) is worth staring at. It encodes the total variance accumulated by the futures between t (now) and T_0 (option expiry), given that delivery is at T. There are three dials:

- σ is the peak instantaneous vol, attained at delivery ($T = u$);
- η is the speed at which long-dated vols decay below that peak;
- the $T - t$ vs $T - T_0$ spacing controls where on the decay curve we are integrating.

Two useful limits:

- $\eta \to 0$**.** Expanding the exponentials, $\Sigma_t^2 \to \sigma^2\,(T_0 - t)$ — ordinary Brownian motion with constant vol. The Samuelson decay vanishes.
- $\eta \to \infty$**.** The exponents push the integrand to essentially zero everywhere except in a tiny sliver just before T, and $\Sigma_t^2 \to 0$: long-dated futures with infinite decay are stationary and have no vol.

Real markets sit in between, with η estimated in the range 0.3 to 2.0 per year for many commodities and interest rates.

9.9.4 8.9.4 Pricing the binary

For the binary call $\varphi(F) = \mathbf{1}_{\{F>K\}}$ at option expiry T_0, with $r = 0$:

$$h(t, F) \;=\; \mathbb{Q}_{t,F}(F_{T_0}(T) > K) \;=\; \mathbb{Q}(F_{T_0} - F_t > K - F_t), \tag{8.42}$$

and writing $F_{T_0} - F_t = \Sigma_t\, Z$ with $Z \sim \mathcal{N}(0, 1)$,

$$h(t, F) \;=\; \mathbb{Q}\left(Z > \frac{K - F}{\Sigma_t}\right) \;=\; \Phi\left(\frac{F - K}{\Sigma_t}\right), \tag{8.43}$$

with Σ_t defined in (8.40).

Structural similarity. Formula (8.43) has the same structural shape as the Bachelier binary formula (8.36): both are Φ of a $(F - K)/\Sigma$ z-score. The Samuelson structure is entirely packaged into the single number Σ_t. This single-number compression is a feature of the arithmetic-Brownian world: no matter how baroque the vol surface of the futures, as long as the SDE is driftless with a deterministic (possibly time-dependent) vol, the terminal distribution is normal with variance equal to the integrated instantaneous variance, and any digital-style payoff prices off Φ of a single z-score.

9.9.5 8.9.5 Practical implication

In a Samuelson world, an option on a long-dated futures contract is dramatically cheaper than an option on a short-dated one of otherwise similar characteristics. The variance Σ_t^2 shrinks as $T - T_0$ grows. Traders who ignore this — who price both options with the same instantaneous vol σ — will systematically overprice the far-dated option and underprice the near-dated one, losing money to anyone running the proper term structure. This asymmetry is the reason commodity options desks and interest-rate desks keep dedicated term-structure models for futures vol rather than relying on a single flat parameter.

The Samuelson effect is a first illustration of a pattern that will recur throughout the guide: *the unconditional vol of an option is an integral over time-dependent instantaneous vols*, and model-free bounds on option prices often follow from convexity arguments applied to this integral. The same algebra reappears in Hull–White (CH12) and in caplet pricing (CH15).

9.9.6 8.9.6 Numerical illustration

Take $\sigma = 0.40$ (40% instantaneous vol at delivery), $\eta = 1.0$ per year, $T = 2$ (delivery at 2 years), $T_0 = 0.25$ (option on the 2-year futures expiring in 3 months), $t = 0$, $F_0 = 50$, $K = 55$. Then

$$\Sigma_0^2 \;=\; 0.40^2\,\frac{e^{-2\cdot(2-0.25)} - e^{-2\cdot 2}}{2} \;=\; 0.16\,\frac{e^{-3.5} - e^{-4}}{2} \;\approx\; 0.16\,\frac{0.0302 - 0.0183}{2} \;\approx\; 0.000952.$$

So $\Sigma_0 \approx 0.0309$ in absolute futures-price units — i.e. the standard deviation of the terminal futures price is about 3.1 cents. Compare this to the analogous ordinary Bachelier model with flat vol $\sigma = 0.40$: the standard deviation would be $0.40\sqrt{0.25} = 0.20$, roughly six times larger. The Samuelson decay has substantially deflated the effective vol because the option is far from the futures' delivery date, and the vol loading at $t \le T_0 = 0.25$ is only $0.40\,e^{-1\cdot 1.75} \approx 0.069$ — about a sixth of the peak vol at delivery.

Consequently, the digital $\Phi((F-K)/\Sigma_0) = \Phi(-5/0.0309) \approx \Phi(-162)$ is effectively zero: the strike is so far above the current futures quote (in units of the tiny Σ_0) that no plausible path can reach it. In the flat-vol Bachelier model with $\Sigma = 0.20$, the analogous $\Phi(-5/0.20) \approx \Phi(-25)$ is also effectively zero — but for different reasons. The moral: in a Samuelson world, option pricing near-to-expiry but far-from-delivery requires genuinely different reasoning than a flat-vol model suggests.

9.9.7 8.9.7 Connection to HJM-style frameworks

The OU futures model of (8.38) is in fact a special case of the Heath–Jarrow–Morton (HJM) framework for modelling the *entire forward curve* as a collection of driftless diffusions. In HJM, each forward price $F(t,T)$ is driftless under the risk-neutral measure but has its own volatility loading $\sigma^F(t,T)$ which can depend on both calendar time t and the contract's delivery T. The Samuelson specification $\sigma^F(t,T) = \sigma\,e^{-\eta(T-t)}$ is the single-factor, exponentially-decaying-vol instance of HJM that is tractable in closed form. Multi-factor HJM models — with several Brownian motions driving different regions of the curve — are standard in commodity and fixed-income practice; we encounter them again in CH13 (multi-factor short-rate models) and CH15 (LIBOR market models).

9.10 8.10 Margrabe Exchange Option — A Numeraire Change Tour de Force

The Margrabe exchange option is the perfect testbed for the numeraire-change machinery of CH05. A frontal assault on it — setting up a two-dimensional PDE in A and B and trying to solve — is painful; a judicious change of numeraire reduces the problem to a one-dimensional Black–Scholes put with no interest-rate discounting visible anywhere. The reduction is near-magical the first time one sees it, and it illustrates a principle that applies far beyond this single example: *if the payoff has a natural unit (dollars, shares of A, shares of B, bonds), price in that unit.* Discounting, drifts, and the multi-dimensional diffusion all collapse when one speaks the right numeraire language.

9.10.1 8.10.1 Setup

Suppose there are two traded assets A and B with dynamics under the physical measure

$$\frac{\mathrm{d}A_t}{A_t} \;=\; \mu^A\,\mathrm{d}t \;+\; \sigma^A\,\mathrm{d}W_t, \qquad \frac{\mathrm{d}B_t}{B_t} \;=\; \mu^B\,\mathrm{d}t \;+\; \sigma^B\,\mathrm{d}W_t \qquad (\mathbb{P}), \tag{8.44}$$

driven (for the moment) by the same scalar Brownian W. The general correlated case is handled by (8.55) below. We value the *Margrabe exchange option* paying

$$\mathcal{C} \;=\; \max(A_T - B_T,\, 0) \tag{8.45}$$

at maturity T. Intuitively this is a call on A struck at B — or, equivalently, a put on B struck at A, since $\max(A-B, 0)$ can be read either way. The striking feature of the Margrabe payoff, as against a vanilla call, is that the *strike is itself a random variable*, moving in lockstep with the other leg. This is the source of all the elegance that follows: if we denominate everything in units of A, the strike becomes the constant 1 (because A_T in units of A_T is simply 1), and the problem collapses to pricing a put on the ratio B/A struck at 1. All the randomness of A has been absorbed into the numeraire.

Margrabe exchange option payoff surface *Margrabe exchange option payoff surface* $\max(A_T - B_T, 0)$ *across* (A_T, B_T).

9.10.2 8.10.2 Step 1 — Risk-neutral pricing under $\mathbb{Q}$

First, change measure from $\mathbb{P}$ to the bank-account measure $\mathbb{Q}$ so that both assets drift at the risk-free rate r:

$$\frac{\mathrm{d}A_t}{A_t} \;=\; r\,\mathrm{d}t \;+\; \sigma^A\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \frac{\mathrm{d}B_t}{B_t} \;=\; r\,\mathrm{d}t \;+\; \sigma^B\,\mathrm{d}W_t^{\mathbb{Q}}. \tag{8.46}$$

(Girsanov's theorem from CH05 guarantees such a measure exists, with drift shifts $\lambda^A = (\mu^A - r)/\sigma^A$ and $\lambda^B = (\mu^B - r)/\sigma^B$. When the same W drives both, consistency requires $\lambda^A = \lambda^B$ — the market-price-of-risk condition of CH06.)

The risk-neutral price is

$$f(t, A, B) \;=\; \mathbb{E}^{\mathbb{Q}}_{t,A,B}\big[\,(A_T - B_T)_+\, e^{-r(T-t)}\,\big]\,. \tag{8.47}$$

Trying to evaluate (8.47) directly requires a two-dimensional log-normal integral in (A_T, B_T) — feasible but not elegant.

9.10.3 8.10.3 Step 2 — Change numeraire to A

Divide both sides of (8.47) by A_t. By the general numeraire-pricing identity of CH05,

$$\frac{f(t, A, B)}{A_t} \;=\; \mathbb{E}^{\mathbb{Q}^A}_{t,A,B}\left[\frac{(A_T - B_T)_+}{A_T}\right] \;=\; \mathbb{E}^{\mathbb{Q}^A}_{t,A,B}\left[\left(1 - \tfrac{B_T}{A_T}\right)_+\right], \tag{8.48}$$

where $\mathbb{Q}^A$ is the measure with numeraire A — under which every traded asset divided by A is a martingale. The Radon–Nikodym density is $\mathrm{d}\mathbb{Q}^A/\mathrm{d}\mathbb{Q} = (A_T/A_0)/(M_T/M_0)$, as derived in CH05.

The move from (8.47) to (8.48) is where the magic happens. Dividing the option price by A_t reinterprets "price" as "how many shares of A is the option worth". Because A is itself a traded asset, the price in units of A must be a $\mathbb{Q}^A$-martingale, which kills the discount factor entirely: a martingale's value today is just its conditional expectation tomorrow, no time-value adjustment required. What remains inside the expectation is a payoff of $(1 - B_T/A_T)_+$ — an ordinary put on the single random variable $X_T = B_T/A_T$ struck at 1. We have turned a two-asset pricing problem into a one-asset pricing problem.

Define

$$X_t \;\equiv\; \frac{B_t}{A_t}, \tag{8.49}$$

which, being the price of the traded asset B expressed in numeraire A, is a $\mathbb{Q}^A$-martingale.

9.10.4 8.10.4 Step 3 — Dynamics of X_t under $\mathbb{Q}^A$

To evaluate (8.48) we need the SDE for X_t under $\mathbb{Q}^A$. The Girsanov shift from $\mathbb{Q}$ to $\mathbb{Q}^A$, derived in CH05, is

$$\mathrm{d}W_t^{\mathbb{Q}} \;=\; \sigma^A\,\mathrm{d}t \;+\; \mathrm{d}W_t^A, \tag{8.50}$$

i.e. the $\mathbb{Q}$-Brownian picks up a drift equal to the volatility loading of the numeraire A when viewed under $\mathbb{Q}^A$.

Compute $\mathrm{d}(1/A_t)$ via Itô on $f(A) = 1/A$, with $f' = -1/A^2$, $f'' = 2/A^3$:

$$\mathrm{d}\left(\frac{1}{A_t}\right) \;=\; -\frac{\mathrm{d}A_t}{A_t^2} \;+\; \tfrac{1}{2}\cdot\frac{2}{A_t^3}\,(\sigma^A A_t)^2\,\mathrm{d}t \;=\; \frac{1}{A_t}\big[((\sigma^A)^2 - r)\,\mathrm{d}t \;-\; \sigma^A\,\mathrm{d}W_t^{\mathbb{Q}}\big]. \tag{8.51}$$

Then, using the Itô product rule on $X_t = B_t \cdot (1/A_t)$ with covariation $\mathrm{d}[B, 1/A]_t = -(B_t/A_t)\,\sigma^A\sigma^B\,\mathrm{d}t$,

$$\mathrm{d}\left(\frac{B_t}{A_t}\right) \;=\; \frac{B_t}{A_t}(r\,\mathrm{d}t + \sigma^B\,\mathrm{d}W_t^{\mathbb{Q}}) \;+\; \frac{B_t}{A_t}\big[((\sigma^A)^2 - r)\,\mathrm{d}t - \sigma^A\,\mathrm{d}W_t^{\mathbb{Q}}\big] \;-\; \frac{B_t}{A_t}\,\sigma^A\sigma^B\,\mathrm{d}t. \tag{8.52}$$

Collecting $\mathrm{d}t$ and $\mathrm{d}W_t^{\mathbb{Q}}$ terms,

$$\mathrm{d}\left(\frac{B_t}{A_t}\right) \;=\; \frac{B_t}{A_t}\Big\{\big[(\sigma^A)^2 - \sigma^A\sigma^B\big]\,\mathrm{d}t \;+\; (\sigma^B - \sigma^A)\,\mathrm{d}W_t^{\mathbb{Q}}\Big\}. \tag{8.53}$$

Now substitute (8.50) $\mathrm{d}W_t^{\mathbb{Q}} = \sigma^A\,\mathrm{d}t + \mathrm{d}W_t^A$:

$$\mathrm{d}\left(\frac{B_t}{A_t}\right) \;=\; \frac{B_t}{A_t}\Big\{\underbrace{\big[(\sigma^A)^2 - \sigma^A\sigma^B + (\sigma^B - \sigma^A)\sigma^A\big]}_{=\,0}\,\mathrm{d}t \;+\; (\sigma^B - \sigma^A)\,\mathrm{d}W_t^A\Big\}, \tag{8.54}$$

confirming that $X_t = B_t/A_t$ is a $\mathbb{Q}^A$-martingale, and

$$\boxed{\frac{\mathrm{d}X_t}{X_t} \;=\; (\sigma^B - \sigma^A)\,\mathrm{d}W_t^A.} \tag{8.55}$$

Despite all the Itô bookkeeping in (8.51)–(8.54), the end result is a geometric Brownian motion for X with *no drift* and a volatility equal to the simple difference $\sigma^B - \sigma^A$. The zero drift is enforced by the numeraire property — it had to come out that way — and the explicit calculation shows every drift term cancelling. The volatility is the *relative* vol: how much B moves relative to A, which is the only vol relevant when we measure everything in units of A. When A and B are driven by the same Brownian with the same vol, the ratio is deterministic and the option has no optionality; (8.55) gives zero vol, correctly.

9.10.5 8.10.5 Correlated case

In the general correlated case with two distinct Brownians $W^{(1)}, W^{(2)}$ of correlation ρ,

$$\frac{\mathrm{d}A_t}{A_t} \;=\; r\,\mathrm{d}t + \sigma^A\,\mathrm{d}W_t^{(1)}, \qquad \frac{\mathrm{d}B_t}{B_t} \;=\; r\,\mathrm{d}t + \sigma^B\,\mathrm{d}W_t^{(2)}, \qquad \mathrm{d}\langle W^{(1)}, W^{(2)}\rangle_t = \rho\,\mathrm{d}t,$$

the same Itô calculation yields $\mathrm{d}X_t/X_t = \hat{\sigma}\,\mathrm{d}W_t^A$ with

$$\boxed{\hat{\sigma}^2 \;=\; (\sigma^A)^2 \;-\; 2\rho\,\sigma^A\sigma^B \;+\; (\sigma^B)^2.} \tag{8.56}$$

The scalar-BM result (8.55) is the special case $\rho = +1$ with both assets driven by the *same* W, which collapses (8.56) to $\hat{\sigma}^2 = (\sigma^B - \sigma^A)^2$ and hence $\hat{\sigma} = |\sigma^B - \sigma^A|$.

Reading (8.56). This is one of those small-looking results that deserves a lot of respect. It says *the only interaction between the two assets that matters for the exchange option is their correlation* — not their individual drifts, not their individual levels, just ρ. Crank ρ up toward $+1$ (assets move in lockstep): $\hat{\sigma}^2$ shrinks toward $(\sigma^A - \sigma^B)^2$, and two assets that always move together cannot drift apart in relative terms — the option is cheap. Crank ρ down toward -1 (assets move oppositely): $\hat{\sigma}^2$ balloons to $(\sigma^A + \sigma^B)^2$, and two maximally anti-correlated assets are maximally divergent in relative terms — the option is expensive. In a Margrabe setting, *correlation is the pricing input* — more so than any individual vol — which is why dispersion traders and index arbitrageurs obsess over correlation term structures.

Margrabe correlation sensitivity *Margrabe price as a function of correlation ρ — option value increases monotonically as ρ decreases, since divergent assets maximise the expected spread.*

9.10.6 8.10.6 Step 4 — The Margrabe formula

Since X_t is a geometric Brownian motion under $\mathbb{Q}^A$ with volatility $\hat{\sigma}$ and *no drift*, the inner expectation in (8.48) is a Black–Scholes put on X struck at 1 with $r = 0$ (no discount under the martingale measure):

$$\frac{f_t}{A_t} \;=\; \Phi(d_+) \;-\; \frac{B_t}{A_t}\,\Phi(d_-), \qquad d_\pm \;=\; \frac{-\ln(B_t/A_t) \pm \frac{1}{2}\hat{\sigma}^2(T-t)}{\hat{\sigma}\sqrt{T-t}}. \tag{8.57}$$

Rearranging into the familiar Margrabe form,

$$\boxed{f_t \;=\; A_t\,\Phi(d_1) \;-\; B_t\,\Phi(d_2),} \qquad d_1 \;=\; \frac{\ln(A_t/B_t) + \frac{1}{2}\hat{\sigma}^2(T-t)}{\hat{\sigma}\sqrt{T-t}}, \quad d_2 \;=\; d_1 - \hat{\sigma}\sqrt{T-t}. \tag{8.58}$$

No discounting appears because we priced under $\mathbb{Q}^A$ and the payoff is denominated in units of A; the numeraire absorbs the interest rate.

9.10.7 8.10.7 Analogy with Black–Scholes

Compare (8.58) to the standard Black–Scholes call formula $C = S\,\Phi(d_1) - K\,e^{-rT}\,\Phi(d_2)$. The two look almost identical, with three substitutions: the stock S becomes A, the discounted strike $K\,e^{-rT}$ becomes B, and the single volatility σ becomes the relative vol $\hat{\sigma}$. The discounted strike in BS plays exactly the role B plays in Margrabe — a "random" strike whose dynamics (growth at rate r in BS, geometric Brownian in Margrabe) have been absorbed into the second term. Once one sees this analogy, Margrabe is not a new formula but Black–Scholes in a different costume.

9.10.8 8.10.8 Relative moneyness, not levels

A practical reading of (8.58): Margrabe does not care about the *levels* of A and B, only their ratio $X = B/A$. The option is at-the-money when $A_t = B_t$. Higher correlation ρ shrinks $\hat{\sigma}$ and therefore the option value — a "best-of" is cheapest when the two underlyings are clones. The limit $\rho = 1$, $\sigma^A = \sigma^B$ gives $\hat{\sigma} = 0$ and the option is worth $\max(A_t - B_t, 0)$ exactly (deterministic, no optionality).

When trading an exchange option, do not look at the absolute levels of the two assets; look at their ratio. If the ratio is far from 1, the option is deep in or deep out of the money, and gamma is low. If the ratio is near 1, the option is ATM and gamma is high. Similarly, do not think about each asset's vol separately; think about the *spread vol* $\hat{\sigma}$, which is the only vol the option cares about. Skew and smile in $\hat{\sigma}$ — to the extent they exist — reflect co-movement of individual skews with correlation dynamics, and they are one of the subtlest risks in dispersion trading.

9.10.9 8.10.9 Worked numerical example

Take $A_0 = B_0 = 100$ (ATM), $\sigma^A = 0.25$, $\sigma^B = 0.30$, $\rho = 0.6$, $T = 1$ year, $r = 3\%$. Then

$$\hat{\sigma}^2 \;=\; 0.25^2 - 2\cdot 0.6\cdot 0.25\cdot 0.30 + 0.30^2 \;=\; 0.0625 - 0.09 + 0.09 \;=\; 0.0625,$$

so $\hat{\sigma} = 0.25$. Note: the spread vol equals σ^A in this specific case — a coincidence of the correlation and vol values, not a general feature. The d-parameters are $d_1 = \frac{1}{2}\hat{\sigma} = 0.125$ and $d_2 = -0.125$, giving $\Phi(d_1) \approx 0.5497$, $\Phi(d_2) \approx 0.4503$. The Margrabe price is

$$f_0 \;=\; 100 \cdot 0.5497 \;-\; 100 \cdot 0.4503 \;=\; 9.94.$$

Notice the absence of any discount factor — this is the numeraire-change magic of (8.58). The price is 9.94% of the reference asset's notional, which is the right-of-magnitude value for an ATM exchange option with a 25% spread vol over one year.

Now flip the correlation to $\rho = -0.6$:

$$\hat{\sigma}^2 \;=\; 0.0625 + 0.09 + 0.09 \;=\; 0.2425, \qquad \hat{\sigma} \;\approx\; 0.4924.$$

The spread vol almost doubles, and the Margrabe price rises to approximately 19.5 — nearly double the positive-correlation case. This is the correlation sensitivity made quantitative: flipping the sign of ρ from $+0.6$ to -0.6 changes the fair price of the exchange option by 100%. Dispersion-trading desks, which systematically sell single-name options and buy index options to monetise the correlation-vs-individual-vol gap, live and die by this relationship.

9.10.10 8.10.10 Generalisations and pitfalls

The Margrabe formula (8.58) has been generalised in several directions that appear elsewhere in the guide:

- **Stochastic volatility.** If either σ^A or σ^B is itself stochastic (a Heston-like model, CH14), the Margrabe formula is an approximation and one falls back on characteristic-function or Monte-Carlo methods.
- **Multiple assets.** The "best-of" option $\max(A_T, B_T, C_T)$ does *not* admit a closed-form generalisation of (8.58) — the best-of requires three-dimensional integration and is typically handled by quasi-Monte-Carlo or semi-analytic methods (Stulz 1982, Johnson 1987).
- **Dividend-paying legs.** If either asset pays a continuous yield, the cash-and-carry growth of A and B is adjusted by subtracting the yield, and the formula remains valid with modified drifts.

A common pitfall: the vol loadings σ^A and σ^B in (8.56) must be the *same Brownian-motion loadings* — i.e. the coefficients appearing in front of the *same* $\mathrm{d}W_t^{(k)}$ components. If one decomposes each asset's volatility into a sum of loadings on common and idiosyncratic factors, the Margrabe spread vol must be computed by appropriately combining the components. For two independent Brownians of volatilities σ^A, σ^B (i.e. $\rho = 0$), the spread vol is $\hat{\sigma} = \sqrt{(\sigma^A)^2 + (\sigma^B)^2}$ — the formula reduces to the Pythagorean sum, as expected for independent vol components.

9.11 8.11 Futures vs Forward — Convexity Adjustment Under Stochastic Rates

We are now in a position to pay off the promise made in §8.2: under stochastic rates, futures and forwards on the same underlying differ by a *convexity adjustment*, whose sign is controlled by the sign of $\mathrm{Cov}^{\mathbb{Q}}(r_t, S_t)$. The intuition is financial, not mathematical, and it is worth building before writing any formulas down.

9.11.1 8.11.1 The intuition

Picture two traders. Alice is long a *forward* on S with maturity T; Bob is long a *futures* on S with the same maturity. Both have the same final exposure to S_T. But between now and T, Bob receives daily mark-to-market cash flows into a margin account, while Alice does nothing. If S goes up, Bob receives cash; if S goes down, Bob pays cash.

Now suppose S and r are positively correlated — a plausible story for an equity index and the short rate during normal macro regimes, or for a bond futures and the short rate (which would be the same rate, making the correlation mechanical). On the days Bob receives cash (up-days for S), rates are also high, so Bob reinvests the cash at a *high* rate. On the days Bob pays cash (down-days for S), rates are low, so Bob finances the losses at a *low* rate. Bob wins on both ends of this timing asymmetry, and the extra value he captures must be compensated at inception by a *higher* futures price relative to the forward. This is the convexity adjustment: $F^{\mathrm{fut}} > F^{\mathrm{fwd}}$ when $\mathrm{Cov}(r, S) > 0$, and the opposite sign when the correlation flips.

Conversely, if r and S are negatively correlated (as they might be for a bond futures and the short rate, depending on the economic regime), Bob reinvests his up-day cash at low rates and finances his down-day losses at high rates. The timing asymmetry now works *against* Bob, and the fair futures price must sit *below* the forward to compensate.

If rates are deterministic — or if S and r are uncorrelated under $\mathbb{Q}$ — the timing asymmetry is gone, and the two prices coincide.

9.11.2 8.11.2 The formula

Under $\mathbb{Q}$ (bank-account numeraire M_t), the futures price $F_t^{\text{fut}}(T)$ is a martingale (§8.3). Under the T-forward measure $\mathbb{Q}^T$ (bond numeraire $P(\cdot, T)$), the forward price $F_t^{\text{fwd}}(T) = S_t/P(t,T)$ is a martingale (CH05). Taking conditional expectations at time t,

$$F^{\text{fut}}(t,T) \;=\; \mathbb{E}_t^{\mathbb{Q}}[S_T], \qquad F^{\text{fwd}}(t,T) \;=\; \mathbb{E}_t^{\mathbb{Q}^T}[S_T] \;=\; \frac{\mathbb{E}_t^{\mathbb{Q}}\Big[S_T\, e^{-\int_t^T r_u\,\mathrm{d}u}\Big]}{P(t,T)}. \tag{8.59}$$

Their difference is the convexity adjustment. By the identity $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ applied to the inner expectation, one obtains

$$F^{\text{fut}}(t,T) \;-\; F^{\text{fwd}}(t,T) \;=\; \frac{1}{P(t,T)}\,\mathrm{Cov}_t^{\mathbb{Q}}\Big(S_T,\, 1 - e^{-\int_t^T r_u\,\mathrm{d}u}\Big), \tag{8.60}$$

which to leading order in the covariance is approximately

$$F^{\text{fut}}(t,T) \;-\; F^{\text{fwd}}(t,T) \;\approx\; \int_t^T\!\!\int_t^T \mathrm{Cov}_t^{\mathbb{Q}}(\mathrm{d}\ln S_u,\, r_v\,\mathrm{d}v). \tag{8.61}$$

9.11.3 8.11.3 The sign rule

Positive correlation between S (hence F^{fut}) and r means MTM gains are reinvested at higher rates while losses are financed at lower rates — this is worth money to the long, so the fair futures price must be *above* the forward. Negative correlation flips it. In summary:

Regime	Relation
r deterministic	$F^{\text{fut}}(t,T) = F^{\text{fwd}}(t,T) = S_t/P(t,T)$
r stochastic, $\mathrm{Cov}^{\mathbb{Q}}(r_t, F_t^{\text{fut}}) = 0$	$F^{\text{fut}} = F^{\text{fwd}}$
r stochastic, positive correlation	$F^{\text{fut}} > F^{\text{fwd}}$
r stochastic, negative correlation	$F^{\text{fut}} < F^{\text{fwd}}$

The two measures — bank-account and T-forward — coincide iff $P(t,T)$ is deterministic. When they coincide, so do the expectations.

Futures-forward basis *Futures-forward basis under stochastic rates — positive r–S correlation pushes the basis positive, negative correlation pushes it negative. The magnitude is controlled by rate volatility and the tenor.*

9.11.4 8.11.4 Magnitudes in practice

How big is the convexity adjustment? A rough rule of thumb:

- **Short-dated eurodollar / SOFR futures.** Here the underlying rate and the reinvestment rate are both tied to the same short-rate curve, so $\rho \approx 1$ and the correlation effect is maximal. The adjustment can reach several basis points and is routinely quoted and hedged by rates desks. Dedicated convexity-adjustment calculators are standard in rates-trading software.
- **Long-dated equity index futures.** Correlation between equity returns and rates is modest, rate vols are low, and the adjustment is typically small (a few basis points on a multi-year contract). Still, for institutional hedgers running multi-billion-dollar exposures, a few basis points can matter.
- **FX futures.** Non-trivial because of the "quanto" correlation between the FX rate and the domestic short rate. The formula (8.61) is still the starting point, but the practitioner's job is to specify a model (typically Hull–White for the short rate, log-normal for FX, with a correlation ρ between them) and evaluate the double integral analytically.

9.11.5 8.11.5 The deeper reading

A subtle but important observation: the convexity adjustment is a *measure-change effect*, not a modelling choice. The two prices F^{fut} and F^{fwd} are *both* martingales, but under *different* measures — the bank-account measure and the T-forward measure — and those two measures assign different weights to states of the world unless rates are deterministic. When rates are deterministic, the bank-account numeraire $M_t = e^{\int r}$ and the bond numeraire $P(t,T)$ differ by a deterministic factor, the two measures coincide, and the two martingale expectations coincide. When rates are stochastic, the numeraires diverge in a state-dependent way, and so do the expectations.

This is the first example of a pattern that pervades the fixed-income chapters of this guide: *convexity adjustments of the same structural form appear wherever a quantity is quoted under one measure but needs to be priced under another*. The classic cases:

- Futures vs forwards on a single asset (this section).
- Convexity adjustments for CMS rates (CH13), where a swap rate expected under the annuity measure differs from its value expected under the T-forward measure.
- Quanto adjustments for FX-linked payoffs, where a dollar payoff proportional to a foreign asset carries a correlation-dependent adjustment.

In every case the adjustment is a covariance term with the same structural shape as (8.60): the difference between two martingale prices is a covariance of state variables under the reference measure. Once the reader internalises this once for futures-vs-forwards, the CH13 convexity-adjustment calculations become routine.

9.11.6 8.11.6 A Hull–White closed-form approximation

Writing down a concrete convexity-adjustment number requires a model for both S and r. The most common textbook choice is to take S log-normal and r Hull–White (CH12), with a constant correlation ρ between the Brownians. Under that model, Hull (2018) gives the approximation

$$\ln F^{\text{fut}}(t,T) - \ln F^{\text{fwd}}(t,T) \;\approx\; \rho\,\sigma_S\,\sigma_r\,\frac{(T-t)^2}{2},$$

where σ_S is the spot volatility and σ_r is the short-rate vol. The quadratic-in-tenor scaling is characteristic of convexity adjustments: over short tenors the effect is negligible, but it grows quickly with maturity. For eurodollar/SOFR futures with $\sigma_r \approx 0.01$ (1% absolute rate vol), $\sigma_S \approx 0.0025$ (own-vol for a short rate), $\rho \approx 1$, and $T - t = 5$ years, the adjustment is roughly $1 \cdot 0.01 \cdot 0.0025 \cdot 12.5 = 0.0003$ or 3 basis points — a number that matches practitioner quotations for multi-year eurodollar convexity adjustments.

9.11.7 8.11.7 Hedging implications

From a portfolio-management standpoint, the futures-vs-forward convexity adjustment has a direct hedging consequence: if a dealer has hedged a forward exposure with futures (or vice versa), and rates subsequently become more volatile, the hedge *slippage* is proportional to the convexity adjustment. Dealers who systematically run futures-vs-forward basis books (the cross between cash-settled swap futures and OTC swaps, for instance) must mark their book daily to the evolving convexity adjustment and adjust hedges to reflect the changing rate volatility. The sensitivity of the basis to rate vol is often called the *vega of the convexity adjustment*, and is one of the diagnostics that rates-trading desks monitor as part of their broader rate-vol exposure.

9.12 8.12 Key Takeaways

1. **Forward price.** The fair forward strike $F_t^{\text{fwd}}(T) = S_t/P(t,T)$ is pinned by cash-and-carry no-arbitrage, independent of volatility or any statistical properties of S. Under constant rates this reduces to $S_t\,e^{r(T-t)}$; with a dividend yield, to $S_t\,e^{(r-\delta)(T-t)}$.

2. **Forwards vs futures.** Forwards settle at maturity; futures settle daily. The futures position is zero-valued at every instant by design (daily MTM absorbs every infinitesimal gain/loss immediately), while the forward's value drifts with the spot until maturity. This zero-value feature is why the futures price is a $\mathbb{Q}$-martingale and why the Black PDE has no first-derivative drift term.

3. **Three drifts under $\mathbb{Q}$.**
   - Traded asset: drift $r\,X_t$.
   - Traded asset with yield δ: drift $(r-\delta)\,X_t$.
   - Futures quote: drift 0 (costless position).
4. **Dynamics of F_t under $\mathbb{Q}$.** $\mathrm{d}F_t(T) = \sigma^F(t,F)\,\mathrm{d}\widehat{W}_t$ with no drift. Specific choices of σ^F yield GBM (constant relative vol, used in Black '76), Bachelier (constant absolute vol, used when F can go negative), and Samuelson/OU (time-decaying vol loading, used for commodity futures).
5. **Black PDE.** For an option $g(t,F)$ on a futures with payoff $\varphi(F_{T_0})$,

$$\partial_t g \;+\; \tfrac{1}{2}\,(\sigma^F)^2\,\partial_{FF} g \;=\; r\,g, \qquad g(T_0,F) = \varphi(F).$$

   The missing $r\,F\,\partial_F g$ term is the signature of martingale-underlying pricing. The $r\,g$ discount term remains because the option's dollar payoff still settles at T_0.
6. **Feynman–Kac cross-check.** The Black PDE has the stochastic representation $g(t,F) = \mathbb{E}^{\mathbb{Q}}[\,e^{-r(T_0-t)}\,\varphi(F_{T_0})\mid F_t = F\,]$ with F_u satisfying the driftless SDE under $\mathbb{Q}$.
7. **Black '76 closed form.** For log-normal futures with constant relative vol σ and a call payoff,

$$g(t,F) \;=\; e^{-r(T_0-t)}\,[F\,\Phi(d_+) \;-\; K\,\Phi(d_-)], \qquad d_\pm \;=\; \frac{\ln(F/K) \pm \frac{1}{2}\sigma^2(T_0-t)}{\sigma\,\sqrt{T_0-t}}.$$

   Structurally, Black '76 is BS with $S \to F$, $r \to 0$ in $d_\pm$, and an overall discount $e^{-r(T_0-t)}$.
8. **Bachelier binary.** Under $\mathrm{d}F_t = \sigma\,\mathrm{d}\widehat{W}_t$, a digital call prices as $\Phi((F-K)/(\sigma\,\sqrt{T_0-t}))$. At the money ($F=K$), the price is exactly $1/2$ — a symmetry unique to the arithmetic model.
9. **Samuelson / OU futures.** Under $\mathrm{d}F_t = \sigma\,e^{-\eta(T-t)}\,\mathrm{d}\widehat{W}_t$, terminal variance is $\Sigma_t^2 = \sigma^2\,(e^{-2\eta(T-T_0)} - e^{-2\eta(T-t)})/(2\eta)$ and a digital call prices as $\Phi((F-K)/\Sigma_t)$. Long-dated futures have smaller Σ_t than short-dated ones — the Samuelson effect.
10. **Margrabe exchange option.** Under numeraire A, the ratio $X_t = B_t/A_t$ is a martingale with vol $\hat{\sigma} = \sqrt{(\sigma^A)^2 - 2\rho\,\sigma^A\sigma^B + (\sigma^B)^2}$, yielding $f_t = A_t\,\Phi(d_1) - B_t\,\Phi(d_2)$. No discount factor appears because the numeraire absorbs interest. Correlation ρ, not individual drifts or levels, determines the option value.
11. **Futures vs forward convexity.** Under stochastic rates, $F^{\mathrm{fut}} - F^{\mathrm{fwd}}$ is a covariance of S_T with the rate-path discount factor, proportional to $\mathrm{Cov}^{\mathbb{Q}}(r,S)$ in the leading-order limit. Positive correlation pushes the futures price above the forward; negative correlation pushes it below. Under deterministic rates, the two prices coincide.
12. **Numeraire is the theme.** The futures price is natural under $\mathbb{Q}$; the forward price is natural under $\mathbb{Q}^T$; the Margrabe exchange option is natural under one of its own legs. Each "naturalness" is an invitation to change measure; accepting the invitation collapses the pricing problem, while refusing it forces one to grind through a PDE that did not need to be set up.

9.13 Appendix — Reference Formulas

Label	Formula
(8.3)	$F_t^{\mathrm{fwd}}(T) = S_t\,e^{r(T-t)}$ (constant rates, no dividend)
(8.4)	$F_t^{\mathrm{fwd}}(T) = S_t/P(t,T)$ (cash-and-carry, any rates)
(8.8)	$\mathbb{E}^{\mathbb{Q}}[F_t(T)\mid\mathcal{F}_s] = F_s(T)$ (futures is a $\mathbb{Q}$-martingale)
(8.9)	$\mathrm{d}F_t(T) = \sigma^F(t,F_t(T))\,\mathrm{d}\widehat{W}_t$ (general futures SDE under $\mathbb{Q}$)
(8.13)	$\mathrm{d}F_t/F_t = \sigma\,\mathrm{d}\widehat{W}_t$ (log-normal futures under $\mathbb{Q}$)

Label	Formula
(8.21)	$\alpha_t = \partial_F g(t, F_t)$ (futures-delta hedge)
(8.24)	$\partial_t g + \frac{1}{2}(\sigma^F)^2\, \partial_{FF} g = r\, g,\ g(T_0, F) = \varphi(F)$ (Black PDE)
(8.26)	$g(t, F) = \mathbb{E}^{\mathbb{Q}}[\, e^{-r(T_0-t)}\, \varphi(F_{T_0}) \,\|\, F_t = F\,]$ (Feynman–Kac)
(8.30)	$g = e^{-r(T_0-t)}[F\,\Phi(d_+) - K\,\Phi(d_-)]$ (Black '76 call)
(8.31)	$d_\pm = [\ln(F/K) \pm \frac{1}{2}\sigma^2(T_0 - t)]/(\sigma\sqrt{T_0 - t})$
(8.32)	$\partial_F g = e^{-r(T_0-t)}\,\Phi(d_+)$ (Black '76 delta)
(8.36)	Bachelier binary: $h(t, F) = \Phi((F - K)/(\sigma\,\sqrt{T_0 - t}))$
(8.40)	Samuelson variance: $\Sigma_t^2 = \sigma^2\,(e^{-2\eta(T-T_0)} - e^{-2\eta(T-t)})/(2\eta)$
(8.43)	OU binary: $h(t, F) = \Phi((F - K)/\Sigma_t)$
(8.55)	$\mathrm{d}X_t/X_t = (\sigma^B - \sigma^A)\,\mathrm{d}W_t^A$ (Margrabe ratio dynamics, scalar BM)
(8.56)	$\hat{\sigma}^2 = (\sigma^A)^2 - 2\rho\,\sigma^A\sigma^B + (\sigma^B)^2$ (Margrabe spread vol)
(8.58)	Margrabe: $f_t = A_t\,\Phi(d_1) - B_t\,\Phi(d_2)$
(8.60)	$F^{\mathrm{fut}} - F^{\mathrm{fwd}} = \mathrm{Cov}_t^{\mathbb{Q}}(S_T,\, 1 - e^{-\int r}) \,/\, P(t, T)$
(8.61)	Convexity $\approx \int_t^T \int_t^T \mathrm{Cov}_t^{\mathbb{Q}}(\mathrm{d}\ln S_u,\, r_v\,\mathrm{d}v)$

10 CH09 — Risk Measures: VaR, CTE, and Hedged Tail Risk

This chapter develops the machinery quantitative risk managers use to summarize the loss distribution of a trading book: Value-at-Risk (VaR), Conditional Tail Expectation (CTE, also called Expected Shortfall), their estimation under historical, parametric, and Monte Carlo paradigms, and the way these measures behave when applied to option books that have been delta- or delta-gamma-hedged. The material is organized so that a reader can move from scalar summary statistics of P&L (§§9.1-9.4), into regulatory and axiomatic context (§§9.5-9.6), into backtesting and stress discipline (§§9.7-9.8), into attribution and liquidity overlays (§§9.9-9.11), and finally into derivative-specific risk arithmetic where Greeks drive the tails (§§9.12-9.14). Cross-references point to CH06 for the underlying GBM dynamics and to CH07 for the Greek definitions themselves.

10.1 9.1 Why Risk Measures Matter

Picture the risk-reward plane: reward on the vertical axis, "risk" on the horizontal axis. Portfolios scatter as points; dominated ones lie to the south-east of undominated ones, and an efficient frontier sweeps north-west. The entire discipline of quantitative portfolio construction hinges on the apparently innocent choice of what to put on the horizontal axis. Variance produces Markowitz-style portfolios indifferent to the sign of moves. Downside semi-variance penalises only losses. VaR cares only about a threshold. CTE cares about the size of losses beyond the threshold. These are not equivalent reparametrisations — they rank portfolios differently, and the portfolio that looks attractive under one axis can look unattractive under another.

Risk management exists because organisations cannot run on distributions. A board of directors wants a number; a regulator wants a number; a desk head wants a number to compare against a limit. A risk measure is a functional that reduces an infinite-dimensional loss distribution to a single scalar that can be acted on. The price of the reduction is information loss, and the *character* of that loss — which features of the distribution the measure emphasises and which it hides — is exactly why the choice of functional matters.

This chapter works exclusively with the *loss* random variable L, whose large positive values correspond to financially painful outcomes. Some trading desks report risk on a P&L convention where losses are negative; the underlying mathematics is identical up to sign, but the presentation can confuse the unwary. Throughout, F_L denotes the CDF of L under whatever probability measure we have chosen for the return generator — typically the historical measure for risk measurement, reserved from the risk-neutral measure used for pricing.

Scope and horizon. The apparatus in this chapter is built for *market risk* — the risk that the mark-to-market value of a traded book changes because of moves in observable market prices. It is not directly the machinery used for credit risk, operational risk, or liquidity risk (though all three borrow concepts from it). Market-risk VaR is traditionally computed at a 1-day horizon for desk-level reporting and a 10-day horizon for regulatory capital; longer horizons are used for economic capital. Confidence levels 95%, 99%, and 99.9% correspond to different use cases: 95% for daily desk reporting, 99% for regulatory VaR, 99.9% for economic capital. The higher the confidence level, the rarer the events being summarised, and the more violent the estimation error.

Risk measurement versus risk management. Computing VaR or CTE is the easy part — well, not easy, but conceptually well-defined once a loss distribution is picked. Doing something useful about the result is harder. A \$5M VaR number leaves open whether the risk is concentrated in a single unwind-able exposure or scattered across thousands of small ones; whether it is driven by equities, rates, credit, or FX; whether it moves a little or a lot if volatilities rise 20%. These are the questions a working risk manager wrestles with, and they require decomposing the aggregate into interpretable pieces — marginal, component, incremental VaR — that we return to in §9.9. Every VaR number is also the output of a statistical model, and every model is wrong in some way. Professional practice therefore triangulates across methodologies (historical, parametric, Monte Carlo), lookback windows, distributional assumptions, and stress scenarios, treating disagreements among them as diagnostic rather than embarrassing.

10.2 9.2 Value-at-Risk: Historical, Parametric, Monte Carlo, Cornish-Fisher

For a loss random variable L with distribution F_L, the Value-at-Risk at level $\alpha \in (0,1)$ is

$$\mathrm{VaR}_\alpha(L) \;=\; \inf\{\ell : \mathbb{P}(L \le \ell) \ge 1-\alpha\}. \tag{9.1}$$

VaR is the $(1-\alpha)$-quantile of L. It answers *"what loss will I not exceed with* $(1-\alpha)$ *confidence?"* and ignores the size of losses beyond the cutoff. A portfolio whose 5%-tail pays $-\$10\,\mathrm{M}$ with probability 0.01 and $-\$1\,\mathrm{M}$ with

probability 0.04 has the same 95%-VaR as a portfolio that pays a flat −\$2 M across the whole tail. Worse, VaR can *decrease* when two risky books are merged — it fails sub-additivity, and therefore fails to be a coherent risk measure in the Artzner sense (§9.6).

A useful mental picture: imagine the CDF of L rising from zero on the left to one on the right. VaR at level $\alpha = 0.05$ is the horizontal line $y = 0.95$ intersected with the CDF and projected onto the x-axis. Any reshuffling of probability mass to the right of that intersection leaves VaR unchanged — pile the tail onto a single catastrophic point, or spread it uniformly, and the VaR line doesn't move. Only the quantile location matters. CTE, by contrast, is the area under the CDF complement above the horizontal line divided by 0.05, and is sensitive precisely to the tail reshaping VaR misses.

The infimum in (9.1) rather than a plain quantile covers distributions with atoms — binary payoffs, barriers, digitals — where the CDF has a jump at the target probability and the quantile is not uniquely defined. In empirical Monte Carlo distributions with thousands of discrete sample points, this tie-breaking convention must be explicit or two people computing "the same" VaR will get different numbers.

Loss distribution with VaR and CTE *Loss distribution: VaR cutoff and CTE mean-loss beyond it.*

10.2.1 9.2.1 Three computational routes to VaR

Three methodologies dominate production risk systems: *historical simulation*, *parametric (variance-covariance)*, and *Monte Carlo*. A well-run risk function uses at least two as cross-checks.

Historical simulation. Collect the last N daily P&L observations (the current portfolio re-priced under historical market moves), sort them worst to best, and pick the $(1-\alpha)$-quantile directly. With $N = 500$ and $\alpha = 0.95$, the 95% VaR is the 25th worst observation. The method is non-parametric: it makes no distributional assumption, letting whatever fat tails, skew, and jumps appeared historically carry through. Strengths: no model risk from mis-specified distributions, trivial to explain. Weaknesses: only as good as its lookback window, assigns equal weight to old and new observations so regime shifts are absorbed with lag, and tail estimation at $\alpha = 0.99$ with $N = 500$ rests on five data points. Refinements include *age-weighted* historical simulation (older observations get less weight), *volatility-scaled* historical simulation (rescale each historical return by current-to-historical vol ratio), and *filtered historical simulation* (fit a GARCH model, scale out historical vol, pick a quantile of residuals, scale back up by current vol).

Parametric / variance-covariance / delta-normal VaR. Assume portfolio returns are Gaussian with zero mean and variance $\sigma_P^2 = w^\top \Sigma w$, where w is the vector of dollar exposures and Σ is the factor covariance matrix. Then

$$\mathrm{VaR}_\alpha = z_{1-\alpha}\,\sigma_P = z_{1-\alpha}\,\sqrt{w^\top \Sigma w}, \tag{9.2}$$

where $z_{1-\alpha}$ is the Gaussian quantile (1.645 at 95%, 2.326 at 99%). Strengths: closed-form, fast even for very large portfolios, decomposable cleanly by factor or asset class. Weaknesses: the Gaussian assumption dramatically underestimates fat-tail risk, and the linear approximation to book P&L fails for option-heavy books. Parametric VaR is sometimes called "delta-normal" precisely because it couples a linear delta approximation of position P&L with a Normal distribution of factor returns.

A partial fix for non-linearity is *delta-gamma parametric VaR*: approximate book P&L by a quadratic function of risk factors,

$$\Delta P \approx \delta^\top \Delta r + \tfrac{1}{2}\,\Delta r^\top \Gamma\,\Delta r, \tag{9.3}$$

where δ is the gradient of P&L with respect to risk factors and Γ is the Hessian. Under multivariate Normal Δr the distribution of ΔP is a quadratic form in Normals — a generalised chi-squared — whose VaR can be obtained by Cornish-Fisher expansion or numerical inversion of its characteristic function. We develop this in full in §9.12; for now, note that delta-gamma parametric VaR improves meaningfully on delta-normal for option-heavy portfolios while retaining the Gaussian factor assumption.

Monte Carlo VaR. Specify a joint distribution for risk factor returns (Gaussian, Student-t, copula-based, stochastic-volatility, full economic-scenario generator), draw simulated paths, reprice the entire portfolio under each, and take the empirical quantile of the simulated P&L distribution. Strengths: flexibility to handle arbitrary distributions, full non-linear pricing including path-dependent payoffs, easy incorporation of complex instruments. Weaknesses: expensive (tens of thousands of full-portfolio revaluations), model-risk-prone (the joint distribution has to be specified

and will be wrong in some way), and slow to converge in the tail. Many large banks run Monte Carlo VaR as the primary regulatory measure and historical simulation as a cross-check.

A sample-size calculation is instructive. The standard error of a quantile estimate based on M i.i.d. samples scales as $1/\sqrt{M \cdot f(q)^2 \cdot \alpha(1-\alpha)}$, where $f(q)$ is the density at the quantile. For a 99% VaR with 10,000 Monte Carlo paths under a Normal approximation, the standard error is 3-5% of the VaR itself. For 99.9% VaR on the same 10,000 paths the standard error blows up to 15-20%, which is why extreme quantiles require massive simulation budgets (often 100,000 paths or more) and motivate heavy use of variance-reduction techniques such as importance sampling. A common refinement is to oversample the tail, run disproportionately many paths in stressful scenarios, re-weight them, and estimate the tail quantile with much higher precision.

Three routes to VaR on a common loss distribution *Historical, parametric (Gaussian), and Monte-Carlo estimates of 95% VaR on the same skewed, jump-prone loss distribution. The Gaussian fit systematically under-reports tail risk because it ignores the jump component; historical and Monte-Carlo agree once the sample is large enough to see the jump events. Running at least two methods in parallel as cross-checks is standard production practice.*

10.2.2 9.2.2 Cornish-Fisher expansion

Between "full Monte Carlo" and "pure Gaussian" sits the *Cornish-Fisher expansion*, a semi-parametric trick that adjusts the Gaussian quantile for skewness and excess kurtosis without a full simulation. Express the true quantile of a non-Gaussian distribution as a series in the standard Gaussian quantile:

$$q_\alpha \approx z_\alpha + \tfrac{1}{6}(z_\alpha^2 - 1)\, s + \tfrac{1}{24}(z_\alpha^3 - 3z_\alpha)\, \kappa - \tfrac{1}{36}(2z_\alpha^3 - 5z_\alpha)\, s^2, \tag{9.4}$$

where z_α is the Gaussian quantile, s is skewness, and κ is excess kurtosis of the target distribution. Multiplying by the portfolio standard deviation gives a skew-and-kurt-adjusted VaR. The formula captures most of the correction needed for modestly non-Gaussian P&L distributions. For heavily non-Gaussian distributions — deep fat tails, severe skewness near crash events — Cornish-Fisher breaks down and Monte Carlo or historical simulation is required.

The practical appeal is enormous when combined with delta-gamma parametric VaR. The quadratic form in (9.3) has skewness and kurtosis computable in closed form from δ, Γ, and Σ. Feeding those moments into (9.4) yields a VaR that incorporates both non-linearity (via Γ) and distributional tail adjustment (via skew/kurt), without a single Monte Carlo path. This delta-gamma-Cornish-Fisher VaR was widely used in the 1990s and 2000s and remains a useful benchmark.

Cornish-Fisher VaR vs Normal *The Cornish-Fisher quantile multiplier at* 99% *confidence as a function of skewness (with excess kurtosis fixed at* $\kappa_{\text{ex}} = 3$*). Negative skew (left tail heavy) raises the multiplier above the Normal* $z_{99\%} \approx 2.33$*; positive skew pulls it down. A quadratic shape in* s *reflects the* s^2 *term in* (9.4).

Two caveats. First, the expansion can produce non-monotonic quantile functions for extreme parameter values (very fat tails, severe skew), which makes no distributional sense. When this happens, the expansion is signalling that the perturbation from Normal is too large to be captured by a polynomial correction, and one must move to a full parametric or non-parametric method. Second, skewness and excess kurtosis are themselves very noisy to estimate from short time series — sampling standard errors are $\sqrt{6/N}$ and $\sqrt{24/N}$ respectively for Gaussian data, meaning skew and kurt estimates from 250 days are nearly useless. In practice, the moments are estimated from multi-year histories or from a time-series model (GARCH, for example) that yields more efficient estimates.

10.2.3 9.2.3 Historical provenance and interpretive tensions

VaR rose to prominence through the J.P. Morgan RiskMetrics framework in the early 1990s and was rapidly adopted by the Basel Committee as the regulatory standard for market-risk capital. The appeal was operational: VaR is easy to compute, easy to explain to non-technical audiences, and easy to aggregate across desks under simple assumptions. Chairman Dennis Weatherstone is famously said to have asked his risk team for a single number summarising how much the bank could lose in the next 24 hours — the "4:15 report" — and VaR was the answer. The sub-additivity failure and the blindness to tail shape were known from the start, but the practical dominance came from digestibility rather than theoretical purity. It took two decades, and the 2008 crisis, for the regulatory consensus to shift meaningfully toward expected shortfall.

An informative framing of VaR is as a *test statistic*: we ask whether actual daily losses exceed the reported VaR more often than advertised, and a model with systematic exceedance mismatch is rejected. VaR's signal-to-noise ratio for

rejecting a bad model is actually good because exceedances are binary events whose count follows a simple binomial distribution. CVaR/CTE, by contrast, requires averaging over exceedances and is statistically harder to backtest. There is a pragmatic tension: CVaR is theoretically superior but statistically harder to verify; VaR is theoretically flawed but easier to subject to empirical discipline. The modern regulatory compromise — running both, backtesting VaR, using CVaR for capital — reflects this tension.

One final practical note. Sell-side VaR reports typically decompose the aggregate figure into contributions from each desk, asset class, region, and risk factor. These decompositions are not unique because of the non-linearity of quantiles; see §9.9 for the full treatment.

10.3 9.3 Conditional Tail Expectation and Expected Shortfall

$$\mathrm{CTE}_\alpha(L) \;=\; \mathrm{ES}_\alpha(L) \;=\; \mathbb{E}\big[\, L \,\big|\, L \geq \mathrm{VaR}_\alpha(L) \,\big]\,. \tag{9.5}$$

CTE ("conditional tail expectation") and ES ("expected shortfall") refer to the same quantity. Where VaR only marks the threshold, CTE integrates the tail *beyond* it, so the two 5%-tail scenarios from §9.2 — one fat, one thin — earn different CTE scores. CTE is coherent: monotone, positively homogeneous, translation-invariant, and — crucially — sub-additive. The sub-additivity is what we need for any risk axis on which diversification is guaranteed to help.

Normal versus fat tail. The thin-tailed Normal has VaR ≈ 1.64 and CTE ≈ 2.06 at 95% — a gap of 0.4. For a Student-t_3 *rescaled to unit variance* (i.e. the base t_3 divided by $\sqrt{\nu/(\nu-2)} = \sqrt{3}$ so both distributions share zero mean and unit variance), the 95% VaR is $\approx 2.35/\sqrt{3}\cdot\sqrt{3} = 2.35$ (retaining the base-t_3 quantile $q_{0.05} \approx 2.353$ for this comparison since the figure plots same-scale densities) and the CTE is ≈ 4.19 — a gap of 1.8, nearly five times larger. (On the unscaled base-t_3 convention, VaR ≈ 2.35 and CTE ≈ 2.34; the qualitative "CTE exceeds VaR by much more in the fat-tailed case" narrative holds under either convention, but the specific 4.19 number uses the unit-variance rescaling so that the comparison with the unit-variance Normal is apples-to-apples.) The practical point: two books can share the same VaR yet have wildly different CTE, which is precisely the pathology CTE was designed to catch. In closed form for Student-t with ν degrees of freedom,

$$\mathrm{CTE}_\alpha^{t_\nu} = \frac{f_{t_\nu}(\mathrm{VaR}_\alpha^{t_\nu})\,(\nu + (\mathrm{VaR}_\alpha^{t_\nu})^2)}{(\nu-1)\,\alpha}, \tag{9.6}$$

where f_{t_ν} is the Student-t density. The key observation is the $(\nu-1)$ denominator: as $\nu \to 1$ (Cauchy), CTE diverges while VaR remains finite. Even at $\nu = 2$, CTE is finite but much larger than VaR; at $\nu = 3$ we get the 1.8-gap quoted above. As $\nu \to \infty$ the distribution approaches Normal and the gap converges to the Normal value of 0.4. The tail-index ν interpolates smoothly between "Normal-like" and "catastrophically fat-tailed", and CTE tracks the deterioration faithfully while VaR only modestly responds.

Why the switch happened. Empirically estimated tail indices for daily equity returns sit in the range 3-5 (4-7 for weekly); emerging-market currencies and distressed credit can drop to 2, making CTE much larger than VaR by a factor of several. Any risk framework that relies on Normal-CTE benchmarks systematically under-reports tail risk in these markets, and the under-reporting compounds during crisis periods when the tail index temporarily shrinks further. Every blow-up in the historical record of hedge funds and investment banks — and a depressing number of insurance companies — can be traced, at least partially, to a VaR that looked fine alongside a CTE that would have screamed. The regulatory transition from 99% VaR charges toward 97.5% CTE charges was a direct response.

The 97.5% CTE calibration deserves explanation. For a Gaussian loss distribution, $\mathrm{CTE}_{97.5\%} \approx \mathrm{VaR}_{99\%}$, so the switch was designed to leave the average capital charge roughly unchanged but to make the charge responsive to tail shape. Fat-tailed portfolios now face genuinely higher capital requirements than thin-tailed portfolios with the same 99% VaR. This is the regulatory incentive-alignment the Basel Committee wanted: portfolios whose tails actually matter are capitalised accordingly.

Estimation is harder. CTE estimation has a statistical subtlety that VaR estimation does not. VaR is a quantile, determined by the local behaviour of the CDF around a single point; estimation depends on the density at that point. CTE is an integral over the tail, so estimation depends on the density *everywhere* in the tail, and density estimates at extreme quantiles have high variance. A 99% CTE from 250 daily returns is the mean of the 2.5 worst returns — so it depends heavily on the single *worst* observation in the sample. If the sample contains a mild outlier like -4% the estimate comes in around -3.5%; if it contains a catastrophic -12% like September 2008, the estimate jumps to

-7% or worse. Two samples drawn from the same underlying distribution can give CTE estimates that differ by a factor of two even when their VaR estimates agree within basis points.

This sensitivity is a blessing and a curse. Blessing: CTE correctly reflects genuine tail events when they occur — the measure is *doing its job* by responding to extreme observations. Curse: CTE estimates have wide confidence intervals, which makes it hard to tell whether a day-over-day change reflects genuine risk change or sampling noise. The pragmatic response is to model the tail parametrically using a Generalised Pareto (GPD) fit to excesses over a threshold — the *peaks-over-threshold* extreme-value technique developed in §9.4. A GPD fit extrapolates the tail smoothly beyond observed data, giving much more stable CTE at the cost of a modest parametric assumption.

Elicitability and model comparison. A risk measure is *elicitable* if it can be expressed as the minimiser of an expected scoring function. VaR is elicitable (via quantile loss); CTE is not elicitable in general. This matters for model comparison: to compare two VaR forecasts we can use average quantile loss, and a lower score is unambiguously better. For CTE no such scoring function exists, and comparing two CTE models rigorously requires more elaborate tests — the Fissler-Ziegel joint elicitability of VaR and CTE, for instance. Regulators continue to debate whether elicitability is essential; the pragmatic view is that it is not, but its absence complicates the model-validation toolkit.

CTE as a portfolio-construction axis. The VaR-minimising portfolio on an efficient frontier is not generally the CTE-minimising portfolio on the same frontier. Empirical studies have found that mean-CTE-optimised portfolios can have 20-40% lower crash losses than mean-VaR-optimised portfolios even when their mean-variance characteristics are similar. This is a second reason for the regulatory shift toward CTE beyond measurement fidelity: CTE actively produces better portfolios when used as an optimisation objective.

VaR vs CTE — thin-tail vs fat-tail *VaR and CTE compared across Normal (thin tail) and Student-t_3 (fat tail) — same quantile can hide very different tail masses.*

10.4 9.4 Thin vs Fat Tails: Student-t and Extreme Value Theory

Normal vs Student-t3 densities with VaR and CTE marked *Same scale, very different tails. The 95% VaR cutoffs differ modestly (Normal* 1.64 *vs Student-t_3* 2.35*), but the CTE values diverge dramatically (*2.06 *vs* 4.19*). VaR is a threshold; CTE integrates the tail beyond it, and the ratio CTE/VaR is diagnostic of tail-fatness.*

The Normal versus Student-t_3 contrast from §9.3 drives much of modern market-risk practice. Under a Normal assumption, losses beyond three standard deviations occur with probability roughly 0.13% per day — one every three years on a 250-day calendar. Under a Student-t_3 with matching scale, the same three-sigma loss occurs with probability roughly 2.5% — about twenty times more often, or six a year. The operating reality of equity indices, emerging-market currencies, corporate credit spreads, and essentially every traded market during crisis periods is much closer to the Student-t picture than to the Normal picture. A desk that models losses as Normal is, in effect, ignoring roughly 95% of the tail mass that actually exists.

This is why Normal-residual VaR is so treacherous. If the volatility estimate is correct but the distributional assumption is wrong, the VaR number still comes out a plausible-looking dollar amount. The problem is that the *next* number up the tail — the CTE — is wildly mis-estimated, and so is the frequency of VaR exceedances. In 2008 and again in March 2020, major sell-side desks reported multi-day runs of 5 , 7 , even 10 moves — events that under the Normal framework should occur roughly once per age of the universe. What had happened was not that the universe grew old in a week; it was that the tail distribution was never Normal to begin with.

The sigma trap. Under a Normal distribution, one sigma captures 68% of observations, two sigma 95%, three sigma 99.7%. Under Student-t_3 the same percentages correspond to different sigma-multiples, and extreme moves are common enough that a single "10-sigma" observation carries essentially no information: it could be a 10-sigma Normal event (probability $\sim 10^{-23}$) or it could be a perfectly representative Student-t_3 tail observation (probability $\sim 10^{-3}$). A risk manager who reports "a 10-sigma move happened today" is implicitly assuming Normal tails; under the more realistic Student-t_3, such moves happen every few years. The appropriate response is not "that's impossible, something must have broken" but "my Normal assumption is wrong, let me fit a fatter-tailed distribution and recompute my risk."

Training risk teams to communicate in tail-index terms rather than raw sigma terms is part of the cultural shift that followed 2008, and it has spread unevenly across the industry. The most sophisticated shops now report "tail-index-adjusted sigma" alongside Normal sigma for crisis diagnostics. When a senior manager asks "was that a 7-sigma event?" the honest answer is "in the tail-index $\nu = 3$ sense it was a 2-sigma event; in the Normal sense it would be 7-sigma but that framework does not describe this market."

10.4.1 9.4.1 Defensive adjustments

Three practical adjustments improve on Normal VaR without requiring a full rewrite of the risk infrastructure.

Parametric fat-tail fit. Fit losses with a Student-t, skew-t, or Generalised Hyperbolic distribution, estimate the degrees of freedom from the data, and recompute VaR and CTE under the fitted distribution. Even a rough fit with $\nu = 5$ or $\nu = 6$ captures most of the relevant fat-tail behaviour. Maximum-likelihood or method-of-moments estimation on a 3-5 year daily history gives reasonably stable parameters; shorter windows give noisy ν estimates that can swing between 3 and 10 over successive weeks.

Empirical historical simulation. Compute VaR and CTE directly from the realised P&L history. This by construction inherits whatever tail the market produced during the lookback window. It is the simplest way to avoid mis-specifying the tail because it does not specify the tail at all — but it equally cannot extrapolate beyond the historical sample.

Explicit stress scenarios. Layer historical stress scenarios (Black Monday 1987, Lehman 2008, COVID 2020) and hypothetical scenarios chosen by the risk committee on top of the parametric tail as a form of risk insurance. The full stress-testing methodology is developed in §9.8.

10.4.2 9.4.2 Extreme Value Theory

A more sophisticated methodology is *Extreme Value Theory (EVT)*, which rests on a fundamental asymptotic result: the Pickands-Balkema-de Haan theorem. For a wide class of loss distributions, the conditional distribution of exceedances over a high threshold u converges to the Generalised Pareto Distribution (GPD) as $u \to \infty$:

$$F_u(y) \;=\; \mathbb{P}(L - u \le y \,|\, L > u) \;\approx\; 1 - \left(1 + \xi \frac{y}{\beta}\right)^{-1/\xi}, \quad y > 0, \tag{9.7}$$

with shape parameter ξ and scale parameter $\beta > 0$. The sign of ξ classifies the tail: $\xi > 0$ is Fréchet (power-law tails, infinite moments above order $1/\xi$), $\xi = 0$ is Gumbel (exponential tails), $\xi < 0$ is Weibull (bounded tails). Equity indices typically give ξ in the range 0.2-0.4; sovereign-rate moves give ξ near 0; commodity returns can give $\xi > 0.5$ during shortage regimes.

The *peaks-over-threshold* (POT) procedure is: pick a threshold u (often the 90th or 95th percentile of observed losses), collect the exceedances $\{L_i - u : L_i > u\}$, fit (ξ, β) by maximum likelihood, and extrapolate the fitted GPD to obtain tail quantiles and conditional tail expectations at confidence levels far beyond the empirical data. A related technique uses the *Hill estimator* to estimate ξ directly from the ratio of the largest order statistics of the sample — simpler to implement but less efficient than MLE.

The advantage of EVT over pure historical simulation is smooth extrapolation beyond the observed data, giving stable estimates even at extreme confidence levels (99.9%, 99.99%) where raw historical data contains only a handful of observations. The disadvantage is that EVT requires a threshold choice (too low and the asymptotic theorem fails to apply well; too high and the fit loses data) and is sensitive to the estimated shape ξ, which has its own uncertainty.

10.4.3 9.4.3 GARCH and conditional dynamics

A complementary framework is the *GARCH* family of volatility models. GARCH (and its variants EGARCH, GJR-GARCH, TGARCH) captures the *clustering* of volatility — high-vol periods persist — and the asymmetry of vol response to signed shocks (the leverage effect, where bad news raises vol more than equally-sized good news). A GARCH-filtered historical simulation takes GARCH residuals (returns divided by conditional volatility), samples from the residual empirical distribution, rescales by the forecast volatility, and computes VaR. This hybrid combines the parametric strength of GARCH in modelling volatility dynamics with the non-parametric strength of empirical residuals in capturing fat tails, and is widely considered a gold-standard methodology for equity and FX risk.

The chapter's emphasis on VaR and CTE as *different* summaries of the same loss distribution is really an emphasis on the limits of any single-number summary. The loss distribution is the primary object; VaR and CTE are two projections of it onto the real line. When the distribution is Normal, VaR and CTE at any level are almost interchangeable. When it is fat-tailed, they diverge dramatically, and the divergence itself is diagnostic: a VaR-to-CTE ratio well above the Gaussian benchmark is a red flag that the tail is mis-modelled.

10.5 9.5 Basel Context and Regulatory Capital

The regulatory use of VaR and CTE deserves its own section because the acronym soup — Basel I, II, III, FRTB, IMA, SA, IRB, SMA — is impenetrable without a map, and because the regulatory framework shapes which risk measures have operational primacy at large banks.

Basel I (1988). The original Basel Accord was a credit-risk framework, introducing risk-weighted assets and minimum capital ratios (8% total capital, 4% Tier 1). It did not address market risk directly.

Basel Market Risk Amendment (1996). Introduced VaR-based capital charges for the trading book. Banks could use an *internal models approach* (IMA), reporting a 10-day 99% VaR subject to regulatory approval and backtesting. The capital charge was a multiple ($k \geq 3$) of the 60-day rolling average VaR, with the multiplier increasing based on backtest exceptions over the preceding year. This "traffic light" framework — green zone up to 4 exceptions, yellow 5-9, red 10+ — tied capital directly to model performance and was the industry's first large-scale exposure to formal backtesting methodology. The amendment ushered in the era of internal VaR models as the standard risk apparatus of large banks.

Basel II (2004). Refined credit risk charges and added operational risk, retaining the market-risk framework largely unchanged. The key innovation was the ability to use internal ratings-based (IRB) models for credit risk, analogous to IMA for market risk.

Basel 2.5 (2009). In direct response to the 2008 crisis, Basel added a *stressed VaR* charge — VaR computed using market data from a stressed historical period (typically 2008) — on top of the regular VaR. The logic: 2007 VaR figures for banks with large mortgage exposures had failed to anticipate the 2008 catastrophe because the 60-day lookback window contained only pre-crisis data. Stressed VaR forces inclusion of an explicitly bad historical period. Basel 2.5 also introduced the *Incremental Risk Charge* for credit migration and default risk in the trading book, and a *Comprehensive Risk Measure* for securitisations.

Basel III (2010/2013). Focused mainly on bank-wide capital adequacy, leverage ratios, and liquidity (LCR, NSFR), but kept the market-risk framework intact pending the fundamental review.

Fundamental Review of the Trading Book (FRTB, 2016/2019). The big one. FRTB replaced 99% VaR with 97.5% expected shortfall as the internal-model risk measure — an explicit regulatory embrace of the CTE framework developed above. FRTB also introduced:

- *Liquidity horizons.* Different capital horizons for different asset classes — 10 days for liquid equities, 20-40 days for investment-grade credit, 60-120 days for illiquid exotic credit and structured products. Capital for illiquid instruments is computed at the longer horizon, capturing the extended exposure to adverse moves during an assumed unwind period.
- *Non-modellable risk factors (NMRF).* Risk factors for which insufficient price data exists to model them internally must be capitalised via a stressed scenario charge rather than through the IMA.
- *P&L attribution test.* Banks' hypothetical P&L (from the risk model using actual market moves) must track actual P&L (from the trading systems) within tight correlation and residual-variance thresholds. A failing P&L attribution forces use of the standardised approach, which typically carries higher charges.

The switch from 99% VaR to 97.5% ES was calibrated so that under a Normal loss distribution the two produce approximately the same capital charge — the relevant benchmark is $\mathrm{CTE}_{97.5\%} \approx \mathrm{VaR}_{99\%}$ for a Gaussian. The point of the switch was not to change the average charge but to make it responsive to tail shape: fat-tailed portfolios now face genuinely higher capital requirements than thin-tailed portfolios with the same 99% VaR, which is exactly the regulatory incentive-alignment the Basel Committee wanted.

Standardised approach (SA). Parallel to IMA, banks can use a standardised approach based on regulator-prescribed risk weights, correlation assumptions, and aggregation formulas. SA is simpler but typically produces higher capital, so well-capitalised banks with sophisticated risk teams use IMA while smaller or less-sophisticated banks use SA. FRTB tightened SA calibration to narrow the gap between SA and IMA and to make SA a credible fallback if a bank's IMA is revoked for poor backtesting.

The regulatory machinery matters because it directly determines which risk measures have operational primacy. A bank whose regulatory capital is driven by CTE/ES at 97.5% over multiple liquidity horizons will build its entire risk infrastructure around those measures — daily reporting, limit-setting, attribution, stress testing, all in CTE/ES terms. Banks still on Basel 2.5 VaR + stressed VaR will continue to think in VaR terms until their jurisdiction

catches up to FRTB. The mathematical treatment in this chapter is equally applicable to either regime; the choice of which measure to track in the daily risk report is regulatory rather than mathematical.

10.6 9.6 The Coherent Risk Framework

A risk measure ρ acting on a space of random losses is *coherent* in the Artzner-Delbaen-Eber-Heath sense if it satisfies four axioms. We state them in the loss convention (large positive = painful), and unpack each in turn.

Monotonicity (M). $L_1 \leq L_2$ almost surely $\Rightarrow \rho(L_1) \leq \rho(L_2)$. A strictly smaller-loss random variable must have a smaller risk measure. This is the bare-minimum sanity requirement and both VaR and CTE satisfy it. (In the *profit* convention used by Artzner et al.'s original paper and some other sources, monotonicity reads $X_1 \leq X_2 \Rightarrow \rho(X_1) \geq \rho(X_2)$ — bigger wealth means smaller risk — so readers cross-referencing should flip the inequality alongside the sign convention.)

Translation invariance (TI). $\rho(L + c) = \rho(L) + c$ for any constant c. Adding a deterministic dollar of loss adds a dollar of risk. Note the sign: adding a deterministic dollar of *profit* (so $c < 0$) reduces the risk by a dollar — this is the "cash can be netted against risk" statement underlying capital-charge accounting.

Positive homogeneity (PH). $\rho(\lambda L) = \lambda\rho(L)$ for $\lambda \geq 0$. Doubling every position doubles the risk. Both VaR and CTE comply. One can argue the axiom is too clean — very large positions face liquidity haircuts and $\rho(10L)$ should probably exceed $10\rho(L)$ — but the classical framework ignores such frictions.

Sub-additivity (S). $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$. This formalises "diversification never hurts." It is what VaR fails and CTE (under mild regularity conditions) obeys.

10.6.1 9.6.1 The sub-additivity failure of VaR

To see the VaR failure concretely, take two independent loans, each of which defaults with probability 4% producing a loss of 100, and otherwise pays zero. Individually, the 95% VaR of each is 0 — a 4% tail event does not breach the 5% confidence line. But the sum can default twice, once, or not at all. The probability of at least one default is roughly 8%, of two defaults about 0.16%, so the 95% VaR of the merged book is around 100, not 0. Merging two risks whose individual VaRs are zero gave a combined VaR of 100.

VaR sub-additivity failure on two independent loans *Monte-Carlo verification of the two-loan thought experiment: each loan has $VaR_{95} = 0$ in isolation, but their sum has $VaR_{95} \approx 100$. CTE is coherent — the merged CTE is bounded by the sum of individual CTEs — and so can serve as a risk-budgeting axis where VaR cannot.*

The pathology is generic to any loss distribution with mass concentrated just inside the VaR threshold. It afflicts credit books particularly badly because individual defaults are exactly such low-probability-high-impact events: an investment-grade bond has a 1-year default probability under 1%, well below the 5% VaR threshold, so its individual VaR is zero. A portfolio of thousands of such bonds has a default frequency of tens per year and generates substantial portfolio VaR. If we used VaR to set capital on individual loans we would allocate zero, yet the portfolio clearly needs substantial capital. The CTE-based alternative assigns positive CTE even to individual low-probability defaults — because the tail expectation averages over the default scenario — and produces a portfolio CTE that equals the sum of individual CTEs minus diversification benefits. A much more coherent story.

A corollary of sub-additivity failure: VaR cannot serve as a direct basis for risk-budget allocation. If desk A has VaR \$1M and desk B has VaR \$1M, the firm's VaR is *not generally* \$2M or anything derivable from the individual VaRs. It could be \$0.5M (perfectly negatively correlated), \$1.4M (independent), \$2M (perfectly correlated), or \$2.2M (the pathological failure). The bank has no way of allocating \$2M of firm-level VaR to desk-level budgets using just the desk VaRs. CTE, being sub-additive, allows clean Euler-style decomposition, and this is part of the technical appeal of using CTE as the desk-level budgeting tool even when VaR is reported to regulators.

10.6.2 9.6.2 The dual representation and spectral measures

The foundational theorem of coherent risk is a *dual representation*: every coherent ρ can be written as a worst-case expectation over a set $\mathcal{Q}$ of probability measures,

$$\rho(L) \;=\; \sup_{\mathbb{Q}\in\mathcal{Q}} \mathbb{E}^{\mathbb{Q}}[L]. \tag{9.8}$$

This unifies many apparently different risk measures under a single umbrella: every coherent risk measure is a worst-case expectation over some class of scenarios. CTE at level α corresponds to taking $\mathcal{Q}$ to be the set of probability measures that are absolutely continuous with respect to the base measure with a Radon-Nikodym derivative bounded by $1/\alpha$. VaR does not admit such a representation because it is not coherent.

A richer parametric family is the class of *spectral risk measures*, each of which can be written as a weighted average of quantiles:

$$\rho_\phi(L) \;=\; \int_0^1 \phi(u)\, F_L^{-1}(u)\, du, \tag{9.9}$$

where $\phi : [0,1] \to [0,\infty)$ is a non-decreasing weighting function with $\int \phi = 1$. VaR at level α corresponds to ϕ being a Dirac delta at $u = 1-\alpha$ — all weight on a single quantile — and is not coherent because the spike violates monotonicity of ϕ. CTE at level α corresponds to uniform weighting $\phi(u) = \mathbf{1}_{u \geq 1-\alpha}/\alpha$ over the tail — coherent because ϕ is non-decreasing. One can imagine ϕ functions that assign even more aggressive weight to the extreme tail, producing risk measures that interpolate between "care only about one quantile" (VaR) and "care linearly about the whole tail mean" (CTE) and beyond into "care hyperbolically about the extreme tail." The desk-level upshot: CTE is a sensible default but not a unique answer; a catastrophe-reinsurance book might prefer a more tail-weighted spectral measure still.

10.6.3 9.6.3 Convex risk measures

A stronger class is *convex risk measures*, which relax positive homogeneity but retain a form of convexity:

$$\rho(\lambda L_1 + (1-\lambda)L_2) \;\leq\; \lambda\rho(L_1) + (1-\lambda)\rho(L_2), \quad \lambda \in [0,1]. \tag{9.10}$$

Convex risk measures can capture diseconomies of scale — $\rho(10L) > 10\rho(L)$ — that reflect liquidity penalties for large positions. The Föllmer-Schied framework generalises the dual representation to convex measures using a penalty function $\alpha : \mathcal{Q} \to \mathbb{R}$:

$$\rho(L) \;=\; \sup_{\mathbb{Q}} \left\{ \mathbb{E}^{\mathbb{Q}}[L] - \alpha(\mathbb{Q}) \right\}. \tag{9.11}$$

Several modern risk measures — *entropic risk measures* (exponential-utility-based), *utility-based shortfall*, and various parametric-family measures — are convex but not coherent. The details are beyond our scope here but worth flagging as a direction of research relevant to liquidity-adjusted risk measurement.

Coherence is a minimum bar, not the last word. The practical punchline is not that coherence is the complete theory — it isn't — but that coherence is a useful minimum bar below which risk measures produce obviously pathological decisions. Any framework that lets you lower your risk by merging two losing books is broken; coherence is the precise formalisation of "not broken." Measures that fail coherence can still be operationally useful in limited contexts (VaR as a communicable summary, for example), but they must be supplemented by a coherent companion (CTE for the tail shape, stress scenarios for the specific catastrophe) before the suite is diagnostically adequate.

10.7 9.7 Backtesting: Kupiec and Christoffersen

Once a VaR model is in production, it must be validated against realised P&L. The framework for this validation is *backtesting*: compare the daily sequence of reported VaRs against the daily sequence of realised P&Ls, count exceedances (days on which the realised loss exceeded the VaR), and apply statistical tests to decide whether the exceedance pattern is consistent with the model.

10.7.1 9.7.1 The exceedance indicator and its expected behaviour

Define the exceedance indicator

$$I_t \;=\; \mathbf{1}\{L_t > \mathrm{VaR}_{\alpha,t}\}. \tag{9.12}$$

If the VaR model is correctly calibrated at confidence level $(1-\alpha)$, then each I_t is a Bernoulli(α) random variable and the sequence $\{I_t\}$ is i.i.d. (independent because past exceedances should carry no information about today, identically distributed because the tail probability is by construction α every day). Both the "correct frequency" claim and the "correct independence" claim are empirically testable.

10.7.2 9.7.2 Kupiec's proportion-of-failures test

The simplest test — targeting frequency — is Kupiec's POF test. Over n days, let $x = \sum_t I_t$ be the number of exceedances. Under the null that the model is correct, $X \sim \text{Binomial}(n, \alpha)$ with mean $n\alpha$ and variance $n\alpha(1-\alpha)$. The likelihood-ratio statistic is

$$\text{LR}_{\text{POF}} \;=\; -2\log\left[\frac{\alpha^x(1-\alpha)^{n-x}}{\hat{p}^x(1-\hat{p})^{n-x}}\right], \qquad \hat{p} = \frac{x}{n}, \tag{9.13}$$

asymptotically χ^2_1 under the null. Reject at 5% significance if $\text{LR}_{\text{POF}} > 3.84$. For a 99% VaR with $n = 250$ days, expected exceedances are $n\alpha = 2.5$; the test rejects above about 7 exceedances or at 0 (which is paradoxically *too few*, suggesting the model is too conservative and over-capitalising the book).

This is the essence of the Basel "traffic light" regime. More than 4 exceedances in a 250-day window puts the model in the yellow zone (capital multiplier k rises from 3.00 by 0.4-0.85 depending on count); more than 9 exceedances puts it in the red zone (multiplier capped at 4.00, pending a full model revision). Capital is directly tied to backtest performance, giving banks a strong financial incentive to build models that actually work.

Kupiec POF acceptance band over a 250-day 99% VaR backtest *Binomial pmf of exceedance counts under the null* $X \sim \text{Binomial}(250, 0.01)$*, coloured green where the Kupiec likelihood-ratio is below the* 5% *critical value* 3.84 *and red where it rejects. Green bars span roughly* 0–6 *exceedances; beyond that the model is rejected as under-capitalised. Zero exceedances is technically also rejected as over-conservative.*

10.7.3 9.7.3 Christoffersen's independence test

Kupiec catches miscalibrated levels but misses *clustering* — a model whose exceedances come bunched together in crisis periods and sparse elsewhere, so the overall count looks right but the temporal pattern is wrong. Clustering indicates the model fails to capture volatility dynamics: on Monday the model reports a VaR that was appropriate for last week's calm conditions, the market blows out, Tuesday's VaR is still calibrated to last week, Wednesday the same, and suddenly you have four exceedances in four days because the dynamic volatility of the market was not reflected in the dynamic VaR of the model.

Christoffersen's test models the exceedance sequence as a two-state Markov chain. Let n_{ij} be the count of transitions from state i to state j (where 0 = no exceedance, 1 = exceedance). Under independence, the transition probability $\mathbb{P}(I_t = 1 \mid I_{t-1} = 1)$ equals the unconditional $\mathbb{P}(I_t = 1) = \alpha$. The likelihood-ratio test against a general Markov alternative is

$$\text{LR}_{\text{IND}} \;=\; -2\log\left[\frac{\hat{p}^x(1-\hat{p})^{n-x}}{\hat{p}_{01}^{n_{01}}(1-\hat{p}_{01})^{n_{00}}\hat{p}_{11}^{n_{11}}(1-\hat{p}_{11})^{n_{10}}}\right] \sim \chi^2_1, \tag{9.14}$$

where $\hat{p}_{i1} = n_{i1}/(n_{i0} + n_{i1})$. Combined with Kupiec, the *conditional coverage* test

$$\text{LR}_{\text{CC}} \;=\; \text{LR}_{\text{POF}} + \text{LR}_{\text{IND}} \;\sim\; \chi^2_2 \tag{9.15}$$

rejects the model if either the overall frequency is wrong or the clustering is significant, giving a comprehensive test of VaR adequacy. A failing LR_{IND} with passing LR_{POF} is a characteristic signature of a model using a slow-moving volatility estimator (rolling 1-year window, for example) in a market with pronounced vol clustering — the fix is typically to move to a GARCH-type conditional-volatility specification.

10.7.4 9.7.4 Backtesting CTE

CTE backtesting is considerably harder because CTE is not elicitable (§9.3). The simplest approach — and the one adopted in FRTB — is to backtest CTE indirectly by backtesting VaR at multiple confidence levels (typically 97.5% and 99%) and examining the distribution of exceedance magnitudes. A model whose VaR is correctly calibrated at both 97.5% and 99% is approximately correctly calibrated for CTE at 97.5%, because CTE is approximately an average over the range between those quantiles.

The *Acerbi-Szekely* framework proposes scoring functions for joint VaR/CTE backtesting. Three families of tests are commonly implemented: (i) a test on the magnitude of exceedances (the average exceedance should match the CTE minus VaR prediction); (ii) a test on the tail-conditional distribution of L relative to the model's predicted tail distribution (a sort of Kolmogorov-Smirnov test in the tail); (iii) a joint test on the first two moments of the exceedance distribution. These are more complex to implement but increasingly standard in banks' model-validation departments.

Practical note: CTE backtests have low statistical power unless the data history is long. On a 1-year backtest at 99% confidence, there are only 2-3 expected exceedances — essentially no usable information about the conditional expectation of the tail. Meaningful CTE backtests require 3-5 year histories or pooling across multiple portfolios, which is why the FRTB compromise of backtesting VaR at two levels and inferring CTE adequacy has dominated implementation.

The broad lesson: backtesting is as much a research area as a compliance exercise. Every risk measure has its own backtesting theory, and practitioners must choose tests that match the quantity being backtested, recognise the limitations of finite-sample power, and interpret results as diagnostic rather than purely pass/fail.

10.8 9.8 Stress Testing

Stress testing sits alongside VaR/CTE as the second major pillar of market-risk measurement. The premise is that statistical models calibrated on historical data fail precisely when history fails to contain the next catastrophe — so one should also ask "what if the stock market falls 30% in a week?" or "what if credit spreads widen to 2008 levels?" regardless of what the statistical model predicts.

10.8.1 9.8.1 Historical stress scenarios

Historical stress scenarios replay a specific historical event against the current portfolio, applying the market moves from that period as instantaneous shocks. Canonical scenarios include Black Monday 1987, the Asian Crisis 1998, LTCM autumn 1998, the Dot-Com crash of 2000-2001, Lehman September 2008, the Flash Crash of May 2010, the EuroZone sovereign crisis 2011-2012, the China devaluation of August 2015, Volmageddon in February 2018, the COVID-19 crash of March 2020, and the UK gilt crisis of September-October 2022.

The appeal of historical scenarios is that they are "real": everyone agrees that Lehman happened, and there is no arguing about the size of the moves. The limitation is that the next crisis will not be the last crisis; hedging against 2008 may not protect against a different structural breakdown. Every historical scenario also requires translation to today's portfolio — one cannot just apply 1987 S&P moves verbatim to today's equity book because today's book contains index futures that did not exist in 1987, options markets whose implied vol surface is unrecognisable from that era, and hedging relationships that behave differently under modern market microstructure. The translation introduces modelling judgement, and different risk teams make different calls.

10.8.2 9.8.2 Hypothetical stress scenarios

Hypothetical stress scenarios are designed by the risk committee to capture imagined but not-yet-realised catastrophes. Examples: "equity down 25%, vol up to 80, credit spreads +500 bp, USD up 10%, oil down 40%"; "Treasury yields rise 300 bp in a week"; "Bitcoin goes to zero overnight, dragging the tech-heavy NASDAQ down 15% on correlation contagion." These scenarios reflect institutional views about what could go wrong and stress the risks that are *not* in the historical record.

Regulatory stress tests — CCAR in the US, EBA stress tests in the EU, Bank of England stress tests in the UK — are hypothetical scenarios imposed on all banks simultaneously, with the bank having to show that its capital remains above regulatory minima even under the imposed stress. These exercises combine a macroeconomic narrative (unemployment rises 5 points, GDP contracts 3%, house prices fall 25%) with specific market-variable shocks implied

by the narrative, and force banks to project their balance sheet, P&L, and capital ratio through the stress horizon (typically 9 quarters).

10.8.3 9.8.3 Reverse stress testing

A third variety, increasingly emphasised by regulators, is *reverse stress testing*: rather than starting with a scenario and computing the loss, start with a loss (typically the failure threshold — the loss that would render the firm insolvent or non-viable) and work backward to identify scenarios that produce it. The output is not a capital number but a catalogue of scenarios whose realisation would destroy the firm, some of which may be obvious and others of which may surprise the risk committee.

Reverse stress testing is mathematically harder than forward stress because there are many scenarios that produce a given total loss — the problem is an inverse problem with a large null space. Practical implementations typically constrain the scenario space by factor structure (what combinations of factor shocks within 5 standard deviations produce failure?) or by qualitative narrative (what operating environments plausibly produce the shocks that produce failure?).

10.8.4 9.8.4 Scenario construction methodology

Building useful stress scenarios is its own craft. Core principles:

- *Internal consistency.* A scenario must be coherent across markets. An equity-down scenario usually has vol up, credit spreads wider, and safe-haven currencies (CHF, JPY, USD) stronger. Applying an equity-down shock in isolation without these co-moves gives an artificially small loss and misses the real tail risk.
- *Magnitude calibration.* Shocks should be large enough to matter but not so large as to be dismissed as absurd. A 20-sigma equity move on a Normal calibration is absurd but on a Student-t_3 calibration it is a routine crisis move. Calibrating shock sizes to fat-tailed distributions gives more credibility and more informative stress results.
- *Coverage of risk types.* The scenario suite should collectively exercise every risk factor the book is exposed to — not just the obvious ones. A scenario suite that stresses equity and rates but ignores FX, credit, vol, and correlation leaves systematic gaps.
- *Narrative plausibility.* Senior management engages with stress tests that come with a story ("oil price shock triggers recession, which triggers credit widening, which forces unwinds in levered funds, which cascades back into equities"). Pure statistical shocks without narrative are harder to action.

10.8.5 9.8.5 Economic capital and solvency horizons

Economic capital is the amount of capital the firm's internal models say is needed to remain solvent over a specified horizon (typically 1 year) at a specified confidence level (typically 99.9% — a 1-in-1000 year event). Economic capital differs from regulatory capital in several ways: it uses the firm's own modelling framework (not the regulator's), it runs at a much longer horizon and higher confidence level, and it aggregates across all risk types (market, credit, operational, business, strategic) whereas regulatory market-risk capital concerns only market risk. Economic capital is the basis for internal return-on-capital calculations, ratings-agency assessments, and internal reinsurance or capital-transfer decisions.

The relationship between stress testing and VaR/CTE is subtle. VaR/CTE report the *expected* catastrophe based on an assumed distribution; stress tests report a *specific* catastrophe imagined or observed. They are complementary, not substitutes. A portfolio with low VaR can have large stress P&L (vulnerable to a specific imagined scenario outside the historical record), and vice versa. A mature risk function reports both, and uses significant discrepancies between them as a diagnostic for hidden non-linearity or for over-reliance on a particular set of historical data.

10.9 9.9 Risk Decomposition

Once an aggregate VaR or CTE has been computed, the operational question is "which positions contribute to it, and by how much?" This is the *risk-decomposition* problem, and it has three principal answers: marginal, component, and incremental. Each answers a slightly different question, and a well-run risk function reports all three.

10.9.1 9.9.1 Marginal VaR

Marginal VaR of a position is the partial derivative of portfolio VaR with respect to the position size. It answers: if I add one more dollar of exposure to this position, how much does my VaR go up? Under the delta-normal model with portfolio volatility $\sigma_P = \sqrt{w^\top \Sigma w}$, the marginal VaR of position i is

$$\mathrm{mVaR}_i \;=\; \frac{\partial \mathrm{VaR}}{\partial w_i} \;=\; z_{1-\alpha}\,\frac{(\Sigma w)_i}{\sigma_P}, \tag{9.16}$$

where $(\Sigma w)_i$ is the covariance of position i's return with the portfolio return. Marginal VaR is natural for incremental trade decisions — it directly measures the risk impact of adding a little more of something. Sign is informative: a position with negative marginal VaR *reduces* portfolio risk when added (it is a net hedge), and the risk team should be happy to see it grow.

10.9.2 9.9.2 Component VaR

Component VaR is the contribution of a position to the portfolio VaR, obtained via Euler allocation:

$$\mathrm{cVaR}_i \;=\; w_i \cdot \mathrm{mVaR}_i, \qquad \sum_i \mathrm{cVaR}_i \;=\; \mathrm{VaR}. \tag{9.17}$$

The additive property is the key. Component VaRs sum to the total, which makes component VaR the natural basis for risk-budget allocation — each desk is given a budget, and the sum of desk-level VaRs equals firm-level VaR without residual. Under the delta-normal framework, component VaR is proportional to the position's covariance with the portfolio, scaled by portfolio size. Euler allocation generalises to any positively homogeneous risk measure, including CTE, so component CTE is defined analogously and sums exactly to portfolio CTE.

10.9.3 9.9.3 Incremental VaR

Incremental VaR of a position is the change in portfolio VaR if the position is removed entirely:

$$\mathrm{iVaR}_i \;=\; \mathrm{VaR}(\text{full portfolio}) - \mathrm{VaR}(\text{portfolio without position } i). \tag{9.18}$$

It is a finite-difference rather than an infinitesimal measure, and for large positions can differ significantly from component VaR. Incremental VaR is natural for unwind decisions — it directly measures "how much risk do I free up if I close this position today?" For small positions, incremental VaR and component VaR agree; for large positions (especially those with non-linear P&L), they diverge by factors of 2 or more. Incremental VaRs do not generally sum to the total, because removing one position changes the risk impact of every other position through the correlation matrix.

10.9.4 9.9.4 A worked contrast

Consider a portfolio consisting of a long stock position and a short call on the same stock — a covered call. Suppose the stock has component VaR \$1M and the short call has component VaR \$0.3M, summing to \$1.3M. But if we remove just the stock, the short call alone has VaR \$0.6M — the incremental VaR of removing the stock is \$0.7M. If we remove just the short call, the long stock alone has VaR \$1.2M — the incremental VaR of removing the call is \$0.1M. The differences reflect the hedging interaction between the two positions: each individually is riskier than its contribution to the pair because each partially hedges the other. The three attributions answer three different questions, and a risk team that reports only one is depriving its traders of meaningful optionality in trade sizing and unwind sequencing.

10.9.5 9.9.5 Non-linear portfolios and CTE decomposition

For portfolios with significant non-linearity (options, convertibles, structured products), the delta-normal formulas above are a first approximation. The Euler-allocation framework extends to CTE and to Monte-Carlo-based risk measures under mild regularity: if ρ is a positively homogeneous risk measure and the portfolio is parameterised by weights w, then

$$\rho(w) \;=\; \sum_i w_i \frac{\partial \rho}{\partial w_i}(w), \tag{9.19}$$

and $w_i \partial\rho/\partial w_i$ is the component contribution of position i. For a Monte Carlo CTE, the component contribution of a position can be estimated by running the simulation with and without an incremental position, or — more efficiently — by computing the conditional expectation of the position's P&L given that the portfolio is in the tail.

Risk decomposition answers the fundamental governance question "where does my risk come from?" Pairing it with the P&L attribution discussed next closes the loop between pre-trade forecasts and post-trade realisations.

10.10 9.10 P&L Attribution

A sibling discipline to risk decomposition is *P&L attribution*: after the trading day closes, decompose the realised P&L into contributions from each Greek and each market move. A canonical Greek-based attribution for a derivatives book reads

$$\Delta V \;=\; \delta \cdot \Delta S + \tfrac{1}{2}\Gamma(\Delta S)^2 + \nu \cdot \Delta\sigma + \theta \cdot \Delta t + \rho \cdot \Delta r + \text{cross-Greeks} + \text{cash flows} + \text{residual}, \tag{9.20}$$

where $\delta, \Gamma, \nu, \theta, \rho$ are the book's Greeks (see CH07 for definitions) and $\Delta S, \Delta\sigma, \Delta t, \Delta r$ are the realised moves in spot, implied vol, time, and rates. The "cross-Greeks" bucket collects contributions from vanna (spot-vol cross-gamma), volga (vol-of-vol), charm (spot-time), and speed (third-order in spot); the "cash flows" bucket captures discrete events such as dividends, coupons, and realised trade P&L from intra-day executions.

The "residual" — the piece the model cannot explain — is the diagnostic. It should be small. A systematically large residual is a red flag that either the Greeks are mis-computed, the book contains instruments not captured by the risk model, or the market moves include components (a basis dislocation, a vol-surface reshaping, a correlation break) that the factor representation does not span.

10.10.1 9.10.1 Hypothetical versus actual P&L

FRTB formalises attribution into a *P&L attribution test*. Two P&L series are produced each day:

- *Hypothetical P&L* — the P&L implied by the risk model using the actual market moves, with positions held constant across the day. This is what the risk model "thinks" the book earned.
- *Risk-theoretical P&L* — the P&L implied by the Greeks and the market moves via equation (9.20) (without the residual). This is the Taylor-expansion approximation to hypothetical P&L.
- *Actual P&L* — the realised P&L from the trading systems, including intra-day trading and any accruals the risk model doesn't capture.

The test compares hypothetical and risk-theoretical P&L and measures the correlation and the variance of the residual. A bank's internal model passes the P&L attribution test if the correlation is high (typically above 80%) and the residual variance is low (typically below some threshold relative to total P&L variance). Failing the test forces the bank to use the standardised approach for the affected desk, which typically carries higher capital charges. The test is a "no surprise" test: if the risk model is genuinely tracking the risk of the book, the hypothetical and risk-theoretical P&L should agree every day within sampling noise.

10.10.2 9.10.2 Attribution disputes and model-maintenance signals

In practice, attribution is where traders and risk managers spend a great deal of time talking past each other. The trader's P&L report (from the front-office system) and the risk desk's P&L report (from the risk engine) will rarely agree to the last dollar — and when they disagree by a material amount, the question is which number to believe. Some differences are legitimate and bookkeepable (FX conversion rates, intraday funding, stale prices); others are diagnostic of a genuine model-data problem (a mispriced exotic, an ignored correlation).

A systematic pattern in the residual is the most valuable piece of information attribution provides. If the residual is consistently positive on up-days and negative on down-days, the risk model is missing a positive gamma exposure somewhere. If the residual is systematically negative near expiry for positions that are being managed to zero, the risk model's decay profile is wrong. If the residual explodes on days with large vol moves, the vega or volga computation

is suspect. Treating attribution residual as pure noise misses the diagnostic signal and lets model errors compound unnoticed.

10.10.3 9.10.3 Greek decomposition versus factor decomposition

Two complementary views of attribution are common. *Greek decomposition* expresses P&L as in (9.20) — one term per Greek, times the corresponding market move. *Factor decomposition* expresses P&L as a sum over underlying risk factors, each factor contributing its net movement against the book's net sensitivity. The two views are mathematically related (a Greek on a derivative is the sensitivity to the factor that drives the derivative) but organisationally distinct: Greek decomposition is natural for a derivatives trader thinking about hedging; factor decomposition is natural for a portfolio manager thinking about exposures and hedges across asset classes.

Together, risk decomposition (pre-trade, "how much risk does this add?") and P&L attribution (post-trade, "where did the P&L come from?") form the bookends of a complete risk-management cycle. The pre-trade forecast and the post-trade realisation should agree on average and diverge only within known bounds; when they diverge outside those bounds, the discrepancy triggers either a trading adjustment (the book is doing something the traders did not intend) or a model revision (the risk framework is missing something).

10.11 9.11 Liquidity-Adjusted Risk

The classical VaR/CTE framework assumes that positions can be liquidated at mid-market prices with no market impact over the VaR horizon. This assumption breaks down for illiquid instruments — distressed credit, emerging-market equities, complex structured products — where selling in a hurry carries material bid-ask cost or even an inability to transact at any price.

10.11.1 9.11.1 Exogenous liquidity cost

The simplest form of liquidity adjustment adds a fixed bid-ask haircut to the VaR:

$$\mathrm{LVaR}_\alpha \;=\; \mathrm{VaR}_\alpha + \tfrac{1}{2}\,|w| \cdot \mathrm{spread}, \tag{9.21}$$

where $|w|$ is the absolute position size and "spread" is the bid-ask spread as a fraction of mid. The factor of one-half reflects that half the spread is the cost of liquidating (selling at the bid from mid) and the other half is the cost of re-entering if the position is repurchased. More refined versions add a term for the *volatility of the spread* itself — widening bid-ask in stressed markets — by treating the spread as a random variable and adding a quantile of its distribution to the VaR.

10.11.2 9.11.2 Endogenous market impact

For large positions in any market, the act of liquidating moves prices. The simplest model is a square-root market-impact law: the price impact of unwinding Q shares over a horizon h is approximately $k\sigma\sqrt{Q/(Vh)}$, where V is average daily volume and k is an asset-class-specific constant (0.5-1.5 typical for equities). Incorporating this into VaR adds a market-impact charge that scales non-linearly with position size, capturing the economic reality that doubling a position more than doubles the unwind cost.

10.11.3 9.11.3 Liquidity horizons in FRTB

FRTB institutionalises liquidity adjustment by prescribing *liquidity horizons* by asset class rather than computing bid-ask directly. The idea is to capture the extended period of price exposure during an assumed unwind rather than the instantaneous cost:

- 10 days: major currencies, large-cap equities, interest rates in G10 currencies.
- 20 days: emerging-market FX and sovereigns, small-cap equities.
- 40 days: investment-grade corporate credit.
- 60 days: high-yield corporate credit, structured products.
- 120 days: exotic credit, private-equity-like structured positions.

Capital for an instrument is computed as a 10-day ES times a liquidity adjustment that scales with the horizon. The formula is constructed so that positions with longer liquidity horizons accumulate capital proportional to their

extended exposure. An illiquid corporate bond whose prescribed liquidity horizon is 60 days carries six times the exposure window of a liquid equity and correspondingly higher capital.

10.11.4 9.11.4 Funding liquidity risk

Another dimension of liquidity risk is *funding liquidity* — the risk that the firm cannot finance its positions at a reasonable cost. A bank whose short-term funding dries up may be forced to unwind positions at fire-sale prices even if the positions themselves are mark-to-market profitable. Funding liquidity risk is usually managed separately from market risk via the *Liquidity Coverage Ratio (LCR)*, the *Net Stable Funding Ratio (NSFR)*, and scenario-based stress analyses, but the two risks interact in crises: a liquidity shock forces unwinds, the unwinds move prices, the price moves trigger further margin calls, and so on.

The Lehman collapse of September 2008 and the COVID liquidity crunch of March 2020 both featured this kind of doom-loop: triple-A rated corporate bonds trading at 70 cents on the dollar in March 2020 not because their credit had deteriorated but because every leveraged fund was simultaneously trying to sell them into a market with no buyers. Modern regulatory frameworks (LCR, NSFR, FRTB's non-modellable risk factor charge) attempt to capture the interaction via explicit scenario-based capital charges, but the fundamental challenge remains: liquidity risk is emergent and correlation-driven, and a single-instrument VaR cannot easily capture it.

10.12 9.12 Delta-Gamma Parametric VaR

The delta-gamma framework (see CH07 for the derivation of the Greeks themselves) delivers a specific computational payoff: *delta-gamma parametric VaR*. This technique was the workhorse of risk systems in the 1990s and 2000s and remains important today as a fast first-pass estimate even when full revaluation is available. It is a concrete application of the quadratic Taylor expansion of book P&L combined with the Cornish-Fisher expansion of §9.2.

10.12.1 9.12.1 The quadratic approximation

For a portfolio of options and hedges on n risk factors $r = (r_1, \dots, r_n)$, expand the portfolio value $V(r)$ to second order around the current factor vector r_0:

$$\Delta V(r) \;\approx\; \delta^\top \Delta r + \tfrac{1}{2}\, \Delta r^\top \Gamma\, \Delta r, \tag{9.22}$$

where $\delta_i = \partial V / \partial r_i$ and $\Gamma_{ij} = \partial^2 V / \partial r_i \partial r_j$ evaluated at r_0. For an equity option book, r includes spots, implied vols, and rates; δ collects the deltas, vegas, and rhos; and Γ collects the gammas, vannas, volgas, and cross-sensitivities. The approximation is accurate for moderate factor moves and degrades for very large moves where cubic and higher-order terms matter.

10.12.2 9.12.2 Moments of the quadratic form under Normal factors

Assume $\Delta r \sim \mathcal{N}(0, \Sigma)$ with covariance matrix Σ. The quadratic form has a closed-form mean, variance, and higher moments. The first two are

$$\mu_P \;=\; \mathbb{E}[\Delta V] \;=\; \tfrac{1}{2}\, \mathrm{tr}(\Gamma\Sigma), \tag{9.23}$$

$$\sigma_P^2 \;=\; \mathrm{Var}[\Delta V] \;=\; \delta^\top \Sigma\, \delta + \tfrac{1}{2}\, \mathrm{tr}((\Gamma\Sigma)^2). \tag{9.24}$$

The mean is non-zero whenever Γ is non-zero: a long-gamma book has positive expected P&L from the gamma convexity even under zero-mean factor moves, and vice versa for a short-gamma book. The variance has a linear-sensitivity piece $\delta^\top \Sigma \delta$ (the delta-normal term) plus a quadratic-sensitivity piece $\frac{1}{2}\mathrm{tr}((\Gamma\Sigma)^2)$ (the gamma contribution).

The third and fourth central moments of ΔV, used for Cornish-Fisher, are

$$m_3 \;=\; \mathrm{tr}((\Gamma\Sigma)^3) + 3\, \delta^\top \Sigma\, \Gamma\, \Sigma\, \delta, \tag{9.25}$$

$$m_4 \;=\; 3 m_3 \frac{m_2}{m_1 \cdot (\text{something complicated})} \dots$$

Rather than wade through the full fourth-moment algebra, note that in practice these are computed numerically — the eigen-decomposition of $\Gamma^{1/2}\Sigma\Gamma^{1/2}$ gives a diagonal representation in which the quadratic form becomes a weighted sum of independent chi-squared random variables, and arbitrary moments follow from sums over the eigenvalues.

10.12.3 9.12.3 Cornish-Fisher VaR

Feed the standardised skewness $s = m_3/\sigma_P^3$ and excess kurtosis $\kappa = m_4/\sigma_P^4 - 3$ into the Cornish-Fisher expansion (9.4):

$$\mathrm{VaR}_\alpha \;\approx\; -\mu_P + \sigma_P \left[z_\alpha + \tfrac{1}{6}(z_\alpha^2 - 1)\, s + \tfrac{1}{24}(z_\alpha^3 - 3z_\alpha)\, \kappa \right]. \tag{9.26}$$

The whole computation is $O(n^2)$ in the number of risk factors for the moment evaluation and $O(1)$ for the Cornish-Fisher adjustment — much cheaper than a full Monte Carlo revaluation. A portfolio with 500 risk factors and 10,000 options can have its delta-gamma-Cornish-Fisher VaR computed in under a second on a modern workstation; a full Monte Carlo revaluation of the same portfolio might take minutes or hours depending on the sophistication of the option pricers. The speed differential is the reason delta-gamma VaR survives even in the full-revaluation era: it is what the intraday risk engine runs, while the overnight batch runs the full Monte Carlo.

10.12.4 9.12.4 Accuracy regime

The accuracy of delta-gamma VaR depends on the magnitude of the stress move relative to the curvature of the payoff:

- *Daily 95% VaR, moderate vol.* The approximation is typically within a few percent of a full-revaluation VaR. Perfectly adequate for intraday limit monitoring and desk-level reporting.
- *Daily 99% VaR, stressed conditions.* Accuracy degrades to 5-15% typical error. Still useful as a first-pass indicator but should be validated against full revaluation before being used for capital.
- *10-day 99% VaR with options near expiry.* Can degrade severely because near-expiry gamma is explosive and the quadratic approximation misses third-order effects. Full revaluation or Monte Carlo is mandatory here.
- *Deeply out-of-the-money options or digital-like payoffs.* Delta-gamma breaks down entirely because the payoff is nearly discontinuous. Full revaluation is required.

Modern risk systems run delta-gamma-Cornish-Fisher VaR for intraday risk updates and full-revaluation Monte Carlo VaR for end-of-day reporting, taking advantage of the speed of the former and the accuracy of the latter. When the two disagree by more than a threshold, the system flags a review.

10.12.5 9.12.5 Moment-matching variants

A refinement of Cornish-Fisher is *moment matching*: fit a parametric distribution (e.g. a mixture of Normals, a Johnson curve, or a shifted log-Normal) whose first four moments match the closed-form moments $\mu_P, \sigma_P^2, m_3, m_4$ of the quadratic form. The quantile of the fitted distribution is taken as the VaR. Moment matching avoids the non-monotonicity problems of Cornish-Fisher at extreme parameter values and tends to be more accurate for moderately skewed distributions. The cost is a slightly more elaborate implementation — the fitting step is not purely closed-form — and in practice Cornish-Fisher and moment-matching give very similar answers except in pathological cases.

An alternative to both is *Imhof's method*: numerically invert the characteristic function of the quadratic form to recover the full distribution of ΔV, then take the exact quantile. Imhof is more accurate than Cornish-Fisher but computationally heavier and is typically reserved for end-of-day validation runs rather than intraday computation. The progression Delta-Normal → Delta-Gamma-Cornish-Fisher → Imhof → full Monte Carlo represents a ladder of increasing accuracy and increasing computational cost, with the risk team picking the appropriate rung for each use case.

10.12.6 9.12.6 Operational role

Delta-gamma parametric VaR is not merely a back-pocket approximation. In the modern risk stack it plays several distinct roles:

- *Intraday limit monitoring.* When a trader executes a new option trade, the risk system can refresh the desk's VaR in milliseconds using the delta-gamma approximation. Full revaluation is too slow for this use case.

- *Sensitivity of VaR to positions.* Because the formula (9.26) is differentiable in δ and Γ, the marginal VaR (§9.9) of any prospective trade can be computed in closed form. This supports pre-trade "what-if" analysis without running Monte Carlo for every candidate trade.
- *Regulatory backup.* If a bank's full-revaluation Monte Carlo engine fails or times out, the delta-gamma approximation provides a fallback VaR that can be used to continue limit monitoring and regulatory reporting while the primary engine is repaired.

Cross-reference: the Greeks δ and Γ used here are the multi-factor generalisations of the scalar deltas and gammas developed in CH07. The single-name case $n = 1$ of (9.22) reduces exactly to the scalar delta-gamma expansion around the Black-Scholes price.

10.13 9.13 Hedged P&L Distribution and Hedged CTE

The input to a VaR calculation for an options desk is the *hedged* P&L distribution, not the unhedged P&L distribution. Hedging reshapes the underlying factor moves into a residual P&L that the desk actually experiences; the risk measure then quantifies the tail of that residual. Hedging decisions and risk-measurement decisions are therefore coupled: a tighter hedge means a smaller residual, which means a smaller VaR, which means lower capital consumption. Conversely, a looser hedge means larger residuals and higher capital. The CFO and the CRO have aligned interests in encouraging tight hedging, subject to the caveat that over-hedging generates transaction-cost bleed that itself shows up in P&L.

10.13.1 9.13.1 The shape of residual hedged P&L

Consider an option position $f(t, S_t)$ hedged with α_t shares and β_t units of money-market account, rebalanced on a grid $t_0 < t_1 < ... < t_N$. Between rebalance times the hedge is stale, and the residual P&L increment over interval (t_{k-1}, t_k) is, to leading order,

$$\Delta R_k \;\approx\; \tfrac{1}{2}\,\Gamma(t_{k-1}, S_{t_{k-1}}) \cdot (S_{t_k} - S_{t_{k-1}})^2 - \theta(t_{k-1}, S_{t_{k-1}}) \cdot \Delta t. \tag{9.27}$$

This is the celebrated Taylor-series identity at the heart of dynamic hedging (see CH07 for the full derivation): the gamma-convexity term pays the holder whenever spot moves, and the theta-decay term charges the holder for the option's time decay. On average, under GBM with the Black-Scholes vol and frictionless rebalancing, the two exactly cancel — that is the content of the replication theorem. In practice, the cancellation is imperfect at every finite rebalance frequency, and the imperfection is the residual P&L.

The distribution of total hedged P&L over the option's life is approximately Gaussian for sufficiently fine rebalancing (by a central-limit argument across the sum of increments) with variance proportional to $1/N$ where N is the number of rebalance steps. More frequent rebalancing gives tighter hedged P&L. This is the source of the scaling law in §9.13.2 below.

10.13.2 9.13.2 Delta versus delta-gamma hedging and CTE reduction

The CTE of a hedged book is typically much smaller than the CTE of the unhedged book, but the reduction ratio depends on whether the hedge is first- or second-order. Rough scaling:

- *Delta-only hedge* reduces residual variance by a factor of approximately $\sigma^2 \Delta t$ per rebalance step. Cumulatively over the life of an option with daily rebalancing, this translates to CTE reductions of 70-90% relative to the unhedged position.
- *Delta-gamma hedge* (using a second traded option to neutralise gamma) reduces residual variance by another factor of approximately $\sigma^2 \Delta t$, giving CTE reductions of 95-99% relative to the unhedged position at realistic daily rebalance frequencies.

The gamma hedge squeezes the residual P&L into the third-order Taylor remainder, which scales as $(\Delta S)^3$ rather than $(\Delta S)^2$. Since $\Delta S \sim \sigma S \sqrt{\Delta t}$, the cube is of order $\sigma^3 S^3 \Delta t^{3/2}$, which is much smaller than the square $\sigma^2 S^2 \Delta t$ for small Δt. Each additional Greek matched reduces the hedging residual by another power of $\sqrt{\Delta t}$, so the improvements compound quickly under frequent rebalancing.

10.13.3 9.13.3 Asymmetry: long-gamma versus short-gamma books

The CTE framework naturally captures an asymmetry between long-gamma and short-gamma positions that VaR misses.

Long-gamma book. Positive Γ means the position profits from large moves in either direction. Residual P&L is bimodal around zero — positive in both large up-moves and large down-moves (gamma profit), negative in small moves (theta bleed). CTE of the *loss* side is modest because the loss distribution is bounded below by the accumulated theta: the worst plausible loss is the sum of theta charges over the rebalance interval, which is a small, bounded quantity. The tail of the *profit* distribution can be very large, but CTE of losses doesn't care about that.

Short-gamma book. Negative Γ means the position loses from large moves. Residual P&L is unbounded below — theoretically catastrophic large-move losses proportional to $\frac{1}{2}\Gamma(\Delta S)^2$ with $\Gamma < 0$. Under fat-tailed underlying distributions, the CTE of the loss side blows up much faster than the VaR because the tail is heavy-tailed in the quadratic of a heavy-tailed variable. This is why short-gamma books are subject to much tighter risk limits than equivalently-sized long-gamma books: the loss profiles are fundamentally different even if the variance (2nd moment) of the two books looks similar. CTE picks up the asymmetry; VaR (a quantile) does not.

A concrete example: consider a short straddle (short gamma) sized so that its 1-day 99% VaR is \$1M. Under Normal factor moves, its 1-day 99% CTE might be \$1.3M — a 30% uplift. Under Student-t_3 factor moves, the same VaR \$1M comes with a CTE of perhaps \$3-5M depending on the vol level. The short-gamma book is exposed to tail events that VaR cannot see, and CTE is the measure that reveals the exposure. A long straddle with identical VaR has a CTE close to the VaR itself (because the loss distribution is bounded below by theta), so the two books with identical VaR have wildly different CTE — and risk limits that target CTE automatically discriminate between them.

10.13.4 9.13.4 Stress-test implications for hedged books

Stress testing a hedged option book requires an extra layer of care. The book's hedge ratio is set using today's Greeks, but a stress scenario will typically move the spot far enough that today's delta is no longer the right hedge. A -25% equity stress on a book that is nominally delta-neutral using today's delta will leave the book with a significant positive delta after the stress (because gamma converts big spot moves into delta moves), and the realised loss in the stress is the quadratic gamma term minus any re-hedging cost inside the stress path.

Realistic stress tests of option books therefore use *scenario-dependent revaluation*: apply the stress shocks to the market data, recompute all option values under the shocked market (possibly with a shocked volatility surface), and take the difference from the pre-stress value. The result captures both the quadratic gamma effect and any second-order vol/correlation effects. This is more expensive than a Taylor-series stress but is the only way to get a meaningful stress P&L for a significantly gamma-exposed book.

10.13.5 9.13.5 Hedged CTE as a risk management target

Running a derivatives desk to minimise hedged CTE is subtly different from running it to minimise hedged variance. Variance minimisation penalises symmetric moves equally in both directions and is content with a long-gamma-heavy book whose upside and downside cancel in the variance. CTE minimisation cares only about the downside and actively prefers positions whose downside is bounded (long gamma, long options) to positions whose downside is open-ended (short gamma, short options). A desk run on a CTE mandate will systematically hold more long optionality than a variance-mandated desk, at the cost of more theta bleed. The trade-off between theta carry and tail safety is the central tension of derivatives portfolio management, and the risk measure chosen to govern the desk is what operationally resolves it.

10.14 9.14 Operational Integration

On a working trading floor the risk-measurement and hedging frameworks are executed by different teams using a shared data infrastructure. Traders manage Greeks (delta, gamma, vega) throughout the day and execute hedges according to the book's policy. Market risk managers compute VaR, CTE, and scenario P&Ls at end of day using the same position data but different analytics. Model-validation teams periodically review the methodologies for both hedging and risk measurement. Regulators audit the whole system and compare its outputs to their prescribed standards.

10.14.1 9.14.1 The data and model layer

The data layer that connects these teams is critical and often where things go wrong in practice.

- *Position data* must be accurate and timely. Every trade executed by the front office must flow to the risk engine before the end of day, with consistent trade identifiers, legal entities, and reference data. A trade that lands in the front-office system but not the risk system creates a silent under-reporting of risk.
- *Market data* (prices, volatilities, rates, correlations, vol surfaces, discount curves) must be sourced from validated vendors with documented quality controls. Stale or bad market data can silently corrupt VaR computations for entire asset classes; robust data systems run automated plausibility checks and stale-price detection.
- *Greek calculations* must be consistent across trading and risk systems. A common pitfall is the "risk model says X, trading system says Y" discrepancy, where the Greeks computed by two different implementations diverge and the firm has conflicting views of its own book. Resolving such discrepancies is a daily operational task.

10.14.2 9.14.2 The end-to-end daily cycle

A typical daily risk cycle on a large derivatives desk looks like this:

1. **Market close.** Prices, vol surfaces, and rates are marked to the close. Reference data (dividends, cash flows, corporate actions) is updated.
2. **Position snapshot.** All trades executed during the day are loaded into the risk engine, netted with previous positions, and validated against front-office trade blotters.
3. **Greek computation.** Position-level and portfolio-level Greeks are computed on the end-of-day market.
4. **Intraday VaR.** A delta-gamma-Cornish-Fisher VaR is computed for quick limit checks and desk-level reporting, often within minutes of close.
5. **End-of-day VaR.** A full-revaluation Monte Carlo VaR runs overnight, using 10,000-100,000 scenarios drawn from the calibrated factor distribution. This is the VaR used for regulatory reporting.
6. **Stress tests.** Historical and hypothetical scenarios are applied to the end-of-day book. Results are compiled into the morning risk report.
7. **P&L attribution.** Actual P&L, hypothetical P&L, and risk-theoretical P&L are computed and reconciled. Residuals are flagged for review.
8. **Backtests.** Daily exceedance counts are updated. If the count enters the yellow zone, a review meeting is scheduled; if red, an immediate escalation to senior risk management occurs.
9. **Morning report.** By start-of-business the next trading day, the risk team delivers an integrated report showing VaR, CTE, stress, attribution, backtest status, and limit utilisation by desk, factor, and region.

10.14.3 9.14.3 Model governance and validation

Model governance is the formal framework that surrounds the technical risk work. Core elements:

- *Model inventory.* Every model used for risk computation is catalogued with its owner, scope, assumptions, and validation history.
- *Model approval.* New models or material changes to existing models require sign-off from an independent Model Validation group, who reviews the mathematical correctness, implementation fidelity, and empirical performance.
- *Annual review.* Every in-production model is re-reviewed at least annually. Backtest performance, market evolution, and any reported issues are all considered.
- *Model risk reserve.* Some banks hold an explicit capital reserve for "model risk" — the risk that their models are wrong in ways that underestimate capital — typically sized as a fraction of the model's capital output.

Reconciling discrepancies between front-office and risk systems monthly in a formal audit is a regulatory requirement; reconciling them daily is an operational best practice. The data and model governance around these flows is sometimes invisible to outsiders but is where much of the actual risk management work happens. A beautifully-specified coherent risk measure is useless if the data feeding it is stale, the positions are mis-captured, or the Greeks are computed inconsistently across systems. Operational integration is the unglamorous bedrock on which the quantitative risk framework stands.

10.15 9.15 Key Takeaways

1. **VaR is a quantile; CTE is a conditional tail mean.** VaR answers "what loss will I not exceed with $(1-\alpha)$ confidence" and ignores the size of losses beyond the cutoff. CTE integrates the tail beyond VaR and captures tail shape. For Normal losses at 95%, CTE exceeds VaR by about 0.4 standard deviations; for Student-t_3 the gap is 1.8 standard deviations — a five-fold amplification that CTE correctly reports and VaR systematically hides.

2. **VaR fails sub-additivity; CTE does not.** Two independent 4%-default loans each have 95% VaR of zero, but their merged 95% VaR is 100 — diversification *increased* reported risk. CTE obeys sub-additivity and is the correct axis for any diversification-aware framework. The VaR-to-CTE ratio is itself diagnostic: well above the Gaussian benchmark signals fat tails.

3. **Three VaR estimation families triangulate.** Historical simulation is non-parametric and model-light but limited by sample size. Parametric (delta-normal, delta-gamma) VaR is fast and decomposable but assumes Gaussian factors. Monte Carlo is flexible but computationally heavy and model-risk-prone. A mature risk function runs multiple methodologies in parallel and treats disagreements as diagnostic.

4. **Cornish-Fisher is the workhorse of fast VaR.** Equation (9.4) adjusts the Gaussian quantile for skewness and kurtosis without a simulation. Combined with the closed-form delta-gamma moments (9.23)–(9.25), it produces a fast and reasonably accurate VaR that captures both non-linearity (via Γ) and tail thickness without a Monte Carlo path.

5. **Coherent risk measures satisfy four axioms.** Monotonicity, translation invariance, positive homogeneity, sub-additivity. Every coherent measure has a dual representation as a worst-case expectation over a set of scenarios. CTE fits; VaR does not. Spectral measures generalise both.

6. **Fat tails are the empirical norm, not the exception.** Daily equity tail indices sit in the range 3-5; EM currencies and distressed credit drop to 2. Any Normal-calibrated risk framework systematically under-reports tail risk in these markets. Defensive adjustments include Student-t fits, filtered historical simulation, GPD-based extreme-value models, and GARCH conditional vol.

7. **Regulation has migrated from VaR to ES.** Basel 1996 introduced 99% VaR; Basel 2.5 added stressed VaR; FRTB (2016/2019) replaced VaR with 97.5% expected shortfall. The percentiles are calibrated so Normal-$\text{ES}_{97.5}$ Normal-VaR_{99}, but ES additionally penalises fat tails. FRTB also prescribes liquidity horizons by asset class, non-modellable risk factor charges, and a P&L attribution test.

8. **Backtesting validates VaR through Kupiec and Christoffersen.** Kupiec's POF test compares exceedance counts to a binomial null. Christoffersen's independence test catches clustered exceedances. The combined conditional-coverage test is the gold standard. CTE is harder to backtest because it is not elicitable; FRTB uses indirect tests via multi-level VaR.

9. **Stress testing complements VaR.** Historical scenarios replay specific events (1987, 2008, 2020); hypothetical scenarios imagine plausible but unrealised catastrophes; reverse stress asks what scenarios produce failure. A risk function that reports VaR without stress, or stress without VaR, is only half equipped.

10. **Risk decomposition answers "where does my risk come from."** Marginal VaR (derivative), component VaR (Euler allocation, sums to total), and incremental VaR (finite difference on removal) give three related but distinct attributions, each useful for different decisions. Component VaR is natural for budget allocation; incremental VaR is natural for unwind decisions.

11. **P&L attribution closes the risk-management loop.** Equation (9.20) decomposes realised P&L into Greek contributions plus a residual. A systematically large residual is diagnostic of model or data problems. FRTB formalises the comparison as a P&L attribution test; failing the test forces the bank to use the standardised approach.

12. **Liquidity-adjusted risk layers in bid-ask and horizon effects.** Simple exogenous versions add a spread haircut; FRTB institutionalises liquidity via asset-class-specific horizons (10 days for major equities, 120 days for exotic credit). Funding liquidity (LCR, NSFR) interacts with market liquidity in crisis regimes — the 2008 and 2020 doom-loops are the canonical cautionary tales.

13. **Delta-gamma parametric VaR is the computational workhorse.** Closed-form moments (9.23)–(9.25) combined with Cornish-Fisher (9.26) deliver VaR in $O(n^2)$ per evaluation — milliseconds for a 500-factor

portfolio. Accurate to a few percent for daily 95% VaR under moderate conditions; degrades for near-expiry options and extreme tails, where full revaluation is mandatory.

14. **Hedging reshapes the loss distribution that risk measures are applied to.** The input to VaR for an options book is the *hedged* P&L distribution. Delta hedging typically reduces CTE by 70-90%; delta-gamma hedging by 95-99%. CTE distinguishes long-gamma (bounded loss tail) from short-gamma (unbounded loss tail) books that may have identical VaR — exactly the asymmetry CTE was designed to capture.

15. **The end-to-end risk cycle is an operational system, not a single calculation.** Position capture, market-data validation, Greek computation, intraday delta-gamma VaR, overnight full-revaluation Monte Carlo, stress tests, P&L attribution, backtests, and morning reports all must reconcile. The mathematical elegance of coherent risk measures is useless if the data pipeline corrupts the inputs. Operational integration is where the framework succeeds or fails in practice.

10.16 9.16 Reference Formulas

10.16.1 Core definitions

Value-at-Risk at confidence $(1-\alpha)$:

$$\mathrm{VaR}_\alpha(L) \;=\; \inf\{\ell : \mathbb{P}(L \le \ell) \ge 1-\alpha\}.$$

Conditional Tail Expectation / Expected Shortfall:

$$\mathrm{CTE}_\alpha(L) \;=\; \mathrm{ES}_\alpha(L) \;=\; \mathbb{E}[L \mid L \ge \mathrm{VaR}_\alpha(L)].$$

10.16.2 Parametric VaR formulas

Delta-normal VaR under Gaussian factors with covariance Σ and exposure vector w:

$$\mathrm{VaR}_\alpha \;=\; z_{1-\alpha}\sqrt{w^\top \Sigma w}.$$

Cornish-Fisher expansion of a non-Gaussian quantile:

$$q_\alpha \;\approx\; z_\alpha + \tfrac{1}{6}(z_\alpha^2 - 1)\, s + \tfrac{1}{24}(z_\alpha^3 - 3z_\alpha)\,\kappa - \tfrac{1}{36}(2z_\alpha^3 - 5z_\alpha)\, s^2,$$

with skewness s and excess kurtosis κ.

10.16.3 Delta-gamma parametric VaR

For book P&L approximated by $\Delta V = \delta^\top \Delta r + \frac{1}{2}\Delta r^\top \Gamma \Delta r$ with $\Delta r \sim \mathcal{N}(0, \Sigma)$:

$$\mu_P \;=\; \tfrac{1}{2}\,\mathrm{tr}(\Gamma\Sigma), \qquad \sigma_P^2 \;=\; \delta^\top \Sigma\, \delta + \tfrac{1}{2}\,\mathrm{tr}((\Gamma\Sigma)^2),$$
$$m_3 \;=\; \mathrm{tr}((\Gamma\Sigma)^3) + 3\,\delta^\top \Sigma\, \Gamma\, \Sigma\, \delta,$$

feeding into Cornish-Fisher:

$$\mathrm{VaR}_\alpha \;\approx\; -\mu_P + \sigma_P\left[z_\alpha + \tfrac{1}{6}(z_\alpha^2 - 1)\, s + \tfrac{1}{24}(z_\alpha^3 - 3z_\alpha)\,\kappa\right].$$

10.16.4 Student-t tail

$$\mathrm{CTE}_\alpha^{t_\nu} \;=\; \frac{f_{t_\nu}(\mathrm{VaR}_\alpha^{t_\nu})\,(\nu + (\mathrm{VaR}_\alpha^{t_\nu})^2)}{(\nu-1)\,\alpha}.$$

10.16.5 Extreme Value Theory — peaks over threshold

Generalised Pareto tail approximation for exceedances over threshold u:

$$F_u(y) \;\approx\; 1 - \left(1 + \xi\frac{y}{\beta}\right)^{-1/\xi}, \qquad y > 0,$$

with shape ξ (positive = Fréchet/power-law, zero = Gumbel/exponential, negative = Weibull/bounded) and scale $\beta > 0$.

10.16.6 Coherence axioms (loss convention)

A risk measure ρ on losses is coherent if for all losses L, L_1, L_2 and constants $c, \lambda \geq 0$:

- **Monotonicity:** $L_1 \leq L_2 \Rightarrow \rho(L_1) \leq \rho(L_2)$.
- **Translation invariance:** $\rho(L + c) = \rho(L) + c$.
- **Positive homogeneity:** $\rho(\lambda L) = \lambda \rho(L)$.
- **Sub-additivity:** $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$.

CTE/ES satisfies all four; VaR fails sub-additivity.

10.16.7 Dual representation of coherent measures

Every coherent ρ equals a worst-case expectation over some set $\mathcal{Q}$ of probability measures:

$$\rho(L) \;=\; \sup_{\mathbb{Q} \in \mathcal{Q}} \mathbb{E}^{\mathbb{Q}}[L].$$

Spectral risk measures:

$$\rho_\phi(L) \;=\; \int_0^1 \phi(u)\, F_L^{-1}(u)\, du, \qquad \phi \geq 0,\ \phi \text{ non-decreasing}, \int \phi = 1.$$

VaR corresponds to $\phi = \delta(u - (1-\alpha))$; CTE corresponds to $\phi(u) = \mathbf{1}_{u \geq 1-\alpha}/\alpha$.

10.16.8 Kupiec proportion-of-failures test

$$\mathrm{LR}_{\mathrm{POF}} \;=\; -2 \log\left[\frac{\alpha^x (1-\alpha)^{n-x}}{(x/n)^x (1 - x/n)^{n-x}}\right] \;\sim\; \chi_1^2,$$

with x = exceedance count over n days.

10.16.9 Christoffersen independence and conditional-coverage tests

$$\mathrm{LR}_{\mathrm{IND}} \;\sim\; \chi_1^2, \qquad \mathrm{LR}_{\mathrm{CC}} \;=\; \mathrm{LR}_{\mathrm{POF}} + \mathrm{LR}_{\mathrm{IND}} \;\sim\; \chi_2^2.$$

10.16.10 Risk decomposition

Under the delta-normal model with $\sigma_P = \sqrt{w^\top \Sigma w}$:

$$\mathrm{mVaR}_i \;=\; z_{1-\alpha} \frac{(\Sigma w)_i}{\sigma_P}, \qquad \mathrm{cVaR}_i \;=\; w_i\, \mathrm{mVaR}_i, \qquad \sum_i \mathrm{cVaR}_i \;=\; \mathrm{VaR}.$$

Incremental VaR of position i:

$$\mathrm{iVaR}_i \;=\; \mathrm{VaR}(w) - \mathrm{VaR}(w_{-i}),$$

where w_{-i} is w with the i-th component set to zero.

10.16.11 P&L attribution

$$\Delta V \;=\; \delta \cdot \Delta S + \tfrac{1}{2}\Gamma(\Delta S)^2 + \nu \cdot \Delta\sigma + \theta \cdot \Delta t + \rho \cdot \Delta r + \text{cross-Greeks} + \text{cash flows} + \text{residual}.$$

10.16.12 Liquidity-adjusted VaR (exogenous spread)

$$\mathrm{LVaR}_\alpha \;=\; \mathrm{VaR}_\alpha + \tfrac{1}{2}\,|w| \cdot \text{spread}.$$

10.16.13 Notation glossary

- L — loss random variable, large positive = painful.
- α — tail probability (small); $(1-\alpha)$ is the confidence level.
- z_α — standard Normal α-quantile.
- F_L, F_L^{-1} — CDF and quantile function of L.
- σ_P — portfolio standard deviation.
- w, Σ — exposure vector and factor covariance matrix.
- δ, Γ — first- and second-order sensitivity vector and matrix (multi-factor Greeks).
- ν, θ, ρ — vega, theta, rho scalar Greeks.
- s, κ — standardised skewness and excess kurtosis.
- ξ, β — GPD shape and scale parameters.
- ϕ — spectral weighting function.

11 Chapter 10 — Monte Carlo, Path-Dependent, and Forward-Starting Options

This chapter is the simulation counterpart to the lattice pricing of Chapter 2 and the PDE-based pricing of Chapter 6. Closed-form formulas like Black–Scholes are beautiful but brittle: they cover a short list of payoffs (European calls, European puts, a handful of digital and barrier variants under continuous monitoring) written on a single-asset geometric Brownian motion with constant volatility. Every realistic trading desk deals daily with claims that lie outside that list — path-dependent payoffs (Asians, lookbacks, cliquets, autocallables), multi-asset baskets, stochastic-volatility models, jump-diffusions, local-volatility surfaces calibrated to a full implied-vol grid. For all of these, the pricing integral remains $V_0 = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\varphi]$ but the analytical evaluation fails, and one must turn to numerical methods. Monte-Carlo simulation is the most general of these methods. It works in any dimension, handles arbitrary payoffs, and is embarrassingly parallel; its only costs are statistical noise and the need for careful variance management.

The chapter arc is narrow and deep rather than wide. We begin by motivating Monte Carlo from the perspective of high-dimensional numerical integration, where quadrature breaks down under the curse of dimensionality. The strong law of large numbers (SLLN) provides the mathematical license for sample-average estimation, and the central limit theorem gives the $M^{-1/2}$ error rate that governs how confidence intervals shrink with sample size. We then build the lognormal GBM path generator — properly derived this time, because Chapter 6 has already given us the Itô solution of the GBM SDE — and apply it to three classes of problem: European claims as a warm-up sanity check against Black–Scholes, forward-starting (cliquet) options as a case where iterated conditional expectation gives a closed form, and barrier options as the prototypical path-dependent payoff that forces a Monte-Carlo implementation. Along the way we encounter a characteristic issue of path-dependent MC — discretisation bias on between-grid events — and the Brownian-bridge correction that removes it. We close with variance-reduction techniques (antithetic variates, control variates) that every production pricer uses, as a bridge to the Heston Monte Carlo of Chapter 14.

A word about how this chapter sits in the prerequisite graph. Chapter 2 gave us lattice pricing and the CRR calibration; we will refer back to the lognormal GBM limit of that chapter as the starting point for the simulation engine. Chapter 6 gave us the Itô derivation of GBM, Girsanov's theorem (via Chapter 5) that fixes the drift under $\mathbb{Q}$ at the risk-free rate, and the Black–Scholes formula that we use as a closed-form benchmark. Everything in this chapter is downstream of that material. Nothing here feeds forward to the foundations; the chapter is a self-contained application, and a reader who has mastered through Chapter 6 can read Chapter 10 immediately without touching Chapters 7–9.

A note on scope. We cover the core Monte-Carlo toolkit — path simulation, sample-mean estimation, standard error, discretisation bias, and two of the canonical variance-reduction techniques. Topics we do not cover here include stratified sampling, importance sampling (beyond a conceptual mention), quasi-Monte-Carlo with low-discrepancy sequences, pathwise and likelihood-ratio Greek estimators, and the Longstaff–Schwartz regression-based scheme for American exotics. Those techniques are discussed briefly where they are natural extensions of the material and are otherwise left as pointers into the literature. The focus is on getting a working production-grade European and path-dependent pricer stood up; the reader who has the machinery presented here can read any of the variance-reduction or American-MC literature without further prerequisites.

11.1 10.1 Why Monte Carlo? Quadrature and the Curse of Dimensionality

The pricing formula $V_0 = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\varphi(A_T, \{A_t\}_{t\le T})]$ is, at its core, an integral over the sample space of the underlying process. In low dimensions this integral can be evaluated by deterministic quadrature — trapezoidal, Simpson's, Gauss–Legendre, or one of the many higher-order schemes — with error scaling as $O(h^p)$ where h is the mesh size and p is the order of the quadrature. For $p = 4$ (Simpson's rule), getting four-significant-figure accuracy on a smooth integrand requires $h \sim 10^{-1}$ or about 10 grid points in one dimension. Quadrature is fast, deterministic, and the textbook first choice when the state space is small.

In high dimensions quadrature breaks down catastrophically. The number of grid points required scales as N^d, where d is the number of independent stochastic factors and N is the number of grid points per factor. For a daily-monitored one-year option on a single underlying, $d = 252$ (one Brownian increment per trading day); even using only $N = 2$ grid points per dimension gives $2^{252} \approx 10^{75}$ total points, vastly more than any computer can enumerate. For a daily-monitored option on a basket of ten underlyings, $d = 2520$, and the number becomes utterly astronomical.

This is the *curse of dimensionality*: deterministic integration techniques whose per-dimension cost is well-controlled nevertheless collapse in high dimensions because the total cost compounds multiplicatively.

Monte-Carlo integration escapes the curse by a simple statistical observation: the sample-mean estimator's error scales as $M^{-1/2}$, *regardless of the dimension of the state space.* For a fixed error tolerance we need a fixed number of paths, and the cost per path scales only linearly in the dimension — each path-step is an $O(d)$ computation. Total cost: $O(Md)$, compared to $O(N^d)$ for quadrature. The break-even dimension is usually between $d = 4$ and $d = 6$: below that, quadrature is faster; above it, Monte Carlo wins decisively. Since every realistic derivative-pricing problem with a daily or finer monitoring grid lives well above $d = 6$, MC is the method of choice for everything except the cleanest European problems on a single underlying.

The flip side of dimension-insensitivity is that Monte Carlo is *statistical*: it produces random error bars, not deterministic error bounds. A practitioner always reports the MC estimate together with its sample standard error, and any downstream use of the estimate — hedging ratios, risk capital, P&L attribution — must account for the Monte-Carlo uncertainty. Deterministic methods produce exact-to-roundoff answers; MC produces confidence intervals. The gap closes as M grows, but never to zero. For a risk manager, this means MC-based prices carry an intrinsic "simulation vega" that must be bounded below desk risk tolerances; for a trader, it means the first question on seeing an MC price is always "what's the standard error?"

Monte-Carlo simulation has a long intellectual history, pre-dating its application to finance by decades. The key idea — that high-dimensional integrals can be approximated by sample averages — was formalised by Metropolis, Ulam, and von Neumann at Los Alamos in the 1940s for neutron-transport calculations, and has since spread to essentially every quantitative discipline. In finance the method is particularly well-suited to derivatives pricing because every pricing formula under $\mathbb{Q}$ is literally an integral (a discounted expected payoff) and because the underlying state space — paths of a diffusion — is naturally high-dimensional. Once the risk-neutral measure $\mathbb{Q}$ is pinned down (by Girsanov, Chapter 5, and the market-price-of-risk shift of Chapter 6), the pricing problem collapses cleanly to "draw paths from $\mathbb{Q}$, evaluate the payoff on each, and average." The conceptual simplicity is matched by the implementation simplicity: modern Monte-Carlo pricers are some of the shortest useful programs in quantitative finance.

The engineering of a production Monte-Carlo pricer involves four distinct sub-problems:

1. **Path generation under $\mathbb{Q}$.** Given the chosen risk-neutral model — GBM, Heston, local-vol, jump-diffusion, etc. — produce paths whose law matches the specification. For GBM the exact lognormal increment of §10.4 gives a bias-free generator; for general SDEs one falls back on Euler–Maruyama or Milstein schemes with their associated discretisation biases.
2. **Variance reduction.** Reduce the sample variance so that fewer paths suffice to achieve a target confidence-interval width. Antithetic variates, control variates, stratified sampling, importance sampling, and quasi-random sequences each contribute.
3. **Path-dependent payoffs.** Many payoffs depend on the whole path: running maximum (lookback), running average (Asian), barrier-crossing indicator (knock-in/out), basket constraints. Efficient data structures and careful handling of between-grid events (Brownian-bridge correction for barriers) are central.
4. **Sensitivity estimation (Greeks).** Delta, gamma, vega, and other Greeks must come out of the same path sample, ideally with pathwise or likelihood-ratio estimators rather than finite-difference bump-and-reprice. We will touch on this briefly but not in depth.

The present chapter concentrates on items (1), (2), and (3). Item (4) — the art of Greek estimation by Monte Carlo — is an entire subdiscipline and appears as a cross-reference at the end of the chapter and in more detail within Chapter 14.

11.2 10.2 The Strong Law of Large Numbers

The mathematical foundation of Monte-Carlo integration is the strong law of large numbers, which says that sample averages almost surely converge to population expectations.

Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{E}[|X|] < +\infty$, and let $X^{(1)}, X^{(2)}, \dots, X^{(M)}$ be independent and identically distributed copies of X. Then

$$\lim_{M \to +\infty} \frac{X^{(1)} + X^{(2)} + \cdots + X^{(M)}}{M} = \mathbb{E}[X] \qquad \text{(strong law, a.s.).} \tag{10.1}$$

The "almost surely" (a.s.) qualifier means that the set of paths along which convergence fails has $\mathbb{P}$-measure zero; for any actual realised sequence of draws the sample average converges. The strong law is the mathematical license for Monte-Carlo pricing: given enough i.i.d. draws from the payoff distribution under $\mathbb{Q}$, the sample average almost surely converges to $\mathbb{E}^{\mathbb{Q}}[\text{payoff}]$, which (times the discount factor) is the price.

A short philosophical aside on the $\mathbb{E}[|X|] < \infty$ hypothesis. Without finite mean the sample average may not converge at all; it can drift to infinity, oscillate, or behave chaotically. For option-pricing problems this is rarely a concern. Bounded payoffs — European puts are bounded by K, digital options by 1, and European calls can be bounded in practice by truncating at far-OTM regions — automatically satisfy the condition. But for *ratios* of random variables, or for some stochastic-volatility models with heavy-tailed variance processes, the integrability check deserves attention. The canonical pathological example is the Cauchy distribution (a ratio of two independent Gaussians), for which the sample average has the same Cauchy distribution regardless of sample size; the Monte-Carlo method fails completely because the variance is infinite and the mean is not even well-defined in the usual sense.

Higher integrability buys strictly more. L^2 integrability (finite variance, $\mathbb{E}[X^2] < \infty$) gives the Central Limit Theorem:

$$\sqrt{M}\,(\widehat{m}_M - \mathbb{E}[X]) \;\xrightarrow{d}\; \mathcal{N}(0,\ \mathbb{V}[X]), \tag{10.2}$$

where $\widehat{m}_M := (1/M)\sum_m X^{(m)}$ is the sample mean. The CLT is the basis for Monte-Carlo confidence intervals: for large M, $\widehat{m}_M$ is approximately normally distributed around $\mathbb{E}[X]$ with standard deviation $\sqrt{\mathbb{V}[X]/M}$, and we can bracket the unknown mean by a conventional $\widehat{m}_M \pm 1.96\,\hat{\sigma}_{m_1}$ 95% interval. L^4 integrability ($\mathbb{E}[X^4] < \infty$) tightens the approximation further via the Berry–Esseen bound, which quantifies the rate at which the CLT distributional approximation becomes good.

In practice the check "does my payoff have finite variance under $\mathbb{Q}$?" is usually done by inspection: polynomially-growing payoffs on a lognormal underlying have finite moments of all orders, so European calls, puts, Asians, look-backs, and barriers on GBM are all fine. The problem cases — which one should be aware of even if they do not arise in this chapter — are payoffs that involve *reciprocals* or *roots* of the underlying, payoffs on stochastic-volatility models near the Feller-condition boundary (see Chapter 14), and payoffs that are integrated over unbounded time horizons.

11.3 10.3 Sample Mean, Sample Variance, and the Monte-Carlo Standard Error

We estimate $m_1 := \mathbb{E}[X]$ by the finite sample

$$\widehat{m}_1 \;=\; \frac{1}{M}\sum_{m=1}^{M} X^{(m)} \qquad \text{(sample mean).} \tag{10.3}$$

The sample mean is unbiased ($\mathbb{E}[\widehat{m}_1] = \mathbb{E}[X]$) because expectation is linear, and has variance

$$\mathbb{V}[\widehat{m}_1] \;=\; \frac{1}{M^2}\sum_{m=1}^{M}\mathbb{V}[X^{(m)}] \;=\; \frac{\mathbb{V}[X]}{M}, \tag{10.4}$$

using independence of the draws. The standard deviation of the sample mean is therefore $\sigma_{m_1} = \mathrm{sd}[X]/\sqrt{M}$. Since σ_{m_1} depends on the unknown population variance, we estimate it by the sample-variance-based standard error

$$\hat{\sigma}_{m_1} \;=\; \frac{1}{\sqrt{M}}\left(\frac{1}{M-1}\sum_{m=1}^{M}\big(X^{(m)} - \widehat{m}_1\big)^2\right)^{1/2}. \tag{10.5}$$

Two small but important points about (10.5). First, the inner bracket is the unbiased sample variance (divisor $M-1$, not M); the $M-1$ correction — Bessel's correction — adjusts for the degree of freedom absorbed in estimating the mean. Second, dividing by $\sqrt{M}$ turns the sample standard deviation of the *underlying variable* into the standard

deviation of the *sample mean*. The Monte-Carlo error bar is the latter, not the former, and they differ by a factor of $\sqrt{M}$ that is easy to confuse.

The $M^{-1/2}$ convergence rate is universal for i.i.d. Monte Carlo with finite variance. Its immediate operational consequence: *halving the error bar costs four times as many paths.* A trader who wants to move an MC estimate from three significant figures to four needs 100× more paths; from four to five, 10,000× more. The law of diminishing returns is brutal, which is exactly why variance reduction matters so much (§10.8).

Monte-Carlo standard error vs sample size *Empirical verification of the $M^{-1/2}$ rate. The standard deviation of the sample-mean estimator across 200 replications of an ATM European-call MC price, plotted against path count M on log-log axes, tracks the $1/\sqrt{M}$ reference line over five decades. Variance reduction (§10.8) changes the constant, not the slope.*

Let us work through a concrete error budget to build intuition. Suppose we are pricing a one-year ATM European call with Black–Scholes parameters $S_0 = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$. The closed-form Black–Scholes value (Chapter 6) is $V_0^{\mathrm{BS}} \approx 10.45$. The payoff $X = e^{-rT}(A_T - K)_+$ has standard deviation roughly $\mathrm{sd}(X) \approx 14$ at these parameters — one can read this from direct moment calculation or from the empirical standard deviation of a pilot MC run. A Monte-Carlo estimate with $M = 10{,}000$ paths therefore has a standard error of $14/\sqrt{10{,}000} = 0.14$, so the 95% confidence interval is approximately

$$10.45 \pm 1.96 \cdot 0.14 \;=\; [10.17,\ 10.73].$$

To push the error below 0.01 (four-digit accuracy), we need $M = (14/0.01)^2 = 1{,}960{,}000$ paths — of order two million. This is entirely feasible in seconds on modern commodity hardware, but expensive enough that variance reduction pays for itself within the first few million paths.

A pair of useful ballpark facts that a trader should carry in their head:

- A standard-error budget of 0.1 on an option worth ~ 10 is typically acceptable for a trading indication. That is $M \sim 20{,}000$ paths for a vanilla European call at typical parameters.
- A standard-error budget of 0.01 on a portfolio-level MC risk calculation is typically required for capital numbers. That is $M \sim 2{,}000{,}000$ paths, an order of magnitude more effort.
- Pathwise sensitivities (deltas, gammas) typically have standard deviations 2–5× higher than the price itself, so Greek estimation at fixed MC budget is 4×–25× noisier than price estimation at the same budget. Practitioners either run more paths for Greeks or use variance-reduction techniques more aggressively on Greeks than on prices.

The standard error is not a guarantee — it is an estimate of an estimate. The sample variance itself has sampling error, and the 95% CI from a Gaussian approximation is only asymptotic. For small samples ($M \lesssim 100$) one should use a Student-t critical value instead of 1.96, and for badly non-Gaussian payoff distributions (heavily OTM or knocked-in barriers with few triggering paths) one should look skeptically at any symmetric confidence interval. When in doubt, bootstrap.

11.4 10.4 The Lognormal GBM Path Generator

With the SLLN in hand and the standard-error formula (10.5) at our disposal, the remaining ingredient for a Monte-Carlo pricer is a path generator — a procedure that, on demand, produces sample paths $\{S_{t_n}\}_{n=0}^{N}$ from the risk-neutral law of the asset. For GBM the generator is *exact*, not approximate; this is one of the two or three SDEs whose discretisation introduces no bias at the grid points. The reason is that Chapter 6 has already solved the GBM SDE in closed form.

GBM sample paths under Q *Thirty risk-neutral GBM paths for $S_0 = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1y$. The red curve is $\mathbb{E}^{\mathbb{Q}}[S_t] = S_0 e^{rt}$, the risk-neutral expected trajectory. Monte-Carlo pricing averages discounted payoffs over such paths; each path contributes one term of the sample mean (10.3).*

11.4.1 10.4.1 Exact lognormal increment

From Chapter 6 we know that under $\mathbb{Q}$ the asset price follows

$$\mathrm{d}S_t \;=\; r\,S_t\,\mathrm{d}t \;+\; \sigma\,S_t\,\mathrm{d}W_t^{\mathbb{Q}}, \tag{10.6}$$

where $W^{\mathbb{Q}}$ is a $\mathbb{Q}$-Brownian motion (the risk-neutral drift r having been installed by the Girsanov shift of Chapter 5). Applying Itô's lemma to $\ln S_t$ (Chapter 3) gives

$$\mathrm{d}\ln S_t \;=\; (r - \tfrac{1}{2}\sigma^2)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W_t^{\mathbb{Q}}, \tag{10.7}$$

and integrating from t_{n-1} to t_n,

$$\ln S_{t_n} - \ln S_{t_{n-1}} \;=\; (r - \tfrac{1}{2}\sigma^2)\,\Delta t_n \;+\; \sigma\,(W_{t_n}^{\mathbb{Q}} - W_{t_{n-1}}^{\mathbb{Q}}), \tag{10.8}$$

with $\Delta t_n := t_n - t_{n-1}$. The Brownian increment $W_{t_n}^{\mathbb{Q}} - W_{t_{n-1}}^{\mathbb{Q}}$ is exactly Gaussian with mean 0 and variance Δt_n, and — crucially — independent of the history up to t_{n-1}. Writing the increment as $\sqrt{\Delta t_n}\,Z_n$ with $Z_n \sim \mathcal{N}(0,1)$ and exponentiating gives the multiplicative path-simulation recursion

$$S_{t_n} \;=\; S_{t_{n-1}}\,\exp\!\Big\{(r - \tfrac{1}{2}\sigma^2)\,\Delta t_n \;+\; \sigma\sqrt{\Delta t_n}\,Z_n\Big\}, \qquad Z_1, Z_2, \ldots \overset{\text{iid}}{\sim}_{\mathbb{Q}} \mathcal{N}(0,1). \tag{10.9}$$

This is the universal GBM path generator. At the grid points $t_0, t_1, \ldots, t_N$ the law of $(S_{t_0}, S_{t_1}, \ldots, S_{t_N})$ produced by (10.9) is *exactly* the law of a GBM sampled at those points — there is no discretisation bias. Between the grid points the discrete path is a log-linear interpolation, which *is* an approximation, and for payoffs that care about the between-grid behaviour (like barriers) we will see below that a correction is needed.

Contrast this with the general case. For a non-GBM SDE $\mathrm{d}X_t = \mu(X_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}W_t^{\mathbb{Q}}$ the Itô increment is Gaussian only in the limit $\Delta t \to 0$, and for finite Δt the Euler–Maruyama scheme

$$X_{t_n} \;\approx\; X_{t_{n-1}} + \mu(X_{t_{n-1}})\,\Delta t_n + \sigma(X_{t_{n-1}})\sqrt{\Delta t_n}\,Z_n$$

introduces a weak error of order Δt and a strong error of order $\sqrt{\Delta t}$. Milstein's scheme adds a second-order correction and upgrades the strong error to Δt. For path-dependent payoffs where individual realisations (not just distributions) matter, the strong-order matters most, and practitioners sometimes pay the price of a higher-order scheme to reduce discretisation bias. For GBM we pay no such price — the exact lognormal increment is available at every grid width.

11.4.2 10.4.2 Monte-Carlo estimator for European claims

For a European claim with payoff $\varphi(S_T)$ the MC estimator is the direct sample-average translation of the SLLN:

$$\widehat{V}_0 \;=\; e^{-rT}\cdot\frac{1}{M}\sum_{m=1}^{M}\varphi\Big(S_0\,e^{(r-\frac{1}{2}\sigma^2)T\,+\,\sigma\sqrt{T}\,Z^{(m)}}\Big), \qquad Z^{(m)} \overset{\text{iid}}{\sim}_{\mathbb{Q}} \mathcal{N}(0,1). \tag{10.10}$$

Only the terminal value S_T enters, so we do not even need a grid — a single Gaussian draw per path suffices. This is the computationally cheapest non-trivial Monte-Carlo pricer and is useful mainly as a sanity check against Black–Scholes and as a warm-up for the path-dependent cases below.

11.4.3 10.4.3 Euler–Maruyama and Milstein for non-GBM SDEs

For the stochastic-volatility, local-volatility, and jump-diffusion models that appear later in the guide (Chapter 14 for Heston, in particular) the exact lognormal increment is *not* available, and one falls back on discretisation schemes. The simplest is Euler–Maruyama:

$$X_{t_n} \;\approx\; X_{t_{n-1}} \;+\; \mu(X_{t_{n-1}})\,\Delta t_n \;+\; \sigma(X_{t_{n-1}})\sqrt{\Delta t_n}\,Z_n.$$

The scheme approximates the Itô increment by freezing the drift and diffusion coefficients at the left endpoint and using a plain Gaussian increment; it has *weak* convergence rate $O(\Delta t)$ (expectations of smooth functions of the

terminal value converge at rate Δt) and *strong* convergence rate $O(\sqrt{\Delta t})$ (individual path realisations converge at the slower $\sqrt{\Delta t}$ rate). The distinction matters for path-dependent payoffs where individual realisations drive the payoff, not just their distributions.

Milstein's scheme augments Euler with a second-order Itô correction:

$$X_{t_n} \;\approx\; X_{t_{n-1}} \;+\; \mu\,\Delta t_n \;+\; \sigma\sqrt{\Delta t_n}\,Z_n \;+\; \tfrac{1}{2}\,\sigma\,\sigma'\,(\Delta t_n)\,(Z_n^2-1),$$

where $\sigma' = \partial\sigma/\partial X$. The correction upgrades the strong convergence rate to $O(\Delta t)$ and the weak rate remains $O(\Delta t)$. For constant diffusion ($\sigma' = 0$) Milstein and Euler coincide. For GBM ($\sigma(X) = \sigma X$, $\sigma' = \sigma$) Milstein's correction is $\frac{1}{2}\sigma^2 X\,(\Delta t)(Z^2-1)$, which matches the first-order Taylor expansion of the exact lognormal increment. For the Heston variance process $\mathrm{d}v_t = \kappa(\theta - v_t)\mathrm{d}t + \xi\sqrt{v_t}\mathrm{d}W_t$ the diffusion $\sigma(v) = \xi\sqrt{v}$ is not differentiable at $v = 0$, and naive Euler can produce negative variance values; the standard fixes are *full-truncation* (set negative values to zero before using them as the next drift) and the exact *QE scheme* of Andersen. Chapter 14 develops these in detail.

For any SDE whose exact solution is known in closed form, one should always prefer the exact simulator to Euler. GBM and OU are the two canonical examples; CIR, Vasicek (as a special case of OU), and a handful of others also admit exact schemes. For everything else, Euler is the pragmatic default, with Milstein applied when the strong-convergence-rate improvement justifies the extra arithmetic cost.

11.4.4 10.4.4 Additive log-space simulation

The exponential-and-multiplicative form of (10.9) is not the only representation. One could instead simulate $\ln S_{t_n}$ additively,

$$\ln S_{t_n} \;=\; \ln S_{t_{n-1}} + (r - \tfrac{1}{2}\sigma^2)\,\Delta t_n \;+\; \sigma\sqrt{\Delta t_n}\,Z_n, \tag{10.11}$$

and exponentiate at the end to recover S_{t_n}. Algebraically the two are identical; numerically they differ in edge cases. The multiplicative form can underflow to zero for very long paths at small S, while the additive form on $\ln S$ is more numerically stable. Production pricers typically work in log-space for this reason, especially when pricing very-long-dated (decades) structures or when the parameters drive the path into the tail. For everyday pricing at typical maturities (months to a few years) the difference is imperceptible.

11.4.5 10.4.5 Correlation of time-slices along a single path

Before using (10.9) to price path-dependent claims, we pause to think carefully about how different time-slices of the same simulated path are correlated. This is both a technical prerequisite for the forward-starting-option derivation of §10.6 and a conceptual anchor for why path-dependent options are fundamentally richer than European options.

For two times $T_1 < T_2$ along the same GBM path, write

$$\mathbb{C}\big[S_{T_1}, S_{T_2}\big] \;=\; \mathrm{Cov}^{\mathbb{Q}}\big(S_{T_1}, S_{T_2}\big), \qquad \rho\big[S_{T_1}, S_{T_2}\big] \;=\; \frac{\mathbb{C}\big[S_{T_1}, S_{T_2}\big]}{\big(\mathbb{V}[S_{T_1}]\,\mathbb{V}[S_{T_2}]\big)^{1/2}}. \tag{10.12}$$

Because $S_{T_2} = S_{T_1}\cdot\exp\{(r-\frac{1}{2}\sigma^2)(T_2-T_1) + \sigma\sqrt{T_2-T_1}\,Z'\}$ with Z' a fresh standard Gaussian independent of $\mathcal{F}_{T_1}$, the two slices S_{T_1} and S_{T_2} are positively correlated but not perfectly correlated — the intervening Brownian shock Z' drives them apart.

For a concrete calculation: under $\mathbb{Q}$, $\ln S_{T_i}$ is Gaussian with mean $(r-\frac{1}{2}\sigma^2)T_i$ and variance $\sigma^2 T_i$. The covariance of the *log*-prices is $\sigma^2\,\min(T_1, T_2) = \sigma^2 T_1$ (because $\ln S_{T_1}$ and $\ln S_{T_2}$ differ by a Brownian increment over $[T_1, T_2]$ that is independent of the increment over $[0, T_1]$). The correlation coefficient of the log-prices is therefore

$$\rho\big[\ln S_{T_1}, \ln S_{T_2}\big] \;=\; \frac{\sigma^2 T_1}{\sqrt{\sigma^2 T_1\cdot\sigma^2 T_2}} \;=\; \sqrt{T_1/T_2}.$$

At $T_1/T_2 = 1/2$ the correlation is $1/\sqrt{2} \approx 0.707$; at $T_1/T_2 = 0.1$ it is $\sqrt{0.1} \approx 0.316$. Higher correlation between nearby times reflects the strong dependence of consecutive Brownian increments. The *levels* S_{T_1} and S_{T_2} are *log*-normally

correlated, not linearly correlated; the two notions differ for skewed distributions, and a practitioner computing correlation of levels from simulated paths should always be clear whether the reported number refers to levels or logs.

The single property that makes path-dependent pricing tractable is *independence of disjoint increments*. The Brownian motion satisfies $W_{T_2} - W_{T_1} \perp \mathcal{F}_{T_1}$; this is the core of the Markov property, and it is what lets us factor joint expectations along time as nested conditional expectations. The next section makes use of this independence in its sharpest form.

11.5 10.5 Sanity Check: European Call by Monte Carlo

Before tackling path-dependent payoffs we work through the simplest non-trivial MC pricer: a European call on a GBM underlying. The point is purely pedagogical — Black–Scholes gives the exact answer in closed form, so there is no need to simulate anything. But the sanity check is worthwhile because it lets us verify the mechanics of (10.10), calibrate our expectations for the $M^{-1/2}$ convergence rate, and provide a reference against which the path-dependent pricers below can be compared.

Take the reference parameters $S_0 = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$. Black–Scholes (Chapter 6) gives $V_0^{\mathrm{BS}} \approx 10.45$ with $d_\pm = (r \pm \frac{1}{2}\sigma^2)T/(\sigma\sqrt{T}) = \{0.35,\, 0.15\}$ and $\Phi(0.35) \approx 0.637$, $\Phi(0.15) \approx 0.560$.

Simulation. Draw $M = 10{,}000$ standard normals $Z^{(m)}$ and compute the payoff on each:

$$S_T^{(m)} \;=\; 100 \cdot e^{0.03+0.2\,Z^{(m)}}, \qquad X^{(m)} \;=\; e^{-0.05}\,(S_T^{(m)} - 100)_+.$$

Average the payoffs: $\widehat{V}_0 = (1/M)\sum_m X^{(m)} \approx 10.43$ (a typical realisation). The sample standard deviation of X is around 14, so the standard error is $\widehat{\sigma}_{m_1} \approx 14/\sqrt{10{,}000} = 0.14$ and the 95% confidence interval is approximately [10.15, 10.71] — it comfortably brackets the closed-form value 10.45. Doubling M to 40,000 halves the standard error to 0.07, illustrating the $M^{-1/2}$ law. Quadrupling to 160,000 halves it again to 0.035.

What the sanity check buys us. A successful MC vs BS cross-check gives assurance on three independently-failable components of the pricer simultaneously: the random-number generator is producing genuine independent Gaussians; the GBM drift $r - \frac{1}{2}\sigma^2$ is applied with the correct sign and magnitude; and the discount factor e^{-rT} is applied exactly once. Each of these is a source of subtle bugs in production code. In particular, the $-\frac{1}{2}\sigma^2$ Itô correction is the single most common bug in home-grown MC pricers — it is easy to forget or get a sign wrong, and in that case the simulation produces a biased estimate whose bias *does not go away with more paths*. The standard-error bars shrink, the estimate converges confidently to a wrong number, and the bug can persist unnoticed for weeks until someone thinks to run a BS sanity check.

The sanity check also gives a cheap calibration of variance-reduction gain. With antithetic variates (§10.8) the standard error at $M = 10{,}000$ typically drops from 0.14 to around 0.10 on this payoff — a variance reduction of about 2. A stock-price control variate (whose expectation is known exactly, $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{rT} = 100e^{0.05} \approx 105.13$) reduces standard error by another factor of 3–5. Combined, these two techniques can bring the standard error at $M = 10{,}000$ down to 0.02, which would otherwise require $M \approx 500{,}000$ paths without variance reduction. That is a factor-of-50 speedup for a handful of extra lines of code.

Finally, the sanity check is where one should worry about edge cases. At extremely low volatility or short maturity, the payoff is nearly deterministic and its variance is nearly zero; MC does not struggle, but the $\widehat{\sigma}_{m_1}$ itself is small and its estimate is noisy, so reported confidence intervals may be misleading. At extremely high volatility or long maturity the payoff has a huge variance, so $M = 10{,}000$ paths give a correspondingly large standard error; one must scale up M accordingly. In both regimes, the relative error $\widehat{\sigma}_{m_1}/\widehat{V}_0$ is a better summary than the absolute standard error.

11.6 10.6 Forward-Starting (Cliquet) Options

Forward-starting options occupy a distinctive place in the exotic-options landscape. They are path-dependent — their payoff depends on the underlying at two different times, not just at maturity — but the dependence is structured enough that iterated conditional expectation gives a closed-form price. The derivation that follows is worth its

weight several times over: it is the atom of cliquet (ratchet) pricing, it illustrates how Brownian independent-increments repackage a correlated two-dimensional problem into a pair of one-dimensional Black–Scholes integrals, and it foreshadows the conditioning arguments that dominate interest-rate and stochastic-volatility pricing in later chapters.

11.6.1 10.6.1 Definition

A forward-starting call is an exotic whose strike is set at an intermediate date T_1 as a fraction α of the then-prevailing spot. The payoff at maturity $T_2 > T_1$ is

$$\varphi \;=\; (S_{T_2} - \alpha\, S_{T_1})_+, \qquad (x)_+ := \max(x, 0). \tag{10.13}$$

The strike $\alpha\, S_{T_1}$ is not known at time 0 — it is stochastic through S_{T_1}.

The economic motivation is protection *relative to wherever the market is when protection kicks in.* An investor who wants "a call whose strike is set to whatever the index is six months from now" is buying a forward-starting call with $T_1 = 0.5$ and $\alpha = 1$. An insurance company that writes annual reset put options on a pension liability is effectively writing a sequence of forward-starting puts. Cliquet options chain forward-starts together: a 12-leg monthly cliquet call consists of twelve forward-starting calls with reset dates at the end of each month, each leg paying the monthly reset-to-reset return above zero, and the total payoff being the sum. Some cliquet structures layer on local caps, global floors, or coupon multipliers; we deal only with the atomic forward-starting call here.

Cliquets are ubiquitous in structured-note design. A *locally-capped globally-floored* cliquet caps each reset-to-reset return at (say) +3% and guarantees a minimum total return of zero over the full life; it is a single-tranche packaging of path-dependent upside, popular with retail insurance customers who want equity-like returns with capital protection. Pricing these involves summing or compounding the atom (10.13) over many reset dates. The deeper reason cliquets are useful in portfolio construction: they hedge *relative* moves rather than *absolute* ones. Because the strikes are not known at trade inception, cliquets are more sensitive to *forward volatility* (vol of vol over future intervals) than vanilla options; they are a natural vehicle for vol-of-vol exposure.

11.6.2 10.6.2 Vanilla Black–Scholes recap

We restate the Black–Scholes formula here because the forward-starting price in §10.6.5 will reuse it as a building block. The cliquet calculation is "Black–Scholes inside Black–Scholes" — an inner BS for the conditional payoff given S_{T_1}, and an outer BS-like integral for the unconditional price. The vanilla European call with strike K and tenor T has price

$$V_0 \;=\; e^{-rT}\, \mathbb{E}^{\mathbb{Q}}\big[(S_T - K)_+\big] \;=\; S_0\, \Phi(d_+) \;-\; K\, e^{-rT}\, \Phi(d_-), \tag{10.14}$$

$$d_\pm \;=\; \frac{\ln(S_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}. \tag{10.15}$$

11.6.3 10.6.3 Reparameterising on two time-slices

The key manipulation is to rewrite the joint law of (S_{T_1}, S_{T_2}) in terms of independent Brownian increments over disjoint intervals. The two slices are correlated (they share a common Brownian path up to T_1), but the *increment* $W^{\mathbb{Q}}_{T_2} - W^{\mathbb{Q}}_{T_1}$ is independent of the history $\mathcal{F}_{T_1}$. This decomposition is what allows iterated conditional expectation to factor the pricing integral into a product of one-dimensional integrals.

Under $\mathbb{Q}$, the asset at T_1 and T_2 admits the direct lognormal representations

$$S_{T_1} \;=\; S_0\, e^{(r - \frac{1}{2}\sigma^2)T_1 \,+\, \sigma\sqrt{T_1}\, Z_1}, \qquad Z_1 \sim_{\mathbb{Q}} \mathcal{N}(0,1), \tag{10.16}$$

$$S_{T_2} \;=\; S_0\, e^{(r - \frac{1}{2}\sigma^2)T_2 \,+\, \sigma\sqrt{T_2}\, Z_2}, \qquad Z_2 \sim_{\mathbb{Q}} \mathcal{N}(0,1), \tag{10.17}$$

with (Z_1, Z_2) correlated. The equivalent independent-increment factorisation is

$$S_{T_1} \;\stackrel{d}{=}\; S_0\, e^{(r-\frac{1}{2}\sigma^2)T_1 \,+\, \sigma\sqrt{T_1}\, Z_1}, \tag{10.18}$$

$$S_{T_2} \;\stackrel{d}{=}\; S_{T_1} \cdot e^{(r-\frac{1}{2}\sigma^2)(T_2-T_1) \,+\, \sigma\sqrt{T_2-T_1}\, Z_3}, \qquad Z_3 \sim_{\mathbb{Q}} \mathcal{N}(0,1), \quad Z_1 \perp Z_3. \tag{10.19}$$

The two Gaussians Z_1 and Z_3 are now independent — they represent the Brownian increments over the disjoint intervals $[0, T_1]$ and $[T_1, T_2]$. The repackaging trades a two-dimensional correlated Gaussian (Z_1, Z_2) for two one-dimensional independent Gaussians, and that decoupling is what makes the inner and outer expectations below factorise cleanly.

11.6.4 10.6.4 Iterated conditional expectation

Starting from the pricing integral and using the tower property $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$:

$$V_0 \;=\; e^{-rT_2}\, \mathbb{E}^{\mathbb{Q}}\Big[\, (S_{T_2} - \alpha\, S_{T_1})_+ \,\Big] \;=\; e^{-rT_2}\, \mathbb{E}^{\mathbb{Q}}\Big[\, \mathbb{E}^{\mathbb{Q}}\Big[\, (S_{T_2} - \alpha\, S_{T_1})_+ \;\Big|\; S_{T_1} \,\Big]\,\Big]\,. \tag{10.20}$$

Separate the two discount factors so that the inner expectation becomes a Black–Scholes pricing problem in its own right, discounted from T_2 back to T_1:

$$V_0 \;=\; e^{-rT_1}\, \mathbb{E}^{\mathbb{Q}}\Big[\, e^{-r(T_2-T_1)}\, \mathbb{E}^{\mathbb{Q}}\Big[\, (S_{T_2} - \alpha\, S_{T_1})_+ \;\Big|\; S_{T_1} \,\Big]\,\Big]\,. \tag{10.21}$$

11.6.5 10.6.5 Inner expectation — Black–Scholes with strike αS_{T_1}

Conditional on S_{T_1}, the forward-starting payoff $(S_{T_2} - \alpha\, S_{T_1})_+$ is a standard European call on S with time-to-expiry $T_2 - T_1$ and strike $\alpha\, S_{T_1}$ (treated as a constant given $\mathcal{F}_{T_1}$). The inner conditional expectation is therefore given by the Black–Scholes formula (10.14)–(10.15), with spot S_{T_1} and strike $\alpha\, S_{T_1}$:

$$e^{-r(T_2-T_1)}\, \mathbb{E}^{\mathbb{Q}}\Big[\, (S_{T_2} - \alpha\, S_{T_1})_+ \;\Big|\; S_{T_1} \,\Big] \;=\; S_{T_1}\, \Phi(d_+) \;-\; \alpha\, S_{T_1}\, e^{-r(T_2-T_1)}\, \Phi(d_-), \tag{10.22}$$

with

$$d_\pm \;=\; \frac{\ln(S_{T_1}/(\alpha\, S_{T_1})) + (r \pm \frac{1}{2}\sigma^2)(T_2 - T_1)}{\sigma\sqrt{T_2 - T_1}} \;=\; \frac{\ln(1/\alpha) + (r \pm \frac{1}{2}\sigma^2)(T_2 - T_1)}{\sigma\sqrt{T_2 - T_1}}. \tag{10.23}$$

Notice what happened: the S_{T_1} factors in the moneyness ratio $S_{T_1}/(\alpha S_{T_1}) = 1/\alpha$ *cancelled*, leaving $d_\pm$ as *constants* — not random variables. This is the defining algebraic property of the forward-starting structure and is what makes closed-form pricing possible. The spot and the strike scale identically with S_{T_1}, so moneyness is always $1/\alpha$ regardless of where S_{T_1} lands. Every dollar increase in S_{T_1} lifts both the (random) spot and the (random) strike by α, leaving the convexity structure invariant.

11.6.6 10.6.6 Outer expectation — linearity and pull-out

Collect the inner result as $S_{T_1} \cdot \eta$ with the deterministic constant

$$\eta \;:=\; \Phi(d_+) \;-\; \alpha\, e^{-r(T_2-T_1)}\, \Phi(d_-). \tag{10.24}$$

Substituting into (10.21),

$$V_0 \;=\; e^{-rT_1}\, \mathbb{E}^{\mathbb{Q}}\Big[\, S_{T_1} \cdot \eta \,\Big] \;=\; e^{-rT_1}\, \eta\, \mathbb{E}^{\mathbb{Q}}[S_{T_1}] \;=\; e^{-rT_1}\, \eta \cdot S_0\, e^{rT_1} \;=\; \eta \cdot S_0, \tag{10.25}$$

where the third equality uses the risk-neutral martingale condition $\mathbb{E}^{\mathbb{Q}}[S_{T_1}] = S_0\, e^{rT_1}$ — a direct consequence of the fact that the discounted spot $e^{-rt}\, S_t$ is a $\mathbb{Q}$-martingale (FTAP from Chapter 2, made rigorous by Girsanov in Chapter

5). The two discount factors e^{-rT_1} and e^{+rT_1} cancel, and the outer expectation collapses to the constant ηS_0. The closed-form forward-starting call price is

$$\boxed{V_0 \;=\; S_0\Big[\,\Phi(d_+) \;-\; \alpha\, e^{-r(T_2-T_1)}\,\Phi(d_-)\,\Big], \qquad d_\pm \;=\; \frac{\ln(1/\alpha) + (r \pm \frac{1}{2}\sigma^2)(T_2-T_1)}{\sigma\sqrt{T_2-T_1}}.} \tag{10.26}$$

11.6.7 10.6.7 Interpretation and a worked example

The forward-starting call price depends on S_0 *only through the overall scale* — the moneyness $d_\pm$ is a function of α and the tenor $T_2 - T_1$ alone, not of S_0. The price therefore scales linearly in S_0, and the whole problem collapses to a Black–Scholes formula with strike-ratio α and effective tenor $T_2 - T_1$. This is why cliquets are sometimes called "relative-moneyness" options: their fair value tracks today's spot rather than any fixed dollar strike, making them natural hedges for wealth held as a percentage of the index.

Worked numerical scenario. Take $S_0 = 100$, $\alpha = 1.0$ (ATM reset), $T_1 = 0.5$, $T_2 = 1.0$, $r = 5\%$, $\sigma = 20\%$. The effective tenor is $T_2 - T_1 = 0.5$. Substituting into (10.26):

$$d_+ \;=\; \frac{0 + 0.07\cdot 0.5}{0.2\sqrt{0.5}} \;=\; \frac{0.035}{0.1414} \;\approx\; 0.2475, \qquad d_- \;=\; \frac{0 + 0.03\cdot 0.5}{0.1414} \;\approx\; 0.1061.$$

Normal CDFs: $\Phi(0.2475) \approx 0.5977$, $\Phi(0.1061) \approx 0.5422$. Discount factor $e^{-0.025} \approx 0.9753$. Then

$$\eta \;=\; 0.5977 - 1.0\cdot 0.9753\cdot 0.5422 \;=\; 0.5977 - 0.5289 \;=\; 0.0688, \qquad V_0 \;=\; 0.0688\cdot 100 \;=\; 6.88.$$

Compare to the vanilla ATM European call at $T = 1$ with these parameters: about 10.45. The forward-starting call is 34% cheaper. Why? Because the first six months of the contract contribute no convexity for the holder — the strike floats with the spot, so the contract is "ATM forever" until the reset. Only the post-reset half-year carries convexity value. A rough rule of thumb: a forward-starting ATM call is worth approximately $\sqrt{(T_2 - T_1)/T_2}$ times the equivalent vanilla ATM call. For our parameters this predicts $10.45 \cdot \sqrt{0.5} \approx 7.39$ — close to the true 6.88, with the residual difference reflecting the $-\frac{1}{2}\sigma^2$ term that does not scale as $\sqrt{T}$.

The linearity in S_0 has two practical implications. First, the *delta* of a forward-starting call is η — a constant — rather than the state-dependent $\Phi(d_+)$ of a vanilla. This makes forward-starts easy to delta-hedge: the delta is static, and no rebalancing is required as S moves (at least until the reset date, at which point the contract becomes a vanilla call and the usual state-dependent delta takes over). Second, the *vega* of the forward-start is concentrated on the *forward volatility* between T_1 and T_2, not on the spot volatility to T_1. This is why forward-start options are used to express views on forward vol — the volatility implied by market prices for a future window — rather than spot vol. A trader long forward-starts is long forward vol; a trader short them is short forward vol.

Monte-Carlo verification. With (10.9) and a grid $\{0, T_1, T_2\}$, a simulation of (10.13) at $M = 10{,}000$ paths typically returns an estimate within 0.1 of the closed-form value 6.88. The MC simulation is a useful check because it verifies the independence of the (Z_1, Z_3) draws — a subtle bug in which the same Z is used for both increments would give a badly wrong price but no obvious error signal. (This is, in practice, the second most common MC bug after the missing $-\frac{1}{2}\sigma^2$ drift correction.)

11.7 10.7 Barrier Options and the Monte-Carlo Time Grid

Barrier options are the canonical path-dependent structure. Their payoff depends not on the final spot alone but on whether the path touches a pre-specified threshold U at *any* point during the contract's life. This seemingly small change — swapping "at maturity" for "at any time before maturity" — transforms the pricing problem fundamentally: we can no longer rely on the terminal distribution of S_T alone; we need the joint distribution of S_T and the running maximum (or minimum) of the path. In return we get a rich family of structured-note primitives, used heavily in retail derivatives packaging and in institutional structured funding.

11.7.1 10.7.1 Taxonomy

Barrier options come in eight canonical flavours: {up, down} × {in, out} × {call, put}. An *up-and-in* call becomes a plain call if the stock ever crosses above U; an *up-and-out* call is a plain call that gets cancelled if the stock ever crosses above U. Symmetric definitions hold for down-and-in, down-and-out, and the put versions. A structurally important identity is *in-out parity*:

$$V^{\text{up-and-in call}} + V^{\text{up-and-out call}} = V^{\text{vanilla call}}, \tag{10.27}$$

since exactly one of the two triggers for every path. In-out parity lets us price one member of the pair in terms of the other; it also serves as a primary source of variance reduction in barrier MC, because the vanilla call price is available in closed form and subtracting a known quantity reduces simulation noise (§10.8). (Implicit assumption: the knock-in and knock-out contracts pay the same — typically zero — rebate on the knock event. If the knock-out pays a cash rebate R at the first-passage time τ_B while the knock-in pays nothing on that event, (10.27) picks up an additional rebate-PV term $\mathbb{E}^{\mathbb{Q}}[e^{-r\tau_B} R\, \mathbf{1}_{\{\tau_B \le T\}}]$ on the right-hand side; the rebate-free version is the one priced throughout this section.)

Barrier options are strictly cheaper than the corresponding vanillas (for knock-out barriers that cancel the contract) or equal-or-cheaper (for knock-ins, which have a chance of never activating and paying zero). They are used extensively in structured products: a "twin-win" note pays the magnitude of index returns unless a down-barrier is breached; a "reverse convertible" pays a coupon unless a down-barrier is breached, in which case the investor is put the stock at a fixed strike. The barrier adds path-dependence to the payoff, preventing closed-form lognormal pricing in general — though the classical Merton reflection-principle formulas *do* give closed forms under continuous monitoring of geometric Brownian motion with constant parameters.

11.7.2 10.7.2 Up-and-in call — definition

The up-and-in call pays

$$\varphi = (S_T - K)_+ \, \mathbf{1}_{\{\max_{0 \le t \le T} S_t \, \ge \, U\}}, \qquad U > S_0, \tag{10.28}$$

where the typical parameter ordering is $U > K > S_0$ (barrier above strike above initial spot). The contract becomes a standard European call with strike K at the first time the path crosses the barrier; otherwise the holder receives nothing. The knock-in event is measurable with respect to the path up to T, not just S_T; this is what makes the payoff path-dependent.

Under continuous monitoring of GBM there is a beautiful closed-form formula (the "reflection-principle" formula). For the up-and-in call with $S_0 < K < U$,

$$V^{\text{UIC}}_{\text{continuous}} = S_0 \Big(\tfrac{U}{S_0}\Big)^{2\lambda} \, \Phi(y_1) \; - \; K\, e^{-rT} \Big(\tfrac{U}{S_0}\Big)^{2\lambda - 2} \, \Phi(y_1 - \sigma\sqrt{T}), \tag{10.29}$$

where $\lambda = (r + \frac{1}{2}\sigma^2)/\sigma^2$ and y_1 is an explicit log-moneyness expression. We do not derive (10.29) here — the reflection principle for Brownian motion and its Girsanov-shifted counterpart for GBM are a standard but lengthy calculation, covered in every classical derivatives text. We use the formula only as an external benchmark: the *continuous-monitoring* closed form gives us a reference number against which discrete-monitoring MC estimates can be calibrated. Under discrete monitoring (say daily fixings), the barrier is crossed only at the fixing dates, not continuously in between; MC implementations naturally approximate discrete monitoring if the grid is chosen as the monitoring dates.

11.7.3 10.7.3 Lattice view

Before moving to Monte Carlo it is worth seeing how a binomial lattice prices barrier options. The lattice view clarifies the discretisation questions that plague the MC implementation and ties the path-dependent pricing back to the backward-induction apparatus of Chapter 2. Propagate known values from the barrier level downward: at any node on or above U, the option is (thereafter) a standard European call with strike K whose value $C^{\text{vanilla}}(S_n, t_n; K, T)$ is already known from the BS formula (10.14)–(10.15). Below the barrier the backward induction is the usual discounted $\mathbb{Q}$-expectation, but with the *known* vanilla-call value pasted in at any node that has touched U along the path to

that node. The tree effectively shrinks to a triangular wedge bounded by the barrier surface, and the barrier acts as a "known-value boundary" analogous to a Dirichlet condition in PDE language.

In the PDE framework, the Black–Scholes equation for a barrier option carries a Dirichlet boundary condition at $S = U$: for a knock-in, $V(U, t) = $ the vanilla-call value at that time; for a knock-out, $V(U, t) = 0$. The lattice backward-induction is solving this PDE numerically, with a two-child backward expectation playing the role of the finite-difference stencil. Analytical PDE solutions via method of images or Green's functions yield the same reflection-principle formulas mentioned above. Different methods — closed form, PDE, lattice, MC — should all agree to within their respective discretisation and variance errors; cross-validation between methods is a standard sanity check in derivatives implementation.

11.7.4 10.7.4 Monte-Carlo pricing with a time grid

The path-simulation recipe (10.9) applies directly. Discretise $[0, T]$ with a grid $t_0 = 0 < t_1 < \cdots < t_N = T$, step sizes $\Delta t_n = t_n - t_{n-1}$, and simulate

$$S_{t_n} \;=\; S_{t_{n-1}}\, e^{(r-\frac{1}{2}\sigma^2)\Delta t_n \,+\, \sigma\sqrt{\Delta t_n}\, Z_n}, \qquad Z_1, Z_2, \ldots \overset{\text{iid}}{\sim}_{\mathbb{Q}} \mathcal{N}(0,1). \tag{10.30}$$

The Monte-Carlo estimator of the up-and-in call price is

$$\widehat{V_0} \;=\; e^{-rT} \cdot \frac{1}{M} \sum_{m=1}^{M} \big(S_T^{(m)} - K\big)_+ \mathbf{1}_{\left\{\, \max_n S_{t_n}^{(m)} \,\geq\, U \,\right\}}. \tag{10.31}$$

Each simulated path contributes to the average only if its discretely-sampled maximum equals or exceeds U; paths that stay below U contribute zero regardless of terminal value.

11.7.5 10.7.5 Discretisation bias

The indicator $\mathbf{1}_{\{\, \max_n S_{t_n} \geq U \,\}}$ in (10.31) samples the path only at grid points; the true knock-in event $\{\, \max_{0\leq t\leq T} S_t \geq U \,\}$ can occur *between* grid points. For a given Brownian path one therefore underestimates the knock-in probability — each grid has some probability of missing a barrier excursion that happened between two of its points. Consequently the barrier-option MC price is *biased downward* at finite N, and the bias vanishes only in the limit $\Delta t_n \to 0$. This is the characteristic issue with MC pricing of path-dependent barrier and lookback contracts; the same issue arises for any payoff whose value depends on a supremum, infimum, or first-hitting time of the path.

The bias can be substantial. For an up-and-in call with $S_0 = 100$, $K = 100$, $U = 110$, $\sigma = 20\%$, $r = 5\%$, $T = 1$: the true continuous-monitoring value from (10.29) is around 6.15. With $N = 50$ monitoring points and $M = 10{,}000$ MC paths, the raw estimate from (10.31) is typically around 5.8 — a 6% downward bias. Refining the grid to $N = 500$ (factor-of-10 more work) without any bias correction gives an estimate near 6.10 — still a small residual bias. Continuing to refine eventually converges to the true value but at substantial computational cost.

11.7.6 10.7.6 Brownian-bridge correction

Between two adjacent grid points t_{n-1} and t_n with known endpoints $S_{t_{n-1}}$ and S_{t_n}, the conditional law of the path is a *Brownian bridge* (under log-coordinates with drift). The conditional distribution of the path maximum $\max_{t\in[t_{n-1},t_n]} S_t$ given the endpoints has a closed-form CDF, derived from the reflection principle:

$$\mathbb{P}\Bigg(\max_{t\in[t_{n-1},t_n]} S_t \;\geq\; U \;\Big|\; S_{t_{n-1}}, S_{t_n}\Bigg) \;=\; \exp\Bigg(-\frac{2\,\ln(U/S_{t_{n-1}})\,\ln(U/S_{t_n})}{\sigma^2\,\Delta t_n}\Bigg) \tag{10.32}$$

for $U > \max(S_{t_{n-1}}, S_{t_n})$; the probability is 1 when the barrier is already at or below either endpoint. Formula (10.32) is the continuous-time analogue of "interpolate between grid points and check if the interpolant crossed": it gives the exact conditional crossing probability under Brownian-bridge dynamics, which is the correct interpolation given the two endpoints.

The Brownian-bridge correction uses (10.32) as follows. For each simulated path that was *not* flagged as knock-in by the grid-point check, compute the per-interval bridge probability (10.31), draw a uniform random number $U_{\text{unif}}^{(m,n)}$ on

$[0,1]$, and flag the path as knock-in if $U_{\text{unif}}^{(m,n)} < p_{\text{bridge}}^{(m,n)}$ in any interval. Equivalently (and more numerically stable), compute the product of non-crossing probabilities over all intervals and subtract from one to get the corrected knock-in probability for the path; use this as the indicator weight. This reduces the bias from $O(\sqrt{\Delta t})$ to $O(\Delta t^{3/2})$ for typical barrier structures — a dramatic improvement.

Concrete numbers. Returning to the up-and-in call with $S_0 = 100$, $K = 100$, $U = 110$, $\sigma = 20\%$, $r = 5\%$, $T = 1$: at $N = 50$ and $M = 10{,}000$, the raw MC estimate is around 5.8; with Brownian-bridge correction applied to the same simulation, the estimate rises to about 6.12, within 0.5% of the true value 6.15. The bridge correction at $N = 50$ is therefore roughly equivalent in accuracy to brute-force refinement to $N \approx 500$ without correction — a factor-of-10 saving in compute. For barriers on high-frequency monitoring the advantage is even more pronounced.

11.7.7 10.7.7 Variance of the barrier estimator

A separate concern from bias is variance. The payoff $X = e^{-rT}(S_T - K)_+ \mathbf{1}_{\{\max S \geq U\}}$ is zero on most paths (those that stay below U). When the parameters put the barrier deep out of the money — U far above S_0, short maturity, low volatility — only a small fraction of paths knock in, and the MC estimator is dominated by a handful of triggering paths. Its sample variance can be very large relative to its mean.

Quantitatively: if only a fraction p_{knock} of paths knock in and each contributing path's payoff has standard deviation σ_X, the overall payoff variance is roughly $p_{\text{knock}}\,\sigma_X^2 + p_{\text{knock}}(1 - p_{\text{knock}})\,\mu_X^2$, and the standard error scales as $\sqrt{(p_{\text{knock}}\,\sigma_X^2)/M}$ — i.e. needing $1/p_{\text{knock}}$ times more paths to achieve the same precision as the vanilla case. For a knock-in probability of 10%, we need 10× more paths; for 1%, 100× more. Rare-event MC is computationally unforgiving.

Two standard remedies. *Importance sampling* shifts the drift of the simulation to make knock-in more likely, then reweights by the Radon–Nikodym derivative (Chapter 5); a well-tuned IS can reduce variance by factors of 10–100 for deeply out-of-the-barrier options. *Control variates* using the vanilla European call (whose expectation is known exactly from (10.14)–(10.15) and whose payoff correlates strongly with the barrier payoff on paths that do knock in) provide another order of magnitude of variance reduction for mildly-barriered options. In production, combining IS with a vanilla-call control and Brownian-bridge correction can drive barrier-MC run times from hours to seconds.

11.7.8 10.7.8 Other path-dependent families

Having priced barriers, we preview the broader landscape of path-dependent options that the Monte-Carlo machinery handles cleanly.

Asian options. Payoffs depend on the *average* of the underlying over a window, such as $(\bar{S}_T - K)_+$ where $\bar{S}_T = (1/N)\sum_{n=1}^{N} S_{t_n}$ (arithmetic average). Arithmetic Asians have no closed form under GBM because the sum of log-normals is not itself log-normal; MC is the standard tool. The corresponding *geometric* average $\bar{S}_T^{\text{geo}} = \prod_{n=1}^{N} S_{t_n}^{1/N}$ *is* log-normal and admits a closed-form Black–Scholes-style pricer, which makes it a natural control variate for arithmetic Asian MC — typical variance reduction factors of 50 to 500. Asian options are popular on commodity markets (oil, gas, metals) because averaging smooths out daily-price noise and resists single-day manipulation of the settlement print.

Lookback options. Payoffs depend on the running extremum, such as $(\max_t S_t - K)_+$ (lookback call on running max) or $(S_T - \min_t S_t)_+$ (lookback put on running min). Lookbacks are the most path-sensitive of the classical exotics and carry prices substantially higher than vanilla equivalents — they capture the best achievable entry point, an upper bound on realisable profit. Closed forms exist under continuous-monitoring GBM (again via the reflection principle), making lookbacks a useful benchmark for MC implementations.

Combinations. Practitioners layer features routinely — an "Asian barrier call" pays $(\bar{S}_T - K)_+$ but only if the running max stays below some upper barrier; a "cliquet with knock-out" terminates the coupon stream if a down-barrier is hit. These combinations have no closed form and are handled by MC with custom payoff functions that walk each path and assemble the appropriate indicator and averaging structure.

Autocallables. Highly structured products that pay coupons on observation dates if the underlying is above a specified trigger, and auto-call (early redemption of principal) once the underlying breaches an upper barrier. Autocallable pricing involves a mix of path-dependence, multiple discrete monitoring dates, and early-termination optionality; MC is the workhorse, typically with quasi-Monte-Carlo low-discrepancy sequences to reduce variance on the discrete fixing dates.

All of these families share the same MC machinery: generate paths under $\mathbb{Q}$, apply the payoff functional to each path, average with discount. The engineering art is in (i) reducing variance through targeted techniques (Asian controls for Asians, bridge corrections for barriers, importance sampling for rare-event payoffs), (ii) handling the exact contractual details correctly (continuous vs. discrete monitoring, fixed vs. floating strikes, local vs. global conditions, business-day conventions), and (iii) extracting Greeks efficiently from the same sample.

11.8 10.8 Variance Reduction: Antithetic and Control Variates

Every production Monte-Carlo pricer uses variance reduction. The $M^{-1/2}$ convergence rate is fixed by the central limit theorem, but the *constant* in front — the standard deviation $\mathrm{sd}(X)$ of the estimator variate — can be driven down by orders of magnitude with modest effort. Two techniques are universal: antithetic variates and control variates. We cover them here as a bridge to the Heston MC of Chapter 14, where they become essential rather than merely convenient.

11.8.1 10.8.1 Antithetic variates

The antithetic-variates trick exploits the symmetry of the Gaussian distribution. For each standard-normal draw $Z^{(m)}$, also use $-Z^{(m)}$. The symmetrised payoff is

$$\widetilde{X}^{(m)} \;=\; \frac{1}{2}\,\big[\,X(Z^{(m)}) + X(-Z^{(m)})\,\big], \tag{10.33}$$

and the antithetic estimator is $\widehat{V}_0^{\mathrm{ant}} = (1/M)\sum_m \widetilde{X}^{(m)}$. The computational cost is $2M$ payoff evaluations, but the effective sample size from the variance-reduction standpoint is *not* $2M$ — the two draws $X(Z)$ and $X(-Z)$ are *negatively* correlated, and the variance of the average is

$$\mathbb{V}[\widetilde{X}^{(m)}] \;=\; \frac{1}{4}\,\big[\,\mathbb{V}[X(Z)] + \mathbb{V}[X(-Z)] + 2\,\mathbb{C}[X(Z), X(-Z)]\,\big] \;=\; \frac{1}{2}\,\mathbb{V}[X]\,(1+\rho), \tag{10.34}$$

where $\rho := \mathrm{Corr}(X(Z), X(-Z))$. If $\rho < 0$, the variance is strictly less than $\mathbb{V}[X]/2$, and antithetic sampling outperforms independent sampling even at equal cost.

When does antithetic sampling work? It works best when the payoff $X(Z)$ is *monotone* in Z — calls are, puts are (after sign flip), digitals are, lookbacks almost are. For a monotone increasing X, $X(Z)$ and $X(-Z)$ have correlation close to -1 in the Gaussian tails, and the variance reduction factor can be substantial. For a European call at typical ATM parameters, antithetic sampling at M pairs typically achieves a variance reduction of about 2 relative to independent sampling at M pairs (i.e., $2M$ independent draws) — equivalent to halving the standard error for the same number of payoff evaluations.

Antithetic vs plain MC standard error at equal compute *At equal payoff-evaluation cost, antithetic sampling lowers the MC standard error below plain sampling across every sample size. Both schemes retain the $1/\sqrt{M}$ slope — variance reduction moves the intercept, not the rate — and the empirical gap matches the theoretical factor of $\sqrt{2}$ for monotone payoffs near ATM.*

Antithetic sampling fails when the payoff is *symmetric in Z* — for example, a straddle $X(Z) = |S_T - K|$ has $X(Z) = X(-Z)$ at strike $K = S_0 e^{rT}$, and the antithetic pair is perfectly positively correlated, giving $\rho = +1$ and *no* variance reduction (actually an increase relative to a single draw, though not relative to two independent draws). A diagnostic: if antithetic sampling does not improve things, try a pilot run with $M = 1000$ independent vs $M = 500$ antithetic pairs and compare the empirical variances. This takes fifteen minutes and can save a day of debugging.

Antithetic extends naturally to path simulation: use the vector of Gaussians $(Z_1, \dots, Z_N)$ on one path and $(-Z_1, \dots, -Z_N)$ on the antithetic pair. The correlated-path structure preserves monotonicity at each time step, and the technique continues to work. For path-dependent payoffs like barriers and Asians the variance reduction is similar to the European case, typically a factor of 2–3.

11.8.2 10.8.2 Control variates

The control-variate trick is more flexible and often more powerful than antithetic sampling. The idea: find an auxiliary variable Y that is strongly correlated with the payoff X and whose expectation $\mathbb{E}[Y]$ is known in closed form. Form the control-variate estimator

$$\widehat{V}_0^{\mathrm{cv}} \;=\; \widehat{m}_X \;+\; \beta\,(\mathbb{E}[Y] - \widehat{m}_Y), \tag{10.35}$$

where $\widehat{m}_X$ and $\widehat{m}_Y$ are the sample means of X and Y on the same M paths, and β is a constant to be chosen. The estimator is unbiased for any β because $\mathbb{E}[\widehat{m}_Y] = \mathbb{E}[Y]$, so the correction term has mean zero; it has variance

$$\mathbb{V}[\widehat{V}_0^{\mathrm{cv}}] \;=\; \frac{1}{M}\,[\,\mathbb{V}[X] \;-\; 2\beta\,\mathbb{C}[X,Y] \;+\; \beta^2\,\mathbb{V}[Y]\,], \tag{10.36}$$

which is minimised at

$$\beta^\star \;=\; \frac{\mathbb{C}[X,Y]}{\mathbb{V}[Y]}, \tag{10.37}$$

giving the minimum variance

$$\mathbb{V}[\widehat{V}_0^{\mathrm{cv}}]\,\big|_{\beta=\beta^\star} \;=\; \frac{\mathbb{V}[X]}{M}\,(1-\rho_{XY}^2), \tag{10.38}$$

where $\rho_{XY} = \mathrm{Corr}(X,Y)$. The variance reduction factor is $1/(1-\rho_{XY}^2)$: for $\rho_{XY} = 0.9$ the reduction is about $5\times$; for 0.99 it is $50\times$; for 0.999 it is $500\times$.

Choosing a good control. The essential question is what auxiliary Y is known in closed form *and* strongly correlated with the target payoff. Several standard choices:

- **The underlying's terminal value** $Y = S_T$, with $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0\,e^{rT}$ from the martingale property. Works well for European calls and puts, whose payoffs are monotone in S_T; typical correlation at ATM parameters is $\rho \approx 0.95$ for calls, giving about a $10\times$ variance reduction.
- **A vanilla Black–Scholes call on the same underlying**, with $\mathbb{E}[Y]$ given by (10.14)–(10.15). Works extremely well for barriers, digitals, and mildly exotic payoffs whose MC error on the vanilla portion would otherwise dominate; typical $\rho \approx 0.99$ for up-and-in calls, giving $50\times$ or more variance reduction.
- **A geometric-average Asian call** for arithmetic Asian pricing, where the geometric version has a closed-form BS-like formula and the arithmetic version does not. Typical $\rho > 0.99$ because the two averages are near-identical on almost every path; 100–500$\times$ variance reduction.
- **The terminal value on a drift-shifted GBM** as a control for stochastic-volatility payoffs, linking the Heston case back to the Black–Scholes benchmark with matched $\mathbb{E}[S_T]$ and approximate variance.

Estimating $\beta^\star$. The optimal $\beta^\star$ depends on the unknown covariance structure. In practice one runs a pilot batch of $M_{\mathrm{pilot}} = 1000$ or so paths, estimates $\widehat{\beta} = \widehat{\mathbb{C}}[X,Y]/\widehat{\mathbb{V}}[Y]$ from the pilot, and then runs the full M batch with that $\widehat{\beta}$ plugged into (10.35). The plug-in estimator is no longer strictly unbiased (because $\widehat{\beta}$ and $\widehat{m}_X$, $\widehat{m}_Y$ are all correlated through the data), but the bias is $O(1/M)$ and negligible relative to the statistical error. More sophisticated implementations use a two-sample scheme: estimate $\widehat{\beta}$ from paths $1,\dots,M_1$, use those paths only for the second-stage estimator on paths $M_1+1,\dots,M$, and aggregate.

11.8.3 10.8.3 Combining techniques

Antithetic and control variates compose. Apply antithetic sampling to generate the path pairs, evaluate both the target payoff X and the control payoff Y on each antithetic pair, take the antithetic average of each, and then apply the control-variate correction (10.35) to the antithetic averages. The variance reductions multiply roughly (not exactly, because the correlations interact), and for a European call with a stock-price control and antithetic draws one routinely achieves total variance reduction factors of 30–50 at essentially no added compute cost.

Beyond these two techniques the next layers — *stratified sampling* (partitioning the Gaussian space into equal-probability strata and drawing one sample per stratum), *importance sampling* (Radon–Nikodym reweighting of the

simulation distribution), *Latin hypercube sampling*, and *quasi-Monte-Carlo with Sobol or Halton low-discrepancy sequences* — each provide additional factors of 2–100 for the right payoff. A production pricer typically stacks two or three of these techniques; a research-grade pricer may stack four or five. The trader's mantra: "MC is universal but wasteful; always be improving the variance."

Variance-reduction techniques are indispensable for Heston Monte Carlo (Chapter 14) where the variance-process simulation itself introduces additional noise on top of the payoff noise, and where the closed-form benchmarks from Black–Scholes are no longer available but *approximate* BS controls based on a matched-moments log-normal still give large reductions. We return to this theme when we take up stochastic volatility.

11.8.4 10.8.4 A quick survey of the wider toolkit

For completeness, a brief summary of the other standard techniques a reader will encounter in the variance-reduction literature. We will not use these in the core chapter but they are the natural next layer for anyone building a production pricer.

Stratified sampling. Partition the Gaussian space into k equal-probability strata $\{S_1, \dots, S_k\}$ and draw M/k samples uniformly within each stratum, then re-weight each stratum by its probability mass (which is $1/k$ by construction). For a one-dimensional draw, use the inverse normal CDF: stratum $S_j = \{Z : \Phi^{-1}((j-1)/k) \le Z < \Phi^{-1}(j/k)\}$. The estimator is unbiased and its variance is always no worse than plain MC; for smooth payoffs it can be orders of magnitude better. The technique scales poorly with dimension — for d-dimensional Gaussian draws, stratifying each dimension into k strata gives k^d strata — so it is typically used only on the *dominant* direction (e.g. the terminal Brownian increment, which drives the bulk of the terminal payoff variance).

Importance sampling. Draw samples from an alternative distribution $\tilde{\mathbb{P}}$ that concentrates probability mass on the region where the payoff is large, then reweight by the Radon–Nikodym derivative $\mathrm{d}\mathbb{P}/\mathrm{d}\tilde{\mathbb{P}}$ from Chapter 5. The unbiased estimator is $\widehat{V}_0 = (1/M)\sum_m (\mathrm{d}\mathbb{P}/\mathrm{d}\tilde{\mathbb{P}})(Z^{(m)}) \cdot X(Z^{(m)})$ with $Z^{(m)} \sim \tilde{\mathbb{P}}$. For a deep-OTM call, shifting the drift upward by μ_{IS} produces more ITM paths, and the RN weight $\exp(-\mu_{\mathrm{IS}} Z - \frac{1}{2}\mu_{\mathrm{IS}}^2 T)$ keeps the estimator unbiased. Optimal μ_{IS} is found by minimising the estimator variance — a separate optimisation that is often done offline or with an adaptive scheme. Variance reductions of 10–1000 are common for rare-event payoffs; this is the technique of choice for deep-OTM options, tail-risk calculations, and credit-portfolio stress tests.

Quasi-Monte-Carlo with low-discrepancy sequences. Replace i.i.d. Gaussian draws with deterministic low-discrepancy sequences — Sobol, Halton, or Niederreiter — transformed by the inverse normal CDF. Low-discrepancy sequences fill the unit cube more uniformly than random samples, which for smooth integrands gives an error bound of $O((\log M)^d/M)$ rather than the probabilistic $O(M^{-1/2})$ of MC. For moderate dimensions ($d \lesssim 20$) the deterministic error bound is much smaller than the MC standard error at equal M. Quasi-MC pairs especially well with scrambling — randomisation of the sequence — to produce error bounds that are both small and unbiased, recovering the best of both worlds.

Multilevel Monte Carlo. For problems where the payoff depends on a discretised path with step Δt, the estimator $\widehat{V}_0 = \mathbb{E}[X_{\Delta t}]$ has bias $O(\Delta t)$ and MC variance $O(\mathrm{Var}[X_{\Delta t}]/M)$. Multilevel MC decomposes $\mathbb{E}[X_{\Delta t_L}] = \mathbb{E}[X_{\Delta t_0}] + \sum_\ell \mathbb{E}[X_{\Delta t_\ell} - X_{\Delta t_{\ell-1}}]$ and estimates each difference on its own grid, distributing the computational budget so that the coarser-grid levels (which are cheaper per path) carry more paths. The overall cost for error ϵ can be reduced from $O(\epsilon^{-3})$ to $O(\epsilon^{-2}(\log\epsilon)^2)$ or $O(\epsilon^{-2})$ depending on the payoff's regularity. Multilevel MC has become the preferred technique in SDE-discretisation-limited problems and is heavily used in academic and research-grade implementations.

Longstaff–Schwartz regression. For American-exercise and Bermudan payoffs, naive MC does not work because the optimal exercise policy depends on the continuation value, which is a conditional expectation that cannot be computed per-path. The Longstaff–Schwartz regression scheme runs MC paths forward, then runs a backward regression of realised continuation values on polynomial basis functions of the state; the fitted regression is used as the proxy continuation-value function. The resulting exercise policy is suboptimal but typically close to optimal, and the MC estimate is a low-biased lower bound on the true price. Combining Longstaff–Schwartz with any of the variance-reduction techniques above is standard practice for exotic American pricing.

A practitioner's rule of thumb. When in doubt, start with antithetic variates (always costs one line of code, typically halves variance). If the payoff has a closed-form correlated cousin, add a control variate (a few more lines, factor of 10–100). If the payoff is deep OTM or rare-event, add importance sampling (non-trivial, factor of 10–1000). If the

dimensionality is moderate and the payoff is smooth, use Sobol with scrambling (moderate effort, factor of 2–100). Stack them; each layer commutes approximately with the others.

11.9 10.9 Key Takeaways

1. **Why MC.** Monte-Carlo integration escapes the curse of dimensionality by achieving $O(M^{-1/2})$ error *independent of dimension*; its cost is $O(Md)$ compared to $O(N^d)$ for deterministic quadrature. Break-even dimension is around $d \approx 5$, and every realistic path-dependent pricing problem lives well above it.

2. **Strong Law of Large Numbers.** For i.i.d. draws of a random variable X with $\mathbb{E}[|X|] < \infty$, the sample mean converges almost surely to $\mathbb{E}[X]$. For finite-variance X, the Central Limit Theorem gives the $M^{-1/2}$ error rate and the basis for confidence intervals.

3. **Standard error.** The MC estimator's standard error is $\hat{\sigma}_{m_1} = \hat{s}/\sqrt{M}$, where $\hat{s}^2 = (1/(M-1)) \sum (X^{(m)} - \widehat{m}_1)^2$ is the unbiased sample variance. Halving the error bar costs 4× more paths.

4. **Exact GBM generator.** Under $\mathbb{Q}$, the lognormal increment $S_{t_n} = S_{t_{n-1}} \exp\{(r - \frac{1}{2}\sigma^2)\Delta t_n + \sigma\sqrt{\Delta t_n}\, Z_n\}$ with $Z_n \sim \mathcal{N}(0,1)$ i.i.d. is *exact at the grid points* — no discretisation bias. The missing-$\frac{1}{2}\sigma^2$ drift correction is the most common bug in home-grown MC pricers.

5. **European MC.** The MC estimator $\widehat{V}_0 = e^{-rT} \cdot (1/M) \sum_m \varphi(S_T^{(m)})$ should cross-check against Black–Scholes as a correctness test. At $M = 10{,}000$ and typical ATM parameters, the standard error is about 0.14 on a call worth about 10.45.

6. **Forward-starting (cliquet) atom.** The payoff $(S_{T_2} - \alpha S_{T_1})_+$ admits a closed form via iterated conditional expectation: $V_0 = S_0\,[\Phi(d_+) - \alpha\, e^{-r(T_2-T_1)}\, \Phi(d_-)]$ with $d_\pm = [\ln(1/\alpha) + (r \pm \frac{1}{2}\sigma^2)(T_2 - T_1)]/(\sigma\sqrt{T_2 - T_1})$. The S_{T_1} factor cancels in the moneyness ratio, making $d_\pm$ deterministic and the price linear in S_0.

7. **Barrier MC.** The up-and-in call payoff $(S_T - K)_+ \mathbf{1}_{\{\max_t S_t \geq U\}}$ is priced by simulation on a time grid with $\max_n S_{t_n}$ substituted for $\max_t S_t$. The substitution introduces a *downward* discretisation bias on the order of $\sqrt{\Delta t}$.

8. **Brownian-bridge correction.** Between two grid points with known endpoints, the conditional barrier-crossing probability has a closed form $\exp(-2\,\ln(U/S_{t_{n-1}})\,\ln(U/S_{t_n})/(\sigma^2 \Delta t_n))$. Applying the correction reduces barrier MC bias from $O(\sqrt{\Delta t})$ to $O(\Delta t^{3/2})$ — equivalent to a factor-of-10 grid refinement.

9. **Rare-event variance.** Barrier payoffs that knock in with small probability give MC estimators with large relative variance. The path count scales inversely with the knock-in probability, making brute-force MC impractical for deeply-barriered structures; importance sampling and control variates are essential.

10. **Antithetic variates.** For each Z draw, also use $-Z$. For monotone payoffs the antithetic pair has strongly negative correlation, and the variance of the average is less than half the variance of a single draw. Typical variance reduction factor ≈ 2 on ATM European calls.

11. **Control variates.** Given an auxiliary Y with known $\mathbb{E}[Y]$, form $\widehat{V}_0^{\mathrm{cv}} = \widehat{m}_X + \beta(\mathbb{E}[Y] - \widehat{m}_Y)$ with $\beta^\star = \mathbb{C}[X,Y]/\mathbb{V}[Y]$. Variance reduction factor $1/(1-\rho_{XY}^2)$. For $\rho = 0.99$, 50×; for $\rho = 0.999$, 500×.

12. **Combining techniques.** Antithetic and control variates compose; production pricers routinely stack two or three techniques for combined variance-reduction factors of 50–500.

13. **Path-dependent families.** Asians, lookbacks, barriers, cliquets, autocallables — all handled by the same MC machinery: generate paths under $\mathbb{Q}$, apply the payoff functional path-by-path, discount and average. The engineering difficulty lies in variance reduction and in handling the exact contractual conventions (continuous vs discrete monitoring, fixed vs floating strikes, business-day conventions).

14. **Bridge to Chapter 14.** Heston Monte Carlo introduces a second simulation dimension (the variance process) and its own set of discretisation biases (the square-root process near zero requires careful handling via the full-truncation Euler or the exact QE scheme of Andersen). The variance-reduction toolkit developed here transfers directly, with BS-based controls replaced by matched-moments lognormal controls.

11.10 10.10 Reference Formulas

Strong law of large numbers.

$$\lim_{M\to+\infty} \frac{1}{M}\sum_{m=1}^{M} X^{(m)} \;=\; \mathbb{E}[X] \quad \text{a.s., provided } \mathbb{E}[|X|]<\infty.$$

Central limit theorem (Monte-Carlo confidence interval).

$$\sqrt{M}\,(\widehat{m}_M - \mathbb{E}[X]) \xrightarrow{d} \mathcal{N}(0,\mathbb{V}[X]), \qquad \widehat{m}_M \pm 1.96\,\hat{\sigma}_{m_1} \text{ is a 95\% CI.}$$

Sample mean and Monte-Carlo standard error.

$$\widehat{m}_1 = \frac{1}{M}\sum_{m=1}^{M} X^{(m)}, \qquad \hat{\sigma}_{m_1} = \frac{1}{\sqrt{M}}\left(\frac{1}{M-1}\sum_{m=1}^{M}(X^{(m)}-\widehat{m}_1)^2\right)^{1/2}.$$

Exact lognormal GBM path simulation.

$$S_{t_n} = S_{t_{n-1}}\,\exp\{(r-\tfrac{1}{2}\sigma^2)\Delta t_n + \sigma\sqrt{\Delta t_n}\,Z_n\}, \quad Z_n \overset{\text{iid}}{\sim}_{\mathbb{Q}} \mathcal{N}(0,1).$$

European MC estimator.

$$\widehat{V}_0 = e^{-rT}\cdot\frac{1}{M}\sum_{m=1}^{M}\varphi\Big(S_0\,e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}\,Z^{(m)}}\Big).$$

Forward-starting (cliquet) call price.

$$V_0 = S_0\,[\Phi(d_+) - \alpha\,e^{-r(T_2-T_1)}\,\Phi(d_-)], \qquad d_\pm = \frac{\ln(1/\alpha)+(r\pm\frac{1}{2}\sigma^2)(T_2-T_1)}{\sigma\sqrt{T_2-T_1}}.$$

Up-and-in call payoff and MC estimator.

$$\varphi = (S_T-K)_+\,\mathbf{1}_{\{\max_{0\le t\le T} S_t \ge U\}}, \qquad \widehat{V}_0 = e^{-rT}\cdot\frac{1}{M}\sum_{m=1}^{M}(S_T^{(m)}-K)_+\,\mathbf{1}_{\{\max_n S_{t_n}^{(m)}\ge U\}}.$$

Brownian-bridge crossing probability (continuous barrier correction).

$$\mathbb{P}\Big(\max_{t\in[t_{n-1},t_n]} S_t \ge U \,\Big|\, S_{t_{n-1}}, S_{t_n}\Big) = \exp\Big(-\frac{2\,\ln(U/S_{t_{n-1}})\,\ln(U/S_{t_n})}{\sigma^2\,\Delta t_n}\Big) \quad \text{for } U>\max(S_{t_{n-1}},S_{t_n}).$$

In-out parity.

$$V^{\text{up-and-in call}} + V^{\text{up-and-out call}} = V^{\text{vanilla call}}.$$

Antithetic variate estimator (single draw).

$$\widetilde{X}^{(m)} = \tfrac{1}{2}\big[X(Z^{(m)}) + X(-Z^{(m)})\big], \qquad \mathbb{V}[\widetilde{X}^{(m)}] = \tfrac{1}{2}\mathbb{V}[X]\,(1+\rho),\ \rho = \mathrm{Corr}(X(Z),X(-Z)).$$

Control variate estimator and optimal coefficient.

$$\widehat{V}_0^{\text{cv}} = \widehat{m}_X + \beta(\mathbb{E}[Y]-\widehat{m}_Y), \qquad \beta^\star = \frac{\mathbb{C}[X,Y]}{\mathbb{V}[Y]}, \qquad \mathbb{V}[\widehat{V}_0^{\text{cv}}]\,\big|_{\beta^\star} = \frac{\mathbb{V}[X]}{M}(1-\rho_{XY}^2).$$

12 Chapter 11 — Calibration: Lattices, Short Rates, and the Yield Curve

12.1 11.0 Motivation: What Calibration Means

The previous part of the course built a continuous-time toolkit. CH03 proved Itô's lemma; CH04 proved Feynman–Kac; CH05 consolidated measure change and Girsanov. Those chapters established that, given a model, we can price: specify the dynamics, choose a numeraire, take an expectation. This chapter inverts the direction of that flow. Given market quotes — option prices, zero-coupon bond yields, swap rates — *how do we choose the model parameters that reproduce what the market is already saying?* That is calibration.

The shift in perspective is worth pausing on before diving into the technical apparatus. In the forward direction, pricing asks a "what if" question: if the world behaves according to these parameters, what price would any given contingent claim command? In the inverse direction, calibration asks a "given that" question: given that the market is quoting these prices right now, what must the world's parameters be? The forward question is mathematically clean — one plugs in numbers and turns a computational crank. The inverse question is philosophically harder, because it forces us to grapple with the fact that the market speaks in a noisy, over-determined, sometimes contradictory voice. Two dealers may quote the same option at slightly different prices; the same option may trade at different prices minutes apart; the same underlying may have liquid quotes for some strikes and maturities but vanishing liquidity for others. Calibration is the discipline of listening to this chorus and extracting a single coherent parameter vector that respects, within tolerance, everything the market has said.

Calibration is the daily bread of any trading desk, structuring team, or risk group. Before a pricing model can be used to quote a fresh trade, hedge a book, or mark positions to market, its parameters must be tuned to whatever the market is already saying. A beautifully specified model with miscalibrated inputs is worse than no model at all — it produces hedges that fail in ways the specification cannot explain, and marks that drift away from any counterparty's reconciliation. A concrete picture helps anchor the stakes. Imagine a desk holding a portfolio of exotic options — cliquets, autocallables, variance swaps — whose value depends on the whole implied-volatility surface and on the interest-rate term structure. Every morning, before trading opens, the pricing system needs to ingest yesterday's closing quotes on several hundred vanillas and several dozen bonds, convert them into a consistent set of model parameters, and use those parameters to mark the exotic book. If the calibration is stable and accurate, the marks are defensible to the counterparty, the risk officer, and the auditor. If the calibration is unstable — if the parameters jump by 20% from Monday to Tuesday despite the market barely moving — the marks wobble too, and the desk spends its day explaining unrealised P&L swings instead of trading. A single bad calibration day can generate a flurry of calls from risk management, custodians, and prime brokers demanding explanations for positions suddenly worth several percent less than yesterday. Stable, disciplined calibration is therefore not an academic nicety; it is an operational necessity.

The Fundamental Theorem of Asset Pricing (FTAP), introduced in CH02 and restated in CH05 after the measure-change machinery was in place, tells us that *no arbitrage* is equivalent to the existence of an equivalent martingale measure $\mathbb{Q}$ relative to a chosen numeraire. Calibration is the inverse problem: pick the $\mathbb{Q}$-parameters so that model-implied prices reproduce the observed prices. If we succeed, we have a model that — by construction — cannot be arbitraged against the specific instruments we calibrated to. Whether it also prices *unobserved* instruments correctly is a different question, and one we will return to repeatedly.

This distinction between "arbitrage-free with respect to calibration instruments" and "arbitrage-free with respect to all instruments" matters more than it might appear. A calibration procedure can make a model arbitrage-free against, say, the twenty vanilla S&P options used as calibration targets, while simultaneously making it deeply inconsistent with the thirty-year swap curve, the VIX futures strip, or correlated foreign-exchange pairs. The model passes a local arbitrage-freeness test but fails the global one. In practice this is almost always the situation — no single-factor or even two-factor model matches every quote in every market simultaneously — so calibration is as much about choosing *which* arbitrages to close as it is about closing them. A desk that trades S&P variance swaps picks calibration instruments that make the S&P variance surface arbitrage-free; it accepts residuals on unrelated markets.

The three faces of a pricing model. Every pricing model leads a triple life. The first face is *descriptive*: it tries to capture the dynamics that generate real price paths (random walks with drift, mean-reverting rates, stochastic volatility, jumps). The second face is *evaluative*: it assigns prices to payoffs not yet traded, by taking risk-neutral expectations. The third face — the one most relevant for this chapter — is *inverse*: it translates observed

market prices back into the parameters the model would need to reproduce them. Calibration is the procedure that mediates between these faces, taking the market's observed prices and turning them into the inputs the descriptive and evaluative faces require. These three faces are not equally respected in practice. The evaluative face is the one customers interact with: they ask for a price and they receive one, and they judge the desk by the tightness and consistency of those prices. The descriptive face is the statistician's preoccupation: does the model generate paths that, after appropriate change of measure, resemble historical data? The inverse face — calibration — is where the two other faces meet, and where the engineering effort is largely spent. A bank's quant analytics group will typically devote a disproportionate share of its development time to calibration infrastructure: solvers, stability monitors, outlier filters, persistence layers. The evaluative engine is often a simpler object sitting downstream of the calibration pipeline.

Why calibration is an inverse problem. A *forward* pricing problem takes parameters Θ (volatility, mean-reversion, risk-neutral drift) and returns a price $C(\Theta)$. An *inverse* problem takes an observed price C^{mkt} and asks: which Θ made it? Unlike the forward direction, which is well-posed by construction, the inverse direction can be ill-posed in three classic ways — *existence* (no Θ reproduces the quote exactly, perhaps because of bid-ask spread noise or model misspecification), *uniqueness* (many Θ's reproduce it, as when two different volatility surfaces imply the same vanilla prices but different exotic prices), and *stability* (small price perturbations cause large parameter jumps, so the calibrated parameters jitter violently from day to day even when the market itself has not moved materially). Any serious calibration effort has to recognise which of these pathologies apply and which technique tames them. Non-existence is typically handled by relaxing the requirement of exact fit and instead minimising a loss function that measures the gap between model and market — weighted least squares being the workhorse, where the weights encode either bid-ask spread tolerance or instrument liquidity. Non-uniqueness is handled by regularisation: adding penalty terms that favour parameter values close to a prior or historical average, so that among equally-well-fitting parameter vectors the one closest to the prior wins. Instability is addressed by a combination of regularisation and smoothing across calibration dates — one does not recalibrate from scratch every day, but rather uses yesterday's calibrated parameters as a warm start, and penalises large deviations unless the data genuinely supports them.

Moment matching — the workhorse of this chapter in the discrete setting — is attractive precisely because it dodges these pathologies: we match a small number of analytically tractable summary statistics (mean, variance, correlation) rather than a full price surface, so the map $\Theta \mapsto$ moments is usually a smooth local diffeomorphism and inversion reduces to solving a low-dimensional algebraic system. The cost is fidelity: matching two moments does not mean matching the full distribution. Two lognormal distributions with the same mean and variance are identical; two skewed-fat-tailed distributions with the same mean and variance can price out-of-the-money options very differently. The price we pay for moment matching's convenience is that the resulting model is only as good as a two-parameter summary of reality. For a variance swap, whose payoff is linear in realised variance, matching the first two moments of log-returns is essentially all the model has to do. For a deep out-of-the-money put, whose payoff lives entirely in the left tail, matching the first two moments is grossly insufficient — you need to match the tail directly. The match between calibration instruments and trading-book exposures is therefore the practitioner's first and most important design choice.

The identifiability question. Can distinct parameter sets produce identical data? On a single-period binomial lattice, (p, c) is identified by matching two moments: given $(\hat{\mu}T, \sigma^2 T)$ there is a unique (p, c) modulo the sign of c. For Vasicek (CH12), (κ, θ, σ) is identified from the yield curve plus one volatility input, but many parameter combinations produce *nearly* identical yield curves (the flat-likelihood direction where κ and σ are only well-identified as the ratio $\sigma^2/(2\kappa)$ in the long yield). Experienced practitioners learn which directions in parameter space are stiff (well-identified) and which are sloppy (nearly flat). Ignoring the sloppy directions is fine for pricing instruments close to the calibration set; it is dangerous for extrapolation. Downstream users who ask "what is κ?" must be told "anywhere in this range; the data is not informative" rather than given a spuriously precise point estimate.

Market-consistent modelling. A second theme running through this chapter is that your model's outputs are other desks' inputs. The yield you quote for a five-year bond will be used, perhaps ten minutes later, by someone pricing a five-year swap or a swaption expiring in two years on a three-year swap. If your calibration produces yields that are inconsistent with observable zero-coupon prices, you have introduced arbitrage into the downstream book. Market-consistent modelling is therefore not a purist's preoccupation: it is a network property of the whole derivatives ecosystem. Calibration is the point at which your model joins the network or introduces an inconsistency into it.

A third theme is *objective function design*. We will move from the naive "minimise squared price error" toward more sophisticated weighting schemes: weighting by vega so that thinly-priced out-of-the-money options do not dominate,

weighting by bid-ask spread so that low-confidence quotes contribute less, weighting by liquidity so that benchmarks dominate the fit. A fourth theme is *numerical method selection*: Nelder–Mead for robustness, Levenberg–Marquardt for speed, BFGS for smooth general objectives, differential evolution for multimodal global search, MCMC for full parameter uncertainty. A well-equipped calibration system uses a hybrid of two or three methods, selected by problem shape and runtime budget.

We work through three settings of increasing operational richness:

1. FTAP on a single-period lattice — recovering q and the step size c from price ratios.
2. Change of measure on a multinomial lattice — recovering the q-weights and handling non-uniqueness via optimisation and regularisation.
3. The interest-rate analogue — zero-coupon bond prices, bootstrapping a short-rate lattice period by period.

The continuous-time Vasicek / Ho–Lee / Hull–White machinery that this discrete story motivates is developed in CH12; its applications (IRS, CDS, bond options, the T-forward measure) appear in CH13.

12.2 11.1 FTAP on a Lattice

12.2.1 11.1.1 Fundamental Theorem of Asset Pricing

The first tool we need is the theorem that connects market data to model structure. Its statement is deceptively simple, but every calibration procedure in this chapter is, at bottom, a specialisation of this single equation to a particular set of observed prices.

FTAP (restated from CH02 / CH05). *No arbitrage $\iff$ there exists an equivalent martingale measure $\mathbb{Q} \sim \mathbb{P}$ such that for every traded asset X:*

$$\frac{X_t}{B_t} = \mathbb{E}^{\mathbb{Q}}\left[\frac{X_T}{B_T}\,\middle|\,\mathcal{F}_t\right] \tag{11.1}$$

where B_t is a traded numeraire with $B_t > 0$ a.s. Any traded asset normalised by the numeraire is a $\mathbb{Q}$-martingale.

Equation (11.1) is the most important equation in this chapter and one of the most important in all of derivatives pricing. It is worth committing to memory, not just as a formula but as a template. Every calibration exercise we will perform is an instance of this template: specify X, specify B_t, observe the left-hand side from market data, parameterise the right-hand side in terms of model parameters, and solve for the parameters. The specific forms will differ — sometimes the right-hand side is a simple discrete expectation, sometimes a continuous-time expectation, sometimes a Gaussian integral — but the structure is universal.

Let us unpack each term carefully. The traded asset X can be a stock, a bond, a futures contract, or a derivative — the theorem is agnostic about type, only that X is a price process quoted in a market where self-financing trading is available. The numeraire B_t is a reference asset against which all other prices are measured; it can be the money-market account (so $B_t = e^{rt}$ in a constant-rate world, or $\exp(\int_0^t r_u\,du)$ with stochastic rates), a zero-coupon bond, a foreign currency, or any other strictly positive traded process. The measure $\mathbb{Q}$ depends on the choice of numeraire: switching numeraires is equivalent to switching measures, as developed canonically in CH05. The conditional expectation $\mathbb{E}^{\mathbb{Q}}[\cdot \mid \mathcal{F}_t]$ is taken under $\mathbb{Q}$ and conditioned on the information available at time t.

The content of the theorem, stripped of formalism, is this: *you can price any claim by taking an expectation, provided you use the right measure.* The right measure is the one under which normalised prices become martingales. The right measure exists precisely when the market admits no arbitrage. And — crucially for us — the right measure is uniquely identified once you know enough about the market. Two subtleties of the statement are worth flagging. First, FTAP in its fullest generality requires care about integrability conditions, stopping times, and the precise notion of "no arbitrage" (no free lunch with vanishing risk is the right technical notion in continuous time). These subtleties matter in pathological cases but are invisible in the calibrations we perform. Second, the uniqueness of $\mathbb{Q}$ holds only in *complete* markets, where every contingent claim can be hedged. In incomplete markets (jumps, stochastic volatility without traded volatility), there are many equivalent martingale measures, each giving different prices for claims that cannot be hedged. Calibration in incomplete markets picks *one* of these measures — the one that matches the observed prices — but the choice is a modelling decision, not a mathematical inevitability.

Reading FTAP as a calibration rule. The theorem is often taught as an existence statement, but operationally it is a *constraint*. For every tradable instrument whose $t = 0$ price we observe, (11.1) gives one equation linking

the unknown $\mathbb{Q}$-probabilities (and unknown lattice parameters) to the observed price. With enough independent instruments and enough unknowns, we get a determined (often over-determined) system. Calibration is then nothing more than *solving this system* — sometimes in closed form (as in §11.1.3 for the binomial moments, or §11.3 for bond lattice bootstrap) and sometimes numerically (least squares against a basket of options). Think of FTAP as giving you a *coordinate system* on the space of market-consistent models. Each observed price is a coordinate; each unknown parameter is an axis. Calibration asks whether the observed coordinates uniquely pick out a point on the axes. If they do, we are done. If they underdetermine the point, we need more quotes. If they overdetermine it, we solve in a least-squares sense and live with residuals — which, in practice, is almost always the situation.

A useful mental exercise: count equations and unknowns at the start of any calibration problem. A single-factor Vasicek has four unknowns $(\kappa, \theta, \sigma, r_0)$. The short-rate observation gives one equation directly. Each bond-price observation gives one more equation. So four bond-price observations are needed to pin down four unknowns, giving an exactly-determined system. In practice we observe far more than four bonds — maybe twenty at various maturities — so the system is over-determined, residuals are non-zero, and calibration is a least-squares exercise. The Heston stochastic-volatility model has five unknowns plus the initial spot; a liquid vanilla market provides hundreds of quotes, so the system is massively over-determined. This disparity between unknowns and observations is typical and is what makes calibration a statistical, not algebraic, problem.

Why numeraire invariance matters. FTAP says the same underlying prices satisfy (11.1) for many choices of numeraire, with correspondingly different measures. This is not a redundancy; it is a powerful tool. In equity derivatives the most common choice is the money-market account, so $\mathbb{Q}$ is the usual risk-neutral measure. For interest-rate derivatives the T-forward measure (numeraire $= P_t(T)$) simplifies expectations that would otherwise involve stochastic discounting; CH13 uses it pervasively. For FX and quanto products, the domestic money market makes sense for some payoffs and the foreign one for others. The practitioner picks the numeraire that *annihilates* the most inconvenient source of randomness from the payoff, and the theorem guarantees the answer is the same regardless.

12.2.2 11.1.2 Single-Period Binomial Lattice

Before we write down the lattice, it is worth pausing on why the binomial structure is chosen rather than, say, a Gaussian step-size distribution. The answer is pedagogical economy: the binomial model has only two free parameters per step (the probability of up and the step size), matching the two pieces of market data we want to extract (drift and volatility). This symmetry between parameters and observations is the cleanest possible setting for learning how moment matching works. Once the binomial case is understood, extending to Gaussian, to multinomial, or to jump-extended processes is a matter of adding structure, not changing logic.

Start with one node A_{n-1} branching to two nodes over Δt:

$$A_{n-1} \;\longrightarrow\; \begin{cases} A_{n-1}\, e^{+c\sqrt{\Delta t}} & \text{(up)} \\ A_{n-1}\, e^{-c\sqrt{\Delta t}} & \text{(down)} \end{cases}$$

More compactly, introduce iid Bernoulli ticks $x_1, x_2, \ldots$ with $\mathbb{P}(x_i = +1) = p$, $\mathbb{P}(x_i = -1) = 1 - p$. Then

$$A_n \;=\; A_{n-1}\, e^{c\, x_n}, \qquad x_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\pm 1). \tag{11.2}$$

The log-return $\ln(A_n/A_{n-1}) = c\, x_n$ is $\pm c$.

The parameter c has a clean interpretation: it is the magnitude of a single-step log-return. If $c = 0.02$ and $\Delta t = 1$ day, then on any given day the asset moves either up by roughly 2% or down by roughly 2% (we use the exponential so that up-and-down produces $e^c e^{-c} = 1$, recovering the original level; multiplicative symmetry matters here because log-returns are what sum across time). The parameter p is the probability that the move is up; under the physical measure $\mathbb{P}$ this is tilted by the expected return, under the risk-neutral measure $\mathbb{Q}$ it is tilted only by the cost of carry.

It is worth pausing on why log-returns, rather than arithmetic returns, are the right quantity to model multiplicatively. Arithmetic returns do not add across time: if an asset is up 10% today and up 10% tomorrow, the two-day return is 21%, not 20%. Log-returns *do* add. This additivity is what makes log-returns amenable to the Central Limit Theorem — the sum of iid log-returns is approximately Gaussian, while the product of iid gross returns is approximately lognormal. The binomial model embraces this by writing each step as $S_n = S_{n-1} e^{c x_n}$ rather than $S_n = S_{n-1}(1 + c x_n)$, making the multi-step composition $S_N = S_0 e^{c(x_1 + x_2 + \cdots + x_N)}$ a clean exponential of the summed shocks.

The Bernoulli shock $x_i \in \{+1, -1\}$ is conventional — it lets us write each step as a scaled shock rather than a case split. In the small-Δt limit, the sum of many iid Bernoulli shocks is approximately Gaussian by the Central Limit Theorem, so the binomial lattice converges to Brownian motion (CH03). This is how the binomial model recovers the Black–Scholes world in the limit, and why the tree can be used as a discrete surrogate for continuous-time pricing. It is also why moment matching gets the right answer: matching two moments of a two-valued random variable fixes its distribution completely, and the CLT does the rest. A practical aside: the binomial tree as described here is *recombining* — the up-then-down path and the down-then-up path land at the same price $S_0 e^c e^{-c} = S_0$. A recombining tree has $n+1$ possible terminal nodes after n steps (rather than 2^n), which makes both pricing and calibration computationally tractable.

12.2.3 11.1.3 Recovering p and c from Observed Data

We want the parameters (p, c) — strictly, (q, c) under $\mathbb{Q}$ — to reflect what the market shows. Match the first two moments of the log-return.

Mean.

$$\mathbb{E}^{\mathbb{P}}\left[\ln \tfrac{A_T}{A_0}\right] \;=\; \mathcal{N}\, c(2p-1) \;\equiv\; \hat{\mu} T \tag{11.3}$$

where $\mathcal{N} = T/\Delta t$ is the number of steps. The equation should be parsed as follows. On each step, the log-return is $c\, x_i$, which has expectation $c\mathbb{E}[x_i] = c(2p-1)$. Summing over $\mathcal{N}$ iid steps gives total log-return $\mathcal{N}c(2p-1)$. Equating this to $\hat{\mu}T$ — the observed physical drift — gives us our first calibration equation. Note the structural parallel with the familiar one-factor Gaussian world: the total drift over T is linear in T, because drift is additive across independent steps.

Where does $\hat{\mu}T$ come from in practice? There are several sources, each with pitfalls. The most direct is historical return data: compute the sample mean of daily log-returns and scale to annual. This gives a point estimate of the physical drift, but the estimate is noisy — a one-year window of daily returns gives a drift estimate with standard error of order $\sigma/\sqrt{T}$, which for $\sigma = 20\%$ and $T = 1$ year is 20% of the true drift. Historical drift estimates are therefore nearly useless at short horizons and only marginal at long ones; this is why risk-neutral measures are so much more widely used in pricing applications — the $\mathbb{Q}$-drift is pinned down by the risk-free rate, which is observable with essentially no noise.

Variance.

$$\mathbb{V}^{\mathbb{P}}\left[\ln \tfrac{A_T}{A_0}\right] \;=\; \mathcal{N}\, c^2(1-(2p-1)^2) \;\equiv\; \sigma^2 T. \tag{11.4}$$

Variance adds across iid steps, so total variance is $\mathcal{N}$ times the single-step variance. The single-step variance is $c^2\mathbb{V}[x_i]$ because c is a deterministic scale factor. And $\mathbb{V}[x_i] = \mathbb{E}[x_i^2] - (\mathbb{E}[x_i])^2 = 1-(2p-1)^2$, using $x_i^2 = 1$ with probability one. The quantity $(2p-1)^2$ is small when p is near $1/2$ — which, as we will see, is the regime relevant for small time-steps — so $\mathbb{V}[x_i] \approx 1$ and the variance is approximately $\mathcal{N}c^2$.

The variance $\sigma^2 T$ is much easier to measure empirically than the drift $\hat{\mu}T$. The standard error of a realised variance estimate scales as $\sigma^2\sqrt{2/T}$ for large T and small sampling intervals, so even a one-year history of daily returns gives a variance estimate accurate to a few percent. This is why volatility estimates from historical data are generally taken seriously in practical pricing, while drift estimates are regarded with considerable skepticism.

Equations (11.3) and (11.4) form a two-equation system in the two unknowns (p, c). The $\mathbb{Q}$-version is structurally identical, only with $\hat{\mu}$ replaced by the risk-neutral drift. This two-for-two structure is a general feature of well-designed moment-matching schemes: the number of free parameters equals the number of moments matched, so the system is square and generically has a unique solution.

12.2.4 11.1.4 Small-Step Asymptotics

Write $\sigma^2 T = \mathcal{N}\, c^2(1-(2p-1)^2)$. For small Δt, $(2p-1)^2 \ll 1$, so

$$\sigma^2 T \;\approx\; \mathcal{N}\, c^2 \;=\; \frac{T}{\Delta t}\, c^2 \quad\Longrightarrow\quad \boxed{c \;\approx\; \sigma\sqrt{\Delta t}} \tag{11.5}$$

Substituting into the mean equation $\hat{\mu}T = c\,\mathcal{N}(2p-1)$ gives $2p-1 \approx (\hat{\mu}/\sigma)\sqrt{\Delta t}$ and hence

$$\boxed{p \;\approx\; \tfrac{1}{2}\Big(1 + \tfrac{\hat{\mu}}{\sigma}\sqrt{\Delta t}\Big)} \tag{11.6}$$

This is the calibration of the physical probability p. Under $\mathbb{Q}$, the same structural formula applies with $\hat{\mu}$ replaced by the drift that makes A_t/B_t a martingale (e.g. $r - \frac{1}{2}\sigma^2$ for a log-normal asset).

Worked example. Take $\sigma = 0.20$, $\hat{\mu} = 0.08$, $\Delta t = 1/252$. Then $c = 0.20\sqrt{1/252} \approx 0.01260$ and $p \approx \frac{1}{2}(1 + 0.4\sqrt{1/252}) \approx \frac{1}{2}(1 + 0.0252) = 0.5126$. An equity with 20% annualised volatility and 8% annualised drift has, on a one-day step, about a 1.26% typical log-return shock and a 51.26% probability of an up-move. The asymmetry between up and down probabilities (just over one percentage point) is small enough that if you looked at a few days of such returns you would see what looks like a fair coin; the asymmetry only becomes clear when you aggregate hundreds of days. Drift is small on short horizons and dominates only over long ones, which is why the "random walk" intuition for short-term stock movements is approximately correct, and why the drift-extractable information in historical returns is notoriously noisy. The Sharpe ratio of $0.08/0.20 = 0.4$ is typical for equity indices — equity risk premium divided by equity volatility — and reappears as the dominant signal in the calibration.

Why the Brownian limit emerges. The deeper reason moment matching recovers the right continuous-time dynamics is the CLT. If on each step we match the conditional mean and variance of log-returns to $\hat{\mu}\Delta t$ and $\sigma^2\Delta t$ respectively, and the shocks are independent, then the sum of $\mathcal{N} = T/\Delta t$ steps is a random variable with mean $\hat{\mu}T$ and variance $\sigma^2 T$. As $\Delta t \to 0$, the CLT guarantees this sum converges in distribution to $\mathcal{N}(\hat{\mu}T, \sigma^2 T)$ — which is exactly the log-return distribution of geometric Brownian motion. The binomial tree is therefore not a separate model from Black–Scholes; it is a discretisation that converges to the same limit, provided the step sizes and probabilities are chosen (i.e. calibrated) in the way (11.5) and (11.6) prescribe. The scaling laws emerge from dimensional analysis alone. Variance scales as $\sigma^2\Delta t$, so the step size scales as $\sigma\sqrt{\Delta t}$. The drift contribution to the log-return scales as $\hat{\mu}\Delta t$, and the probability tilt must therefore scale as $\hat{\mu}\Delta t/(\sigma\sqrt{\Delta t}) = (\hat{\mu}/\sigma)\sqrt{\Delta t}$.

From physical to risk-neutral: where the Sharpe ratio goes. Under $\mathbb{Q}$, the drift that enters (11.6) is not $\hat{\mu}$ but the martingale drift — for a stock paying no dividends in a constant-rate world, $r - \frac{1}{2}\sigma^2$. The Sharpe ratio of the *risk premium* is therefore $(\hat{\mu} - r)/\sigma$, and this is precisely the quantity that disappears when we pass to $\mathbb{Q}$. The formula structure is unchanged; the numerator in the tilt shifts from $\hat{\mu}$ to $r - \frac{1}{2}\sigma^2$, reflecting the replacement of realised returns by risk-free returns. This is the lattice incarnation of Girsanov's theorem (CH05).

Calibration as an inverse problem, restated. The binomial calibration is formally an *exactly-determined* inverse problem, not an ill-posed one. Two moments and two unknowns; a smooth, monotonic map between moments and parameters as long as $\sigma > 0$; closed-form inversion via (11.5)–(11.6); straightforward stability. All three of the usual pathologies of inverse problems — non-existence, non-uniqueness, instability — are absent. This is why the binomial case is pedagogically so useful: it lets us see the moment-matching machinery at its cleanest, before the complications of real-world calibration sink their teeth in. The other calibration problems in this chapter and the next — the bond-lattice bootstrap in §11.3, the Vasicek yield-curve fit in CH12, and the implied-vol surface fits in CH14 — are all, to varying degrees, ill-posed, and each will require one or more of the three remedies: weighted least squares, regularisation, and cross-date smoothing.

12.3 11.2 Change of Measure on Multinomial Trees

12.3.1 11.2.1 Setup: Why Multinomial

Consider a richer lattice where a single node A_0 branches to multiple children (e.g. three children A_1, B_1, C_1 each with its own sub-tree). Pricing under $\mathbb{P}$ uses probabilities $\{p^A, p^B, p^C\}$; pricing under $\mathbb{Q}$ uses $\{q^A, q^B, q^C\}$. We want to express the q's in terms of the p's (and of observed price ratios).

Why bother with multinomial trees? The binomial tree is adequate for simple equity options, but as soon as the state space has more than one non-trivial dimension — say, an equity *and* an interest rate, or an asset with both a continuous price component and a discrete jump component — we need lattices with more than two branches per step. Trinomial trees also have a useful practical feature: the middle branch allows the state to hold roughly constant, which is convenient for modelling slow processes like mean-reverting interest rates. The calibration logic below is cleanest in the binomial case but generalises naturally.

A concrete example makes the case for multinomial lattices vivid. Consider a callable bond: the issuer has the right to redeem the bond at par on specified dates. Pricing this requires a short-rate lattice, because the call decision depends on where rates go. If rates spike up, the bond price drops, and the issuer has no incentive to call (they

can reborrow more cheaply by not calling). If rates drop, the bond price rises, and the issuer calls to refinance. A binomial short-rate tree is clumsy for this: the two branches move the state up or down with equal probability, but mean-reverting rates typically have three economic regimes — rising, falling, or flat — and the flat regime is where most of the probability mass sits. A trinomial tree with up, middle, and down branches captures this more naturally: the middle branch lets the rate stay roughly where it is (consistent with a typical short-rate half-life of several years), and the up and down branches capture the tails. The price of a callable bond computed on a trinomial lattice converges to the true continuous-time price faster than on a binomial lattice with the same number of total nodes. Another example: jump-diffusion equity models. If the underlying can either diffuse continuously or jump to a distressed level, a single binomial branch cannot capture both dynamics. A trinomial tree with "up diffusion", "down diffusion", and "jump" branches handles this cleanly, at the cost of needing to calibrate three probabilities and three step sizes rather than two.

The martingale identity (11.1) applied to numeraire-normalised values gives, for each asset C (traded) and numeraire A:

$$\frac{C_0}{A_0} = q^A \frac{C_u}{A_u} + (1 - q^A) \frac{C_d}{A_d} \tag{11.7}$$

$$\frac{C_0}{B_0} = q^B \frac{C_u}{B_u} + (1 - q^B) \frac{C_d}{B_d} \tag{11.8}$$

Here C_u, C_d are the values of C in the up and down states; A_u, A_d, B_u, B_d are similarly the values of the two numeraire candidates. The superscripts on q track *which* numeraire the measure is defined relative to — $\mathbb{Q}^A$ is the measure that makes things a martingale in units of A, and correspondingly for $\mathbb{Q}^B$. Different numeraires correspond to different martingale measures, even though they price the same underlying claims; CH05 treated the continuous-time version of this relationship exhaustively.

12.3.2 11.2.2 Solving for q

From (11.7):

$$q^A = \frac{\dfrac{C_0}{A_0} - \dfrac{C_d}{A_d}}{\dfrac{C_u}{A_u} - \dfrac{C_d}{A_d}}, \qquad q^B = \frac{\dfrac{C_0}{B_0} - \dfrac{C_d}{B_d}}{\dfrac{C_u}{B_u} - \dfrac{C_d}{B_d}}. \tag{11.9}$$

Formulas (11.9) are simply the no-arbitrage pricing rule inverted. Read them as follows: the risk-neutral probability of an up-move is the "lever arm" that makes the current normalised price C_0/A_0 an exact convex combination of the up- and down-state normalised prices. If the current price is above the midpoint of the up and down possibilities, $q^A > 1/2$; if below, $q^A < 1/2$. Crucially, if the current price is *outside* the range $[C_d/A_d, C_u/A_u]$, there is no valid $q^A \in [0, 1]$ and the market is arbitrageable. This last observation is why FTAP links the existence of $\mathbb{Q}$ to the absence of arbitrage: when an arbitrage exists, no consistent martingale measure can reproduce the quotes.

A worked scenario makes this concrete. Suppose at time zero we observe $C_0 = 100$, $A_0 = 98$, and at time Δt the two possible states are $(C_u, A_u) = (110, 102)$ and $(C_d, A_d) = (90, 98)$. The current ratio $C_0/A_0 = 100/98 \approx 1.0204$; the up-state ratio $C_u/A_u = 110/102 \approx 1.0784$; the down-state ratio $C_d/A_d = 90/98 \approx 0.9184$. The current ratio sits comfortably between the two future ratios, so the system is arbitrage-free, and we compute $q^A = (1.0204 - 0.9184)/(1.0784 - 0.9184) = 0.1020/0.1600 = 0.6375$. The $\mathbb{Q}^A$-probability of an up-move is 63.75%; the down-probability is 36.25%. These probabilities are the market's *implicit* forward-looking assessment, under numeraire A, of the likelihood of each future state; they reflect both genuine subjective probabilities and risk preferences, compressed into a single no-arbitrage quantity. Now suppose instead that C_0 is observed at 112 rather than 100. The current ratio becomes $112/98 \approx 1.1429$, which is *above* the up-state ratio 1.0784. There is no $q^A \in [0, 1]$ that can reconcile this — the current price is higher than even the best future outcome, so an arbitrageur could short C, buy A, and collect the spread. The system is explicitly arbitrageable, and the calibration fails. Calibration is not an optional step that improves the model — it is an *arbitrage-detection* step. If calibration produces well-defined parameters, the market is (locally) arbitrage-free; if it cannot, there is a profitable trade sitting on the screen. In practice, this situation is rare but not unheard of: stale quotes, manual data-entry errors, or genuine market dislocations can all produce momentary violations. A disciplined calibration system flags such quotes and either excludes them or alerts a human operator.

12.3.3 11.2.3 Radon–Nikodym on a Finite Tree

Express the same ratio C_0/B_0 using numeraire A and convert:

$$\frac{C_0}{B_0} \;=\; q^A\,\frac{A_0}{B_0}\cdot\frac{C_u}{A_u} + (1-q^A)\,\frac{A_0}{B_0}\cdot\frac{C_d}{A_d}$$

Rewriting with B-ratios,

$$\frac{C_0}{B_0} \;=\; \underbrace{q^A\,\frac{A_0}{B_0}\cdot\frac{B_u}{A_u}}_{>0}\cdot\frac{C_u}{B_u} \;+\; \underbrace{(1-q^A)\,\frac{A_0}{B_0}\cdot\frac{B_d}{A_d}}_{>0}\cdot\frac{C_d}{B_d}. \tag{11.10}$$

The key trick here is multiplying top and bottom by appropriate ratios so that C appears normalised by B (as on the left) rather than by A. What matters is the *form* of the result: C_0/B_0 is expressed as a convex combination of C_u/B_u and C_d/B_d, with new weights that differ from q^A. Those new weights are the $\mathbb{Q}^B$-probabilities, and they must equal $q^B, 1-q^B$ because (11.8) is the unique such representation. Compare term-by-term with (11.8):

$$q^B \;=\; q^A\cdot\frac{B_u/A_u}{B_0/A_0}, \qquad (1-q^B) \;=\; (1-q^A)\cdot\frac{B_d/A_d}{B_0/A_0}. \tag{11.11}$$

Consistency check. Summing the two pieces must give 1:

$$q^B + (1-q^B) \;=\; \frac{A_0}{B_0}\left[q^A\,\frac{B_u}{A_u} + (1-q^A)\,\frac{B_d}{A_d}\right] \;=\; \frac{A_0}{B_0}\cdot\frac{B_0}{A_0} \;=\; 1.\ \checkmark \tag{11.12}$$

The bracket equals B_0/A_0 because B/A is itself a $\mathbb{Q}^A$-martingale. This consistency check is not a triviality; it is the embodiment of the martingale property. If B/A is not a $\mathbb{Q}^A$-martingale, the calculation fails — the two new weights do not sum to one, meaning there is no valid measure $\mathbb{Q}^B$. So the existence of an alternative pricing measure is equivalent to the original one correctly pricing the *new numeraire* as a tradable asset.

Equations (11.11) are exactly the Radon–Nikodym density on the lattice:

$$\boxed{\frac{d\mathbb{Q}^B}{d\mathbb{Q}^A}(\omega) \;=\; \frac{B_\cdot(\omega)/B_0}{A_\cdot(\omega)/A_0}} \tag{11.13}$$

i.e. the density is the *ratio of numeraire-returns*. In shorthand, writing $\tilde{A} = A/A_0$ and $\tilde{B} = B/B_0$:

$$\frac{d\mathbb{Q}^B}{d\mathbb{Q}^A} \;=\; \frac{\tilde{B}}{\tilde{A}}. \tag{11.14}$$

This is the discrete-time, finite-state incarnation of the Radon–Nikodym derivative that CH05 developed in continuous time. Under $\mathbb{Q}^A$ all prices are "measured in units of A"; switching numeraires to B is a relabelling of the unit, so scenarios where B has over-performed A get up-weighted by their realised relative return $B_\cdot/A_\cdot$ (normalised by the $t=0$ ratio so that the density has mean one). The Radon–Nikodym density is therefore a tradable object — the "relative performance" of the two numeraires — not a mysterious abstraction. A probability measure has to integrate to one, so any valid Radon–Nikodym density must have $\mathbb{E}^{\mathbb{Q}^A}[d\mathbb{Q}^B/d\mathbb{Q}^A] = 1$. From (11.14): under $\mathbb{Q}^A$, $\mathbb{E}^{\mathbb{Q}^A}[\tilde{B}_T/\tilde{A}_T] = \tilde{B}_0/\tilde{A}_0 = 1$, using that B/A is a $\mathbb{Q}^A$-martingale. So indeed the density has mean one, and $\mathbb{Q}^B$ is a well-defined probability measure.

Multi-step extension. If the tree branches further at the up-node (children with sub-probabilities $q_u, r_u, \ldots$) and similarly at the down-node, each local change-of-measure is governed by the same ratio. The overall change of measure from $t=0$ to $t=T$ is the composition of all these local operations along every path. In the continuous-time analogue (CH05), this becomes a stochastic exponential; in the discrete case it is a telescoping product of local ratios. A calibrated tree can be diagnosed: at every node, one can check that the local martingale property holds for each numeraire pair. If any of these local checks fails, the calibration has introduced a local arbitrage — which is a bug.

Production calibration systems routinely run these checks as unit tests, catching implementation errors that would otherwise manifest as mysterious mispricings downstream.

Radon-Nikodym density across states from multinomial calibration *Five-state discrete world: physical probabilities p_i (blue) are uniform, while calibrated risk-neutral probabilities q_i (red) tilt mass toward the central "stress" states that pricing data implicates. The RN density $dQ/dP = q_i/p_i$ (purple markers) records that tilt state-by-state and is the discrete-tree analogue of the continuous-time Girsanov density of CH05.*

12.3.4 11.2.4 Calibration as Optimisation: The Least-Squares Setup

Before turning to the interest-rate case, we pause to lay out the calibration problem as it is actually formulated in practical systems. Given N observed market instruments with prices $\{C_i^{\text{mkt}}\}_{i=1}^N$, a model with parameter vector $\Theta \in \mathbb{R}^d$, and a model pricing function $C_i^{\text{model}}(\Theta)$, the simplest calibration objective is

$$\Theta^\star \;=\; \arg\min_{\Theta} \sum_{i=1}^{N} \big(C_i^{\text{model}}(\Theta) - C_i^{\text{mkt}}\big)^2. \tag{11.15}$$

This is the ordinary least-squares (OLS) objective. Minimising (11.15) gives the parameter vector that, on average, best reproduces the market prices. The word "best" hides multitudes — "best" in what sense, under what weighting, with what penalty for over-fit — and the next several paragraphs unpack these choices.

The first and most important refinement is *weighting*. Not all market prices are equally informative. A deep in-the-money call option that trades with a tight bid-ask is a tighter constraint on the model than a deep out-of-the-money put trading with a wide bid-ask and small absolute price. Equal weighting in price space treats them as equally important; weighting each residual by the inverse bid-ask squared downweights the noisy quote and lets the clean quote dominate. The weighted least-squares objective is

$$\Theta^\star \;=\; \arg\min_{\Theta} \sum_{i=1}^{N} w_i \big(C_i^{\text{model}}(\Theta) - C_i^{\text{mkt}}\big)^2, \tag{11.16}$$

with $w_i = 1/\text{spread}_i^2$ or some related quantity. The algebraic form is the same as (11.15); the economic content is very different.

A second common weighting scheme is *vega weighting*. For options, the natural trader-friendly quantity is implied volatility, not price. Two options with the same implied volatility but different strikes have very different prices, yet carry the same economic information. Converting the least-squares fit from price space to implied-volatility space is equivalent to weighting each price residual by $1/\text{vega}_i$, because $\text{vega} = \partial C/\partial \sigma_{\text{imp}}$ is exactly the local conversion factor. The vega-weighted objective is

$$\Theta^\star \;=\; \arg\min_{\Theta} \sum_{i=1}^{N} \frac{\big(C_i^{\text{model}}(\Theta) - C_i^{\text{mkt}}\big)^2}{\text{vega}_i^2}, \tag{11.17}$$

which is equivalent (to first order, near the market-implied volatility) to

$$\Theta^\star \;=\; \arg\min_{\Theta} \sum_{i=1}^{N} \big(\sigma_i^{\text{model}}(\Theta) - \sigma_i^{\text{imp,mkt}}\big)^2. \tag{11.18}$$

Vega is the option's sensitivity to implied volatility; near-the-money options have high vega, deep OTM options have low vega. In price space, a 1% error in an at-the-money price might correspond to a 0.1-vol error, while a 1% error in a deep OTM price might correspond to a 5-vol error. An unweighted fit would treat these as equally bad; a vega-weighted fit treats the deep OTM error as 50 times worse — as a trader quoting in vol space would insist. A subtle caveat: vega-weighting is a *linearisation*; it breaks down for very deep OTM options where the vega is vanishingly small and a literal $1/\text{vega}^2$ weight gives that strike enormous, usually spurious, importance. A practical weighting scheme caps the weight, $w_i = \min(1/\text{vega}_i^2, w_{\max})$, or combines vega and spread weighting via a harmonic mean. Tuning these caps is part of the craft of building a robust calibration.

12.3.5 11.2.5 Global vs Local Minima

The objective function (11.15) is, in general, a non-convex function of the parameters Θ. For the binomial lattice it happens to be quadratic (the moment-matching equations are linear in parameters after log-transformation), but for more general models — stochastic volatility, jump diffusion, multi-factor rates — the landscape has multiple local minima. Which minimum the optimiser finds depends on where it starts. This is a serious practical concern. An equity trader who runs a stochastic-volatility model on Monday morning and gets parameters ($\kappa = 2, \theta = 0.04, \sigma_v = 0.3, \rho = -0.7$) may rerun the same calibration on Tuesday with yesterday's minimum as a starting point and get ($\kappa = 2.1, \theta = 0.041, \sigma_v = 0.31, \rho = -0.71$) — a small, economically sensible shift. But if the starting point is reset to a default every day, Tuesday's run might find a completely different local minimum, say ($\kappa = 0.5, \theta = 0.06, \sigma_v = 0.5, \rho = -0.9$), with similar fit quality but very different economic implications. The two parameter vectors produce nearly identical prices on the calibration basket but very different prices on instruments *outside* the basket. If the bank uses the Monday parameters to price exotics on Tuesday but recalibrates for the blotter, the exotics will be valued one way while the hedging positions use a different calibration — a recipe for P&L surprise.

The standard defenses against this problem are:

1. **Warm starts.** Always initialise today's calibration from yesterday's minimum. This enforces continuity across dates and typically keeps the optimiser in the same local basin even if another basin is marginally deeper. The cost is that the calibration may become "stuck" in a stale minimum if the market has genuinely moved to a new regime; the remedy is to add periodic cold starts to check for regime shifts.
2. **Multi-start optimisation.** Run the optimiser from several starting points (a small random grid, or structured starts from economically plausible prior values) and take the best. If multiple starts converge to distinct minima of comparable depth, the problem is flagged as non-identifiable and a human operator is consulted.
3. **Global optimisation methods.** Use algorithms designed to escape local minima — simulated annealing, differential evolution, genetic algorithms, basin-hopping. These are slower than gradient-based local methods but more reliably find global minima.
4. **Problem reformulation.** Often the landscape can be smoothed by changing variables. Calibrating Heston (CH14) in $(\kappa, \theta, \sigma_v, v_0, \rho)$ is notoriously multimodal; reparameterising in terms of more orthogonal quantities — at-the-money variance, skew, curvature, mean-reversion speed — can make the landscape nearly convex. This is sometimes called "calibrating in the right coordinates".

12.3.6 11.2.6 Numerical Methods

The optimisation problems above are nonlinear least-squares in a moderate-dimensional space. Several algorithmic families are in common use. *Nelder–Mead* maintains a simplex of $d + 1$ points and iteratively reflects, expands, contracts, and shrinks it; it requires only function evaluations (no gradients), so it is robust to objectives that are noisy, discontinuous, or expensive to differentiate, at the cost of slow convergence. *Levenberg–Marquardt* is the gold standard for nonlinear least-squares: it exploits the sum-of-squares structure to approximate the Hessian from Jacobians alone, and blends Gauss–Newton with gradient-descent via a damping parameter λ. It typically converges in tens of iterations on well-posed problems; for implied-vol-surface fits with $N \sim 100$ instruments and $d \sim 5$ parameters, it converges in milliseconds. *BFGS* and *L-BFGS* are quasi-Newton methods building up a Hessian approximation iteratively — the full version for moderate d, the limited-memory version for thousands of parameters. *Differential evolution* and related global methods maintain a population of candidate parameter vectors and evolve them via mutation, crossover, and selection; they are slower (seconds to minutes) but reliably find global minima.

Hybrid strategies. A production calibration system rarely uses a single method. A typical setup: start with a coarse differential-evolution search to find a plausible global minimum; refine with Levenberg–Marquardt for tight local convergence; verify stability with a Nelder–Mead polish. Practical runtime targets: a single-factor rates model against 20 bond quotes under 10 ms; Heston against 100 vanilla options under 100 ms; a multi-factor Libor market model against 500 caplet/swaption quotes under 10 seconds. Faster is always better, but stability and correctness trump speed by a wide margin in production.

12.3.7 11.2.7 Regularisation and Well-Posedness

When the calibration problem is under-determined — more parameters than informative observations, or observations that leave a sloppy direction unpinned — regularisation adds a penalty term to the objective function:

$$\Theta^\star \;=\; \arg\min_{\Theta}\Big[\sum_i w_i\big(C_i^{\text{model}}(\Theta) - C_i^{\text{mkt}}\big)^2 + \lambda\,\mathcal{R}(\Theta)\Big]. \tag{11.19}$$

The penalty $\mathcal{R}(\Theta)$ encodes prior knowledge: smoothness ($\|\nabla^2\Theta\|^2$ for curve fits), proximity to a historical mean ($\|\Theta - \Theta_{\text{prior}}\|^2$), sparsity ($\|\Theta\|_1$), or any other property the modeller wants to encourage. The weight λ controls how much regularisation is applied: small λ trusts the data, large λ trusts the prior. Choosing λ is itself a subproblem, usually handled by cross-validation (try several λ values and pick the one that minimises out-of-sample error) or by Bayesian hierarchical modelling (treat λ as a hyperparameter to be integrated over).

The Bayesian interpretation of regularisation is worth holding in mind. A penalty $\|\Theta - \Theta_{\text{prior}}\|^2$ is equivalent to a Gaussian prior on Θ centred at Θ_{prior}; a penalty $\|\Theta\|_1$ is equivalent to a Laplace prior at zero; a penalty $\|\nabla^2\Theta\|^2$ encodes a prior that parameter values change smoothly along some ordering (useful for term-structure calibration where we expect neighbouring maturities to have similar parameters). Regularisation is the frequentist's way of sneaking Bayesian priors into a point-estimate framework.

The fit-vs-extrapolation tradeoff. Strong regularisation (λ large) produces parameter estimates that are close to the prior, insensitive to the data, and therefore stable but potentially biased. Weak regularisation (λ small) produces parameter estimates that fit the data tightly, sensitive to every wiggle, and therefore unbiased in expectation but high-variance in any given run.

L-curve: fit residual vs solution norm across lambda *L-curve diagnostic for a Tikhonov-regularised linear inverse problem. Each point is one value of λ; the corner (elbow, marked) is the classical heuristic choice — the regularisation strength beyond which further stiffening raises residual without lowering solution norm. Small-λ end: data-fitting, overfit regime. Large-λ end: prior-dominated, underfit regime.* The sweet spot depends on the application: a conservatively-marked book wants stability (high λ); an aggressive trading book wants responsiveness (low λ). There is no universal right answer, and the choice is part of the desk's risk-appetite statement. A related perspective: in-sample fit is a strictly decreasing function of λ (regularisation makes the model stiffer, so it fits the training data less well), while out-of-sample fit is typically U-shaped (zero regularisation overfits, infinite regularisation underfits, and the optimum is somewhere in between). Cross-validation finds this optimum empirically.

Training vs out-of-sample RMSE across regularisation strength *The U-shape of out-of-sample error (red) in λ contrasts with the monotone rise of training error (blue). Small λ overfits — training error shrinks but OOS error spikes; large λ underfits — both errors rise. The optimum λ is the OOS minimiser, found in practice by cross-validation.*

Out-of-sample stability. A calibrated model's in-sample performance is tautological: by construction, the model matches its calibration instruments to within the optimiser's tolerance. The real test is *out-of-sample*: does the model correctly price instruments that were not in the calibration basket? There are two flavours of out-of-sample test. The first is *cross-sectional*: calibrate using a subset of instruments (say, vanilla options at strikes $K_1, \dots, K_{10}$) and check the fit on a held-out strike K_{11}. If the model is correctly specified and the calibration is well-regularised, the held-out fit should be nearly as good as the in-sample fit. If the held-out fit is markedly worse, the model is overfitting — squeezing the in-sample error to zero at the cost of off-sample generalisation. The second flavour is *temporal*: calibrate today, price a hold-out instrument today, then hold the model parameters fixed and re-price the same instrument tomorrow. Compare against tomorrow's market price. If parameters are stable day-to-day, the deviation should be small; if parameters wobble, the deviation will be dominated by recalibration noise rather than genuine market movement. Both tests are routinely neglected in practice — the in-sample fit is visually appealing and easy to report to management, while out-of-sample tests require more infrastructure. But any model going into production should be subjected to both.

12.4 11.3 Short-Rate Bootstrap on a Bond Lattice

12.4.1 11.3.1 Zero-Coupon Bonds as Calibration Instruments

Let $P_t(T)$ denote the zero-coupon bond price at time t paying 1 at maturity T. On a short-rate lattice $\{r_0, r_{11}, r_{22}, \dots\}$ (a recombining tree of realised short rates indexed by time step and node-within-step), the one-period bond satisfies

$$P_0(1) \;=\; \frac{1}{1+r_0} \qquad \Longleftrightarrow \qquad r_0 \;=\; \frac{1}{P_0(1)} - 1. \tag{11.20}$$

(Convention note. The discount factor $1/(1+r_t)$ used throughout §11.3 is the *simple-compounding* version; elsewhere in the guide — and in the CRR lattice conventions of §11.1.2 — we use exponential discounting $e^{-r_t\,\Delta t}$. The two agree to leading order in Δt — $1/(1+r_t\Delta t) = e^{-r_t\Delta t}(1+O(\Delta t^2))$ — so the sequential-bootstrapping arithmetic below is unaffected at the orders of accuracy we care about. We keep the simple-compounding form in §11.3 because it matches the market quoting convention for short-dated zero coupons, which is how the bootstrap inputs are usually presented in practice.)

Two-period bonds satisfy, at each child node,

$$P_{11}(2) \;=\; \frac{1}{1+r_{11}}, \qquad P_{22}(2) \;=\; \frac{1}{1+r_{22}}. \tag{11.21}$$

Rolling back to $t=0$ under the risk-neutral measure $\mathbb{Q}$:

$$P_0(2) \;=\; \frac{1}{1+r_0}\left[\,q\,P_{11}(2) + (1-q)\,P_{22}(2)\right]. \tag{11.22}$$

Zero-coupon bonds are the canonical calibration instruments for interest-rate models because their payoffs are entirely deterministic — they pay 1 at maturity, with no optionality to muddy the waters. Any uncertainty in the bond's current price therefore comes purely from the uncertainty of the path of short rates between now and maturity. Observing $P_0(T)$ for many maturities pins down the *term structure* of the short-rate risk-neutral dynamics, which is exactly what a short-rate model needs.

A useful intuition: a zero-coupon bond's yield is the constant rate at which, if the future were certain, the bond would compound to face value. When the future is uncertain, the yield is the *certainty-equivalent* rate — the deterministic rate that, if applied over the bond's life, would produce the same discounted payoff as the actual stochastic rate path. Because Jensen's inequality makes the expectation of the discount factor bigger than the discount factor at the expected rate, the certainty-equivalent yield is slightly below the expected average rate. This is the convexity correction that CH12 derives rigorously for the Vasicek model; at this stage, the point is only that every bond quote is a statement about the certainty-equivalent of a stochastic path, and each independent maturity provides one independent equation relating parameters to data.

Equation (11.20) is the simplest imaginable calibration equation: knowing today's one-period bond price directly gives today's short rate. Equation (11.21) is equally simple, but at the next time step — at each possible future short-rate state, the corresponding one-period bond price is fixed. The interesting work happens in (11.22), which enforces no-arbitrage between the $t=0$ quote and the $t=\Delta t$ states. Here both q and the short-rate states are, in principle, unknowns. The progression from (11.20) to (11.22) illustrates a key feature of tree-based calibration: each successive equation adds information about one more future time slice. (11.20) pins the $t=0$ node; (11.22) pins the $t=\Delta t$ node (given a vol input); the analogous three-step equation pins $t=2\Delta t$; and so on. The calibration proceeds recursively, one time slice at a time, building up a discrete yield-curve-consistent tree. This is the same recursive logic as in the Black–Derman–Toy (BDT) model or the Ho–Lee model — both of which are, at bottom, schemes for backing out tree parameters from bond-price ratios, one time slice at a time, in a way that exactly reproduces the observed yield curve. CH12 develops the continuous-time limits of these constructions; here we focus on the discrete bootstrap.

12.4.2 11.3.2 Recovering q and the Short Rates

Given $P_0(1)$ (one-year bond price) and $P_0(2)$ (two-year bond price), plus a volatility assumption that fixes the spread between r_{11} and r_{22} (e.g. $r_{22} = r_{11}\,e^{-2c\sqrt{\Delta t}}$), equations (11.20)–(11.22) give a system of two equations in (q, r_{11}) that can be solved numerically. A simple illustrative case takes $q=1/2$, which then delivers r_{11} explicitly from (11.22): r_{11} is chosen such that $\frac{1}{1+r_{11}}$ and $\frac{1}{1+r_{22}}$, mixed half-half, reproduce $P_0(2)(1+r_0)$. In practice the tree is built *sequentially* from the observed yield curve $\{P_0(1), P_0(2), P_0(3), ...\}$ and a term structure of volatilities — at each new level we solve for the fresh short-rate node that reprices the newly observed bond.

Why sequential construction works. The recursive nature of bond pricing — today's bond price is a discount of a convex combination of tomorrow's bond prices, which are themselves discounts of further-future bond prices — means calibration can be done layer by layer. Once the $t=0$ node is pinned down (trivially from (11.20)), the $t=\Delta t$ layer is pinned down by the two-period bond (11.22) plus a volatility input. The three-period bond $P_0(3)$ then pins

down the $t = 2\Delta t$ layer, and so on. This is the classical Ho–Lee / Hull–White / BDT construction, and it produces a tree that exactly reproduces the input yield curve by design. The tree is then said to be *bootstrapped*.

The role of the volatility input. Without a volatility input, the system is underdetermined: for a binomial tree at step $t = \Delta t$ there are two rate states r_{11}, r_{22} and a transition probability q, but only one bond quote $P_0(2)$ — three unknowns and one equation. We need extra information to pin down the spread between r_{11} and r_{22}. In practice this comes from one of three sources: (i) an explicit volatility assumption (e.g. "short rates have 1.2% annualised vol"), (ii) a market quote on an interest-rate option (caplet, swaption) whose price encodes the volatility, or (iii) a direct measurement of historical volatility of the short rate. Each source gives a different calibration flavour; market-consistency purists prefer (ii) because it is the only source that is consistent with tradable instruments.

Interpolation vs extrapolation. The bootstrapped tree reproduces the calibration instruments *exactly*. That is a property of construction, not a property of truth. For maturities *within* the calibration set, the model is interpolating between observed quotes, and we are entitled to a measured trust in its outputs. For maturities *beyond* the calibration set — or for instruments that depend on joint dynamics we have not fit — the model is extrapolating, and its outputs are only as good as the extrapolation. This distinction matters enormously in practice: a Vasicek fit calibrated to the 1-month, 3-month, 6-month, and 1-year bonds will reproduce those four quotes exactly (if it has four free parameters) but may produce wildly different 30-year yields from a Vasicek fit calibrated to 10-year, 20-year, and 30-year bonds. Both fits are "correct" in the interpolating region; neither is trustworthy in the extrapolating region.

The practical implication is that a calibration's range of trustworthiness is determined by its calibration set, not by the model's parametric form. A Vasicek model calibrated to short maturities will give short-maturity answers you can trust, but its long-maturity extrapolation is model-dependent in ways the short-maturity data cannot validate. Similarly, an implied-vol surface calibrated to near-the-money strikes will give near-the-money prices you can trust, but deep out-of-the-money extrapolation is driven by the model's tail behaviour, which the near-the-money data does not constrain. Recognising and reporting this range-of-trustworthiness is a hallmark of mature modelling practice. It is often the first question a risk committee asks before approving the use of a model: "for what range of instruments and market conditions has this model been validated?"

> **Key point.** The tree expectation is taken under the *bond-numeraire* risk-neutral measure, i.e. payoffs are discounted by $\prod(1 + r_{m-1}\Delta t)$ along each path, which is the discrete analogue of discounting by $\exp(\int_0^T r_u\, du)$ under $\mathbb{Q}^M$. CH05 formalised this numeraire-by-numeraire correspondence in continuous time.

Recovered short-rate tree *Recovered short-rate tree — given two bond quotes and a volatility spread c, the pair (r_0, r_{11}) is pinned down by the no-arbitrage mixing relation (11.22).*

Calibrated four-step short-rate tree *Layered extension: a four-step recombining short-rate tree whose node rates have been bootstrapped from a sequence of bond quotes $P_0(1), \ldots, P_0(4)$ and a term structure of volatilities. Each time-slice is pinned down by the freshly observed bond; the tree is arbitrage-free by construction and exactly reprices its calibration instruments.*

Bond-price lattice with calibrated rates at each node *The same calibrated tree, now annotated with the implied one-period bond value $P = 1/(1 + r\,\Delta t)$ at each node. This is the object pricing routines roll back over: each step multiplies the weighted average of child-node bond values by the parent's discount factor, and matching this roll-back to the observed $P_0(T)$ quotes pins down the calibration.*

12.4.3 11.3.3 Bid-Ask, Stale Quotes, and Data Hygiene

Bid-ask spreads and stale quotes. In real markets, $P_0(T)$ is not a single number but a bid-ask pair $(P^{\text{bid}}, P^{\text{ask}})$, and between transactions the quote may be stale — indicating only an indicative level rather than a tradable price. Calibration that targets an arbitrary mid-price within the bid-ask can give nonsensical parameter wobble: if the spread is 2 basis points and calibration is trying to fit to 0.1 basis point precision, the resulting parameters will move randomly within a band set by the spread even when no economic information has changed. A disciplined calibration treats the bid-ask as a confidence interval: fit the mid, but accept any parameters that produce model prices inside the bid-ask as equally valid. Stale quotes are even more pernicious; if a bond hasn't traded in a week, its "price" is an echo, and calibrating to it pollutes the rest of the curve.

An instructive exercise is to simulate the impact of bid-ask noise on a calibrated parameter vector. Take a realistic fit — say, a Vasicek calibration to ten bond quotes — and perturb each input yield by a uniform random amount in $[-\text{spread}/2, +\text{spread}/2]$. Rerun the calibration for many such perturbations, and plot the resulting distribution

of calibrated parameters. If the spread is 2 basis points and the parameter distribution spans 10% of the point estimate, the calibration is reporting precision far in excess of what the data supports. A well-calibrated system's reported precision should roughly match the width of this simulated perturbation distribution — any tighter is over-confidence, any looser is under-utilisation of the data. This exercise is worth running periodically as a calibration sanity check; it catches both model misspecification and implementation bugs that compress or inflate the reported precision artificially.

Liquidity weighting. A related practical refinement: weight each quote in the calibration objective by its liquidity, proxied by trading volume or quote frequency. A quote for an on-the-run 10-year Treasury is updated every second and traded millions of times a day; a quote for an off-the-run 17-year Treasury may be updated only a few times a day and traded in small size. Both quotes are nominally "mid prices", but the 10-year mid is a much more trustworthy signal. Weighting by volume (or by the inverse of some proxy for staleness) is one way to ensure the calibration reflects the information in the market rather than the information in yesterday's stale print. A common scheme: weight each residual by the instrument's proportion of recent trading volume, normalised so that the weights sum to one.

Data hygiene. Before calibration runs, a disciplined pipeline filters quotes according to several rules: drop quotes older than a threshold (typically 30 minutes for rates, shorter for equities); drop quotes with bid-ask spreads wider than some multiple of the instrument's typical spread; drop outliers that disagree with nearby quotes by more than an error bar. These hygiene steps are mundane but they account for a substantial share of the "why did the calibration jump" incidents that quantitative groups fire-drill around. A good calibration system logs exactly which quotes were accepted and which were rejected, so that post-hoc investigations can reconstruct the decision chain. The lattice bootstrap is a good place to enforce this discipline because its layer-by-layer structure makes it easy to localise a bad quote to a specific maturity: if the fit blows up at year 7, the culprit is typically a stale or cross-dealer-inconsistent 7-year quote, and an operator can verify this in seconds.

12.5 11.4 Roadmap: From Lattices to Continuous-Time Models

The three settings we have worked through — the single-period binomial, the multinomial lattice with optimisation and regularisation, the recursive bond-lattice bootstrap — share a common skeleton. Each instance begins with FTAP (11.1), translates observed market prices into equations for unknown parameters, and solves the resulting system either in closed form or by optimisation. The lattice structure is a pedagogical convenience: it makes the equations transparent, the arbitrage conditions explicit, and the Radon–Nikodym density computable by hand. But no real trading desk actually runs its production calibration on a binomial tree for a complex product. The next step — the one this part of the course is about to take — is to replace these discrete lattices with their continuous-time limits, keeping the calibration logic but upgrading the dynamics.

The pattern of that upgrade is already visible in the binomial case. Equations (11.5) and (11.6) said the step size scales as $\sigma\sqrt{\Delta t}$ and the drift-tilt scales as $(\hat{\mu}/\sigma)\sqrt{\Delta t}$. Taking $\Delta t \to 0$ under this scaling, the sum of iid Bernoulli log-returns converges by the CLT to Brownian motion with drift $\hat{\mu}$ and diffusion σ. That is the Donsker limit. In continuous time the asset becomes a geometric Brownian motion; its log is a Brownian motion with drift; Girsanov (CH05) gives the change of measure from $\mathbb{P}$ to $\mathbb{Q}$ explicitly as a shift in drift. The same argument, applied to a short-rate lattice with a mean-reverting drift, produces the Vasicek SDE $dr_t = \kappa(\theta - r_t)\,dt + \sigma\,dW_t$ in the limit. That SDE is the subject of **CH12**, where we derive its explicit solution, the distribution of r_T, the distribution of $\int_0^T r_u\,du$, and the closed-form bond price. We also pick up the Ho–Lee continuous limit and the Hull–White time-dependent extension, all developed from a single canonical Vasicek backbone. The three separate continuous-time short-rate treatments that originally lived in the old guide are consolidated there.

The applications of those continuous-time short-rate models — interest-rate swaps, swap rates, credit default swaps, callable bonds, European bond options, the T-forward measure in use rather than in preview — are the subject of **CH13**. The T-forward measure is particularly important: by choosing the T-maturity zero-coupon bond as numeraire, the bond price itself becomes a tradable whose discounted process is a martingale, and the stochastic discounting that clutters expectations under the money-market measure $\mathbb{Q}^M$ is absorbed into the numeraire. The result is a pricing machinery that produces Black-type closed forms for caplets and bond options. CH13 derives all of these applications by cross-reference to CH05 (for Girsanov and measure change) and CH12 (for the short-rate dynamics); it does not re-derive Vasicek, and it does not re-derive Girsanov. The calibration patterns from the present chapter carry over directly: every application in CH13 is, ultimately, another instance of FTAP (11.1) applied to a specific payoff and numeraire.

Two themes thread through the rest of the course. The first is that *calibration is a thin wrapper around pricing*: once the forward pricing function $\Theta \mapsto C^{\text{model}}(\Theta)$ is known, the calibration optimisation (11.15)–(11.19) is a generic numerical procedure. CH12 and CH13 contribute the forward pricing functions for the rates products; CH11 has contributed the optimisation and regularisation machinery. The second theme is that *the lattice is the right mental model even when the production code is continuous-time*. When a Vasicek calibration misbehaves in CH12, the diagnostic question is always "what does this failure look like on a coarsened trinomial lattice?" The tree makes the arbitrage conditions visible; the SDE hides them inside an Itô calculus. A calibration engineer who can translate freely between the two representations has both the rigour of the continuous-time machinery and the intuition of the lattice.

CH14 (Heston stochastic volatility) and CH15 (caps, floors, swaptions) will use every piece of the calibration machinery developed here: market-consistent objectives, vega weighting, regularisation, warm starts, non-identifiability diagnostics. The Heston calibration in particular is a notoriously ill-posed problem — the parameters $(\kappa, \theta, \sigma_v, v_0, \rho)$ have strong pairwise correlations in the likelihood surface, the objective is multimodal, and naive optimisers wander between local minima. Every remedy we have described here (reparameterisation in orthogonal coordinates, multi-start optimisation, regularisation toward a historical prior, cross-validation on held-out strikes) is used in practice. By the time the reader reaches CH14, the calibration apparatus should feel routine; only the Heston-specific pricing machinery (Fourier inversion of the characteristic function, Riccati ODEs for the transform) will be new.

12.6 11.5 Governance and the Sociology of Calibration

Before the chapter closes, a few perspectives that cut across the technical apparatus above but deserve standalone treatment. Calibration is a technical discipline; it is also an institutional one, with stakeholders, governance structures, and failure modes that no amount of mathematical sophistication alone can address.

12.6.1 11.5.1 The Sociology of Calibration

Quantitative finance is often taught as pure mathematics, but calibration is a profoundly social activity. The calibration team in a bank interacts daily with traders (who want their marks to reflect where they think the market is — even if that disagrees with the quoted mid), risk managers (who want parameters to be conservative, stable, and auditable), sales and structuring (who want competitive quotes on new trades, which means tight fits), operations (who want the calibration to complete reliably overnight without human intervention), and auditors (who want the whole process to be documented, reproducible, and governed). These stakeholders have genuinely conflicting priorities. A trader wants a tight fit; a risk manager wants stability. Sales wants aggressive quotes; operations wants simple code. Resolving these tensions is the real work of the quantitative calibration function, and it requires technical tools (the regularisation and weighting schemes we have surveyed) plus governance structures (change-management processes, model validation, override logs). A calibration system is not just a piece of code; it is a sociotechnical artifact that embodies the bank's approach to these tradeoffs.

12.6.2 11.5.2 Calibration vs Estimation: A Terminological Aside

The words "calibration" and "estimation" are sometimes used interchangeably but should not be. *Estimation* is a statistical procedure: given data generated by an unknown process, infer the process's parameters. *Calibration* is a pricing procedure: given market quotes generated by a pricing process, infer the parameters that reproduce the quotes.

The distinction matters because the objectives differ. Estimation minimises some notion of statistical discrepancy — likelihood, L^2 distance from empirical moments, whatever. Calibration minimises *pricing* discrepancy — the difference between model-implied prices and observed market prices. In the Vasicek case developed in CH12, the calibration objective minimises squared yield errors; this is not quite the same as the maximum-likelihood estimator of the Vasicek parameters from historical short-rate data, though in well-behaved cases they should produce similar answers.

A practical consequence: parameters "calibrated" to market quotes are under the *risk-neutral* measure $\mathbb{Q}$, while parameters "estimated" from historical data are under the *physical* measure $\mathbb{P}$. The two measures differ by the market price of risk (Girsanov's theorem, CH05), and failing to distinguish them is a common cause of confusion and error. For pricing applications, use $\mathbb{Q}$-calibrated parameters. For risk management (e.g. computing historical VaR, CH09), use $\mathbb{P}$-estimated parameters. Mixing the two will produce systematic biases.

12.6.3 11.5.3 The Role of Hedging in Calibration Selection

The choice of calibration instruments should be driven by the desk's hedging strategy. If you plan to hedge an exotic option using a basket of vanilla options at specific strikes and maturities, calibrate your model to exactly that basket. This ensures the model reproduces the hedge-instrument prices and therefore produces consistent hedge ratios. Calibrating to a different basket, even a superset, can produce incorrect hedge ratios, leading to hedge slippage and P&L surprises.

A concrete example: suppose you hedge a five-year cliquet using variance swaps at $1, 2, 3, 4, 5$ year maturities. Calibrate your model to exactly these variance swap prices. The calibrated model will price your cliquet in a way consistent with the hedge portfolio, so the sum of (cliquet mark + hedge portfolio mark) is stable under market moves to first order. Calibrate instead to a fine grid of vanilla options — which contains *more* information than the variance swap quotes — and the cliquet mark may drift relative to the hedge, producing spurious P&L. This principle is sometimes called "calibrate to what you hedge with". It is one of the most important practical lessons of derivatives pricing, and it explains why there is no universal best calibration recipe — the calibration that is right for your desk depends on the specific products and hedge mix.

12.6.4 11.5.4 The Choice of Calibration Frequency

How often should a model be recalibrated? The answer depends on the volatility of the market, the cost of computation, the stability properties of the model, and the business's tolerance for mark shifts. Market-making desks may recalibrate implicitly every time they receive a quote, using fast local methods (Levenberg–Marquardt with warm starts) to re-fit in milliseconds; end-of-day calibration against closing quotes is the standard for marking the book and computing regulatory metrics; slower-moving parameters (e.g. mean-reversion speeds in rates models) may be recalibrated only weekly or monthly, producing more stable marks at the cost of slower adaptation to regime changes; event-driven calibration triggered by a central-bank announcement, a volatility spike, or a residual outside a threshold combines stability with responsiveness. A well-designed calibration system supports multiple frequencies simultaneously, with different models tied to different frequencies as appropriate.

Calibration frequency trade-off: stability vs responsiveness *A synthetic parameter that drifts smoothly through a regime break at day 140.* ***Daily*** *recalibration (faint red) is responsive but noisy;* ***monthly*** *recalibration (blue) is stable but lags the regime shift by up to a month;* ***weekly*** *recalibration (orange) sits in between. No frequency dominates — the trade-off is real and the choice belongs to the desk's risk-appetite statement.*

12.6.5 11.5.5 The Limits of Market Consistency

We have emphasised market consistency throughout this chapter, but it has limits. A market-consistent model reproduces observed prices by construction. It does *not* predict future prices; it cannot anticipate regime shifts; it cannot distinguish an informed quote from a stale one; and it cannot correct for market mispricings that your analysis has identified.

A value-oriented investor or discretionary trader who has an independent view about what prices *should be* will deliberately diverge from market-consistency. They will use models calibrated to *their* view of fair value, not to the market's quoted mid. This is a different modelling paradigm, and it is appropriate for certain seats (prop trading, asset management, hedge funds) but not others (market making, flow derivatives). For the market maker, consistency with the market is a must; for the prop trader, the whole point is to be inconsistent with the market in defensible directions. Calibration, in the market-consistent sense we have been developing, is therefore a *specific* modelling discipline associated with *specific* business functions.

12.6.6 11.5.6 Connections to Machine Learning

In recent years, there has been growing interest in using machine-learning techniques for calibration. The motivation: neural networks can learn the model pricing function (the map from parameters to prices) once, and then calibration reduces to inverting this learned function — which can be orders of magnitude faster than re-solving the pricing problem at every optimiser step. A typical setup: generate a large training dataset of (parameters, prices) pairs by running the model forward on randomly sampled parameters, train a neural network to approximate the pricing function, and at calibration time solve an inverse problem over the network using gradient-based methods that backpropagate through it. This approach has been applied successfully to Heston calibration (CH14), Bermudan options, and rough-vol models. The downsides: the network has to be retrained when the model specification changes; accuracy in extreme parameter regions is limited by training-set coverage; and the system inherits the stability issues

of neural networks including adversarial sensitivity and out-of-distribution failures. For stable well-studied models, the learning-accelerated calibration is often worth the complexity; for experimental or custom models, a traditional iterative calibration is often simpler.

12.6.7 11.5.7 Benchmarking Calibration Performance

How do you know your calibration is doing a good job? Several benchmarks are worth tracking on a daily basis. *In-sample fit quality* — the average residual across calibration instruments — should be comparable to the bid-ask spread, not dramatically smaller (a sign of overfitting) or larger (a sign of model misspecification). *Out-of-sample fit quality* on instruments not in the calibration basket should be comparable to in-sample fit; a large gap indicates overfitting. *Temporal stability* — the day-over-day change in calibrated parameters on a calm day — should be small; large daily jumps suggest either an unstable calibration or a genuine market move, and careful attribution is needed to distinguish them. *Consistency across product lines*: if the same parameters price multiple products, residuals on each product's validation instruments should be comparable. *Predictive power*: for models used in forward-looking analysis, does the simulated distribution of future prices match what actually happens? A mature calibration function tracks all of these metrics on dashboards, alerts on unusual patterns, and escalates to human review when any threshold is breached.

12.6.8 11.5.8 Calibration Governance and Model Risk

Calibration is a regulated activity. Post-2008, banks are required to maintain formal model-risk-management frameworks that cover calibration procedures. The typical requirements include documentation of the calibration methodology (choice of instruments, weighting schemes, regularisation, numerical methods), independent validation of statistical properties and out-of-sample performance, monitoring of calibration outputs over time with alerts for unusual residuals or parameter jumps, override tracking of any manual adjustments with documented rationale, and periodic review of the methodology to ensure it remains fit for purpose under changing market conditions. These requirements are not bureaucratic overhead; they are defenses against the failure modes we have discussed throughout the chapter. A well-calibrated model that no one monitors will silently drift into a bad regime; a well-monitored model with bad calibration methodology will produce a documented trail of bad answers. Good governance combines technical quality with organisational discipline.

12.6.9 11.5.9 A Final Pragmatic Perspective

Calibration sits at the intersection of mathematics, statistics, software engineering, and institutional judgement. It is a deeply practical discipline, yet it rests on theoretical foundations (FTAP, change of measure, affine models) as elegant as any in quantitative finance. The practitioner who masters calibration is mastering the translation layer between the market's abstract signal and the model's concrete prediction — a translation performed dozens of times a day, across every desk and every product, at every institution that trades derivatives. Calibration is an inverse problem; it is often ill-posed in subtle ways; and the three classical pathologies (non-existence, non-uniqueness, instability) each call for distinct remedies. Moment matching is the cleanest tool for lattice calibration; change-of-numeraire machinery is a practical tool, not an abstract indulgence; and identifiability is never guaranteed. Bid-ask spreads set a precision floor. With these tools and mindsets in place, calibration becomes not a mysterious black art but a disciplined craft — one where mathematics, code, and judgement combine to extract useful signal from a noisy, over-determined market, in a way that is auditable, reproducible, and defensible.

12.7 11.6 Key Takeaways

1. **FTAP is the calibration engine.** Every model parameter is pinned down by requiring (11.1) to hold at observed prices. Market quotes are equations; parameters are unknowns; calibration is just solving the system.
2. **Calibration is an inverse problem.** It can be ill-posed in three classic ways: non-existence (no Θ reproduces the quotes exactly), non-uniqueness (many Θ's reproduce them), and instability (small price perturbations cause large parameter jumps). Each pathology has a distinct remedy: weighted least squares, regularisation, and warm-start smoothing respectively.
3. **Binomial scaling.** $c \sim \sigma\sqrt{\Delta t}$ and $p \sim \frac{1}{2}(1+(\hat{\mu}/\sigma)\sqrt{\Delta t})$ — moment matching on the lattice. This recovers the Black–Scholes limit because matching mean and variance per step is exactly what the CLT needs to produce Brownian motion. CH03 proved the corresponding continuous-time construction; CH06 uses it to derive the BS PDE.

4. **Change of numeraire on a lattice.** $d\mathbb{Q}^B/d\mathbb{Q}^A = (B/B_0)/(A/A_0)$. The same payoff can be priced under any consistent (measure, numeraire) pair — pick the numeraire that simplifies the expectation. CH05 developed the continuous-time canonical derivation.
5. **Bond-price ratios recover (q, r_0).** $r_0 = 1/P_0(1) - 1$ and the two-period relation (11.22) recovers the next tree layer given a volatility input. Sequential bootstrapping builds the full tree. CH12 develops the continuous-time Vasicek / Ho–Lee / Hull–White models whose lattice discretisation this construction is.
6. **Identifiability matters.** Some parameter directions are stiff (well-pinned by the data), others are sloppy (nearly flat likelihood). For single-factor Vasicek, κ and σ are jointly identified as the ratio σ^2/κ^2 from the long yield — the individual values are not pinned by bond quotes alone. To break the degeneracy, include volatility-sensitive instruments (caplets, swaptions from CH13) in the calibration set.
7. **Bid-ask and stale quotes set the precision floor.** Never calibrate tighter than the market's own resolution. Overfitting inside the bid-ask band produces spurious parameter instability. A disciplined pipeline filters out stale quotes and outliers before calibration runs.
8. **Interpolation vs extrapolation.** A calibrated model reproduces its inputs by construction; that does not make its outputs trustworthy outside the calibration range. Always distinguish between "what the market told us" and "what the model guesses".
9. **Calibration objective design is load-bearing.** The choice between unweighted, spread-weighted, vega-weighted, and regularised objectives is not cosmetic — different choices lead to different parameter estimates, different stability properties, and different out-of-sample behaviours. Make the choice explicitly and revisit it when market conditions change.
10. **Numerical methods must match the problem shape.** Nelder–Mead for noisy low-dimensional fits; Levenberg–Marquardt for smooth nonlinear least squares; BFGS for smooth general objectives; differential evolution for multimodal global search. A production system uses hybrids with warm starts and periodic multi-start checks.
11. **Out-of-sample testing is mandatory.** A model that fits in-sample beautifully but fails on held-out data is overfitting; production use of such a model is dangerous. Cross-validation and held-out strikes/maturities are essential discipline.
12. **Regime-awareness protects against silent calibration failure.** Monitor residuals, parameter-change magnitudes, and fit quality over time; escalate unusual patterns to human judgment rather than letting the system keep running silently.
13. **Calibrate vs estimate are not synonyms.** $\mathbb{Q}$-parameters (from market quotes) price derivatives; $\mathbb{P}$-parameters (from historical data) drive risk scenarios. Do not mix them.
14. **Calibrate to what you hedge with.** The right calibration basket is the one whose instruments you plan to use as hedges; fitting a richer basket than the hedge set can produce consistent prices but inconsistent hedge ratios.

12.8 11.7 Reference Formulas

12.8.1 11.7.1 Binomial lattice

$$A_n = A_{n-1} e^{c\,x_n}, \quad x_n \overset{\text{iid}}{\sim} \text{Bernoulli}(\pm 1)$$

$$c \;\approx\; \sigma\sqrt{\Delta t}, \qquad p \;\approx\; \tfrac{1}{2}\Big(1 + \tfrac{\hat{\mu}}{\sigma}\sqrt{\Delta t}\Big)$$

Risk-neutral version: replace $\hat{\mu} \to r - \frac{1}{2}\sigma^2$ for a lognormal asset.

12.8.2 11.7.2 Change of measure on a lattice

$$q^B = q^A\,\frac{B_u/A_u}{B_0/A_0}, \qquad 1 - q^B = (1 - q^A)\,\frac{B_d/A_d}{B_0/A_0}$$

$$\frac{d\mathbb{Q}^B}{d\mathbb{Q}^A}(\omega) \;=\; \frac{B(\omega)/B_0}{A(\omega)/A_0}$$

12.8.3 11.7.3 Zero-coupon bonds on a lattice

$$P_t(T) \;=\; \frac{1}{1+r_t}\,\mathbb{E}_t^{\mathbb{Q}}[P_{t+\Delta t}(T)]$$

$$r_0 \;=\; \tfrac{1}{P_0(1)} - 1, \qquad M_T \;=\; \prod_{m=1}^{T/\Delta t}(1 + r_{m-1}\Delta t)$$

In the continuous-time limit (developed in CH12),

$$P_t(T) \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_u\,du\right)\right]$$

12.8.4 11.7.4 Calibration objective functions

Unweighted least squares:

$$\Theta^\star \;=\; \arg\min_{\Theta} \sum_i \left(C_i^{\text{model}} - C_i^{\text{mkt}}\right)^2$$

Weighted least squares (bid-ask or liquidity):

$$\Theta^\star \;=\; \arg\min_{\Theta} \sum_i w_i\left(C_i^{\text{model}} - C_i^{\text{mkt}}\right)^2, \quad w_i = 1/\text{spread}_i^2$$

Vega-weighted (implied-vol-space):

$$\Theta^\star \;=\; \arg\min_{\Theta} \sum_i \left(\sigma_i^{\text{model}} - \sigma_i^{\text{imp,mkt}}\right)^2$$

Regularised:

$$\Theta^\star \;=\; \arg\min_{\Theta} \left[\sum_i w_i\left(C_i^{\text{model}} - C_i^{\text{mkt}}\right)^2 + \lambda\,\|\Theta - \Theta_{\text{prior}}\|^2\right]$$

12.8.5 11.7.5 Identifiability diagnostics

Hessian condition number:

$$\kappa(H) \;=\; \sigma_{\max}(H)/\sigma_{\min}(H)$$

- $\kappa(H) \sim 10$: well-posed
- $\kappa(H) \sim 10^3$: marginal; consider regularisation
- $\kappa(H) \sim 10^6$: ill-posed; restructure problem

12.8.6 11.7.6 Stepwise calibration checklist

1. Load and filter market quotes; drop stale and outlier observations.
2. Convert to calibration targets (yields for bonds, implied vols for options).
3. Set weights (bid-ask, vega, or liquidity based).
4. Initialise from yesterday's parameters (warm start) or from a default prior (cold start).
5. Run local optimiser (Levenberg–Marquardt or BFGS) to convergence.
6. Check convergence diagnostics: residuals, condition number, profile likelihood.
7. Sanity check parameters against historical ranges and economic plausibility.
8. Verify out-of-sample fit on held-out instruments.
9. Persist calibrated parameters with metadata (timestamp, residuals, diagnostics).

Cross-references. CH02 (FTAP on the multi-period binomial) is the direct predecessor of §11.1; CH05 (Radon–Nikodym, Girsanov, density process) is the canonical treatment of the measure-change machinery whose discrete-lattice instance is §11.2. CH12 develops the continuous-time Vasicek / Ho–Lee / Hull–White short-rate models whose lattice discretisation this chapter has been calibrating. CH13 deploys those continuous-time models in applications (IRS, CDS, bond options, callable bonds, T-forward measure). CH14 (Heston) applies the ill-posed-calibration remedies surveyed in §11.2 to a stochastic-volatility setting with notoriously correlated parameter directions.

13 CH12 — Short-Rate Models: Vasicek, Ho-Lee, Hull-White, Affine

Part IV — Interest-Rate Models. Depends on CH03 (Itô, OU solution), CH04 (Feynman-Kac for bond pricing), CH05 (Girsanov for the passage from the physical measure $\mathbb{P}$ to the risk-neutral measure $\mathbb{Q}$), and CH11 (calibration on lattices).

This is the chapter in which the yield curve becomes an object we can *model* rather than merely observe. The calibration chapter (CH11) built short-rate lattices by matching observed bond prices term by term; here we replace the lattice with a continuous-time stochastic differential equation and derive closed-form bond prices, yield curves, and volatility structures from the dynamics themselves. Three models — Vasicek, Ho-Lee, and Hull-White — are developed as a single narrative, with Vasicek as the canonical derivation that the other two specialise or extend. The applications (interest-rate swaps, CDS, bond options, callable bonds) live in Chapter 13; the caps, floors, and swaptions capstone lives in Chapter 15. This chapter's single job is to build the model.

13.1 12.1 Motivation: why short-rate models

The philosophical shift from equity option pricing to interest-rate option pricing is worth making explicit at the outset. In the Black–Scholes world there is one underlying asset (a stock) and one derivative price to compute at a time; discounting is done by a constant or deterministic rate and plays no interesting role. In interest-rate modelling the situation is inverted: the "underlying" is the entire term structure — a whole surface of bond prices indexed by maturity T — and the quantity being discounted is itself stochastic. The core ambition of short-rate modelling is the dramatic compression this entails: we posit a single scalar state variable r_t, specify its dynamics, and from that one equation recover the *entire* yield curve along with consistent prices for every rate-sensitive derivative written on it.

To appreciate the ambition, consider the scale of the object the model is trying to describe. A full yield curve has continuously many points (maturities from one day to fifty years), each carrying a price. The market simultaneously quotes prices for thousands of bonds, swaps, swaptions, caps, floors, and exotic rate derivatives, all written on this curve. Any internally consistent price system for all these products must respect no-arbitrage relationships between them — too many to enumerate by hand. The short-rate programme proposes that all of this structure is implicit in a single stochastic process for r_t, and that once r_t's dynamics are specified, every other price follows. This is an extraordinary claim, reminiscent of the classical ambition of physics to derive all of macroscopic phenomena from a small set of microscopic laws. In rates, the short-rate SDE plays the role of the fundamental law, and the whole machinery of this chapter is the derivation of its consequences.

The compression succeeds because the required structural ingredients — linear drift, linear-or-state-dependent squared volatility, Markovian state — are exactly the conditions for the affine-pricing apparatus to work. A non-affine short-rate model would fail to produce tractable bond prices (not because of any physical objection but because the mathematics would not close in finite form), and would therefore fail to support the whole edifice of rate-derivative pricing. Affine structure is both the secret of the model's success and the source of its limitations: real rate dynamics have features (regime shifts, heavy tails, non-affine macro dependence) that cannot be captured in a pure affine framework. We will return to these limitations at several points.

A second thing worth noting at the outset: short-rate models are *reduced-form*. They do not ask where interest rates come from (monetary policy, inflation expectations, term-premium decomposition) but instead posit a parsimonious stochastic process and calibrate its parameters to observed market prices. This is a feature, not a bug. For trading and hedging purposes one needs consistency with market quotes and analytical tractability; deep macro structure is left to other models. The Vasicek class we develop here is the simplest non-trivial example, and the techniques transfer almost verbatim to its descendants (Cox–Ingersoll–Ross, Hull–White, multi-factor affine, quadratic-Gaussian).

A final orienting remark: the whole programme sits at the intersection of three disciplines — stochastic analysis (Itô calculus, Girsanov, martingale representation), partial differential equations (Feynman–Kac, parabolic PDEs with source terms), and economics (no-arbitrage, equilibrium pricing, the stochastic discount factor). Each supplies a tool, none of them is the whole story. The stochastic analyst sees an Ornstein–Uhlenbeck process and writes down moment-generating functions; the PDE theorist sees a linear parabolic operator and solves with an affine ansatz; the economist sees a no-arbitrage price and asks what constraints consistency imposes. When they all converge on the same bond-price formula $P_t(T) = e^{A_t(T)-B_t(T)r_t}$, each feels vindicated. In truth, the formula lives at the intersection and is richer for it.

Short-rate models are *not* forecasting tools. A Vasicek model, fed with today's yield curve and today's implied vols,

will say nothing useful about whether rates will rise or fall — it will only say, given a family of possible future rate paths, what price today's arbitrage conditions impose on a specific derivative. That distinction is subtle but fundamental, and newcomers to rates often get it wrong. More precisely: a Vasicek calibrated to today's curve produces expectations of future rates under $\mathbb{Q}$, not under $\mathbb{P}$. The difference between these two expectations is the term premium — a quantity of genuine economic interest but one that short-rate models treat as a free parameter rather than something to be explained. If you want to ask "what will rates actually do next year?" you need macroeconomics, survey data, or factor models, not Vasicek. If you want to ask "what is today's arbitrage-free price for a swaption?" you need Vasicek (or a cousin), not macroeconomics. Confusing these two questions is the most common conceptual mistake in rates modelling, and clarity on the split is essential before any formulas get written down. Every time a trader talks about "the market's implied path of rates," they are implicitly working under a specific measure; losing track of which measure is a recipe for expensive mistakes.

13.1.1 12.1.1 Warm-up: rate trees and the non-uniqueness of $\mathbb{Q}$

Before writing a continuous-time model, it pays to recall what went right on a recombining *equity* tree and what goes subtly different on a *rate* tree. On an equity tree the risk-neutral probability q is *pinned down* by no-arbitrage once u, d, r are fixed: $q = (e^{r\Delta t} - d)/(u - d)$. There is no choice. On a rate tree the corresponding q is *not* unique. This is the conceptual pivot of the whole chapter, and it motivates both the time-dependent drift in Hull-White and the notion of "calibration" itself.

Why the difference? Because the Fundamental Theorem of Asset Pricing applies to *tradables*. The short rate r_t is not a tradable: you cannot, at time t, walk into a market and "buy one unit of r_t" the way you can buy a share of Apple. What you can buy is the money-market account (which earns r_t instantaneously) or a zero-coupon bond (which pays \$1 at a specific future date). So arbitrage pins down the measure $\mathbb{Q}$ through its action on bonds, not through its action on r_t. Different short-rate dynamics can reproduce the same observed bond prices, and arbitrage has no way to distinguish between them. Resolving this freedom is what "calibration" does.

Concretely, imagine a two-step rate tree with r_0 pinned by $P_0(1)$. For any choice of $q_0 \in (0, 1)$, and any choice of (r_{1u}, r_{1d}), the tree prices $P_0(2)$ as some function of (r_{1u}, r_{1d}, q_0). With three free parameters and one observation, there is a two-parameter family of consistent choices. Adding $P_0(3)$ adds one more equation but also three more free r_{2*} values plus q_1, leaving the system perpetually under-determined: at every step we add one observation and more than one free parameter.

> **Slogan.** *The short-rate tree is not the price of a traded asset. Therefore q is not uniquely determined by no-arbitrage. We must add external information — the market bond prices — to pin down θ.*

This is the discrete-time shadow of the continuous-time statement "Hull-White has infinitely many $\theta(t)$ choices for any given (κ, σ), and only the curve-fit picks one out."

Two further consequences. First, calibration of a rate model requires strictly more inputs than calibration of a stock model. A stock model needs only the spot and (maybe) one vol surface. A rate model needs an entire yield curve (to pin down θ or θ_t), at least one vol instrument (to pin down σ), and a choice of mean-reversion (to pin down κ). This reflects the deeper fact that rate derivatives live on a surface in (t, T) space, while stock derivatives live on a curve in T alone.

Second, the non-uniqueness is what gives short-rate modelling its freedom and, simultaneously, its burden. Every modeller picks a family — Ho-Lee, Vasicek, Hull-White, CIR, Black-Karasinski — and within that family parameters are calibrated to the market. Different families produce different pricings for off-curve instruments (caps, swaptions, CDS) even when they agree on the calibration bonds. The choice of family is a genuine modelling decision, not algebra.

13.1.2 12.1.2 The traded primitives: M_t and $P_t(T)$

Short rates are not traded. What is traded are:

- the money-market (bank) account

$$\mathrm{d}M_t = r_t\, M_t\, \mathrm{d}t, \qquad M_0 = 1, \tag{12.1}$$

equivalently $M_t = \exp(\int_0^t r_s\, \mathrm{d}s)$; - zero-coupon bonds of various maturities T, with price process $\big(P_t(T)\big)_{0 \le t \le T}$ and

terminal condition $P_T(T) = 1$. Bonds are contingent claims on the short-rate *path* — every bond pays \$1 at maturity, so its value today is determined by the law of $\int_0^T r_s \, \mathrm{d}s$.

Think of the contrast with Black-Scholes. There, the stock S_t is itself the traded asset: no-arbitrage directly pins its $\mathbb{Q}$-drift to be r, end of story. There is no market-price-of-risk freedom to worry about because the state variable coincides with the tradable. In short-rate modelling, the state variable (r_t) and the tradables (M_t, bonds) are *different*, and the mathematical relationship between them must be specified. That relationship is the market price of risk, and it is an additional *modelling* choice on top of the SDE for r_t. Short-rate models have a layer of structure that equity models do not; they are correspondingly more flexible.

The bank account M_t plays the role of a "price of forgetting." Every instant, M_t grows by $r_t \, \mathrm{d}t$, regardless of whether r_t is moving randomly or staying fixed. It is a non-decreasing adapted process that encodes the total discount accumulated from 0 to t. Dividing other prices by M_t factors out this accumulated discount and is what produces $\mathbb{Q}$-martingales: it is the cleanest way of separating the deterministic exponential growth due to compounding from the stochastic information embedded in prices themselves. Every subsequent numeraire change in this chapter and the next is a variation on this theme — pick a reference asset, divide, and study the resulting relative prices.

13.2 12.2 Discrete motivation: AR(1) and the continuous limit

Why does an AR(1) belong here at all? Because empirical short-rate series (Fed funds, EONIA, SONIA, SOFR) look remarkably unlike random walks and remarkably like stationary mean-reverting processes. Plot a multi-decade history of a policy rate and it moves in slow waves that hover around levels broadly set by trend inflation and potential growth; it does not diffuse toward infinity the way a geometric Brownian motion would. The simplest model that captures this is a linear combination of yesterday's rate, a drift toward a mean, and a Gaussian shock — the discrete AR(1):

$$r_n \;=\; \alpha \, r_{n-1} \;+\; \beta_n \;+\; \sigma\sqrt{\Delta t}\, \varepsilon_n, \qquad \varepsilon_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1). \tag{12.2}$$

Rewriting (12.2) in "innovation form,"

$$r_n - r_{n-1} \;=\; \kappa(\theta - r_{n-1})\, \Delta t \;+\; \sigma\sqrt{\Delta t}\, \varepsilon_n, \tag{12.3}$$

makes the correspondence with a continuous mean-reverting SDE transparent. Identifying $\alpha \leftrightarrow 1-\kappa\Delta t$, $\beta_n \leftrightarrow \kappa\theta\Delta t$, and $\varepsilon_n\sqrt{\Delta t} \leftrightarrow \mathrm{d}W_t$, and sending $\Delta t \to 0$, recovers the Vasicek SDE

$$\mathrm{d}r_t \;=\; \kappa(\theta - r_t)\, \mathrm{d}t \;+\; \sigma \, \mathrm{d}W_t. \tag{12.4}$$

The persistence α and the mean-reversion speed κ are two views of the same quantity: $\alpha = e^{-\kappa\Delta t}$ in exact correspondence, and for small Δt this is well approximated by $1 - \kappa\Delta t$. A daily AR(1) with $\alpha = 0.998$ corresponds to $\kappa = -\ln(0.998)/(1/252) \approx 0.50$ — a roughly-annual mean-reversion speed consistent with empirical policy rates.

The other useful intuition from the AR(1) view is that the continuous-time process r_t has *memory*. Under geometric Brownian motion (the Black-Scholes stock model) today's price contains no information about tomorrow's trajectory beyond its own level; under Vasicek, today's rate contains information about the *deviation from the mean* that partially persists. This memory is what makes mean reversion predictable at longer horizons — not because we can forecast the noise, but because we know the drift will systematically pull the process back toward θ at an estimable rate. For derivatives pricing this matters: an option that pays off a function of the integrated rate over time is fundamentally a bet on these mean-reverting dynamics, and its price depends sensitively on how much memory the process has.

13.2.1 12.2.1 Why continuous time?

The continuous limit is not just aesthetic. In discrete time, pricing a T-year zero-coupon bond means computing an expectation of a product of $T/\Delta t$ correlated random discount factors — a high-dimensional integral with no closed form. In continuous time, by contrast, the same object becomes an expectation of the exponential of a Gaussian integral, whose moment-generating function is explicit. Continuous time is what unlocks the closed-form bond price $P_t(T) = e^{A_t(T) - B_t(T) r_t}$ that makes the calibration and hedging machinery workable.

The gap between a continuous-time Vasicek and a sensible daily discrete approximation is, in practice, negligible compared with model-specification error, parameter-estimation error, or liquidity effects. Nobody should lose sleep over the "continuous vs discrete" dichotomy once calibration is done. What matters is that continuous time gives us a tractable language in which to express and solve pricing problems; the language is a tool, not a claim about ontology.

There is also a pedagogical virtue to continuous time. In discrete time, the various random-walk and AR(1) models look superficially similar — they differ only in the fine print of their coefficients — and it is easy to lose sight of which model is adequate for a given problem. In continuous time the functional forms of drifts and diffusions are starker and more revealing: a constant drift is obviously different from a linear-in-state drift, a constant diffusion is obviously different from a $\sqrt{r}$ diffusion. The continuous-time formulation makes the modelling choices *legible*, and legibility is a precondition for careful thinking.

13.2.2 12.2.1.b Estimating AR(1) Vasicek from historical data

The AR(1) representation (12.3) is useful not only pedagogically but for parameter estimation. Given a time series of observations $\{r_0, r_1, \dots, r_N\}$ on a Δt grid, ordinary least squares regression of r_n on r_{n-1} yields estimates of the slope $\beta = 1 - \kappa\Delta t$ and intercept $\kappa\theta\Delta t$, from which $\kappa = (1-\beta)/\Delta t$ and $\theta = \alpha/(1-\beta)$ follow directly. The residuals from this regression have variance $\sigma^2\Delta t$, giving an estimate of σ. This simple estimation procedure is the physical-measure ($\mathbb{P}$) calibration of Vasicek, and it can be implemented in a few lines of linear algebra — a reminder that many "exotic" quantitative finance tools reduce to well-known statistical procedures once the algebra is peeled away.

The sample autocorrelation at lag h — $\mathrm{Corr}(r_s, r_{s+h})$ — should follow the theoretical form $e^{-\kappa h}$ in stationary equilibrium. Plotting empirical autocorrelation against lag on a log-linear scale and fitting a straight line is a quick diagnostic of whether the data are well-described by an OU / AR(1) process. If the empirical autocorrelation decays exponentially, the AR(1) is a good fit; if it decays faster (over-damped) or slower (long memory), the data are telling you something the simple model misses. Long memory is a particularly well-studied deviation; models with multiple mean-reversion speeds (G2++, three-factor affine) capture it by having one slow and one fast factor.

Two important identifiability warnings. First, estimates of κ are notoriously noisy in short samples: the "half-life of a shock" is a slow-moving feature, and short-window data contains limited information about it. Robust estimates typically require decades of daily data, or at least years of high-frequency data. Second, the physical-measure estimates of $\kappa^{\mathbb{P}}$ and $\theta^{\mathbb{P}}$ differ systematically from their risk-neutral counterparts $\kappa^{\mathbb{Q}}$ and $\theta^{\mathbb{Q}}$ by the market price of risk. The volatility σ is measure-invariant (Girsanov changes drift but not diffusion), so it should theoretically agree across historical and calibration estimates — a useful sanity check when running both.

13.2.3 12.2.2 From $\mathbb{P}$ to $\mathbb{Q}$ (cross-reference CH05)

Under the physical measure $\mathbb{P}$, write the short rate as

$$\mathrm{d}r_t \;=\; \mu(t, r_t)\,\mathrm{d}t \;+\; \sigma(t, r_t)\,\mathrm{d}W_t. \tag{12.5}$$

By Girsanov's theorem (Chapter 5), the change of measure to the risk-neutral $\mathbb{Q}$ shifts the drift by $-\lambda_t\,\sigma(t, r_t)$, where λ_t is the *market price of interest-rate risk*:

$$\mathrm{d}r_t \;=\; \big(\mu(t, r_t) - \lambda_t\,\sigma(t, r_t)\big)\,\mathrm{d}t \;+\; \sigma(t, r_t)\,\mathrm{d}\widehat{W}_t. \tag{12.6}$$

We do not re-derive Girsanov here — that happened in Chapter 5, where the density process, Doléans-Dade exponential, and two-numeraire switching were developed as the canonical treatment. What matters for this chapter is the consequence. Three standard choices of λ_t preserve the affine structure:

- Constant $ _t = $ const: only shifts θ (the long-run mean under $\mathbb{Q}$ differs from that under $\mathbb{P}$).
- Affine in r_t: $\lambda_t = a + b\,r_t$: shifts both θ and κ.
- Deterministic function of time $\lambda_t = a_t + b_t\,r_t$: yields Hull-White with time-dependent θ_t.

All three preserve the linear drift, and under $\mathbb{Q}$ the short rate continues to satisfy

$$\mathrm{d}r_t \;=\; \kappa(\theta_t - r_t)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}\widehat{W}_t, \tag{12.7}$$

where θ_t is (in general) a deterministic function of time. The diffusion σ does *not* change under Girsanov: Brownian motions have the same quadratic variation under equivalent measures, so calibrated volatilities carry over directly across measure specifications without needing adjustment. Only drifts change when measures change.

The economic content of λ_t is worth a pause. Under $\mathbb{P}$ rate shocks are perceived as risky, and investors demand compensation for holding bonds whose prices move inversely to those shocks. That compensation is the term premium, and it biases the $\mathbb{P}$-drift of r_t relative to what a risk-neutral agent would perceive. Because we never directly observe λ_t — we only see prices, and prices aggregate $\mathbb{P}$-dynamics with λ_t into a single $\mathbb{Q}$-drift — the modelling convention is to work entirely under $\mathbb{Q}$ and let historical data inform the shape (if any) of λ_t separately. In short-rate modelling for pricing, it is almost universal to specify dynamics directly under $\mathbb{Q}$ and never write $\mathbb{P}$-dynamics explicitly.

13.2.4 12.2.3 The fundamental bond-pricing formula

By the general theory of risk-neutral pricing (Chapter 5), relative prices of all traded assets are $\mathbb{Q}$-martingales once we divide by the numeraire M_t. For a zero-coupon bond with terminal payoff 1,

$$\frac{P_t(T)}{M_t} \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\frac{1}{M_T}\right] \qquad \Longrightarrow \qquad \boxed{P_t(T) \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_s\,\mathrm{d}s}\right].} \tag{12.8}$$

Equation (12.8) is the master formula of short-rate modelling. Read carefully, it says that today's bond price is *exactly* the risk-neutral expectation of the continuously-compounded discount factor over the bond's life. This sentence contains a lot: it simultaneously asserts no-arbitrage (the price is an expectation), identifies the right measure ($\mathbb{Q}$), specifies the right numeraire (M_t), and provides a concrete algorithm (integrate r_s along simulated paths and average). Every bond price in this chapter — and every yield, forward rate, swap rate, caplet, and swaption derived from bond prices in Chapter 13 — is a consequence of this formula.

A useful consistency check: when $r_t = r$ is deterministic and constant, (12.8) reduces to $P_t(T) = e^{-r(T-t)}$. When r_t is deterministic but time-varying, it reduces to $P_t(T) = e^{-\int_t^T r_s\,\mathrm{d}s}$. Only when r_t is stochastic does the expectation become non-trivial; the formula generalises the deterministic cases smoothly and recovers them as special limits.

The algorithmic reading of (12.8) is equally important. If we cannot evaluate the expectation analytically — say, because the model is too complex for a closed form — we can always simulate: generate many Monte Carlo paths of r_t from t to T under $\mathbb{Q}$, compute the realised discount factor along each path, and average. The bond price converges in probability to the average as the number of paths grows. The entire business of finding closed forms (the affine trick we are about to execute) is a way of *avoiding* Monte Carlo by exploiting special structure; when that structure is absent, Monte Carlo is there to catch us.

13.3 12.3 Vasicek — the canonical derivation

This is the technical heart of the chapter. Take (12.7) with constant θ, which gives the original Vasicek (1977) model:

$$\boxed{\mathrm{d}r_t \;=\; \kappa(\theta - r_t)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}\widehat{W}_t, \qquad \kappa, \theta, \sigma > 0.} \tag{12.9}$$

The drift is linear-restoring, the diffusion is constant, and the solution is an Ornstein-Uhlenbeck process: a mean-reverting Gaussian process with long-run mean θ and mean-reversion speed κ.

Intuition — mean reversion. Whenever r_t wanders above θ, the drift pulls it *down* in proportion to the deviation; when it drifts below, the drift pushes it *up*. Economically this mirrors the behaviour of real short rates — central banks tighten when rates are "too low" (inflation pressure) and ease when they are "too high" (recession risk), so rates orbit around a neutral level rather than random-walking. The speed κ controls how fast this happens: the *half-life* of deviations is $\ln 2/\kappa$. With $\kappa = 0.3$ a shock decays by half in about 2.3 years; with $\kappa = 1.0$, in about 8 months. The stationary distribution is $\mathcal{N}(\theta, \sigma^2/(2\kappa))$ — larger κ (stronger pull) or smaller σ (less noise) narrows the long-run distribution around θ.

The half-life interpretation deserves emphasis because it is how practitioners think about κ. Nobody calibrating reasons directly in units of "inverse time" — they reason in half-lives, because a half-life has immediately physical

meaning. "A κ of 0.3 corresponds to a rate-shock half-life of 2.3 years" is a statement any rates trader can sanity-check against intuition about how long a Fed-funds deviation persists before the next cutting or hiking cycle unwinds it. If a calibrated κ implies a half-life that is wildly too short (weeks) or absurdly too long (decades), something is wrong.

The Gaussian structure is the other defining feature of Vasicek. It is what makes every subsequent computation tractable: a linear combination of Gaussian increments is Gaussian, an integral of Gaussians is Gaussian, and the exponential of a Gaussian has a closed-form mean (its moment-generating function). The whole affine bond-price formula ultimately rests on this one fact. The price paid is that r_t can, in principle, go negative: the Gaussian distribution is supported on all of $\mathbb{R}$. Before the 2008-2020 era of unconventional monetary policy, this was viewed as the model's most conspicuous defect. After 2014 several European and Japanese policy rates did go meaningfully negative, and suddenly Vasicek's ability to accommodate negative rates became a *feature*, not a bug. No other equally tractable short-rate model handles negative rates so naturally.

13.3.1 12.3.1 The SDE and the explicit solution

Solve (12.9) by the standard OU integrating factor. Set $r_t = e^{-\kappa t} g_t$ (equivalently $g_t = e^{\kappa t} r_t$). By Itô's lemma (Chapter 3),

$$\mathrm{d}r_t = -\kappa\, e^{-\kappa t} g_t\, \mathrm{d}t \;+\; e^{-\kappa t}\, \mathrm{d}g_t = -\kappa r_t\, \mathrm{d}t + e^{-\kappa t}\, \mathrm{d}g_t. \tag{12.10}$$

Matching (12.10) against (12.9),

$$\mathrm{d}g_t \;=\; \kappa\, e^{\kappa t}\, \theta\, \mathrm{d}t \;+\; \sigma\, e^{\kappa t}\, \mathrm{d}\widehat{W}_t. \tag{12.11}$$

Integrate from t to T and rearrange using $g_T - g_t = e^{\kappa T} r_T - e^{\kappa t} r_t$:

$$\boxed{r_T \;=\; e^{-\kappa(T-t)}\, r_t \;+\; \theta\big(1 - e^{-\kappa(T-t)}\big) \;+\; \sigma \int_t^T e^{-\kappa(T-s)}\, \mathrm{d}\widehat{W}_s.} \tag{12.12}$$

The integrating-factor trick used here generalises to essentially every linear SDE. The key move is multiplying through by $e^{\kappa t}$, which converts the mean-reverting term $-\kappa r_t\, \mathrm{d}t$ into a pure differential. Once that is done, the remaining drift and diffusion have no r_t dependence and can be integrated directly — the non-trivial feedback has been algebraically removed. The price of admission is that the final expression involves a weighted stochastic integral against the kernel $e^{-\kappa(T-s)}$, with weights that decay exponentially as you look further back. These decaying weights have a beautiful physical meaning: the current short rate is a weighted average of all past shocks, with older shocks discounted more heavily because the mean-reverting drift has had more time to "forget" them. This is exactly the continuous-time analogue of a discrete AR(1) with autoregressive coefficient $e^{-\kappa \Delta t}$.

The structural similarity between (12.12) and the Ornstein-Uhlenbeck result in physics is worth noting: there an OU process describes the velocity of a particle undergoing Brownian motion in a viscous fluid. The mean-reverting drift $\kappa(\theta - r_t)$ is analogous to a frictional force pulling the velocity toward equilibrium; the diffusion $\sigma\, \mathrm{d}\widehat{W}_t$ is analogous to random molecular collisions. The exponential memory of past shocks is the mathematical signature of frictional damping. All classical physical intuition about coloured noise, spectral densities, and correlation times carries directly across to interest-rate modelling.

Read (12.12) piece by piece: the first term $e^{-\kappa(T-t)} r_t$ is the *deterministic* decay of the current rate toward the mean; the second term $\theta(1 - e^{-\kappa(T-t)})$ is the deterministic buildup of the mean-reversion level; the third term is a mean-zero Gaussian random variable whose variance grows from 0 (at $T = t$) toward the stationary value $\sigma^2/(2\kappa)$ (as $T \to \infty$). Three effects neatly separated: where we are now, how fast we revert, how noisy the journey is. Every intuition about OU processes — and therefore about Vasicek, Hull-White, and half the short-rate literature — can be read off this one equation.

13.3.2 12.3.1.b Joint distribution over multiple times

A property of (12.12) that matters in practice: the joint distribution of $(r_{t_1}, r_{t_2}, \dots, r_{t_n})$ at any finite collection of times is multivariate Gaussian. This follows because each r_{t_i} is an affine function of the same Brownian motion, and

linear combinations of a Brownian path are jointly Gaussian. The mean and covariance of this multivariate Gaussian can be computed explicitly:

$$\mathrm{Cov}(r_s, r_t) \;=\; \sigma^2\, e^{-\kappa(s+t)} \int_0^{\min(s,t)} e^{2\kappa u}\,\mathrm{d}u \;=\; \frac{\sigma^2}{2\kappa}\big(e^{-\kappa|t-s|} - e^{-\kappa(t+s)}\big)\,. \tag{12.12a}$$

The dominant term is the exponentially decaying autocorrelation $e^{-\kappa|t-s|}$; the subtracted term accounts for the boundary effect near $t = s = 0$. In the stationary limit ($s, t \gg 1/\kappa$) the covariance reduces to $\frac{\sigma^2}{2\kappa}e^{-\kappa|t-s|}$, exactly matching the classical OU correlation structure.

This multivariate-Gaussian structure is what makes *exact* Monte Carlo simulation of Vasicek paths so efficient. If you need samples at specific times $\{t_1, \dots, t_n\}$, you can compute the exact covariance matrix, Cholesky-factor it, and draw correlated Gaussians in one shot — no Euler discretisation error. For Monte Carlo of payoffs that depend only on terminal rates (e.g. European bond options) you do not even need the full covariance matrix; a single Gaussian draw for r_T suffices. CIR, by contrast, does not admit this luxury: its non-central chi-squared distribution is exact but more awkward to sample. The analytical tractability of Vasicek therefore extends beyond pricing into simulation efficiency — a reason to prefer Vasicek when it fits.

13.3.3 12.3.2 Distribution of r_T

From (12.12), r_T conditional on $\mathcal{F}_t$ is Gaussian with

$$\mathbb{E}_t^{\mathbb{Q}}[r_T] \;=\; e^{-\kappa(T-t)}\, r_t + \theta(1 - e^{-\kappa(T-t)}), \tag{12.13}$$

$$\mathbb{V}_t^{\mathbb{Q}}[r_T] \;=\; \sigma^2 \int_t^T e^{-2\kappa(T-s)}\,\mathrm{d}s \;=\; \frac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa(T-t)}\big)\,. \tag{12.14}$$

The variance starts at 0 at $T = t$ (no time, no uncertainty) and saturates at $\sigma^2/(2\kappa)$ as $T \to \infty$ (the stationary variance). The half-saturation time is $\ln 2/(2\kappa)$, so half of the stationary variance accumulates in $\ln 2/(2\kappa)$ years. With $\kappa = 0.3$ this is about 1.15 years — meaning over a 1-year horizon, rates have accumulated about half their long-run uncertainty. Beyond about $5/(2\kappa) \approx 8$ years the distribution is essentially stationary.

A worked example sharpens this. Take $\kappa = 0.3$ (half-life about 2.3 years) and $\sigma = 0.01$ (annualised 100 bp vol). The stationary standard deviation is $\sigma/\sqrt{2\kappa} = 0.01/\sqrt{0.6} \approx 0.0129$, or 129 basis points. A normal distribution with this standard deviation would put roughly two-thirds of its mass within $\theta \pm 129$ bp and 95% within $\theta \pm 258$ bp. For $\theta = 0.04$ that gives a plausible long-run range of roughly 1% to 7% for the short rate — consistent with the long-run range of actual policy rates in mature economies. Had we taken $\sigma = 0.03$ (a highly volatile regime), the stationary sd would be 387 bp and the 95% band would extend from -3.7% to $+11.7\%$, clearly including negative rates. This explicit scaling is one of Vasicek's virtues: it lets you read off the implications of parameter choices in two-second mental math.

The ratio $(1-e^{-2\kappa(T-t)})$ is a useful "mixing fraction" that tells you how close to stationarity the process is at horizon $T - t$. At $T - t = 1/\kappa$ (one mean-reversion time) the mixing fraction is $1 - e^{-2} \approx 0.865$. At $T - t = 2/\kappa$ it is about 0.982. For practical purposes one can treat any horizon beyond $3/\kappa$ as effectively stationary, and this is how simulation studies typically validate their "burn-in" periods before extracting stationary statistics.

Vasicek paths with mean reversion *Vasicek paths from three starting points all pull toward θ (dashed line), with dispersion widening then stabilising at $\sigma/\sqrt{2\kappa}$.*

There is a visual intuition worth internalising: draw the (t, r)-plane with a horizontal line at $r = \theta$. The rate process starts at $(0, r_0)$. In the deterministic case (set $\sigma = 0$), the path is an exponential curve approaching θ asymptotically:

$$r_t \;=\; r_0\, e^{-\kappa t} \;+\; \theta(1 - e^{-\kappa t}). \tag{12.13a}$$

In the stochastic case, this curve is the "centreline" around which sample paths fluctuate, with Gaussian noise amplitude growing from zero (at $t = 0$ the rate is deterministically r_0) toward the stationary value $\sqrt{\sigma^2/(2\kappa)}$ as

$t \to \infty$. Drawing many sample paths overlaid on the same axes produces an "hourglass" pattern: tight at $t = 0$, fanning out over time, eventually settling into a constant-width band centred at θ.

Concrete numbers: with $\kappa = 0.15$, $\theta = 4\%$, $r_0 = 2\%$, the deterministic rate path rises from 2% toward 4% along an exponential with half-life $\ln 2/0.15 \approx 4.6$ years. $\mathbb{E}[r_1] \approx 2.28\%$, $\mathbb{E}[r_5] \approx 3.06\%$, $\mathbb{E}[r_{20}] \approx 3.90\%$. The approach is exponential: the gap to θ halves every 4.6 years. If instead $r_0 = 6\%$ (above target), the trajectory is inverted — an exponential decay from 6% down toward 4%, same timescales. In either case the mean rate at time t is a specific weighted average of starting point and target, with weights determined entirely by κt.

Intuition for the stationary variance $\sigma^2/(2\kappa)$. Each incremental shock injects variance $\sigma^2 \Delta t$ into the process; each unit of time, mean reversion exponentially erodes that variance at rate 2κ (the factor of 2 because variance is a second moment). In equilibrium, the rate at which new variance is injected equals the rate at which old variance is damped: $\sigma^2 = 2\kappa V_{\text{eq}}$, giving $V_{\text{eq}} = \sigma^2/(2\kappa)$. This balance between injection and decay is the mechanism behind the ergodic variance. Imagine water being poured into a leaky bucket: constant inflow at rate σ^2, outflow proportional to the current water level at rate $2\kappa V$. In equilibrium, the water level settles at $V = \sigma^2/(2\kappa)$, where inflow equals outflow.

The κ-σ identifiability problem, re-posed. The stationary variance depends only on the ratio σ^2/κ, so any two parameter vectors with the same ratio (but different individual σ and κ) produce identical equilibrium distributions. The difference shows up only in the *transient* behaviour: how quickly the distribution approaches equilibrium from a non-equilibrium start. A high-κ, high-σ combination has both strong mean reversion and high volatility; it converges quickly. A low-κ, low-σ combination converges slowly. On a yield curve close to equilibrium, the two parameter sets fit equally well, and the calibration is non-identifiable. On a yield curve far from equilibrium, the transient differs and data becomes informative.

13.3.4 12.3.3 Distribution of the integrated rate $\int_t^T r_u \, \mathrm{d}u$

To compute the bond price $P_t(T) = \mathbb{E}_t^{\mathbb{Q}}[e^{-\int_t^T r_s \, \mathrm{d}s}]$ we need the law of $\int_t^T r_s \, \mathrm{d}s$. Substitute (12.12) and apply Fubini to swap the time integral with the stochastic integral:

$$\int_t^T r_s \, \mathrm{d}s \;=\; \frac{1 - e^{-\kappa(T-t)}}{\kappa} r_t \;+\; \theta \int_t^T (1 - e^{-\kappa(T-s)}) \, \mathrm{d}s \;+\; \frac{\sigma}{\kappa} \int_t^T (1 - e^{-\kappa(T-s)}) \, \mathrm{d}\widehat{W}_s. \tag{12.15}$$

The kernel $(1 - e^{-\kappa(T-s)})/\kappa$ that appears throughout (12.15) has a name worth remembering — it is the quantity we will call $B_s(T)$. Its role here is to tell us how much each shock $\mathrm{d}\widehat{W}_s$ at time s contributes to the integrated short rate through maturity T. A shock at time $s \ll T$ has almost time $T - s$ for its effect to propagate through the mean-reverting dynamics, so its contribution is nearly $1/\kappa$. A shock at $s \approx T$ barely has time to act, so its contribution tends to zero. This exponential down-weighting of late-arriving shocks is the same phenomenon that gives bonds their characteristic convexity behaviour.

The Fubini step deserves a brief justification. Technically, Fubini for stochastic integrals requires enough integrability of the integrand; in the OU case the kernel $e^{-\kappa(T-s)}$ is bounded uniformly in $s \in [t, T]$, so all the required integrability conditions are trivially satisfied. In richer models (say, a state-dependent-diffusion CIR process) the justification is more delicate. For Vasicek, however, the swap goes through with no fuss — another small benefit of working in the Gaussian world. (See Chapter 3 for the Itô isometry that underpins this.)

Another useful perspective: $B_s(T)$ is exactly what one would compute if asked, "how much does a point shock to r_s raise the integrated short rate $\int_t^T r_u \, \mathrm{d}u$?" A shock of size δ at time s means that, subsequently, r_u for $u > s$ is elevated by $\delta e^{-\kappa(u-s)}$ (the deterministic OU response to an impulse). Integrating from s to T gives $\delta \cdot (1 - e^{-\kappa(T-s)})/\kappa = \delta B_s(T)$. So $B_s(T)$ is the impulse-response function of the integrated short rate with respect to a shock at time s. This impulse-response interpretation links the bond-pricing formula to the broader literature on linear systems, transfer functions, and spectral methods.

Conditional on $\mathcal{F}_t$, (12.15) is therefore a Gaussian random variable $\int_t^T r_s \, \mathrm{d}s \,|\, \mathcal{F}_t \sim \mathcal{N}(\mu_*, \Sigma_*^2)$ with

$$\mu_* \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\int_t^T r_s \, \mathrm{d}s\right] \;=\; \frac{1 - e^{-\kappa(T-t)}}{\kappa} r_t \;+\; \theta\left[(T - t) - \frac{1 - e^{-\kappa(T-t)}}{\kappa}\right], \tag{12.16}$$

$$\Sigma_*^2 \;=\; \mathbb{V}_t^{\mathbb{Q}}\left[\int_t^T r_s\,\mathrm{d}s\right] \;=\; \frac{\sigma^2}{\kappa^2}\int_t^T \left(1-e^{-\kappa(T-s)}\right)^2 \mathrm{d}s. \tag{12.17}$$

The Gaussianity of $\int_t^T r_s\,\mathrm{d}s$ is the crux. It follows because (i) under Vasicek, r_s is a linear combination of Gaussian shocks through (12.12), (ii) integration against a deterministic kernel preserves Gaussianity (a linear operation on a Gaussian vector), and (iii) Fubini lets us legitimately swap the time integral and the expectation. That the *integral* of a Gaussian process is itself Gaussian is not obvious — for a non-linear function of a Gaussian process it would fail — and this is precisely why Gaussian short-rate models are so disproportionately tractable compared to alternatives.

To see why the CIR model cannot easily replicate this step, notice that under CIR $r_s =$ (something involving $\sqrt{r_{s-}}$), which is *not* a linear combination of Gaussian noise — it is a non-central chi-squared process whose integral has no simple closed-form distribution. In the CIR bond-pricing derivation one still obtains an affine formula $P_t(T) = e^{A_t(T)-B_t(T)r_t}$, but the derivation goes through the PDE route (§12.3.5 below) rather than the direct-expectation route. The PDE approach generalises; the direct-expectation approach is a Gaussian-specific luxury.

The explicit variance formula (12.17), after expanding the square and integrating term-by-term, becomes

$$\Sigma_*^2 \;=\; \frac{\sigma^2}{\kappa^2}\left[(T-t) \;-\; \frac{2(1-e^{-\kappa(T-t)})}{\kappa} \;+\; \frac{1-e^{-2\kappa(T-t)}}{2\kappa}\right]. \tag{12.18}$$

For typical parameters ($\kappa = 0.3$, $\sigma = 0.01$, $T-t = 5$ years), $B_t(T) = (1-e^{-1.5})/0.3 \approx 2.59$, and the bracketed expression evaluates to approximately $5 - 5.18 + 1.59 = 1.41$. Multiplying by $\sigma^2/\kappa^2 \approx 1.11\times 10^{-3}$ gives $\Sigma_*^2 \approx 1.57\times 10^{-3}$, so the standard deviation of the integrated rate is about 4%. Worth internalising these numerics: they are what the formulas translate into with real inputs.

13.3.5 12.3.4 Closed-form bond price

Because $\int_t^T r_s\,\mathrm{d}s$ is Gaussian with mean μ_* and variance Σ_*^2, its moment-generating function at -1 gives the bond price directly:

$$P_t(T) \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^T r_s\,\mathrm{d}s}\right] \;=\; \exp\{-\mu_* + \tfrac{1}{2}\Sigma_*^2\}\,. \tag{12.19}$$

Equation (12.19) is the single most important step in the derivation if you want to understand *why* the bond price has the form it does. The identity $\mathbb{E}[e^X] = e^{\mathbb{E}X+\frac{1}{2}\mathbb{V}X}$ for Gaussian X is the engine. Without Gaussianity this identity fails (one would need the full characteristic function or a Laplace transform), and the resulting bond price would have no closed form.

The $+\frac{1}{2}\Sigma_*^2$ term is the *convexity correction*: because the exponential is convex, a random discount factor is worth *more* on average than the exponential of the mean rate, and the gap is exactly one-half the variance. Practitioners sometimes call this "Jensen's mountain" — it is the reason bond yields are below expected future short rates even in the absence of a term premium.

To make the Jensen effect vivid, consider a toy case: imagine the short rate over the next year will be either 2% or 6% with equal probability, so the mean rate is 4%. The "naive" discount factor computed from the mean rate is $e^{-0.04} = 0.9608$. The correct discount factor is $\frac{1}{2}(e^{-0.02}+e^{-0.06}) = \frac{1}{2}(0.9802+0.9418) = 0.9610$. The difference is small (2 bp in yield terms) but always in the same direction. Now change the rates to 0% and 8% (higher variance, same mean): correct discount $\frac{1}{2}(1+e^{-0.08}) = 0.9616$. The gap has widened to 8 bp. The gap scales with the variance of the rate distribution, exactly as the $+\frac{1}{2}\Sigma_*^2$ term predicts. This is Jensen's inequality in action.

The convexity correction has profound practical implications. It explains why forward rates are typically *higher* than expected future spot rates (the convexity premium baked into forwards); why a long-dated bond yield can be below the long-run mean of the short rate even in the absence of any risk premium; why swaps convexity-adjusted to remove the bias have a specific bid-offer structure tied to σ^2. A rates trader who does not internalise the convexity correction is doomed to misprice every structured rate product; the $\frac{1}{2}\mathbb{V}$ term is where the money hides.

Plugging (12.16) and (12.17) into (12.19) gives, after some algebra, the *affine term structure*:

$$\boxed{P_t(T) \;=\; e^{A_t(T) - B_t(T)\, r_t},} \tag{12.20}$$

with

$$B_t(T) \;=\; \frac{1 - e^{-\kappa(T-t)}}{\kappa}, \tag{12.21}$$

$$A_t(T) \;=\; -\theta\big((T-t) - B_t(T)\big) \;+\; \frac{\sigma^2}{2\kappa^2}\int_t^T \big(1 - e^{-\kappa(T-s)}\big)^2\, \mathrm{d}s. \tag{12.22}$$

(Here s is the dummy integration variable running across the bond's life, not to be confused with the current time t; the reference formulas at §12.9 reuse the same s label.) The structure $P = e^{A-Br}$ — with A, B deterministic functions of (t, T) independent of r — is called an affine term structure.

Vasicek bond-price surface *Vasicek bond-price surface $P(t, T)$ as a function of the current short rate and maturity.*

Let us decode (12.20). The bond price is a log-linear function of the current short rate: $\ln P_t(T) = A_t(T) - B_t(T) r_t$. The coefficient $B_t(T)$ is the bond's *duration-like* sensitivity: $-\partial_{r_t} \ln P_t(T) = B_t(T)$ is the modified duration with respect to short-rate shocks. For short maturities ($T \to t$), $B_t(T) \to 0$ and the bond is insensitive to r_t. For long maturities ($T \to \infty$), $B_t(T) \to 1/\kappa$, meaning a very long bond has a finite duration of $1/\kappa$ years — *not* unbounded, because mean reversion caps the effect. A Vasicek with $\kappa = 0.1$ (very slow reversion) gives a long-bond duration of 10 years; with $\kappa = 0.5$, 2 years. This is a radically different picture from a random-walk interest rate, under which long-bond duration would grow without bound.

The finite long-bond duration is one of Vasicek's most distinctive — and economically sensible — predictions. Actual 30-year bonds do not have three times the sensitivity of 10-year bonds to short-rate moves. The reason is mean reversion: a short-rate shock today does not permanently change expected rates fifty years from now — it is absorbed by the mean-reverting dynamics over a few $1/\kappa$ years. So the marginal duration contribution of each additional year beyond a few $1/\kappa$ years is essentially zero. The bond-duration cap at $1/\kappa$ is the quantitative expression of this economic insight.

Boundary conditions: $B_T(T) = 0$, $A_T(T) = 0$, so $P_T(T) = 1$. These encode the defining property of a zero-coupon bond: at maturity it pays exactly one unit of currency regardless of the state of the economy.

Affine coefficients A(t,T) and B(t,T) $B_t(T)$ *rises from 0 to the asymptote $1/\kappa$, encoding rate sensitivity of long bonds; $A_t(T)$ is negative and grows in magnitude with maturity, reflecting the drift and convexity adjustments.*

13.3.6 12.3.5 Affine ansatz and the PDE route

There is a second, independent route to (12.20) — and it is the one that generalises most smoothly to more complex models (CIR, quadratic-Gaussian, multi-factor affine). Fix T and view $P_t(T) = g(t, r_t)$ as a function of the state r_t. The Feynman-Kac representation (Chapter 4) says

$$P_t(T) \;=\; \mathbb{E}_t^{\mathbb{Q}}\Big[e^{-\int_t^T r_s\, \mathrm{d}s}\Big], \quad \mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}\widehat{W}_t, \tag{12.23}$$

and therefore $g(t, r)$ satisfies the bond-pricing PDE

$$(\partial_t + \mathcal{L})\, g(t, r) \;=\; r\, g(t, r), \qquad g(T, r) = 1, \tag{12.24}$$

with infinitesimal generator

$$\mathcal{L} \;=\; \kappa(\theta - r)\,\partial_r + \tfrac{1}{2}\sigma^2\,\partial_{rr}. \tag{12.25}$$

Why does the PDE carry a source term $-rg$ rather than being sourceless? Because we are not working with discounted prices directly; we are working with undiscounted bond prices. As time advances by $\mathrm{d}t$, the bond's value is reduced by a factor of $e^{-r_t\,\mathrm{d}t} \approx 1 - r_t\,\mathrm{d}t$, which contributes $-r_t g\,\mathrm{d}t$ to the dynamics. Multiplying g by the discount factor

$e^{-\int r}$ would give a process satisfying a sourceless PDE; the sourced PDE here is for the un-discounted price. A bond is a traded asset, so its discounted price is a $\mathbb{Q}$-martingale; its undiscounted price is not (it drifts at r_t), which is why (12.24) has a non-zero source.

Affine ansatz. Guess

$$g(t,r) \;=\; e^{A_t - B_t r}, \qquad A_T = 0,\; B_T = 0. \tag{12.26}$$

Derivatives: $\partial_t g = (\dot{A}_t - \dot{B}_t r)g$, $\partial_r g = -B_t g$, $\partial_{rr} g = B_t^2 g$. Substitute into (12.24) and divide by g:

$$\left(\dot{A}_t - \kappa B_t \theta + \tfrac{1}{2}\sigma^2 B_t^2\right) \;+\; \left(-\dot{B}_t + \kappa B_t - 1\right) r \;=\; 0. \tag{12.27}$$

Equation (12.27) is doing something subtle. The left-hand side is a polynomial in r of degree one — an r^0 coefficient and an r^1 coefficient. For this polynomial to vanish for *all* values of r, both coefficients must individually be zero. This is the *matching of coefficients* move, and it is the step that turns one PDE into two ODEs. It works only because the affine ansatz was the right guess; had we tried a quadratic ansatz $\log P = A - Br - Cr^2$, we would have picked up an r^2 coefficient that did not naturally vanish, and the ansatz would have failed. The class of models for which coefficient-matching succeeds is precisely the class of affine models — hence the name.

Coefficient matching gives the *affine term-structure ODE system*:

$$\boxed{\begin{cases} \dot{A}_t - \kappa B_t\,\theta + \frac{1}{2}\sigma^2 B_t^2 \;=\; 0, & \text{(A-ODE)} \\ \dot{B}_t - \kappa B_t + 1 \;=\; 0, & \text{(B-ODE)} \end{cases} \qquad A_T = 0,\; B_T = 0.} \tag{12.28}$$

The ODE system (12.28) is structurally beautiful: the B-equation is autonomous (involves only B and constants), while the A-equation depends on B as an input. Solve for B first, then plug the result into the A-equation and integrate. In richer models the B-equation becomes a Riccati ODE (quadratic in B); here in Vasicek it is merely linear, because the diffusion is constant rather than state-dependent.

Solving B-ODE. Equation $\dot{B}_t - \kappa B_t + 1 = 0$ in standard form is $\dot{B}_t = \kappa B_t - 1$ with $B_T = 0$. Integrating factor $e^{-\kappa t}$ gives

$$B_t(T) \;=\; \frac{1 - e^{-\kappa(T-t)}}{\kappa}, \tag{12.29}$$

matching (12.21) exactly. This function depends only on $\tau = T - t$, not separately on t and T — a consequence of the constant-coefficient structure.

Solving A-ODE. Plug $B_t(T)$ into the A-ODE and integrate from t to T using $A_T = 0$:

$$A_t(T) \;=\; -\int_t^T \kappa\, B_s(T)\,\theta\,\mathrm{d}s \;+\; \frac{\sigma^2}{2}\int_t^T B_s(T)^2\,\mathrm{d}s. \tag{12.30}$$

Substituting $\kappa B_s(T) = 1 - e^{-\kappa(T-s)}$ and $\kappa^2 B_s(T)^2 = (1 - e^{-\kappa(T-s)})^2$, (12.30) becomes exactly (12.22), providing a cross-check between the two derivation routes.

The two derivations — moment-generating-function (§12.3.4) and PDE / affine ansatz (§12.3.5) — arrive at the same answer via genuinely different routes. The former is analytically direct but only works because Gaussianity makes the MGF explicit. The latter is abstractly more general (it depends only on affine structure, not on Gaussianity) and therefore extends to CIR, quadratic-Gaussian models, and affine jump-diffusions. In research the PDE approach is more often the starting point because it generalises more cleanly; in pedagogy the direct-expectation route is more transparent for the Vasicek special case we are working through.

Some authors prefer yet a third derivation: the martingale-based approach, where one guesses the bond-price formula and verifies that the discounted price $P_t(T)/M_t$ is a martingale under $\mathbb{Q}$. This is the most elegant route when it works, because it directly confirms the no-arbitrage condition rather than deriving the price and verifying arbitrage-freeness as a corollary. The three approaches — expectation, PDE, martingale — form a triad that characterises risk-neutral

pricing from three complementary angles: operational (how to compute it), analytic (what equation does it satisfy), and economic (why is it arbitrage-free). Mastering all three is the mark of a mature quant.

A more general methodological lesson from this cross-check: in quantitative finance, having two independent derivations of the same result is not redundancy — it is insurance. Each derivation exposes a different aspect of the mathematical structure, and each is vulnerable to a different class of errors. The direct-expectation approach is prone to integration mistakes and Fubini-justification gaps; the PDE approach is prone to ansatz-guessing errors and boundary-condition confusion. When both converge on the same formula, one gains high confidence that neither class of errors has been made.

13.4 12.4 The yield curve under Vasicek

13.4.1 12.4.1 Yields and instantaneous forward rates

The continuously-compounded yield for maturity T at time t is

$$y_t(T) \;=\; -\frac{\ln P_t(T)}{T-t} \;=\; \frac{-A_t(T) + B_t(T)\, r_t}{T-t}. \tag{12.31}$$

Yields, not bond prices, are what practitioners and the press typically quote, so (12.31) is probably the most practically used formula derived from Vasicek. The conversion from bond prices to yields is conceptually a simple logarithm-and-divide; the practical utility is that yields are approximately stationary over long periods, while bond prices drift toward 1 at maturity.

Under Vasicek, the yield decomposition $y_t(T) = -A_t(T)/(T-t) + B_t(T)/(T-t) \cdot r_t$ shows that yields are themselves affine in r_t. The coefficient $B_t(T)/(T-t)$ is the yield's sensitivity to the short rate; it approaches 1 at short maturities (so short-maturity yields essentially equal r_t) and approaches 0 at long maturities (so long-maturity yields are almost insensitive to r_t, a consequence of mean reversion). The intercept $-A_t(T)/(T-t)$ is the "term-structure contribution" — the part of the yield that does not depend on today's rate.

The instantaneous forward rate at maturity T is $f_t(T) = -\partial_T \ln P_t(T)$. It is the rate at which an investor could lock in a forward contract today starting at T for an infinitesimal period. Equivalently, it is the expected T-instantaneous short rate under the T-forward measure (see Chapter 13). Bond prices and forward rates are related by $\ln P_t(T) = -\int_t^T f_t(s)\,\mathrm{d}s$, and the short rate itself satisfies $r_t = f_t(t)$.

A common trap: the forward rate $f_t(T)$ is *not* in general equal to the expected future short rate $\mathbb{E}[r_T]$. Under the T-forward measure, $f_t(T) = \widetilde{\mathbb{E}}^T[r_T]$; under the risk-neutral measure, $\mathbb{E}^{\mathbb{Q}}[r_T]$ differs by a convexity adjustment; under the physical measure, there is an additional risk-premium correction. When a commentator says "the market expects rates to be X% in five years" and cites a five-year forward rate, that statement is true under $\widetilde{\mathbb{Q}}^5$ but not under $\mathbb{P}$ — and the wedge between them can be substantial, particularly at longer horizons. Being precise about *which* measure one is discussing is a hallmark of careful rates analysis.

13.4.2 12.4.2 Yield curve shapes

For Vasicek:

- As $T \to t$: $y_t(T) \to r_t$ (the short rate itself).
- As $T \to \infty$: $y_t(T) \to \theta - \frac{1}{2}\,\sigma^2/\kappa^2$ (the long yield — constant in r_t).
- In between: the curve can be monotone, humped, or inverted depending on r_t vs θ.

The long-yield limit $\theta - \frac{1}{2}\sigma^2/\kappa^2$ deserves a pause. The first term, θ, is the long-run mean of the short rate — what you would expect if you simply averaged rates over the infinite future. The second term is a *subtraction*: the long yield is *below* the expected long-run short rate. This negative gap is the convexity premium: because the discount factor is a convex function of the integrated rate, random variation in future rates raises the expected discount factor above the deterministic benchmark, and therefore lowers the implied yield. The magnitude $\sigma^2/(2\kappa^2)$ is proportional to the stationary variance of r_t — more volatile or less mean-reverting short rates give larger convexity premia. This is a genuine economic prediction: in high-volatility environments, long yields should trade *below* consensus expectations of the long-run short rate, and the gap should widen with volatility. Empirically this is reasonably well supported.

Quantifying in a concrete case: with $\sigma = 0.01$ and $\kappa = 0.3$, the premium is $(0.01)^2/(2 \cdot 0.09) = 5.56 \times 10^{-4}$, or about 5.6 basis points. With higher volatility $\sigma = 0.02$ it quadruples to 22 bp. With slower reversion $\kappa = 0.15$,

the denominator halves-squared to 0.0225 and the premium is 22.2 bp. So the convexity premium is most visible in slow-reverting, high-volatility environments — exactly the sort of regime that tends to produce inverted or unusually flat yield curves.

Vasicek yield-curve shapes *Vasicek yield-curve shapes for three starting short rates, showing the classical monotone-rising, humped, and inverted regimes.*

An illustrative calibration. Take $\kappa = 0.3$, $\theta = 0.04$, $\sigma = 0.015$ and consider three starting rates: $r_0 \in \{0.01, 0.04, 0.08\}$. For each compute $y_0(T)$ across $T \in \{1, 2, 5, 10, 30\}$ years.

For $r_0 = 0.01$ (well below θ): the short end is near 1%, rises through about 2.5% at 2 years, 3.2% at 5, 3.6% at 10, and approaches the long-end asymptote $\theta - \sigma^2/(2\kappa^2) = 0.04 - 0.00125 \approx 3.88\%$ at 30. A monotonically rising curve — the classical "normal" yield curve in a below-neutral environment.

For $r_0 = 0.04$ (at θ): the curve is nearly flat around 3.9%, with only the convexity correction producing a slight downward drift at longer maturities.

For $r_0 = 0.08$ (well above θ): the short end is at 8%, falling through about 6.6% at 2 years, 5.0% at 5, 4.3% at 10, and 3.88% at 30. An inverted curve — typical of late-cycle monetary tightening when short rates are elevated but the market expects them to fall.

These three scenarios show that Vasicek produces economically sensible curve shapes across the full range of initial conditions, entirely through the mean-reversion parameter κ and the long-run mean θ. The deterministic convergence $r_t \to \theta$ drives the shape, and the convexity correction fine-tunes the long end.

13.4.3 12.4.3 The 1-factor limitation

Crucially, all shifts of the curve under Vasicek are driven by a single factor r_t. Any change in the short rate must propagate through the *entire* yield curve in a fixed, deterministic way dictated by $B_t(T)$. So if the short rate rises by 25 bp, the 2-year yield rises by some specific amount, the 10-year by another, the 30-year by yet another — but the *ratios* among these moves are fixed by the model. Reality is not so rigid: actual yield-curve moves are dominated by three roughly orthogonal factors (level, slope, curvature) that move independently. Principal-components analysis of daily yield-curve changes consistently shows that three factors explain 95-99% of the variance.

To capture this, practitioners turn to multi-factor affine models — typically two-factor Hull-White or three-factor affine with independent Ornstein-Uhlenbeck factors. Each factor contributes its own affine loading, so a linear combination can match observed yield-curve dynamics much more faithfully. The cost is higher calibration complexity and more parameters to stabilise; the benefit is hedges that actually work under real curve reshapes.

A few common multi-factor specifications deserve mention.

First, the **two-factor Gaussian model (G2++)**: two independent (or correlated) Ornstein-Uhlenbeck processes x_t and y_t, with the short rate defined as $r_t = x_t + y_t + \phi(t)$ where $\phi(t)$ is a deterministic shift calibrated to today's curve. With two mean-reversion speeds κ_x, κ_y and two volatilities σ_x, σ_y plus a correlation ρ, the model has enough freedom to match level-plus-slope curve dynamics and a reasonable vol surface. G2++ is the most widely used multi-factor rates model in derivative pricing libraries.

Second, the **three-factor affine model**: three OU factors, allowing separate calibration to level, slope, and curvature. The bond-price formula remains exponential-affine in the three-factor state vector, and the matrix Riccati ODE for $B(t, T)$ can be solved explicitly under mild assumptions (diagonal mean-reversion matrix).

Third, the **Libor Market Model (LMM)**: a radical departure that models individual forward rates directly rather than a short rate. LMM can match any yield curve by construction (forward rates are model inputs, not model outputs) and any swaption vol cube by suitable specification of forward-rate volatilities and correlations. The price for this flexibility is giving up the low-dimensional affine structure: LMM is high-dimensional, state-vector-based, and typically requires Monte Carlo for complex payoffs.

Which multi-factor model to use is ultimately a portfolio-specific decision. For vanilla cap-floor-swaption books, G2++ is usually sufficient. For Bermudan or callable products, LMM or extended affine models may be needed. For CVA/XVA computations across a huge book, the calibration speed and hedging accuracy of affine models make them attractive even when LMM would be "more accurate" per trade.

13.5 12.5 Ho-Lee — the canonical derivation

Vasicek with constant θ has three parameters (κ, θ, σ) and can fit only three features of a yield curve — a level, a slope, and a curvature, roughly. A real observed curve has dozens of quoted maturities and cannot be reproduced by any three-parameter model. The natural fix is to let the drift schedule vary with time. The simplest such extension drops the mean-reversion altogether and retains only the time-varying drift: this is the Ho-Lee model, which we now derive from first principles. It serves a dual purpose — it is historically the first example of exact yield-curve calibration, and it is mathematically the $\kappa \to 0$ limit of Hull-White that we develop in §12.6.

13.5.1 12.5.1 Discrete Ho-Lee

The discrete Ho-Lee dynamics on a Δt grid:

$$\boxed{r_n \;=\; r_{n-1} \;+\; \theta_{n-1}\,\Delta t \;+\; \sigma\sqrt{\Delta t}\,x_n, \qquad x_n \overset{\text{iid}}{\sim} \pm 1 \text{ Bernoulli},\ \mathbb{Q}(x_n = \pm 1) = \tfrac{1}{2}.} \tag{12.32}$$

Read (12.32) aloud: "tomorrow's short rate equals today's short rate, plus a deterministic drift over the step, plus a symmetric random coin-flip scaled to the right standard deviation." The drift θ_{n-1} is what curves the rate process to match the market; the shock $\sigma\sqrt{\Delta t}\,x_n$ is the noise that lets the rate move up or down. Because the shock is additive (not multiplicative) and symmetric, the up-move and down-move at each node have equal magnitude. This produces a recombining tree — two steps up and one step down lands at the same node as one up and two down, reshuffled.

Key features.

- *Arithmetic, not geometric.* Rates are added, not scaled — rates can go negative. Historically this was Ho-Lee's greatest weakness; post-2008, actually a feature when pricing negative-rate regimes.
- *No mean reversion.* The Ho-Lee dynamic is Vasicek with $\kappa = 0$; only the drift θ_n shapes the curve. A pure random walk in rates (with drift) means that far out in maturity the distribution of r_T becomes increasingly diffuse and wide.
- *Recombining tree.* With a symmetric ± 1 shock and constant σ, up-then-down and down-then-up return to the same node. The recombination reduces the node count at step n from 2^n to $n+1$, making calibration linear-time rather than exponential. This is the single largest computational advantage of Ho-Lee and a property preserved by its continuous-time limit.

Calibration goal. Find $\theta_0, \theta_1, \theta_2, \ldots$ so that the model matches the market bond prices $P_0(1), P_0(2), P_0(3), \ldots$. The bootstrap is triangular: θ_0 depends only on $P_0(1)$ and $P_0(2)$, θ_1 on $P_0(3)$ given θ_0, and so on. We never solve a nonlinear system. At every step, each new market bond price adds exactly one new equation in exactly one new unknown θ_n — one of the cleanest pieces of structure in quantitative finance.

With $q = 1/2$ inside the tree probability measure, bond prices are

$$P_0(1) \;=\; e^{-r_0\,\Delta t}, \qquad P_0(2) \;=\; e^{-r_0 \Delta t}\left[\tfrac{1}{2}e^{-r_{1u}\Delta t} \;+\; \tfrac{1}{2}e^{-r_{1d}\Delta t}\right], \tag{12.33}$$

and similarly $P_0(3)$ depends on θ_0, θ_1, giving us one equation in θ_1 once θ_0 is known. The philosophical takeaway: calibration replaces the abstract non-uniqueness of $\mathbb{Q}$ (§12.1.1) with a specific choice of θ-schedule that makes model-implied bond prices equal market-observed bond prices. Once that schedule is fixed, every other derivative prices off the resulting tree without further calibration.

13.5.2 12.5.2 Continuous-time limit (direct Donsker argument)

The cleanest path to the continuous-time Ho-Lee model: start from (12.32), sum the recursion, and pass $N \to \infty$ directly, bypassing the Vasicek/Hull-White machinery. Fix a horizon T, set $\Delta t = T/N$ with $t_n = n\Delta t$, and iterate (12.32):

$$r_N \;=\; r_0 \;+\; \sum_{n=1}^{N} \theta_{n-1}\,\Delta t \;+\; \sigma\sqrt{\Delta t}\sum_{n=1}^{N} x_n. \tag{12.34}$$

This is a pure telescoping sum: starting level r_0, cumulative drift, cumulative shock. The moments are immediate:

$$\mathbb{E}^{\mathbb{Q}}[r_T] \;\to\; r_0 \;+\; \int_0^T \theta_u \,\mathrm{d}u, \qquad \mathbb{V}^{\mathbb{Q}}[r_T] \;=\; \sigma^2 T. \tag{12.35}$$

By Donsker's Invariance Principle — a sum of iid ± 1 shocks, scaled by $\sqrt{\Delta t}$, converges in distribution to Brownian motion — we have

$$\boxed{r_T \;\stackrel{d}{=}\; r_0 \;+\; \int_0^T \theta_u \,\mathrm{d}u \;+\; \sigma\sqrt{T}\, Z, \qquad Z \sim \mathcal{N}(0,1),} \tag{12.36}$$

equivalently the SDE

$$\mathrm{d}r_t \;=\; \theta_t \,\mathrm{d}t \;+\; \sigma \,\mathrm{d}W_t. \tag{12.37}$$

The $\sqrt{\Delta t}$ scaling on the shock is the fundamental scaling of Brownian motion, which has quadratic variation $\mathrm{d}t$. Had we scaled by Δt instead, the limit would have been deterministic (shock vanishing); by 1, it would have been infinite (shock exploding). The "right" scaling $\sqrt{\Delta t}$ is exactly what keeps the variance of the sum of N shocks finite. This scaling shows up in everything from heat diffusion to stock returns to interest-rate innovations; it is the signature of a process driven by "many small independent shocks" (Chapter 3's construction of Brownian motion reviews this in detail).

Negative rates are explicit: $r_T \sim \mathcal{N}(r_0 + \int \theta, \sigma^2 T)$, so $\mathbb{P}(r_T < 0) = \Phi(-M/\sqrt{V}) > 0$ where M is the mean and V the variance. For $T = 10$ years, $\sigma = 1\%$, $M = 3\%$, that probability is $\Phi(-0.95) \approx 17\%$ — substantial. For $T = 1$ year it drops to $\approx 2\%$. Negative rates thus bite mainly at long horizons, where variance has had time to accumulate.

Distribution of the integrated rate. A shock at time t affects *all* future r_u for $u \in [t, T]$ (the additive structure with no reversion has infinite memory), contributing with lever-arm weight $T - t$ to the integrated rate. Working through the double-sum calculation,

$$\int_0^T r_u \,\mathrm{d}u \;\stackrel{d}{=}\; r_0 T \;+\; \int_0^T \int_0^u \theta_s \,\mathrm{d}s \,\mathrm{d}u \;+\; \left(\frac{\sigma^2 T^3}{3}\right)^{1/2} Z. \tag{12.38}$$

The cubic growth $T^3/3$ of the variance — as opposed to T for the process itself — is Ho-Lee's signature of missing mean reversion. Physically: $\mathrm{Var}[\int_0^T r_u \,\mathrm{d}u] = \int_0^T (T-t)^2 \,\mathrm{d}t \cdot \sigma^2 = \sigma^2 T^3/3$. A 30-year Ho-Lee at $\sigma = 1\%$ would give $\sigma^2 T^3/3 = 10^{-4} \cdot 9000 = 0.9$, an integrated-rate standard deviation of 94%. This is absurdly large — translated into bond-price uncertainty, it implies extreme price sensitivity to model parameters at long horizons. Mean reversion (Vasicek/HW) fixes this by making variance grow only linearly in T at long horizons. This is the single biggest practical reason Ho-Lee is rarely used for long-dated instruments.

13.5.3 12.5.2.b The meaning of the $\sqrt{\Delta t}$ scaling

The scaling $\sqrt{\Delta t}$ on the shock is not arbitrary. Brownian motion was first introduced by Einstein (1905) to model pollen grains in water — particles buffeted by countless small random molecular collisions. Einstein's analysis showed that for such a process, the position's variance grows linearly in time: $\mathbb{V}[X_t] \propto t$. Linear variance growth in continuous time translates to $\sqrt{\Delta t}$ scaling of discrete increments (because variance of a sum of N independent increments equals N times the per-increment variance, and if N times some number equals T, then the per-increment standard deviation is $\sqrt{T/N} = \sqrt{\Delta t}$). This scaling shows up in heat diffusion, stock returns, interest-rate innovations, and countless other physical processes. It is the signature of "many small independent shocks."

The $\sqrt{\Delta t}$ scaling also underpins the famous quadratic- variation result: $(\mathrm{d}W_t)^2 = \mathrm{d}t$ in the Itô-calculus limit. This is not an equality in the ordinary sense (both sides are random) but a precise statement about how squared infinitesimal increments behave in L^2. That $(\mathrm{d}W_t)^2$ is deterministic (equal to $\mathrm{d}t$) is the reason Brownian calculus has its peculiar features — Itô's lemma includes a second-order term proportional to variance, reflecting the nonzero quadratic variation. In rate pricing, this shows up as the convexity term C_T; in equity pricing, as the extra $\frac{1}{2}\sigma^2$ drift in the geometric- Brownian solution.

If we relax the "finite variance" constraint and allow shocks to have heavy tails with infinite variance, the limit becomes a stable Lévy process instead of Brownian motion. Stable Lévy processes have jumps; they are discontinuous. This is the mathematical basis for jump- diffusion models of rates, equity prices, and credit spreads. Such models are sometimes used when heavy tails are important (e.g. pricing equity index options during tail events), but they are not standard for rates. The Gaussian limit is the default because it matches observed rate-change distributions reasonably well at the level of daily moves, and the mathematical benefits are substantial.

13.5.4 12.5.3 Bond price and calibration under continuous Ho-Lee

Because $I_T = \int_0^T r_u \, \mathrm{d}u$ is Gaussian, $\mathbb{E}^{\mathbb{Q}}[e^{-I_T}] = \exp(-\mathbb{E}I_T + \frac{1}{2}\mathbb{V}I_T)$, giving

$$\boxed{\ln P_0(T) \;=\; -r_0\,T \;-\; \int_0^T \int_0^u \theta_s \, \mathrm{d}s \, \mathrm{d}u \;+\; \frac{\sigma^2 T^3}{6}.} \tag{12.39}$$

The $+\sigma^2 T^3/6$ is the Ho-Lee convexity adjustment. Read in context of (12.19): $+\frac{1}{2}\Sigma_*^2 = +\frac{1}{2}\sigma^2 T^3/3 = \sigma^2 T^3/6$ — Jensen's mountain applied to a cubically growing variance.

Calibration via differentiation. Since (12.39) expresses $\ln P_0(T)$ as a double integral of θ plus known terms, differentiating twice with respect to T pulls out θ_T pointwise.

First derivative:

$$\partial_T \ln P_0^*(T) \;=\; -r_0 \;-\; \int_0^T \theta_s \, \mathrm{d}s \;+\; \frac{\sigma^2 T^2}{2}. \tag{12.40}$$

Since $\partial_T \ln P_0^*(T) = -f_0(T)$, this rearranges to the Ho-Lee forward-curve identity $f_0(T) = r_0 + \int_0^T \theta_s \, \mathrm{d}s - \sigma^2 T^2/2$.

Second derivative:

$$\partial_T^2 \ln P_0^*(T) \;=\; -\theta_T \;+\; \sigma^2 T, \tag{12.41}$$

and therefore

$$\boxed{\theta_T \;=\; -\,\partial_T^2 \ln P_0^*(T) \;+\; \sigma^2 T.} \tag{12.42}$$

Equation (12.42) is the classical Ho-Lee calibration identity. Read it intuitively: the drift target θ_T at time T is "the slope of the forward curve" plus "a convexity bump proportional to T." If the forward curve is flat, the drift target is just the convexity bump. If the forward curve is rising, θ_T is positive and the model drifts rates upward on average.

In practice one fits a smooth parametric form (Nelson-Siegel, Svensson, cubic splines) to the observed log-bond curve, differentiates the fit twice, adds the convexity bump, and reads off θ_T at every grid point. The choice of curve-smoother affects the resulting $\theta(t)$ qualitatively: noisy second derivatives translate into noisy $\theta(t)$, which makes the rate model unstable and gives erratic prices for non-calibration instruments.

13.5.5 12.5.4 Ho-Lee as the $\kappa \to 0$ limit of Hull-White

Ho-Lee is precisely the $\kappa = 0$ specialisation of the Hull-White dynamics (below):

$$\underbrace{\mathrm{d}r_t = \kappa(\theta_t - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t}_{\text{Hull-White}} \xrightarrow{\kappa \to 0} \underbrace{\mathrm{d}r_t = \theta_t\,\mathrm{d}t + \sigma\,\mathrm{d}W_t}_{\text{Ho-Lee}}$$

with the understanding that the Hull-White θ_t must diverge as $1/\kappa$ for the limit to be well-defined — the joint limit $\kappa\theta_t \to \theta_{\mathrm{HL}}$ is what survives. Conceptually the three models nest: Ho-Lee is the simplest thing; Vasicek adds a pull-to-constant-target; Hull-White makes that target time-varying. The calibration machinery for Hull-White specialises to Vasicek (constant θ) and to Ho-Lee ($\kappa = 0$) with minimal changes. This pedagogical nesting is why we present them in this order.

13.6 12.6 Hull-White — the canonical derivation

Vasicek captures mean reversion but has only a constant θ, so it can fit only a single feature of the observed term structure (roughly, the long-run mean). Ho-Lee captures exact yield-curve matching but has no mean reversion, so its variance grows cubically and the model breaks at long horizons. Hull-White is the best of both worlds: Vasicek dynamics with a *time-dependent* drift target $\theta(t)$. The function $\theta(t)$ is calibrated so the model exactly reproduces today's yield curve, while the mean-reversion parameter κ tames variance growth.

13.6.1 12.6.1 Discrete Hull-White

Promote the constant θ in (12.9) to a deterministic function θ_t and replace ± 1 Bernoullis with Gaussian innovations:

$$\boxed{r_{t_n} - r_{t_{n-1}} \;=\; \kappa(\theta_{t_{n-1}} - r_{t_{n-1}})\,\Delta t \;+\; \sigma\sqrt{\Delta t}\,\chi_n, \qquad \chi_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1).} \tag{12.43}$$

The move from constant θ to time-varying $\theta(t)$ is the conceptual leap. In Vasicek the process always reverts toward the same number, so you can only fit one long-run level. In Hull-White the process reverts toward a *moving target* — the target itself is a calibrated function of time — so every part of the yield curve can be independently matched. The price is infinite-dimensional calibration (one θ per time point), but as we will see this reduces to a linear bootstrap exactly as in Ho-Lee.

What the parameters mean.

- κ is a *speed*: the deterministic part of (12.43) pulls r toward θ_t at exponential rate κ. Large $\kappa \Rightarrow$ very "sticky" short rate around θ; small $\kappa \Rightarrow$ near-random-walk (Ho-Lee). Typical calibrated values live in 0.05-0.3 per year, giving shock half-lives of 2-15 years.
- σ sets the innovation scale. The unconditional variance of r_T approaches $\sigma^2/(2\kappa)$ as $T \to \infty$. Typical σ values are 0.005-0.02 in absolute rate terms (50-200 bp per year root-volatility).
- $\theta(t)$ is the *drift shaper*: the knob tuned to the market. In the limit it is essentially $f_0(t) + \sigma^2 \cdot (\text{convexity})$, where $f_0(t)$ is the instantaneous forward rate.

It is worth pausing on the division of labour. κ and σ are the *volatility-structure* parameters — they determine how rates move, how persistent shocks are, how volatile caps and swaptions will be. These are typically calibrated *jointly* against vol-sensitive instruments (ATM swaption prices). $\theta(t)$ is the *curve-fitting* parameter, chosen conditionally on (κ, σ) to exactly match the discount curve. So calibration has two stages: first pick (κ, σ) to match vol-sensitive instruments, then bootstrap $\theta(t)$ to match bond prices. This two-stage structure is characteristic of rates calibration in practice.

In production systems the two stages are often iterated. A first pass with historically-sensible (κ, σ) produces a $\theta(t)$ schedule. The resulting model is then used to price a set of swaptions, and the (κ, σ) are adjusted to reduce swaption-price errors. Then $\theta(t)$ is re-bootstrapped with the updated (κ, σ), and so on until convergence. The inner bootstrap (curve) takes microseconds; the outer optimisation (swaption fit) takes seconds to minutes. The separation of concerns is crucial: the fast bootstrap runs inside the slow optimiser.

Hull-White simulated short-rate paths *Hull-White simulated short-rate paths reverting to time-dependent $\theta(t)$.*

13.6.2 12.6.2 Summed integral and continuous limit

Unrolling (12.43) recursively with $\beta = 1 - \kappa\Delta t$ and $\alpha_{n-1} = \kappa\theta_{t_{n-1}}\Delta t$ yields the same geometric structure as the Vasicek unrolling, with the single difference that the intercept α varies with n:

$$r_T \;=\; r_0\,\beta^N \;+\; \sum_{n=1}^{N} \alpha_{n-1}\,\beta^{N-n} \;+\; \sigma\sqrt{\Delta t}\sum_{n=1}^{N} x_n\,\beta^{N-n}. \tag{12.44}$$

Taking $N \to \infty$ with $\Delta t = T/N$, the standard Donsker-type identifications give:

- $(1 - \kappa T/N)^{N-n} \to e^{-\kappa(T-u)}$,
- $(1 - \kappa T/N)^{N} \to e^{-\kappa T}$,
- $\sum_n (\cdot)\,\Delta t \to \int_0^T (\cdot)\,\mathrm{d}u$,
- $\sigma\sqrt{\Delta t}\sum_n x_n\,\beta^{N-n} \to \sigma\int_0^T e^{-\kappa(T-u)}\,\mathrm{d}W_u$.

Each of these is a standard stochastic-calculus limit (Chapter 3): the first two use $\lim_N (1+x/N)^N = e^x$; the third is the Riemann-sum-to-integral limit; the fourth is Itô's stochastic-integral construction for a deterministic integrand.

The Wiener integral $\int_0^T e^{-\kappa(T-u)}\,\mathrm{d}W_u$ is a Gaussian random variable with mean zero and variance $\int_0^T e^{-2\kappa(T-u)}\,\mathrm{d}u = (1-e^{-2\kappa T})/(2\kappa)$ — exactly the Vasicek stationary-variance formula. This clean Gaussian structure is the source of the closed-form bond price. Had the integrand been state-dependent (like $\sqrt{r_u}$ in CIR), the integral would be a true Itô integral, not a Wiener integral, and its distribution would not be Gaussian.

The limit of (12.44) is the explicit Hull-White solution:

$$r_T \;=\; r_0\, e^{-\kappa T} \;+\; \kappa \int_0^T e^{-\kappa(T-u)}\,\theta_u\,\mathrm{d}u \;+\; \sigma \int_0^T e^{-\kappa(T-u)}\,\mathrm{d}W_u. \tag{12.45}$$

The pattern is identical to Vasicek with constant θ (equation (12.12)); only the deterministic drift contribution has been generalised from a function of $\tau = T - t$ alone to a true integral against the time-varying θ_u.

13.6.3 12.6.3 Hull-White bond price

The integrated rate, by the same Fubini manipulation as in §12.3.3:

$$\int_0^T r_u\,\mathrm{d}u \;=\; r_0\, B_T \;+\; A_T \;+\; \frac{\sigma}{\kappa}\int_0^T \big(1-e^{-\kappa(T-u)}\big)\,\mathrm{d}W_u, \tag{12.46}$$

where

$$B_T \;=\; \frac{1-e^{-\kappa T}}{\kappa}, \qquad A_T \;=\; \int_0^T \theta_u\,\big(1-e^{-\kappa(T-u)}\big)\,\mathrm{d}u. \tag{12.47}$$

B_T is worth a moment because it is the single most-used quantity in Hull-White computations: the "effective duration" that the current short rate r_0 exerts on the integrated rate. In the $\kappa \to 0$ limit, $B_T \to T$ (no reversion means the current rate affects the entire horizon). In the $\kappa \to \infty$ limit, $B_T \to 0$ (infinite reversion instantly forgets the current rate). For typical $\kappa \in [0.05, 0.3]$, the ratio B_T/T ranges from near 1 at short T down to something much smaller at long T. For $\kappa = 0.15$ and $T = 30$: $B_T = (1-e^{-4.5})/0.15 \approx 6.6$, so $B_T/T \approx 0.22$. The current rate has 22% as much influence on the 30-year bond price as it would under no mean reversion.

Because the integrated rate is Gaussian with mean $r_0 B_T + A_T$ and variance

$$\mathbb{V}\left[\int_0^T r_u\,\mathrm{d}u\right] \;=\; \frac{\sigma^2}{\kappa^2}\int_0^T \big(1-e^{-\kappa(T-u)}\big)^2\,\mathrm{d}u \;\equiv\; 2\,C_T, \tag{12.48}$$

the Hull-White bond price comes out directly from the Gaussian moment-generating function:

$$\boxed{P_0(T) \;=\; \exp\{-r_0\,B_T \;-\; A_T \;+\; C_T\}\,,} \tag{12.49}$$

with convexity adjustment

$$C_T \;=\; \frac{\sigma^2}{2\kappa^2}\int_0^T \big(1-e^{-\kappa(T-u)}\big)^2\,\mathrm{d}u. \tag{12.50}$$

Reading (12.49), three pieces:

- $r_0 B_T$: current-level contribution, decaying at rate κ because a rate shock "today" is damped over the life of the bond.
- A_T: the drift contribution from the θ path. All the market information about future forward rates lives here.

- $+C_T$: positive convexity. Jensen's inequality makes the bond strictly *more* valuable than naive plug-in discounting suggests. This is the Hull-White analogue of the $\sigma^2 T^3/6$ term in Ho-Lee and of the $\frac{1}{2}\Sigma_*^2$ term in constant-θ Vasicek.

Sanity checks. As $\kappa \to 0$ (Taylor-expanding $(1-e^{-\kappa(T-u)})^2 \approx \kappa^2(T-u)^2$), we have $B_T \to T$, $A_T \to \int_0^T \theta_u \, \mathrm{d}u$, and $C_T \to \sigma^2 T^3/6$. Formula (12.49) collapses to the Ho-Lee formula (12.39). Setting $\theta(t) = \theta$ constant, $A_T = \theta(T - B_T)$ and (12.49) specialises to Vasicek. The nesting is clean and complete: Ho-Lee $\subset$ Vasicek $\subset$ Hull-White.

The κ^2 cancellation in the C_T limit is a specific instance of a more general phenomenon: many quantities in Hull-White approach their Ho-Lee limits smoothly as $\kappa \to 0$, because the mean-reversion structure enters quadratically in the variance terms and linearly in the mean terms. This smoothness of limits lets us calibrate Vasicek/Hull-White parameters to low κ values without numerical difficulty.

Worked magnitude. For $T = 10$, $\sigma = 0.01$, $\kappa = 0.1$: the integral $\int_0^{10}(1-e^{-0.1(10-u)})^2 \, \mathrm{d}u = 10 - 2(1-e^{-1})/0.1 + (1-e^{-2})/0.2 = 10 - 12.64 + 4.32 = 1.68$, so $C_{10} = (10^{-4}/0.02) \cdot 1.68 = 0.0084$ — an 84-bp bonus in log-bond. By comparison, Ho-Lee with the same σ at $T = 10$ gives $C_{10} = \sigma^2 T^3/6 = 0.0167$, or 167 bp — roughly double. Mean reversion halves the 10-year convexity bonus and the gap grows exponentially at longer tenors. At $T = 30$ the Ho-Lee number is $\sigma^2 T^3/6 = 0.45$ (4500 bp!) while Hull-White ($\kappa = 0.1$) comes in around 600 bp — an order of magnitude smaller. This is why mean reversion matters for long-dated pricing: Ho-Lee's unbounded convexity growth is unrealistic; Hull-White's bounded growth matches market observations.

Model-implied zero-coupon bond curve *Model-implied zero-coupon bond / discount curve* $P_0(T)$.

13.6.4 12.6.3.b Precomputation and numerical considerations

For pricing efficiency it is convenient to precompute the "basis functions" once per maturity T: $1 - e^{-\kappa(T-t)}$, $(T-t) - B_t(T)$, and $(T-t) - 2B_t(T) + (1-e^{-2\kappa(T-t)})/(2\kappa)$. Then $A_t(T)$ and $B_t(T)$ are simple linear combinations with coefficients θ, $\sigma^2/(2\kappa^2)$, etc. In production rates libraries, the ODEs for A and B are solved once at model-initialisation time and cached; during pricing, the pre-computed $A(t,T)$ and $B(t,T)$ are simply evaluated at the appropriate (t,T) pair. This makes bond pricing essentially instantaneous — a table lookup rather than an integration. The same caching strategy applies to multi-factor affine models and is part of why these models scale so gracefully to large trading books.

The finite long-bond duration $1/\kappa$ is the quantitative expression of mean reversion, and it is a useful diagnostic. A κ of 0.02 implies a long-bond duration cap of 50 years, which is physically implausible — no market participant believes that short-rate shocks persist for half a century. A κ of 2.0 implies a cap of 6 months, which is implausibly small. The reasonable range $\kappa \in [0.1, 0.5]$ maps to long-bond durations between 2 and 10 years, matching empirical observation. When a calibration algorithm returns a κ outside this range, something is likely wrong: over-fitting, compensating misspecification, or an identification issue.

13.6.5 12.6.4 Calibrating $\theta(u)$ to the market curve

In practice one observes market bond prices $P_0^*(T)$ (or yields) at $t = 0$ for all maturities T. The idea of Hull-White: choose the deterministic function θ_t so the model reproduces $P_0^*(T)$ exactly. Why is this so important? Because a rates trader using a model that doesn't match today's curve is effectively betting that today's market is mispriced — almost never a useful default. With $\theta(t)$ free, the model has enough functional freedom to fit any observed curve exactly; the remaining parameters (κ, σ) are then fit to options on bonds.

From (12.49), taking logs at $t = 0$,

$$\ln P_0^*(T) \;=\; -r_0 \, B_T \;-\; \int_0^T \theta_u \, (1 - e^{-\kappa(T-u)}) \, \mathrm{d}u \;+\; C_T. \tag{12.51}$$

Closed-form inversion. Differentiating twice in T gives the pointwise Hull-White calibration identity:

$$\boxed{\theta(T) \;=\; \frac{1}{\kappa} \, \partial_T f_0(T) \;+\; f_0(T) \;+\; \frac{\sigma^2}{2\kappa^2}(1 - e^{-2\kappa T}) \, ,} \tag{12.52}$$

where $f_0(T) = -\partial_T \ln P_0^*(T)$ is the market instantaneous forward rate. The first term is the slope of the forward curve scaled by $1/\kappa$, the second is a level match to the forward itself, and the third is a maturity-dependent convexity adjustment that saturates at $\sigma^2/(2\kappa^2)$ for large T. In the $\kappa \to 0$ limit, $\kappa\,\theta(T)$ — not $\theta(T)$ itself — must remain finite (see §12.5.4): multiplying (12.52) through by κ gives $\kappa\theta(T) = \partial_T f_0(T) + \kappa f_0(T) + (\sigma^2/2\kappa)(1 - e^{-2\kappa T})$, and taking $\kappa \to 0$ the middle term vanishes while the last approaches $\sigma^2 T$, so $\kappa\theta \to \partial_T f_0 + \sigma^2 T = \theta_{\mathrm{HL}}$, recovering the Ho-Lee formula (12.42).

Bucket bootstrap. In implementation, (12.52) is rarely used literally because it requires a second derivative of the observed forward curve, which is numerically ill-conditioned. Production implementations parameterise $\theta(t)$ as piecewise constant on the bond-maturity grid, and solve for the parameter values that exactly reproduce observed bond prices. Splitting the integral in (12.51) into bucket contributions with piecewise-constant $\theta^{(m)}$ on $[T_{m-1}, T_m]$:

$$A_{T_k} \;=\; \sum_{m=1}^{k} \theta^{(m)}\,\alpha^{(m,k)}, \qquad \alpha^{(m,k)} \;=\; (T_m - T_{m-1}) \;-\; \frac{e^{-\kappa(T_k - T_m)} - e^{-\kappa(T_k - T_{m-1})}}{\kappa}. \tag{12.53}$$

Solving (12.51) for the current bucket $\theta^{(k)}$ yields the bootstrap recursion:

$$\boxed{\theta^{(k)} \;=\; \frac{T_k\, y_k^* \;-\; r_0\, B_{T_k} \;+\; C_{T_k} \;-\; \sum_{m=1}^{k-1} \theta^{(m)}\,\alpha^{(m,k)}}{\alpha^{(k,k)}}.} \tag{12.54}$$

This is a simple linear bootstrap, exactly analogous to Ho-Lee and to yield-curve stripping: each tenor T_k determines $\theta^{(k)}$ given the already-calibrated earlier buckets. No optimisation, no iteration, no nonlinearity. Just a sequence of divisions.

The $\alpha^{(m,k)}$ are "overlap weights" — how much of bucket m's drift influences the integrated rate up to horizon T_k. They are pure geometry of the exponential kernel and do not depend on market data; production implementations precompute them once per κ and cache. Their key property is causality: $\alpha^{(m,k)} = 0$ for $m > k$, which is what makes the bootstrap triangular. A $\theta^{(m)}$ for $m > k$ has no effect on $P_0(T_k)$ (it refers to rates in the future relative to T_k), so adding tenor T_k determines only $\theta^{(k)}$ and leaves earlier buckets fixed.

Worked example. Take $\kappa = 0.15$, $\sigma = 0.012$, $r_0 = 0.04$, market yields $y_1^* = 0.042$ at $T_1 = 1$ and $y_2^* = 0.045$ at $T_2 = 2$. Compute $B_{T_1} = (1 - e^{-0.15})/0.15 \approx 0.9286$ and $C_{T_1} \approx 3.4 \times 10^{-5}$. With $k = 1$ the sum is empty and $\alpha^{(1,1)} = T_1 - B_{T_1} \approx 0.0714$. Then

$$\theta^{(1)} \;\approx\; \frac{1 \cdot 0.042 - 0.04 \cdot 0.9286 + 3.4\mathrm{e}{-5}}{0.0714} \;\approx\; 0.0691.$$

The calibrated drift target is about 6.9% — well above the current 4% level. This is the "overshoot": mean reversion only partially converges the rate toward $\theta^{(1)}$ over one year ($\mathbb{E}[r_1] = 0.04 \cdot e^{-0.15} + \theta^{(1)}(1 - e^{-0.15}) \approx 0.04 \cdot 0.861 + \theta^{(1)} \cdot 0.139$), so $\theta^{(1)}$ must overshoot the target rate level to achieve the market-implied integrated-rate value.

theta(t) vs market forward curve $\theta(t)$ *sits slightly above the market forward curve because of the convexity adjustment* C_T.

For a trader using this model, the practical consequence is that the calibrated drift schedule $\theta(t)$ should *not* be interpreted as "what the market expects the rate to be at time t." That would only be true in the $\kappa \to 0$ limit. With mean reversion, $\theta(t)$ is instead "what the target has to be so that the expected rate path matches the market's forward curve," which is different. The expected rate path itself is $\mathbb{E}[r_t] = r_0 e^{-\kappa t} + \kappa \int_0^t e^{-\kappa(t-u)}\theta_u\,\mathrm{d}u$, which is smoother than $\theta(t)$.

Philosophical note. By adjusting $\theta(t)$ to match today's curve exactly, Hull-White gives up the possibility of *explaining* the curve in terms of a small number of stationary parameters. The curve is now an input, not a prediction. This is a feature for pricing (you want your model to agree with market quotes) but a bug for economic insight (you can't test your $\mathbb{Q}$-dynamics against curve data you haven't already used to calibrate them). In this sense Hull-White is a member of a broad class of "calibrated" models in finance whose primary virtue is practical rather than explanatory.

There is a deeper parallel to the Black-Scholes world. In equity, the "implied volatility" surface is what the market uses to quote options; it is not a model parameter but a derived quantity. In rates, the forward curve $f_0(\cdot)$ is what the market quotes; $\theta(t)$ is the corresponding derived quantity. Both are "market-implied" model inputs computed from quoted prices via specific formulas.

13.6.6 12.6.5 Practical implementation considerations

The calibration described above is conceptually simple but has several practical wrinkles worth flagging for someone who will actually implement it.

Curve smoothing. The pointwise formula (12.52) requires a first and second derivative of the market forward curve. Real market data (swap quotes, futures, bond prices) are discrete and noisy; naive numerical differentiation will produce erratic $\theta(t)$ that oscillates between buckets and destabilises the pricing engine. Production implementations fit a smooth parametric form to the log- discount curve before differentiating. Common choices include cubic splines (flexible, but can oscillate near endpoints), Nelson-Siegel (four parameters capturing level, slope, curvature, and one shape parameter), Svensson (Nelson-Siegel plus two more parameters for improved curvature fit), and tension splines (controlled smoothness). The choice trades off exact fit against parameter stability; over-smoothed curves produce stale prices, under-smoothed curves produce noisy prices. The sweet spot is chosen based on the desk's risk appetite and is re-evaluated periodically.

Bucket stability. Piecewise-constant $\theta(t)$ has a subtle defect: the drift is discontinuous at bucket boundaries, producing small kinks in the implied forward curve. These kinks are usually invisible in pricing but can show up as spurious "vol spikes" in certain risk metrics. Smoother representations (piecewise linear, cubic splines) avoid the kinks at the cost of more calibration parameters per bucket. A compromise is to use piecewise-constant $\theta(t)$ plus a post-processing smoothing step that redistributes drift mass across adjacent buckets while preserving exact bond-price fit. Some practitioners prefer to parameterise the cumulative integral $\Theta(t) = \int_0^t \theta(u)\,\mathrm{d}u$ directly, which is smoother than θ itself.

Chicken-and-egg (κ, σ, θ) joint calibration. The Hull-White convexity term C_T depends on (κ, σ), so bootstrapping $\theta(t)$ requires knowing these first. But (κ, σ) are typically determined by fitting to caplet and swaption prices, which themselves depend on the already-extracted $\theta(t)$. The standard resolution is iterative: pick initial (κ, σ) from historical estimates; extract $\theta(t)$ via (12.54); fit (κ, σ) to option prices; re-extract $\theta(t)$; iterate until convergence. In practice convergence is fast (a few iterations) because the interaction between $\theta(t)$ and option prices is weak for typical parameters. Damping, regularisation, and convergence diagnostics are standard components of serious implementations.

Instrument set selection. Which market instruments should the model be calibrated to? For vanilla books, the practical answer is: (i) the short-end of the curve from overnight rates and short-dated futures; (ii) the intermediate-dated part from medium-term swaps and futures; (iii) the long end from long-dated swaps; (iv) a strip of ATM caplets or swaptions for (κ, σ). For more exotic portfolios (callable bonds, Bermudan swaptions) additional volatility instruments may be needed. The calibration error across each instrument set is typically monitored as a diagnostic; deviations signal either a model limitation or a data issue.

Real-time re-calibration. Because each calibration step is linear-time, production systems can re-calibrate every few seconds or faster as the market ticks. Whether they *should* is a deeper question: constantly-shifting $\theta(t)$ muddies P&L attribution ("market moved" vs "model re-fit") and creates numerical noise. Most desks re-calibrate at a few natural cadences (morning open, mid-day, end of day, plus on large market moves). The choice of cadence is a trade-off between model freshness and signal-to-noise.

13.7 12.7 Affine term structure and the duality of short-rate models

The word "affine" denotes a very specific class of models with a precise structural signature: the drift and the squared diffusion of the state variable are both linear (affine) functions of the state. Vasicek sits at the simplest end of this class (constant diffusion, linear drift). CIR complicates matters by allowing the diffusion $\sigma\sqrt{r_t}$ to depend on r_t — this makes the squared diffusion $\sigma^2 r_t$ linear in r_t, so the model is still affine but the ODE for B becomes a genuine Riccati equation. Multi-factor extensions replace r_t with a vector state and the scalar ODE for B with a matrix Riccati ODE. In every case the bond price remains exponential-affine in the state vector, and that is the single structural property that makes the entire edifice tractable.

The affine class has been systematically characterised in a celebrated 1996 paper by Duffie and Kan. They showed

that if one posits an exponential-affine bond price $P(t,T,x) = \exp(A(t,T) + B(t,T)^\top x)$ for some state vector x, then the underlying state must have a very restricted form of dynamics: the drift must be affine in x, the instantaneous covariance matrix must be affine in x (with coefficients specifying exactly which linear combinations of state variables can drive which diagonal entries of the covariance), and the structure must be compatible across maturities. Moreover, Duffie and Kan classified the admissible affine models by their "signature" — whether each state variable can take all real values or must be non-negative. Vasicek is a pure "Gaussian" affine model (all state variables unconstrained); CIR is a "square-root" affine model (one state variable positive, with $\sqrt{\cdot}$ diffusion); mixed affine models combine these. The Duffie-Kan classification underlies essentially every multi-factor bond-pricing library in practice.

Outside the affine class, exponential-affine bond prices are impossible: if the dynamics of the state are not affine, the bond price will not have this form. What do you get instead? Typically, semi-closed-form approximations (via characteristic functions, transform inversions, or perturbation expansions), Monte Carlo, or numerical PDE methods. These techniques work, but they are slower and less transparent than the affine closed form. For this reason, most practical rates libraries implement pricing as "affine-first, with plug-ins for non-affine corrections."

13.7.1 12.7.1 The three models in one diagram

The three models derived in this chapter occupy three corners of a small parameter space:

Model	κ	θ	SDE / bond price
Ho-Lee	0	$\theta(t)$	$\mathrm{d}r_t = \theta(t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$; $\ln P_0(T)$ has $+\sigma^2 T^3/6$ convexity
Vasicek	$\kappa > 0$	constant	$\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$; $P_t(T) = e^{A_t(T) - B_t(T) r_t}$ affine
Hull-White	$\kappa > 0$	$\theta(t)$	Vasicek dynamics with θ_t calibrated to market curve

The nesting is transparent: Hull-White $\supset$ Vasicek $\supset$ Ho-Lee, with each inclusion obtained by freezing one more parameter. Vasicek is Hull-White with θ_t held constant; Ho-Lee is Hull-White with $\kappa \to 0$ (and the $\kappa\theta_t$ product held fixed). In production use, Hull-White is almost always the model of choice because it is the only one of the three that fits the observed yield curve exactly.

13.7.2 12.7.2 Duality: three perspectives on the same formula

The bond price $P_t(T) = e^{A_t(T) - B_t(T) r_t}$ can be derived by three independent routes, each shining light on a different aspect:

1. **Direct expectation (§12.3.4, §12.5.3, §12.6.3).** Use the explicit Gaussian solution for r_t, integrate to get the Gaussian law of $\int_t^T r_u\,\mathrm{d}u$, apply the moment-generating function. This route makes the Jensen-convexity origin of the formula transparent and generalises to any Gaussian affine model.
2. **Affine ansatz + PDE (§12.3.5).** Use Feynman-Kac to convert the expectation into a parabolic PDE with source term $-rg$; guess an exponential-affine solution; match coefficients to obtain a pair of ODEs for A and B. This route makes the affine structure of the model explicit and generalises to any affine model including CIR and quadratic-Gaussian.
3. **Martingale verification (discussed but not executed).** Postulate the formula $P = e^{A - Br}$, compute $\mathrm{d}(P/M)$ using Itô, verify that the drift is zero under $\mathbb{Q}$. This route directly confirms the no-arbitrage condition and is the most elegant but requires pre-knowledge of the formula.

That all three converge on the same answer is both a consistency check and a window onto the structural richness of affine models. Each route carries some information the others don't: the direct expectation carries the probabilistic intuition (Jensen, Gaussian MGF); the PDE carries the analytic intuition (elliptic operators, Riccati ODEs); the martingale verification carries the economic intuition (no-arbitrage as martingality). Mastering all three is the difference between pricing a bond and understanding what its price means.

13.7.3 12.7.2.b Historical context and the negative-rates saga

The three models derived in this chapter have a combined history worth a brief reflection. Vasicek (1977) was the first major analytically- tractable short-rate model. Its introduction of the affine framework foreshadowed decades of work that would eventually produce the Duffie- Kan classification and the entire affine-pricing industry. For pedagogical purposes, it is the natural starting point: three parameters, a clean OU solution, a closed-form bond.

Ho and Lee (1986) shifted the emphasis from model structure to model *calibration*: they proposed not a new class of dynamics but a recognition that discrete drift parameters should be tuned to match observed bond prices bucket by bucket. This was a conceptual advance as much as a mathematical one. It reframed modelling as a two-layer endeavour — posit a structure, then calibrate its free parameters to market data — that has dominated quantitative finance ever since.

Hull and White (1990) fused the two: they took Vasicek's mean-reverting structure and Ho-Lee's time-dependent drift, obtaining a model that combined analytical tractability with exact yield-curve calibration. The result was so immediately useful that Hull-White rapidly became the industry default, and it remains the base-case rates model in most pricing libraries today. The Black-Karasinski (1991) and Black-Derman-Toy (1990) alternatives, which work with $\ln r_t$ rather than r_t to enforce positivity, are sometimes preferred for their non-negativity property but pay for it in lost affine structure.

The saga of negative rates deserves attention because it illustrates how modelling orthodoxies get disrupted by market events. For decades the CIR (1985) model — Vasicek-with-$\sqrt{r}$-diffusion — was considered the "principled" choice for rates modelling because it keeps $r_t \geq 0$. Regulators sometimes even required it, on the logic that a model permitting nonsensical outputs must be wrong. Then came 2014: the ECB's deposit facility rate went to -0.10%, then -0.20%, eventually -0.50%; Swiss and Danish rates went deeply negative; Japanese rates followed. Suddenly the "nonsensical" output of Vasicek was the *actual* output of the market, and CIR was stranded on the wrong side of reality, unable to express the current state of the world. Hull-White's Gaussian structure — permitting negative rates as part of the state space — became a feature, not a bug.

The lesson is that modelling choices are not timeless; they reflect economic conditions, and conditions change. A rates quant in 2026 who cannot express a negative interest rate in their model is a rates quant who cannot price a five-year EUR receiver swaption, and that is a crippling limitation. The sensible response is modellers' pluralism: understand multiple models, know which is appropriate in which regime, and be prepared to switch when the world changes. Because the mathematical techniques for affine models are largely shared across the family, the cost of keeping both in one's toolkit is modest; the cost of having only one is potentially catastrophic.

13.7.4 12.7.3 Where this chapter leaves the reader

By the end of §12.6 the reader can:

- Derive Vasicek from first principles, arriving at the bond-price formula (12.20)-(12.22) by both the direct-expectation and PDE routes.
- Extend to Hull-White by letting $\theta \to \theta(t)$, and bootstrap $\theta(t)$ to exactly reproduce an observed yield curve.
- Specialise to Ho-Lee by setting $\kappa \to 0$, with the understanding that the resulting cubic variance growth limits the model to short-dated products.
- Read off the yield curve, forward curve, and convexity adjustment as analytical by-products.

What is *not* in this chapter: applications. The IRS par-rate formula, the Jamshidian bond-option formula, the T-forward measure, callable bonds, CDS, and multi-factor / LMM / SABR all live in Chapter 13 (rate derivatives) and Chapter 15 (caps and swaptions). The division of labour is: this chapter *builds the model*; the next two *use it*.

13.8 12.8 Key Takeaways

1. **Short rates are not traded; bonds are.** The Fundamental Theorem of Asset Pricing applies to tradables, so the risk-neutral measure $\mathbb{Q}$ is defined by its action on M_t and $P_t(T)$, not on r_t. This creates a genuine degree of freedom (the market price of risk λ_t) that must be fixed externally — typically by calibration to observed bond prices.

2. **The master pricing formula is $P_t(T) = \mathbb{E}_t^{\mathbb{Q}}[e^{-\int_t^T r_s \, \mathrm{d}s}]$.** It is model-agnostic — every short-rate model computes the same conditional expectation, differing only in how the expectation is evaluated. Affine models

evaluate it in closed form.

3. **Vasicek: three parameters, one bond-price formula.** The SDE $\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ yields, via either the direct-expectation route or the PDE / affine-ansatz route, the closed-form bond price $P_t(T) = e^{A_t(T) - B_t(T) r_t}$ with $B_t(T) = (1 - e^{-\kappa(T-t)})/\kappa$.
4. **Bonds are convex in rates (Jensen's mountain).** The $+\frac{1}{2}\Sigma_*^2$ term in the Gaussian MGF is the convexity adjustment that makes long bonds worth *more* than naive plug-in discounting suggests. It explains why long yields trade below expected long-run short rates even without a term premium.
5. **Long-bond duration is capped at $1/\kappa$.** This is Vasicek's most economically distinctive prediction: unlike a random-walk rate model, mean reversion means long bonds have finite sensitivity to the current short rate. Any calibrated κ outside the range $[0.1, 0.5]$ is suspect.
6. **Long yield $= \theta - \frac{1}{2}\sigma^2/\kappa^2$.** The long end of the yield curve is *below* the long-run mean of the short rate by a convexity premium. Larger in high-vol or slow-reverting regimes.
7. **Ho-Lee is the $\kappa \to 0$ limit.** Pure random walk with time-varying drift. Bond price has the simple form $\ln P_0(T) = -r_0 T - \iint \theta + \sigma^2 T^3/6$; drift pointwise calibration $\theta_T = -\partial_T^2 \ln P_0^*(T) + \sigma^2 T$. The cubic T^3 variance growth means Ho-Lee breaks at long maturities.
8. **Hull-White = Vasicek with time-varying $\theta(t)$.** Exact yield-curve fit by construction, via a linear triangular bootstrap (12.54) analogous to Ho-Lee's. Two-stage calibration in practice: (κ, σ) fit to swaption vols, then $\theta(t)$ bootstrapped to match bonds.
9. **One-factor is a limitation.** Vasicek/HW can only produce parallel-shift curve moves (up to the $B_t(T)$ loading). Real curve dynamics have three factors (level, slope, curvature); for hedging and pricing curve-twist-sensitive products, move to G2++ or multi-factor affine or LMM.
10. **Affine = tractable.** The structural property "drift linear, squared diffusion linear in the state" is what makes coefficient- matching work in the PDE ansatz, reduces the infinite-dimensional pricing problem to two ODEs, and ultimately gives closed-form bond prices. Ho-Lee, Vasicek, HW, CIR, G2++, and the Duffie-Kan affine family all share this property; Black-Karasinski (log-rate) and stochastic-vol rates models do not.

13.9 12.9 Reference Formulas

13.9.1 Core dynamics

$$\mathrm{d}r_t = \kappa(\theta(t) - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}\widehat{W}_t$$

- \$ (t) = \$ const gives Vasicek; $\kappa = 0$ gives Ho-Lee.

13.9.2 Explicit solution

$$r_T = e^{-\kappa(T-t)}\,r_t + \kappa\int_t^T e^{-\kappa(T-s)}\theta(s)\,\mathrm{d}s + \sigma\int_t^T e^{-\kappa(T-s)}\,\mathrm{d}\widehat{W}_s$$

13.9.3 Distribution of r_T (Vasicek, const θ)

$$r_T \;\sim\; \mathcal{N}\Big(\theta + e^{-\kappa(T-t)}(r_t - \theta),\;\; \tfrac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa(T-t)}\big)\Big)$$

Stationary limit: $\mathcal{N}(\theta, \sigma^2/(2\kappa))$.

13.9.4 Integrated rate (Vasicek)

$$\int_t^T r_s\,\mathrm{d}s \;=\; B_t(T)\,r_t + \theta\big((T-t) - B_t(T)\big) + \tfrac{\sigma}{\kappa}\int_t^T \big(1 - e^{-\kappa(T-s)}\big)\,\mathrm{d}\widehat{W}_s$$

Gaussian with mean and variance in (12.16)-(12.18).

13.9.5 Vasicek affine bond price

$$P_t(T) = e^{A_t(T) - B_t(T)\, r_t}, \qquad B_t(T) = \frac{1 - e^{-\kappa(T-t)}}{\kappa}$$

$$A_t(T) = -\theta\big((T-t) - B_t(T)\big) + \tfrac{\sigma^2}{2\kappa^2} \int_t^T \big(1 - e^{-\kappa(T-s)}\big)^2 \, \mathrm{d}s$$

13.9.6 Affine ansatz ODE system

$$\begin{cases} \dot{A}_t - \kappa B_t\, \theta(t) + \frac{1}{2}\sigma^2 B_t^2 = 0 \\ \dot{B}_t - \kappa B_t + 1 = 0 \end{cases} \qquad A_T = B_T = 0$$

13.9.7 Vasicek yield curve

$$y_t(T) = \frac{B_t(T)\, r_t - A_t(T)}{T - t}$$

Limits: $y_t(t) = r_t$; $y_t(\infty) = \theta - \frac{1}{2}\sigma^2/\kappa^2$.

13.9.8 Ho-Lee bond price

$$\ln P_0(T) \;=\; -r_0\, T \;-\; \int_0^T \int_0^u \theta_s \, \mathrm{d}s \, \mathrm{d}u \;+\; \frac{\sigma^2 T^3}{6}$$

13.9.9 Ho-Lee calibration identity

$$\theta_T \;=\; -\,\partial_T^2 \ln P_0^*(T) \;+\; \sigma^2 T \;=\; \partial_T f_0(T) + \sigma^2 T$$

13.9.10 Hull-White bond price

$$P_0(T) = \exp\{-r_0\, B_T - A_T + C_T\}$$

$$B_T = \frac{1 - e^{-\kappa T}}{\kappa}, \quad A_T = \int_0^T \theta_u (1 - e^{-\kappa(T-u)}) \, \mathrm{d}u, \quad C_T = \frac{\sigma^2}{2\kappa^2} \int_0^T \big(1 - e^{-\kappa(T-u)}\big)^2 \, \mathrm{d}u$$

13.9.11 Hull-White pointwise calibration

$$\theta(T) \;=\; \frac{1}{\kappa}\, \partial_T f_0(T) + f_0(T) + \frac{\sigma^2}{2\kappa^2}(1 - e^{-2\kappa T})$$

13.9.12 Hull-White bucket bootstrap

$$\theta^{(k)} \;=\; \frac{T_k\, y_k^* - r_0\, B_{T_k} + C_{T_k} - \sum_{m=1}^{k-1} \theta^{(m)}\, \alpha^{(m,k)}}{\alpha^{(k,k)}}$$

with

$$\alpha^{(m,k)} \;=\; (T_m - T_{m-1}) \;-\; \frac{e^{-\kappa(T_k - T_m)} - e^{-\kappa(T_k - T_{m-1})}}{\kappa}.$$

Model	κ	θ	Long-run $\mathbb{V}[r_T]$	Convexity C_T as $T \to \infty$

13.9.13 Three-model nesting

Model	κ	θ	Long-run $\mathbb{V}[r_T]$	Convexity C_T as $T \to \infty$
Ho-Lee	0	$\theta(t)$	$\sigma^2 T$	$\sigma^2 T^3/6$ (cubic)
Vasicek	$\kappa > 0$	const	$\sigma^2/(2\kappa)$	bounded growth
Hull-White	$\kappa > 0$	$\theta(t)$	$\sigma^2/(2\kappa)$	$\sigma^2 T/(2\kappa^2)$ (linear)

13.9.14 Half-life

$$\tau_{1/2} = \frac{\ln 2}{\kappa}$$

Typical calibrated $\kappa \in [0.1, 0.5]$ corresponds to rate-shock half-lives of 1.4 to 7 years.

14 CH13 — Rate-Derivative Applications: Swaps, CDS, Bond Options, Callables

14.1 13.1 From short-rate dynamics to products that depend on them

CH12 built the Vasicek, Ho-Lee, and Hull-White short-rate models — the SDEs, the integrated-rate distributions, the closed-form bond prices, the affine-ODE system, the calibration of θ_t to today's curve. All of that machinery was developed with one pay-off in mind: *use it to price rate-dependent products.* This chapter is that pay-off. We take the calibrated short-rate model from CH12 as given and work outward to the four product classes that dominate rates desks: **interest-rate swaps (IRS)**, **callable bonds**, **credit default swaps (CDS)**, and **European bond options** (together with their capstone application, swaptions via the Jamshidian decomposition). We close with a tour of where the single-factor Gaussian framework starts to break and which extensions (HJM, LMM, SABR, multi-factor Gaussian) handle the residual.

The chapter is organised by how much of the short-rate model each product actually uses. Swaps are almost model-free: only the discount curve enters, so Vasicek versus Hull-White versus any other calibrated-curve model gives the same swap value. Callable bonds are genuinely model-dependent: the embedded American option on a bond requires a backward induction on a short-rate tree, and both the level of rates and the variance of the short rate matter. CDS depend on the calibrated discount curve plus a separate hazard-rate curve; the two combine into a risky discount factor. European bond options under Vasicek/Hull-White admit a clean closed-form (the Jamshidian / Vasicek-Black formula), and the derivation is the canonical application of the **T-forward measure** — a specific change of numeraire that makes the forward bond price a martingale with deterministic volatility. Crucially, we will *not* re-derive Girsanov's theorem or the T-forward measure here: CH05 already did that once. This chapter cites the result from CH05 and uses it.

A roadmap to the rest of the chapter. §13.2 prices an IRS and extracts the par swap rate — a model-free calculation that exercises only the discount curve. §13.3 tackles callable bonds on a Hull-White trinomial tree, introducing the backward-induction-plus-early-exercise pattern that appears again and again in rate-derivative pricing. §13.4 handles CDS with constant and time-varying hazard, deriving the "credit triangle" $S \approx (1-R)\lambda$ and its corrections. §13.5 takes a visual pause on the yield surface — a 3D object that is the natural state of the rates-trading book. §13.6 derives the Vasicek-Black bond-option formula via the T-forward measure, citing CH05 for the Girsanov apparatus. §13.7 walks through worked examples of that formula and introduces the Jamshidian decomposition that reduces a swaption to a portfolio of bond options. §13.8 previews where we go beyond single-factor Vasicek: the multi-factor Gaussian G2++, HJM, LMM, and SABR. §13.9 closes with the PDE-versus-Monte-Carlo trade-offs that dictate which numerical method a rates desk deploys per product. §13.10 summarises the chapter; §13.11 is the reference-formula appendix.

The conceptual thread running through all five products is this: **every rate derivative is a functional of the future yield curve, and pricing it reduces to computing a conditional expectation of that functional under some convenient martingale measure**. The specific measure depends on when the payoff settles and what numeraire most naturally absorbs its stochastic discount factor. For swaps we use no measure machinery at all — the linearity-in-rates of the payoff lets us replicate and price pathwise. For CDS we use $\mathbb{Q}$ with a hazard-rate extension. For bond options we use the T-forward measure. For swaptions we use the annuity measure. Each choice is motivated by the same principle: *match the numeraire to the payoff's natural scale, and the pricing collapses to a Black-type lognormal expectation.* That principle — one sentence — is the take-away of the entire chapter.

One notational convention before we start. All bond prices $P_t(T)$, affine coefficients $A_t(T)$ and $B_t(T)$, and the short-rate parameters $(\kappa, \theta_t, \sigma)$ are as in CH12. The $\mathbb{Q}$-Brownian motion is $\widehat{W}_t$; the T-forward Brownian motion (introduced in §13.6 via CH05 cross-reference) is $\widetilde{W}_t^T$. The discount factor from t to T under the money-market numeraire is $M_t/M_T = e^{-\int_t^T r_u \, \mathrm{d}u}$; we use $P_t(T)$ when we want the deterministic (time-t-conditional expectation) bond price.

14.2 13.2 Interest-rate swaps and the par swap rate

14.2.1 13.2.1 Swap cash flows and definitions

A plain-vanilla interest-rate swap exchanges a fixed leg against a floating leg on a notional N at reset dates $t_0 < t_1 < \cdots < t_n$, with day-count $\Delta t_k = t_k - t_{k-1}$. Counterparty A pays fixed coupon F at each t_k for $k = 1, \ldots, n$ and receives a floating coupon ℓ_k that was *set in arrears* at the prior reset date t_{k-1} (this is the standard "fix at t_{k-1}, pay at t_k"

convention). Counterparty B is the mirror image: receives fixed, pays floating. There is **no exchange of notional** — only the periodic coupon differences — which is why swap-market notional figures (hundreds of trillions of dollars of cleared IRS outstanding) vastly exceed the bond market.

```
            Fixed coupon F (annualised, × Δt_k · N)


        A                                       B


            Floating coupon _k   (set at t_{k-1}, paid at t_k)

  A = fixed-payer (long the "pay-fixed / receive-float" leg).
  B = fixed-receiver (mirror image). Notional N never changes hands -
  only the netted coupon difference (F - _k)·Δt_k·N flows each period.
```

The payment schedule lives on the interval grid $t_0 < t_1 < \cdots < t_n$ with "first-reset" at t_0 (rate fixed there) and "first-payment" at t_1; subsequent coupons at $t_2, \ldots, t_n$.

Why swaps exist. An IRS converts a floating liability into a fixed one (from the fixed-payer's viewpoint) or vice versa. A corporate treasurer who borrowed at 3-month SOFR + spread can lock in a fixed funding cost by paying fixed / receiving floating on a swap — the floating receipts offset the floating loan interest, leaving only the fixed swap payment as net cost. A bank with fixed-rate mortgages in its book can hedge the rate risk by receiving fixed / paying floating — now the bank's net position is floating in/out, matching its floating-rate funding. Swaps are the workhorse instrument of interest-rate risk management, and this dual use (corporate hedging vs bank asset-liability management) explains their enormous scale.

To put the size in perspective: global GDP is around \$100 trillion; global bonds outstanding around \$130 trillion; cleared IRS notional is roughly \$400–500 trillion with another \$100+ trillion in OTC non-cleared IRS. Swaps are the single most actively used derivative class by any measure. Even a 0.1 basis-point mispricing on a trillion-dollar swap portfolio is worth \$10 million, so the pricing machinery matters.

14.2.2 13.2.2 Pricing from the discount curve

Fixed-leg value. Paying $F \cdot N \cdot \Delta t_m$ at each t_m has time-0 value

$$V_0^{\text{fix}} \;=\; \sum_{m=1}^{n} \Delta t_m \, F \, N \, P_0(t_m) \tag{13.1}$$

The factor $\sum_{m=1}^{n} \Delta t_m P_0(t_m)$ is the **annuity** (or PVBP per unit rate). At time t before the swap matures, the fixed leg value is $V_t^{\text{fix}} = \sum_m F \Delta t_m P_t(t_m)$.

The annuity interpretation is worth dwelling on. Think of $\mathcal{A}_0 \equiv \sum_m \Delta t_m P_0(t_m)$ as the present value of a stream of \$1 payments per year, on the coupon schedule, discounted to today. It functions as the "unit of account" for swap rates: the par swap rate F is quoted as the number such that $F \cdot \mathcal{A}_0$ equals the floating-leg value. Multiplying or dividing by the annuity converts between dollars and per-annum rate units. The annuity is also the natural numeraire for changing measure in swaption pricing: under the "annuity measure" (the probability measure under which the annuity is the numeraire), the par swap rate S_t becomes a martingale, and the swaption collapses to $V_0 = \mathcal{A}_0 \cdot \mathbb{E}^{\mathcal{A}}[(S_T - K)_+]$ — a direct lognormal expectation in Black form.

Floating-leg value via "buy \$1 and roll". The floating coupon ℓ_k is locked in at each reset t_{k-1} from the simple yield on a Δt_k-bond:

$$P_{t_{k-1}}(t_k) \;=\; (1 + \Delta t_k \, \ell_k)^{-1} \;\Longrightarrow\; \ell_k \;=\; \frac{1}{\Delta t_k}\left(\frac{1}{P_{t_{k-1}}(t_k)} - 1\right). \tag{13.2}$$

The reset mechanism works like this: at each reset date t_{k-1}, the market quotes a simply-compounded rate for the upcoming period $[t_{k-1}, t_k]$, and that rate is fixed as the floating coupon for that period. The coupon $\ell_k \Delta t_k N$ is then paid at t_k. So floating coupons are *set in arrears*, meaning the k-th coupon is known at the $(k-1)$-th reset and

paid at t_k. Standard LIBOR (and its SOFR successor) follow this "set-in-advance, paid-in-arrears" convention, and (13.2) is the algebraic identity that makes replication of the floating leg possible.

Claim: $V_t^{\text{fl}} = (P_t(t_0) - P_t(t_n))N$.

Financial argument (self-financing replication). Consider the trader who at t_0 is handed \$1 and asked to replicate the floating-leg coupons:

- At t_0: buy \$1-worth of t_1-bonds — holdings $1/P_{t_0}(t_1)$. These mature to $\$1/P_{t_0}(t_1)$ at t_1.
- At t_1: take \$1 of that and buy t_2-bonds: $1/P_{t_1}(t_2)$ units; pocket the remainder $(1/P_{t_0}(t_1) - 1) = \ell_1 \Delta t_1$.
- Continue to t_n, pocketing $\ell_k \Delta t_k$ each period and holding \$1 of fresh short-dated bonds.

Let us verify with numbers. Suppose $P_{t_0}(t_1) = 0.98$, so at t_0 we buy $1/0.98 \approx 1.0204$ units of t_1-bond. At t_1 these pay $1.0204 \cdot \$1 = \$1.0204$. Take \$1 to reinvest, pocket \$0.0204. The floating coupon is $\ell_1 \Delta t_1 = (1/0.98 - 1) \approx 0.0204$ — match. The surplus is precisely the first floating coupon. No model, no arbitrage argument about distributions, just the identity "a bond maturing to par grew at the rate ℓ over its tenor."

The cash-flows generated by this self-financing strategy are exactly the floating-leg coupons, plus a terminal \$1 repayment that we return. The cost of the strategy is the \$1 received at t_0 minus the \$1 returned at t_n, priced today:

$$V_0^{\text{fl}} \;=\; (P_0(t_0) - P_0(t_n))\, N \tag{13.3}$$

This is a striking fact: the floating leg is **model-free**. Neither κ nor σ nor $\theta(t)$ enters (13.3). The randomness cancels pathwise in the replication strategy. Contrast with a cap or swaption, where the payoff is a max function and no such cancellation occurs. The distinction "linear in rates (swap) vs nonlinear (cap)" is what separates model-free from model-dependent products.

Post-2008, a practical caveat: (13.3) assumes the floating index perfectly tracks the discount curve. Pre-2008 LIBOR contained credit and liquidity premium that violated this assumption, so modern swap pricing uses "OIS discounting" (discounting at the overnight index rate) while accruing floating coupons at SOFR or LIBOR. This creates a "tenor basis" that requires a separate adjustment. Interpreting $P_0(t)$ as the OIS discount factor keeps (13.3) approximately correct in a multi-curve world.

14.2.3 13.2.3 The par (fair) swap rate

Setting $V_0^{\text{fix}} = V_0^{\text{fl}}$ gives the rate $F^* \equiv S_0$ that makes the swap worth zero at inception:

$$\boxed{F \;=\; \frac{P_0(t_0) - P_0(t_n)}{\displaystyle\sum_{m=1}^{n} \Delta t_m\, P_0(t_m)} \;\equiv\; S_0 \quad \text{(initial par swap rate)}} \tag{13.4}$$

The denominator is the annuity. Since both legs are priced off the discount curve alone, the par swap rate is a ratio of discount factors — **no model required**.

The par swap rate S_0 is the market's standard quote for "the n-year swap rate." When a Bloomberg screen shows "10y USD swap is 3.5%," that 3.5% is S_0 from (13.4) with n chosen to give a 10-year maturity. The swap curve — the set of par swap rates across tenors — is directly observable and widely used as a reference interest-rate curve for mortgages, corporate borrowing, and internal bank funding.

Swap rate ≡ par coupon yield. Up to convexity corrections, S_0 equals the par coupon yield of a bond with the same tenor. A par coupon bond with coupon c has $\text{PV} = c \cdot \mathcal{A}_0 + P_0(t_n)$, which equals par (notional) exactly when $c = (1 - P_0(t_n))/\mathcal{A}_0 = S_0$. So "a swap is a bond with no principal" — the coupon-rate structure is identical; only the principal exchange differs.

Forward-start / ongoing swap rate. For a swap with payment dates $\{t_\ell\}_{\ell: t_\ell > t}$ viewed at time t (some resets already happened), the current fair-value swap rate is

$$S_t \;=\; \frac{P_t(t_{k_t}) - P_t(t_n)}{\displaystyle\sum_{m=k_t+1}^{n} \Delta t_m \, P_t(t_m)} \tag{13.5}$$

where k_t is the index of the most recent reset before t. If $t < t_0$ (spot-start), $P_t(t_{k_t}) = P_t(t_0)$ and (13.5) reduces to (13.4) with P_t replacing P_0.

Because discount factors are stochastic, S_t is a stochastic process. Plotting it alongside the short rate r_t reveals very different dynamics: S_t is smooth and moves with the yield curve, while r_t is volatile and determines the floating coupons. In a sense, S_t is a low-pass-filtered version of r_t, weighted toward the long end of the curve. Empirically, the 10-year swap rate might move 5 bp on a given day while the overnight rate moves 25 bp — a 5:1 ratio, matching the Hull-White prediction that long-tenor yields move less than short-tenor yields in response to short-rate shocks.

14.2.4 13.2.4 Mark-to-market and swap DV01

For off-par swaps (where the fixed coupon F differs from S_0), the swap has non-zero initial PV. If $F = 4.00\%$ versus a par of $S_0 = 4.49\%$, the fixed-payer pays below market so the swap has positive PV to the fixed-payer:

$$V_0^{\text{fp}} \;=\; (S_0 - F) \cdot \mathcal{A}_0 \cdot N \tag{13.6}$$

In the example, $V_0^{\text{fp}} = (4.49\% - 4.00\%) \cdot 2.7532 \cdot 100\text{M} \approx \$1.35\text{M}$. This is the "off-market adjustment" paid upfront in a typical trade. Market convention is to trade at-par (no upfront) for most swaps; off-market swaps arise for customised deals or for collapsing multiple existing positions into a single trade.

DV01. The "01" (pronounced "oh-one" or DV01) is the change in PV for a 1-bp parallel shift in rates. For a receiver swap, DV01 $\approx \mathcal{A} \times 0.0001 \times N$. For a 10-year \$100M swap with annuity $\mathcal{A} = 8.5$:

$$\text{DV01} \;\approx\; 100\text{M} \times 8.5 \times 0.0001 \;=\; \$85{,}000 \text{ per basis point.}$$

This is the single most important risk metric for IRS trading, and it is directly the annuity-weighted notional. DV01 decomposes by tenor ("bucket DV01") to reveal which parts of the yield curve the portfolio is exposed to. Hedging a swap portfolio requires matching the bucket DV01s via positions in benchmark instruments (typically government-bond futures, short-end deposits, and other swaps at standard tenors). Mismatched DV01s lead to residual P&L as the yield curve reshapes.

Related quantities: BPV (basis-point value, same as DV01), PV01 (present value of a 1-bp change, same), 01 (trade-floor shorthand). The important distinction is between "parallel-shift" DV01 (all tenors up 1 bp together) and "bucket" DV01 (only one tenor moves 1 bp). Parallel DV01 is a portfolio-level summary; bucket DV01 is the detailed breakdown needed for hedging.

14.2.5 13.2.5 Worked example: a 3-year par swap

With $P_0(1) = 0.9592$, $P_0(2) = 0.9177$, $P_0(3) = 0.8763$ and $\Delta t_k = 1$:

$$S_0 \;=\; \frac{1 - 0.8763}{0.9592 + 0.9177 + 0.8763} \;=\; \frac{0.1237}{2.7532} \;\approx\; 4.49\%.$$

The 3-year discount falls from 1.0 at $t = 0$ to 0.8763 at $t = 3$, so roughly 12.4 cents are "lost" to discounting over three years. Spread over three annual coupons with annuity 2.75, that works out to 4.49% per annum. This exactly matches the yield-to-maturity of a 3-year par coupon bond with coupon 4.49% — the swap rate *is* the par coupon yield.

Cross-check via zero yields. $P_0(1) = 0.9592 = e^{-y_1}$ gives $y_1 = -\ln 0.9592 \approx 4.17\%$. Similarly $y_2 \approx 4.30\%$, $y_3 \approx 4.40\%$. So the zero-coupon curve is 4.17% at 1y, 4.30% at 2y, 4.40% at 3y — gently upward-sloping. The par swap rate (4.49%) is slightly above the 3-year zero yield (4.40%) because of the convexity implicit in the coupon structure: a coupon bond receives coupons as well as principal, and because coupons are smaller than the principal,

their yield must be slightly higher to equate PVs. This "convexity premium" between coupon bonds and zero-coupon bonds is a standard bond-math result.

IRS cashflow ladder *Fixed-payer sees outgoing blue fixed coupons and incoming orange floating coupons; net PV is zero at the par swap rate S_0.*

14.3 13.3 Callable bonds on a short-rate tree

14.3.1 13.3.1 Callable bond cash flows

IRS pricing is essentially model-free, but a **callable bond** is a genuinely model-dependent rate structure and the canonical application of Hull-White numerical pricing. A callable bond is a fixed-coupon bond that the issuer has the right (but not obligation) to redeem early at a specified call price on one or more call dates. It is the bond-market equivalent of a mortgage with a prepayment option — the issuer can refinance into a lower rate if rates fall.

Economically, a callable bond equals a straight (non-callable) bond minus the issuer's call option:

$$V_t^{\text{callable}} = V_t^{\text{straight}} - V_t^{\text{embedded call}}. \tag{13.7}$$

The issuer benefits if rates fall (they refinance at lower rates); the bondholder loses the upside. So callable bonds trade at a *higher* yield than comparable straight bonds, with the spread representing the option value. A 10-year callable with a 5-year first call might trade 30–80 bps above a comparable 10-year straight, with the spread widening when rate volatility is high (because a more volatile world makes the call option more valuable).

The yield differential is called the **option-adjusted spread (OAS)**. It is the spread earned per year for being "short the call" as a bondholder. If OAS is 50 bps, you are paid 50 bps per year to bear the call-risk exposure. A useful way to think about the position: start with a straight bond with clean duration-based pricing. Add in the issuer's right to call (subtract the call option value). Now the bond's price is lower and its *effective* duration is lower — because the bond will likely be called early in a falling-rate scenario, shortening its life. The **option-adjusted duration (OAD)** captures this shortened effective life and is the correct hedge quantity for managing the interest-rate risk of a callable-bond portfolio.

Pricing requires evaluating a max function at each call date:

$$V_{t_c}^{\text{call}} = \min\big(\text{call price},\ V_{t_c}^{\text{continuation}}\big). \tag{13.8}$$

The issuer will call if the continuation value exceeds the call price — they pay off the bond for less than its market value. This is **American-exercise optionality**, embedded in a rate-derivative structure. No closed form exists in general; we must evaluate numerically via a tree, lattice, or Monte Carlo with regression-based exercise.

14.3.2 13.3.2 Backward induction on a Hull-White trinomial tree

The standard approach is a Hull-White trinomial lattice. Build a tree discretising r_t at each time step t_n, with transition probabilities calibrated to match the Hull-White moments. At each call date, apply (13.8) at each node. Then roll back to t_0 with standard backward induction. The tree gives a complete price and, as byproducts, the OAS (the yield spread over the discount curve that equates the tree price to the market price) and the option-adjusted duration and convexity.

Construction sketch. At each time t_n, the short-rate state is discretised to $r_n^{(i)}$ for $i = -m, \dots, +m$ (the tree has $2m + 1$ possible rate states). Transition probabilities to the three successor states (up, middle, down) at t_{n+1} are calibrated to match the first two conditional moments of $r_{n+1} \mid r_n$ under the Hull-White SDE. The mean reversion is handled by tweaking the *geometry* of the tree near the mean: at nodes where the standard branching would require negative probabilities, the branching is tilted (up-up-middle instead of up-middle-down, or the mirror) so all probabilities stay in $[0, 1]$. This is the Hull-White (1994) trinomial scheme — robust, fast-converging, and standard in every fixed-income pricing library.

A key feature is that the scheme handles the time-dependent $\theta(t)$ exactly: at each time step, the tree's drift is set to match the calibrated $\theta(t)$ at that time. So the tree automatically incorporates the full curve-fit from the θ bootstrap from CH12, without any additional work. This is the computational payoff of the affine structure: calibration and pricing both leverage the same underlying Gaussian geometry.

Backward-induction algorithm. Let $V_n^{(i)}$ denote the callable-bond value at node (n, i). Work backward from maturity T:

1. **Terminal condition** ($n = N$): $V_N^{(i)} = \text{face} + \text{final coupon}$ at every node.
2. **Continuation value** ($n < N$): discount the expectation of the three successor values:

$$V_n^{(i),\text{cont}} \;=\; e^{-r_n^{(i)}\Delta t}(p_u V_{n+1}^{(i+1)} + p_m V_{n+1}^{(i)} + p_d V_{n+1}^{(i-1)}) + c_n$$

   where c_n is the coupon paid at t_n (if any).
3. **Call test** (if t_n is a call date): apply $V_n^{(i)} = \min(\text{call price}_n,\ V_n^{(i),\text{cont}})$. Otherwise $V_n^{(i)} = V_n^{(i),\text{cont}}$.
4. Continue to $n = 0$; the callable price is $V_0^{(0)}$ at the root.

The algorithm is structurally identical to American-option pricing on a stock tree, with the discount factor now state-dependent via $r_n^{(i)}$ rather than a constant r. Each step is $O(\text{nodes at that time})$; total cost is $O(N \cdot m)$ in space and time, which is very fast even for dense calendars (quarterly calls over 30 years).

14.3.3 13.3.3 Decomposition into straight bond minus embedded call

The accounting identity (13.7) can be verified numerically by running the backward induction twice: once with the call test enabled (giving the callable price), and once with the call test disabled (giving the straight bond price). The difference is the value of the embedded call option, which the issuer is short and the bondholder is long (as seen from the issuer's liability side).

This decomposition is useful for *Greeks*. The straight bond's Greeks (duration, convexity) are well understood. The embedded call's Greeks are what make callables complex — and can be extracted from the tree by finite-differencing the call-option value with respect to the underlying rate curve, volatility, or time. The net callable Greeks are then the straight-bond Greeks minus the call-option Greeks, mirroring the value identity.

Negative convexity. A defining feature of callable bonds is that OAD is *non-monotonic* in the underlying rate: near the call strike, OAD is low (because the bond is expected to be called soon), while far from the call strike, OAD approaches the straight-bond duration. This concavity in OAD is the "negative convexity" of callable bonds. It creates two issues:

1. A portfolio manager holding callable bonds has duration that *falls* when rates fall — the opposite of a straight bond, whose duration also falls when rates fall but by less. This asymmetric duration makes hedging callable bonds tricky: a hedge calibrated for one rate environment may not work in another. Dynamic rebalancing is required, and the cost of this dynamic hedging eats into the OAS the portfolio earns.
2. Market-wide, when a large pool of callable-like debt (mortgage-backed securities) experiences a rate drop, the implied duration of the pool suddenly shortens. Fixed-duration mandates (pension funds, insurance companies) have to buy more duration to compensate, typically via Treasury purchases. This "duration-chasing" creates feedback loops in the rate market and can amplify rate moves.

14.3.4 13.3.4 Numerical example

Take a 10-year bond with 5% annual coupon, face value \$100, callable annually at par (\$100) after year 3. Initial rate $r_0 = 3\%$, Hull-White parameters $\kappa = 0.1$, $\sigma = 0.015$, flat yield curve at 3% for simplicity. Build a trinomial tree of r values at each year $0, 1, \ldots, 10$ with transition probabilities calibrated to Hull-White. At year 10, each node has $V = \$105$ (principal + last coupon). Roll back year-by-year, testing the call condition at years 3 through 9.

For parameters consistent with moderate vol, the callable bond might price at \$105.20, compared to a straight bond at \$108.30. The embedded call has value \$3.10, representing roughly 3% of bond value. As σ increases the call becomes more valuable and the callable bond price falls further below the straight. This is the option-value sensitivity: callable bonds are **short vega**, because they hold a short call option.

Interpretation of the price: the callable's yield-to-worst is about 4.9% (compared to a straight-bond yield of 4.1%), so the callable holder earns an extra 80 bps per year for bearing the call risk. In a world where rates stay high, the issuer will not call (continuation value stays below par), and the callable holder earns all the coupons. In a world where rates fall, the issuer calls at par, and the holder loses the future coupon stream. The 80 bps spread is the probabilistically-weighted compensation for bearing this downside.

Related products. *Putable bonds* are callable bonds with swapped counterparties: the *holder* owns the put option. Pricing is analogous — backward induction with a put test at each decision node — and the holder earns a yield

lower than a comparable straight bond (because they own the put option value). *Convertible bonds* are fixed-coupon bonds the holder can convert into a specified number of issuer shares; pricing requires a joint equity/rate model but the core machinery is the same. *Mortgage-backed securities (MBS)* are the largest class of callable-like structures by notional (US residential MBS exceeds $10T). Residential mortgages have an embedded prepayment option economically similar to a callable bond, but the "rational-exercise" Hull-White model overprices prepayment because borrowers are slow to refinance. MBS pricing uses empirical prepayment models (SMM, PSA curves) on top of the lattice; the backward-induction skeleton remains the same.

Calibration consistency. The calibrated (κ, σ) from the swaption market (see §13.6–§13.7 below) should be used for callable pricing. If these parameters are too low (understating vol), the embedded call is undervalued and the callable bond is overvalued — leading to trades that look attractive on screen but are actually losers. Getting these parameters right is the bread and butter of callable-bond trading.

Bridge to §13.4. Before developing CDS, note the structural parallel: just as a callable bond is a straight bond minus an option, a corporate bond is a riskless bond minus a default option (the firm defaults, the holder loses the remaining coupons and principal). CDS prices the default option explicitly, and the same discounting machinery — Hull-White or its equivalents — supplies the rate side.

14.4 13.4 Credit default swaps

14.4.1 13.4.1 Default intensity, survival, and the protection leg

Setup. Protection buyer C pays periodic premium S (annualised spread) to seller A on the reference name B until default time τ. If $\tau \leq t_n$, the seller pays loss-given-default $(1-R)\,N$ at (or just after) τ, where R is the recovery rate and N the notional. We model the default time as the first jump of a Cox process; for this chapter $\tau \sim \mathrm{Exp}(\lambda)$ with constant intensity λ, independent of the short rate, unless noted otherwise.

```
        premium  S · Δt_k · N  (until default or maturity)
C                     A

        contingent leg  (1 - R) · N       paid iff    t_n
                                          (  = default time of ref. B)


              reference name     ← defaults at   ~ Exp( )
                    B


C = protection buyer;  A = protection seller.
C pays a stream of spreads; A pays loss-given-default once, at  .
```

Economic picture. A CDS is insurance against default. The protection buyer pays a steady premium — expressed as an annualised spread S times the notional — in exchange for a payout that arrives if and only if the reference entity defaults before maturity. This is economically equivalent to "shorting the credit of the reference name": a long position in the bond plus a purchased CDS has its default risk hedged out, leaving only rate-sensitivity exposure.

Concrete example. Own $100M of "Acme Corp" bonds and want to eliminate default risk. Buy $100M CDS protection on Acme for 5 years. Pay $100M × spread (say 200 bps = 2%) per year quarterly, for five years or until Acme defaults. If Acme stays solvent, keep paying, and at year 5 the CDS expires worthless. If Acme defaults (say in year 3), the CDS seller pays $(1-R)\cdot\$100\text{M}$ where R is the recovery rate on Acme's debt (typically 40% for senior unsecured). You receive $60M at default, compensating for the $60M loss on your bond (now trading at $40M — the recovery). Net: you are insured.

Intensity, survival, and the hazard interpretation. The intensity λ has a sharp probabilistic meaning: for an infinitesimal time interval $\mathrm{d}t$, the probability of defaulting in $[t, t+\mathrm{d}t]$ given no prior default is exactly $\lambda\,\mathrm{d}t$. So λ is the instantaneous default rate, the **hazard rate**. In a flat-intensity world, the survival probability is

$$\mathbb{Q}(\tau > t) \;=\; e^{-\lambda t} \tag{13.9}$$

and the default distribution is exponential with mean $1/\lambda$. A typical investment-grade bond might have $\lambda \approx 0.005$

per year (50 bps hazard, 200-year expected survival); a junk bond might have $\lambda \approx 0.05$ (5% hazard, 20-year expected survival).

The hazard-rate approach is a direct analog of the instantaneous-rate approach to interest-rate modelling. Both use "intensity" as the fundamental quantity. Both integrate the intensity to get the cumulative quantity (integrated rate $\to$ discount factor; integrated hazard $\to$ survival probability). Both admit time-varying intensities (Hull-White $\theta(t)$ for rates; stochastic hazard $\lambda(t)$ for credit) for realistic calibration.

14.4.2 13.4.2 Premium and default legs; the par CDS spread

Premium leg. Premium payments $S \cdot \Delta t_\ell \cdot N$ are paid on the schedule up to (and including the accrual period containing) default:

$$V_0^{\text{prem}} \;=\; S \cdot N \cdot \sum_{\ell=1}^{n} \Delta t_\ell \, \mathbb{E}^{\mathbb{Q}}\Big[\mathbf{1}_{\{\tau > t_\ell\}} \, e^{-\int_0^{t_\ell} r_u \, \mathrm{d}u}\Big] \;=\; S \cdot N \cdot \sum_{\ell=1}^{n} \Delta t_\ell \, \widetilde{P}_0(t_\ell) \tag{13.10}$$

where $\widetilde{P}_0(t_\ell) = e^{-\lambda t_\ell} P_0(t_\ell)$ is the **risky discount factor** (survival $\times$ riskless ZCB). The sum $\sum_\ell \Delta t_\ell \widetilde{P}_0(t_\ell)$ is the **risky annuity**. Under independence of hazard and rate under $\mathbb{Q}$, the expectation factorises cleanly: otherwise $\widetilde{P}_0(t_\ell) = \mathbb{E}^{\mathbb{Q}}[e^{-\int_0^{t_\ell}(r_u+\lambda_u)\,\mathrm{d}u}]$ which, for joint-Gaussian $r + \lambda$, again admits an affine closed form.

A heuristic: a risky bond earns the "hazard-adjusted rate" $r + \lambda$ instead of just r. The "credit spread" λ is effectively an additional rate that the bond earns to compensate for default risk. A corporate bond with a yield 150 bp above comparable Treasury is earning $r + 0.015$ in a flat-rate, flat-hazard world. The 150 bp is the credit spread, which the credit triangle (below) tells us equals $(1 - R)\lambda$ approximately.

Default leg. At default time τ, the payoff $(1 - R)N$ is paid, discounted to $t = 0$ via the riskless discount to τ. Under independence of τ and r:

$$V_0^{\text{D}} \;=\; \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_0^{\tau} r_s \, \mathrm{d}s} \, (1 - R) \, N \, \mathbf{1}_{\tau \le t_n}\Big] \;=\; (1 - R) \, N \int_0^{t_n} \lambda \, e^{-\lambda \tau} \, P_0(\tau) \, \mathrm{d}\tau. \tag{13.11}$$

The default density $\lambda e^{-\lambda\tau}$ is the probability of defaulting in $[\tau, \tau + \mathrm{d}\tau]$ per unit $\mathrm{d}\tau$; the present value of the loss is $(1 - R)NP_0(\tau)$; the integration bound $[0, t_n]$ reflects that the default leg pays only if default occurs before the contract maturity.

The Markov property of Hull-White means that, conditional on r_τ, the future bond price $P_\tau(T)$ depends only on $(r_\tau, \tau, T, \kappa, \sigma, \theta)$ — not on history. So the post-default discount $P_\tau(T)$ is the Hull-White bond-price formula from CH12 evaluated at shifted origin τ and state r_τ. Under independence of r and λ, default timing is a purely exponential process — it does not care what the rate is doing — so integrating over default times first and then over rate paths gives the same answer as doing them simultaneously. The pricing factorises.

Par CDS spread. Setting $V_0^{\text{prem}} = V_0^{\text{D}}$ at inception gives

$$\boxed{S_0^{\text{CDS}} \;=\; \frac{(1 - R) \displaystyle\int_0^{t_n} \lambda \, e^{-\lambda\tau} \, P_0(\tau) \, \mathrm{d}\tau}{\displaystyle\sum_{\ell=1}^{n} \Delta t_\ell \, \widetilde{P}_0(t_\ell)}} \tag{13.12}$$

Structurally this is (13.4) dressed up with survival probabilities: the numerator is the PV of loss-given-default weighted by the default density, the denominator is the risky annuity. The parallel between (13.4) and (13.12) is not accidental: both are par-rate formulas for insurance-like instruments. The structural similarity "ratio of value-generating leg to annuity-like leg" recurs throughout rates and credit markets.

14.4.3 13.4.3 The credit triangle and worked numbers

The credit triangle. In the limit of low hazard and short maturity (so both survival $e^{-\lambda t}$ and discount $P_0(t)$ are close to 1), the numerator of (13.12) integrates to approximately $(1-R)\lambda t_n$ and the denominator to approximately t_n, so

$$S_0^{\mathrm{CDS}} \;\approx\; (1-R)\,\lambda \quad\Longleftrightarrow\quad \lambda \;\approx\; \frac{S_0^{\mathrm{CDS}}}{1-R}. \tag{13.13}$$

This is the single most important formula in credit trading. Given a quoted CDS spread, extract the implied hazard rate by dividing by the loss-given-default. With $R = 40\%$ and a 100 bp CDS spread: $\lambda \approx 100/(1-0.4) = 167$ bp, implying a 1-year default probability of $1 - e^{-0.0167} \approx 1.65\%$. So a 100 bp CDS is pricing in about 1.65% default odds over the next year, 40% recovery if it happens. Every credit trader internalises the triangle: *CDS spread maps to hazard maps to default probability maps to rating-equivalent.*

Worked numerical example. Take $\lambda = 0.02$ (2% hazard, typical BBB), $R = 0.4$ (40% recovery), flat riskless curve at 3% for $t_n = 5$ years with quarterly coupons ($\Delta t_\ell = 0.25$).

- Risky discount factors: $\widetilde{P}_0(t_\ell) = e^{-(0.03+0.02)t_\ell} = e^{-0.05t_\ell}$.
- Risky annuity: $\sum_{\ell=1}^{20} 0.25 \cdot e^{-0.05\cdot 0.25\ell} \approx 4.42$.
- Default-leg integrand: $\int_0^5 0.02 \cdot e^{-0.05\tau}\,\mathrm{d}\tau \approx 0.0885$. Multiply by $(1-R) = 0.6$: 0.053.
- Par spread: $0.053/4.42 \approx 120$ bp.
- Credit-triangle approximation: $(1-R)\lambda = 0.6 \cdot 0.02 = 120$ bp. Exact match for low hazard.

Try a higher hazard: $\lambda = 0.10$ (distressed). Risky discount factor $\widetilde{P}_0(\tau) = e^{-0.13\,\tau}$. Risky annuity $\sum_{\ell=1}^{20} 0.25 \cdot e^{-0.13\cdot 0.25\,\ell} \approx 3.62$; default-leg integrand $\int_0^5 0.10 \cdot e^{-0.13\tau}\,\mathrm{d}\tau \approx 0.368$, multiplied by $(1-R) = 0.6$ gives default-leg PV ≈ 0.221; par spread $\approx 0.221/3.62 \approx 610$ bp. Credit triangle: $0.6 \cdot 0.10 = 600$ bp. The exact formula gives a slightly higher spread than the triangle (about 10 bp more) because the high-hazard exponential damps the annuity faster than the triangle assumes. This "discounting correction" stays modest at $\lambda = 0.10$ but grows materially for more distressed credits.

14.4.4 13.4.4 Bootstrapping the hazard-rate curve

Real-world hazards vary with the firm's health and the macro cycle. Time- varying hazard $\lambda(t)$ changes the survival formula to $S(t) = \exp(-\int_0^t \lambda(u)\,\mathrm{d}u)$, and the default-leg integral generalises to

$$V_0^{\mathrm{D}} \;=\; (1-R)\,N \int_0^{t_n} \lambda(\tau)\,S(\tau)\,P_0(\tau)\,\mathrm{d}\tau \tag{13.14}$$

Still a one-dimensional quadrature — CDS pricing remains essentially instantaneous on modern hardware.

Typical calibration. Given quoted CDS spreads at standard tenors (1y, 3y, 5y, 7y, 10y), bootstrap a piecewise-constant hazard curve $\lambda(t)$ to match each spread exactly. This is the credit analog of the Hull-White $\theta(t)$ bootstrap from CH12, and the algorithm is analogous:

1. Start at the shortest tenor ($t_n = 1$y) and pick λ_1 so that (13.12) with constant $\lambda = \lambda_1$ matches the 1y quote.
2. Move to the next tenor ($t_n = 3$y). With $\lambda(t) = \lambda_1$ on $[0,1]$ and $\lambda(t) = \lambda_2$ on $[1,3]$, pick λ_2 so that (13.14) matches the 3y quote.
3. Continue tenor by tenor, building the full hazard schedule.

The bootstrapped hazard curve is directly interpretable: λ_k is the market-implied hazard for the bucket ending at tenor k. Upward- sloping credit curves (5y > 1y) imply increasing hazard over time and are the norm for investment-grade names; inverted curves (5y < 1y) signal near-term distress.

The $\mathbb{Q}$**-recovery** in the triangle is the risk-neutral expected recovery, which may differ from the actual observed average recovery (under $\mathbb{P}$). Typically $\mathbb{Q}$-recovery is a bit lower than $\mathbb{P}$-recovery because investors demand a risk premium for bearing recovery uncertainty. Moreover, historical default rates for a credit bucket are substantially lower than the CDS-implied hazards — the ratio is called the **credit risk-premium multiplier** and is typically 2-4x for investment-grade names, higher for distressed. This risk premium is the credit analog of the equity risk

premium and has direct implications for structured-credit pricing (CDO tranches priced under $\mathbb{Q}$ give risk-neutral expected losses, not historical ones; pricing a CDO under $\mathbb{Q}$ and interpreting as an expected $\mathbb{P}$-loss is a classical mistake).

14.4.5 13.4.5 CDS mark-to-market and DV01

For off-par CDS (market spread has moved since inception), the mark-to- market is the difference between premium and default legs evaluated at current spread and current discount/hazard curves. ISDA conventions since 2009 fix the *coupon* at a standard 100 bp or 500 bp and settle the spread-vs-coupon difference via an upfront payment:

$$\text{Upfront} \;=\; (\text{market spread} - \text{fixed coupon}) \cdot \text{risky annuity} \cdot N. \tag{13.15}$$

Standardised fixed-coupon CDS with upfront payments make the CDS fungible and centrally clearable, the post-2009 market structure.

CDS DV01 and "credit-DV01". Two sensitivities matter:

- **Rate DV01** (interest rate): sensitivity to a 1 bp parallel shift in the discount curve. Typically small (seconds-to-tens of dollars per \$1M notional) because rate effects largely cancel between the two legs.
- **Credit-DV01** (spread): sensitivity to a 1 bp parallel shift in the hazard curve (or equivalently a 1 bp shift in the quoted spread). The dominant risk metric, equal to roughly $(1-R) \cdot \text{risky annuity} \cdot N \cdot 0.0001$.

A \$10M 5y CDS with risky annuity 4.5 and 40% recovery has credit-DV01 $\approx$ \$10M $\cdot\, 4.5 \cdot 0.6 \cdot 0.0001 =$ \$2,700 per basis point. A portfolio of dozens of names requires bucket credit-DV01 by name and tenor, analogous to IRS bucket DV01 by tenor.

CDS premium leg vs default leg *The premium leg (left, blue) is a risky annuity. The default leg (right, orange) is the probability-weighted loss density. At the par spread the two areas match up to a factor of S_0^{CDS}; the credit triangle $(1-R)\lambda$ is the near-flat-curve limit.*

Note on accrued premium. The formulas above handle premium payments "up to the last scheduled coupon before default" but ignore the fact that actual CDS contracts pay accrued premium at default. When Acme defaults between coupon dates, the seller owes a partial coupon covering the accrual from the last coupon date to the default date. The correction is $S \cdot N \cdot \int_0^{t_n} (\tau - t_{\text{last}(\tau)}) \lambda e^{-\lambda\tau} P_0(\tau)\, \mathrm{d}\tau$ — typically a few-percent adjustment to the premium leg, translating into a few-bps adjustment to the par spread. Production pricing engines handle it carefully; we suppress it here for pedagogical clarity.

14.5 13.5 The yield surface

14.5.1 13.5.1 Spot, forward, and par-swap surfaces

Before moving to bond options, we pause to visualise the object that unifies everything the chapter has done so far: the **yield surface** $y_t(T)$, a 3D surface with axes $(t, T, y_t(T))$ showing how the entire yield curve evolves over time. Under a calibrated Hull-White model, at each sample date t we compute $P_t(T)$ for a range of T via the CH12 bond-price formula evaluated at the sampled r_t, then invert $y_t(T) = -\ln P_t(T)/(T-t)$.

The yield surface is the natural 3D object for a rates trader because it makes explicit a distinction often blurred in text: the difference between **calendar time** t (what the clock does) and **maturity time** T (the endpoint of the bond). A 10-year yield today and a 10-year yield one year from now are *different bonds*: one matures in 10 years, the other in 11 years from today's vantage. Organising all this into a surface separates the two time dimensions cleanly.

Features visible on the surface:

- Along the T-axis at fixed t: the usual upward-sloping yield curve (in a below-neutral rate regime).
- Along the t-axis at fixed T: a time series of the T-yield, evolving as r_t fluctuates.
- A red curve $y_t(T)$ traces out the current yield curve; a red dot r_t marks the front end ($T \to t$ limit, $y_t(t) = r_t$ by l'Hôpital).
- The surface is **smooth in** T (yields are model outputs from the affine exponent, which is a smooth function of maturity) but **rough in** t (driven by the stochastic r_t).

This contrast — smooth in T, rough in t — is a fundamental signature of affine term-structure models. In the affine exponent $-r_t B_{T-t} - A_{T-t,t} + C_{T-t,t}$, all three components are smooth functions of $(T-t)$; so the yield is smooth in

T. But r_t itself follows a Brownian-driven process, whose sample paths are continuous but not differentiable. When we plot the whole surface over time, it looks like a smoothly-curving sheet being shaken by vertical stochastic motion.

14.5.2 13.5.2 One-factor signature and its limits

A one-factor model like Hull-White has a specific signature: all points of the yield curve move together in response to a shock to r_t, but not by the same amount. A positive shock δr to r_t raises the short end by the full shock magnitude and raises longer maturities by the damped amount:

$$\delta y_t(T) \;=\; \frac{B_{T-t}}{T-t} \cdot \delta r \tag{13.16}$$

For $T-t$ short, $B_{T-t}/(T-t) \approx 1$ and the yield moves almost 1-for-1 with the shock. For $T-t$ long, $B_{T-t}/(T-t) \approx 1/(\kappa(T-t))$ and the yield moves by a damped factor. With $\kappa = 0.15$ and $T-t = 30$ years, the damping factor is $1/(0.15 \cdot 30) = 0.22$: a 1 bp shock to r_t moves the 30-year yield by only 0.22 bp in the model. This is a specific prediction of the one-factor structure.

In reality, the 30-year yield responds to short-rate shocks more vigorously than one-factor Hull-White predicts — typically by a factor of 2-4×. Real-world rate shocks are often not just level shocks; they come with concurrent slope shifts (inflation expectations, policy stance). A pure level shock does damp out at the long end, but a level-plus-slope shock can move the long end substantially. This is the fundamental limitation of one-factor models and the motivation for multi-factor extensions like G2++ and beyond (§13.8).

Trader's frame. For a fixed-income trader, the yield surface is the natural frame for visualising portfolio exposure. A swap's value is a functional of the surface; a cap's value depends on both the surface and its volatility; a swaption's value depends on the expected future surface. Every rate instrument has some dependence on this surface, and good risk management requires understanding which portions of the surface matter most for each position. A "rate trade" can be: long 5-year, short 10-year (a slope trade); long 2-year, short 5-year, long 10-year (a butterfly); long 10-year, short 10y-into-5y swaption (an implied-vol trade). Each is a different exposure to a different dimension of the surface.

No-arbitrage constraints. The surface is not arbitrary. The condition that $e^{-\int_0^t r_u \,\mathrm{d}u} P_t(T)$ is a $\mathbb{Q}$-martingale imposes a drift condition that ties $\theta(t)$ to the initial forward curve — the **HJM drift condition** in its generalised form (§13.8). Any yield surface that fails this condition admits arbitrage: borrow one maturity and lend another, riding the discrepancy for risk-free profit. The Hull-White surface is no-arbitrage by construction, derived from a single short-rate process under $\mathbb{Q}$. This is why consistent modelling matters; it is also why $\theta(t)$ calibration (CH12) is not a cosmetic choice but a no-arbitrage necessity.

Bridge to §13.6. The yield surface sets the stage for pricing *options* on the surface — specifically, **European bond options**, whose payoff depends on $P_{T_0}(T_1)$, a single spot-yield draw from the future surface at time T_0. Pricing these options is where the T-forward measure earns its keep.

14.6 13.6 European bond options and the T-forward measure

14.6.1 13.6.1 Why change to the T-forward measure

Setup. A European call on a bond has payoff $\left(P_{T_0}(T_1) - K\right)_+$ paid at T_0, where $t < T_0 < T_1$. Let $\mathfrak{g}_t$ denote the call price process. Risk-neutral pricing under the money-market numeraire M_t:

$$\mathfrak{g}_t \;=\; \mathbb{E}_t^{\mathbb{Q}}\Big[e^{-\int_t^{T_0} r_s \,\mathrm{d}s} \left(P_{T_0}(T_1) - K\right)_+ \Big]. \tag{13.17}$$

Under Vasicek or Hull-White, $P_{T_0}(T_1) = \exp(A_{T_0}(T_1) - B_{T_0}(T_1)\, r_{T_0})$, so the integrand in (13.17) couples *two* random variables driven by the same short-rate path: the discount factor $e^{-\int_t^{T_0} r_s \,\mathrm{d}s}$ and the terminal bond value $P_{T_0}(T_1)$. Both are nonlinear functions of r_{T_0} and both are highly correlated (when rates are high at T_0, the discount factor is small *and* the bond is cheap).

A direct computation would require evaluating a 2-dimensional Gaussian integral involving the product of an exponential, an indicator function (from the max), and another exponential. Feasible, but ugly. The **numeraire-change trick** below eliminates the stochastic discount factor entirely, reducing the problem to a clean 1-dimensional Gaussian integral that is precisely the Black-76 setup.

14.6.2 13.6.2 The T-forward measure (summary from CH05)

CH05 derived the T-forward measure once as the canonical application of Girsanov's theorem: we will not re-derive it here. The result we need is this.

(Result, from CH05.) Let $P_t(T_0)$ be a positive traded numeraire. Define the **T-forward measure** $\widetilde{\mathbb{Q}}_{T_0}$ by the Radon-Nikodym derivative

$$\left.\frac{\mathrm{d}\widetilde{\mathbb{Q}}_{T_0}}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_t} = \frac{P_t(T_0)/M_t}{P_0(T_0)/M_0}. \tag{13.18}$$

Under $\widetilde{\mathbb{Q}}_{T_0}$, every traded asset divided by $P_t(T_0)$ is a martingale:

$$\frac{\mathfrak{g}_t}{P_t(T_0)} = \tilde{\mathbb{E}}_t^{T_0}\left[\frac{\mathfrak{g}_{T_0}}{P_{T_0}(T_0)}\right] = \tilde{\mathbb{E}}_t^{T_0}\left[\mathfrak{g}_{T_0}\right] \quad (\text{since } P_{T_0}(T_0) = 1) \tag{13.19}$$

and the $\mathbb{Q}$-Brownian motion $\widehat{W}_t$ is transformed to $\widetilde{W}_t^0$ via

$$\mathrm{d}\widetilde{W}_t^0 = \mathrm{d}\widehat{W}_t + B_t(T_0)\,\sigma\,\mathrm{d}t. \tag{13.20}$$

The drift shift $B_t(T_0)\sigma$ is the $\mathbb{Q}$-volatility of the numeraire $P_t(T_0)$ (with a sign flip, because bond prices move *inversely* to rates). This is the **convexity adjustment** from changing measure, and it is the standard Girsanov prescription ("shift by the numeraire's volatility") derived in CH05.

Hence

$$\boxed{\mathfrak{g}_t = P_t(T_0)\,\tilde{\mathbb{E}}_t^{T_0}\left[\left(P_{T_0}(T_1) - K\right)_+\right]} \tag{13.21}$$

The awkward stochastic discount factor $e^{-\int r\,\mathrm{d}s}$ is gone: it has been absorbed into the numeraire $P_t(T_0)$, which is now a deterministic quantity outside the expectation.

Intuition — why Black's formula works here. Switching numeraire to $P_t(T_0)$ — the bond that matures exactly when the option pays — solves the problem in one stroke:

- $P_{T_0}(T_0) = 1$ by definition, so at option expiry the numeraire equals 1 and the "discount factor" at that instant simply vanishes.
- The **forward bond** $X_t \equiv P_t(T_1)/P_t(T_0)$ is a ratio of two traded assets divided by the numeraire — hence a $\widetilde{\mathbb{Q}}_{T_0}$-martingale by construction.
- In Vasicek/Hull-White, the ratio of two exponential-affine bond prices is itself exponential-affine in a *deterministic* combination $B_t(T_0) - B_t(T_1)$, so X_t has deterministic volatility and is exactly a driftless geometric Brownian motion.

Geometric Brownian motion + driftless + deterministic volatility is precisely the setup of Black-76 (Black's 1976 commodity-futures formula). The bond-option price collapses to a Black-type $\Phi(d_\pm)$ expression with the "forward" X_t, the "discount" $P_t(T_0)$, and the "total variance" $\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s$. The numeraire choice is not cosmetic — it is what makes the problem lognormal.

The deeper lesson: *every derivative pricing problem has a preferred numeraire in which the pricing equation is simplest.* For vanilla equity options it is the bank account (giving Black-Scholes). For bond options it is the bond maturing at the option's expiry (giving Black-76). For swaptions it is the annuity associated with the underlying swap (giving the swap-market-model formula). The general recipe: *match your numeraire to your payoff, and the payoff's associated asset becomes a driftless martingale under the corresponding measure — the Black setup.*

14.6.3 13.6.3 Forward bond price dynamics under $\widetilde{\mathbb{Q}}_{T_0}$

By Itô applied to $P_t(T_1) = e^{A_t(T_1) - B_t(T_1) r_t}$ using the Hull-White SDE $\mathrm{d}r_t = \kappa(\theta_t - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}\widehat{W}_t$ from CH12:

$$\frac{\mathrm{d}P_t(T_1)}{P_t(T_1)} \;=\; r_t\,\mathrm{d}t \;-\; \sigma\,B_t(T_1)\,\mathrm{d}\widehat{W}_t. \tag{13.22}$$

Every bond has $\mathbb{Q}$-drift r_t (consistent with being a traded asset under the risk-neutral measure), with diffusion coefficient $-\sigma B_t(T_1)$. The minus sign reflects the inverse relationship between bond prices and rates; the magnitude $\sigma B_t(T_1)$ scales with both the short-rate vol σ and the bond's duration $B_t(T_1)$. For a short bond ($B \approx 0$) the diffusion coefficient is tiny; for a long bond it saturates at σ/κ. This is the Vasicek analogue of the duration-based bond-price volatility heuristic.

Critically, in Vasicek/Hull-White, the bond's diffusion coefficient $-\sigma B_t(T_1)$ is **deterministic** — it depends on time and maturity but not on any random state variable. This is special to Gaussian models. In CIR, the short-rate volatility $\sigma\sqrt{r_t}$ is state-dependent, and so is the bond's diffusion coefficient; the bond is no longer lognormal under any measure change, and the resulting bond-option formula is not of Black form (it involves a non-central chi-squared distribution). The deterministic-volatility property is what makes Vasicek bond options admit a Black-type closed form.

Under $\widetilde{\mathbb{Q}}_{T_0}$, using (13.20) to substitute $\mathrm{d}\widehat{W}_t = \mathrm{d}\widetilde{W}_t^{\,0} - B_t(T_0)\sigma\,\mathrm{d}t$, the T_1-bond's drift changes:

$$\frac{\mathrm{d}P_t(T_1)}{P_t(T_1)} \;=\; \big(r_t + \sigma^2 B_t(T_0)\,B_t(T_1)\big)\,\mathrm{d}t \;-\; \sigma\,B_t(T_1)\,\mathrm{d}\widetilde{W}_t^{\,0}. \tag{13.23}$$

The drift correction $+\sigma^2 B_t(T_0)B_t(T_1)$ is the **convexity adjustment** from changing measure. Under $\mathbb{Q}$, the bond drifts at r_t. Under $\widetilde{\mathbb{Q}}_{T_0}$, it drifts at a slightly higher rate, by an amount proportional to the product of the two bonds' durations. This extra drift keeps the *ratio* $P_t(T_1)/P_t(T_0)$ a martingale, as required by the new numeraire choice.

Whenever you change numeraire, you shift drifts; whenever drifts shift, martingales reorganise; and whenever martingales reorganise, your closed-form pricing formulas change. The specific Vasicek expression $B_t(T_0)\sigma$ in (13.20) is just the bond's diffusion coefficient from (13.22); in a different model (CIR, say) the numeraire's volatility would differ, and the drift shift would change accordingly. But the *form* of the adjustment — "shift by the numeraire's volatility" — is universal across continuous diffusion models.

Rate distribution under Q vs T-forward measure *Changing numeraire from M_t to $P_t(T_0)$ shifts the drift of r_t downward by $-\sigma^2 B_t(T_0)$: the conditional law of r_{T_0} under the forward measure is a Gaussian with the same variance but a lower mean than under $\mathbb{Q}$. This shift is the convexity adjustment relating forward rates to expected spot rates.*

14.6.4 13.6.4 Forward bond price X_t is a $\widetilde{\mathbb{Q}}_{T_0}$-martingale

Introduce the **forward bond price**

$$X_t \;=\; \frac{P_t(T_1)}{P_t(T_0)}. \tag{13.24}$$

Key boundary property:

$$X_{T_0} \;=\; \frac{P_{T_0}(T_1)}{P_{T_0}(T_0)} \;=\; P_{T_0}(T_1), \tag{13.25}$$

so the option's payoff at T_0 is exactly $(X_{T_0} - K)_+$. X_t is a $\widetilde{\mathbb{Q}}_{T_0}$-martingale (ratio of traded asset to numeraire).

By Itô on the ratio (both bonds driven by the same $\widetilde{W}^{\,0}$):

$$\frac{\mathrm{d}X_t}{X_t} \;=\; \sigma\,\big(B_t(T_0) - B_t(T_1)\big)\,\mathrm{d}\widetilde{W}_t^{\,0} \;\equiv\; \Sigma_t\,\mathrm{d}\widetilde{W}_t^{\,0}, \tag{13.26}$$

where the deterministic **forward-bond volatility** is

$$\Sigma_t \;=\; \sigma\,(B_t(T_0) - B_t(T_1)) \;=\; -\frac{\sigma}{\kappa}\,e^{-\kappa T_0}\big(1 - e^{-\kappa(T_1-T_0)}\big)\,e^{\kappa t}. \tag{13.27}$$

The sign/convention is immaterial in what follows — only Σ_t^2 appears. The formula $\Sigma_t = \sigma(B_t(T_0) - B_t(T_1))$ has a beautifully intuitive reading: the forward bond is a long position in the T_1-bond and a short position in the T_0-bond (that is what the ratio encodes). Each bond has its own exposure to rate shocks: $\sigma B_t(T_1)$ for the long leg, $\sigma B_t(T_0)$ for the short leg. The forward bond's net exposure is the difference — negative because the longer bond has higher duration, so the net effect of a rate shock is a loss on the long leg larger than the gain on the short leg. The key point: this net exposure is **deterministic**, which is precisely why Vasicek admits a Black-type bond-option formula.

Because Σ_t is deterministic, X_t is a geometric Brownian motion with deterministic volatility, and

$$\boxed{X_T \;=\; X_t\,\exp\Big\{-\tfrac{1}{2}\int_t^T \Sigma_s^2\,\mathrm{d}s \;+\; \int_t^T \Sigma_s\,\mathrm{d}\widetilde{W}_s^{\,0}\Big\}} \tag{13.28}$$

so $\ln(X_T/X_t)$ is Gaussian under $\widetilde{\mathbb{Q}}_{T_0}$ — a lognormal forward bond price.

Forward-bond vol Sigma_t and integrated variance $|\Sigma_t|$ *grows as* $t \to T_0$ *(the pull-to-par of* $P_t(T_0)$ *becomes less effective at absorbing rate noise); the cumulative total variance* $\int_0^t \Sigma_s^2\,\mathrm{d}s$ *is what appears in the Black* $d_\pm$ *denominator.*

14.6.5 13.6.5 Closed-form European bond call (Jamshidian / Vasicek-Black)

Plugging into (13.21):

$$\mathfrak{g}_t \;=\; P_t(T_0)\,\tilde{\mathbb{E}}_t^{T_0}\big[\,(X_{T_0} - K)_+\,\big]. \tag{13.29}$$

This is a Black-style lognormal expectation with "volatility" $\sqrt{\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s}$. Evaluating the Gaussian integral in the standard way:

$$\boxed{\mathfrak{g}_t \;=\; P_t(T_0)\,\{\,X_t\,\Phi(d_+) \;-\; K\,\Phi(d_-)\,\}} \tag{13.30}$$

with

$$\boxed{d_\pm \;=\; \frac{\ln(X_t/K) \;\pm\; \frac{1}{2}\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s}{\sqrt{\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s}}} \tag{13.31}$$

This is the **Vasicek-Black bond-option formula**: a Black-Scholes-type expression where the "stock" is the forward bond X_t, the "discount factor" is $P_t(T_0)$, and the total variance is the deterministic integral $\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s$.

Boundary-condition check. At expiry $t = T_0$: the integral $\int_{T_0}^{T_0} \Sigma_s^2\,\mathrm{d}s = 0$, so $d_\pm = \pm\infty \cdot \mathrm{sgn}(\ln(X_{T_0}/K))$. Hence $\Phi(d_+) - \Phi(d_-) = \mathbf{1}\{X_{T_0} > K\}$, and the formula reduces to $(X_{T_0} - K)_+ = (P_{T_0}(T_1) - K)_+$, matching the contractual payoff. At deep OTM strikes ($K \gg X_t$): both $\Phi(d_+)$ and $\Phi(d_-) \to 0$, and the price goes to zero. At deep ITM ($K \ll X_t$): $\Phi(d_+) \to 1$, $\Phi(d_-) \to 1$, and the price approaches $P_t(T_0)(X_t - K) = P_t(T_1) - K P_t(T_0)$, the intrinsic value (forward value of bond minus forward value of strike). All expected limits hold.

Total-variance factorisation. The integral admits a clean closed form:

$$\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s \;=\; \frac{\sigma^2}{\kappa^2}\,\big(1 - e^{-\kappa(T_1-T_0)}\big)^2 \;\cdot\; \frac{1 - e^{-2\kappa(T_0-t)}}{2\kappa}. \tag{13.32}$$

The three factors have clean physical interpretations:

- σ^2/κ^2: the overall *scale*, roughly the stationary variance of r_t times 2.
- $(1-e^{-\kappa(T_1-T_0)})^2$: the *tenor-sensitivity* factor — how much longer T_1 is than T_0.
- $(1-e^{-2\kappa(T_0-t)})/(2\kappa)$: the *time-to-expiry weighting*, saturating at $1/(2\kappa)$ as $T_0-t\to\infty$.

Once these three numbers are computed for the relevant parameter grid, pricing any Vasicek bond option is three multiplications plus a Black formula — a core rule-of-thumb for rates traders.

14.6.6 13.6.6 Put-call parity for bond options

$$\mathfrak{g}_t^{\text{call}} - \mathfrak{g}_t^{\text{put}} \;=\; P_t(T_1) - K\,P_t(T_0). \tag{13.33}$$

Parity is structurally the same as for equity options: call minus put equals forward value of the underlying minus forward value of the strike. Here the "forward value" is $P_t(T_1)/P_t(T_0) = X_t$; multiplying by $P_t(T_0)$ gives $P_t(T_1) - KP_t(T_0)$. This is a pure no-arbitrage statement that must hold in *any* consistent framework. If your pricing library's puts and calls for a given strike do not satisfy (13.33), something is wrong. Parity is the single cheapest sanity check in rates option pricing, and it is also the relation used in practice to translate between calls and puts (dealers typically price calls analytically and derive puts via parity, or vice versa, depending on which is numerically more stable at the relevant strike).

Digital bond options — options paying \$1 if $P_{T_0}(T_1) > K$ and 0 otherwise — collapse to $\mathfrak{g}_t^{\text{digital}} = P_t(T_0)\cdot\Phi(d_-)$. This is the Black-76 digital formula in disguise, useful for structuring bespoke rate payoffs.

14.7 13.7 Worked examples: Jamshidian decomposition and Vasicek-Black

14.7.1 13.7.1 Worked example — forward bond volatility for a 1×3 option

Suppose $\sigma = 0.01$, $\kappa = 0.3$, $T_0 = 1$, $T_1 = 3$. From (13.32):

- $(1-e^{-\kappa(T_1-T_0)})^2 = (1-e^{-0.6})^2 = (0.4512)^2 = 0.2036$
- $(1-e^{-2\kappa T_0})/(2\kappa) = (1-e^{-0.6})/(0.6) = 0.7520$
- $\sigma^2/\kappa^2 = (0.01)^2/0.09 = 1.111\times10^{-3}$

Total variance $\approx 1.111\times10^{-3}\cdot0.2036\cdot0.7520 = 1.70\times10^{-4}$, so effective lognormal vol $\approx\sqrt{1.70\times10^{-4}} = 1.30\%$.

Plug into (13.30)–(13.31) with $X_0 = 0.94$ (forward bond price) and strike $K = 0.94$ (ATM). Then $\ln(X_0/K) = 0$, $d_\pm = \pm\sqrt{1.70\times10^{-4}}/2 = \pm0.00652$, $\Phi(0.00652) = 0.5026$, $\Phi(-0.00652) = 0.4974$. Call price:

$$\mathfrak{g}_0 \;=\; P_0(1)\cdot(0.94\cdot0.5026 - 0.94\cdot0.4974) \;=\; P_0(1)\cdot0.94\cdot0.0052 \;=\; 0.00489\cdot P_0(1).$$

For $P_0(1)\approx0.97$, the dollar ATM call price is about 0.00474, or 47 bps of notional. Small premium — consistent with short expiry and modest vol — but the right order of magnitude for a short-dated ATM bond option.

ATM approximation. Compare to $\mathfrak{g}\approx P_0(1)\cdot X_0\cdot\sigma_{\text{eff}}/\sqrt{2\pi}$ with $\sigma_{\text{eff}} = 1.30\%$: $0.97\cdot0.94\cdot0.0130/\sqrt{6.28} = 0.00473$. Matches the exact Black result to within 0.2%. The ATM approximation is strikingly accurate for near-the-money strikes — a trader's rule of thumb. The $1/\sqrt{2\pi}\approx0.3989$ factor is a universal constant in normal-distribution- based option pricing (it appears in Black-Scholes, Black-76, Bachelier). Memorising it as "roughly 0.4" lets you compute ATM Black prices to about 10% accuracy in your head.

14.7.2 13.7.2 ITM and OTM strikes

Returning to the 1×3 bond option with $X_0 = 0.94$, consider an ITM call ($K = 0.90$) and an OTM call ($K = 0.98$).

ITM ($K = 0.90$). $\ln(X_0/K) = \ln(0.94/0.90) = 0.0435$. With total variance 1.70×10^{-4}, $\sqrt{\cdot} = 0.01304$. Then $d_+ = (0.0435 + 8.5\times10^{-5})/0.01304 = 3.34$, $d_- = 3.33$. $\Phi(3.34)\approx0.9996$. Call price (per unit $P_0(1)$) $= 0.94\cdot0.9996 - 0.90\cdot0.9996 = 0.04$. This is essentially the intrinsic value ($X_0 - K = 0.04$); time value is tiny at such a deep-ITM strike and short expiry.

OTM ($K = 0.98$). $\ln(X_0/K) = -0.0417$. $d_+ = -3.19$, $d_- = -3.20$. $\Phi(-3.19)\approx0.00071$. Call price $\approx X_0\cdot0.00071 - K\cdot0.00069\approx0$. The option is so deep OTM that the forward bond has essentially no chance of reaching the strike in one year.

The asymmetry between ITM (near-intrinsic) and OTM (near-zero) is why bond options trade like delta-one when far ITM and like pure gamma plays when near ATM. Structured trades across strikes can achieve specific exposure profiles: an ATM straddle gives pure gamma/vega; a wide-strike strangle gives tail-event exposure.

14.7.3 13.7.3 Sensitivity to mean-reversion speed

Fix $\sigma = 0.01$, $T_0 = 1$, $T_1 = 5$, and vary κ:

- $\kappa = 0.10$: $\sigma^2/\kappa^2 = 0.01$, $(1 - e^{-0.4})^2 = 0.1087$, $(1 - e^{-0.2})/0.2 = 0.9063$. Total var $= 9.85 \times 10^{-4}$, effective vol $\approx 3.14\%$.
- $\kappa = 0.30$: $\sigma^2/\kappa^2 = 1.11 \times 10^{-3}$, $(1 - e^{-1.2})^2 = 0.4968$, $(1 - e^{-0.6})/0.6 = 0.7520$. Total var $= 4.15 \times 10^{-4}$, effective vol $\approx 2.04\%$.
- $\kappa = 1.00$: $\sigma^2/\kappa^2 = 10^{-4}$, $(1 - e^{-4})^2 = 0.9640$, $(1 - e^{-2})/2 = 0.4323$. Total var $= 4.17 \times 10^{-5}$, effective vol $\approx 0.646\%$.

Doubling κ from 0.10 to 0.30 reduces the effective vol by about a third; raising κ to 1.0 reduces it by almost 80%. The reason is clear from the structure: σ^2/κ^2 drops like $1/\kappa^2$, the tenor factor grows with κ (offsetting partially), and the time-to-expiry factor shrinks in relative terms. The dominant effect is the first — rapid mean reversion compresses rate uncertainty into a narrow band, and narrower rate bands mean smaller bond-option vols.

Calibration implication. If the market quotes a bond-option implied vol that is much higher than what Vasicek produces for a plausible κ, two levers are available: raise σ to push vol up, or lower κ to slow mean reversion. These two knobs interact: a given implied vol can be matched by many (κ, σ) pairs, and disentangling them requires data at multiple option maturities. Short- dated options are more sensitive to σ; long-dated options are more sensitive to κ. A careful calibration uses the full term structure of implied vols to identify both parameters, often with regularisation to prevent over-fitting.

Bond-option price surface vs strike and T_0 *Left: call price surface $\mathfrak{g}_0(T_0, K)$ on a 5-year zero via the Black-type Vasicek formula. Right: Black slices at fixed T_0 — the dotted verticals mark the forward bond price $X_0 = P_0(T_1)/P_0(T_0)$ (ATM).*

14.7.4 13.7.4 Jamshidian's decomposition: swaption as a portfolio of bond options

A European **swaption** gives the holder the right to enter a fixed-for- floating swap at some future date T_0. Economically, it is an option on a portfolio of cash flows: at each payment date $T_1 < T_2 < \cdots < T_N$ after T_0, the swap holder receives either a fixed cash flow cK (with c the fixed rate and K the notional-accrual factor) or pays a floating cash flow. The swap value at T_0 can be rewritten as a sum of zero-coupon bond prices, so the swaption is an option on a *sum* of bond prices — a much harder object than an option on a *single* bond.

Jamshidian's 1989 insight. For a one-factor Gaussian short-rate model, the sum of bond prices is a **monotonically decreasing** function of r_{T_0}. So there exists a unique "critical rate" r^* at which the sum equals the swaption's strike, and the sum exceeds the strike if and only if $r_{T_0} < r^*$. This means the option on the sum is equivalent to a portfolio of options on individual bonds, each struck at the critical-rate-implied strike:

$$\text{swaption}_0 \;=\; \sum_{i=1}^{N} c_i \cdot \text{BondOption}(T_0, T_i, K_i) \tag{13.34}$$

where c_i are the swap coefficients (tenors and notional), $K_i = P_{T_0}(T_i)\big|_{r=r^*}$ are effective strikes computed via the Vasicek/Hull-White bond-price formula at $r = r^*$, and r^* is the critical rate obtained by solving a scalar equation.

Procedure.

1. **Set up the critical-rate equation.** The swaption payoff at T_0 is $(V_{T_0}^{\text{swap}} - K)_+$ where $V_{T_0}^{\text{swap}} = \sum_i c_i P_{T_0}(T_i)$. Each $P_{T_0}(T_i) = \exp(A_{T_0}(T_i) - B_{T_0}(T_i) r_{T_0})$ is a monotonically decreasing function of r_{T_0}. So $V_{T_0}^{\text{swap}}$ is monotonically decreasing in r_{T_0}.
2. **Solve for r^*.** Find the unique root of $\sum_i c_i \exp(A_{T_0}(T_i) - B_{T_0}(T_i) r^*) = K$ — a one-dimensional monotone root-find (bisection or Newton).
3. **Compute effective strikes.** For each i, set $K_i = \exp(A_{T_0}(T_i) - B_{T_0}(T_i) r^*)$.
4. **Price a portfolio of bond options.** Each term $\text{BondOption}(T_0, T_i, K_i)$ is a Vasicek-Black call on the T_i-bond with strike K_i, priced by (13.30)-(13.31).

The Jamshidian decomposition is one of the most elegant results in rates quant: it reduces a high-dimensional option-on-portfolio problem to a 1D root-find plus a handful of scalar option evaluations. It works *exactly* for Vasicek and Hull-White (any one-factor Gaussian affine model). For multi-factor models it is only an approximation; but even there it is often the first step in a computational pipeline — solve the 1D Jamshidian to get a decent starting point, then correct with Monte Carlo or other numerics.

A side-effect of the Jamshidian decomposition is that the swaption's implied-volatility "smile" under Vasicek is effectively flat in each tenor bucket — because each component bond option has a Black formula with a single total-variance. Reality shows substantial swaption smile (OTM payers trade above OTM receivers), which Vasicek cannot reproduce. Capturing smile requires stochastic vol (SABR-rate extensions), jump components, or multi-factor correlations — motivating §13.8.

14.7.5 13.7.5 Calibration pipeline

The pipeline that ties the whole chapter together:

1. **Fit** (κ, σ) from swaption and/or cap quotes by minimising weighted least-squares between model-implied and market prices across the vol cube.
2. **Bootstrap** θ_t from the initial bond curve (CH12) so that $P_0^{\text{model}}(T) = P_0^{\text{market}}(T)$ for all T.
3. **Price other rate derivatives** — bond options via (13.30), swaptions via Jamshidian (13.34), callables via the Hull-White lattice (§13.3), caps/floors via strips of caplets (each a put on a zero-bond).

Step 1's calibration is typically not unique: (κ, σ) combinations trade off "persistent long-rate shocks" (high κ, high σ) against "temporary short-rate shocks" (low κ, low σ). Short-expiry swaptions carry more information about σ; long-expiry about κ. To resolve degeneracy, practitioners often add long-dated cap vols or fix κ at a historical value (say 0.1) and only calibrate σ.

14.8 13.8 Multi-factor preview: HJM, LMM, and SABR

14.8.1 13.8.1 Why one factor is not enough

The single-factor Vasicek model is the cleanest entry point into short- rate modelling, but it is the start of a journey, not the destination. Four structural limitations drive the move to richer models:

1. **Perfect correlation across maturities.** In a one-factor model, all bonds are driven by the same Brownian driver $\widehat{W}_t$, so they are perfectly correlated — the 2y-vs-10y spread always moves in a fixed proportion to the 10y move, with no independent randomness. Empirically, the short-long spread has substantial idiosyncratic variation driven by monetary-policy timing relative to economic expectations.
2. **Limited yield-curve shapes.** One-factor Hull-White produces a specific family of curve shapes (shifts and mean-reverting twists). Real curves exhibit richer dynamics (parallel, slope, curvature — three independent principal components explaining ~99% of curve variance, not one).
3. **Flat swaption smile.** The Jamshidian decomposition gives flat-in- strike swaption vol for a given expiry-tenor, but markets show clear skews and smiles (OTM payers rich of OTM receivers).
4. **Long-end calibration.** The damping factor $B_t(T)/T \to 1/(\kappa T)$ for long T means one-factor models under-predict long-end volatility by factors of 2-4× in practice. Multi-factor models add explicit slope-like factors to capture this.

Each structural limitation has a corresponding remedy.

14.8.2 13.8.2 Two-factor Gaussian: G2++

The most common multi-factor extension of Vasicek is **G2++**, where the short rate is

$$r_t \;=\; x_t + y_t + \phi(t) \tag{13.35}$$

with x_t and y_t two correlated Ornstein-Uhlenbeck factors:

$$\begin{aligned} \mathrm{d}x_t &= -a\, x_t\, \mathrm{d}t + \sigma_1\, \mathrm{d}W_t^x, \\ \mathrm{d}y_t &= -b\, y_t\, \mathrm{d}t + \sigma_2\, \mathrm{d}W_t^y, \\ \langle W^x, W^y \rangle &= \rho\, t, \end{aligned} \tag{13.36}$$

and $\phi(t)$ a deterministic shift calibrated to today's yield curve. The two factors have different mean-reversion speeds a and b; typically one is "fast" (capturing short-end variability) and one is "slow" (capturing long-end persistence). The correlation ρ lets the factors partially offset or reinforce each other, modulating the yield- curve dynamics.

The bond-price formula in G2++ remains exponential-affine in both factors:

$$P_t(T) \;=\; A(t,T)\exp\!\big(-B_1(t,T)\,x_t - B_2(t,T)\,y_t\big) \tag{13.37}$$

with B_1, B_2 analogous to Vasicek's B (just with a or b as the reversion speed), and $A(t,T)$ absorbing the shift ϕ and the convexity corrections. The bond-option formula is still Black-type under the forward measure, with total variance now depending on $\sigma_1, \sigma_2, \rho, a, b$. Jamshidian decomposition does *not* apply exactly (the sum-of-bonds is not monotone in a single state), but an approximate 2D Jamshidian can be used, or the swaption can be priced by numerical integration over the 2-factor state space.

Calibration payoff. With two vol parameters and a correlation, G2++ can fit the term structure of cap vols *and* the swaption vol cube at a given strike — something 1-factor Hull-White cannot do with any reasonable parameter choice. The correlation ρ carries much of the pricing information: negative ρ produces "humped" cap vol term structures, while positive ρ produces monotone profiles. Real cap markets typically show humped profiles, which is why negative- correlation G2++ calibrations are common.

14.8.3 13.8.3 HJM: modelling the entire forward curve

Short-rate models take r_t as the primitive state variable and derive forward rates and bond prices as consequences. An alternative framework, developed by **Heath, Jarrow, and Morton (HJM)** in 1992, takes the entire *forward-rate curve* $f_t(T)$ as the primitive, modelling its dynamics directly:

$$\mathrm{d}f_t(T) \;=\; \alpha(t,T)\,\mathrm{d}t \;+\; \sigma(t,T)\,\mathrm{d}W_t. \tag{13.38}$$

For this specification to be arbitrage-free, HJM showed that the drift $\alpha(t,T)$ is *not* a free parameter: it must equal

$$\alpha(t,T) \;=\; \sigma(t,T)\int_t^T \sigma(t,s)\,\mathrm{d}s \tag{13.39}$$

— a "no-arbitrage drift" uniquely determined by the volatility structure. This is the **HJM drift condition**, and it is the framework's central result. By choosing $\sigma(t,T)$ we fully specify the arbitrage-free dynamics of the whole forward-rate curve.

The HJM framework is much more general than short-rate models. Vasicek corresponds to $\sigma(t,T) = \sigma e^{-\kappa(T-t)}$ (exponentially- decaying vol kernel); G2++ corresponds to a sum of two such kernels. General specifications — including those with random vol or state- dependence — are allowed. In exchange for this generality, HJM gives up analytic tractability: forward rates $f_t(T)$ are rarely Markovian in a low-dimensional state (except for special kernel choices corresponding to affine short-rate models), so pricing most exotics requires Monte Carlo.

Where HJM wins. For pricing products whose payoffs are naturally expressed in terms of forward rates — caps, floors, European swaptions, many Asian-style rate products — HJM is conceptually cleaner than a short-rate model. For products where discounting matters (exotic callable structures, path-dependent), the short-rate framework may be easier. Choosing between them is a matter of which set of primitives most naturally aligns with the pricing problem.

14.8.4 13.8.4 LMM: the discrete-tenor LIBOR market model

The **LIBOR Market Model (LMM)**, developed by Brace-Gatarek-Musiela (BGM) and independently by Jamshidian, is the discrete-tenor version of HJM. Instead of modelling an infinite-dimensional forward curve, LMM models a finite collection of *forward LIBOR rates* $\{L_t(T_{i-1},T_i)\}_{i=1}^N$ — typically the LIBOR rates that underlie the most liquid caps and swaptions. Each forward LIBOR is a (driftless under its own forward measure) lognormal process:

$$\mathrm{d}L_t(T_{i-1},T_i) \;=\; L_t(T_{i-1},T_i)\,\sigma_i(t)\,\mathrm{d}W_t^{(i)}, \tag{13.40}$$

where $W^{(i)}$ is a Brownian under the T_i-forward measure. Under a common measure (typically the spot or terminal measure), the LMM drifts are determined by no-arbitrage — specifically, by the covariation between the relevant forwards.

LMM's great strength. By construction it reproduces Black's caplet/swaption formula exactly for the modelled forwards. Calibration to cap prices becomes trivial (read off $\sigma_i(t)$ from market implied vols); calibration to swaption prices requires an additional step that approximates the swap rate as a linear combination of LIBOR rates. The resulting calibrated model prices any exotic whose payoff depends on these LIBOR rates — Bermudan swaptions, callable range accruals, target- redemption notes — at the cost of Monte Carlo simulation (the state space is high-dimensional).

Post-LIBOR. The transition to risk-free rates (SOFR, €STR, SONIA, TONA) has prompted a shift to "risk-free-rate market models" (RFR LMM). The mathematical structure is analogous (compounded overnight rates replacing LIBOR rates), but specifics of compounding and in-arrears vs in-advance payments add complexity. For 2026-era rates modelling, the RFR variant is the active one; classical LMM is legacy.

14.8.5 13.8.5 SABR and rate-vol smile

The models above (Vasicek, G2++, HJM, LMM) all produce a *single* implied volatility for each option — no smile. Reality shows substantial implied- vol smile in caps, floors, and swaptions: OTM options trade rich of ATM, and the richness varies with strike in specific ways (typically a negative skew for payers, reflecting the asymmetry of rate distributions). Capturing this smile requires stochastic-volatility or jump components.

The industry-standard overlay is **SABR** (Stochastic Alpha Beta Rho), developed by Hagan and co-authors:

$$\mathrm{d}F_t \;=\; \alpha_t\, F_t^{\beta}\, \mathrm{d}W_t, \qquad \mathrm{d}\alpha_t \;=\; \nu\, \alpha_t\, \mathrm{d}Z_t, \qquad \langle W, Z\rangle \;=\; \rho\, t. \tag{13.41}$$

The parameter $\beta \in [0,1]$ controls the "backbone" (how ATM vol moves with the level of rates); ν is vol-of-vol; ρ is the correlation between rate and vol shocks (the skew). SABR gives a simple closed-form asymptotic implied-vol formula (the **Hagan expansion**) that can be calibrated to a smile of prices in one or two iterations. It has become the industry standard for quoting rate-vol surfaces across strikes.

SABR is not a full term-structure model — it is a single-strike, single- expiry vol model, applied to each option separately. A fully consistent term-structure plus smile model is harder: variants like Stochastic- Alpha Hull-White, SABR-LMM, and Cheyette models extend the framework, each with its own trade-offs between tractability and flexibility. For a 2026 trading desk, the typical stack is:

- **Hull-White or G2++** for the underlying rate model.
- **SABR** or its descendants for the smile overlay per expiry-tenor.
- **Careful calibration procedures** that fit both consistently to cap/ swaption quotes.

14.8.6 13.8.6 Where to use what

A rough practitioners' guide:

Product	Recommended model
Vanilla caps, floors, European swaptions	1-factor Hull-White or G2++; SABR overlay per strike
Bermudan swaptions, callable bonds	G2++ or LMM; Jamshidian-like decomposition + MC for paths
CMS, LIBOR-in-arrears, convexity products	Multi-factor affine with explicit convexity corrections
Path-dependent exotics (range accruals)	LMM or extended HJM, with Monte Carlo
XVA (CVA/DVA/FVA)	G2++ for efficiency; hybrid with equity/credit factors
Negative-rate environments	Vasicek/Hull-White/G2++ (Gaussian); shifted-lognormal LMM

No single model dominates, and serious trading desks maintain multiple models in their libraries, switching per product, regime, and computational budget. The common thread is the mathematical apparatus we have developed across CH03-CH13: SDEs (CH03), Itô calculus (CH03), Feynman-Kac (CH04), Girsanov (CH05), no-arbitrage via numeraire change (CH05, §13.6), affine closed forms where they apply (CH12), PDE-vs-MC trade-offs where they don't (§13.9). Master these techniques in the Vasicek context and the rest is a matter of algebra.

14.9 13.9 PDE versus Monte Carlo for rate products

When closed forms are unavailable — which is most of the time outside vanilla caps, floors, and European swaptions — one must resort to numerical methods. The two main candidates are **PDE** (finite-difference or finite-element) solvers and **Monte Carlo** simulation. Each has strengths and weaknesses and the choice is product-specific.

14.9.1 13.9.1 When a PDE wins

PDE methods discretise the Feynman-Kac PDE from CH04 on a grid in (t, r) space (or $(t, x_1, x_2, \dots, x_n)$ for multi-factor models) and solve backward from the terminal condition to the current time. The workhorse algorithms are the **Crank-Nicolson scheme** (for parabolic PDEs) and its implicit cousins (for stability with stiff source terms). The output is the *function* $g(t, r)$ at every grid point, from which prices, deltas, gammas, and thetas can all be read off simultaneously.

Strengths of PDE.

- **Free Greeks.** Deltas, gammas, and thetas are numerical derivatives of the gridded price surface — computed as a byproduct of the solve.
- **Early-exercise naturally handled.** At each time step, replace the solution with $\max(g, \text{exercise payoff})$. No special machinery for American/Bermudan options (contrast MC, where Longstaff-Schwartz regression is needed).
- **Deterministic.** The same grid gives the same answer every time; no Monte Carlo noise. Reproducibility matters for regulatory filings and audit trails.

Weaknesses of PDE.

- **Dimensional curse.** A grid in 1D has N points, in 2D N^2, in 3D N^3, and so on. Beyond 3-4 dimensions, PDE becomes infeasible.
- **Boundary conditions.** Must be specified at the edges of the grid, and poor boundary choices contaminate the interior solution. For rate models with negative-rate possibility, the lower boundary needs special care.
- **Path dependence.** Asian, lookback, and similar path-dependent payoffs require additional state variables, pushing dimensionality up fast.

14.9.2 13.9.2 When Monte Carlo wins

Monte Carlo simulates paths of the state variable (or the forward- rate curve in HJM/LMM), computes the payoff along each path, and averages.

Strengths of Monte Carlo.

- **Dimensional scaling.** A 10-factor LMM is no harder (per path) than a 1-factor Vasicek. The curse of dimensionality is broken — variance in the estimator grows with dimension, but the per-path cost does not.
- **Path-dependent payoffs.** Natural fit: the path is generated in full, and any functional of the path can be computed (barrier, Asian, TARN, lookback).
- **Parallelism.** Paths are independent, so GPUs and cluster machines deliver near-linear speedup. A cloud-based pricing farm can produce a CVA number in minutes that would take hours or days via PDE.

Weaknesses of Monte Carlo.

- **Slow convergence.** $O(1/\sqrt{N})$ in the number of paths. Achieving one basis-point precision typically needs 10^6 to 10^8 paths.
- **Greeks.** Harder than PDE. Options include pathwise differentiation (where payoffs are smooth), likelihood-ratio methods (where the payoff is discontinuous), and perturbed re-simulation (computationally expensive but simple). Each method has its subtleties and failure modes.
- **Early exercise.** Subtle: the natural approach is Longstaff-Schwartz regression, where continuation values are estimated by regressing future-path values on current state variables. Convergence of early- exercise values is slower than European values, and the method is sensitive to basis-function choice.

14.9.3 13.9.3 Hybrid schemes and American-style rate products

Practical guidance. The choice between PDE and MC is dictated by dimensionality and payoff structure:

- **1-factor models (Vasicek, CIR, Hull-White) with vanilla products.** PDE.

- **Multi-factor models (G2++) with mildly exotic products.** PDE still feasible in 2-3 dimensions but MC becomes competitive.
- **LMM, HJM, or high-dimensional models with path-dependent payoffs.** Monte Carlo.
- **XVA on a large book.** Monte Carlo (often with American MC or Least-Squares MC for callable products).

Hybrid approaches exist and are important in practice:

- **PDE-Monte Carlo.** The continuation value for American options is computed via a nested PDE within each MC path. Accurate for exercise- value estimation but computationally expensive.
- **Quasi-Monte Carlo.** Uses low-discrepancy sequences (Sobol, Halton, Faure) instead of pseudo-random numbers to accelerate convergence from $O(1/\sqrt{N})$ toward $O(\log^d N/N)$ in dimension d.
- **Multi-level Monte Carlo (MLMC).** Combines coarse-grid and fine-grid simulations for variance reduction. Especially effective for SDE discretisation error, which dominates in high-dimensional MC.

The state of the art in rates-derivative pricing is a **tool chest** of these methods, chosen per-product based on dimensionality, smoothness, and early-exercise features. A production rates desk in 2026 will have PDE solvers for 1-2 factor vanilla pricing, GPU Monte Carlo for large multi-factor books, and specialised libraries (American MC, quasi-MC, MLMC) for the edge cases. The underlying mathematics — SDEs, Itô, Girsanov, Feynman-Kac, the affine structure we have developed across CH03-CH13 — underlies every one of these methods. The difference is the numerical implementation, not the theory.

14.10 13.10 Key takeaways

1. **Swaps are essentially model-free.** The floating leg self-replicates via "buy \$1 and roll": $V_0^{\text{fl}} = (P_0(t_0) - P_0(t_n))N$ — no short-rate model required, only the discount curve. The par swap rate $S_0 = (P_0(t_0) - P_0(t_n))/\mathcal{A}_0$ is the ratio of that self-replicating value to the annuity; neither κ nor σ nor θ_t appears.

2. **Callable bonds are genuinely model-dependent.** The embedded American option requires backward induction on a Hull-White trinomial tree with a call-value test at each exercise date: $V^{\text{call}} = \min(\text{call price}, V^{\text{continuation}})$. OAS, OAD, and negative convexity emerge from the tree. Calibrated (κ, σ) from the swaption market feed into the tree directly.

3. **CDS factor into a risky annuity and a default density.** The par CDS spread is

$$S_0^{\text{CDS}} = \frac{(1-R)\int_0^{t_n} \lambda e^{-\lambda\tau} P_0(\tau)\,\mathrm{d}\tau}{\sum_\ell \Delta t_\ell \widetilde{P}_0(t_\ell)}$$

   with $\widetilde{P}_0(t) = e^{-\lambda t} P_0(t)$ the risky discount. The **credit triangle** $S \approx (1-R)\lambda$ is the low-hazard short-tenor limit and the most important formula in credit trading.

4. **T-forward measure eliminates stochastic discounting.** Changing numeraire from M_t to $P_t(T_0)$ makes the discount factor deterministic at expiry (since $P_{T_0}(T_0) = 1$) and turns the forward bond $X_t = P_t(T_1)/P_t(T_0)$ into a martingale with deterministic volatility $\Sigma_t = \sigma(B_t(T_0) - B_t(T_1))$.

5. **Vasicek-Black bond-option formula.** A European call on a zero- coupon bond has the Black-76 form

$$\mathfrak{g}_t = P_t(T_0)\{X_t \Phi(d_+) - K\Phi(d_-)\}$$

   with $d_\pm$ computed from the deterministic total variance $\int_t^{T_0} \Sigma_s^2\,\mathrm{d}s = (\sigma^2/\kappa^2)(1 - e^{-\kappa(T_1 - T_0)})^2 \cdot (1 - e^{-2\kappa(T_0 - t)})/(2\kappa)$.

6. **Jamshidian decomposition reduces a swaption to a portfolio of bond options.** Exploit monotonicity of $\sum_i c_i P_{T_0}(T_i)$ in r_{T_0}: find the critical rate r^*, compute effective strikes K_i, and price each bond option by Black-76. Exact for one-factor Gaussian models; approximate but still useful for multi-factor.

7. **One-factor Vasicek is the start, not the destination.** G2++ for curve richness, HJM for forward-curve primacy, LMM for direct forward-rate modelling, SABR for vol smile. Each extension fixes a specific limitation of the 1-factor Gaussian framework.

8. **PDE versus MC is a dimensionality trade-off.** PDE wins for 1-2 factor vanilla products; MC wins for high-dimensional and path- dependent payoffs; hybrids handle edge cases. The underlying mathematics is unchanged — only the numerical implementation differs.

9. **The chapter's unifying principle.** Every rate derivative prices as a conditional expectation of a payoff functional under the most convenient martingale measure. Match the numeraire to the payoff's natural scale, and the pricing collapses to a Black-type lognormal expectation. That sentence — one idea — ties every formula in this chapter together.

14.11 13.11 Reference formulas

14.11.1 Swaps

$$
\begin{aligned}
V_0^{\text{fix}} &= \sum_{m=1}^{n} \Delta t_m F N P_0(t_m) = F \cdot \mathcal{A}_0 \cdot N \\
V_0^{\text{fl}} &= (P_0(t_0) - P_0(t_n))N \\
S_0 &= \frac{P_0(t_0) - P_0(t_n)}{\sum_m \Delta t_m P_0(t_m)} = \frac{P_0(t_0) - P_0(t_n)}{\mathcal{A}_0} \\
S_t &= \frac{P_t(t_{k_t}) - P_t(t_n)}{\sum_{m=k_t+1}^{n} \Delta t_m P_t(t_m)} \\
\text{DV01} &\approx \mathcal{A} \cdot 0.0001 \cdot N
\end{aligned}
$$

14.11.2 Callable bonds (Hull-White lattice)

Backward induction with call test:

$$
V_n^{(i)} = \begin{cases} \min(\text{call price}_n, V_n^{(i),\text{cont}}) & \text{if } t_n \text{ is call date} \\ V_n^{(i),\text{cont}} & \text{otherwise} \end{cases}
$$

where $V_n^{(i),\text{cont}} = e^{-r_n^{(i)}\Delta t}(p_u V_{n+1}^{(i+1)} + p_m V_{n+1}^{(i)} + p_d V_{n+1}^{(i-1)}) + c_n$.
Value decomposition: $V^{\text{callable}} = V^{\text{straight}} - V^{\text{embedded call}}$.

14.11.3 CDS (constant hazard λ under $\mathbb{Q}$, independent of r)

$$
\begin{aligned}
\widetilde{P}_0(t) &= e^{-\lambda t} P_0(t) \quad \text{(risky discount)} \\
V_0^{\text{prem}} &= S \cdot N \cdot \sum_{\ell} \Delta t_\ell \widetilde{P}_0(t_\ell) \\
V_0^{\text{D}} &= (1-R) \cdot N \cdot \int_0^{t_n} \lambda e^{-\lambda\tau} P_0(\tau)\,\mathrm{d}\tau \\
S_0^{\text{CDS}} &= \frac{(1-R)\int_0^{t_n} \lambda e^{-\lambda\tau} P_0(\tau)\,\mathrm{d}\tau}{\sum_\ell \Delta t_\ell \widetilde{P}_0(t_\ell)} \\
S &\approx (1-R)\lambda \quad \text{(credit triangle, low } \lambda)
\end{aligned}
$$

Time-varying hazard: replace $e^{-\lambda t}$ with $S(t) = \exp(-\int_0^t \lambda(u)\,\mathrm{d}u)$ and $\lambda e^{-\lambda\tau}$ with $\lambda(\tau)S(\tau)$ inside the default-leg integral.

14.11.4 T-forward measure (cross-reference to CH05 for Girsanov derivation)

$$
\left.\frac{\mathrm{d}\widetilde{\mathbb{Q}}_{T_0}}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_t} = \frac{P_t(T_0)/M_t}{P_0(T_0)/M_0} \qquad \mathrm{d}\widetilde{W}_t^0 = \mathrm{d}\widehat{W}_t + B_t(T_0)\sigma\,\mathrm{d}t
$$

14.11.5 Bond SDE and forward bond

$$\frac{\mathrm{d}P_t(T)}{P_t(T)} = r_t\,\mathrm{d}t - \sigma B_t(T)\,\mathrm{d}\widehat{W}_t \quad (\text{under } \mathbb{Q})$$

$$X_t = \frac{P_t(T_1)}{P_t(T_0)}, \qquad X_{T_0} = P_{T_0}(T_1)$$

$$\frac{\mathrm{d}X_t}{X_t} = \Sigma_t\,\mathrm{d}\widetilde{W}_t^0, \qquad \Sigma_t = \sigma(B_t(T_0) - B_t(T_1))$$

14.11.6 European bond call (Vasicek-Black)

$$\mathfrak{g}_t = P_t(T_0)\{X_t\Phi(d_+) - K\Phi(d_-)\} \qquad d_\pm = \frac{\ln(X_t/K) \pm \frac{1}{2}\int_t^{T_0}\Sigma_s^2\,\mathrm{d}s}{\sqrt{\int_t^{T_0}\Sigma_s^2\,\mathrm{d}s}}$$

$$\int_t^{T_0}\Sigma_s^2\,\mathrm{d}s = \frac{\sigma^2}{\kappa^2}(1 - e^{-\kappa(T_1-T_0)})^2 \cdot \frac{1 - e^{-2\kappa(T_0-t)}}{2\kappa}$$

14.11.7 Put-call parity for bond options

$$\mathfrak{g}_t^{\text{call}} - \mathfrak{g}_t^{\text{put}} = P_t(T_1) - K \cdot P_t(T_0)$$

14.11.8 Jamshidian decomposition (swaption → portfolio of bond options)

$$\text{swaption}_0 = \sum_{i=1}^{N} c_i \cdot \text{BondOption}(T_0, T_i, K_i), \qquad K_i = P_{T_0}(T_i)\big|_{r=r^*}$$

with r^* solving $\sum_i c_i \exp(A_{T_0}(T_i) - B_{T_0}(T_i)r^*) = K$.

14.11.9 Greeks for the bond call

$$\Delta = P_t(T_0)\Phi(d_+), \qquad \Gamma = \frac{P_t(T_0)\phi(d_+)}{X_t\sqrt{\int_t^{T_0}\Sigma_s^2\,\mathrm{d}s}}$$

$$\text{Vega} = \frac{\partial\mathfrak{g}_t}{\partial\sigma} = P_t(T_0)X_t\phi(d_+)\sqrt{\int_t^{T_0}\Sigma_s^2\,\mathrm{d}s}/\sigma$$

14.11.10 Multi-factor and HJM

$$\begin{aligned}
\text{G2++:}\quad & r_t = x_t + y_t + \phi(t), \qquad \mathrm{d}x_t = -ax_t\,\mathrm{d}t + \sigma_1\,\mathrm{d}W_t^x \\
& \mathrm{d}y_t = -by_t\,\mathrm{d}t + \sigma_2\,\mathrm{d}W_t^y, \qquad \langle W^x, W^y\rangle = \rho\,t \\
\text{HJM:}\quad & \mathrm{d}f_t(T) = \alpha(t,T)\,\mathrm{d}t + \sigma(t,T)\,\mathrm{d}W_t \\
& \alpha(t,T) = \sigma(t,T)\int_t^T \sigma(t,s)\,\mathrm{d}s \quad (\text{no-arb drift}) \\
\text{LMM:}\quad & \mathrm{d}L_t(T_{i-1},T_i) = L_t(T_{i-1},T_i)\sigma_i(t)\,\mathrm{d}W_t^{(i)} \\
\text{SABR:}\quad & \mathrm{d}F_t = \alpha_t F_t^\beta\,\mathrm{d}W_t, \qquad \mathrm{d}\alpha_t = \nu\alpha_t\,\mathrm{d}Z_t
\end{aligned}$$

14.11.11 Numerical methods — choosing between PDE and MC

Regime	Method
1-factor vanilla	PDE (Crank-Nicolson)
2-3 factor vanilla/mild exotic	PDE or MC
High-dimensional (LMM, HJM)	Monte Carlo

Regime	Method
American/Bermudan low-dim	PDE with min-test
American high-dim	Longstaff-Schwartz MC
XVA on large book	Monte Carlo (GPU)
Path-dependent exotic	Monte Carlo

15 Chapter 14 — Heston Stochastic Volatility

Part V opens with the equity-side capstone. CH06 handed us the Black–Scholes PDE under the assumption that instantaneous variance σ^2 is a known constant; every chapter since has carried that assumption along. The market, of course, does not. Implied-volatility surfaces slope, twist, and deform in ways that a constant-variance GBM cannot reproduce. Heston's 1993 model is the cleanest fix: promote the variance to its own mean-reverting stochastic process, keep tractability by choosing a square-root (CIR) dynamics for the variance, and pay for the extra degree of freedom with a second Brownian motion that is correlated with the underlying. The reward is a semi-closed pricing formula via Fourier inversion, a genuine stochastic smile, and a hedging framework that admits volatility as a traded risk factor.

This chapter leans on the full Part II toolkit. Two-dimensional Itô's lemma (CH03) is used in every derivation; multi-dimensional Girsanov (CH05) is how we move from the physical bivariate SDE to the risk-neutral dynamics; Feynman–Kac (CH04) underlies the Riccati ODE system for the characteristic function; and the GBM / Black–Scholes machinery of CH06 is the benchmark we compare against. Monte Carlo variance-reduction techniques introduced in CH10 resurface when we discuss simulation-based Heston pricing. The chapter is almost entirely self-contained analytically, but pedagogically it is the payoff for Parts II and III.

15.1 Chapter map

1. **§14.1 Bivariate SDE** — motivation, specification, and what each parameter controls.
2. **§14.2 Feller condition** — when variance stays strictly positive.
3. **§14.3 Change of measure** — market price of variance risk, risk-neutral drift adjustments, uniqueness failure and calibration implications.
4. **§14.4 Implied volatility** — why constant-σ GBM is inadequate and what shapes a stochastic-vol model must generate.
5. **§14.5 Characteristic function** — Riccati ODE derivation for the log-price characteristic function.
6. **§14.6 Fourier-inversion pricing** — recovering P_1 and P_2 and assembling the call price.
7. **§14.7 Greeks and calibration** — delta, vega, the two extra vol-of-vol and correlation sensitivities, calibration workflow.
8. **§14.8 Variance swaps** — a pure-variance product that prices by expectation under $\mathbb{Q}$.
9. **§14.9 Worked example** — a numerical Heston vs. Black–Scholes comparison.
10. **§14.10–§14.10C Limitations, comparisons, extensions, and summary.**
11. **§14.11 Key takeaways** and **Appendix A** (reference formulas) plus **Appendix B** (intuition checklist).

15.2 14.1 Bivariate SDE — motivation

The Black–Scholes world is complete: a constant volatility σ plus the bank account span every contingent claim. Empirically, option markets violate this — implied vol exhibits a smile / skew across strike and a term structure across maturity. The Heston (1993) model is the canonical fix: promote the variance v_t to its own mean-reverting square-root SDE, correlate it with the spot, and retain semi-closed-form pricing via a characteristic function. This chapter rebuilds the derivation from first principles — bivariate SDE → Feller condition → change of measure with market price of variance risk → characteristic-function ansatz → Riccati ODEs → Fourier-inversion pricing (the P_1, P_2 probabilities) — and finishes with calibration / Greeks notes and variance swaps.

To understand why Heston matters so deeply in modern quantitative finance, you must first understand why practitioners spent the better part of two decades trying to fix Black–Scholes before settling on stochastic volatility. The 1987 crash was the watershed moment: in the space of a single trading session, the S&P 500 dropped more than twenty percent, and the implied volatility of out-of-the-money puts — already slightly elevated relative to at-the-money — exploded upward and never came back down. Before the crash, equity implied-vol surfaces were reasonably flat; after, they developed the persistent left skew that defines them to this day. The market learned, in a matter of hours, that tail risk was systematically underpriced, and the institutional memory of that lesson has kept OTM puts bid ever since. Black–Scholes, with its single-σ assumption, was suddenly seen to be systematically wrong in a way that could not be papered over with small parameter tweaks. A new framework was needed.

The options industry responded with a cascade of models. Dupire's local volatility (1994) provided an exact fit to any arbitrage-free implied-vol surface by allowing σ to be a deterministic function of spot and time — $\sigma(S, t)$. This solved the pricing problem for vanillas but failed catastrophically on forward-smile dynamics: the forward smile predicted by local vol flattens unrealistically with horizon, which makes it a poor model for path-dependent exotics like cliquets, forward-starts, and variance contracts. Heston (1993) took a different route: instead of making volatility a deterministic function of state, make it an independent stochastic process with its own SDE. The result is richer dynamics, realistic forward smiles, and — crucially — a tractable characteristic function via the affine structure. Later, SABR (Hagan et al. 2002) offered a different stochastic-vol parameterisation optimised for interest-rate vol; Bates (1996) added jumps; Heston-with-time-dependent-parameters extended the basic model to the full term structure; and more recently, rough volatility (Gatheral–Jaisson–Rosenbaum 2014) showed that Heston's Markovian-diffusion assumption itself is too smooth compared to realised vol, pointing toward fractional Brownian motion as the correct driving noise. Heston sits at the centre of this timeline — the first tractable stochastic-vol model, still the industry default for many applications, and a pedagogical gateway to the richer models that succeeded it.

Before any mathematics, it is worth pausing on what the word "volatility" even means in a Black–Scholes context. In that framework σ is a single scalar, a number, fed once into the pricing formula and never revisited. The pricing engine does not care which option you are valuing: the same σ governs the one-week at-the-money call, the six-month deep out-of-the-money put, and every contract in between. When traders invert Black–Scholes against actual market prices, however, they discover that no single σ fits the whole board. An at-the-money three-month option might imply 18%, while a three-month put struck twenty percent below spot implies 26%, and a call struck twenty percent above spot implies 16%. There is no arithmetic error at work; the implied vol surface is simply not flat. That surface has a shape — a smile, a skew, a term structure — and the shape is persistent, economically meaningful, and structurally incompatible with Black–Scholes' single-number world.

The economic content of that surface shape is what makes stochastic volatility models necessary. Consider what it means for an OTM put to trade at 26% implied vol while the ATM option trades at 18%. Interpreting both through Black–Scholes, the market is quoting two different "volatilities" for the same underlying. Of course the underlying has only one realised volatility path, so what the market is really telling you is that Black–Scholes is the wrong pricing lens and the extra premium on the OTM put encodes information that Black–Scholes cannot represent. The most important piece of that information is the joint behaviour of spot and vol: when markets crash, vol spikes, which makes OTM puts pay off on two fronts simultaneously. The 8-point spread between 26% and 18% implied vol is the market's measured compensation for the joint-distribution risk that Black–Scholes ignores. To price that spread correctly, you need a model in which vol itself is stochastic and correlated with spot. That is exactly what Heston delivers.

The persistence of the smile deserves attention too. If the smile were a short-term arbitrage opportunity, sophisticated traders would arbitrage it away within days. Instead, the smile has persisted for nearly four decades, through multiple volatility regimes, Fed tightening cycles, recessions, and policy shocks. The only reasonable conclusion is that the smile represents a genuine risk premium — the compensation demanded by option sellers for accepting tail risk — and that premium is not going away because the underlying risk is real and ongoing. Any model that hopes to price or hedge options meaningfully must therefore reproduce this premium structurally. Heston does it via the interaction of ρ (spot-vol correlation) and α (vol-of-vol), and the calibration of those parameters tells you, quantitatively, how much tail risk the market is currently pricing in.

15.2.1 14.1.1 Why stochastic vol produces a smile

A Black–Scholes lognormal has a single scale parameter σ: the risk-neutral density of $\ln X_T$ is a tidy Gaussian with one variance. Under Heston the conditional density (conditioning on the whole path of v_t) is still lognormal — but the mixture over the v-path puts weight on very quiet and very loud regimes, producing fatter tails than a single lognormal. Fatter tails $\rightarrow$ deep OTM options are worth more than BS would suggest $\rightarrow$ inverted BS gives higher implied vols at the wings. The non-zero ρ then asymmetrises that mixture: when spot falls, the vol tends to spike (for $\rho < 0$), which loads the left tail more than the right. This is the skew. In short: α (vol-of-vol) controls the wings (curvature / smile); ρ controls the slope (skew); κ controls how quickly the term-structure of smile flattens; θ anchors long-dated ATM; v_0 anchors short-dated ATM.

The "mixture of lognormals" intuition deserves a worked example. Suppose we toss a fair coin once. On heads, the stock trades with $\sigma = 10\%$ for six months; on tails, with $\sigma = 40\%$ for six months. Each regime is Black–Scholes-lognormal, but the mixture — the density you see before the coin is tossed — is bimodal-ish, with a narrow peak from the 10% regime and a wide bell from the 40% regime. The wide-regime contribution dominates at large deviations,

making the mixture fatter-tailed than either component alone. If you now compute European option prices off this mixture and invert them through Black–Scholes, you get implied vols that rise as you move away from at-the-money — the wings of the mixture are fatter than any single lognormal, so the BS-inversion has to fatten σ to match. That is a smile, generated by mixing just two regimes. Heston does the same thing, but with a continuum of regimes indexed by the path of v_t, and with a structurally motivated way of weighting those regimes via the CIR variance dynamics.

A second useful analogy: think of Heston as Black–Scholes with a stochastic "volatility dial" that the market can turn. At each instant, the dial has a setting v_t; the spot evolves with that instantaneous variance; and simultaneously the dial itself wiggles randomly, mean-reverts toward a target level, and is mechanically coupled to the spot. The market's expectation of option payoffs is then the average over all possible dial trajectories. The Black–Scholes world is the special case where the dial is welded at a single setting — no fluctuation, no mean reversion, no coupling. Heston admits a family of dial trajectories, and the parameters $(\kappa, \theta, \alpha, \rho, v_0)$ shape that family. The richness of the smile follows directly.

A helpful mental picture is this. Imagine you could not observe spot directly but could only see a noisy thermometer attached to it — one whose reading fluctuated, drifted, and occasionally spiked or collapsed. Black–Scholes says the thermometer is broken and always reads the same number; Heston says the thermometer is itself a dynamic object with its own inertia, its own overshoot, its own long-run comfort zone, and a tendency to rise when the patient's temperature falls. The market, watching the thermometer, prices options as if they pay off on the joint behaviour of patient and thermometer. The richness of the implied vol surface is what the market reveals about the joint dynamics it expects.

One useful way to sharpen this picture is to think about what it would take to "hedge the thermometer" in a Heston world. In Black–Scholes, the broken thermometer means you only need to hedge the patient — the spot — and a single delta hedge leaves you perfectly replicated. In Heston, the thermometer moves independently, so you must hedge both the spot and the variance exposure. But here is the twist: you cannot trade instantaneous variance directly. There is no exchange-listed "variance contract" that pays off on v_t alone. The closest instruments are variance swaps (which pay on realised variance over a window, not instantaneous variance) and liquid options themselves. So the practitioner must hedge variance exposure using other options — typically ATM options with the same or nearby maturities — because those are the most sensitive to v_t. This is why vega hedging in a Heston world is called "vol hedging with options," and why exotic desks carry complicated option inventories to neutralise their vol exposures. The incomplete-market structure of Heston is not an abstract theorem; it translates directly into the messiness of real-world vol hedging.

A final framing that proves useful in practice is thinking of Heston as a two-factor model where one factor (spot) is observed and tradable and the other (variance) is partially observed and only indirectly tradable. Variance is "observed" in the sense that a long record of high-frequency returns gives you a noisy estimate of current instantaneous variance. It is "not directly observed" in that you cannot read v_t off a screen — you have to estimate it. The natural estimator is a rolling quadratic variation of intraday log-returns, and good estimators can achieve remarkable precision over short windows (think days, not weeks). But the estimator has a lag, it has measurement noise, and it is subject to its own model assumptions (e.g. about intraday seasonality, overnight gaps, and jump activity). The upshot is that the Heston state (F_t, v_t) is only partially observable, and any honest implementation has to deal with the resulting filtering problem — usually by treating v_0 as an extra calibration parameter that is fit from options rather than from time-series data.

The chapter is organised to follow the natural arc from intuition to machinery to payoff. We begin by writing down the bivariate SDE and unpacking what each parameter does physically. We pause at the Feller condition because it controls whether the variance process is well-posed, and because in practice it is the first diagnostic any calibrator checks. We then execute the change of measure, being careful about the market price of variance risk, because risk-neutral pricing in an incomplete market is not automatic — we must choose a measure rather than simply derive one. The characteristic function drops out of an affine ansatz and a pair of Riccati ODEs. Fourier inversion delivers the call price as two Black–Scholes-like probabilities. The chapter closes with calibration practice, Greeks in the Heston sense, variance swaps, and a worked numerical example.

One last framing before we dive in. You should read the derivation that follows as a case study in affine modelling — the broad family of models where drift and diffusion covariance are affine in the state. Affine models have an astonishingly clean mathematical structure: their characteristic functions solve finite-dimensional ODE systems regardless of state-space dimension, which makes them computationally tractable even when the state is high-dimensional. Heston is the two-dimensional, continuous-state archetype; the same techniques extend to multi-factor short-rate models

(Cox–Ingersoll–Ross, affine term-structure models — see CH12), to jump-diffusions (Duffie–Pan–Singleton, Bates), and to credit models (CIR-style intensity processes). Mastering the Heston derivation therefore teaches you not just how to price Heston options, but how to exploit affine structure generally — a technique that shows up throughout modern quantitative finance. The Riccati ODEs we derive below are the tip of a very large iceberg.

A pedagogical suggestion before proceeding. As you work through this chapter, keep two mental images side-by-side: the pricing image (option values derived by Fourier inversion of the Heston characteristic function) and the dynamics image (the bivariate SDE driving spot and variance over time). These two images correspond to the same model, but one focuses on terminal distributions (what the option pays at maturity) and the other focuses on paths (how the underlying gets there). Much of the elegance of Heston — and much of the subtlety of the derivations below — lies in the interplay between these two views. The SDE gives you dynamics; the characteristic function gives you pricing; the Riccati ODE system is the bridge between them, encoding the evolution of expectations under the dynamics. When you can flip between dynamics view and pricing view at will, you have understood Heston deeply.

Another complementary framing: Heston is a canonical example of "structural" modelling in quantitative finance, as opposed to "reduced-form" modelling. A structural model specifies the underlying dynamics — here, the bivariate SDE for (F, v) — and derives prices as consequences of those dynamics. A reduced-form model specifies prices directly as functions of observables, without necessarily committing to a consistent underlying process. Black–Scholes is structural (constant-vol diffusion); local vol is reduced-form (arbitrary function $\sigma(S, t)$); Heston is structural (bivariate diffusion); SABR is structural (different SDE); rough vol is structural (fractional SDE). Structural models carry more theoretical weight because they respect dynamical consistency, but reduced-form models can be more flexible in fitting specific observables. For teaching and for deep understanding, structural models are preferable; for production pricing at maximum accuracy, hybrid structural/reduced-form (local-stoch-vol) is often the right choice. Heston is the textbook structural model for stochastic vol, and mastering it sets up the framework for everything that follows.

A final note on the organisation of what follows. Each section ends with one or more figures that visualise the mathematical content. These figures are not decoration; they are the most direct way to build intuition about how parameters shape behaviour. Treat each figure as a piece of content to be studied, not just an illustration to be glanced at. The patterns that emerge from the figures — the smile shape, the skew rotation, the term-structure decay — are the observable manifestations of the parameters we calibrate. Being able to look at a smile and estimate its Heston parameters by eye is a skill worth developing, and the figures are where that skill is forged.

15.2.2 14.1.2 FTAP recap and incompleteness

No arbitrage $\iff$ there exists a measure $\mathbb{Q} \sim \mathbb{P}$ such that all tradeables f_t/M_t are $\mathbb{Q}$-martingales. This holds always. The measure is unique only if the market is complete; in incomplete markets (like stochastic vol) there is no unique $\mathbb{Q}$.

This single sentence compresses two of the deepest results in mathematical finance, so it is worth unpacking. The First Fundamental Theorem of Asset Pricing (FTAP I) states that a financial market is arbitrage-free if and only if there exists an equivalent martingale measure — a probability measure $\mathbb{Q}$ that is equivalent to the physical measure $\mathbb{P}$ (meaning they agree on which events have probability zero) and under which every tradeable asset, once discounted by the numeraire, follows a martingale. This is not a computational trick; it is a statement about the structure of arbitrage. If such a measure exists, no self-financing trading strategy can produce a riskless profit from zero initial wealth, because under $\mathbb{Q}$ every discounted portfolio is a martingale — its expectation is its current value — and expectations cannot be bent upward for free. Conversely, if no such measure exists, one can construct an explicit arbitrage. The theorem is an iff: existence of $\mathbb{Q}$ equals absence of arbitrage.

The Second Fundamental Theorem (FTAP II) then asks whether $\mathbb{Q}$ is unique. The answer: $\mathbb{Q}$ is unique if and only if the market is complete — meaning every contingent claim can be replicated by dynamic trading in the underlying assets. Completeness, in turn, requires that the number of independent risk factors equals the number of tradeable assets minus one (accounting for the numeraire). In Black–Scholes we have one risk factor (the Brownian motion driving spot) and two tradeables (spot and bank), so the count matches, the market is complete, and $\mathbb{Q}$ is unique. In Heston we have two risk factors (the Brownian motions driving spot and variance) and still only two tradeables, so the count is off, the market is incomplete, and $\mathbb{Q}$ is not unique. This is the mathematical origin of the market-price-of-variance-risk parameter that will appear shortly. It is not a modelling artefact; it is an inescapable consequence of the incompleteness that stochastic vol introduces. The canonical treatment of FTAP and the Radon–Nikodym machinery that underpins it is in CH05; in what follows we use those results without re-deriving them.

Understanding incompleteness this way helps you interpret the Heston pricing framework correctly. We are not "deriving" the risk-neutral measure from the physical measure; we are choosing one out of infinitely many candidates. That choice must be pinned down by market data — specifically, by the prices of vol-sensitive instruments like options and variance swaps. A Heston calibration is therefore not "estimating the true parameters" but rather "identifying which risk-neutral measure is currently consistent with observed option prices." The physical- measure parameters are a related but distinct set, estimated from returns time-series, and the gap between the two parameter sets is the variance risk premium — a systematically observed phenomenon, not a modelling choice.

Incompleteness is the conceptual price we pay for realism. In Black–Scholes, the single risk factor is the Brownian motion driving the spot, and the spot is tradable; any contingent claim can be replicated by dynamic trading in spot and bank. In Heston, there are two risk factors — the Brownian motion driving the spot and a second Brownian motion driving variance — but there is still only one tradable asset (the spot or its futures contract) plus bank. The variance process is not directly tradable; you cannot buy "instantaneous variance" on an exchange the way you buy a stock. This rank mismatch between risk factors and hedging instruments is precisely what makes the market incomplete. The consequence is that many different risk-neutral measures price the traded assets consistently, and choosing among them requires an additional input — the market price of variance risk — which the data must pin down.

To make this concrete, consider what "hedging" even means in an incomplete market. In Black–Scholes, for any contingent claim $f(S_T)$, there exists a self-financing strategy (Δ_t, β_t) in spot and bank that replicates f exactly at every time and in every scenario — zero residual risk. You can therefore price f by the no-arbitrage principle: its price must equal the cost of constructing the replicating portfolio. In Heston, no such exact replication exists using spot and bank alone. If you run a delta-hedge based on spot only, you will be exposed to variance innovations; your hedged portfolio will have residual P&L proportional to the vega times the variance shock. The best you can do with spot alone is minimise some residual-risk metric — variance-of-P&L, say — but you cannot eliminate it. To eliminate it, you must add another hedging instrument that is sensitive to variance — typically another option — but then you need that option to be priced, which requires a model, which requires a $\mathbb{Q}$, and we are back to the choice problem.

The resolution in practice is to assume (or impose) a specific choice of $\mathbb{Q}$, calibrate it to observed option prices (so that the market's implied λ^v is respected by construction), and then perform hedging in a model-consistent way. This is called "hedging under the calibrated measure" and it is what every vol desk does implicitly. The hedges are not perfect — they are model hedges — and the residual P&L reflects model risk rather than pure incompleteness-risk. But the framework at least gives you a coherent way to think about hedging and risk.

There is a philosophical point here that is easy to miss. In a complete market, the "price" of a contingent claim is unambiguous — it is the replication cost. In an incomplete market, "price" is an economic concept, determined by demand and supply, risk preferences, and market frictions. The model gives you a framework for representing those ingredients compactly (via the choice of $\mathbb{Q}$), but the model does not determine the price on its own. This is why calibration, rather than pure derivation, is the central activity of a stochastic vol practitioner: the prices exist because people trade, and the model is merely the language in which those prices are expressed.

15.2.3 14.1.3 From Black–Scholes drift to the bivariate SDE

Under $\mathbb{P}$, a risky asset X_t that is traded satisfies

$$\mathrm{d}X_t = \mu_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}\tilde{W}_t^{\mathbb{P}}, \qquad \frac{f_t}{M_t} = \mathbb{E}^{\mathbb{Q}}\left[\frac{f_T}{M_T}\right], \tag{14.1}$$

and the $\mathbb{P}$-Brownian motion is linked to the $\mathbb{Q}$-Brownian motion by

$$\mathrm{d}\tilde{W}_t = \lambda_t \,\mathrm{d}t + \mathrm{d}W_t^{\mathbb{Q}}. \tag{14.2}$$

Substituting,

$$\mathrm{d}X_t = (\mu_t - \lambda_t \sigma_t)\,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t^{\mathbb{Q}}, \tag{14.3}$$

where λ_t is the market price of risk. If X is itself traded, then under $\mathbb{Q}$ its drift must be rX_t, pinning down $\mu_t - \lambda_t \sigma_t = rX_t$. This is precisely the Girsanov shift derived in CH05 — the Radon–Nikodym density $Z_t = \exp\!\big(-\int_0^t \lambda_s \,\mathrm{d}\tilde{W}_s^{\mathbb{P}} - \frac{1}{2}\int_0^t \lambda_s^2 \,\mathrm{d}s\big)$ is what converts the $\mathbb{P}$-drift into the $\mathbb{Q}$-drift, state by state. The reader who wants the full derivation should consult CH05 §5; we use the conclusion here.

Read this shift intuitively. Under the physical measure $\mathbb{P}$, the asset drifts at its expected return, which includes a risk premium over the risk-free rate — that is how risky assets compensate their holders for bearing risk. The change of measure does not change any actual path; it rebalances probability weight toward the states that have been systematically overweighted by risk aversion. Under $\mathbb{Q}$, the asset appears to drift at r because the premium has been re-absorbed into the measure. The same logic applies — but now in two dimensions — when we introduce stochastic variance.

Here is another way to see it. Imagine running Monte Carlo simulation of the spot under $\mathbb{P}$ and under $\mathbb{Q}$. The individual paths look similar — in fact, on any individual path, the realisation of the Brownian motion is the same (Girsanov tells us that the path spaces are essentially isomorphic). What differs is the weight that each path carries in the expectation. Under $\mathbb{P}$, the "good" paths (with high realised returns) have high probability; under $\mathbb{Q}$, those good paths are weighted less because risk-averse investors have bid up their prices (making them seem less attractive from the valuation standpoint). The effect is that $\mathbb{Q}$-expectations come out lower for linear payoffs on risky assets than $\mathbb{P}$-expectations, which is exactly the "risk-free rate" drift you would expect. None of this is about the trajectory of any particular path; it is all about the measure.

The famous economic interpretation is in terms of pricing kernels or stochastic discount factors. The Radon–Nikodym derivative $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ is essentially proportional to a representative investor's marginal utility — a number that is high in bad states of the world (when investors value an extra dollar most) and low in good states. Discounting payoffs by this kernel and taking a $\mathbb{P}$-expectation is equivalent to taking a $\mathbb{Q}$-expectation and discounting by the risk-free rate. Derivatives pricing in incomplete markets is thus fundamentally linked to asset-pricing theory: different measures correspond to different assumptions about the representative investor's risk aversion and preferences. When you calibrate Heston to market prices, you are implicitly inferring what marginal utility looks like in the vol dimension — at least in the part of the state space where liquid options exist to probe.

Futures. For a futures price F_t, the Black (1976) argument (CH08) gives under $\mathbb{Q}$

$$\frac{\mathrm{d}F_t}{F_t} = \sigma\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \text{(Black model for futures prices).} \tag{14.4}$$

Under $\mathbb{P}$,

$$\frac{\mathrm{d}F_t}{F_t} = \mu_t\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}. \tag{14.5}$$

We work in the futures frame partly out of convenience — futures are martingales under the risk-neutral measure with no drift to carry around — and partly because it highlights the key point that the volatility structure is what distinguishes Heston from Black, not the drift. The drift in the risk-neutral world is pinned down by arbitrage; the interesting degree of freedom is the stochastic behaviour of the diffusion coefficient.

The practical virtue of the futures-price framework goes beyond cleanliness. For index and equity markets, listed options actually expire on forward-contract-settled underliers or near-dated futures; for FX options, the relevant underlier is a forward rate; for commodity options, the entire pricing paradigm is built on futures rather than on any elusive "spot" price. So Black (1976) is not a toy simplification — it is the framework that actually prices the bulk of liquid options traded globally. Heston's original 1993 paper followed the Black tradition (although Heston expressed things in terms of spot with carry; the two are isomorphic), and the industry has largely standardised on the futures-frame parameterisation. When a practitioner says "$\mathrm{d}F/F = \sqrt{v}\,\mathrm{d}W$," they are stating the Heston–Black benchmark dynamics that underlie essentially every vanilla calibration you will encounter. Even when the underlying is a spot rather than a futures, the drift term $r - q$ (rate minus dividend yield) can be absorbed into the forward price, and the residual dynamics look like (14.4) with no drift. In other words: by working in the futures frame, you focus the analysis exactly on what makes Heston different from Black–Scholes — the diffusion $\sqrt{v_t}$ — and defer all drift-related complications to a thin wrapper around the core SDE.

15.2.4 14.1.4 The Heston bivariate system

We want to correct for stochastic volatility. Replace the constant σ by $\sqrt{v_t}$ and give v_t its own SDE:

$$\boxed{\frac{\mathrm{d}F_t}{F_t} = \mu_t\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^{\mathbb{P},F}} \tag{14.6}$$

$$\boxed{\mathrm{d}v_t = \kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - v_t)\,\mathrm{d}t + \alpha\sqrt{v_t}\,\mathrm{d}W_t^{\mathbb{P},v}} \tag{14.7}$$

with

$$\mathrm{d}W_t^{\mathbb{P},F} \cdot \mathrm{d}W_t^{\mathbb{P},v} = \rho\,\mathrm{d}t, \qquad (W^{\mathbb{P},F}, W^{\mathbb{P},v}) \text{ correlated.} \tag{14.8}$$

Look carefully at the choice $\sqrt{v_t}$ for the diffusion coefficient. It is not incidental. Modelling in the variance rather than in the volatility pays two dividends. First, the variance is the natural additive quantity: variances of independent returns add, volatilities do not, and any equation that treats variance linearly will be cleaner than the equivalent vol-level equation. Second, the square-root CIR diffusion $\alpha\sqrt{v_t}\,\mathrm{d}W^v$ has the beautiful property that its diffusion coefficient vanishes as $v_t \to 0$, which acts as a soft floor: a variance that wanders toward zero stops diffusing and is pulled back up by the mean-reversion drift. This is how Heston constructs a process that stays non-negative under mild parameter conditions, without resorting to reflecting barriers or hard truncations.

Contrast this with the SABR alternative. SABR models the instantaneous vol (not variance) with lognormal dynamics: $\mathrm{d}\sigma_t = \alpha\sigma_t\,\mathrm{d}W^\sigma$. Lognormal vol automatically stays positive, which is appealing, but the tradeoff is that SABR dynamics are not affine, and SABR does not admit a closed-form characteristic function. Instead, SABR is typically priced via Hagan's asymptotic expansion — an approximation that works brilliantly for short maturities and ATM moneyness but degrades at wings and long dates. Heston gives up lognormal positivity in exchange for the affine structure, and the square-root floor on the CIR process is the mathematical trick that makes this exchange work. Either choice is defensible; both dominate naive alternatives like an arithmetic Brownian motion on vol (which can go negative and cannot be rescued by any floor).

A subtle point about (14.6): the spot is driven by $\sqrt{v_t}\,\mathrm{d}W^F$, not by $\sqrt{v_t}\,\mathrm{d}t$ plus $\mathrm{d}W^F$ with some scaling. The diffusion coefficient is a random function of v_t, which is itself stochastic. This is what makes the spot a "stochastic-volatility" process rather than a "time-changed" Brownian motion. Equivalently, you can think of spot's log-returns as Brownian motion scaled at each instant by $\sqrt{v_t}$: when v_t is high, returns are more volatile; when v_t is low, returns are calmer. The picture to keep in mind is a Brownian motion whose clock speed is itself random, fluctuating between slow and fast phases. This is not a metaphor — there is a formal equivalence between stochastic vol processes and time-changed Brownian motions, where the time change is the integrated variance. Ocone's representation of continuous local martingales makes this precise. For our purposes, it means that many results about Brownian motion carry over to Heston's log-price after a random time change.

Reading off (14.7):

- $\kappa^{\mathbb{P}}$ — mean-reversion rate of variance.
- $\theta^{\mathbb{P}}$ — mean-reversion level (long-run variance).
- α — vol-of-vol.
- ρ — spot/vol correlation (empirically negative for equity indices — the "leverage effect").
- The $\sqrt{v_t}$ diffusion is the CIR / square-root form: such processes are called Feller processes.

Each of these has an economic interpretation worth lingering over. The mean-reversion rate κ answers the question, "if volatility is currently elevated, how quickly does the market expect it to return to normal?" A large κ corresponds to a rubber-band variance that snaps back quickly — news-driven spikes fade within days. A small κ corresponds to a sticky variance that can stay elevated for quarters or years. The long-run level θ is what volatility reverts to — the "cruising altitude" of the market. For broad equity indices, θ calibrated over long histories tends to land in the territory of 20%–25% annualised vol, which is roughly the long-run average of realised equity vol. The vol-of-vol α governs how chaotic the variance itself is — whether the vol process is a gentle drift or a ragged, spiky trajectory. It is essentially a measure of how confident the market is in its current vol estimate; higher α means the market allows more probability mass on extreme vol regimes.

To calibrate these parameters against historical intuition: typical calibrated values for a liquid equity index are $\kappa \approx 2$–6 year^{-1}, giving a half-life of vol shocks of roughly one to four months; $\theta \approx 0.04$–0.06, corresponding to a long-run ATM vol of 20%–25%; $\alpha \approx 0.4$–1.0, varying with market regime; $\rho \approx -0.6$ to -0.8, stubbornly negative; and v_0 matching whatever ATM implied vol is currently trading. These are broad ranges; different asset classes and different volatility regimes produce quite different calibrated values. Single-stock options typically show larger α and θ than index options because single names have more idiosyncratic vol. Sector ETFs fall in between. Currency pairs have much smaller α and less negative (sometimes near-zero) ρ. Commodities vary wildly by underlier; storable commodities like gold behave more like FX, while seasonal commodities like natural gas can exhibit seasonal patterns in θ itself.

The timescale interpretation of κ deserves a second look. Because κ has units of inverse time, the product $\kappa\tau$ is a dimensionless "time to mean revert." When $\kappa\tau \ll 1$ (short maturity relative to mean-reversion timescale), the variance has not had time to converge toward θ, and the option's pricing is dominated by the starting variance v_0.

When $\kappa\tau \gg 1$ (long maturity), the variance has relaxed to its long-run mean θ, and the option's pricing is dominated by θ rather than v_0. The crossover between these two regimes happens around $\kappa\tau \approx 1$, which for $\kappa = 2$ is roughly six months. This is why short-dated and long-dated options carry quite different parameter-sensitivity fingerprints: short-dated options are sensitive to v_0 and α; long-dated options are sensitive to θ and κ. A good calibration uses this structural difference to separate the parameters cleanly.

The correlation ρ is perhaps the most interesting parameter. For equity indices it is strongly negative — typically in the range -0.6 to -0.8 after calibration — because of what practitioners call the leverage effect: as the stock price falls, the debt-to-equity ratio of the underlying firm rises mechanically, making the equity more levered and therefore more volatile. There is also a behavioural layer: falling markets trigger risk-off flows, forced deleveraging, and volatility spikes independent of any firm-level balance sheet. The result is that equity vol and equity spot move in opposite directions with remarkable consistency. This negative correlation is the engine that drives the equity smile's downward skew: puts become more valuable because the market attaches probability to the joint event of a falling price and rising vol, which is precisely the tail that out-of-the-money puts pay off on.

To understand the leverage effect quantitatively, consider a simple capital structure. A firm has assets with value V, debt with face value D, and equity with value $E = V - D$. The equity volatility is approximately $\sigma_E \approx \sigma_V \cdot V/E = \sigma_V \cdot (1 + D/E)$, where D/E is the debt-to-equity ratio and σ_V is the asset-value volatility. If equity falls by 20%, then (holding debt roughly constant) D/E rises by a factor of V/E minus one — which for a moderately leveraged firm could be 15% or more. That percentage rise in debt-to-equity translates directly into a rise in equity volatility. For the S&P 500 index, which aggregates hundreds of firms with heterogeneous leverage, the effect is diluted but still real. Empirically the leverage channel accounts for perhaps 10–20% of observed negative ρ; the rest is driven by risk aversion, volatility contagion, and forced deleveraging.

The behavioural and institutional mechanisms are probably more important in practice. Risk managers across the industry run VaR-based limits (CH09); when vol rises, VaR rises, and limits are breached; breached limits force position cuts, which drive spot down; falling spot produces more vol, which produces more VaR breaches, and a feedback loop develops. This mechanism is the engine of volatility clustering — vol begets more vol — and it manifests as a highly negative ρ in stochastic vol models. There is also a purely psychological layer: traders fear losses more than they value gains (prospect theory), so they demand higher implied vols for OTM puts even absent any leverage-induced physical channel. All of these effects sum into the single observed $\rho \approx -0.7$ that calibrators typically report.

Interestingly, ρ is the most stable calibrated parameter across time. If you calibrate Heston daily to SPX over a long history, you will find that $(\kappa, \theta, \alpha, v_0)$ wander quite a bit as the vol regime changes, but ρ sits in a narrow band around -0.7 with surprisingly low variance. This structural stability reflects the deep economic persistence of the leverage effect: the mechanisms that make spot and vol negatively correlated do not go away, even as the magnitude of vol itself varies. For this reason, some practitioners anchor ρ to a fixed value during calibration and only fit the other four parameters. This imposes discipline on the optimiser and rules out spurious local minima with wrong-signed ρ.

For other asset classes, ρ tells different stories. In some commodity markets — particularly those with tight physical supply, such as natural gas in winter or crude oil during geopolitical tension — ρ can be positive: rising prices signal scarcity and volatility together, so the smile tilts the other way. In foreign exchange, ρ is often close to zero or modest in magnitude, reflecting the relatively symmetric nature of currency pair dynamics. This is not a detail; it means the Heston parameters are not universal constants, they must be calibrated per asset and per regime.

A few representative examples help build intuition. In FX, USD/JPY tends to show a slightly negative ρ — the yen strengthens (USD/JPY falls) during risk-off episodes, which often coincide with vol spikes — but the magnitude is modest, perhaps -0.2. EUR/USD tends toward $\rho \approx 0$ during normal times but swings wildly during crises when cross-asset correlations realign. GBP/USD exhibits Brexit-era shocks that briefly pushed ρ strongly negative. In rates, SOFR or Treasury yield vol is often priced with near-zero ρ in normal regimes but with strongly positive ρ in inflation-fear regimes (rising rates produce rising rate-vol). In equities, the single-stock vs index distinction matters: an individual high-beta tech stock might show $\rho \approx -0.5$ to -0.7, while an index like SPX averages across many names and ends up closer to -0.75. These patterns are not random; they reflect the underlying economic dynamics of each asset class.

It is worth stating explicitly that Heston's five-parameter structure imposes a functional form on the smile. The model cannot represent every conceivable smile shape. In particular, Heston smiles are monotonic in convexity as you move away from the skew — you cannot construct a Heston smile with a "W" shape, with two local minima or a kink at some particular moneyness. Real smiles occasionally show such features, especially at short maturities

where event risk creates localised anomalies. When they do, Heston simply cannot fit them, and the calibrator will produce a best-fit smile that smooths over the irregularities. This is a structural limitation, not a bug, and it points toward the richer model families (local-stoch vol, jump processes, rough vol) needed to capture the full richness of observed smiles.

15.3 14.2 Feller condition

The variance process (14.7) can hit zero. For v_t to be strictly positive (i.e. to avoid hitting zero and getting stuck / reflecting) we need

$$\boxed{2\,\kappa^{\mathbb{P}}\,\theta^{\mathbb{P}} \;\geq\; \alpha^2} \tag{14.9}$$

the Feller condition. Intuition: $2\kappa\theta$ is twice the pull back toward the mean at the origin (the drift at $v = 0$ is $\kappa\theta$), while α^2 is the diffusion push that drives v around. When the pull dominates, the origin is an entrance-only boundary and v_t is > 0 almost surely. When the push dominates, v_t can touch zero (and in continuous time will, repeatedly). In practice calibrated equity parameters often violate it mildly; simulation schemes then use full-truncation (take $v^+ = \max(v, 0)$ in the diffusion coefficient) or reflection. Closed-form pricing is unaffected — the characteristic function is analytic in the parameters regardless of Feller.

Think of the Feller condition as a tug-of-war between two forces at the origin. On one side, the mean-reversion drift $\kappa(\theta - v)$ evaluated at $v = 0$ equals $\kappa\theta$, a strictly positive pull away from the boundary. On the other side, the volatility of variance α^2 represents the intensity with which random fluctuations can knock the process around. If the pull is strong enough — specifically, at least half the kick — the process never reaches the boundary. If it is weaker, the boundary is attainable and the process spends infinitesimal but non-zero time there. In continuous time this subtlety matters because any time the variance touches zero, the local dynamics degenerate: the diffusion term vanishes, leaving only the drift, which then pushes the process back into the interior. This is not a catastrophe — the process is still well-defined and has a proper distribution — but it causes havoc for naive discretisation schemes.

The origin of the Feller condition is in the theory of one-dimensional diffusions on a half-line. Feller classified boundary points for such diffusions into four categories: natural (unattainable, the process does not reach it in finite time), entrance (reachable from outside but not from inside), exit (reachable from inside but absorbing), and regular (reachable in both directions). For the CIR process, the origin is entrance if $2\kappa\theta \geq \alpha^2$, regular if $2\kappa\theta < \alpha^2$. An entrance boundary means the process started at the boundary can move into the interior, but a process started in the interior never reaches the boundary. A regular boundary means the process reaches it, and you must specify what happens there — reflect back into the interior, or absorb. For the Heston variance, we want the process to stay non-negative and continue, so reflection is the natural choice when Feller is violated.

A calculable consequence of Feller violation: the stationary density of v_t becomes unbounded at zero. Specifically, the CIR process has a gamma-distributed stationary density $v_\infty \sim \Gamma(2\kappa\theta/\alpha^2, \alpha^2/(2\kappa))$. When $2\kappa\theta \geq \alpha^2$ (Feller satisfied), the shape parameter is ≥ 1 and the density is bounded near zero. When $2\kappa\theta < \alpha^2$ (violated), the shape parameter is < 1 and the density diverges at zero like $v^{2\kappa\theta/\alpha^2 - 1}$. The divergence is integrable — the overall probability is still one — but it means the process spends substantial time near zero, and any numerical scheme that evaluates $\sqrt{v}$ needs to handle this carefully. In practice, Feller-violated Heston calibrations produce a variance process that dips down to tiny positive values often, a signature that shows up clearly in Monte Carlo simulation.

A further remark on the non-tradability of the variance. The Feller condition applies to both the $\mathbb{P}$-dynamics and the $\mathbb{Q}$-dynamics of variance, but with different parameter values in general (because λ^v shifts κ and θ between the two measures). It is possible for the $\mathbb{P}$-dynamics to satisfy Feller while the $\mathbb{Q}$-dynamics violate it, or vice versa. This can create puzzling situations where the "same" variance process seems to have different boundary behaviour under different measures. The resolution: the boundary classification is a property of the SDE coefficients, and different SDEs — even two that describe the "same" process under different measures — can have different classifications. For pricing we care only about the $\mathbb{Q}$-dynamics and their Feller status; for physical interpretation and historical time-series estimation we care about the $\mathbb{P}$-dynamics.

The practical consequence is subtle but important. If you calibrate Heston to an equity smile, you will typically find parameters that violate Feller by a small amount. This is not a pathology of your calibration; it is what the market is telling you. The market demands more wing richness than a Feller-compliant Heston can produce with reasonable (κ, θ), so the calibrator cranks up α to get the curvature right and pushes the system across the Feller boundary. The closed-form characteristic function does not mind: the Riccati integration produces a valid complex-analytic function

of ϕ regardless. What does mind is your Monte Carlo scheme. A naive Euler step on (14.7) will occasionally produce $v_{n+1} < 0$, at which point $\sqrt{v_{n+1}}$ is imaginary and the next step blows up. The standard fix is full-truncation: clamp the variance at zero whenever it tries to go negative, use the clamped value in the diffusion coefficient, and let the mean-reversion drift (which remains $\kappa\theta$ at zero) bring it back. Reflection — absolute value rather than truncation — is another common choice. Exact simulation via the non-central chi-squared transition density is the gold standard but is computationally more expensive.

The hierarchy of Heston simulation schemes is worth cataloguing explicitly because the quality of your Monte Carlo output depends directly on the scheme you pick. From fastest and crudest to slowest and most accurate:

First, Euler with absolute-value correction. Simple, fast, badly biased. The bias comes from treating negative v values as their absolute value rather than zero, which artificially inflates the diffusion contribution near the boundary. Not recommended for production use, although still common in academic papers for historical reasons.

Second, Euler with full truncation (Lord, Koekkoek, Van Dijk, 2010). Slightly more biased at the boundary than reflection in theory, but simpler to analyse and typically preferred in practice. Clamp v at zero in the diffusion coefficient; let the drift carry you back. The bias decays linearly in Δt, and for $\Delta t = 1/252$ (daily) the bias on ATM options is typically under 0.5%.

Third, the Quadratic-Exponential (QE) scheme of Leif Andersen (2007). The workhorse production scheme. QE exploits the fact that the CIR transition density is non-central chi-squared, and approximates it by either a quadratic function of a normal (for large variance) or a mixture with a point mass at zero (for small variance). Biased, but the bias is much smaller than Euler. The QE scheme is roughly 2–3× slower than Euler per step but often 10× more accurate, so net it is a win. Andersen's paper also contains a "QE martingale correction" that fixes an additional bias in the coupled spot-variance simulation; this is worth implementing if you care about bias below a few basis points.

Fourth, exact simulation (Broadie–Kaya, 2006). Uses the known transition density of the CIR process (non-central chi-squared) and a conditional-on-variance treatment of the spot. Zero bias but expensive: each time step requires drawing from a non-central chi-squared, which typically involves an inverse-CDF evaluation that is itself an iterative procedure. For benchmarking Heston prices when you absolutely need to pin down the answer, this is the gold standard. For routine pricing it is overkill.

A fifth approach, often underappreciated, is not to Monte Carlo at all — use the closed-form Heston call price for European options and lean on variance-reduction for exotics. The Fourier method gives you the full distribution of X_T via the characteristic function, and from there you can integrate any European payoff without simulating a single path. For American and path-dependent payoffs Monte Carlo is unavoidable, but for vanillas the Fourier route is strictly better. CH10 developed the variance-reduction toolkit (antithetic variates, control variates, stratified sampling) that is standardly applied when one does Monte-Carlo a Heston path engine; we invoke those techniques here rather than re-deriving them.

Feller boundary in (kappa, alpha) plane *Feller boundary $\alpha = \sqrt{2\kappa\theta}$ for four different long-run variance levels. Below each curve the condition $2\kappa\theta \geq \alpha^2$ holds and $v_t > 0$ almost surely; above it v_t can touch zero and a full-truncation / reflection scheme is needed in Monte Carlo. The SPX-like calibrated point $(\kappa = 2, \alpha = 0.5, \theta = 0.04)$ sits just above its $\theta = 0.04$ curve — a textbook mild violation.*

The takeaway from the figure: real calibrations live close to the Feller boundary rather than comfortably below it. A rough rule of thumb is that if your calibrated α is smaller than about 0.3, you are likely Feller-safe for reasonable (κ, θ); if it climbs above 0.5, you are almost certainly violating. That does not invalidate the calibration; it is simply a trigger to be disciplined about the simulation scheme and, if the violation is severe, to question whether the smile you are fitting is really a smile or whether it contains short-dated features (jumps, idiosyncratic ATM dislocations) that Heston structurally cannot capture.

There is one more practical diagnostic to keep in mind. Heston sometimes produces calibrated parameters that are very near the Feller boundary but on the "safe" side, with α artificially constrained during optimisation to satisfy the condition. If your optimiser imposes Feller as a hard constraint and the unconstrained optimum would have wanted to exceed it, you will see the constraint binding at the calibrated solution — $\alpha^2 = 2\kappa\theta$ exactly. This typically degrades the fit quality (residual sum of squares is larger than it needs to be) and suggests that Heston's structure is simply too restrictive for the surface you are fitting. The correct response is usually to drop the Feller constraint, accept the violation, and use a careful simulation scheme. Enforcing Feller purely for simulation-convenience reasons, at the cost of market-fit quality, is rarely worthwhile.

A philosophical note. Feller is a mathematical condition on continuous-time dynamics. Real markets are discrete-time — quotes update in milliseconds, trades in fractions of seconds — and the continuous-time idealisation is a useful fiction. In a truly discrete-time world, variance is always non-negative by definition (it is a sum of squared increments), and the Feller condition is essentially moot. The only reason we care about Feller is that we are using a continuous-time SDE as a pricing model, and the SDE has to be well-posed for the pricing mathematics to make sense. So when you see a calibrated Feller-violating Heston, do not be alarmed by the nominal violation — the real underlying process is still well-behaved. The violation is a property of the mathematical idealisation, not of the market itself.

15.4 14.3 Change of measure — market price of variance risk

Going from $\mathbb{P}$ to $\mathbb{Q}$, (14.6) loses its drift (futures are martingales),

$$\frac{\mathrm{d}F_t}{F_t} = \sqrt{v_t}\,\mathrm{d}W_t^F, \qquad \text{corr. } (\rho). \tag{14.10}$$

For the variance, write the Girsanov shift — the multi-dimensional version derived in CH05 §5.4, specialised here to two correlated Brownian motions:

$$\mathrm{d}W_t^F = \frac{\mu_t}{\sqrt{v_t}}\,\mathrm{d}t + \mathrm{d}W_t^{\mathbb{P},F}, \qquad \mathrm{d}W_t^v = \lambda_t^v\,\mathrm{d}t + \mathrm{d}W_t^{\mathbb{P},v}. \tag{14.11}$$

Then the $\mathbb{Q}$-dynamics of variance become

$$\mathrm{d}v_t = \Big(\kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - v_t) \;-\; \alpha\,\lambda_t^v\sqrt{v_t}\Big)\mathrm{d}t \;+\; \alpha\sqrt{v_t}\,\mathrm{d}W_t^v, \tag{14.12}$$

with λ_t^v the market price of variance (vol) risk. Note that the two-dimensional Girsanov we are using is the vector-valued version: we apply independent density processes to two correlated Brownian motions by first rotating into an independent pair via Cholesky, shifting each, and rotating back. The net effect on the correlated pair (W^F, W^v) is recorded in (14.11); the correlation ρ is preserved by the change of measure, as the Girsanov shift is a pure drift adjustment and does not affect the quadratic covariation.

Here is the crucial point. The drift of variance under $\mathbb{P}$ is whatever we think it is empirically — it reflects the long-run mean of realised variance, the speed at which variance reverts after shocks, and so on. But for option pricing, we do not price under $\mathbb{P}$; we price under $\mathbb{Q}$. And $\mathbb{Q}$ attaches a different drift to the variance process, because market participants demand (or offer) a premium for bearing variance risk. Empirically, buyers of variance — people long volatility through options or variance swaps — pay a premium on average, meaning realised variance comes in lower than implied variance on average. This is the variance risk premium, and it is captured in the model by λ^v. A negative λ^v for equity markets would reflect the fact that long-vol positions earn a negative premium — they are hedges, and hedges cost something. The variance risk premium is persistent, documented over long samples, and is arguably the single most important empirical feature driving the wedge between historical and implied vol.

To put numbers on this: over long samples of SPX options, the average difference between implied variance (VIX^2-scaled) and subsequently realised variance is about 3–5 percentage points in annualised-vol units. In other words, if VIX is 20, realised vol over the next 30 days averages about 17. Variance-swap investors — the sellers of variance — earn this 3–5 point wedge on average, net of trading costs, and this is the explicit compensation for bearing tail-risk exposure. (The wedge is not risk-free, obviously; it can invert during crises, and the shorts can take large losses. Long-vol positions are attractive when vol spikes, which is exactly the "tail hedge" use case that dominates quant-institutional flows.) When we write Heston with a specific λ^v, we are implicitly asserting a particular functional form for this premium — one that is compatible with the CIR dynamics. Different asset classes and different regimes have different premium structures; Heston's affine-preserving choices capture the main cases cleanly.

A pedagogical note on terminology. The "market price of variance risk" is sometimes written λ^v and sometimes with other symbols; the economic quantity it represents is the Sharpe-ratio-like premium demanded per unit of variance-factor exposure. If the variance risk is priced (i.e. $\lambda^v \neq 0$), then the $\mathbb{Q}$- dynamics of v differ from the $\mathbb{P}$-dynamics, which means the risk-neutral expected path of variance differs from the physical expected path. Practitioners often summarise this by saying "implied vol is higher than expected realised vol on average, by the variance risk premium." That one sentence captures essentially the entire economics of the λ^v parameter.

Two natural choices keep the SDE affine (preserving Heston form):

Choice 1. $\lambda_t^v = c\sqrt{v_t}$. Then

$$\lambda_t^v \cdot \alpha\sqrt{v_t} = \alpha c\, v_t,$$

so the drift becomes

$$\kappa^{\mathbb{P}}\theta^{\mathbb{P}} - (\kappa^{\mathbb{P}} + \alpha c)\, v_t = \kappa\left(\frac{\kappa^{\mathbb{P}}\theta^{\mathbb{P}}}{\kappa} - v_t\right), \tag{14.13}$$

with $\kappa = \kappa^{\mathbb{P}} + \alpha c$ and $\theta = \kappa^{\mathbb{P}}\theta^{\mathbb{P}}/\kappa$.

Choice 2. $\lambda_t^v = \dfrac{\ell}{\sqrt{v_t}} + c\sqrt{v_t}$. Then

$$\lambda_t^v \cdot \alpha\sqrt{v_t} = \alpha\ell + \alpha c\, v_t,$$

so both level and slope of the mean-reversion shift: $\kappa = \kappa^{\mathbb{P}} + \alpha c$ and $\theta = (\kappa^{\mathbb{P}}\theta^{\mathbb{P}} - \alpha\ell)/\kappa$.

Notice what (14.13) achieves: the shape of the variance SDE under $\mathbb{Q}$ is algebraically identical to its shape under $\mathbb{P}$ — the same $\kappa(\theta - v)\,\mathrm{d}t + \alpha\sqrt{v}\,\mathrm{d}W$ form — with (κ, θ) reshuffled. This is why Heston is so clean. A wrong choice of λ^v (say, a constant λ) would break the affine structure: the change of measure would contribute $\alpha\lambda\sqrt{v}$ to the drift, which is not linear in v, and the Riccati tractability would collapse. The two "affine-preserving" choices above are the ones that respect Heston's algebraic skeleton, and they cover the usual practitioner parametrisations.

This is a subtle but deep point. In most finance contexts, the choice of λ^v is driven by economic or statistical arguments: we want the risk-neutral measure that best fits observed data, or that has some desired theoretical property. In Heston, there is an additional computational constraint: only certain functional forms of λ^v preserve the affine structure, and only affine structure gives us the closed-form characteristic function. If we chose a λ^v that broke affine-ness, the Riccati ODEs would become general nonlinear ODEs (without closed form), the characteristic function would require numerical PDE solving at every evaluation, and the computational cost of Heston pricing would increase by two or three orders of magnitude. For this reason, the affine-preserving λ^v choices are not just theoretically elegant — they are computationally essential. They are the reason we can calibrate Heston to a full vol surface in seconds rather than hours.

Another way to see this: the "Heston model under $\mathbb{Q}$" and the "Heston model under $\mathbb{P}$" are two instances of the same algebraic family, parameterised differently. Pricing derivatives uses the $\mathbb{Q}$-instance; estimating historical dynamics uses the $\mathbb{P}$-instance; the affine-preserving change-of-measure provides the bridge between them. If you wanted to do joint estimation — fitting both physical time-series data and risk-neutral option data simultaneously — you would use this bridge to write down the joint likelihood, with λ^v as the parameter linking the two. Some sophisticated calibration frameworks do exactly this, treating the variance risk premium as a directly estimable parameter. Most practitioner implementations, however, calibrate under $\mathbb{Q}$ only and infer the risk premium ex post by comparing calibrated parameters to historical estimates.

From the option trader's perspective, this means something striking. The risk-neutral $(\kappa, \theta, \alpha, \rho, v_0)$ that we calibrate from option prices are not the same as the physical-measure $(\kappa^{\mathbb{P}}, \theta^{\mathbb{P}}, \alpha^{\mathbb{P}}, \rho^{\mathbb{P}}, v_0^{\mathbb{P}})$ that we might estimate from historical returns. The two sets of parameters differ precisely by λ^v, the variance risk premium. Typically $\theta > \theta^{\mathbb{P}}$ — the risk-neutral long-run variance is higher than the historical long-run variance — because buyers of variance swaps must be compensated. The gap is the variance risk premium in disguise, and it is paid for by the option buyer and collected by the option seller.

The implications for trading are concrete. A strategy of systematically selling variance (selling options, selling variance swaps, running a "vol carry" book) earns the premium on average but takes on significant tail risk: in a vol spike the P&L can be punishingly negative. The Sharpe ratio of such a strategy is positive but not spectacular — typically in the range 0.5 to 1.0 for liquid markets over long samples — and the drawdowns are deep. LTCM's 1998 blowup, the 2018 "Volmageddon" that wiped out XIV, and the March 2020 vol spike are all examples of this strategy's tail risk materialising. Understanding the origin of the premium via Heston's λ^v helps demystify the strategy: the premium is compensation for exactly the tail exposure that occasionally bites hard. There is no free lunch; just a structural risk-premium for those willing to bear the risk.

It is also worth noting that the risk-neutral α can differ from the physical $\alpha^{\mathbb{P}}$, and the risk-neutral ρ can differ from $\rho^{\mathbb{P}}$, depending on the precise form of λ^v (and on whether there is a price of correlation-risk in addition to vol-risk). In the simplest affine-preserving choice (Choice 1 above), α and ρ are preserved across measures; only κ and θ shift. In richer parameterisations, all four can shift. Practitioners usually take the simplest case and then check empirically

whether the calibrated α and ρ are consistent with time-series estimates. Significant discrepancies suggest either model misspecification or a richer risk-premium structure than the simple model captures.

Under $\mathbb{Q}$ (market price of risk absorbed, with Heston's canonical choice), we get the standard Heston system:

$$\boxed{\frac{\mathrm{d}F_t}{F_t} = \sqrt{v_t}\,\mathrm{d}W_t^F} \tag{14.14}$$

$$\boxed{\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \alpha\sqrt{v_t}\,\mathrm{d}W_t^v} \tag{14.15}$$

with $\mathrm{d}W_t^F \cdot \mathrm{d}W_t^v = \rho\,\mathrm{d}t$.

From here on, $(\kappa, \theta, \alpha, \rho, v_0)$ are the five risk-neutral Heston parameters.

Five parameters is a modest parameter count for a model that fits an entire implied vol surface. By comparison, some practitioner "local vol" or "local-stochastic vol" models carry dozens or even hundreds of parameters. The minimalism of Heston is one of its virtues: each parameter has a clean economic meaning, calibration is a well-posed optimisation in low dimension, and the fitted parameters can be compared across dates and assets without the apples-to-oranges problem that plagues over-parameterised models.

The parsimony has a second benefit that is sometimes overlooked: generalisation. A five-parameter model fitted to (say) ATM vols and the 25-delta risk reversal and the 25-delta butterfly at a single maturity has four effective calibration targets and five parameters, so the fit is nearly exact. But those same five parameters then extrapolate — via the Heston dynamics — to every other strike and every other maturity. The extrapolation is not arbitrary; it is dictated by the structural form of the Heston SDE. This means the model can price exotic and off-the-grid strikes in a way that is internally consistent with the calibration points, and that is one of the reasons Heston is a favourite choice for pricing mildly exotic products like barriers, forward-starts, and cliquets. A dense local-vol model with hundreds of parameters can fit the grid exactly, but extrapolates poorly off the grid because there are no dynamics constraining what happens between quoted points. Heston, with five parameters, is forced to make structural assumptions that constrain the extrapolation — often to the benefit of the exotic pricing.

Compare this to regime-switching models, which can have many parameters per regime (shape of transition matrix, regime-specific vols, regime-specific jump intensities) and easily bloat to 15+ parameters. Those models offer richer dynamics but are less stable across calibrations and harder to interpret. Heston's five-parameter structure sits in the sweet spot: rich enough to capture the major features of vol dynamics (mean reversion, vol-of-vol, leverage), parsimonious enough to calibrate cleanly and be interpreted economically. This sweet spot is a big part of why Heston has endured as the industry default.

15.5 14.4 Implied volatility — why we need this model

If we price a call in the Heston world at (K, T) and invert Black–Scholes to back out the volatility that matches, we get the implied vol surface:

$$V^{\text{Heston}}(K,T) \;=\; V^{\text{BS}}(r,\, \sigma^{\text{impvol}}(K,T);\, K,T). \tag{14.16}$$

The Heston model produces a volatility smile in K and a non-trivial term structure in T. Graphically: out-of-the-money puts (low K) and calls (high K) price richer than BS, bending the flat BS line into a smile/skew. The PDF of $\ln X_T$ under Heston is fatter-tailed and skewed relative to the BS log-normal. Payoff $(X-K)_+$ is sensitive to those tails.

Step back and consider what is actually happening when we talk about "implied vol." Implied vol is not a property of the option market; it is a property of the translation between option prices and Black–Scholes. When we say "the 3-month 90-strike put implies 22%," what we mean is that the Black–Scholes formula, fed $\sigma = 0.22$, produces exactly the observed market price. It is a unit conversion from price to something more intuitive. The puzzle — the whole reason this chapter exists — is that the number we get from this conversion is not the same across strikes and maturities. It is a surface, not a number. A model that correctly describes the risk-neutral distribution of future underlying prices will reproduce that surface automatically, without us having to "tune" σ by strike.

The act of "inverting Black–Scholes" to get implied vol is, in a sense, a deliberate decision by the industry to keep quoting in BS units even though everyone knows BS is wrong. Why not quote prices directly? Because prices are hard to compare — a 90-strike put on SPX might cost \$5 and a 110-strike call might cost \$1, and the meaningful

comparison is not dollar-for-dollar but normalised in some way. Implied vol is the standard normalisation: it is the "monetary value per unit of time and unit of moneyness risk" in BS terms. Traders exchange vol quotes, risk reversals are vol-differences, butterflies are vol-convexity measures, and the whole taxonomy of vol trading is built on the BS-implied vol system. This conventional language is so entrenched that even when a pricing model is non-BS (like Heston), the model outputs are usually inverted back to BS implied vols for quoting and comparison. Heston is a pricing engine; BS-implied vol is the display format.

Another consequence: implied vol is monotonic in option price (for a given strike, maturity, and rate), so there is a bijection between prices and implied vols. This is why it is safe to move back and forth between the two — there is no information loss. The BS formula itself is never invoked as a model; it is used purely as a bijection. Modern derivatives trading is, in a sense, BS-free but BS-formula- dependent. The formula lives on as a coordinate chart; the model behind it has been replaced by Heston and its richer cousins. The canonical constant-σ BS formula and its PDE derivation are in CH06; we treat it here as the coordinate chart rather than as a pricing model in its own right.

The Heston density is a mixture. Conditional on the entire path of variance $\{v_s : 0 \leq s \leq T\}$, the distribution of $\ln F_T$ is essentially lognormal — the integrated variance $\int_0^T v_s \, \mathrm{d}s$ plays the role of $\sigma^2 T$ in Black–Scholes. The unconditional distribution is a weighted average of these lognormals, with the weights given by the probability of each variance path. A calm path of v produces a narrow lognormal; a turbulent path produces a wide lognormal. The mixture of narrow and wide lognormals is fatter-tailed than any single lognormal — the wide components dominate in the extremes, the narrow components dominate in the centre. This is the origin of Heston's excess kurtosis, and it is why deep out-of-the-money options price richer under Heston than under Black–Scholes at the same ATM vol.

To make the mixing picture precise, let $I_T \equiv \int_0^T v_s \, \mathrm{d}s$ be the integrated variance. Then, conditional on I_T (and assuming $\rho = 0$ for simplicity), $\ln F_T$ is $\mathcal{N}(\ln F_0 - I_T/2, I_T)$. Unconditionally, it is a continuous mixture of normals indexed by I_T. The density of $\ln F_T$ is therefore $p(x) = \int p_{\mathcal{N}}(x \mid I) \, q(I) \, \mathrm{d}I$, where q is the density of I_T and $p_{\mathcal{N}}$ is the conditional normal. The resulting p is a variance-gamma-like mixture with fatter tails than any single normal. The kurtosis of $\ln F_T$ in the $\rho = 0$ case is roughly $3 + 6 \, \mathrm{Var}(I_T)/\mathbb{E}[I_T]^2$, increasing in the variability of integrated variance. Larger α means more variable I_T means higher kurtosis means fatter smile at the wings. This is the quantitative content of "α drives wings."

When $\rho \neq 0$, the mixing becomes asymmetric: low-I_T paths are correlated with specific realisations of the terminal spot (via the shared innovation structure), and the mixture skews. With $\rho < 0$, the paths that produce low F_T tend to be those with high integrated variance, so the left tail of $\ln F_T$ is fattened disproportionately. This asymmetric mixing is the origin of skew. It is also what makes the analytics more involved: with $\rho = 0$ the characteristic function factorises cleanly as the product of a terminal-spot and integrated-variance contribution; with $\rho \neq 0$ the factorisation is coupled and the affine ansatz is needed to handle the coupling.

The characteristic-function framework handles all of this implicitly. You never need to explicitly write down the mixture or the density; the characteristic function encodes them, and Fourier inversion extracts whatever moments or probabilities you need. This is one of the reasons Heston is elegant: the mixture structure of the density is completely obscure analytically, but the characteristic function is transparent.

The skew — the asymmetry of the smile — has a different origin. Skew comes from the correlation ρ between spot moves and variance moves. If $\rho < 0$, then the scenarios in which F_T ends up very low are precisely the scenarios in which variance was high along the way, pumping up the left tail even further. Conversely, the scenarios in which F_T ends up very high are scenarios in which variance was low (boring path, no interesting moves), compressing the right tail. The result is a density that leans left: the left tail is fatter than the right. Out-of-the-money puts, which live in the left tail, become rich; out-of-the-money calls, which live in the right tail, become cheap. The implied vol surface tilts down to the right — classical equity skew.

There is a useful mnemonic for how ρ manifests in the density shape. Consider the joint realisation of spot and vol along a path. Path A: spot rises gradually, vol falls toward θ. Path B: spot falls sharply, vol spikes up. Under $\rho < 0$, the two paths together are more likely than two alternative paths where spot and vol move the same direction. Path B is exactly the "tail scenario" that OTM puts pay off on — both the move in spot (into the put's payoff region) and the move in vol (making the path more extreme) conspire to increase the put value. This joint-tail effect is precisely what makes OTM puts trade richer under Heston with $\rho < 0$ than under BS with the same ATM vol. The market prices this joint tail at a premium; ρ is the knob that controls the premium size.

A helpful picture: imagine plotting the Heston density of $\ln F_T$ against the BS-lognormal at the same ATM vol. At

$\rho = 0$, the two densities have the same centre but Heston has fatter symmetric tails — the smile is a "smile" in the classical sense, with both wings lifted. As ρ becomes more negative, the Heston density shifts mass to the left: the left tail fattens further, and the right tail thins. The smile evolves from symmetric smile (at $\rho = 0$) to tilted skew (at $\rho = -0.7$) to a pure downward-sloping line (at $\rho \to -1$, the "perfect correlation" limit where the vol move is mechanically determined by the spot move). The progression is continuous and visually intuitive, and it maps directly onto the range of behaviours observed in real equity-smile markets.

Heston-implied IV smile *Heston-implied IV smile*

15.5.1 14.4.1 Simulation scheme

Set $X_t = \ln F_t$. Itô (two-dimensional, CH03 §3.10) gives

$$\mathrm{d}X_t = -\tfrac{1}{2}\,\gamma^2 v_t\,\mathrm{d}t + \gamma\sqrt{v_t}\,\mathrm{d}W_t^F, \tag{14.17}$$

(with $\gamma = 1$ for the pure Heston; we keep γ to track units). Euler-discretise with step Δt:

$$X_{t_n} - X_{t_{n-1}} = -\tfrac{1}{2}\gamma^2 v_{t_{n-1}}^+\,\Delta t + \gamma\sqrt{v_{t_{n-1}}^+}\,\sqrt{\Delta t}\,Z_{1,n}, \tag{14.18}$$

$$v_{t_n} - v_{t_{n-1}} = \kappa(\theta - v_{t_{n-1}}^+)\,\Delta t + \alpha\sqrt{v_{t_{n-1}}^+}\,\sqrt{\Delta t}\left(\rho\,Z_{1,n} + \sqrt{1-\rho^2}\,Z_{2,n}\right), \tag{14.19}$$

where $Z_{1,n}, Z_{2,n}$ are i.i.d. $\mathcal{N}(0,1)$ and

$$v_t^+ \;=\; \max(v_t,\,0) \tag{14.20}$$

is the full-truncation fix (Lord–Koekkoek–Van Dijk) that prevents $\sqrt{v}$ from going complex when Feller is violated.

The scheme as written has a number of subtle features worth appreciating. The two innovations Z_1, Z_2 are independent standard normals; the linear combination $\rho Z_1 + \sqrt{1-\rho^2}\,Z_2$ produces a normal with variance one that is correlated ρ with Z_1. This is the standard trick for sampling from a bivariate normal by Cholesky factorisation. The use of v^+ in both the drift and the diffusion of the variance update ensures that even if a previous step dragged v below zero, we do not propagate that negativity: the next step treats v as if it were zero, allowing the mean-reversion drift $\kappa\theta$ to lift it back into positive territory at the next time step. Full truncation is biased — it systematically underestimates the variance of v near zero — but the bias shrinks as $\Delta t \to 0$ and is widely accepted as the best trade-off between simplicity and accuracy for standard use cases.

A practical note on step size. For daily-step Heston simulation ($\Delta t = 1/252$), the full-truncation Euler scheme introduces bias on European option prices of a few basis points typically — perfectly acceptable for most risk-management purposes. For path-dependent options (barriers, Asians, forward-starts), the bias can be larger, and finer time steps or better schemes (QE) become necessary. For very short-dated options (days or less), $\Delta t = 1/(252 \cdot N)$ with N in the range 10–100 intraday sub-steps is often needed. The computational cost scales linearly with the number of steps, so the choice of N is a cost-accuracy tradeoff that should be benchmarked against the closed-form Heston price before committing to a production scheme.

A related practical note: use antithetic variates to reduce MC variance. The key insight is that Heston's dynamics are symmetric in $(Z_1, Z_2) \to (-Z_1, -Z_2)$ — flipping the signs of both innovations produces an equally-probable path. Pricing the option on both the original and the flipped path and averaging the two gives a zero-bias variance reduction that typically cuts MC standard error by a factor of 1.5–2× at no extra computational cost (since the variance update for the flipped path is essentially free). This and other variance-reduction techniques (control variates based on the BS or Heston closed-form price, importance sampling) are standard tools developed in CH10 and routinely used in production vol desks.

Yet another subtle but important consideration is the discretisation of the spot-variance coupling. In the Euler scheme as written, the spot update uses $v_{t_{n-1}}^+$ — the variance at the start of the interval — as the diffusion coefficient. A more accurate scheme would average the variance over the interval, e.g. $(v_{t_{n-1}}^+ + v_{t_n}^+)/2$, but doing so introduces an implicit coupling that requires iteration. Andersen's QE scheme handles this coupling more carefully; naive Euler does not. The result is a small bias in the joint spot-variance distribution that most implementations accept as the cost of simplicity.

Heston sample paths (S and sqrt(v)) *Heston sample paths (S and sqrt(v))*

The figure shows two kinds of dynamics happening simultaneously. The top panel traces the spot price, which looks at first glance like a standard diffusion: it wanders, drifts, has typical diffusion jaggedness. But the local volatility of

that wander is not constant — compare periods where spot is oscillating gently with periods where it swings wildly. The bottom panel shows the source of that varying agitation: it plots $\sqrt{v_t}$, the instantaneous volatility. Periods where $\sqrt{v_t}$ spikes correspond to the rough episodes on the spot panel; periods where $\sqrt{v_t}$ is low produce the calm stretches above. Notice also the anticorrelation: when spot drops sharply, $\sqrt{v_t}$ tends to spike up, visible as the mirror-image texture between the two panels. This is exactly what $\rho < 0$ encodes, and it is the microstructural fingerprint of equity markets.

Look, too, at the mean-reversion visible in the bottom panel. The $\sqrt{v_t}$ trace is not a random walk — it oscillates around a roughly horizontal baseline (the square root of θ), with upward spikes that decay back toward baseline over a few months. This is the κ-rate relaxation in action. A random walk in $\sqrt{v}$ would drift further and further from any baseline; Heston's mean-reverting variance creates a "homing" tendency that keeps the long-run distribution bounded. Without mean reversion, the integrated variance $\int_0^T v\,\mathrm{d}s$ would grow too variable at long horizons, and long-dated option pricing would become unstable. Mean reversion is the mathematical device that keeps Heston well-behaved at all horizons.

A subtle but important visual feature is the asymmetry of vol dynamics. Real markets (and Heston with $\rho < 0$) exhibit "sharp spikes up, slow decays down" in vol — vol rises quickly during panics, falls gradually as calm returns. This is not a property of the CIR process by itself, which is symmetric in its innovation structure; it is a property of the joint (F, v) dynamics. When spot crashes, the coupled vol spike happens all at once; when spot drifts sideways, vol decays at the leisurely rate κ. The resulting asymmetry in the vol trace is a visible signature of $\rho < 0$, and it matches the empirical behaviour of VIX quite closely.

15.6 14.5 Characteristic function via Riccati ODEs

Define the conditional characteristic function of $X_T = \ln F_T$:

$$g_t \;\equiv\; \mathbb{E}_t^{\mathbb{Q}}\big[\, e^{\omega(X_T - K)} \,\big] \;=\; g(t, X_t, v_t), \tag{14.21}$$

for $\omega \in \mathbb{C}$ on some admissible strip. (Setting $\omega = i\phi$ gives the usual Fourier transform in X_T.) We seek a PDE for g and then an exponential-affine ansatz.

Here is why we bother with the characteristic function at all. The density $p(X_T)$ of the log-futures price under Heston does not have a closed form. It is a mixture of lognormals, indexed by the unobserved path of variance, and the mixture weights are themselves complicated. We have no hope of writing p down. The characteristic function, on the other hand, turns out to be elementary — a neat exponential-affine expression in the state — and it contains exactly the same information as the density. Every moment of the distribution can be read off from derivatives of the characteristic function at zero; every expectation of the form $\mathbb{E}[f(X_T)]$ can be computed by Fourier inversion, provided f has a well-defined Fourier transform. European option prices are linear functionals of the density, so the Fourier route is not only convenient — it is the natural one.

The philosophical shift from density-first to characteristic-function-first thinking is one of the most important re-orientations in modern derivatives pricing. Classical probability texts teach you to work with densities because densities are concrete — you can plot them, integrate against them, compute moments by direct integration. But densities for complicated processes are often impossible to write down in closed form, while characteristic functions for the same processes are often much simpler. This is not an accident; it reflects the fact that taking exponential expectations of Gaussian-linked processes produces relatively clean algebra, while writing down the density requires inverting that exponential expectation via Fourier inversion — which is a separate operation that may or may not yield a closed-form answer. For many practical problems, you do not need the density at all; you just need some expectation, and the characteristic function gets you there directly via Fourier inversion or via moment formulas.

Another way to see why characteristic functions are natural here: they are the multiplicative objects of probability theory. When you add two independent random variables, their densities convolve — a multiplicative operation in the characteristic-function domain via the inverse Fourier relation. When you scale or shift, the characteristic function transforms by simple algebraic operations. For stochastic processes that are linear combinations of independent increments, the characteristic function structure is essentially combinatorial. Heston's integrated variance is such a construction, and its characteristic function reflects this combinatorial structure cleanly.

15.6.1 14.5.1 Itô PDE for g

Applying two-dimensional Itô to $g(t, X_t, v_t)$ (CH03 §3.10): the bivariate SDE under $\mathbb{Q}$ is (14.17) + (14.15) with $\mathrm{d}X \cdot \mathrm{d}v = \rho\,\alpha\,v_t\,\mathrm{d}t$. Then

$$\mathrm{d}g_t = \partial_t g\,\mathrm{d}t + \partial_x g\,\mathrm{d}X_t + \partial_v g\,\mathrm{d}v_t + \tfrac{1}{2}\partial_{xx} g\,(\mathrm{d}X_t)^2 + \tfrac{1}{2}\partial_{vv} g\,(\mathrm{d}v_t)^2 + \partial_{xv} g\,(\mathrm{d}X_t\,\mathrm{d}v_t). \tag{14.22}$$

Substituting the drifts and quadratic variations $(\mathrm{d}X)^2 = v\,\mathrm{d}t$, $(\mathrm{d}v)^2 = \alpha^2 v\,\mathrm{d}t$, $\mathrm{d}X\,\mathrm{d}v = \rho\alpha v\,\mathrm{d}t$,

$$\mathrm{d}g_t = \mathrm{d}t\,\{\partial_t g + \left(-\tfrac{1}{2}\gamma^2 v_t\right)\partial_x g + \kappa(\theta - v_t)\,\partial_v g + \tfrac{1}{2}\partial_{xx} g\,\gamma^2 v_t + \tfrac{1}{2}\partial_{vv} g\,\alpha^2 v_t + \partial_{xv} g\,\rho\,\gamma\,\alpha\,v_t\} + \text{mart}. \tag{14.23}$$

By tower g_t is a $\mathbb{Q}$-martingale (this is the Feynman–Kac connection from CH04: g is the conditional expectation of a terminal functional and therefore solves the Kolmogorov backward PDE associated with the generator of the bivariate diffusion), so the drift vanishes for all (t, x, v). Dropping $\gamma = 1$:

$$\boxed{\partial_t g + \left(-\tfrac{1}{2}v\right)\partial_x g + \tfrac{1}{2}v\,\partial_{xx} g + \kappa(\theta - v)\,\partial_v g + \tfrac{1}{2}\alpha^2 v\,\partial_{vv} g + \rho\,\alpha\,v\,\partial_{xv} g \;=\; 0} \tag{14.24}$$

with the terminal condition

$$g(T, x, v) = e^{\omega(x-K)}. \tag{14.25}$$

The PDE (14.24) is the Kolmogorov backward equation for the Heston two-factor diffusion. It says that g, viewed as a function of state (x, v) and running time t, is conserved in expectation along the Heston dynamics — its generator annihilates it. The structure of the equation is revealing: every coefficient of a second-derivative term is linear in v, and the drift coefficients are affine in v. This linearity in v is the algebraic fingerprint of the broader class of affine processes, and it is exactly what enables the exponential-affine ansatz below.

Affine processes are the grand generalisation of what we are seeing here. A diffusion is affine if its drift and covariance matrix are both affine (linear plus constant) functions of the state. The theory developed by Duffie, Pan, and Singleton in a series of papers in the early 2000s shows that every affine process has an exponential-affine characteristic function, with coefficients solving a system of generalised Riccati ODEs whose number equals the dimension of the state space. Heston is two-dimensional and therefore produces two ODEs (A for the constant, B for the coefficient of v; X does not contribute to the state-dependent drift because X's drift is affine in v, not in X). For higher-dimensional affine factor models — affine short-rate models with multiple factors (CH12), multivariate stochastic vol models, affine credit models — the number of ODEs grows with dimension but the overall structure remains tractable. The technique of "write the PDE, plug in the affine ansatz, get a system of Riccati ODEs, solve the ODEs in closed form or by numerical integration" is a general recipe that applies throughout affine modelling.

A note on the mixed partial $\partial_{xv} g$: it appears in (14.24) because of the $\rho\alpha v$ quadratic-covariation term, which in turn comes from the correlation between the spot and variance Brownian motions. Without correlation ($\rho = 0$), the mixed-partial term vanishes, the PDE decouples into a spot equation and a variance equation that can be treated separately, and the characteristic function factorises into a product of a spot contribution and a variance contribution. With $\rho \neq 0$, the coupling is genuine and the exponential-affine form is the cleanest way to handle it. This is why Heston with zero correlation is "easy" — a relatively straightforward exercise in conditioning on the variance path — while Heston with nonzero correlation requires the affine machinery.

15.6.2 14.5.2 Affine ansatz and Riccati ODEs

Because the PDE is linear in v and the terminal is exponential-affine in x, try

$$g(t, x, v) \;=\; \exp\Big\{\,\omega(x - K) + A(\tau) + B(\tau)\,v\,\Big\}, \qquad \tau = T - t. \tag{14.26}$$

Compute the derivatives:

$$\partial_t g = -(A'(\tau) + B'(\tau)v)\,g, \quad \partial_x g = \omega g, \quad \partial_{xx} g = \omega^2 g,$$
$$\partial_v g = B(\tau)g, \quad \partial_{vv} g = B(\tau)^2 g, \quad \partial_{xv} g = \omega B(\tau) g.$$

Substituting into (14.24) and dividing by g:

$$-(A' + B'v) - \tfrac{1}{2}v\,\omega + \tfrac{1}{2}v\,\omega^2 + \kappa(\theta - v)\,B + \tfrac{1}{2}\alpha^2 v\,B^2 + \rho\alpha v\,\omega B = 0. \tag{14.27}$$

Separate the coefficient of v and the constant term:

$$[\text{const}]: \qquad -A'(\tau) + \kappa\theta\,B(\tau) = 0, \tag{14.28}$$

$$[\text{coef of } v]: \qquad -B'(\tau) + \tfrac{1}{2}\omega(\omega-1) - \kappa B(\tau) + \tfrac{1}{2}\alpha^2 B(\tau)^2 + \rho\alpha\omega B(\tau) = 0. \tag{14.29}$$

Pause and appreciate what just happened. A partial differential equation in two spatial dimensions plus time has been reduced to two ordinary differential equations in a single variable τ. The dimensional reduction is dramatic and not accidental: it is a direct consequence of the affine structure. The exponential-affine ansatz converted the PDE into an algebraic identity that had to hold for all v, and separating by powers of v produced one ODE for the constant term (A) and one ODE for the coefficient of v (B). This reduction generalises beyond Heston — it works for any affine diffusion in any dimension — and it is one of the cornerstones of modern affine jump-diffusion theory.

The reduction is a computational gift, not a theoretical accident. The original PDE (14.24) lives on a three-dimensional manifold (two state dimensions plus time), and solving it numerically by finite-difference or finite-element methods would require discretising that manifold — storing a grid of values and propagating them backward in time step by step. That is an expensive computation, and it must be redone for every change in parameters or initial conditions. The ODE system, by contrast, involves two scalar functions of one variable. Solving them numerically (or in our case, in closed form) is essentially free. The lesson is general: whenever you have affine structure, hunt for the exponential-affine ansatz, because it always reduces the computational complexity by a full dimensional order.

The method is sometimes called "transform analysis" and it is a cornerstone technique. It shows up in yield-curve modelling (Vasicek, CIR, and multi-factor affine term-structure models — see CH12), in credit-risk modelling (affine intensity processes), in commodity modelling (Schwartz and two-factor Gibson–Schwartz models), and in non-Gaussian time-series modelling (affine volatility models, stochastic intensity Poisson processes). Anywhere you see an affine drift and covariance, the transform method applies. Mastering the Heston derivation equips you to derive closed-form characteristic functions in all these other settings as well. This is why educators spend so much time on Heston despite it being just one of many stochastic vol models — the technique is the point, not just the model.

Rewrite (14.29) in standard Riccati form:

$$\boxed{B'(\tau) = \tfrac{1}{2}\alpha^2 B^2 + (\rho\alpha\omega - \kappa)B + \tfrac{1}{2}\omega(\omega-1), \qquad B(0) = 0} \tag{14.30}$$

$$\boxed{A'(\tau) = \kappa\theta\, B(\tau), \qquad A(0) = 0} \tag{14.31}$$

Terminal matching at $\tau = 0$ gives $g(T, x, v) = \exp\{\omega(x - K)\}$, so $A(0) = B(0) = 0$, as stated.

The Riccati equation (14.30) is quadratic in B. Riccati equations are the standard nonlinear ODE you meet in control theory, linear-quadratic optimisation, and — here — affine pricing models. They have the pleasant property of being linearisable: a Riccati of the form $B' = aB^2 + bB + c$ can be transformed into a linear second-order ODE, which then admits closed-form solutions in terms of exponentials when the coefficients are constant. In our case a, b, c depend on ω but are constant in τ, so we get a clean closed-form answer below. Once $B(\tau)$ is in hand, $A(\tau)$ is a simple antiderivative: $A(\tau) = \kappa\theta \int_0^\tau B(s)\,\mathrm{d}s$.

The Riccati linearisation trick is worth knowing because it recurs. Set $B(\tau) = -U'(\tau)/(\frac{1}{2}\alpha^2\, U(\tau))$. Then B satisfies a Riccati if and only if U satisfies the linear second-order ODE $U'' - (\rho\alpha\omega - \kappa)U' - \frac{1}{4}\alpha^2\omega(\omega-1)U = 0$, which has constant coefficients and therefore admits solutions as exponentials $U = C_1 e^{r_+\tau} + C_2 e^{r_-\tau}$ with $r_\pm$ the roots of the characteristic polynomial. Back-substituting and applying $B(0) = 0$ produces the formula for $B(\tau)$ that we report below. This linearisation works for any quadratic Riccati with constant coefficients and is the standard technique in affine pricing theory.

A nuance: the Riccati ODE depends on ω (through the coefficients), so for each Fourier frequency ϕ we evaluate at $\omega = i\phi$, we get a different $B(\tau; i\phi)$ and $A(\tau; i\phi)$. The dependence on ω is analytic (the coefficients are polynomials in ω), so the resulting B, A are analytic functions of ω. This analyticity is what makes the Fourier inversion work — we can continue the characteristic function into the complex plane and deform contours as needed for numerical stability. It also means that the "little trap" of branch choices is a genuine issue at isolated points where the argument of a square root or logarithm can cross a branch cut; careful implementation avoids this.

15.6.3 14.5.3 Closed-form Riccati solution

Let

$$d(\omega) = \sqrt{(\rho\alpha\omega - \kappa)^2 - \alpha^2\,\omega(\omega-1)}, \tag{14.32}$$

$$g_*(\omega) = \frac{\kappa - \rho\alpha\omega - d}{\kappa - \rho\alpha\omega + d}. \tag{14.33}$$

Then

$$B(\tau) = \frac{\kappa - \rho\alpha\omega - d}{\alpha^2} \cdot \frac{1 - e^{-d\tau}}{1 - g_* \, e^{-d\tau}}, \tag{14.34}$$

$$A(\tau) = \frac{\kappa\theta}{\alpha^2} \left\{ (\kappa - \rho\alpha\omega - d)\,\tau - 2\ln\left[\frac{1 - g_* \, e^{-d\tau}}{1 - g_*}\right] \right\}. \tag{14.35}$$

(Heston's original form; the "little trap" of Albrecher et al.
recommends flipping to $\tilde{g} = 1/g_*$ for branch-cut stability. Either works numerically, both are solved analytically.)

The "little trap" is worth flagging for anyone who plans to actually implement these formulas. The complex square root $d(\omega)$ and the complex logarithm in $A(\tau)$ both involve branch choices. On the original form of the equations, the branch of the logarithm can jump as ω sweeps along the real axis of Fourier inversion, producing a discontinuity in A and, consequently, a nonsensical characteristic function. The remedy — routinely applied in production code — is to use an equivalent algebraic rewriting (with $\tilde{g} = 1/g_*$) that keeps the logarithm on a continuous branch. This is not a deep issue; it is a numerical housekeeping step. But forget it, and your Fourier integrals give garbage.

A useful debugging trick: evaluate $\varphi(\phi)$ at $\phi = 0$. Since φ is a proper characteristic function (a probability-weighted exponential), it must equal exactly 1 at the origin. If your implementation returns something other than 1 at $\phi = 0$ (or nearby, for numerical reasons), you have a bug — most likely a branch-cut issue. Also check that $|\varphi(\phi)|$ decays as $\phi \to \infty$; for Heston, the decay is exponential in ϕ^2 for large ϕ, and any implementation that produces a non-decaying or growing $|\varphi|$ is wrong. These two sanity checks — $\varphi(0) = 1$ and decaying magnitude — catch essentially all implementation bugs quickly.

Production implementations also often compute $\ln \varphi$ rather than φ itself, to avoid underflow at large $|\phi|$ where φ can be very small. Working in the log domain preserves accuracy throughout the Fourier integration. At the very end, one exponentiates to get the integrand of the Gil–Pelaez formula. This is a standard numerical trick that goes beyond just Heston; it is the right way to handle any exponential-family quantity in floating-point arithmetic.

So the Riccati system can be solved analytically — this is the punchline that makes Heston useful.

15.6.4 14.5.4 Why a characteristic function but not the density

The joint SDE (X_t, v_t) is affine: drift and diffusion covariance are linear functions of v. For affine state-space models, a theorem of Duffie–Pan–Singleton guarantees that $\mathbb{E}[e^{\omega X_T + u v_T}]$ is exponential-affine in the current state — i.e. the characteristic function $\varphi(\phi)$ solves a finite-dimensional ODE system. The density $p(X_T)$ is the inverse Fourier transform of φ and has no closed form — it is a mixture of log-normals indexed by the unobserved variance path. We can price options because European payoffs are linear functionals of the density that Fourier-invert cleanly; we cannot write down p.

This distinction — having the characteristic function but not the density — deserves a moment of reflection. In probability theory we usually reach for the density when we want to compute expectations. But the characteristic function carries exactly the same information, and in some cases it is the cleaner object. For Heston, the mixture structure of the density is messy enough that there is no analytical handle on it, but the exponential-affine form of the characteristic function exposes its structure cleanly. The fact that we can price options without ever seeing the density is not a work-around; it is a genuinely different way of organising the computation. Fourier methods are now standard across modern derivatives — Heston was one of the first models to exploit this elegantly, and the technique generalises to jump models, stochastic interest rates, Lévy processes, and affine factor models.

To appreciate how far the characteristic-function approach extends, consider the Bates model (Heston plus log-normal jumps in spot). The characteristic function of a Lévy process is exponential in τ by the Lévy–Khintchine formula, and for a compound Poisson jump it has a neat closed form in terms of the jump distribution. In Bates, the Heston characteristic function is multiplied by a Lévy jump contribution, yielding a still-exponential-affine form that the Fourier inversion handles as easily as pure Heston. In the SVJJ model (stochastic vol with jumps in both spot and variance), the same machinery works with one more ODE for the jump contribution to the variance process. In Carr–Madan–Yor (CGMY) Lévy models, the characteristic function is explicit and Fourier inversion is trivial. The entire zoo of modern derivatives pricing models is built around characteristic-function structure, and Heston was the breakthrough paper that made this architecture popular.

For the rough volatility models that have emerged in the last decade, the characteristic function is no longer exponential-affine — the state space is infinite-dimensional — but the Fourier-based pricing approach still works, with the characteristic function computed by different methods (Laplace transforms of fractional integrals, numerical

quadrature in abstract function spaces). The spirit of the approach — inverting a characteristic function to price European options — has survived the shift to rough vol, even though the specific ODE structure has not. Heston's legacy is the framework; the specific model has been superseded in many applications but the pricing paradigm it established remains central.

Finally, the conditional characteristic function of $X_T = \ln F_T$ given $(X_t, v_t) = (x, v)$:

$$\boxed{\varphi(\phi;\, t, x, v) \;\equiv\; \mathbb{E}_t^{\mathbb{Q}}\big[\, e^{i\phi X_T}\,\big] \;=\; \exp\Big\{\, i\phi x + A(\tau; i\phi) + B(\tau; i\phi)\, v \,\Big\}} \tag{14.36}$$

with A, B from (14.34)–(14.35) evaluated at $\omega = i\phi$.

15.7 14.6 Fourier-inversion pricing — P_1 and P_2

A European call on the futures struck at K, maturity T, is

$$C(t, F_t, v_t) = e^{-r\tau}\, \mathbb{E}_t^{\mathbb{Q}}[(F_T - K)^+] = e^{-r\tau}\Big\{\mathbb{E}_t^{\mathbb{Q}}[F_T\, \mathbf{1}_{F_T > K}] - K\, \mathbb{Q}_t(F_T > K)\Big\}. \tag{14.37}$$

Heston's trick: split into two "Black–Scholes-like" pieces,

$$C = e^{-r\tau}\, F_t\, P_1 \;-\; e^{-r\tau}\, K\, P_2, \tag{14.38}$$

where

$$P_2 = \mathbb{Q}_t(\, X_T > \ln K\,), \tag{14.39}$$

and P_1 is the same probability under the share measure $\tilde{\mathbb{Q}}$ (using F as numeraire), i.e.

$$P_1 = \tilde{\mathbb{Q}}_t(\, X_T > \ln K\,). \tag{14.40}$$

This decomposition mirrors the Black–Scholes formula (CH06) exactly. In Black–Scholes, $C = F\, N(d_1) - K\, N(d_2)$, and the two Gaussian probabilities $N(d_1)$ and $N(d_2)$ are exactly the probabilities of $F_T > K$ under two different measures: the share measure (for $N(d_1)$) and the domestic risk-neutral measure (for $N(d_2)$). The two measures differ only by a numeraire change — F versus M — which is CH05's two-numeraire switching specialised to this call-on- futures problem. In Heston, the same two probabilities show up, but they are no longer Gaussian; they are Heston-distribution tails, and we compute them by Fourier inversion from the characteristic function.

The deep reason this decomposition exists is a consequence of the call payoff structure, not of the pricing model. $(F_T - K)_+ = F_T \mathbf{1}_{F_T > K} - K \mathbf{1}_{F_T > K}$, and the two terms are expectations of (delta-function-weighted indicator) and (simple indicator) respectively. The first is the first moment of F_T conditional on exercise; under the share measure it becomes a pure probability. The second is a pure risk-neutral probability. So P_1 and P_2 are not Heston-specific objects — they are universal to any call-price formula, under any underlying dynamics. What Heston gives us is a computable form for these two probabilities via the characteristic function, which otherwise would have to be obtained by direct density integration (which is not feasible for Heston).

The two-measure framing also clarifies why Fourier inversion is so natural. Both P_1 and P_2 are CDFs of a random variable (the log-spot at maturity) under different measures. CDFs are integrals of densities, which are Fourier transforms of characteristic functions. The Gil–Pelaez formula lets us skip the density step: go directly from characteristic function to CDF via a single integral. This streamlines the computation substantially — we never need to invert the characteristic function to get a density, only to compute the CDF at the single point $\ln K$.

Each probability is recovered from the characteristic function by the Gil–Pelaez inversion:

$$\boxed{P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-i\phi \ln K}\, \varphi_j(\phi)}{i\phi}\right] \mathrm{d}\phi, \qquad j = 1, 2} \tag{14.41}$$

The Gil–Pelaez formula is one of those classical results that deserves to be better known. It says that the probability that a random variable exceeds a given level can be computed purely from its characteristic function via a single integral — no density required. The 1/2 comes from the fact that an unbiased symmetric Brownian motion assigns

equal probability to being above or below its mean; the integral corrects for the actual asymmetry of the distribution as encoded in the imaginary part of the characteristic function. Numerically, the integrand decays like $1/\phi$ modulated by the decay of $|\varphi_j(\phi)|$, which for Heston is Gaussian-like in ϕ and shrinks rapidly. Most of the contribution to the integral comes from $\phi < 50$ or so, and Gauss–Laguerre quadrature with 64 nodes is more than adequate.

The Gil–Pelaez formula was derived in 1951 in a different context — it was originally stated for probability distribution functions rather than for financial applications — and it sat relatively dormant until stochastic-vol modelling made it the workhorse of Heston pricing. Its derivation follows from a straightforward Fourier-inversion argument applied to the characteristic function. If you write $F(x) = \mathbb{P}[X \le x] = \int_{-\infty}^{x} p(y)\,\mathrm{d}y$ and express p as an inverse Fourier transform of φ, the order of integration can be swapped (with some care at the integration limits) to give the Gil–Pelaez expression. The complex exponential $e^{-i\phi x}/(i\phi)$ is the Fourier transform of the step function $\mathbf{1}_{y>x}$, so the formula is essentially "the probability that $X > x$ equals the expectation of the step function, which equals (by Parseval) the Fourier integral of the product of the characteristic function and the Fourier transform of the step."

A practical numerical point about the Gil–Pelaez integral: the integrand is oscillatory, which can cause convergence issues if you use naive uniform quadrature. Gauss–Laguerre quadrature handles the exponential decay well; alternative approaches include the COS method (Fang and Oosterlee), which uses Fourier cosine series expansions and can be even faster for Heston than Gauss–Laguerre. For production systems, the COS method is increasingly the default because it allows very efficient pricing of non-European payoffs (barriers, Bermudans) using the same characteristic function, via backward recursion in the COS coefficients.

Heston characteristic function Re/Im/|phi| *Re, Im and magnitude of* $\varphi(\phi)$ *for the base Heston parameters. The rapid decay in* $|\varphi|$ *is what guarantees the Gil–Pelaez integrand* $\mathrm{Re}[e^{-i\phi \ln K}\varphi_j(\phi)/(i\phi)]$ *is absolutely integrable and hence Fourier-invertible.*

with

$$\varphi_2(\phi) = \varphi(\phi;\, t, x, v), \qquad \varphi_1(\phi) = \frac{\varphi(\phi - i;\, t, x, v)}{\varphi(-i;\, t, x, v)}. \tag{14.42}$$

(Division by $\varphi(-i) = F_t$ normalises the density for the share measure. With $X_T = \ln F_T$, $\varphi(-i) = \mathbb{E}_t[e^{-i\cdot iX_T}] = \mathbb{E}_t[e^{X_T}] = \mathbb{E}_t[F_T] = F_t$ because F is a $\mathbb{Q}$-martingale in the futures frame, so the ratio $F_t/F_t = 1$ drops out cleanly inside the Gil-Pelaez integrand. In the spot frame under $\mathbb{Q}$ the analogous normaliser would instead be $S_t\, e^{r\tau}$, but we have chosen the futures frame throughout this chapter.)

The shift by $-i$ in φ_1 is the Esscher-transform fingerprint of the measure change to the share measure. Under the share measure, F is the numeraire, and densities get reweighted by $F_T/(F_t e^{r\tau})$. On the characteristic function, this reweighting translates exactly to a complex argument shift of $-i$. This is a general trick: any "forward-measure" or "share-measure" probability can be computed from the original characteristic function by shifting ϕ appropriately, without ever redoing the derivation under the new measure. The underlying mechanism is the same numeraire-change machinery introduced in CH05: the Radon–Nikodym density process is $F_T/(F_t e^{r\tau})$, so multiplying by $e^{i\phi X_T}$ and renormalising amounts to shifting the Fourier argument by $-i$.

The Esscher transform is named after Swedish actuary Fredrik Esscher, who introduced it in 1932 in the context of insurance risk. It has deep connections to exponential tilting in statistics and to change-of-measure arguments in finance. The general idea: given a characteristic function $\varphi(\phi)$ of X, the tilted characteristic function $\varphi(\phi+i\alpha)/\varphi(i\alpha)$ corresponds to a new measure under which X's density is multiplied by $e^{\alpha X}$ and renormalised. For $\alpha = 1$ this is the share-measure in finance; for other α values it is the Esscher principle for pricing under a risk-neutral measure chosen to match a specific expectation. Whenever you see a characteristic- function shift by $\pm i$ or $\pm i\alpha$, there is a change of measure hiding in it.

Why is this useful? Because the Heston characteristic function was derived once, for $X = \ln F_T$, under the $\mathbb{Q}$ measure. The complex shift lets us re-use that one characteristic function to compute probabilities under a family of related measures (share measure, forward measure, etc.) at zero additional derivation cost. This is a significant computational benefit: the Heston characteristic function evaluation dominates the runtime of Fourier-inversion pricing, so being able to re-use it under multiple measures saves substantial work.

The Heston call price.

$$\boxed{C^{\mathrm{Heston}}(t, F_t, v_t; K, T) \;=\; e^{-r\tau}\big[F_t\, P_1 \;-\; K\, P_2\big]} \tag{14.43}$$

Put is by put-call parity. One integral per probability, computed by Gauss–Laguerre / FFT / Carr–Madan in practice.

The "one integral per probability" line is important for performance. In a large calibration problem you might price several thousand options — every strike and maturity on a dense grid — and each one requires two Gil–Pelaez integrals. With 64 quadrature nodes per integral, that is over a million characteristic function evaluations per calibration iteration. This sounds expensive, but each evaluation is essentially a handful of complex exponentials, and the whole calibration completes in under a second on modern hardware. Carr–Madan FFT collapses the computation further by pricing an entire log-strike curve in a single FFT, dropping the cost per option to effectively zero. This is why FFT-based Heston calibration is the industry default.

The Carr–Madan idea is worth savouring. If you write the call price as a Fourier integral over strike, you notice that the integrand depends on strike only through $e^{-i\phi \ln K}$. This means that fixing the quadrature points in ϕ and then scanning strike is just a Fourier transform of a weighted characteristic function evaluated at those ϕ-points. The Fast Fourier Transform (FFT) accomplishes exactly this — it transforms a sequence of function values into a sequence of Fourier coefficients (or vice versa) in $O(N \log N)$ time, where N is the number of points. By sampling ϕ on a regular grid with appropriate spacing and performing an FFT, Carr and Madan produce the entire log-strike curve of call prices in a single pass. The implementation is elegant: choose an η damping, compute $\psi(\phi)$ at the grid of ϕ-points, FFT to get the log-strike curve, read off prices at desired strikes by interpolation. Total cost: one N-point FFT (a few milliseconds for $N = 4096$) regardless of how many strikes are needed.

Before Carr–Madan (1999), pricing a full smile for a single maturity required one Gil–Pelaez integral per strike per probability — typically 20 strikes $\times$ 2 probabilities $\times$ 64 quadrature nodes = 2560 characteristic-function evaluations per smile. Carr–Madan reduced this to a single FFT with 4096 characteristic-function evaluations, which priced the entire smile at once. The net speedup is modest on a single smile but compounds dramatically in calibration where you price thousands of options per iteration. Modern Heston calibration is Carr–Madan plus a Levenberg–Marquardt optimiser, and this combination reliably calibrates SPX to within basis-point RMS error in under a second.

Carr–Madan alternative. Dampen the payoff by $e^{\eta x}$ and transform:

$$C = \frac{e^{-\eta \ln K - r\tau}}{\pi} \int_0^\infty \mathrm{Re}\big[e^{-i\phi \ln K}\, \psi(\phi)\big]\, \mathrm{d}\phi, \qquad \psi(\phi) = \frac{\varphi(\phi - (\eta+1)i)}{\eta^2 + \eta - \phi^2 + i(2\eta+1)\phi}. \tag{14.44}$$

This is a single FFT over a log-strike grid — the workhorse for calibration.

The damping parameter $\eta > 0$ is a free knob that practitioners tune (usually $\eta \in [0.5, 2]$) to trade off between accuracy at different strikes. A larger η improves accuracy for deep out-of-the-money calls but hurts the ATM region; a smaller η is more uniform across strikes. Rule of thumb: $\eta = 0.75$ is a reasonable default for most equity calibrations. Pick a bad η and your far-OTM prices will be inaccurate to several percent, which is enough to throw off the wings of your calibrated smile.

The need for damping exists because the raw call-price integrand is not absolutely integrable as a function of ϕ — it has an $O(1/\phi)$ decay that is borderline for convergence. Multiplying by $e^{\eta \ln K} = K^\eta$ (equivalently, damping the payoff by $e^{\eta x}$ in the strike domain) shifts the decay to exponential and makes the integral convergent. The catch is that the damping shifts weight toward high-K regions — deep OTM calls — and away from low-K regions. Asymmetric damping is the price we pay for using a single-sided call-price transform. An alternative approach, sometimes called the "generalised FFT," uses a symmetric combination of call and put prices and eliminates the η dependence, but at the cost of more bookkeeping. Most practitioners stick with Carr–Madan and tune η for their typical strike range.

A related numerical note: the FFT log-strike grid spacing and the ϕ grid spacing are reciprocally related via the FFT Nyquist constraint. If you want strike resolution $\Delta \ln K = 0.01$ (roughly 1% moneyness resolution), you need ϕ grid extending to $2\pi/\Delta \ln K \approx 628$. With 4096 FFT points, the ϕ spacing is roughly 0.15, and the integration captures the characteristic-function contribution down to ϕ roughly equal to that spacing. For Heston, the characteristic function decays rapidly above $\phi \sim 10$, so these grid parameters produce negligible truncation error. Finer grids are used for deep OTM calibration or for improved accuracy in the extreme tails.

15.8 14.7 Greeks and calibration notes

Five risk-neutral parameters $(\kappa, \theta, \alpha, \rho, v_0)$ plus r. Each has a clean economic role:

Parameter	Role	Empirical sign (equity)
v_0	instantaneous variance	matches ATM level
θ	long-run variance	matches long-dated ATM
κ	mean-reversion rate	term-structure slope
α	vol-of-vol	smile curvature (convexity)
ρ	spot/vol corr.	smile skew (negative $\rho \Rightarrow$ left skew)

The parameter identification story is crisp, and this is one of Heston's greatest practical virtues. Different features of the implied vol surface map cleanly to different parameters, so the calibration problem is well-posed and the fitted parameters are stable across days. If the overall level of your vol surface rises by 2 points, v_0 and θ rise; if the skew steepens, ρ becomes more negative; if the wings puff out, α increases. Rarely does moving one parameter require compensating moves in another to stay on-surface, and when it does, the required compensation is modest.

Here is the quantitative version of the parameter-sensitivity story, useful for debugging calibration. The Heston ATM implied vol at maturity T is approximately $\sqrt{m_T}$ where $m_T = \mathbb{E}[v_T]$ follows the exponential relaxation formula. Differentiating ATM vol with respect to parameters: $\partial\text{ATM}/\partial v_0$ is nonzero but decays with T (short-dated ATM sensitive to v_0, long-dated not); $\partial\text{ATM}/\partial\theta$ grows with T (long-dated ATM tracks θ); $\partial\text{ATM}/\partial\kappa$ is the blend-rate sensitivity. The short-maturity ATM skew (slope of smile at the money) is approximately $\rho\alpha/(4\sqrt{v_0})$ — a *finite* constant as $T \to 0$ — so ρ and α both enter multiplicatively. The ATM convexity (curvature of smile) is approximately $\alpha^2 T/(v_0\kappa)$ for moderate T, so this is primarily an α sensitivity. These formulas are approximations, but they capture the dominant structure and are extremely useful for parameter identification.

When calibration is well-posed, the Jacobian matrix of calibration residuals with respect to parameters is well-conditioned, and Levenberg–Marquardt converges rapidly. When it is ill-posed, the Jacobian has near-zero eigenvalues corresponding to parameter combinations that are under-determined. A common sign of ill-posedness is α and ρ drifting together in a correlated way — indicating that the calibration cannot distinguish between them on the data. This is most likely to happen when only ATM quotes are available (no skew information) or when the maturity range is too narrow. The remedy is to include more smile points across strikes and more maturities in the calibration.

Calibration targets the smile, not the term structure. This is an important practitioner truth. Heston has a five-parameter structure and is trying to fit what is typically a two-dimensional surface — strikes across maturities. In practice, the model captures the smile (variation across strikes for a fixed maturity) much better than it captures the full term structure of the smile (how the smile changes with maturity). If you calibrate Heston to the full surface, you get parameters that are a compromise — they neither fit short-dated smiles well nor long-dated smiles well. The standard practitioner workflow is to calibrate to a single representative maturity (say three months), accept that shorter and longer maturities will be imperfect, and use a richer model (or a term-dependent extension) when the term structure matters. The flat-term-structure failing is one of Heston's blind spots, and we come back to it below.

The term-structure fit problem has a specific structure that is worth understanding. At very short maturity, Heston's smile is controlled mostly by v_0 (the starting variance) and the short-time asymptotics of the Riccati solution. The ATM vol matches $\sqrt{v_0}$ and the short-time ATM skew is $\rho\alpha/(4\sqrt{v_0})$ (Durrleman 2003; Gatheral 2006) — a bounded constant as $T \to 0$. Real short-dated smiles, in contrast, exhibit a skew that *grows* as maturity shortens (empirically like T^{-H} with $H \in (0, 1/2)$), so Heston *understates* the short-dated skew relative to market. At long maturity, the skew decays faster than $1/\sqrt{T}$ because the variance has time to mean-revert and average out the correlation effect. The full term-structure shape implied by Heston is "steep skew at short dates, flat skew at long dates" — which matches the qualitative pattern of real smiles but rarely the quantitative shape. Markets typically have "steep skew at short dates, less-steep but still-steep skew at intermediate dates, and flatter skew at very long dates" — a more gradual decay that Heston cannot match with constant parameters.

This is where term-dependent Heston comes in. Let κ, θ, α, ρ all be piecewise-constant functions of time, with breakpoints at the maturities of traded options. The characteristic function can still be computed (by integrating the Riccati ODE piecewise in time), and the model can now match the full term structure exactly. The downside is parameter proliferation — now you have $5 \times N$ parameters for N maturity buckets — and calibration becomes more delicate. But for desks that need to match long-dated exotic books to the traded surface exactly, term-dependent Heston is the workhorse. Some implementations go further and treat v_0 as a "state" while calibrating $\kappa(t), \theta(t), \alpha(t), \rho(t)$ as "parameter functions" — this maintains Markov consistency while adding flexibility.

Heston smile vs rho *Heston smile for $\rho \in \{-0.9, -0.5, 0, +0.5\}$, holding $(\kappa, \theta, \alpha, v_0)$ fixed. Negative ρ tilts the smile*

down-right (equity left-skew); positive ρ tilts it up-right (FX or commodity inverted-skew). The ATM point barely moves — ρ is a pure slope control.

Notice how remarkably clean the rotation is: as ρ sweeps from negative through zero to positive, the smile pivots around its ATM point. This is the cleanest visual demonstration of what it means to say "ρ controls the skew." Notice also that the ATM level is essentially invariant — changing ρ by ± 0.3 moves the 25-delta implied vols meaningfully but leaves the 50-delta ATM vol essentially unchanged. This clean decomposition is what makes Heston calibration well-posed.

In the language of risk reversals, the 25-delta risk reversal — defined as the difference between 25-delta call vol and 25-delta put vol — is essentially proportional to ρ at fixed maturity. A risk reversal of zero corresponds to $\rho = 0$; a positive risk reversal (calls dearer than puts) corresponds to $\rho > 0$; a negative risk reversal corresponds to $\rho < 0$. The approximate proportionality coefficient is $\alpha/\sqrt{v_0}\sqrt{T}$, so at fixed α and v_0, the risk reversal decays as $1/\sqrt{T}$. Real markets show a similar pattern, which is part of why Heston matches the qualitative term structure of risk reversals reasonably well. The cleanness of the ρ-to-risk-reversal mapping is what lets traders calibrate ρ almost by inspection of the risk reversal quote for a representative maturity.

The visual symmetry of the ρ-sweep also demystifies what is sometimes called "sticky-delta" vs "sticky-strike" behaviour. In a Heston world with $\rho < 0$, the smile tilts downward to the right; when spot moves, the delta coordinates of the options shift, and the vols at fixed delta stay roughly the same (sticky-delta behaviour). Heston therefore naturally generates sticky-delta dynamics, which is the empirical behaviour observed in FX and less-systematically in equities. Models with different dynamics (pure local vol) tend to generate sticky-strike behaviour (vols at fixed strike stay the same), which is less empirically supported. This is an underappreciated advantage of stochastic vol over local vol for forward-smile modelling.

Heston smile vs alpha *Smile curvature grows with α. Small α (weak vol-of-vol) flattens the smile toward the BS line; large α lifts both wings — deep puts and deep calls both gain, symmetrically around the mild $\rho = -0.3$ tilt.*

The complementary picture for α: increase vol-of-vol and both wings lift together, preserving the skew that ρ encoded. Think of α as the "wingspan" parameter. A flat Black–Scholes smile corresponds to $\alpha = 0$ (deterministic variance). As α grows, the mixture of lognormals becomes more heterogeneous — some paths are much more volatile than others — and the resulting unconditional distribution has fatter tails, hence higher implied vols at the wings. In the limit of very large α (and Feller severely violated), the smile becomes so steep at the wings that the density takes on heavy-tail features approaching Student-t behaviour.

The 25-delta butterfly — the average of 25-delta call and put vols minus ATM vol — is the market quote that directly reflects α. A butterfly of zero means a flat smile, consistent with $\alpha = 0$. A positive butterfly means both wings are richer than ATM, which maps to $\alpha > 0$. In the approximate asymptotic formula, the butterfly scales roughly as $\alpha^2\kappa^{-1}$ with a mild T-dependence. The cleanness of this mapping means that, like ρ-to-risk-reversal, the α-to-butterfly mapping can be read off approximately by eyeballing the butterfly at a representative maturity.

When calibrators describe a surface with "rich wings," they mean a high butterfly. This maps to high calibrated α, which in turn triggers Feller-violation diagnostics. Hence the indirect but real connection between rich-wing smiles, high vol-of-vol, and Feller-boundary behaviour. Surfaces with near-flat wings (low butterfly, typical of calm FX markets) calibrate to $\alpha \approx 0.3$ or less and are comfortably Feller-safe. Surfaces with very rich wings (high butterfly, typical of stressed equity markets) calibrate to $\alpha > 0.7$ and may violate Feller aggressively. The relationship between market feature (butterfly), calibrated parameter (α), and diagnostic (Feller violation) is clean and traceable.

15.8.1 14.7.1 Greeks

Because $C = e^{-r\tau}(F\,P_1 - K\,P_2)$ and P_1, P_2 depend on v only through $B(\tau)\,v$ in φ:

$$\Delta_F = \frac{\partial C}{\partial F} = e^{-r\tau} P_1, \tag{14.45}$$

$$\mathcal{V}_v = \frac{\partial C}{\partial v_0} = e^{-r\tau}\left(F\,\frac{\partial P_1}{\partial v} - K\,\frac{\partial P_2}{\partial v}\right), \qquad \frac{\partial P_j}{\partial v} \text{ via differentiating (14.41) under the integral.} \tag{14.46}$$

Vega in the Heston sense is with respect to v_0, not BS σ.

This is a subtle point worth emphasising. When a practitioner says "vega," they typically mean Black–Scholes vega — the sensitivity of the option price to the Black–Scholes implied volatility. In Heston, the natural vega is with respect to v_0, the current instantaneous variance. These two vegas are related but not identical; they differ by a

Jacobian that depends on the full set of Heston parameters. If your hedging is driven by Heston, you should hedge $\mathcal{V}_v$; if your hedging is driven by BS conventions (e.g. if you quote in BS vega terms), you can convert back via the chain rule. A deeper practitioner insight: in Heston, vega is only one of many vol-sensitivity parameters. You also have sensitivities to κ, θ, α, and ρ. Each of these is a degree of freedom in the model that can move, and a complete hedging programme would neutralise each of them — the so-called model-risk hedge, as distinct from the delta/vega hedge.

In practice, delta-vega hedging in Heston is close enough to BS-delta-vega hedging that most desks treat them interchangeably for vanilla books. The real divergence shows up for path-dependent products. Consider a barrier option. In BS, the barrier risk is hedged by matching the terminal payoff; in Heston, the barrier risk is coupled to vol dynamics because vol spikes can push spot through the barrier. A Heston-consistent barrier hedge uses not just delta and vega but also vanna (the cross sensitivity) and volga (the second vega). These higher-order Greeks are straightforward to compute in Heston via numerical differentiation of the characteristic-function price, and their management is a major focus of exotic desks. For a vanilla book, higher-order Greeks matter less but still exist; ignoring them is a source of hedge slippage over time.

A related concept: the concept of "Heston vega" as distinct from "BS vega" becomes important when you discuss the cost of a vega hedge. In BS, the vega hedge is obtained by buying or selling a liquid option to offset the position's BS-vega. In Heston, a vega hedge must offset the sensitivity to v_0, which depends on the Heston parameters. The quantitative difference is typically small but can be meaningful for large positions or for positions with significant vol convexity. Many production systems compute both BS-vega (for P&L attribution) and Heston-vega (for hedging) simultaneously and report both. The gap between them is the "model-convexity" adjustment.

One more consideration: in a Heston world, the natural vol risk to hedge is often the variance-swap risk, not the v_0 risk. The variance swap has a linear payoff in realised variance and provides a clean way to hedge the integrated-variance exposure that options carry. Many exotic desks run a hybrid hedging programme: delta on spot, v_0-vega on options (for short-term risk), variance-swap hedges (for long-term integrated-variance risk), and periodic rebalancing as the vol regime changes. This layered hedging approach is more robust than pure BS-vega hedging and is a direct consequence of thinking in Heston terms rather than BS terms.

15.8.2 14.7.2 Calibration workflow (practitioner)

1. Snap an implied-vol surface (liquid listed options).
2. Choose a loss $L(\kappa, \theta, \alpha, \rho, v_0) = \sum_i w_i \left(\sigma_i^{\text{mkt}} - \sigma_i^{\text{Heston}}\right)^2$ (IV RMS) or price-RMS.
3. Minimise by Levenberg–Marquardt / differential evolution. Start at $\rho < 0$, $v_0 \approx (\sigma^{\text{ATM}})^2$, $\theta \approx v_0$, $\kappa \in [1, 5]$, $\alpha \in [0.3, 1.0]$.
4. Check the Feller condition $2\kappa\theta \geq \alpha^2$; if violated, use full-truncation Monte Carlo for path-dependent payoffs.
5. Re-calibrate daily; watch for ρ jumps (data artefact) or α blowups (ill-posed surface).

A few additional practitioner notes on calibration. First, the choice of loss function matters more than one might expect. Fitting in implied-vol space (IV RMS) gives equal weight to each quote regardless of moneyness; fitting in price space gives more weight to ATM options (which have larger prices) and less to OTM. Most calibrators fit in IV space because it aligns better with trader intuition, but some use price-weighted IV, where each IV is weighted by its vega — this is closest to maximum-likelihood if the mispricings are homoscedastic in price units. Second, you want good starting values. Random starts often land in local minima that have the wrong sign of ρ or a nonsensical κ. The heuristic starts given above (ATM vol for v_0, equal to θ, a moderate κ and α) put you within basin-of-attraction range for the global optimum on almost any reasonable surface.

A subtler but equally important consideration is the choice of calibration targets. If your final use is pricing short-dated options, calibrate on short-dated quotes with heavy weights on ATM; the calibrated parameters will reproduce the short-dated surface faithfully at the cost of mild errors on long-dated options. If your final use is pricing long-dated cliquets, calibrate on long-dated quotes and be tolerant of short-dated error. The "universal calibration" that targets the entire surface equally is rarely the right answer for any specific application. Practitioners often run multiple calibrations with different weighting schemes and deploy each for a specific product class. This means you might have three or four "flavours" of Heston parameters on any given day, each tailored to a different product, rather than a single calibrated set.

Third, be disciplined about the choice of the option grid. Include enough strikes to pin down the smile shape (at least five quotes: 10-delta put, 25-delta put, ATM, 25-delta call, 10-delta call) and enough maturities to span the term structure you care about. Weekly quotes can be noisy and short-lived; prefer monthly and quarterly quotes

as calibration targets. Avoid strikes so deep OTM that the quotes are stale or illiquid; these carry low information value and high noise. A good target grid for SPX calibration might be 20 strikes $\times$ 8 maturities = 160 quotes, which is plenty for a clean Heston fit without over-fitting to noise.

Fourth, store your calibration history. Day-to-day parameter drift is a diagnostic; a multi-month history of calibrated parameters gives you a sense of the model's stability and lets you detect regime changes or data glitches. Many desks display plots of $(\kappa_t, \theta_t, \alpha_t, \rho_t, v_{0,t})$ as time-series and use them for model risk management. Unusual jumps in any parameter are worth investigating; small drifts are expected and benign.

Fifth, watch for parameter drift day to day. Stable market conditions should produce stable calibrated parameters — especially ρ and κ, which capture structural features of vol dynamics. If your ρ jumps from -0.7 to $+0.3$ overnight, something is wrong: either your data has a stale quote, your optimiser found a local minimum, or a genuine regime shift happened and you should investigate. A practice that many desks adopt is to calibrate with penalty toward yesterday's parameters: add a term $\lambda \sum_i (p_i - p_i^{\text{yest}})^2$ to the loss to encourage smoothness through time. The penalty weight λ is tuned by trial.

A related refinement is to penalise parameters toward reasonable prior ranges ("Bayesian calibration"). Add a term $\sum_i (p_i - p_i^{\text{prior}})^2/\sigma_i^2$ with prior means and standard deviations derived from cross-sectional or historical experience. This regularisation discourages the optimiser from landing on parameters that are numerically permissible but economically implausible (e.g. $\kappa = 50$, which corresponds to a week-long half-life — unphysically fast for equity vol). Bayesian calibration is particularly useful when data is noisy or sparse, e.g. in illiquid single-stock markets where a clean smile is hard to observe. In these settings, the prior effectively replaces missing information and produces a more stable and interpretable calibrated set.

Finally, many practitioners use a two-stage calibration. Stage one: fit a simpler BS-smile parameterisation (e.g. SVI, SABR) to the observed quotes to get a smooth arbitrage-free approximation of the surface. Stage two: calibrate Heston to the smoothed surface rather than to the raw quotes. This decouples the "smooth-the-data" problem from the "fit-the-model" problem and often produces cleaner Heston parameters. The SVI intermediate step is particularly popular because SVI can fit essentially any observed smile exactly, so stage two becomes a pure Heston-fitting exercise without noise. The downside is that stage one's SVI parameters do not directly inform Heston; they just produce a smoothed target.

Sixth, the calibrated parameters depend on what you are calibrating to. A surface that includes only short-dated options (weeklies, monthlies) will calibrate to a very different (κ, θ) than a surface that includes long-dated LEAPS. This is not pathological; it is a consequence of Heston's inability to simultaneously fit short and long dates — a topic we return to in the blind-spots discussion below.

One should also note that Heston (as a continuous-time diffusion) has zero probability of discrete jumps. Real markets have jumps — around earnings, Fed announcements, major geopolitical events. If your calibration window includes any such event, the short-dated smile right before the event will show an anomaly (a "kink" or localised lift at near-money strikes) that Heston cannot fit. The calibrator will try to spread this anomaly across the smile by boosting α, which distorts the calibration. Some desks exclude the last few days before a known event from calibration, or explicitly model the event as a jump-diffusion overlay (Bates model) and calibrate the residual smoothed surface with Heston. Handling these event-driven dynamics properly is one of the harder parts of real-world calibration, and no single solution works across all cases.

Sample implied-vol surface (calibrated Heston, 3-D plot in the notes): strike $\in [0.5, 1.5]$, term $\in [0, 1.5]$, impvol $\in [0.20, 0.40]$ — shows the classic left skew short-dated, flattening long-dated equity shape.

One final word on calibration: the residual errors tell a story. After the best-fit Heston calibration, inspect the residuals — the gap between calibrated model IVs and market quotes — across strikes and maturities. Clean residuals should be noise-like, without systematic patterns. If you see a smile-shaped residual (model flat, market smiling) at the short end and flat residuals at the long end, your calibration is targeting long-dated features at the expense of short-dated. If you see positive residuals at the wings and negative at the middle across all maturities, your α is too small. If you see negative residuals on the put side and positive on the call side, your ρ is too negative (or not negative enough). These diagnostic patterns are how experienced calibrators diagnose and tune their fits. A good residual inspection tool is as important as the optimiser itself for production calibration.

15.9 14.8 Variance swaps

A variance swap pays realised variance minus a strike at T. The realised variance is

$$\text{"realised variance"} \triangleq \sum_{n=1}^{N} \left(\frac{F_{t_n}}{F_{t_{n-1}}} - 1 \right)^2 = R, \tag{14.47}$$

pay-off at T:

$$\text{Pays } (R - K) \text{ @ } T. \tag{14.48}$$

A variance swap is a spectacularly clean instrument conceptually. Unlike an option, whose payoff depends on the terminal underlying price, a variance swap's payoff depends only on the total amount of variability the underlying exhibited between now and maturity — the sum of squared daily returns. Realised variance is what traders actually pay when they delta-hedge an option position, so a variance swap lets you trade the very quantity that BS theory says you are paying for when you buy or sell vol. In effect, a variance swap is "pure vol exposure" — delta-neutral by construction, vega-one by construction, and a cleaner hedging instrument than options themselves when the goal is vol exposure.

The history of variance swaps is worth knowing. They were introduced in the 1990s by Demeterfi, Derman, Kamal, and Zou at Goldman Sachs, who showed that a variance swap can be replicated by a static portfolio of options plus dynamic delta hedging. The replication result is a linchpin of modern vol trading: it guarantees that variance swap prices are arbitrage-consistent with option prices, without any model assumption. The "log-contract" representation — where a variance swap's fair strike is derived from a weighted integral of call and put prices across strikes — is explicit and model-free. This is why VIX (the implied variance of SPX over 30 days) is computed directly from option prices without needing any stochastic-vol model; the VIX formula is essentially a discrete approximation to the replication integral.

Variance swaps became a major institutional product in the 2000s, peaking in popularity around 2007. The 2008 crisis exposed a concerning feature: the replication portfolio is vulnerable to jumps in the underlying. A large jump introduces a "jump-contract" term that is not captured by the log-contract replication, and variance swap P&Ls deviated sharply from model predictions during the crisis. This motivated the shift toward "capped" variance swaps (which bound the maximum payoff) and toward more robust variance products like VIX futures and VIX options. The cleaner theoretical structure of variance swaps remains a gold standard for vol analysis, but the practical preference has shifted toward capped instruments that bound tail risk.

Use $\frac{F_{t_n}}{F_{t_{n-1}}} - 1 \approx \frac{\mathrm{d}F_t}{F_t}$ on a fine grid, so

$$R \simeq \sum_{n=1}^{N} \left(\frac{\mathrm{d}F_{t_n}}{F_{t_n}} \right)^2. \tag{14.49}$$

Now using (14.14) $\frac{\mathrm{d}F_t}{F_t} = \gamma \sqrt{v_t}\, \mathrm{d}W_t$ (with $\gamma = 1$),

$$R \simeq \sum_{n=1}^{N} (\gamma \sqrt{v_{t_{n-1}}})^2 (W_{t_n} - W_{t_{n-1}})^2. \tag{14.50}$$

Taking expectation, as $N \to \infty$, the Itô isometry (CH03) turns the sum of squared Brownian increments into a time integral of the squared diffusion coefficient:

$$\mathbb{E}[R] \longrightarrow \mathbb{E}\left[\int_0^T \gamma^2 v_s \, \mathrm{d}s \right] = \gamma^2 \int_0^T \mathbb{E}[v_s] \, \mathrm{d}s. \tag{14.51}$$

The limit operation turns a discrete sum into a continuous integral, and the Itô isometry (or equivalently, the quadratic variation of a semimartingale) converts the sum of squared Brownian increments into a time integral of the squared diffusion coefficient. The result is fundamental: in the infinite-sampling limit, the expected realised variance is the time-integral of the expected instantaneous variance. This is a model-free identity — it holds for any stochastic vol model, not just Heston — but Heston makes the right-hand side computable in closed form.

The distinction between "expected realised variance" and "variance of the expectation" is worth nailing down. Expected realised variance is the risk-neutral average of what the realised variance will turn out to be — an integral

over all possible paths of the variance process. The variance of the expectation is the uncertainty in our forecast of the variance, which is a quite different object. For pricing a variance swap, we care only about the expected realised variance, because the swap pays on the first moment. For pricing options on variance swaps or variance futures, we care about higher moments of realised variance, which Heston also makes computable but in a more elaborate way.

In discrete-time practice, the sum $\sum(F_{t_n}/F_{t_{n-1}}-1)^2$ uses simple returns, not log returns. The distinction matters only at second order but occasionally produces observable differences. Variance swap contracts typically specify exactly how the realised variance is computed — most standard contracts use simple returns scaled by an annualisation factor, with specific handling of dividend ex-dates and weekends. When you price a variance swap, you should match the contract's specific realisation formula rather than assuming generic quadratic variation.

15.9.1 14.8.1 Expected variance under Heston

Compute $\mathbb{E}[v_s]$ under the Heston SDE. From (14.15),

$$v_t - v_0 = \int_0^t \kappa(\theta - v_s)\,\mathrm{d}s + \alpha \int_0^t \sqrt{v_s}\,\mathrm{d}W_s^v. \tag{14.52}$$

Taking expectations (the Itô integral is a martingale, expectation zero — the martingale property of Itô integrals proved in CH03 §3.7),

$$\mathbb{E}[v_t] - v_0 = \int_0^t \kappa(\theta - \mathbb{E}[v_s])\,\mathrm{d}s + 0. \tag{14.53}$$

Let $m_t \triangleq \mathbb{E}[v_t]$. Then

$$m_t - m_0 = \int_0^t \kappa(\theta - m_s)\,\mathrm{d}s, \tag{14.54}$$

$$\begin{cases} \dot{m}_t = \kappa(\theta - m_t), \\ m_0 = v_0, \end{cases} \qquad \Longrightarrow \qquad \boxed{m_t = v_0\, e^{-\kappa t} + \theta\,(1 - e^{-\kappa t})} \tag{14.55}$$

An exponential relaxation from v_0 to θ at rate κ.

This is a beautiful intermediate result. The expected variance at time t is a convex combination of the current variance v_0 and the long-run variance θ, with weights that decay exponentially in t. At $t = 0$ the weight on v_0 is one; as $t \to \infty$ the weight on v_0 goes to zero and the weight on θ goes to one. The rate of this transition is precisely κ — which is now seen to be a "time constant" in the engineering sense, with units of inverse time. A mean-reversion rate $\kappa = 2$ corresponds to a half-life of $\ln 2/2 \approx 0.35$ years, or roughly four months. This is the timescale over which expected variance shocks decay.

The formula $m_t = v_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t})$ is the Heston analogue of the term structure of expected variance. If you substitute this into (14.56) and differentiate with respect to T, you get the "forward variance" — the instantaneous rate of variance accumulation at time T. The forward variance at $T = 0$ is v_0; the forward variance at $T \to \infty$ is θ; the transition between the two happens at rate κ. The forward variance curve is what VIX-like indices measure: VIX implicitly prices the forward variance curve over a 30-day horizon, and VIX futures price forward-variance contracts at longer horizons. So m_T is the theoretical object that these market instruments are measuring.

Understanding the forward variance curve also clarifies why Heston has a rising or falling VIX term structure. If $v_0 < \theta$ (currently calm), the forward variance is low at short horizons and approaches θ from below at long horizons — a rising VIX term structure, which is the typical "contango" shape. If $v_0 > \theta$ (currently stressed), the forward variance starts high and decays toward θ — a falling VIX term structure, or "backwardation," which occurs around vol spikes like 2008 and 2020. The slope of the VIX term structure therefore carries a direct Heston-parameter interpretation: it reflects the ratio of current to long-run variance, weighted by the mean-reversion rate.

15.9.2 14.8.2 Fair variance strike

The fair strike K is the one making $\mathbb{E}[R - K] = 0$:

$$\boxed{K \;=\; \mathbb{E}[R] \;=\; \gamma^2 \int_0^T m_s\,\mathrm{d}s \;=\; \gamma^2 \Big[\theta\, T + \frac{v_0 - \theta}{\kappa}\,(1 - e^{-\kappa T})\Big]} \tag{14.56}$$

Remarkably, the fair variance strike depends only on (κ, θ, v_0) — not on α and not on ρ. The vol-of-vol and correlation enter the distribution's higher moments but not the mean.

This is a deep point. Variance swaps price on the mean of realised variance, and the mean of Heston's variance process depends only on its drift parameters (κ, θ, v_0), not on its diffusion parameters (α, ρ). Options, in contrast, price on the whole distribution, and the higher moments of the Heston distribution do depend on α and ρ. The upshot: variance swaps and options provide complementary information about Heston parameters. Variance swaps pin down (κ, θ, v_0) directly; options then pin down (α, ρ) given those. A joint calibration to variance-swap term structure and an option smile is often more stable than a calibration to either alone.

The practical procedure, then, for joint calibration: first, use the variance-swap term structure (or VIX futures, or long-dated ATM implied vols as a proxy) to fit (κ, θ, v_0). These three parameters have clean identification from the variance-swap curve alone: v_0 pins down the short end, θ pins down the long end, κ pins down the transition rate. Second, holding those three fixed, fit (α, ρ) to the smile shape — ideally at a representative maturity where the smile is well-resolved. This two-stage calibration decouples the parameter identification problem and typically converges faster and more robustly than a joint five-parameter fit.

The ability to separate mean dynamics from shape dynamics is specific to Heston's particular drift/diffusion structure. In other stochastic vol models (SABR, for instance, which lacks mean reversion), this separation does not apply. Heston's clean factorisation of mean vs. shape control is one of the reasons it remains popular despite its known limitations.

Another consequence: the fair variance-swap strike is model-free in the sense that it depends only on mean-reverting dynamics, not on specific distributional assumptions. This is why variance-swap markets are relatively standardised — participants do not need to agree on a full model to price, they just need to agree on the mean of realised variance. Exchange-traded variance futures (like VIX futures) rely on this structural feature, which is why they can be priced with relatively model-agnostic methods.

The VIX connection deserves its own paragraph. VIX is, by construction, the square root of a 30-day variance-swap strike on SPX, computed directly from observed option prices via the log-contract replication formula. In Heston terms, $\mathrm{VIX}^2 \cdot T = \int_0^T \mathbb{E}[v_s]\,\mathrm{d}s$ with $T = 30/365$. This is just (14.56) with the specific maturity plugged in. VIX therefore has a clean Heston interpretation: it is the forward-looking 30-day expected integrated variance, scaled to an annualised volatility. When VIX is 20, the market is pricing 20% realised vol over the next 30 days (on average); when VIX is 40, it is pricing 40%. VIX futures are contracts on this same measure at forward dates, priced by Heston via (14.56) with the appropriate maturity. VIX options are options on VIX futures, which under Heston become options on m_T — a more elaborate pricing problem but still tractable via the characteristic function framework.

A deeper structural point: in Heston, the full VIX dynamics are endogenous. You do not need a separate "VIX model" on top of Heston; the variance process v_t determines VIX_t through the convolution with the exponential decay kernel. Any change in v_t translates instantly to a change in VIX, with the coupling coefficient determined by κ. This is a clean and parsimonious way to think about VIX dynamics, and it is the basis for "internally consistent" pricing of VIX derivatives alongside SPX options. Many desks use calibrated Heston (or richer models like double Heston or Heston++) to simultaneously price SPX vanillas, variance swaps, VIX futures, and VIX options as a single consistent book.

15.9.3 14.8.3 Variance swap as tradable

Assume the swap itself is quoted with price dynamics

$$\frac{\mathrm{d}R_t}{R_t} = \sigma(v_t, t)\,\mathrm{d}W_t, \qquad R_0 = \hat{V}_0(T) \quad \text{— fair variance swap strike of } T\text{-maturity.} \tag{14.57}$$

Payoff $(R_T - K)_+$ at T on this tradable now becomes a standard option that can be solved analytically against the Heston $\hat{V}_0(T)$ curve.

This layered construction — model vanilla options with Heston, then use the fair variance-swap curve as a tradable to price options on variance — is how modern volatility desks handle the full suite of vol products. Variance swaps become inputs to the calibration; options on variance (like VIX options) become outputs of the calibrated model; and the whole system is internally consistent at the level of Heston dynamics.

One more connection to flesh out: the relationship between Heston and VIX drives a lot of the cross-market trading activity in the vol complex. If SPX smile calibration suggests one set of Heston parameters, and VIX futures

suggest another, the inconsistency creates a trade: sell the over-priced side, buy the under-priced side, and wait for convergence. This "VIX-SPX basis" trade is a staple of vol arbitrage strategies, and it depends structurally on the Heston linkage between variance dynamics and VIX. Traders who understand the Heston framework can spot these inconsistencies cleanly; traders who do not rely on cruder heuristics. The Heston framework therefore serves not just for pricing but for relative-value analysis across the entire vol complex.

15.10 14.9 Worked example — Heston price vs. Black–Scholes

Setup. $F_0 = 100$, $r = 0$, $T = 0.5$, $v_0 = 0.04$, $\theta = 0.04$, $\kappa = 2.0$, $\alpha = 0.5$, $\rho = -0.7$. Feller: $2\kappa\theta = 0.16 \geq 0.25 = \alpha^2$ — violated (by a hair); use full-truncation sim, but closed form still fine.

Grounding these parameter choices: $v_0 = \theta = 0.04$ means the instantaneous vol starts at and reverts to $\sqrt{0.04} = 20\%$, a realistic equity-ATM-vol level. The mean-reversion rate $\kappa = 2$ gives a six-month half-life for vol shocks, again realistic for equity indices. The vol-of-vol $\alpha = 0.5$ is in the upper range of typical calibrated values and mildly violates Feller. The correlation $\rho = -0.7$ captures a strong leverage effect. These numbers produce a visible but not extreme equity smile at six-month maturity.

The Feller check is a useful sanity diagnostic. $2\kappa\theta = 2 \times 2 \times 0.04 = 0.16$, and $\alpha^2 = 0.5^2 = 0.25$. The ratio is $0.16/0.25 = 0.64$, so the condition fails by a factor of $1/0.64 \approx 1.56$ — a mild violation. In the stationary distribution, the variance process has a gamma density with shape parameter $2\kappa\theta/\alpha^2 = 0.64$, which means the density is infinite at zero. This is a signature of meaningful but not catastrophic Feller violation. Full-truncation MC will work; exact MC via non-central chi-squared will be slightly more accurate.

Note also that the choice $v_0 = \theta$ is deliberately clean for pedagogy. In real calibrations, v_0 rarely equals θ exactly; usually there is a meaningful gap that reflects whether the market is currently above or below its long-run baseline. If $v_0 = 0.06$ and $\theta = 0.04$ (currently stressed), the variance-swap curve slopes downward (backwardation); if $v_0 = 0.02$ and $\theta = 0.04$ (currently calm), the curve slopes upward (contango). The VIX term structure carries this information directly and is a useful input to calibration.

Step 1 — Riccati coefficients at $\phi = 1$ (illustration, for $\omega = i$):

$$d(i) = \sqrt{(\rho\alpha i - \kappa)^2 - \alpha^2\, i(i-1)} = \sqrt{(-0.35i - 2)^2 - 0.25\, i(i-1)}.$$

Compute $(-0.35i - 2)^2 = 4 + 1.4i + 0.1225i^2 = 4 - 0.1225 + 1.4i = 3.8775 + 1.4i$. And $0.25\, i(i-1) = 0.25(i^2 - i) = -0.25 - 0.25i$. So

$$d(i)^2 = 3.8775 + 1.4i - (-0.25 - 0.25i) = 4.1275 + 1.65i.$$

Take the principal square root: $|d(i)^2| = \sqrt{4.1275^2 + 1.65^2} \approx 4.4451$, $\arg(d(i)^2) = \arctan(1.65/4.1275) \approx 0.3804$ rad; hence $|d(i)| = \sqrt{4.4451} \approx 2.1083$, $\arg d(i) \approx 0.1902$ rad, giving $d(i) \approx 2.070 + 0.399i$.

Step 2 — B, A at $\tau = 0.5$ follow (14.34)–(14.35).

Step 3 — P_1, P_2. Numerically integrate (14.41) via Gauss–Laguerre (64 nodes) for each strike.

Step 4 — compare. For ATM $K = 100$: Heston IV ≈ 0.200, matches v_0. For $K = 90$: Heston IV ≈ 0.224 vs BS-flat 0.200 — the skew. For $K = 110$: Heston IV ≈ 0.186 — thinner right tail due to $\rho < 0$.

Look at those three numbers in concert. The ATM implied vol matches $\sqrt{v_0}$, which is what we would expect: at the money, the option is so close to delta-neutral that the tail behaviour of the density does not affect its price much, and the Black–Scholes-comparable vol simply reads off the level of the Heston density's standard deviation. Ten percent out of the money on the put side (K = 90), implied vol has lifted by about 2.4 points — that is the combined effect of fatter tails (α contribution) and the leverage skew (ρ contribution). Ten percent out of the money on the call side (K = 110), implied vol has fallen by about 1.4 points — the fatter-tail effect is more than offset by the skew pulling right-side prices down. The asymmetry between +1.4 (down on calls) and +2.4 (up on puts) is the signature of $\rho = -0.7$: the skew dominates the smile on the downside.

To quantify this more carefully, consider the 25-delta risk reversal. At six months, with these parameters, the 25-delta put would be around K = 88 and the 25-delta call around K = 113. The implied vols at these strikes from Heston are approximately 23.0% and 18.8% respectively, so the risk reversal is 23.0 - 18.8 = 4.2 points. The 25-delta butterfly is [(23.0 + 18.8)/2 - 20.0] = 0.9 points. These numbers are close to what you would actually see on a six-month SPX surface on a typical trading day, so the choice of Heston parameters $(v_0, \theta, \kappa, \alpha, \rho) = (0.04, 0.04, 2, 0.5, -0.7)$

produces a realistic surface. This is not a coincidence — these parameters are broadly consistent with typical SPX calibration results.

A useful exercise is to perturb each parameter individually and observe the effect. Move ρ from -0.7 to -0.4: the put IV at K=90 falls from 22.4% toward 21.3% (less skew), and the call IV at K=110 rises from 18.6% toward 19.1%, narrowing the risk reversal from 3.8 to 2.2 points. Move α from 0.5 to 0.7: both wings lift, the ATM vol stays essentially put, and the butterfly rises from 0.9 to 1.6 points. Move κ from 2 to 4: the term structure of smile (how quickly it flattens with maturity) speeds up. Move θ from 0.04 to 0.06: the long-dated ATM rises, and the short-dated smile shifts up modestly because v_t is drifting toward a higher target. These sensitivities are the quantitative content of "what each Heston parameter does," and working through them by hand is the best way to internalise the model.

Variance-swap fair strike for this parameter set:

$$K_{\mathrm{VS}} = \theta T + \frac{v_0 - \theta}{\kappa}(1 - e^{-\kappa T}) = 0.04 \cdot 0.5 + 0 = 0.020 \quad (= 0.5 \times \theta, \text{ since } v_0 = \theta).$$

Annualised vol: $\sqrt{K_{\mathrm{VS}}/T} = \sqrt{0.04} = 0.200$ — as expected when $v_0 = \theta$.

When $v_0 = \theta$, the variance process starts at its long-run level; there is no transient decay, and the expected integrated variance over $[0, T]$ is simply θT. Annualising by dividing by T recovers θ itself, and taking the square root gives the vol level $\sqrt{\theta}$. This is a useful sanity check: if your Heston calibration gives a variance-swap fair strike that disagrees with the term ATM vol, something is off. The two should agree when $v_0 \approx \theta$, and should deviate predictably when they differ.

Working through the numerics one more time: $T = 0.5$, $\theta = 0.04$, $v_0 = 0.04$, $\kappa = 2$. Formula gives $K_{VS} = \theta T + (v_0 - \theta)(1 - e^{-\kappa T})/\kappa = 0.04 \times 0.5 + 0 = 0.02$. Annualised: $K_{VS}/T = 0.04$; square root: 0.2. So the fair variance-swap strike implies a 20% annualised realised vol over the 6-month horizon, consistent with starting vol and target vol both being 20%. If the market were quoting a fair 6-month variance-swap strike of 22%, Heston would need a higher θ (or a higher v_0) to match. If it were quoting 18%, Heston would need lower parameters. Either way, the variance-swap curve is a clean target for θ and v_0 calibration.

For longer horizons, the convergence rate to θ becomes the critical feature. At $T = 2$ years, with $\kappa = 2$, the mean reversion is essentially complete ($e^{-\kappa T} = e^{-4} \approx 0.018$), so $K_{VS}/T \approx \theta$ almost exactly. At $T = 0.25$ years (three months), only partial mean reversion has occurred ($e^{-\kappa T} \approx 0.607$), so the fair strike reflects a weighted average of v_0 and θ. If $v_0 \neq \theta$, the weighting creates a specific term-structure shape that identifies κ from the curvature of the variance-swap curve. This is exactly how practitioners calibrate κ in practice when variance-swap markets are liquid enough to provide multiple maturities.

15.11 14.10 Heston's blind spots

No model is perfect, and no textbook treatment of Heston is complete without an honest inventory of what it cannot do. Heston has three structural blind spots that every practitioner should know.

First, Heston struggles to fit short-dated smiles. An implied vol smile at very short maturity — think weeklies or overnight options — is typically much steeper and more convex than Heston can produce. The reason is structural: Heston is a pure diffusion, and diffusions accumulate small increments over time. At very short horizons, the density of $\ln F_T$ under any diffusion is nearly Gaussian (central limit behaviour), and the implied-vol smile is therefore nearly flat. Real markets show very steep smiles at one-week maturity — driven by jump risk, overnight gaps, event risk around earnings, and microstructural liquidity effects — that no continuous diffusion model can capture. To fit short-dated smiles, you need jumps in either the spot or the variance process, or ideally both. The Bates model (Heston + jumps in spot) and the SVJJ model (Heston + jumps in both) are the standard extensions.

The short-maturity asymptotic of the ATM vol skew in Heston is $\partial\sigma/\partial \ln K \approx \rho\alpha/(4\sqrt{v_0})$ — a constant, independent of T. But real equity surfaces show skew that grows as $T \to 0$, often scaling as $T^{-1/2}$ or faster. This divergent short-end skew is a feature of markets with jumps or rough vol, not continuous vol. Heston produces a finite short-end skew, which flattens the observed steep short-dated surfaces. The gap between Heston prediction and market reality is often several points of implied vol at the short wings.

The rough vol successor to Heston takes this observation seriously. Empirical work by Gatheral, Jaisson, and Rosenbaum (2014) showed that the realised volatility process is fractional Brownian motion with Hurst exponent

$H \approx 0.1$ — much rougher than Brownian motion's $H = 0.5$. A rough-vol SDE driven by fractional Brownian motion produces smile dynamics with divergent short-end skew that matches the data. The trade-off: rough-vol models are not Markovian in the usual sense (the state space is infinite-dimensional), they do not admit exponential-affine characteristic functions, and their pricing requires more elaborate numerical methods. The rBergomi model (Bayer, Friz, Gatheral 2016) is the most popular rough-vol specification; its characteristic function can be computed but requires integration over auxiliary variables. Heston is the Markovian, rational cousin of rough vol; rough vol is Heston corrected for the observed roughness of realised volatility. Both have their place, and modern practice often uses Heston for short-to-medium maturities and rough vol for very short maturities where the roughness effect dominates.

Second, Heston cannot simultaneously fit the full term structure of the smile. Heston has five parameters; the implied-vol surface has thousands of points (strike $\times$ maturity). It is impossible for five parameters to match thousands of quotes exactly. What Heston does in practice is fit one representative cross-section well (say three months) at the expense of others. Short-dated smiles are underfit (too flat), and very long-dated smiles are often mispriced too. To fit the full term structure, you need a Heston extension with time-dependent parameters — sometimes called piecewise Heston or term-dependent Heston — or a richer model altogether (like a local-stochastic vol hybrid).

A concrete comparison helps here. Consider SPX on a typical trading day with 20 strikes across 8 maturities — 160 quotes. A standard Heston fit will produce RMS implied-vol error around 50–100 basis points across the surface, with the largest errors at the short (3-month and below) and long (18-month and above) ends. A local vol fit can achieve sub-basis-point error by construction (Dupire's formula matches every quote exactly), but the local vol function itself is wildly jagged and produces unrealistic forward smile dynamics. A local-stochastic vol (LSV) hybrid can achieve both: exact surface fit via the local-vol part and realistic forward dynamics via the Heston part. LSV is therefore the practitioner's default for exotics; Heston is the default for vanillas where forward dynamics matter more than exact surface fit.

The tradeoff between Heston and LSV maps to a broader pricing-vs-hedging tradeoff. For book valuation (what are my positions worth today?), exact surface fit matters because traders are quoting against the observed market. For hedging and risk management (how do my positions change under plausible market scenarios?), realistic dynamics matter more. A pure local-vol book has slightly better valuations but worse hedges; a pure Heston book has slightly worse valuations but better hedges. LSV tries to give you both. The engineering complexity of LSV, however, is substantially higher than Heston, which is why Heston remains the default workhorse for most purposes.

Third, Heston's kurtosis is controlled, but not independently of its skew. In real markets, the smile can have independent skew and convexity — a change in one does not necessitate a change in the other. In Heston, ρ controls skew and α controls convexity, but they are not quite orthogonal: changing α also shifts the ATM level modestly, and changing ρ influences the effective kurtosis. This limited orthogonality means that some smile shapes are simply not in Heston's attainable set. A smile with very high kurtosis but zero skew (an ATM-centred smile), or a smile with strong negative skew but little convexity, can be difficult to fit cleanly. Again, the workaround is richer models.

The space of Heston-attainable smile shapes is a five-dimensional manifold embedded in the infinite-dimensional space of all smooth smile functions. There are many smile shapes that are not on this manifold, and a Heston calibrator landing on the nearest-on-manifold point will produce residuals that reveal the gap. In practice, the gap is usually small — a few basis points RMS — for well-behaved liquid surfaces. It can be larger for stressed surfaces (2008-type regimes), for single-name options with idiosyncratic features, or for illiquid strikes where the observed "quotes" are really just market-maker estimates.

Beyond the big three blind spots, Heston has some subtler limitations worth noting. The correlation parameter ρ is constant — it does not vary with vol level — but empirically the correlation between spot and vol moves often tightens during high-vol regimes and loosens during calm regimes. Heston cannot capture this conditional correlation dynamics. Similarly, Heston's vol-of-vol is constant — the variance process has uniform chaotic behaviour — but empirically the vol of vol itself varies over time (a "vol of vol of vol" effect). Richer models with stochastic correlation or stochastic vol-of-vol (double Heston, 3/2 model, etc.) address these limitations at the cost of more parameters and more numerical complexity.

There is also the question of parameter stability across market regimes. In calm markets, Heston calibration is well-posed and stable. In high-vol regimes (2020, 2008), the smile shape changes qualitatively — wings become richer, skew steepens — and calibrated Heston parameters shift accordingly. This is not a bug; it is the model adapting to

new data. But the shift in parameters means that option prices implied by the old calibration can be wildly off from the new calibration, and any risk-management scheme based on fixed Heston parameters will be exposed to model risk. Continuous re-calibration is the practical remedy, but it requires liquid enough option markets to support daily or intraday calibration.

Despite these blind spots, Heston remains the workhorse stochastic- vol model in industry. Its strengths — closed-form characteristic function, fast Fourier-based calibration, clean parameter interpretation, well-posed calibration problem — outweigh its weaknesses for the bulk of vanilla-option pricing and hedging applications. For exotic payoffs that require modelling the full joint dynamics of spot and vol, Heston is often the first model used, and then the outputs are compared against richer models only if the Heston results look suspect.

A comparative taxonomy helps place Heston in context. Black–Scholes: one parameter, analytical, no smile. Local vol (Dupire): infinite parameters (a function), fits any smile exactly, poor forward dynamics. Stochastic vol (Heston, SABR, 3/2): five or six parameters, realistic forward dynamics, imperfect surface fit. Local-stochastic vol (LSV): combination, exact surface fit plus realistic forward dynamics, high numerical cost. Stochastic vol with jumps (Bates, SVJJ): stochastic vol plus Poisson jumps, handles short-dated smiles and crash risk, more parameters. Rough volatility (rBergomi, rough Heston): fractional driving noise, matches short-dated skew, non-Markovian. Stochastic correlation or stochastic vol-of-vol (double Heston, 3/2): richer vol dynamics, many parameters, specialised use. Each step up the hierarchy adds capability at the cost of complexity, calibration difficulty, and compute time. The right choice depends on the product being priced and the desired accuracy.

For vanilla options on liquid indices, Heston is usually sufficient. For weeklies and event-driven products, Bates or SVJJ is usually needed. For path-dependent exotics (barriers, cliquets, forward-starts), LSV or stochastic-vol-with-jumps is standard. For very-short-dated trades at the cutting edge of accuracy, rough vol is increasingly deployed. Heston sits in the middle of this hierarchy — neither the simplest nor the most complex — and for most applications it is the right trade-off. Know when to reach for the richer models, but do not reflexively dismiss Heston; for bread-and-butter vanilla pricing and hedging, it remains the best tool for the job.

15.12 14.10A Heston vs. SABR vs. local volatility

A subsection devoted to comparison is warranted because practitioners frequently have to choose between these three frameworks, and the choice depends on the application. All three are stochastic-vol-adjacent models that aim to fit the implied volatility smile, but they differ in structure, strengths, and weaknesses.

SABR (Stochastic Alpha Beta Rho) is a stochastic vol model that directly parameterises the evolution of vol itself (not variance), with lognormal dynamics. The SDE is $\mathrm{d}F_t = \sigma_t F_t^\beta \,\mathrm{d}W^F$, $\mathrm{d}\sigma_t = \alpha\sigma_t \,\mathrm{d}W^\sigma$, with $\mathrm{d}W^F \,\mathrm{d}W^\sigma = \rho \,\mathrm{d}t$. Notice the CEV-exponent β on the spot and the lognormal (rather than CIR) vol. SABR has four parameters — $(\sigma_0, \alpha, \rho, \beta)$ — and is famously used by rates desks because of Hagan's asymptotic formula, which gives a closed-form approximation of the implied vol smile as a function of $(\sigma_0, \alpha, \rho, \beta, F, K, T)$ and is extraordinarily fast to evaluate. SABR matches the smile at short maturity cleanly and provides simple intuition: β shapes the skew coming from leverage on spot ($\beta = 1$ is lognormal, $\beta = 0$ is normal), ρ shapes the skew coming from spot-vol correlation, and α controls curvature. SABR's main limitation: no mean reversion in vol. This means SABR's term structure of smile is less flexible than Heston's, and long-dated calibration is poor. SABR also has issues with arbitrage at extreme strikes (negative densities can arise), which has motivated fixes like shifted-SABR and Heston-SABR hybrids.

Local volatility (Dupire) goes the opposite direction: instead of adding a second stochastic factor, it makes σ a deterministic function of spot and time, $\sigma(S, t)$. Given any arbitrage-free implied vol surface, Dupire's formula produces the unique local vol function that reproduces the surface exactly:

$$\sigma^2_{\mathrm{loc}}(K, T) = \frac{2\partial_T C + 2(r - q)K\partial_K C}{K^2 \partial^2_{KK} C}.$$

The beauty is that local vol fits any surface exactly by construction. The curse is that the forward smile — the implied vol surface seen at a future date, conditional on spot reaching a particular level — flattens unrealistically with horizon. This makes local vol poor for forward-starting options, cliquets, and any product where the forward smile dynamics matter. Local vol is therefore the "snapshot" model: great for today's valuation, bad for tomorrow's dynamics. Most practitioners use local vol as a valuation workhorse for vanillas and switch to stochastic vol (Heston) or LSV for exotics.

Heston sits in the middle. Its continuous-state stochastic-vol structure produces more realistic forward dynamics than local vol, its closed-form characteristic function makes calibration fast, and its parameter interpretation is clean. Heston is less flexible than local vol (cannot fit any surface exactly) but more flexible than SABR (has mean reversion, handles term structure better). For most vanilla applications, Heston is the right default; for specific asset classes with specific conventions (rates markets using SABR, equity books running LSV), alternatives dominate. The art of derivative pricing is picking the right model for the problem at hand, and the three models in this comparison are the three most common tools in that toolkit.

A practical point: when a desk commits to Heston for vanilla pricing, they typically also commit to one or two "rich models" as backups — usually Bates (for short-dated products) and LSV (for path-dependent products). Model risk is managed by pricing positions under multiple models and examining the spread. A position whose valuation is tightly consistent across Heston, Bates, and LSV is low model risk; a position where the three models disagree substantially is high model risk and may need a wider bid-ask spread or a model-risk reserve. This multi-model approach is standard at top-tier derivatives desks and acknowledges that no single model is perfect for all products.

15.13 14.10B Extensions and successors

The modern stochastic-vol landscape extends well beyond pure Heston. It is worth cataloguing the major extensions because they all build on Heston's foundation and understanding them deepens one's appreciation for what Heston does well and where it falls short.

Bates (1996). Heston plus Poisson jumps in spot: $\mathrm{d}F/F = \sqrt{v}\,\mathrm{d}W^F + (e^J - 1)\,\mathrm{d}N - \lambda\mathbb{E}[e^J - 1]\,\mathrm{d}t$, where N is a Poisson process with intensity λ and J is a random jump size (typically Gaussian). The model adds three parameters (jump intensity, jump mean, jump standard deviation) and produces much richer short-dated smile shapes. The characteristic function has a simple additive adjustment: the Heston $A(\tau), B(\tau)$ are augmented by a jump contribution $\lambda\tau(\phi_J(\phi) - 1)$, where ϕ_J is the characteristic function of the jump distribution. Bates is the natural extension when short-dated smiles are too steep for Heston alone.

SVJJ (Duffie–Pan–Singleton, 2000). Adds simultaneous jumps in both spot and variance: when a jump occurs, spot drops (typically) and variance spikes simultaneously. This captures the crash dynamics where a large negative spot move coincides with a variance spike, beyond what pure diffusive correlation can achieve. SVJJ has more parameters than Bates but produces realistic crash dynamics and is commonly used in exotic pricing for products sensitive to joint tail events.

Double Heston. Two independent CIR variance factors driving spot: one fast (short-timescale) and one slow (long-timescale). The characteristic function is a product of the two Heston-like factors, and calibration remains tractable. Double Heston handles the term structure of smile better than single Heston because the two factors span different maturities.

3/2 model. Variance follows $\mathrm{d}v = \kappa v(\theta - v)\,\mathrm{d}t + \alpha v^{3/2}\,\mathrm{d}W$. The $v^{3/2}$ diffusion replaces the $\sqrt{v}$ of CIR. The 3/2 model is sometimes preferred because it has fatter tails than Heston in the variance distribution itself, which produces more realistic VIX option pricing (Heston underprices VIX calls, 3/2 is closer to market).

rBergomi and rough Heston. Driven by fractional Brownian motion with Hurst exponent $H < 0.5$, producing "rough" paths that are more chaotic than ordinary Brownian motion. Rough models match observed short-end skew divergence and match the empirical roughness of realised vol. Trade-off: non-Markovian, no closed-form characteristic function, more elaborate pricing methods.

Hybrid models (LSV). Local-vol times stochastic-vol. The local vol component handles the exact surface fit; the stochastic vol component handles the dynamics. Parameterised as $\mathrm{d}S/S = L(S,t)\sqrt{v}\,\mathrm{d}W^S + \cdots$ with L chosen so that the marginal distribution of S_T matches the observed surface. LSV pricing is Monte Carlo or PDE, not Fourier. Calibration of the local vol function L while maintaining the Heston-like stochastic component is a non-trivial numerical exercise (Gyöngy's Markovian projection theorem and related techniques). LSV is the gold standard for exotics pricing in many banks today.

Each of these extensions addresses a specific Heston limitation. Bates fixes short-end smile; SVJJ fixes crash dynamics; double Heston and term-dependent Heston fix term structure; rough vol fixes short-end scaling; LSV fixes surface fit. But each extension comes with additional complexity and model risk. The practitioner's craft is knowing which extension is worth the cost for a given application.

15.14 14.10C Summary of Heston as a pricing and hedging framework

Putting together the threads of this chapter, Heston can be evaluated as a framework along several dimensions:

Pricing vanillas: excellent. The Fourier method is fast, accurate, and handles a full surface of quotes. Heston matches ATM and near-money smile cleanly; wings are approximately right; long-dated extrapolation is reasonable.

Pricing short-dated wings and weeklies: poor. The continuous-diffusion structure cannot match the divergent short-end skew. Bates or rough-vol extensions are needed.

Pricing long-dated LEAPS: moderate. The mean-reversion machinery gives reasonable long-dated ATM, but the long-dated smile shape is often oversimplified. Term-dependent Heston helps.

Pricing barriers and touch options: moderate. Heston captures the key fact that barrier breach probability depends on vol dynamics, but quantitative accuracy depends on how well-calibrated the particular instance is. LSV is often preferred for exact barrier matching.

Pricing forward-start and cliquet products: good. Heston's realistic forward smile dynamics (from mean reversion and vol-of-vol) produce sensible forward-start prices. Local vol alone fails here.

Pricing variance and volatility derivatives (VIX): excellent. The internal consistency between Heston variance dynamics and VIX structure makes calibrating to both simultaneously a natural and clean exercise.

Hedging vanillas: good. Delta-vega hedging under Heston is close to BS but refined by the v_0-based vega; adding α and ρ sensitivity hedges reduces model risk further.

Hedging exotics: moderate. For exotics, the full set of Heston sensitivities matters, and the incomplete-market nature of Heston means no hedge is perfect. Multi-model hedging is typical.

Risk management and stress testing: good. Heston's clean parameter interpretation lets risk managers think about vol regime shifts in terms of parameter changes (higher θ, higher α, more negative ρ), and stress tests can be specified cleanly in these terms.

Overall, Heston is the workhorse stochastic-vol framework precisely because it sits in the sweet spot of power, cleanness, and tractability. It is the first model every quant learns in stochastic vol, and the fallback model when richer alternatives fail or are computationally impractical. Even as rough-vol models gain traction in research and some niche applications, Heston's role as the pedagogical and practical baseline seems secure.

15.15 14.11 Key takeaways

1. Heston replaces BS's constant vol (CH06) with a mean-reverting CIR variance correlated to spot; this incomplete-market model has five risk-neutral parameters $(\kappa, \theta, \alpha, \rho, v_0)$.
2. The Feller condition $2\kappa\theta \geq \alpha^2$ keeps $v_t > 0$. Calibrated equity markets often sit near or below it; use full-truncation for MC.
3. The change-of-measure (CH05 multi-dimensional Girsanov) absorbs the market price of variance risk λ^v; affine choices ($\lambda^v = c\sqrt{v_t}$ or $\ell/\sqrt{v_t} + c\sqrt{v_t}$) preserve Heston form and just reshuffle (κ, θ) between $\mathbb{P}$ and $\mathbb{Q}$.
4. The characteristic function of $\ln F_T$ is exponential-affine in (x, v): $\varphi = \exp\{i\phi x + A(\tau) + B(\tau)v\}$, with A, B solving a pair of Riccati ODEs that admit a closed-form solution.
5. Call prices are recovered by Fourier inversion (Gil–Pelaez): $C = e^{-r\tau}(F\,P_1 - K\,P_2)$, one integral per probability. Carr–Madan FFT is the production variant.
6. Heston reproduces implied-vol smiles and skews: ρ controls skew, α controls curvature, κ controls term-structure decay.
7. The fair variance-swap strike depends only on (κ, θ, v_0) — α and ρ do not enter the mean.
8. Heston's blind spots: short-dated smiles are underfit (need jumps), full term structure cannot be fit simultaneously (need time-dependent parameters), and skew-kurtosis decomposition is only approximately orthogonal.

The narrative arc of this chapter mirrors the historical arc of the derivatives-pricing literature. Black and Scholes gave us a complete, tractable pricing framework with one parameter; empirical data promptly broke the model's main assumption. Heston restored tractability at the cost of incompleteness, and for three decades served as the

industry baseline. Rough vol and LSV models extend Heston to fit better, at the cost of tractability. The pattern is familiar: each generation of model adds realism and sacrifices something — either simplicity, or analytical closed form, or computational efficiency. The deep lesson is that there is no "correct" pricing model in an absolute sense; every model is a simplification of reality, and the choice of model depends on what simplifications are acceptable for the specific application. Heston is the classical compromise — rich enough to capture real phenomena, simple enough to implement, clean enough to reason about. Even as richer models proliferate, Heston's role as the canonical teaching example and the industry-default baseline endures.

A final reflection: the elegance of Heston is not just mathematical. It is conceptual. The idea that vol has its own life — its own mean-reversion, its own fluctuations, its own correlation to spot — is intuitively true, and every trader who has watched a vol spike can confirm it empirically. Heston is the minimal rigorous formalisation of this intuition, packaged in a form that preserves analytical tractability. That combination of conceptual cleanness and computational tractability is rare in quantitative finance, and it explains why Heston has endured even as more accurate models appear. Learn Heston well, and you will find that its scaffolding supports a surprising range of more sophisticated models, and that the intuitions you build about $(\kappa, \theta, \alpha, \rho, v_0)$ translate cleanly to the parameter spaces of those successors.

15.16 Appendix A — Reference formulas

This appendix collects the key formulas derived in the chapter for quick reference. Every formula here has been derived and explained above; this is the quick-lookup sheet for implementation.

Risk-neutral SDE.

$$\frac{\mathrm{d}F_t}{F_t} = \sqrt{v_t}\,\mathrm{d}W_t^F, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \alpha\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}W^F\,\mathrm{d}W^v = \rho\,\mathrm{d}t.$$

Feller condition. $2\kappa\theta \geq \alpha^2$. When satisfied, the variance process is strictly positive almost surely. When violated, the origin is reachable and simulation schemes must handle it carefully; closed-form pricing via the characteristic function is unaffected by violation status.

Riccati ODEs.

$$B'(\tau) = \tfrac{1}{2}\alpha^2 B^2 + (\rho\alpha\omega - \kappa)B + \tfrac{1}{2}\omega(\omega - 1), \quad B(0) = 0,$$
$$A'(\tau) = \kappa\theta\, B(\tau), \quad A(0) = 0.$$

Closed form ($\omega = i\phi$):

$$d = \sqrt{(\rho\alpha\omega - \kappa)^2 - \alpha^2\omega(\omega - 1)}, \quad g_* = \frac{\kappa - \rho\alpha\omega - d}{\kappa - \rho\alpha\omega + d},$$

The quantities $d(\omega)$ and $g_*(\omega)$ are the two auxiliary functions through which the Heston closed form is most compactly expressed. Both are complex-valued for complex ω, and both require careful branch-cut handling when evaluated numerically. The practical rule of thumb: if your implementation produces non-decaying $|\varphi|$ as $\phi \to \infty$, the branch of d is wrong; if your implementation produces discontinuities in A as ϕ sweeps the real axis, the branch of the logarithm in A is wrong. Both errors are cured by the Albrecher-type rewriting that flips g_* and keeps the logarithm on a continuous branch.

$$B(\tau) = \frac{\kappa - \rho\alpha\omega - d}{\alpha^2}\,\frac{1 - e^{-d\tau}}{1 - g_* e^{-d\tau}},$$

$$A(\tau) = \frac{\kappa\theta}{\alpha^2}\left[(\kappa - \rho\alpha\omega - d)\tau - 2\ln\frac{1 - g_* e^{-d\tau}}{1 - g_*}\right].$$

Characteristic function. $\varphi(\phi) = \exp\{i\phi\, x + A(\tau; i\phi) + B(\tau; i\phi)\, v\}$.

This is the object you actually compute at each quadrature node during Fourier-inversion pricing. Every calibration iteration involves thousands of evaluations of φ; making it fast and numerically stable is essential to good calibration performance. Production implementations typically cache intermediate quantities (the square root d, the ratio g_*) so that only the parts that change with τ are recomputed, and pre-compute the Riccati solution at grid points for re-use across strikes.

Gil–Pelaez / Heston call. For a European call at strike K and maturity T, with the forward price F_t at time t and risk-free rate r:

$$P_j = \tfrac{1}{2} + \tfrac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-i\phi \ln K}\varphi_j(\phi)}{i\phi}\right] \mathrm{d}\phi,$$

$$C = e^{-r\tau}\,(F_t\,P_1 - K\,P_2).$$

Variance swap fair strike.

$$K_{\mathrm{VS}} = \theta\,T + \frac{v_0 - \theta}{\kappa}(1 - e^{-\kappa T}).$$

This formula is the Heston analogue of the VIX calculation. VIX itself is roughly $100 \times \sqrt{K_{\mathrm{VS}}(T = 30/365)/(30/365)}$, which, once you plug in typical Heston parameters, gives a reasonable approximation to observed VIX levels. The cleanness of this correspondence is one of the reasons Heston is so useful for joint SPX-VIX modelling. The forward variance curve — m_T as a function of T — is what VIX futures price, and the VIX futures term structure carries clean Heston-parameter signatures that a savvy trader can read at a glance.

Simulation (full-truncation Euler). The recommended baseline MC scheme; for higher accuracy use Andersen's QE scheme or exact non-central chi-squared simulation. With $v^+ = \max(v, 0)$,

$$X_n = X_{n-1} - \tfrac{1}{2}v_{n-1}^+\Delta t + \sqrt{v_{n-1}^+\,\Delta t}\,Z_1,$$

$$v_n = v_{n-1} + \kappa(\theta - v_{n-1}^+)\Delta t + \alpha\sqrt{v_{n-1}^+\,\Delta t}\,(\rho Z_1 + \sqrt{1-\rho^2}\,Z_2).$$

15.17 Appendix B — Intuition checklist and exam-style questions

The following checklist summarises the core intuitions one should be able to recite fluently after working through this chapter. Treat each as a question; if the answer is not immediate, revisit the relevant section.

Why does constant-vol Black–Scholes (CH06) fail empirically? Because real implied-vol surfaces are not flat across strike and maturity. The smile, skew, and term structure are persistent features of option markets and are incompatible with a single-parameter model.

What does the CIR process model, and why is $\sqrt{v}$ the right diffusion coefficient? The CIR process models a non-negative state variable with mean reversion; the $\sqrt{v}$ diffusion vanishes at zero, providing a soft floor that keeps the process non-negative under mild conditions and preserves affine structure.

What is the Feller condition and why does it matter? $2\kappa\theta \geq \alpha^2$ ensures the variance process stays strictly positive. Calibrated equity Heston typically violates it mildly; closed-form pricing is unaffected, but Monte Carlo schemes need careful handling (full-truncation, QE, or exact simulation).

What is the economic meaning of negative ρ in equity markets? The leverage effect: falling spot mechanically raises leverage and vol, plus behavioural and institutional factors (VaR deleveraging, risk-off flows). The result is a persistent negative correlation between spot and vol moves, which generates the downward-sloping equity skew.

What is the market price of variance risk λ^v? The premium risk-averse investors demand for bearing variance risk. Empirically, implied vol exceeds expected realised vol by a few percentage points on average; this wedge is the variance risk premium, and it underpins the distinction between physical-measure and risk-neutral Heston parameters.

What is the characteristic function and why is it useful? The Fourier transform of the density. For Heston it has a clean exponential-affine form $\varphi = \exp\{i\phi x + A + Bv\}$ with A, B solving Riccati ODEs. The density itself is a mixture of lognormals and has no closed form.

How does Gil–Pelaez let us price options? Gil–Pelaez expresses the CDF of a random variable as a single Fourier integral of its characteristic function. For a Heston call, we need two CDFs (under different measures — see CH05's two-numeraire switching) and the call price is a combination.

How does Carr–Madan FFT speed up calibration? By computing the whole log-strike smile in a single FFT, reducing pricing cost from $O(N_K)$ characteristic-function evaluations per strike to $O(N_\phi \log N_\phi)$ for the entire smile.

What determines the fair variance-swap strike and which Heston parameters enter? The fair strike depends only on (κ, θ, v_0) — the drift parameters. Diffusion parameters α and ρ drive higher moments but not the mean, so they do not appear in the variance-swap strike formula.

What are Heston's three main blind spots? Short-dated smiles (need jumps), full term-structure fit (need time-dependent parameters), and limited orthogonality between skew and kurtosis.

When should you use Heston, SABR, local vol, LSV, or rough vol? Heston as the default; SABR for rates markets; local vol for quick vanilla valuation with exact surface fit; LSV for exotics where both exact fit and realistic forward dynamics matter; rough vol when short-end skew divergence matters.

If you can answer all of these without hesitation, you have internalised the core of stochastic-vol pricing. If any answer feels uncertain, the corresponding section above is worth re-reading.

End of Chapter 14. Chapter 15 moves from the equity-side stoch-vol capstone to the rates-side capstone: caps, floors, and swaptions, where the T-forward and annuity measures from CH05 and the short-rate dynamics from CH12 come together in the final fixed-income applications.

16 Chapter 15 — Caps, Floors, and Swaptions

16.1 15.0 The rates-side capstone

This chapter is the final application of the numeraire-change machinery built earlier in the guide. In Chapter 5 we constructed the Radon-Nikodym / Girsanov apparatus in full generality: every strictly positive tradable defines a measure, forward measures in particular are generated by zero-coupon bonds, and the annuity measure arises as a portfolio-of-bonds numeraire. In Chapter 12 we built the short-rate models — Vasicek, Ho-Lee, Hull-White, and the general affine framework — that give us closed-form bond prices $P_t(T) = e^{A_t(T) - B_t(T)\, r_t}$ and explicit bond volatilities $B_t(T)\,\sigma$. In Chapter 13 we priced interest-rate swaps, CDS, callable bonds, and European bond options, using the T-forward measure as a cross-reference back to Chapter 5.

Caps, floors, and swaptions are the product set that ties all of this together. A cap decomposes exactly, with no cross terms, into a strip of single-date caplets, each priced as a Black-76 call under its own T-forward measure $\mathbb{Q}^{T_i}$. A swaption is irreducibly multi-date, but under the annuity measure $\mathbb{Q}^X$ the par swap rate S_t becomes a single scalar martingale and the whole n-dimensional bond-price problem collapses into a single Black call on S. Both moves are instances of the same general principle — pick a numeraire under which the underlying becomes a martingale and the payoff factors cleanly — and both are immediate applications of the Girsanov theorem from Chapter 5. We do not re-derive the T-forward measure or the change-of-Brownian-motion formula; we cite them. What is new in this chapter is how they get used.

Interest-rate derivatives — caps, caplets, floors, and swaptions — form the backbone of the modern rate-options market. They exist because the floating reference rate a borrower actually pays each quarter (historically LIBOR, increasingly SOFR, EURIBOR, TONA, SONIA) is random, and every participant in the interest-rate universe — a homeowner with a variable-rate mortgage, a corporate treasurer rolling short-term funding, a pension plan hedging long-dated liabilities, a swap dealer running a matched book — ends up needing convex protection against moves in that rate. A cap is the clean way to buy a ceiling on floating cost; a floor is the clean way to buy a floor on floating income; a swaption is the option to enter a swap at a pre-agreed fixed rate. Together they span a huge fraction of the daily volume in interest-rate vega.

Unlike equity options, where the natural numeraire is the money-market account $B(t)$ and everything lives on a continuous-time stock process, rate options live on a discrete tenor grid $\{t_0 < t_1 < \cdots < t_n\}$ — the fixing and payment dates — and each payoff depends on a rate that is only ever observed on those dates. This discrete scaffolding looks clumsy at first, but it is actually a gift: by choosing the right numeraire for each tenor point, the pricing problem decomposes cleanly into a strip of independent Black-76-style calls. The trick is to price each caplet under the T_i-forward measure $\mathbb{Q}^{T_i}$ from Chapter 5, whose numeraire is the zero-coupon bond $P(t, T_i)$. Under $\mathbb{Q}^{T_i}$ the simple forward LIBOR $L(t, T_{i-1}, T_i)$ is a martingale — so if we additionally assume it is log-normal, we recover Black-76 formulas for the individual caplets, and the cap is simply $\sum_i$ caplets with no cross-terms. That decomposition is the single most important fact in this chapter; everything else is either setup or consequence.

The rest of this chapter builds that picture carefully. We start from the definition of LIBOR via zero-coupon bond accrual, show how a cap arises naturally as the optional part of a swap (the swap itself was priced in Chapter 13), change numeraires to simplify the pricing expectation (using the Girsanov machinery from Chapter 5), derive Black-76 for each caplet, then extend the same machinery to swaptions via the annuity numeraire. We end with worked numerical examples and a discussion of practical issues: flat cap vols versus stripped caplet vol term structures, what happens in negative-rate regimes (Bachelier rather than Black), and the post-2008 transition from LIBOR to SOFR and its overnight-compounded cousins.

A word on mindset before we begin. For a practitioner, the art of rate options is much less about the Black formula itself — which is one-liner code once you know what goes into it — and much more about *knowing which numeraire to use when.* Every product has a numeraire under which its pricing problem collapses to a scalar Black expectation. Finding that numeraire is the whole game. Caplets want the t_k-bond; swaptions want the annuity; bond options (Chapter 13) want the delivery-date bond; constant-maturity-swap-flavored products need a convexity-adjusted measure; and so on. Once you internalize the principle — pick a numeraire such that the underlying becomes a martingale and the payoff factors cleanly — the formulas flow. The rest is implementation detail and bookkeeping.

16.2 15.1 Caps, Caplets, and the Interest-Rate Swap

A cap is an interest-rate option made up of caplets. Caps/floors are close cousins of swaptions (swap options). The easiest way to see where a cap comes from is to start with its underlying — a plain vanilla interest-rate swap, as priced in Chapter 13 — and ask what happens when you put an option on the floating leg.

Consider a standard interest-rate swap (IRS) between counterparties A and B on notional N, with fixed leg F and floating leg ℓ:

$$A \underset{\ell}{\overset{F}{\rightleftharpoons}} B \qquad \text{(IRS)} \tag{15.1}$$

In words, A pays fixed and receives floating; B pays floating and receives fixed. The notional N never actually changes hands — it is simply a scale factor that multiplies the rate spread on each payment date. Both legs are quoted as annualized rates, but each period's payment applies only for the year-fraction Δt_k between consecutive fixing/payment dates.

On the tenor grid $t_0 < t_1 < t_2 < \cdots < t_n$:

- Fixed leg pays $F\,\Delta t_k$ at each t_k, $k = 1, \ldots, n$.
- Floating leg pays $L_{t_{k-1}}(t_k)\,\Delta t_k$ at t_k, where the rate is set at t_{k-1} and paid at t_k.

The timing convention — fix at t_{k-1}, pay at t_k — is not a quirk but the defining feature of the instrument. It is the same thing a bank does when it quotes you a three-month floating-rate loan: on the reset date, it looks up the prevailing rate, locks it in for the coming quarter, and hands you the bill at the end of the quarter. In options-pricing language this has a subtle consequence: the payoff is known one period before it is paid, so we need to discount it from the payment date t_k even though randomness is resolved at the fixing date t_{k-1}. That small dislocation is exactly what the t_k-forward measure is built to handle.

Here $L_{t_{k-1}}(t_k) \equiv L(t_{k-1}, t_{k-1}, t_k)$ is the simple (LIBOR) rate for the period $[t_{k-1}, t_k]$ fixed at t_{k-1}. It is defined through the zero-coupon bond:

$$P_{t_{k-1}}(t_k) \;=\; \left(1 + \Delta t_k\, L_{t_{k-1}}(t_k)\right)^{-1} \quad \Longrightarrow \quad L_{t_{k-1}}(t_k) \;=\; \frac{1}{\Delta t_k}\left[\frac{1}{P_{t_{k-1}}(t_k)} - 1\right] \tag{15.2}$$

so that investing $P_{t_{k-1}}(t_k)$ at t_{k-1} yields \$1 at t_k. This is the single most important definition in the chapter. LIBOR (and every simple forward rate) is not a primitive object floating in the air — it is a *derived quantity*, built out of zero-coupon bond prices by the simple-accrual formula above. The market does not pick a number and call it the rate; it observes bond prices and implies a rate. That ordering matters. When we later want to model LIBOR as log-normal under the t_k-forward measure, we are really making a modelling choice about the *ratio* $P_t(t_{k-1})/P_t(t_k)$, because that is what the rate is. Every martingale statement, every change-of-measure drift, and every Black-76 formula in this chapter ultimately rests on the fact that L is a ratio of tradables — the generic sufficient condition for a $\mathbb{Q}^N$-martingale under a numeraire N, as established in Chapter 5.

Intuitively, one dollar at t_{k-1} invested at the simple rate L for the fraction Δt_k grows to $1 + \Delta t_k L$ at t_k. Equivalently, the bond $P_{t_{k-1}}(t_k)$ is what you must pay at t_{k-1} to receive exactly one dollar at t_k, and by no-arbitrage its reciprocal equals the growth factor $1 + \Delta t_k L$. Any deviation and you can arbitrage a lender or borrower. So simple-rate accrual and the zero-coupon bond are two views of the same object.

16.2.1 Net cash flow to A and the cap payoff

A's net flow at t_k is $(L_{t_{k-1}}(t_k) - F)\Delta t_k$ — linear in the realized rate ℓ. If the realized LIBOR comes in above the fixed leg F, A (who is paying fixed) makes money on that period; if LIBOR comes in below F, A loses on the period. The swap is symmetric and linear — there is no optionality.

A cap caps the floating cost at strike K: instead of paying F, A enters a contract C that delivers the payoff

$$\Delta t_k \left(L_{t_{k-1}}(t_k) - K\right)_+ \qquad \text{at each } t_k,\ k = 1, \ldots, n. \tag{15.3}$$

Each of these n period payoffs is called a caplet. The function $(\cdot)_+$ makes it clear that A is only ever compensated when LIBOR exceeds K — the downside is truncated at zero. Economically, the caplet is A's insurance against a single bad fixing; the cap is the bundled insurance across all future fixings of the underlying swap. Graphically the net flow is hockey-stick shaped in ℓ with kink at K:

```
caplet payoff  Δt_k · (  - K)_+




0                        →    (reset LIBOR)
                           K

Hockey-stick in  : zero below the strike, slope Δt_k above.
Convex in      cap is long vega in the forward-rate vol.
```

So the cap is a strip of caplets, each one a call option on the forward LIBOR rate fixed at t_{k-1} and paid at t_k. The hockey-stick diagram above also reveals why a cap is convex in the rate: every caplet is, and the sum of convex functions is convex. Convexity in the floating rate translates directly into positive vega — a cap always benefits from a higher realized volatility of forward rates, holding everything else fixed. This is the structural reason cap markets trade a vol surface in the first place.

Caplet payoff vs reset rate *Caplet payoff vs reset rate*

A brief note on naming. A "1y1y" caplet resets in one year and pays at approximately one-year-and-one-tenor. A "5y cap" is a strip of caplets running from the first reset out to five years. Swaption quotes use a two-dimensional grid, expiry × tenor, e.g. "2y5y" is a 2-year option to enter a 5-year swap. Once you internalize the tenor grid, all this nomenclature becomes self-explanatory.

16.3 15.2 Caplet Price under the Risk-Neutral Measure

Our job is to price a caplet today, at time t. There are two natural candidates for the numeraire we use to discount the random future payoff: the money-market account B_t, which is universal and always available, or the zero-coupon bond $P_t(t_k)$ maturing on the payment date, which is more bespoke but (as we will see) much cleaner. Let us try the money-market account first, and see why it fails.

Let $g_t^{(k)}$ denote the price at time t of a caplet maturing at t_k (payment date). Under the risk-neutral measure $\mathbb{Q}$ with bank-account numeraire $B_t = \exp\int_0^t r_s \, \mathrm{d}s$:

$$\frac{g_t^{(k)}}{B_t} \;=\; \mathbb{E}_t^{\mathbb{Q}}\left[\frac{\Delta t_k \left(L_{t_{k-1}}(t_k) - K\right)_+}{B_{t_k}}\right]. \tag{15.4}$$

This is awkward because B_{t_k} is random and correlated with $L_{t_{k-1}}(t_k)$. Remedy: use the t_k-bond as numeraire (the T-forward measure from Chapter 5).

> Intuition. B_{t_k} depends on the entire short-rate path from t to t_k, and high rates imply both a high LIBOR fixing *and* heavy discounting — these two effects do not separate inside the expectation. Switching to $P_t(t_k)$ as numeraire effectively "pre-pays" the discount at time t, so the only random thing left inside the expectation is the payoff itself. The cost is a change of measure: the driving Brownian acquires a drift $B_t(t_k)\sigma$ (the generic Girsanov shift of Chapter 5), but that drift vanishes for the one quantity we care about — the forward rate — because forward rates are, by construction, ratios of tradables to the new numeraire.

Before we actually execute that change, it is worth pausing on the intuition about why the bank-account numeraire is awkward. In an equity option, the underlying (a stock price) and the numeraire (the bank account) are almost

independent in practice — rates are not the primary driver of stock variance, and for short-dated equity options you can usually treat them as constants. In a rate option, the underlying *is* a rate, and the bank account is driven by the *same* rate. When LIBOR realizes high, the money-market account has also grown fast, so the discount factor $1/B_{t_k}$ has shrunk. The payoff goes up, the discount goes down, and the two are negatively correlated inside the same expectation. You cannot factor them, and you cannot pull one out. This is exactly the situation a good change of numeraire is designed to fix: rather than fight the correlation, we absorb the discounting into the numeraire by choosing a numeraire whose value on the payment date exactly cancels the discount.

16.4 15.3 Change of Numeraire — the T_i-Forward Measure

Under the t_k-forward measure $\mathbb{Q}^{t_k}$, whose numeraire is $P_t(t_k) = P(t, t_k)$ (see Chapter 5 for the general density-process construction and Girsanov drift shift):

$$\frac{g_t^{(k)}}{P_t(t_k)} \;=\; \mathbb{E}_t^{\mathbb{Q}^{t_k}}\left[\frac{\left(L_{t_{k-1}}(t_k) - K\right)_+}{P_{t_k}(t_k)}\right]\Delta t_k \;=\; \mathbb{E}_t^{\mathbb{Q}^{t_k}}\left[\left(L_{t_{k-1}}(t_k) - K\right)_+\right]\Delta t_k, \tag{15.5}$$

since $P_{t_k}(t_k) = 1$. Read (15.5) carefully. The discount factor $1/B_{t_k}$ that plagued (15.4) has been replaced by $1/P_{t_k}(t_k) = 1$, because the t_k-bond is worth exactly one dollar on its own maturity. All of the discounting has been absorbed into the $P_t(t_k)$ that sits outside the expectation. What is left inside the expectation is just the clean payoff. This is the payoff of a textbook call option on $L_{t_{k-1}}(t_k)$, evaluated under the measure $\mathbb{Q}^{t_k}$, with no discount to worry about.

Plugging in (15.2), $L_{t_{k-1}}(t_k) = \frac{1}{\Delta t_k}\left[\frac{1}{P_{t_{k-1}}(t_k)} - 1\right]$.

Define the likelihood ratio (key ratio that becomes a martingale under $\mathbb{Q}^{t_k}$):

$$X_t \;=\; \frac{P_t(t_{k-1})}{P_t(t_k)}, \qquad X_{t_{k-1}} = \frac{1}{P_{t_{k-1}}(t_k)}. \tag{15.6}$$

Why is X_t the "right" object? Because X_t is a ratio of two tradable zero-coupon bonds, one of which is the numeraire. That is the generic template for a $\mathbb{Q}^{t_k}$-martingale: any tradable divided by the numeraire is a martingale under the measure associated with that numeraire (Chapter 5). No drifts to calculate, no Itô acrobatics. The caplet payoff is an affine function of $X_{t_{k-1}}$, which is itself an affine function of $L_{t_{k-1}}(t_k)$, so the whole pricing problem boils down to the distribution of X (or equivalently L) at the fixing date.

16.4.1 Vasicek illustration

To make this concrete, let us do the change of measure inside a specific short-rate model where bond dynamics are explicit. The Vasicek model (Chapter 12) is the cleanest choice because bond prices are affine-exponential in the short rate, which makes every calculation transparent.

In the Vasicek model bond prices are affine (Chapter 12):

$$P_t(T) \;=\; e^{A_t(T) - B_t(T)\, r_t}, \qquad \frac{\mathrm{d}P_t(T)}{P_t(T)} \;=\; r_t\,\mathrm{d}t \;-\; B_t(T)\,\sigma\,\mathrm{d}\widetilde{W}_t, \tag{15.7}$$

by Itô (the drift is the short rate, and $B_t(T)\sigma$ is the bond volatility). Notice the intuitive content: bond prices drift upward at the short rate r_t under the risk-neutral measure (that is the "bonds earn the riskless rate on average" statement), and the volatility is $B_t(T)\sigma$ — bigger for longer-dated bonds, because a longer bond has more duration to be shaken around by the short-rate shock.

Taking $\mathrm{d}\ln$ of X_t:

$$\frac{\mathrm{d}X_t}{X_t} \;=\; (\mathrm{vol}\ t_{k-1} - \mathrm{vol}\ t_k) \;=\; \left(-B_t(t_{k-1}) + B_t(t_k)\right)\sigma\,\mathrm{d}\widetilde{W}_t^{(k)}, \tag{15.8}$$

where $\widetilde{W}_t^{(k)}$ is a Brownian motion under $\mathbb{Q}^{t_k}$. The drift has cancelled — X_t is a $\mathbb{Q}^{t_k}$-martingale, consistent with being a ratio of a tradable ($P_t(t_{k-1})$) and the numeraire ($P_t(t_k)$). This cancellation is not an accident of the Vasicek model; it is a general fact about changes of numeraire (Chapter 5), of which Vasicek is just a concrete instance.

Write the bond-vol differential as

$$\Sigma_t \;\equiv\; -B_t(t_{k-1}) + B_t(t_k), \qquad \mathrm{d}\widetilde{W}_t^{(k)} = B_t(t_k)\,\sigma\,\mathrm{d}t + \mathrm{d}\widetilde{W}_t. \tag{15.9}$$

The second equation is the key change-of-measure statement: when you switch the numeraire from the bank account B_t to the t_k-bond $P_t(t_k)$, the Brownian motion that drives everything gets a drift $B_t(t_k)\sigma$. This drift is exactly the bond's volatility, which is exactly the amount by which the new numeraire differs from the old. That is Girsanov talking (Chapter 5); the story is the same whatever the driving model, as long as the numeraire is a strictly positive tradable.

Then

$$g_t^{(k)} \;=\; P_t(t_k)\,\mathbb{E}_t^{\mathbb{Q}^{t_k}}\Big[\,(X_{t_{k-1}} - \mathcal{K})_+\,\Big]\,\Delta t_k, \qquad \mathcal{K} \equiv 1 + \Delta t_k\,K. \tag{15.10}$$

Notice the strike transformation. In terms of LIBOR the strike was K; in terms of $X = 1/P_{t_{k-1}}(t_k)$ it becomes $\mathcal{K} = 1 + \Delta t_k K$. This is just a re-expression of the same kink in the payoff function; the kink sits at whatever value of the underlying makes the pre-$(\cdot)_+$ expression vanish. Because X and L are affinely related, an option on one is exactly an option on the other, with a re-scaled strike and a re-scaled notional. No information lost.

Because X_t is a log-normal $\mathbb{Q}^{t_k}$-martingale (in the Gaussian Vasicek case), this is exactly a Black-Scholes expectation:

$$\boxed{g_t^{(k)} \;=\; P_t(t_k)\Big[\,X_t\,\Phi(d_+) - \mathcal{K}\,\Phi(d_-)\,\Big]} \tag{15.11}$$

$$d_\pm \;=\; \frac{\ln(X_t/\mathcal{K}) \;\pm\; \frac{1}{2}\int_t^{t_{k-1}} \Sigma_s^2\,\mathrm{d}s}{\Big(\int_t^{t_{k-1}} \Sigma_s^2\,\mathrm{d}s\Big)^{1/2}}. \tag{15.12}$$

The term $\int_t^{t_{k-1}} \Sigma_s^2\,\mathrm{d}s$ is the cumulative quadratic variation of $\ln X$ up to the fixing date. In a stationary Black-76 framing it will simply become $\sigma^2(t_{k-1} - t)$ — a single vol times time to fix — but in a more general model it can be time-dependent. Either way, the structure is Black-Scholes: log-moneyness divided by a standard deviation.

16.4.2 Cap as a sum of caplets

$$\mathrm{Cap}_t \;=\; \sum_{k=1}^{n} g_t^{(k)}. \tag{15.13}$$

This single equation is the whole point of the change of numeraire. We got here because the cap payoff is a *sum* of single-date payoffs; each single-date payoff depends on a fixing at a single date; and each fixing can be priced under its *own* forward measure. There is no need for a joint model of the whole forward-rate curve — you only need one marginal distribution per fixing date. That is a huge reduction in modelling burden, and it is the structural reason caps are vastly easier to price than path-dependent products.

16.5 15.4 Black Implied Vols — the Market Convention

The Vasicek derivation gave us a Black-style formula, but the market standard is to skip the short-rate model and assume directly that each forward LIBOR is log-normal under its own t_k-forward measure (GBM for forward rates). This is the LIBOR market model (LMM / BGM) seed.

Why skip the short-rate model? Because practitioners care about cap and swaption prices, not about calibrating some hidden short-rate process. If we can directly parameterize the quantity that appears in the payoff — the

forward rate — with a log-normal distribution, and use the market to back out the one free parameter (the Black caplet vol), then we have a quoting convention that is both unambiguous and model-free in spirit. The short-rate model is a means to an end; the Black caplet vol is the end itself.

Let $L_{t_{k-1}}(t_k) \stackrel{d}{=} L_t(t_{k-1}, t_k) \cdot Y_{t_{k-1}}$, where $L_t(t_{k-1}, t_k)$ is today's forward rate for the $[t_{k-1}, t_k]$ period:

$$L_t(t_{k-1}, t_k) \;=\; \frac{1}{\Delta t_k}\left[\frac{P_t(t_{k-1})}{P_t(t_k)} - 1\right]. \tag{15.14}$$

Equation (15.14) says the forward rate is directly readable off the zero-coupon curve. There is no need to observe it in some separate market; once you have bond prices, you have forward rates. Conversely, forward rates and bond prices are two languages for the same object — if you quote the curve in forward-rate language, you can rebuild every bond; if you quote it in bond language, you can rebuild every forward. This dual view matters because caps and floors are effectively bets on the *forward-rate* representation, while bond desks tend to think in the *bond-price* representation. The conversion is mechanical but worth internalizing.

One checks, using $P_t(t_{k-1}) = P_t(t_k)(1 + \Delta t_k L_t(t_{k-1}, t_k))$, that $L_t(t_{k-1}, t_k)$ is a ratio of tradables to the numeraire $P_t(t_k)$, hence a $\mathbb{Q}^{t_k}$-martingale. To see the tradable identification explicitly, note that $P_t(t_{k-1})/P_t(t_k) - 1$ can be interpreted as the time-t price of a portfolio: "buy one t_{k-1}-bond, short one t_k-bond." That portfolio's price divided by the numeraire $P_t(t_k)$ is a $\mathbb{Q}^{t_k}$-martingale by the generic tradable-over-numeraire rule (Chapter 5). The constant $1/\Delta t_k$ just rescales the units.

Forward-rate strip under three curve shapes *Three stylized zero-curve regimes — normal (upward-sloping), flat, and inverted (downward-sloping) — each imply a very different forward-LIBOR strip* $\{L_0(t_{k-1}, t_k)\}_k$.

In practice the shape of the forward strip is one of the first things a rates trader looks at. An upward-sloping curve ("normal") says the market expects future short rates to rise, so out-in-the-future caplets will start life closer to the money than the first caplet. A flat curve says the market is pricing roughly constant expected short rates. An inverted curve signals expected easing. All three can coexist with the same spot rate, and the caplet strip will behave very differently under each.

Postulate driftless GBM under that measure:

$$\frac{\mathrm{d}L_t^{(k)}}{L_t^{(k)}} \;=\; \sigma\,\mathrm{d}W_t^{t_k}, \qquad L_t^{(k)} \equiv L_t(t_{k-1}, t_k). \tag{15.15}$$

"$\mathbb{Q}$-Brownian motion" here means under $\mathbb{Q}^{t_k}$. This is the direct postulate: forward LIBOR is a log-normal martingale under its own forward measure. The only free parameter is the volatility σ; by assumption the drift is zero (which is forced, because $L^{(k)}$ is a martingale under $\mathbb{Q}^{t_k}$). There is nothing to solve — given σ, the distribution of $L_{t_{k-1}}^{(k)}$ is pinned down.

Starting at t:

$$L_{t_{k-1}}^{(k)} \;=\; L_t(t_{k-1}, t_k)\,\exp\!\Big(-\tfrac{1}{2}\sigma^2(t_{k-1} - t) \;+\; \sigma(W_{t_{k-1}}^{t_k} - W_t^{t_k})\Big). \tag{15.16}$$

This is the textbook GBM solution: log-mean $-\frac{1}{2}\sigma^2(t_{k-1} - t)$, log-variance $\sigma^2(t_{k-1} - t)$, multiplicative shift by the forward rate today. The log-normality is what keeps forward LIBOR strictly positive — a feature that made perfect sense in the regime before the 2008 financial crisis, and that causes trouble afterwards (see §15.4b below).

16.5.1 Black's formula for a caplet

Plug (15.16) into (15.5). Since $L_{t_{k-1}}^{(k)}$ is log-normal with mean $L_t^{(k)}$ under $\mathbb{Q}^{t_k}$:

$$g_t^{(k)} \;=\; P_t(t_k)\,\mathbb{E}^{\mathbb{Q}^{t_k}}\Big[\,(L_{t_{k-1}}^{(k)} - K)_+\,\Big]\,\Delta t_k,$$

$$\boxed{g_t^{B(k)} \;=\; P_t(t_k)\Big\{\,L_t(t_{k-1}, t_k)\,\Phi(d_+) \;-\; K\,\Phi(d_-)\,\Big\}\,\Delta t_k} \tag{15.17}$$

$$d_{\pm} \;=\; \frac{\ln\!\big(L_t(t_{k-1},t_k)/K\big) \;\pm\; \frac{1}{2}\sigma^2(t_{k-1}-t)}{\sigma\,(t_{k-1}-t)^{1/2}}. \tag{15.18}$$

This is the Black-76 caplet formula. Each caplet has its own vol $\sigma_k^{B\ell}$ (the "Black caplet vol") — there is one implied vol per caplet. Structurally, (15.17) looks identical to a Black-Scholes call on a non-dividend-paying stock with spot L_t, strike K, volatility σ, time to expiry $t_{k-1}-t$, with one crucial difference: the discount factor outside the bracket is $P_t(t_k)$, which uses the payment date t_k, not the fixing date t_{k-1}. That subtle dislocation — pay at t_k, fix at t_{k-1} — is exactly why the forward measure is valuable. The discount uses the right number, even though the randomness is resolved one period earlier.

The Black-76 framing is also what makes caps and floors trade as a vol market rather than a price market. Dealers quote Black caplet vols (or flat cap vols); buyers and sellers negotiate in vol units; prices are computed from vols on the fly. This is directly analogous to the equity-option market, with one important distinction: each caplet has its own vol, because each one is struck against a different forward rate at a different maturity. The object that gets quoted is therefore not a single vol number but a *term structure* of caplet vols.

Alternative derivation: caplet as a put on a zero-coupon bond. There is a second, model-independent route to the caplet that does not require a measure change. Start from the caplet payoff at t_{k-1} (i.e., one period before payment): $\Delta t_k(L_{t_{k-1}}(t_k)-K)_+ = (1-(1+K\Delta t_k)P_{t_{k-1}}(t_k))_+$, using $1+L_{t_{k-1}}(t_k)\Delta t_k = 1/P_{t_{k-1}}(t_k)$. Multiplying through by $(1+K\Delta t_k)$ and rearranging,

$$(1+K\Delta t_k)\cdot\Delta t_k\,(L_{t_{k-1}}(t_k)-K)_+ \;=\; \big(K_{\text{bond}}-P_{t_{k-1}}(t_k)\big)_+\,, \qquad K_{\text{bond}} := \frac{1}{1+K\Delta t_k}.$$

Discounting from t_k back to t under the money-market measure, the caplet is equivalent (up to the scalar $(1+K\Delta t_k)$) to a European *put* on the zero-coupon bond $P_{\cdot}(t_k)$ with expiry t_{k-1} and strike K_{bond}. Any short-rate model that gives a closed-form European bond-option formula — the Vasicek-Black put of CH13 §13.6 is the canonical example — therefore prices the caplet without a measure change. This alternative is pedagogically useful because it links caplets directly to the bond-option technology of CH13 and provides a clean consistency check between the Black-76 caplet formula (forward-measure route) and the Vasicek-Black bond-put formula (money-market-measure route) — agreement of the two routes is a good diagnostic for any rates-model implementation.

Cap price vs strike (Black model) *Cap price vs strike (Black model)*

The cap price is monotonically decreasing in the strike K (higher strike, less in-the-money, less intrinsic, less option value). For very low strikes the cap behaves like a forward swap (all caplets deep in the money, optionality worth almost nothing on top of intrinsic). For very high strikes the cap behaves like a lottery ticket (all caplets deep out of the money, value almost entirely time value). The curvature between these extremes is driven by the Black caplet vol term structure; that is what makes the cap surface genuinely interesting.

16.5.2 Cap in terms of Black vols

$$\sum_{k=1}^{n} g_t^{B(k)} \;=\; \sum_{k=1}^{n} g_t^{B(k)}(\sigma_k^{B\ell}), \tag{15.19}$$

with $\sigma_k^{B\ell}$ the k-th caplet Black implied vol. A flat cap vol $\sigma_n^{C\ell}$ is the single σ such that $\sum_k g^B(\sigma) = \text{Cap}_t^{\text{mkt}}$ — conceptually useful but coarser than the per-caplet term structure:

```
caplet-vol strip  (each row = one market-quoted cap)
  _1^{B }                                  1-period cap  (1 caplet)
  _2^{B }                              2-period cap  (3 caplets)
  _3^{B }                          3-period cap  (5 caplets)
                            → reset date
              t_0  t_1  t_2  t_3  t_4 …

Stripping: solve for _k^{B } recursively so each cap row re-prices
its quoted market premium. Adding one tenor adds one unknown vol.
```

One "strips" caplet vols recursively from quoted cap prices with increasing maturity. The stripping procedure is simple in principle and sometimes fiddly in practice. You start at the shortest-dated cap — say, a 1-year cap with four quarterly caplets — and find the single flat vol that reprices it. That flat vol is imposed as the common vol for caplets 1–4 (or, more defensibly, the first caplet inherits its vol and the later three share a common number). You then take the next cap (the 2-year), subtract the known value of caplets 1–4 from its market price, and the residual must equal the value of caplets 5–8 under *their* common vol. Solve one-dimensionally for that vol. Repeat out the curve. The result is a term structure $\sigma_k^{B\ell}$ defined pointwise at each tenor, usually interpolated smoothly in between.

One subtlety: the market actually quotes caps at a few standard strikes (ATM plus a handful of fixed-strike grids), and the stripping must be done at each strike separately, giving a full two-dimensional caplet-vol surface indexed by (expiry, strike). Traders then fit parametric forms (SABR being the workhorse) through the strike dimension at each expiry, creating a smooth implied-vol surface that can be interpolated and extrapolated.

Caplet-vol term structure vs flat cap vol *A stylized upward-humped caplet-vol curve $\sigma_k^{B\ell}$ — typical of cap markets — is shown alongside the flat cap vol $\bar{\sigma}^{C\ell}$ that reprices the whole 5-y cap. The flat cap vol averages the strip (roughly in a vega-weighted sense), which is why traders prefer the per-caplet term structure for hedging.*

The humped shape in the figure is common in real markets. Very short-dated caplets tend to have lower vol because the first fixing is close and the near-term rate path is well-anchored by central-bank policy. Very long-dated caplets also tend to have lower vol because long-run rate expectations are less volatile than near-term dislocations. The peak tends to sit around the 2–3 year point, where central-bank uncertainty and term-structure dynamics combine to produce the widest distribution of outcomes. Flat cap vols smear this shape into a single number, and while they are handy for quick quoting, no serious hedger uses them to compute vega: you want the sensitivity to each individual caplet vol, not to the weighted average.

> Why cap = sum of caplets exactly (no cross-terms). Each caplet pays $\Delta t_k(L_{t_{k-1}}(t_k) - K)_+$ on a *single* fixed date t_k. The cap's payoff is the pointwise sum over k, and expectation is linear, so $\mathrm{Cap}_t = \sum_k g_t^{(k)}$ — no covariance or correlation terms ever appear. This is what makes caps structurally simpler than, say, a swaption (which is an option on a *sum* of rates and so does carry cross-sectional covariance). It also means each caplet can be priced under its *own* forward measure $\mathbb{Q}^{t_k}$ with its *own* Black vol $\sigma_k^{B\ell}$ — the chapter's central decomposition.

The italicized distinction in that callout is worth extra emphasis. In a cap, the nonlinearity acts caplet-by-caplet: each caplet takes a max, and then you add. In a swaption, the nonlinearity acts on the *already-summed* stream of cashflows: you add up the swap values first, and only then take a max. This difference is what forces swaptions onto a different numeraire (the annuity) and makes them sensitive to forward-rate correlations that caps are not. It is also why caps and swaptions, although both built from the same underlying rate dynamics, carry different information: cap vols reflect the marginal distribution of each forward rate, while swaption vols additionally reflect the covariance structure across forward rates. A full rates model must calibrate to both simultaneously to be internally consistent.

16.6 15.4b Bachelier vs Black in a Negative-Rate World

The Black-76 framework above rests on one assumption that is uncontroversial in most of modern financial history: that forward rates are strictly positive. Log-normality forces this, since $L_{t_{k-1}}^{(k)} = L_t^{(k)} \exp(\text{something})$ and the exponent is always strictly positive. For decades this was a reasonable view of the world — a negative short rate was almost unthinkable, because banks could always hold physical cash instead of lending at a loss, and so the zero lower bound seemed binding by arbitrage.

The post-2008 experience blew that up. The European Central Bank, the Swiss National Bank, the Bank of Japan, and others spent multiple years with deposit rates set explicitly below zero, and market forward rates on EURIBOR and JPY curves traded visibly negative across a wide range of tenors. Suddenly Black-76 could not even *represent* a market quote — you cannot compute $\ln(L/K)$ if L is negative, and the model assigns zero probability to outcomes that were happening in real time.

Market practice migrated in two directions. The first was the *shifted-Black* model: write $L \to L + \alpha$ for some positive shift α, assume that $L + \alpha$ is log-normal under $\mathbb{Q}^{t_k}$, and price Black-76 with the shifted variables. This is a cheap fix — it preserves the closed-form and the market intuition of a Black vol, while permitting negative realizations of L itself as long as they stay above $-\alpha$.

The second, more structural fix was the *Bachelier* or *normal* model: assume that the forward rate follows arithmetic Brownian motion rather than geometric, so

$$\mathrm{d}L_t^{(k)} \;=\; \sigma_N \,\mathrm{d}W_t^{t_k},$$

with σ_N the *normal* (basis-point) volatility. The caplet price under Bachelier is another closed-form expression involving Φ and the normal density, with a moneyness $(L_t - K)/(\sigma_N\sqrt{t_{k-1} - t})$. The normal model prices forward rates as a Gaussian distribution centered on the current forward, with no lower bound — perfectly happy to assign positive probability to negative rates, because that is exactly what happened in reality.

The two conventions are related by a rough rule of thumb: at the money, $\sigma_N \approx \sigma_B \cdot L$, where σ_B is the Black vol and L is the underlying forward rate. So a 25% Black vol on a 4% forward is roughly equivalent to a 100 bp normal vol. Away from ATM they diverge, and the choice of quoting convention matters. Modern rate-derivative desks now routinely switch between conventions, with normal vols dominating European markets (especially EUR and CHF) and shifted-Black or straight Black vols still common in USD markets where the zero bound has remained in force for most of history.

The same observations apply to swaptions — normal swaption vols are now the industry standard in much of Europe, and a senior quant cannot afford to be sloppy about which convention a given quote is in.

16.7 15.5 Swaptions

So far we have priced an option on a *single* fixing of a forward rate. A swaption is an option on an entire *swap* — a bundled set of fixings. The payoff structure is similar in spirit but different in a crucial detail: the max is taken at the swap level, not at the caplet level. This has real consequences for what modelling assumptions we need and what numeraire makes the problem tractable.

A swaption (swap option) gives A the right, at t_0, to enter a payer swap with fixed leg F, reset/payment dates $t_1, \dots, t_n$, floating rates $L_{t_0}(t_1), L_{t_1}(t_2), \dots, L_{t_{n-1}}(t_n)$. So on the exercise date t_0, A looks at the then-current swap value: if the floating leg is worth more than the fixed leg (i.e. rates have risen above F on average), A exercises and enters the swap at the pre-agreed fixed rate, locking in a favorable position. If rates have fallen, A lets the option expire.

16.7.1 Payoff at t_0

$$C_{t_0} \;=\; \big(V_{t_0}^{\ell} - V_{t_0}^{\text{fixed}}\big)_+, \tag{15.20}$$

where

$$V_{t_0}^{\text{fixed}} \;=\; F\sum_{k=1}^{n} P_{t_0}(t_k)\,\Delta t_k, \qquad V_{t_0}^{\ell} \;=\; 1 - P_{t_0}(t_n). \tag{15.21}$$

(The floating leg's PV at t_0 is $1 - P_{t_0}(t_n)$ — a standard "par" identity from Chapter 13.) The par identity says: receive a stream of LIBOR fixings from t_0 to t_n is, in present-value terms, the same as receiving \$1 at t_0 and paying back \$1 at t_n. This is because each floating cashflow "resets" the invested principal at the then-current rate, so the floating leg is PV-equivalent to principal in minus principal out, which is $1 - P_{t_0}(t_n)$.

Hence

$$C_{t_0} \;=\; \Big(1 - P_{t_0}(t_n) \;-\; F\sum_{k=1}^{n} P_{t_0}(t_k)\,\Delta t_k\Big)_+. \tag{15.22}$$

Now the max operates on a nonlinear combination of $n+1$ bond prices — $P_{t_0}(t_1), \dots, P_{t_0}(t_n)$ all enter through the fixed leg, and $P_{t_0}(t_n)$ also enters through the floating leg. Under a multi-factor short-rate model these bond prices are generically correlated, and the max cannot be decomposed caplet-style. This is the structural reason swaptions are harder than caps.

16.7.2 In an affine (Vasicek) model: find r^*

There is a slick trick available in one-factor affine models (recall the affine ansatz and ODE system of Chapter 12). Because all bond prices are monotone (decreasing) in the single state variable r_{t_0}, the swaption's intrinsic value is monotone in r_{t_0} too. So the exercise region $\{C_{t_0} > 0\}$ is a half-line in r_{t_0}, and there is a single critical value r^* that separates "in the money" from "out of the money."

In an affine short-rate model, $P_t(T) = e^{A_t(T) - B_t(T)\, r_t}$. At t_0, the swaption is in-the-money iff a threshold condition on r_{t_0} is satisfied. Find r^* such that

$$\sum_{k=1}^{n} c_k\, P_{t_0}(t_k; r^*) \;=\; 1, \qquad c_k = F\,\Delta t_k \;(k < n), \;\; c_n = 1 + \Delta t_n F. \tag{15.23}$$

Then exercise region is $\{r_{t_0} > r^*\}$ (bonds monotone $\downarrow$ in r), and

$$C_{t_0} \;=\; \Big(1 - \sum_k c_k\, P_{t_0}(t_k)\Big)\mathbf{1}_{r_{t_0} > r^*} \;=\; \mathbf{1}_{r_{t_0} > r^*} - \sum_k c_k\, P_{t_0}(t_k)\, \mathbf{1}_{r_{t_0} > r^*}. \tag{15.24}$$

Using $\mathbf{1}_{r_{t_0} > r^*} = \mathbf{1}_{P_{t_0}(t_k) < P_k^*}$ with $P_k^* \equiv P_{t_0}(t_k; r^*)$ (monotone in r), so

$$-B_{t_0}(t_k)\, r_{t_0} \;<\; -B_{t_0}(t_k)\, r^* \;\iff\; A_{t_0}(t_k) - B_{t_0}(t_k)\, r_{t_0} \;<\; A_{t_0}(t_k) - B_{t_0}(t_k)\, r^*. \tag{15.25}$$

This is the Jamshidian trick in miniature (compare Chapter 13's European bond-option derivation): a joint indicator over all n bond prices reduces, in a one-factor model, to a single scalar condition on r_{t_0}. Each bond-price indicator can then be handled separately, and the swaption decomposes into a portfolio of bond options — each one priced with its own Black-style formula as in Chapter 13.

16.7.3 Risk-neutral pricing of the swaption

$$\begin{aligned} g_t \;&=\; \mathbb{E}_t^{\mathbb{Q}}\left[e^{-\int_t^{t_0} r_s \mathrm{d}s} \big(\mathbf{1}_{r_{t_0} > r^*} - \sum_k c_k\, P_{t_0}(t_k)\, \mathbf{1}_{r_{t_0} > r^*}\big) \right] \\ &=\; \mathbb{E}_t^{\mathbb{Q}}\Big[e^{-\int_t^{t_0} r_s \mathrm{d}s}\, \mathbf{1}_{r_{t_0} > r^*} \Big] \;-\; \sum_k c_k\, \mathbb{E}_t^{\mathbb{Q}}\Big[e^{-\int_t^{t_0} r_s \mathrm{d}s}\, P_{t_0}(t_k)\, \mathbf{1}_{P_{t_0}(t_k) < P_k^*} \Big]. \end{aligned} \tag{15.26}$$

Each of these expectations can now be handled by an appropriate change of numeraire. The first term uses $P_t(t_0)$ (the t_0-bond), which converts the integral-times-indicator into a pure $\mathbb{Q}^{t_0}$-probability. The sum of later terms uses $P_t(t_k)$ for each k, again turning each summand into a $\mathbb{Q}^{t_k}$-probability times a discount factor. This is the affine-model workhorse.

Use $P_t(t_k)$ as the numeraire for the k-th term (change from $\mathbb{Q}$ to $\mathbb{Q}^{t_k}$):

$$h_t^k \;\equiv\; \mathbb{E}_t^{\mathbb{Q}}\Big[e^{-\int_t^{t_0} r_s \mathrm{d}s}\, P_{t_0}(t_k)\, \mathbf{1}_{P_{t_0}(t_k) < P_k^*} \Big], \qquad \frac{h_t^k}{P_t(t_k)} = \mathbb{E}_t^{\mathbb{Q}^{t_k}}\big[\mathbf{1}_{P_{t_0}(t_k) < P_k^*}\big]. \tag{15.27}$$

Hence

$$h_t^k \;=\; P_t(t_k)\, \mathbb{Q}^{t_k}(P_{t_0}(t_k) < P_k^*). \tag{15.28}$$

Under $\mathbb{Q}^{t_k}$ the bond dynamics carry the numeraire adjustment (Girsanov drift shift, Chapter 5):

$$\begin{aligned} \frac{\mathrm{d}P_t(t_n)}{P_t(t_n)} \;&=\; r_t\, \mathrm{d}t - B_t(t_n)\, \sigma\, \mathrm{d}\widetilde{W}_t \;=\; (r_t + B_t^2(t_k)\, \sigma^2)\mathrm{d}t - B_t(t_n)\, \sigma\, \mathrm{d}\widetilde{W}_t^{\,k}, \\ \mathrm{d}\widetilde{W}_t^{\,k} \;&=\; B_t(t_k)\, \sigma\, \mathrm{d}t \;+\; \mathrm{d}\widetilde{W}_t. \end{aligned} \tag{15.29}$$

The remaining term is a Gaussian tail:

$$\mathbb{Q}^{t_k}(P_{t_0}(t_k)^{-1} > P_k^{*-1}) \;=\; \mathbb{Q}^{t_k}(X_{t_0} > P_k^{*-1}), \qquad X_t = \frac{P_t(t_0)}{P_t(t_k)}, \quad t \geq t_0, \tag{15.30}$$

a $\mathbb{Q}^{t_k}$-martingale (same structure as §15.3). So each term collapses to a Gaussian tail probability, computable in closed form once the model's bond-vol structure is pinned down. The whole swaption price is then a finite sum of Φ's, each weighted by an appropriate bond price.

This Vasicek approach is historically important and still pedagogically clean, but it is not how modern desks price swaptions at their core. Modern practice uses the annuity measure, described next, which gives a one-shot Black formula directly on the par swap rate — no Jamshidian decomposition, no need for a calibrated short-rate model.

16.8 15.6 Swaption as Black on the Swap Rate

The Jamshidian route above works, but is ugly — it introduces a model-dependent threshold r^*, requires an affine short-rate specification, and does not directly give a Black formula on the swap rate, which is what the market actually quotes. There is a cleaner approach: choose a numeraire that matches the *structure* of the payoff. The apparatus is exactly the general change-of-numeraire of Chapter 5; we just pick a portfolio numeraire instead of a single-bond numeraire.

Collect terms using the annuity numeraire:

$$C_{t_0} \;=\; \Big(1 - P_{t_0}(t_n) - F\sum_k \Delta t_k\, P_{t_0}(t_k)\Big)_+, \qquad X_t \;\equiv\; \sum_k \Delta t_k\, P_t(t_k) \quad \text{(annuity)}. \tag{15.31}$$

The annuity $X_t = \sum_k \Delta t_k P_t(t_k)$ is a strictly positive portfolio of zero-coupon bonds — exactly the present value of \$1 received on every payment date of the swap's fixed leg. Because it is a portfolio of tradables, it is itself tradable and strictly positive, so it qualifies as a numeraire (Chapter 5).

Assume X_t is a valid numeraire:

$$\frac{\mathrm{d}X_t}{X_t} \;=\; r_t\,\mathrm{d}t + (\cdots)\,\mathrm{d}\widetilde{W}_t, \qquad X_t > 0 \text{ a.s.} \tag{15.32}$$

Change to $\mathbb{Q}^X$ with numeraire X_t:

$$\begin{aligned}\frac{g_t}{X_t} \;&=\; \mathbb{E}_t^{\mathbb{Q}^X}\left[\frac{(1 - P_{t_0}(t_n) - F\sum_k \Delta t_k P_{t_0}(t_k))_+}{\sum_k \Delta t_k P_{t_0}(t_k)}\right] \\ &=\; \mathbb{E}_t^{\mathbb{Q}^X}\left[\left(\frac{1 - P_{t_0}(t_n)}{\sum_k \Delta t_k P_{t_0}(t_k)} - F\right)_+\right] \;=\; \mathbb{E}_t^{\mathbb{Q}^X}[(S_{t_0} - F)_+],\end{aligned} \tag{15.33}$$

where the par swap rate is (recall Chapter 13's IRS section)

$$\boxed{S_t \;=\; \frac{1 - P_t(t_n)}{\sum_k \Delta t_k\, P_t(t_k)}} \tag{15.34}$$

— and S_t is a $\mathbb{Q}^X$-martingale (it is a ratio of the tradable $1 - P_t(t_n)$ to the numeraire X_t). This is the critical insight. The par swap rate is defined as the value of the fixed rate that makes the two legs of the swap equal on net at inception. By construction, S_t = (floating PV)/(annuity PV). Both numerator and denominator are tradables, and the denominator is the numeraire, so the ratio is automatically a $\mathbb{Q}^X$-martingale by the tradable-over-numeraire principle (Chapter 5).

Hence

$$g_t \;=\; X_t\, \mathbb{E}_t^{\mathbb{Q}^X}\big[(S_{t_0} - F)_+\big]. \tag{15.35}$$

All of the complicated n-dimensional bond-price structure has collapsed into a scalar call on the single random variable S_{t_0}. If we posit a distribution for S_{t_0} under $\mathbb{Q}^X$, the entire expectation is tractable.

16.8.1 Log-normal swap-rate model (LSM)

Assume

$$\frac{\mathrm{d}S_t}{S_t} \;=\; \sigma\, \mathrm{d}\widetilde{W}_t^X \;\Longrightarrow\; \text{Black formula for swaption.} \tag{15.36}$$

Then (under GBM with mean S_t):

$$\boxed{g_t \;=\; X_t\,\big(\, S_t\,\Phi(d_+) \;-\; F\,\Phi(d_-)\,\big)} \tag{15.37}$$

$$d_\pm \;=\; \frac{\ln(S_t/F) \;\pm\; \frac{1}{2}\sigma^2(t_0 - t)}{\sigma\,(t_0 - t)^{1/2}}. \tag{15.38}$$

This is the Black swaption formula: a call on the swap rate S_t, with the annuity X_t as discount factor. Exactly the same structure as Black-Scholes on an equity, exactly the same structure as the Black caplet, with the right numeraire (X_t) replacing the bank account or the single t_k-bond.

Swaption price surface over expiry x tenor *The Black swaption price from (15.37) varies smoothly over the expiry × tenor grid — the standard market quoting convention. With flat forward rates and flat vol, longer tenor (larger annuity X_t) dominates, while longer expiry adds Black-time value.*

The swaption market therefore trades a *two-dimensional* vol surface: expiry (how long until the option decision) times tenor (how long is the underlying swap). A 2y5y swaption is the 2-year option to enter a 5-year swap. Dealers quote vols for all relevant pairs, and a senior vol trader will typically run hedges along both axes simultaneously. The caplet surface, by contrast, is one-dimensional in tenor (the expiry and the fixing date are tied together by the tenor grid).

Consistency between the caplet surface and the swaption surface is a nontrivial constraint on a full rates model. Caps give you the marginal vol of each forward rate individually; swaptions give you the vol of a weighted sum of forward rates, which depends on the correlations among them. A model that calibrates to caps alone is underdetermined on the correlation structure; a model that calibrates to swaptions alone is underdetermined on the marginals. Full-scale production models (LMM, SABR-LMM, stochastic-vol variants) try to fit both simultaneously.

> What the annuity numeraire does. The swaption payoff $\big(1 - P_{t_0}(t_n) - F\sum_k \Delta t_k P_{t_0}(t_k)\big)_+$ mixes $n+1$ bond prices through a *nonlinear* max operator. Under $\mathbb{Q}$ or any single $\mathbb{Q}^{t_k}$, none of these terms is a martingale, so no clean closed form exists. Dividing the payoff by the annuity $X_t = \sum_k \Delta t_k P_t(t_k)$ — itself a positive portfolio of tradables — produces a clean call on a single scalar, the par swap rate $S_t = (1 - P_t(t_n))/X_t$. Because $1 - P_t(t_n)$ is itself a (self-financing) tradable (buy a t_n-bond, fund with a rolling float), S_t is a ratio of a tradable to the numeraire and hence a $\mathbb{Q}^X$-martingale. The whole problem collapses to Black on one variable. This is the same trick as a caplet, just with a bigger numeraire.

16.9 15.7 Worked Example — Pricing a 1-into-1 Caplet

Concrete numbers are the best way to anchor the formulas. Let us price a single quarterly caplet using the Black-76 formula and check that the answer makes sense.

Suppose today is $t = 0$, with:

- $t_{k-1} = 1$ yr, $t_k = 1.25$ yr, $\Delta t_k = 0.25$ yr.

- Today's forward rate $L_0(1, 1.25) = 4.00\%$.
- Strike $K = 3.50\%$.
- Black caplet vol $\sigma = 25\%$.
- Discount $P_0(1.25) = 0.955$.

We are therefore looking at a caplet that fixes 1 year from now and pays 1.25 years from now, on a forward rate that is currently 4% against a strike of 3.5%. The caplet is in the money in the forward sense (forward rate above strike), so intrinsic value is positive. Time to fix is 1 year and the vol is 25%, so there is a meaningful time value component as well.

Compute:

$$d_+ = \frac{\ln(0.04/0.035) + 0.5(0.25)^2(1)}{0.25\sqrt{1}} = \frac{0.1335 + 0.03125}{0.25} = 0.659,$$
$$d_- = d_+ - 0.25 = 0.409.$$

With $\Phi(0.659) \approx 0.745$, $\Phi(0.409) \approx 0.659$:

$$g_0 = 0.955 \times (0.04 \times 0.745 - 0.035 \times 0.659) \times 0.25$$
$$= 0.955 \times (0.02980 - 0.02307) \times 0.25 = 0.955 \times 0.006734 \times 0.25 \approx 0.001607$$

i.e. about 16.07 bp of notional — the caplet PV. To see that this number is in the right ballpark, compare to the rough intrinsic value: $0.25 \times (0.04 - 0.035) \times 0.955 \approx 0.001194$, or about 11.94 bp. The Black price of 16.07 bp is above intrinsic by about 4 bp, which is the time-value contribution. That looks right: 1-year time to fix at 25% Black vol on a rate close to the money should buy you a few bp of convexity value.

Rough sanity checks like this are indispensable at a trading desk. If you compute a Black caplet value and the answer comes out *below* intrinsic, something is wrong. If the answer comes out many multiples above intrinsic on a caplet that is deep in the money, something is wrong. Time value should be non-negative, bounded by the product of vol and $\sqrt{T}$, and roughly symmetric around the ATM log-moneyness. These heuristics catch a huge fraction of data errors.

ATM vs OTM caplet decomposition *For a 1y caplet, the Black price splits into intrinsic* $P_t(t_k)\Delta t_k(L_t - K)_+$ *and time value* $g_t^{B(k)}$ *– intrinsic. The time component peaks near ATM and decays symmetrically in* $\ln(L/K)$ *— the classic log-moneyness curvature that drives cap vega.*

16.10 15.8 Worked Example — Stripping Caplet Vols from a 2-Year Cap

A harder and more realistic task is to back out caplet vols from market cap quotes. Assume we see market prices for quarterly-settled caps at two tenors, and we want to produce a caplet-vol term structure consistent with both. The mechanics of the stripping procedure are easy once you see them laid out.

Suppose quarterly caps:

- 1-yr cap (4 caplets) trades at C_4^{mkt}.
- 2-yr cap (8 caplets) trades at C_8^{mkt}.

Given already-stripped $\{\sigma_1^{B\ell}, \dots, \sigma_4^{B\ell}\}$ from the 1-yr cap, solve

$$C_8^{\text{mkt}} - \sum_{k=1}^{4} g^{B(k)}(\sigma_k^{B\ell}) \;=\; \sum_{k=5}^{8} g^{B(k)}(\sigma^\star)$$

for the common forward caplet vol $\sigma^\star$ over the fifth-through-eighth periods (one-dimensional root find). This is the bootstrap logic behind caplet-vol term structures. The left-hand side is what the market is telling you about "extra" option value beyond the 1-year cap; the right-hand side is what the Black formula gives for caplets 5–8 at an unknown common vol σ^*. One equation, one unknown, monotone in the unknown, so the root-find is straightforward.

Repeating this bootstrap at progressively longer tenors — 3-year, 4-year, 5-year, 7-year, 10-year — builds out a caplet-vol term structure. The key thing to notice is that each bootstrap step only requires the *already-stripped*

earlier caplet vols plus one additional market cap price; it is a recursive procedure that rolls forward in tenor. If any single market quote is stale or mis-recorded, the error propagates into all later vols — so real desks run multiple integrity checks as part of the stripping pipeline.

In practice, caps are quoted at several strikes simultaneously (ATM, ATM $\pm$ 25 bp, ATM $\pm$ 50 bp, and further), and the stripping must be done at each strike. The result is a two-dimensional (tenor, strike) caplet-vol surface. Practitioners typically fit a parametric smile model — SABR being the standard choice — through the strike dimension at each tenor, yielding a smooth and arbitrage-aware surface that can be interpolated and extrapolated.

16.11 15.9 LIBOR to SOFR and Why It Matters

The entire Black-76 caplet framework was built in a world where the reference floating rate was LIBOR — the London Interbank Offered Rate. LIBOR had two features that made it a natural fit for this pricing technology: it was forward-looking (quoted at the start of an accrual period for that period's rate), and it was a credit-sensitive unsecured bank rate (which gave the curve a term premium and a nonzero credit spread over a truly riskless reference).

Following the rate-fixing scandals of the 2010s, LIBOR has been substantially withdrawn. USD LIBOR for most major tenors ceased publication; GBP LIBOR is gone; EUR is largely replaced by €STR; JPY by TONA; CHF by SARON. The replacements — SOFR in the US, SONIA in the UK, €STR in the euro area, TONA in Japan — are overnight risk-free rates. They are *backward-looking* rather than forward-looking: the rate for an accrual period is computed by compounding realized overnight fixings, so you only know the period's rate at the *end* of the period, not the beginning.

This timing inversion changes several things in our framework. First, the defining identity (15.2) no longer applies cleanly — the compounded SOFR rate is not a single bond-based accrual but a product of overnight factors. Second, the forward rate concept is now subtler: you can still define a forward SOFR rate, but the convexity adjustment between the in-arrears-compounded fixing and a hypothetical in-advance fixing is not zero, and careful dealers track it. Third, the caplet/floorlet conventions have been redefined to handle backward-looking fixings — SOFR caplets pay on the payment date based on a compounded in-arrears rate, which introduces a small Black-model bias that is typically accommodated by a small adjustment to the effective fix time.

The cap market has adapted: SOFR caps now trade actively in the US, with quoting conventions analogous to the old LIBOR cap market but adapted for the in-arrears mechanics. SOFR caplet vol stripping works similarly, with the appropriate fix-time adjustments. The fundamental architecture — per-caplet Black vols under per-caplet forward measures — survives essentially intact, which is testimony to how structurally the framework is built. What changed was the identity of the underlying rate, not the pricing framework.

A senior quant in 2026 should be fully bilingual in LIBOR and SOFR conventions, particularly on books that still contain legacy LIBOR-linked instruments (some long-dated swaps and structured products will persist for years). The transition has also highlighted how important it is not to hard-code "LIBOR" anywhere in a risk system; the abstraction layer is "a simple forward rate under its own forward measure," and that abstraction is robust to the index choice.

There is one more subtlety worth mentioning about the LIBOR-SOFR transition. Because SOFR is a compounded overnight rate and carries no bank-credit component, the SOFR term structure sits systematically below the historical LIBOR curve by a basis spread that varies with credit conditions. For legacy LIBOR-linked contracts that were converted to SOFR, the industry adopted a standard fallback: the reference rate becomes "SOFR + fallback spread," where the spread is the historical median LIBOR-SOFR basis over a five-year lookback window. The cap-and-floor framework above treats that fallback spread as an additive shift to the forward curve, and the Black-76 pricing proceeds as before. Dealers hedging legacy books must track the basis risk — a cap struck against the fallback-adjusted SOFR curve has slightly different sensitivities than an otherwise-identical cap struck against a clean SOFR curve, and the difference shows up in P&L when the basis moves. This is the kind of nuance that separates a practitioner who merely knows the formulas from one who actually runs a rate-vol book.

16.12 15.10 Key Takeaways

1. Cap = strip of caplets. Each caplet is a European call on a forward LIBOR rate, fixed at t_{k-1} and paid at t_k. The "cap = sum of caplets" decomposition is exact (no cross-terms), because the max acts on each caplet's fixing separately.
2. Pricing numeraire: switching from B_t to $P_t(t_k)$ kills the correlation between discount and payoff. Under $\mathbb{Q}^{t_k}$ the forward LIBOR $L_t(t_{k-1}, t_k)$ is a martingale, because it is a ratio of the tradable $(P_t(t_{k-1}) - P_t(t_k))/\Delta t_k$ to the numeraire $P_t(t_k)$.
3. Change of measure (Chapter 5): $\mathrm{d}\widetilde{W}_t^{(k)} = B_t(t_k)\sigma\,\mathrm{d}t + \mathrm{d}\widetilde{W}_t$ in the Vasicek model of Chapter 12; more generally, drift compensation absorbs the bond-vol. This is just Girsanov applied to the $P_t(t_k)$-numeraire.
4. Black-76 caplet falls out when we postulate log-normal forward rates under $\mathbb{Q}^{t_k}$ — one implied vol per caplet, not one per cap. Quoting flat cap vols is a convenience; hedging is always in terms of the per-caplet term structure.
5. In negative-rate regimes, Black-76 fails outright because $\ln L$ is undefined. Shifted-Black and Bachelier models are the two practical fixes; normal (Bachelier) vols dominate European quoting conventions today.
6. Swaptions are priced analogously using the annuity numeraire $X_t = \sum_k \Delta t_k P_t(t_k)$, under which the par swap rate $S_t = (1 - P_t(t_n))/X_t$ is a martingale. Log-normal S_t gives the Black swaption formula. The Jamshidian route of Chapter 13 gives an alternative derivation in one-factor affine models.
7. Calibration: strip caplet vols from cap quotes in order of maturity; quote swaption vols on a (expiry, tenor) grid. Consistent full-curve models (LMM variants) calibrate to both simultaneously.
8. LIBOR is largely retired; SOFR (and other overnight-compounded risk-free rates) has taken over. The pricing architecture survives; the mechanical details of fixing and compounding change. Risk systems should abstract "LIBOR" to "simple forward rate under its own forward measure."

16.13 15.11 Reference Formulas

16.13.1 Forward rate / bond relationship

$$P_t(t_k) = P_t(t_{k-1})\big(1 + \Delta t_k\, L_t(t_{k-1}, t_k)\big)^{-1}, \qquad L_t(t_{k-1}, t_k) = \frac{1}{\Delta t_k}\left[\frac{P_t(t_{k-1})}{P_t(t_k)} - 1\right].$$

16.13.2 Caplet payoff (paid at t_k)

$$\text{payoff}_{t_k} = \Delta t_k \left(L_{t_{k-1}}(t_k) - K\right)_+.$$

16.13.3 Caplet price — Black-76

$$g_t^{(k)} = P_t(t_k)\,\Delta t_k\,[L_t(t_{k-1}, t_k)\,\Phi(d_+) - K\,\Phi(d_-)],$$
$$d_\pm = \frac{\ln(L_t/K) \pm \frac{1}{2}\sigma_k^2(t_{k-1} - t)}{\sigma_k\sqrt{t_{k-1} - t}}.$$

16.13.4 Cap

$$\text{Cap}_t = \sum_{k=1}^{n} g_t^{(k)}.$$

16.13.5 Swap rate (par)

$$S_t = \frac{1 - P_t(t_n)}{\sum_{k=1}^{n} \Delta t_k\, P_t(t_k)}.$$

16.13.6 Swaption — Black

$$g_t = X_t\,[S_t\,\Phi(d_+) - F\,\Phi(d_-)], \qquad X_t = \sum_k \Delta t_k P_t(t_k),$$
$$d_\pm = \frac{\ln(S_t/F) \pm \frac{1}{2}\sigma^2(t_0 - t)}{\sigma\sqrt{t_0 - t}}.$$

16.13.7 Change-of-measure driver under $\mathbb{Q}^{t_k}$ (Vasicek, Chapter 12 bond dynamics + Chapter 5 Girsanov)

$$\mathrm{d}\widetilde{W}_t^{(k)} = B_t(t_k)\,\sigma\,\mathrm{d}t + \mathrm{d}\widetilde{W}_t, \qquad \frac{\mathrm{d}P_t(T)}{P_t(T)} = r_t\,\mathrm{d}t - B_t(T)\,\sigma\,\mathrm{d}\widetilde{W}_t.$$

16.13.8 Bachelier / normal caplet (for negative-rate regimes)

$$\mathrm{d}L_t^{(k)} = \sigma_N\,\mathrm{d}W_t^{t_k},$$

$$g_t^{N(k)} = P_t(t_k)\,\Delta t_k\,[(L_t - K)\Phi(d) + \sigma_N\sqrt{t_{k-1} - t}\,\varphi(d)], \qquad d = \frac{L_t - K}{\sigma_N\sqrt{t_{k-1} - t}}.$$